Farmer's Handbook on Basic Agriculture

A holistic perspective of scientific agriculture

A joint initiative to impart farmers with technical knowledge on basic agriculture.

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Farmer's Handbook on Basic Agriculture

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Higher demand for agricultural raw material is now anticipated and agriculture is not any more about producing farm products and selling them exclusively at the local market. Instead farmers today have a world market to serve. But the new chances bring new challenges. Farmers and agricultural enterprises, willing to be part of the new expanding world market, not only have to take into consideration customers' preferences whom they want to serve, but also adhere to international trade regulations set by WTO and comply with high production and quality standards required by the importing countries.

Agriculture contributes around 17% to GDP and continues to be among the most important and successful sectors in India. Around 58% of the Indian population depend on agriculture for their livelihood. Apart from delivering the local industries with top quality raw materials for processing, agriculture provides almost 10% of total export earnings. However, to support the impressive Indian economic growth in the coming years, agriculture will have to contribute more towards value addition, productivity enhancement, high quality products and trained manpower to successfully tackle these challenges.

The states of Gujarat and Maharashtra have competitive advantages for the production of several commodities. However, productivity and competitiveness remains low. Rising quality requirements of export and domestic markets require an up-scaling of the production which is only feasible with educated farmers and skilled workers.

Desai Fruits and Vegetables (DFV)in cooperation with the Deutsche Gesellschaft für Internationale Zusammenarbeit(GIZ) GmbH on behalf of the German Federal Ministry for Economic Cooperation and Development (BMZ) takes up the existing education gap by implementing a development partnership called "Partnership Farming India".

The goal of Partnership Farming India (PFI) is to enable farmers to be self-sufficient decision-makers, "agripreneurs", which allows for a more flexible production system and highlights farming as profession by choice and not by inheritance.

Furthermore, PFI strengthens farmers' and farm workers' management skills to adopt modern agricultural practices and concepts and enhance the international competitiveness of smallholders' agricultural produce by giving farmers and workers in Gujarat and Maharashtra access to practical agricultural education. Therefore, DFV and GIZ in close cooperation with the National Institute of Agricultural Extension Management (MANAGE, an organization of Ministry of Agriculture, Government of India) developed the training material on basic agricultural knowledge and skills.

The states of Gujarat and Maharashtra will serve as an example on how to establish long term successful and trustful business relationships by combining small scale production in the field with large scale processing and marketing. I am confident that this effort will serve the Indian agriculture as a replicable model make lasting contributions towards sustainable agriculture and prosperous farmers.

I would like to express my sincere gratitude to the people and institutions namely MANAGE, DFV and GIZ,which supported this project and enabled making information available. This is a useful source of information for farmers, trainers, and other interested persons to improve not only the agriculture but also the livelihood of the farming community.

Mrs. Sabine Preuss
GIZ-India Programme Director
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Preface

Agriculture is an important sector of Indian Economy as more than half of its population relies on Agriculture as principle source of income. Research and Extension systems play major role in generation and dissemination of Agricultural technologies aiming at enhancing the income of farmers. The extension system adopts series of extension methods such as Training, demonstration, exposure visit to transfer the technologies from lab to land. Majority of these extension efforts mainly focus on location and crop specific technologies, and mostly on solution to problem basis. However, there is a need for equipping the farmers with Basic knowledge of Agriculture in order to create a better knowledge platform at farmer level for taking appropriate farm management decisions and to absorb modern technologies.

In view of this, Desai Fruits and Vegetables Pvt. Ltd. (DFV), India, in cooperation with the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH on behalf of the German Federal Ministry for Economic Cooperation and Development (BMZ) in close cooperation with National Institute of Agricultural Extension Management (MANAGE- An Organization of Ministry of Agriculture, Government of India) brought out Farmer's Handbook on Basic Agriculture to impart technical knowledge on Basic Agriculture to farmers to provide holistic perspective of scientific Agriculture.

This handbook is a product of series of brainstorming workshops and consultative meetings with various stakeholders such as Researchers, Academicians, Extension Functionaries, Agripreneurs, Master Trainers and Farmers. Based on the identified needs, the topics were prioritized and contents were developed with the help of experts. The farmer-friendly content has been written in simple language, using more pictures with practical examples for the benefit of farmers.

The book contains six chapters, each focusing on a particular topic. The first chapter, "General conditions for cultivation of crops", talks about the basic needs of farmers and farming sector, by providing basic knowledge on Good Agricultural Practices (GAP), enhancing the awareness of farmers on critical factors in selection of crops and cropping patterns, judicious use of natural resources such as soil and water, and emphasizing the importance of mechanization in the field of agriculture.

The second chapter "Soil and Plant Nutrition" is aimed at increasing the awareness and understanding of farmers about soil, it's structure, physical, chemical, biological properties, soil fertility and managing the soil fertility in an economically and environmentally sustainable manner. It also focuses on the need for soil testing, plant nutrition requirement, organic & inorganic fertilizers, and Integrated Nutrient Management (INM) for efficient, economic and sustainable production of crops.

The third chapter of the book is about Pest Management, and focuses on enhancing the awareness of and understanding among farmers about the crop pests, diseases and weed management through Integrated Pest Management. It also aims at sensitizing farmers on safe handling of chemicals and plant protection equipments as also elaborated further in the fifth chapter on "Occupational health and safety of farmers". It creates awareness about causes, preventive measures of health hazards, risks & fatalities in agriculture, and use of first aid in emergencies. It further includes safety tips and care to reduce the risk of injuries and fatalities while handling machineries and pesticides by farmers.

Time and resources management is an integral part of each and every activity, be it service sector, business or day-to-day activities of life. Farming sector too has not remained untouched by it. Therefore, the fourth chapter of the book is devoted to "Farm Management". It is to educate and equip the farmers to make proper plans, take appropriate decisions and also to take advantage of the improved technologies to increase production, assure food security for the family and market opportunities to increase income considering available resources, anticipated risks, including market fluctuations.

"Farmer's access to services", the last chapter of the book, aims at enhancing awareness among farmers about sources of extension, information and services, public and private extension services, agricultural credit, insurance and legal aspects through Information & Communication Technologies. The content is useful not only for farmers but also for other stakeholders involved in farm advisory services such as Agri input dealers, Agripreneurs, Kisan Call Centers and extension functionaries working at grass roots level.

We trust that this Handbook will benefit maximum number of farmers to make farming economically and environmentally more sustainable.

B. Srinivas, IAS
Director General
MANAGE

Contents

1. General Conditions for Cultivation of Crops

1.1. Objectives of the session

- To enhance awareness of farmers on critical factors in selection of crops and cropping patterns.
- To create an understanding on judicious use of natural resources such as soil and water.
- To provide basic knowledge on seed and cropping systems.
- To emphasize the importance of mechanization.
- To sensitize the farmers on Good Agricultural Practices (GAP).

1.2. What do we know at the end of the session

- Critical factors in selection of crops and cropping patterns
- Judicious use of natural resources such as soil and water
- Basic knowledge on seed
- Cropping systems
- Mechanization
- Good Agricultural Practices (GAP)

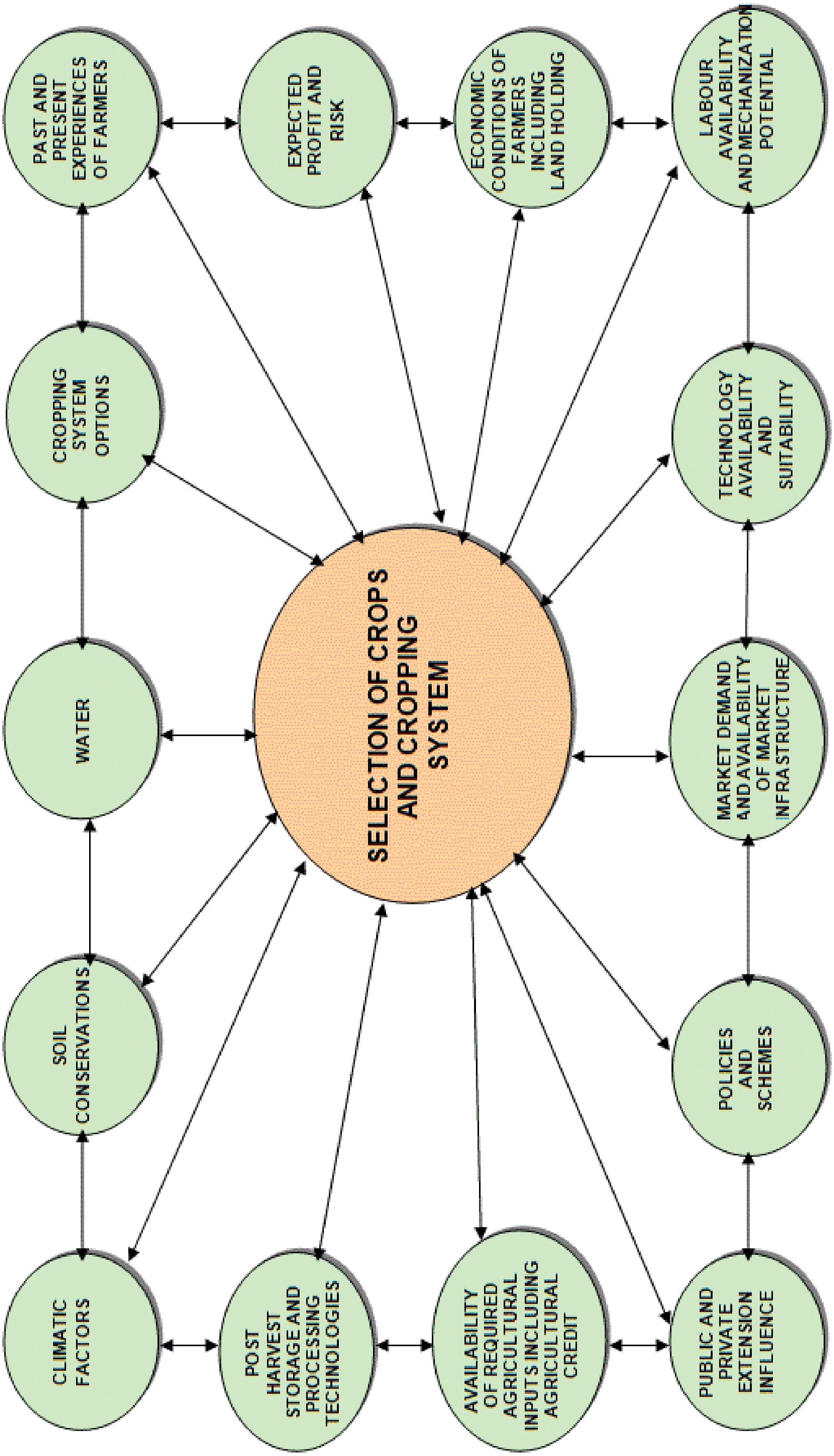
Agricultural Universities, Research Institutes, Krishi Vigyan Kendras have been generating ample technologies to improve the productivity and profitability of the farmers. How many of these technologies are reaching the farmers? Baseline Situation Assessment conducted by Partnership farming in India, in Gujarat and Maharashtra, clearly indicated that farmers with access to technical knowledge on agriculture realized better income compared to others. Fifty one percent of sample farmers who were part of partnership farming India had knowledge of soil testing compared to only 28% of control group. Mulching and intercropping as a practice were not widely adopted by control group of farmers. They were less aware about other organic fertilizers. Fertigation as a method of application of fertilizers was not widely practiced by control group. Farmers who accessed information from agricultural universities and magazines were less in number in the control group. The treatment farmers had an average yield of 35.65 tons and control farmers had yield of 22.36 tons of banana per acre. The average net income of the treatment farmers was Rs 93,822 and for the control farmers Rs.81,659. More than 85% of the farmers wanted basic education on agriculture and crop production and ready to pay for undergoing such basic education and training. There was a clear interest in the farmers to improve their skill and knowledge and they were ready to pay for the service.

The above study clearly indicates that the knowledge gap is prevailing among farmers and those who have access to knowledge harvested better profits.

Increase in productivity and profitability can be achieved through:

- Blending practical knowledge with scientific technologies
- Efficient use of natural resources
- Adopting time specific management practices
- Giving priority for quality driven production
- Adopting suitable farming systems
- Adoption of location specific technology
- Market demand driven production
- Adopting low cost and no cost technologies

Factors influencing decisions on the selection of crops and cropping system



1.3. Factors influencing decisions on the selection of crops and cropping system

Climatic factors

Is the crop/cropping system suitable for local weather parameters such as temperature, rainfall, sun shine hours, relative humidity, wind velocity, wind direction, seasons and agro-ecological situations?

Soil conditions

Is the crop/cropping system suitable for local soil type, pH and soil fertility?

Water

- Do you have adequate water source like a tanks, wells, dams, etc.?
- Do you receive adequate rainfall?
- Is the distribution of rainfall suitable to grow identified crops?
- Is the water quality suitable?
- Is electricity available for lifting the water?
- Do you have pump sets, micro irrigation systems?

Cropping system options

- Do you have the opportunity to go for intercropping, mixed cropping, multi-storeyed cropping, relay cropping, crop rotation, etc.?
- Do you have the knowledge on cropping systems management?

Past and present experiences of farmers

- What were your previous experiences with regard to the crop/cropping systems that you are planning to choose?
- What is the opinion of your friends, relatives and neighbours on proposed crop/cropping systems?

Expected profit and risk

- How much profit are you expecting from the proposed crop/cropping system?
- Whether this profit is better than the existing crop/cropping system?
- What are the risks you are anticipating in the proposed crop/cropping system?
- Do you have the solution? Can you manage the risks?
- Is it worth to take the risks for anticipated profits?

Economic conditions of farmers including land holding

- Are the proposed crop/cropping systems suitable for your size of land holding?
- Are your financial resources adequate to manage the proposed crop/cropping system?
- If not, can you mobilize financial resources through alternative routes?

Labour availability and mechanization potential

- Can you manage the proposed crop/cropping system through your family labour?
- If not, do you have adequate labours to manage the same?
- Is family/hired labour equipped to handle the proposed crop/cropping system?
- Are there any mechanization options to substitute the labour?
- Is machinery available? Affordable? Cost effective?
- Is family/hired labour equipped to handle the machinery?

Technology availability and suitability

- Is the proposed crop/cropping system suitable?
- Do you have technologies for the proposed crop/cropping system?
- Do you have extension access to get the technologies?
- Are technologies economically feasible and technically viable?
- Are technologies complex or user-friendly?

Market demand and availability of market infrastructure

- Are the crops proposed in market demand?
- Do you have market infrastructure to sell your produce?
- Do you have organized marketing system to reduce the intermediaries?
- Do you have answers for questions such as

where to sell? When to sell? Whom to sell to? What form to sell in? What price to sell for?
- Do you get real time market information and market intelligence on proposed crops?

Policies and schemes

- Do Government policies favour your crops?
- Is there any existing scheme which incentivises your crop?
- Are you eligible to avail those benefits?

Public and private extension influence

Do you have access to Agricultural Technology Management Agency (ATMA)/ Departmental extension functionaries to get advisory?

- Do you know Kissan Call Center?
- Do you have access to KVKs, Agricultural Universities and ICAR organizations?
- Do you subscribe agricultural magazines?
- Do you read agricultural articles in newspapers?
- Do you get any support from input dealers, Agribusiness Companies, NGOs, Agriclinics and Agribusiness Centers?

Availability of required agricultural inputs including agricultural credit

- Do you get adequate agricultural inputs such as seeds, fertilizers, pesticides, and implements in time?
- Do you have access to institutional credit?

Post harvest storage and processing technologies

- Do you have your own storage facility?
- If not, do you have access to such facility?
- Do you have access to primary processing facility?
- Do you know technologies for value addition of your crop?
- Do you have market linkage for value added products?
- Are you aware about required quality standards of value added products of proposed crops?

Farmers need to answer all the above questions while making decisions for choosing a crop/ cropping pattern. During this decision making process, farmer cross check the suitability of proposed crop/cropping systems with his existing resources and other conditions. Thereby, they justify choosing or rejecting a crop/cropping systems. This process enables the farmers to undertake a SWOT analysis internally which in turn guides them to take an appropriate decision.

1.4. Climatic factors

Climate and agriculture

- Monsoon is a key source of water in agriculture
- Most of our rivers are seasonal fed by the monsoon; even irrigated agriculture depends on monsoon.
- Cropping pattern has evolved over years based on climate.
- Market forces influence cropping patterns in recent times.

Climatic factors and crops

- Rainfall drives water availability and determines sowing time (rainfed crops).
- Temperature drives crop growth, duration and influences milk production in animals.
- Temperature and relative humidity influence pest and diseases incidence on crops, livestock and poultry.
- Wet and dry spells cause significant impact on standing crops, physiology, loss of economic products (e.g. fruit drop).
- Extreme events (e.g. high rainfall, floods, heat / cold wave, cyclone, hail, frost) cause enormous losses of standing crops, livestock and fisheries.

Climate and seasons

- Rainy (June-September) season also known as Kharif, supports most of the rainfed crops (coarse cereals, pulses, oilseeds, etc.).
- Post-rainy (October-February) season also known as Rabi, supports the irrigated or stored moisture grown crops (wheat, mustard, chickpea, etc.).
- Summer season (March-May) supports short duration pulses and vegetables.
- Rabi production is more assured, has a higher yield and reduces pest and disease related problems.
- Over time, with irrigation development, the contribution of Kharif is declining and Rabi is increasing.

Climate, cropping pattern and agricultural production issues

- Cropping patterns based on climate and land capability are sustainable but market forces and farmers' aspirations are forcing unsustainable systems.
- Farmers must innovate in producing more even from less endowed areas by adopting suitable technologies to cope with changing climate.
- Climate change will likely to cause further problems in our crop production and is likely to become the most important environmental issue in the 21st century.

Important agricultural related factors responsible for climate change

- Deforestation and forest degradation
- Burning of fuel and farm waste
- Water logged condition
- Excessive use of external input
- Large-scale conversion of land for non-agricultural purpose

Impact of climate change in India

- **Rainfall:** No long-term trend noted. However, regional variations seen, increased summer rainfall and less number of rainy days.
- **Temperature**: About 0.6 °C rise in surface temperature during 100 years. Projected to increase 3.5 to 5 °C by 2100.
- **Carbon dioxide:** Increasing at the rate of 1.9 ppm per year and expected to reach 550 ppm by 2050 and 700 ppm by 2100.
- **Extreme events:** Increased frequency of heat wave, cold wave, droughts and floods observed during last decade.
- **Rising sea level:** Rise of 2.5 mm/year since 1950.
- **Glaciers:** Rapid melting of the glaciers in the Himalayas.
- **Rainfall distribution:** Shift in peak rainfall distribution also noticed in some parts of country.

Expected impact of climate change on agriculture

- Due to increase in temperature, crop may require more water.
- Yield may be reduced in cereal crops especially in Rabi; i.e. wheat.

Impact of Drought

Impact of Flood

Heat Wave on Maize

Cold wave damage to chana harvest

Change in pest and disease scenario due to climate change

- **Due to increase in rainfall:** Pests like bollworm, red hairy caterpillar and leaf spot diseases may increase.Due to increase in temperature: Sucking pests such as mites and leaf miner may increase.
- **Due to variation in rainfall and temperature:** Pest and diseases of crops to be altered because of more enhanced pathogen and vector development, rapid pathogen transmission and increased host susceptibility. Sometimes a minor pest may become a major pest.
- Agricultural biodiversity is also threatened by decreased rainfall and increased temperature, sea level rise and increased frequency and severity of drought, cyclone and flood. Quality of farm products such as fruits, vegetables, tea, coffee, aromatic and medicinal plants may be affected.

Water

- Demand for irrigation to increase with increased temperature and higher amount of evapo-transpiration. This may result in lowering of groundwater table at some places.
- The melting of glaciers in the Himalayas will increase water availability in the Ganga, Brahmaputra and their tributaries in the short-run but in the long-run the availability of water will decrease considerably.
- A significant increase in runoff is projected in the rainy season, however, may not be very beneficial unless storage infrastructure could be vastly expanded. This extra water in the rainy season, on the other hand, may lead to increase in frequency and duration of floods.
- The water balance in different parts of India will be disturbed and the quality of ground water along costal track will be more affected due to intrusion of sea water.

Soil

- Organic matter content, which is already quite low in Indian soil, would become even lower. Quality of soil organic matter may be affected.
- Reduction in rate of decomposition and nutrient supply.
- Increase in soil temperature may reduce Nitrogen availability due to volatilization and denitrification.
- Change in rainfall volume and frequency as well as wind may alter the severity, frequency and extent of soil erosion.
- Rise in sea level may lead to salt water entry in the coastal lands turning them less suitable for conventional agriculture.

Livestock

- Affect feed production and nutrition of livestock. Increased temperature would reduce digestibility. Increased water scarcity would also decrease the food and fodder production.
- Major impacts on vector-borne diseases through expansion of vector populations during rainy years, leading to large outbreaks of diseases.
- Increase water, shelter, and energy requirement of livestock for meeting projected milk demands.
- Climate change is likely to aggravate heat stress in dairy animals, adversely affecting their reproductive performance.

Fishery

- Increased sea and river water temperature is likely to affect fish breeding, migration and harvest.
- Impacts of increased temperature and tropical cyclonic activity would affect capture, production and marketing costs of the marine fish.

Coping options for farmers

Access to information

- Progressive Farmers
- ATMA extension functionaries – Block Technology Manager, SMS, farmer friend, Farm School
- Trained input dealers
- Agri Clinics and Agribusiness Centers
- KVK
- Agricultural Research Stations
- Agricultural Universities
- ICAR Organisations
- Kissan Call Centers (Toll free no.1551 or 1800 – 180 – 1551)
- Concerned NGOs
- Agribusiness Companies
- Radio, TV, Agricultural Magzines, Community Radio, Newspapers, Agricultural Websites etc.

Coping options for farmers

Enlarging the Food Basket

- Diversifying the livelihood sources.
- Changing cropping patterns.
- Increased traditional coping strategies.
- Change to a mixed cropping pattern.

E.g: Crop Mixture-Nutri Millets, Pulses and Oilseed

Integrated Farming System

- Increased share of non-agricultural activities
  E.g: **Type of Integrated Farming Systems**
  Agriculture +vegetable cultivation
  Agriculture + animal husbandry

Neem, Mulberry & Cowpea

- Planting more drought tolerant crops and increased agro-forestry practices.
- Agro-forestry systems to provide more stable incomes during years of extreme weather events.

Mango, Pumpkin, maize mixed cropping

Mixed farming/Multi level farming

Coping options for farmers continued....

Lucerne & Sunhemp for green manuring & fodder

Farm Pond

Conservation Furrow

- Improved on-farm soil & water conservation.
- Adopting scientific water management, nutrient management and cultural practices.

Vegetative Barriers

Percolation Tanks

Coping options for farmers continued....

Contour trenching for runoff collection

Conventional Raised Bed Planting

- 20-25% Saving in irrigation water

Shelterbelts

- Shelterbelts reduce wind velocity.
- Moderate temperature.
- Reduce evaporative loss and conserve soil moisture.

Straw Thatching

- Protecting young seedlings against cold by covering with straw thatching.

Frost Protection

1.5. Soil and Water Conservation

Soil and water are our precious heritage. Hence, it is obligatory on our part to protect and hand over these resources to further generations. It is estimated that about 50% of the cultivated area in India suffers from severe soil erosion and requires remedial measures.

- Water resources are essential for increasing and stabilizing crop production.
- Wind erosion has been responsible for destroying the valuable top soil.

1.5.1. Degradation of soil and water takes place with water and wind erosion

- The main cause of water erosion is unmanaged runoff.
- Runoff is the portion of the rainfall or irrigation water applied which leaves a field either as surface or as subsurface flow.

Several factors are responsible for runoff

- **Climatic factors:** Precipitation characteristics - duration, intensity, distribution, direction, temperature, humidity, wind velocity.
- **Watershed characteristics:** Geological shape of the catchments, size and shape of the catchments, topography, drainage pattern.
- **Barren land without vegetation**
- **Soil types:**
  - **Sandy soil:** Average rain – no problem of erosion. High intensity – More serious of less binding material i.e. fine soil particle.
  - **Clay soil:** Ordinary rain – more runoff in moderate and steep slopes but high water holding capacity.
  - **Silt loam, loamy and fine sandy loam:** More desirable soils from the point of view of minimizing soil erosion.

How vegetation reduces runoff

- Interception of rainfall
- Root structure
- Biological influences
- Transpiration effects
- Intercept, absorb the impact of raindrop
- Hindrance to runoff water slows down the rate at which travels down the slope
- Knitting and binding effect aggregates the soil into granules
- Die and decay increase pore space and water holding capacity
- One cubic meter of soil has several kilometres of root fibre
- More vegetative cover, most active soil fauna, channels of earth worm, beetles and other life
- Vegetation increases the storage capacity of the soil for rainfall by the transpiration of large quantities of moistures from the soil

Soil erosion

Soil erosion is the detachment and transportation of soil material from one place to another through the action of wind, water in motion or by the hitting action of the rain drops.

- When the vegetation is removed and land is put under cultivation the natural equilibrium between soil building and soil removal is disturbed.
- The removal of surface soil takes place at a much faster rate than it can be built up by the soil forming process.

Erosion by water: Known as water erosion, is the removal of soil from the lands surface by water in motion.

Sheet erosion: The removal of a thin relatively uniform layer of soil particles by the action of rainfall and runoff.

- Extremely harmful
- Usually so slow that the farmer is not conscious of its existence
- Common on lands having a gentle uniform slope
- Results in the uniform removal of the cream of the top soil with every heavy rain
- Shallow top soil overlies a tight sub soil are most susceptible to sheet erosion
- Movement of soil by rain drop splash is the primary cause of sheet erosion
- Sheet erosion has damaged millions of hectares of slopping land throughout the India

Rill erosion is the removal of soil by running water with the formation of shallow channels that can be smoothed out completely by normal cultivation.

- There is no sharp lines of demarcation where sheet erosion and rill erosion begins but rill erosion is more readily apparent than sheet erosion.

Sheet and Rill Erosion

Gully Erosion

Landslide

Shelterbelts for Moderating microclimate

- Rills develop when there is a concentration of runoff water which, if neglected, grow into large gullies.
- More serious in soils having a loose shallow top soil.
- Transition stage between sheet erosion and gullying.

Gully erosion: Removal of soil by running water with the formation of channels that cannot be smoothed out completely by cultivation.

- Advance stage of rill erosion.
- Any concentration of surface runoff is a potential source of gullying.
- Cattle paths, cart tracks, dead furrows, tillage furrows or other small depression down a slope favour concentration of flow.
- Unattended rills deepen and widen every year and begin to attain the form of gullies.
- Unattended gullies may result over a few years for an entire landscape to be filled with a network of gullies.
- More spectacular than other type of erosions.

Stream channel erosion: Erosion caused by stream flow.

- Closely resembles rill erosion.
- Intensive channel erosion areas are on the outside of lands where flow shear stresses are high.

Mass movement: Enmass movement of soil.

- Landslides, land slips, soil and mudflows are various forms of mass movement.

Wind erosion: Movement of soil particles is caused by wind force exerted against or parallel to surface of the ground.

1.5.2. Conservation

Conservation is the utilization without wastage of resources is required to ensure a high level of production.

Important soil conservation measures are

- Conservation Tillage
- Minimum tillage
- Zero tillage
- Stubble mulching
- Trash farming

Conservation farming

- Farming across the slope
- Strip cropping
- Rotations
- Mixed cropping and intercropping
- Surface mulching
- Timely farm operations
- Improved water user efficiency
- Land levelling
- Providing safe drainage
- Intermittent terraces
- Growing vegetation on the bunds

Vegetation and vegetative management

- Strip cropping
- Stubble mulching
- Mulching

Wind erosion management

- Protect the soil surface with a cover of vegetation or vegetative residues.
- Produce or bring to the surface soil aggregates or clods which are large enough to resist the wind force.
- Roughen the land surface to reduce wind velocity and trap drifting soil.
- Establish barriers or trap strips at intervals to reduce wind velocity and soil drifting.

Best practices to control soil blowing

- Deep ploughing
- Summer ploughing
- Surface roughness
- Conserving moisture
- Wind breaks and shelterbelts
- Mechanical or vegetative barriers

For instance: Shelterbelts for moderating microclimate

- Shelterbelts reduce wind velocity
- Moderate temperature
- Reduce evaporative loss and conserve soil moisture

Water erosion can be managed by

- In situ water harvesting
- Summer ploughing

Overland flow management

- Contour bund
- Graded bund
- Broad based bund
- Bench terrace
- Water harvesting and recycling

Zero tillage

- Several practices are in use such as zero tillage, minimum tillage and direct seeding.
- Planting crops in previously untilled soil by opening a narrow slot, trench or band only of sufficient width and depth to obtain seed coverage. No other soil tillage is done.

Advantages of zero tillage farming

- Erosion control: Retained stubble and crop residue reduces soil erosion and enhances soil fertility
- Moisture conservation: Stubble traps water, reduce runoff water, better infiltration leading to improved soil moisture condition
- Higher nitrogen availability
- Seedling protection: Stubbles protects young seedling from wind and heat
- Crop yields will be on par with traditional tillage system. However good yield can be harvested during dry years
- Reduce labour and save time
- Savings on equipment cost
- Savings on oil/fuel cost

Mulching: Benefits of crop residue mulching are

- Increased availability of water and organic matter
- Less erosion
- Environment protection

Additional benefit to farmers

- Less drought susceptibility
- Improved soil quality and fertilizer efficiency
- Minimises long term dependency on external inputs

1.6. Irrigation

An adequate water supply is important for plant growth. When rainfall is not sufficient, the plants must receive additional water from irrigation.

Points consider for irrigation decisions

- Land suitability for irrigation like slope
- Effective rainfall: Part of the total rain is useful for crop production
- When to irrigate: Decide based on soil, crop and climatic condition
- How much to irrigate: Decide based on crop water requirement
- How to irrigate: Select appropriate method for irrigation
- Quality of irrigation water

1.6.1. Various methods can be used to supply irrigation water to the plants

- Surface irrigation:
  - Basin irrigation
  - Furrow irrigation
- Sprinkler irrigation
- Drip irrigation

Surface Irrigation

Surface irrigation is the application of water by gravity flow to the surface of the field.

- Either the entire field is flooded (Basin Irrigation) or the water is fed into small channels (furrows) or strips of land (borders).

Basin Irrigation

- Basins are flat areas of land, surrounded by low bunds.
- The bunds prevent the water from flowing to the adjacent fields.
- Basin irrigation is commonly used for rice grown on flat lands or in terraces on hillsides. Paddy grows best when its roots are submerged in water. Hence, basin irrigation is the best method to use for this kind of crop.
- Trees can also be grown in basins, where one tree is usually located in the middle of a small basin.
- In general, the basin method is suitable for crops that are no affected by standing in water for longer periods.
- Basin irrigation is suitable for many field crops.
- Crops suitable for basin irrigation include pastures, citrus, banana and crops that are broadcasted such as cereals and to some extent row crops such as tobacco.
- Basin irrigation is generally not suited to crops, which cannot stand in wet or waterlogged conditions for periods longer than 24 hours; eg: potatoes, beet root and carrots
- The flatter the land surface, the easier it is to construct basins.
- It is also possible to construct basins on sloping land, even when the slope is quite steep. Level basins, called terraces, can be constructed like the steps of a staircase.
- Soils suitable for basin irrigation depend on the crop grown.

Basin should be small if the:

- Slope of the land is steep
- Soil is sandy
- Stream size to the basin is small
- Required depth of the irrigation application is small
- Field preparation is done by hand or animal power

Basin can be large if the:

- Slope of the land is gentle or flat
- Soil is clay
- Stream size to the basin is large
- Required depth of the irrigation application is large
- Field preparation is mechanized
- The land slope, the soil type, the available stream size, the required depth of the irrigation application and farming practices mainly determine the shape and size of basins

- If the land slope is steep, the basin should be narrow; otherwise too much earth movement will be needed to obtain level basins.
- Three other factors, which may affect basin width, are depth of fertile soil, method of basin construction, agricultural practices.
- **There are two methods to supply irrigation water to basins:** (i) The direct method: Irrigation water is led directly from the field channel into the basin through siphons,

or bund breaks. (ii) The cascade method: irrigation water is supplied to the highest terrace, and then allowed to flow to a lower terrace and so on.

Maintenance of basins

- Bunds are susceptible to erosion. This may be caused by, for example, rainfall, flood or the passing of people when used as foot-paths.
- Rats may dig holes in the sides of the bunds.
- Therefore, it is important to check the bunds regularly, notice defects and repair them instantly, before greater damage is done.

Advantages of basin irrigation

- Conservation of rainfall and reduction in soil erosion.
- High water application and distribution efficiencies.
- Useful in leaching of salts.
- Suitable to all close growing crops, row crops and orchards.

Basin Irrigation

Furrow irrigation

- Furrows are small channels, which carry water down the land slope between the crop rows.
- Water infiltrates into the soil as it moves along the slope.
- The crop is usually grown on the ridges between the furrows.
- This method is suitable for all row crops and for crops that cannot stand in water for long periods. Crops such as maize, sunflower, sugarcane, and soybean can be irrigated by furrow irrigation.
- Crops that would be damaged by inundation, such as tomatoes, vegetables, potatoes, beans; fruit trees like citrus and grape as well as broadcasted crops like wheat.
- Irrigation water flows from the field channel into the furrows by opening up the bank of the channel or by means of siphons or spiles.
- Furrows must be on consonance with the slope, soil type, stream size, irrigation depth, cultivation practice and field length.
- Uniform flat or gentle slopes are preferred for furrow irrigation.
- On undulating land, furrows should follow the land contours.

Advantages of furrow irrigation

- Suitable for row crops and vegetables.
- Suitable for soils in which the infiltration rates vary between 0.5 and 2.5 cm/hr.
- Ideal for slopes varying from 0.2 to 0.5 per cent and a stream size of 1-2 liters/sec.
- In areas requiring surface drainage or prone to temporary water logging, furrows are very effective.
- In areas where water for irrigation purposes. is scarce, the practice of alternate or skip furrow irrigation can save considerable quantity of water without significantly affecting yields.

Furrow Irrigation

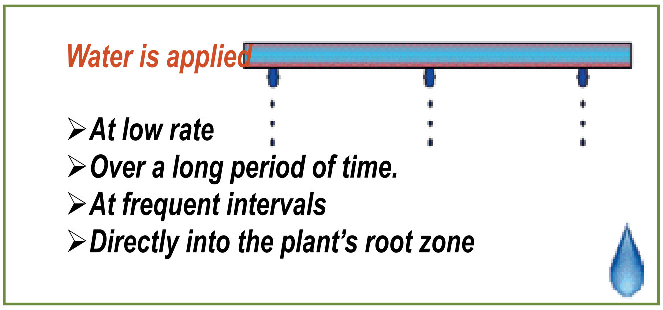
Sprinkler irrigation

Water is pumped through a pipe system and then sprayed onto the crops through sprinkler heads.

Advantages

- Water conservation
- Soil conservation
- Efficient use of water
- Saving of labour
- Early seed germination
- Fertigations
- Soil amendments
- Frost protection

- Cooling of crops
- Higher pro+ductivity of crops



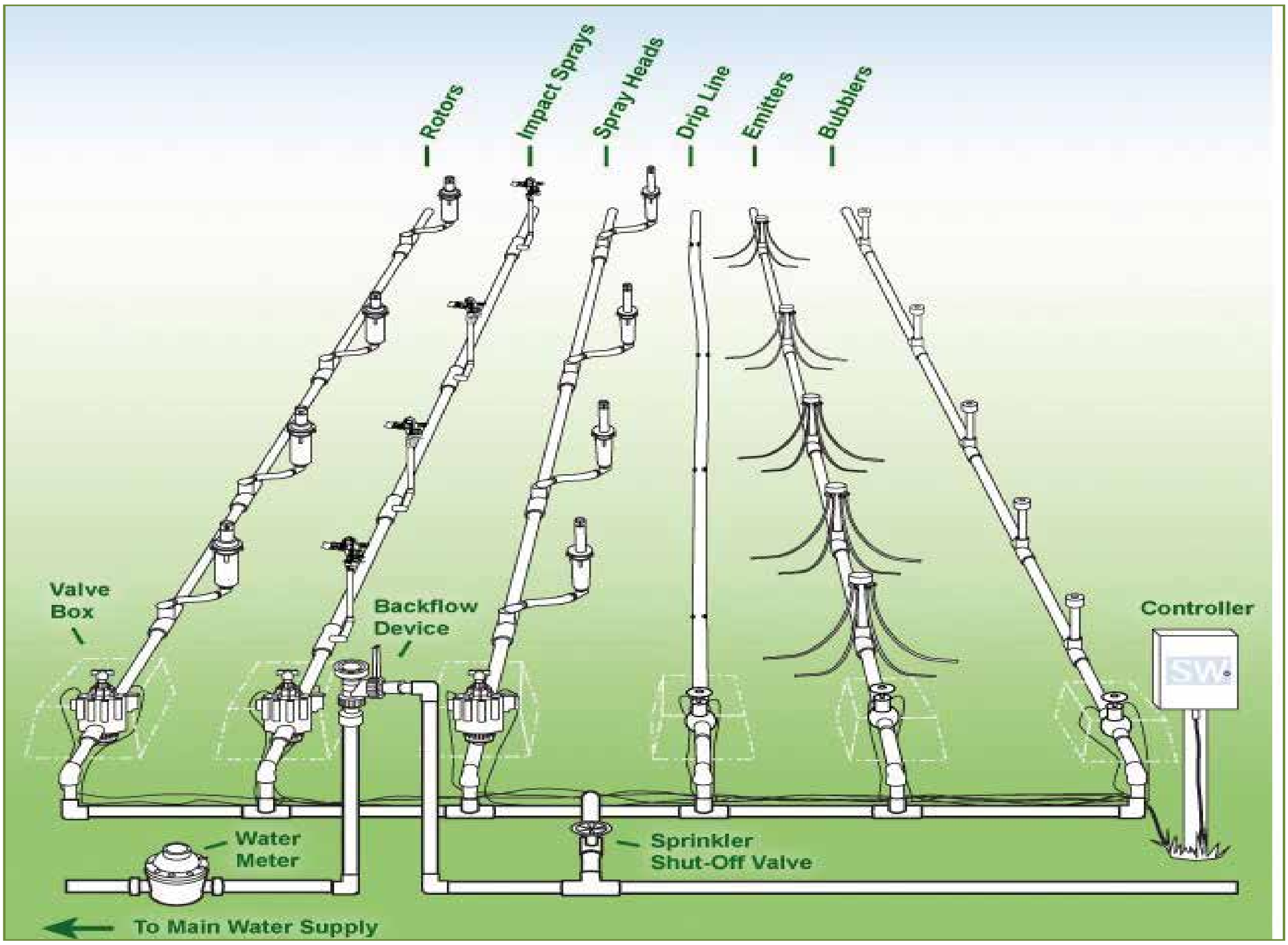
Drip Irrigation

Use of Sprinklers for different crops

Crop Type	Crop Example
Cereals	Maize, Sorghum, Wheat, Jowar
Flowers	Carnation, Jasmine, Marigold
Oilseeds	Groundnut, Mustard, Sunflower
Vegetables	Onion, Potato, Radish, Carrot
Fodders	Asparagus, Pastures
Pulses	Gram, Pigeon pea, Beans
Plantation	Coffee, Rubber, Tamarind
Fibre	Cotton, Sesame
Spices	Cardamom

Response of different crops to sprinkler irrigation

Crop	Water saving (%)	Yield increase (%)
Bajra	56	19
Barley	56	16
Bhendi	28	23
Cabbage	40	3
Cauliflower	35	12
Chillies	33	24
Cotton	36	50
Cowpea	19	3
Fenugreek	29	25
Garlic	28	6
Gram	69	57
Groundnut	20	40
Jowar	55	34
Lucerne	16	27
Maize	41	36
Onion	33	23
Potato	46	4
Sunflower	33	20
Wheat	35	24



Lay out of Sprinkler Irrigation System

Drip irrigation

Water is conveyed under pressure through a pipe system to the fields, from where it is discharged slowly or at a pre designed rate. The latter can be matched to the soil infiltration capacity through emitters or drippers that are located close to the root zone of the plants.

A typical drip irrigation system consists of the following components:

- Pump unit
- Control unit
- Filtering unit
- Mainline and sub mainlines
- Laterals
- Emitters

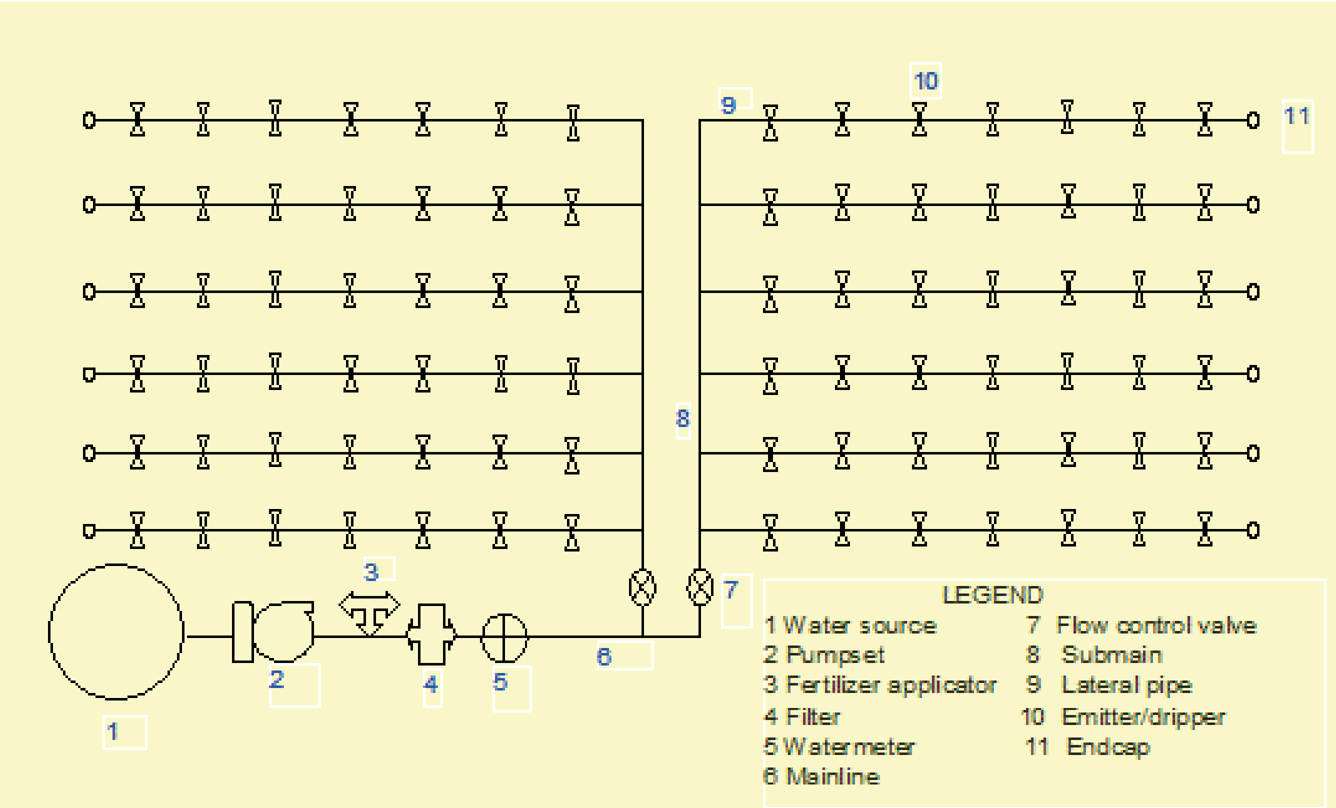
Head Control Unit

Critical stages for irrigation in different crops

Name of the Crop	Critical Stages
Cereals	
Rice/Paddy	Tillering, Panicle Initiation, Heading and Flowering
Wheat	Crown Root Initiation, Tillering to Booting
Sorghum	Booting, Blooming and Milky Dough Stage
Maize	Silking and Tasseling to Dough Stage
Pearl millet	Heading and Flowering
Finger millet	Primordial Initiation and Flowering
Pulses	
Chickpea	Late Vegetative Phage
Black gram	Flowering and Pod Setting
Green gram	Flowering and Pod Setting
Beans	Flowering and Pod Setting
Peas	Flowering and Early Pod Formation
Alfalfa	After Cutting and Flowering

Name of the Crop	Critical Stages
Oil Seeds	
Ground nut	Flowering, Peg Formation and Pod Development
Sesame	Blooming to Maturity
Sunflower	Pre-flowering to Post-flowering
Soybean	Blooming and Seed Formation
Vegetables	
Onion	Bulb Formation and Pre-maturity
Tomato	Flowering and Fruit Setting
Chilies	Flowering and Fruit Setting
Cabbage	Head Formation
Potato	Tuber Initiation to Maturity
Carrot	Root Enlargement
Others	
Cotton	Flowering and Boll Formation
Citrus	Flowering, Fruit Setting and Fruit Enlargement
Mango	Pre-flowering and Fruit Setting

Layout of micro irrigation system



Benefits of drip irrigation over surface irrigation

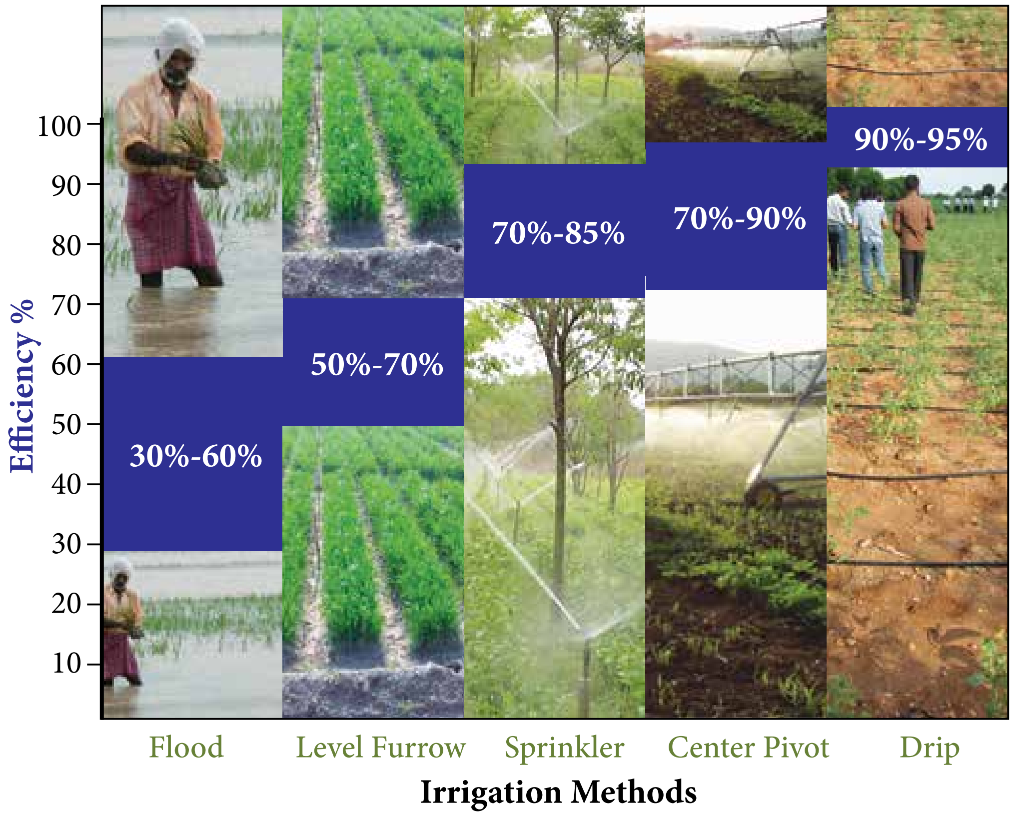
Crop	Yield increase (%)	Water saving (%)
Mango	80.0	34.8
Banana	52.0	45.0
Grapevine	23.0	48.0

Crop	Yield increase (%)	Water saving (%)
Pomegranate	98.0	45.0
Tomato	50.0	39.0
Watermelon	88.0	36.0

Benefits of drip irrigation over surface irrigation continued....

Crop	Yield increase (%)	Water saving (%)	Crop	Yield increase (%)	Water saving (%)
Lady's finger	16.0	40.0	**Sugarcane**	133.3	49.3
Brinjal	14.0	53.0	Cotton	88.0	46.6
Chillies	44.0	62.0	**Onion**	53.8	46.1
Papaya	75.0	68.0	**Potato**	79.5	54.1

On-farm irrigation efficiency of different irrigation methods



1.6.2. Centrally sponsored micro irrigation scheme

It is clear from the above diagram that drip irrigation is the most efficient irrigation in terms of water use efficiency compared to all other methods. Flood irrigation method is found to be the most uneconomical irrigation method in terms of water use efficiency when compared to all other methods.

In order to popularize micro irrigation, the Govt. of India is implementing the Micro Irrigation Scheme through which interested farmers be supported. The farmers can approach nearest extension functionary. The details are as follows:

Name of Scheme	Micro Irrigation
Type	Centrally Sponsored Scheme (CSS)
Year of Commencement	2005-06
Objectives	To increase the area under efficient methods of irrigation viz. drip and sprinkler irrigation as these methods have been recognized as the only alternative for efficient use of surface as well as ground water resources.

Salient Features

- Out of the total cost of the Micro Irrigation (MI) System, 40% will be borne by the Central Government, 10% by the State Government and the remaining 50% will be borne by the beneficiary either through his/her own resources or soft loan from financial institutions.
- Assistance to farmers will be for covering a maximum area of 5 hectare per beneficiary family.
- Assistance for drip and sprinkler demonstration will be 75% of the cost for a maximum area of 0.5 ha per beneficiary, which will be met entirely by the Central Government.
- The Panchayati Raj Institutions (PRIs) will be involved in selecting the beneficiaries.
- All categories of farmers are covered under the Scheme. However, it needs to be ensured that at least 25% of the beneficiaries are small and marginal farmers.
- The scheme includes both drip and sprinkler irrigation. However, sprinkler irrigation will be applicable only for those crops where drip irrigation is uneconomical.
- There will be a strong HRD input for the farmers, field functionaries and other stakeholders at different levels.
- Moreover, there will be publicity campaigns, seminars/workshops at extensive locations to develop skills and improve awareness among farmers about importance of water conservation and management.
- The Precision Farming Development Centres (PFDCs) will provide research and technical support for implementing the scheme.
- Supply of good quality system both for drip and sprinkler irrigation having BIS marking, proper after sales services to the satisfaction of the farmer is paramount.

Subsidy Pattern: Assistance is provided @ 50% (40% by the Government of India and 10% by the State Government) for drip/sprinkler Irrigation System. Assistance to the extent of 75% of the cost of demonstration is provided up to a limit of 0.5 ha.

Structure of Scheme

- At the National level, National Committee on Plasticulture Application in Horticulture (NCPAH) will be responsible for coordinating the Scheme, while the Executive Committee of NCPAH will approve the Action Plan. At the State level the State Micro Irrigation Committee will coordinate the programme, while at the District level the District Micro Irrigation Committee will oversee the programme.
- The Scheme will be implemented by an Implementing Agency (IA), appointed by the State Government, which will be the District Rural Development Agencies (DRDAs) or any identified Agency, to whom funds will be released to directly on the basis of approved district plans for each year.
- The IA shall prepare Annual Action Plan for the District which will be forwarded by the DMIC and SMIC for approval by the Executive Committee (EC) of NCPAH.

Funding Pattern

80:20 by the Centre and States

Eligibility	As indicated in column 5 above.
Area of Operation	The focus will be on horticultural crops being covered under the National Horticulture Mission in 24 States/UTs. A cluster approach will be adopted. The focus has also been extended to non horticultural crops.
Procedure to Apply	Project proposals are submitted through the State Government for release of assistance.

1.6.3. Drainage

Drainage it is a removal of water from the field as a moisture control mechanism.

- Drainage and irrigation are important aspects to be understood by the farmers
- Drainage provides desirable environment in the crop root zone
- Necessity of drainage is felt when there is excess water in root zone
- Source of excess water are
- Uncontrolled irrigation
- Seepage loss from an unlined channel
- Ground water moving from a shallow aquifers
- Non maintenance of natural drainage system

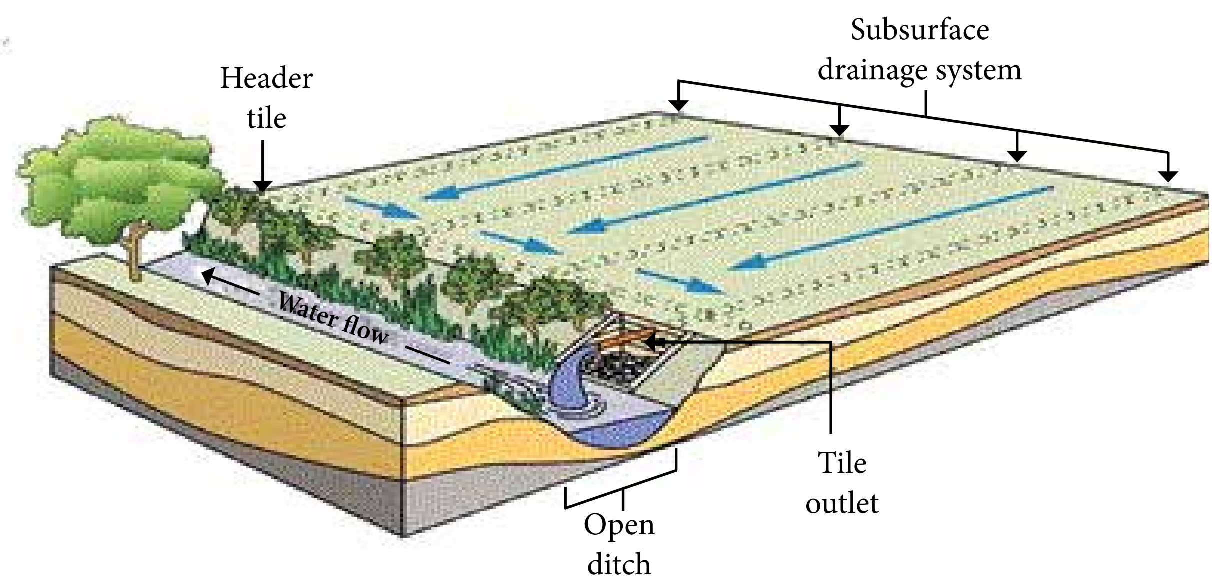
Generally two types of drainage systems are adopted based on techno-economic feasibility:

Surface drainage: Can be achieved by following any one of the below method based on the need and intensity of the problem.

- Land forming
- Land smoothening
- Land grading or levelling
- Bedding system
- Open ditches

Sub surface drainage: Can be achieved by following any one of the below method based on the need and intensity of problem.

- Horizontal sub surface drains
- Vertical drainage
- Other methods like
    - Mole drainage
    - Seepage intercepting farm pond
- Bio drains



1.7. Seed

A 'seed' (in some plants, referred to as a 'kernel') is a small embryonic 'plant' enclosed in a covering called the seed coat, usually with some 'stored food'. Seeds fundamentally are a means of reproduction and most seeds are the product of 'sexual reproduction', which remixes genetic material and 'phenotype variability' that 'natural selection' acts upon.

The seed is the basic input in agriculture upon which other inputs are applied. A good vigorous seed utilizes all the resources and realizes a reasonable output to the grower. It is wealth to the farmer since yesterday's harvest is tomorrow's hope. Good seed in good soil realizes a good yield. Moreover, it is the link between two generations.

Functions of seeds

- Nourishment of the embryo
- Dispersal to a new location
- Dormancy during unfavourable conditions

Characteristics of good seed

- Genetically pure
    - Breeder /Nucleus - 100%
    - Foundation seed - 99.5%
    - Certified seed - 99.0%
- Required level of physical purity for certification
    - All crops - 98%
    - Carrot - 95%
- High pure seed percentage
    - Bhendi - 99.0 %
    - Sesame, soybean & jute - 97.0 %
    - Ground nut - 96.0 %
- Free from other crop seeds
- Free from designated diseases like loose smut in wheat
- Free from objectionable weed seed like wild paddy in paddy
- Have good shape, size, colour, etc. according to specifications of variety
- Have high physical soundness and weight
- Posses high physiological vigour and stamina
- Posses high longevity and shelf life
- Have optimum moisture content for storage
    - Long term storage: 8% and below
    - Short term storage: 10-13%
- Have high market value

Seed types and characteristics

Seed Type	Characteristics	Genetic Purity	Tag Colour
Nucleus Seed	Produced by the breeder and it is genetically pure seed	100%	-
Breeder Seed	Produced by the breeder from nucleus seed	100%	Yellow
Foundation Seed	Produced by the breeder seed under the supervision of the concerned seed certification agency	99.5%	White
Certified Seed	Certified seed is the progeny of foundation seed and its production is supervised and approved by certification agency. The seed of this class is normally produced by the State and National Seeds Corporation and Private Seed Companies on the farms of progressive growers. This is the commercial seed which is available to the farmers.	99.0%	Azar Blue

Seed treatment

Seed treatment is usages of specific products and specific techniques to improve the growth environment for the seed, seedlings and young plants. It ranges from a basic dressing to coating and pelleting.

Seed dressing: This is the most common method of seed treatment. The seed is dressed with either a dry formulation or wet treated with a slurry or liquid formulation. Dressings can be applied at both, the farm and industries. Low cost earthen pots can be used for mixing pesticides with seed or seed can be spread on a polythene sheet. The required quantity of chemical can be sprinkled on the seed lot and mixed mechanically by the farmers.

Seed coating: A special binder is used with a formulation to enhance adherence to the seed.

Seed pelleting: The most sophisticated Seed Treatment Technology changes the physical shape of a seed to enhance pelletability and handling. Pelleting requires specialized application machinery and techniques and is the most expensive application.

The farmer must take care of the following while buying the seeds

- When purchasing the seed farmer should obtain a bill/cash memo wherein the lot number and seed tag number is mentioned.
- After purchasing the seed, empty bag/packet (pouches) and receipt should be kept safely.
- Out of purchased seed, 100 seeds are taken from each purchased variety to test them for germination before sowing in the field. Knowing the germination percentage, the farmer can decide the seed rate when sowing in the field.


Seed Dressing


Pelleted Onion Seed

Recommendation of seed treatment for different crops contiued...

Name of Crop	Pest / Disease	Seed Treatment	Remarks
Sugarcane	Root rot, wilt	Trichoderma spp. 4-6 gm/kg seed	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Rice	Root rot disease other insects /pests Bacterial sheath blight	Trichoderma 5-10 gm/kg seed (before transplanting) Pseudomonas flourescens 0.5% W.P. 10 gm/kg.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

Recommendation of seed treatment for different crops contiued...

Name of Crop	Pest / Disease	Seed Treatment	Remarks
Chillies	Anthracnose spp. Damping off Soil borne infection of fungal disease Jassid, aphid, thrips	Seed treatment with Trichoderma viride 4g/kg Trichoderma viride @ 2 gm/kg. seed and Pseudomonas flourescens@10gm/kg Captan 75 WS @ 1.5 to 2.5 gm a.i./litre for soil drenching. Imidacloprid 70 WS @ 10-15 gm a.i./kg seed (To be used in proper doses under guidance of an agriculture expert)	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Pigeon pea	Wilt, Blight and Root rot	Trichoderma spp. @ 4 gm/kg. Seed	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Pea	Root rot White rot	Seed treatment with 1. Bacillus subtilis 2. Pseudomonas fluorescens Soil application @ 2.5 – 5 kg in 100kg FYM	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Bhendi	Root knot nematode	Paecilomyces lilacinus and Pseudomonas fluorescens @ 10 gm/kg as seed dresser.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Tomato	Soil borne infection of fungal disease Early blight Damping off Wilt	T. viride @ 2 gm/100gm seed. Pseudomonas fluorescens and V. clamydosporium @ 10gm/kg as seed dresser.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Sunflower	Seed rot Jassids, Whitefly	Trichoderma viride @ 6 gm/kg seed. Imidacloprid 48FS @ 5-9 gm a.i. per kg. Seed (To be used in proper doses under guidance of an agriculture expert) Imidacloprid 70WS @ 7 gm a.i. per kg. Seed (To be used in proper doses under guidance of an agriculture expert)	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

1.8. Cropping systems

Farmers resort to cultivation of a number of crops and rotate particular crop combinations. More than 250 cropping systems are being followed in India, of which 30 cropping systems are more prevalent. Some of the important cropping systems are:

1. Sequential cropping system:

Growing crops in sequence within a crop year, one crop being sown after the harvest of the other. For example, rice followed by pigeonpea, pigeonpea followed by wheat.

2. Intercropping System:

Growing more than one crop in the same area in rows of definite proportion and pattern.

Cereals + Legumes

The following intercropping practices were found to be remunerative in India's groundnut growing states.

State	Crop combination
Maharashtra	Groundnut + Red gram (6:1/4:1) Groundnut + Soybean (6:2) Groundnut + Sunflower (6:2/3:1)
Gujarat	Groundnut + Castor (9:2/3:1) Groundnut + Sunflower (3:1/2:1) Groundnut + Red gram (4:1)

Alley cropping

Is an agroforestry practice in which perennial, preferably leguminous, trees or shrubs are grown simultaneously with an arable crop. The trees, managed as hedgerows, are grown in wide rows and the crop is planted in the interspace or 'alley' between the tree rows.

During the cropping phase, the trees are pruned. Prunings are used as green manure or mulch on the crop to improve the organic matter status of the soil and to provide nutrients, particularly nitrogen, to the crop.

Alley Cropping and Silvipasture

a. Season based cropping system

i. Kharif rice based cropping system
ii. Kharif maize based cropping system
iii. Kharif sorghum based cropping system
iv. Kharif millet based cropping system
v. Kharif groundnut based cropping system
vi. Winter wheat and chickpea based crop ping system
vii. Rabi sorghum based cropping system

b. Mixed cropping

In order to minimise the risk and uncertainty of mono cropping and to have sustainable yield and income, farmers are advised to go for mixed cropping.

Mixed Cropping

Integrated farming System (IFS)

To feed ever-increasing population of the country, extensive cropping system give ways to intensive cropping which are exploiting natural resources. Therefore in future more thrust will be on efficient natural resource management and sustainable production system. This encompasses an animal component, an perennial and annual crop component, aqua culture, agro based production and processing units. Integrated farming system typically involves:

- Many enterprises including animal component
- Planning is based on resource available
- It is purely location specific/farmer/holding specific activity plan
- Very high resource use efficiency
- Sustainable farming

IFS - Duck & Fish rearing

Objectives of IFS

- To compliment and maximize use of by products
- To provide useful employment to all the family members
- Maximizing land use
- Value addition
- Self sustainability
- Less dependence on external resources

Crop production in IFS

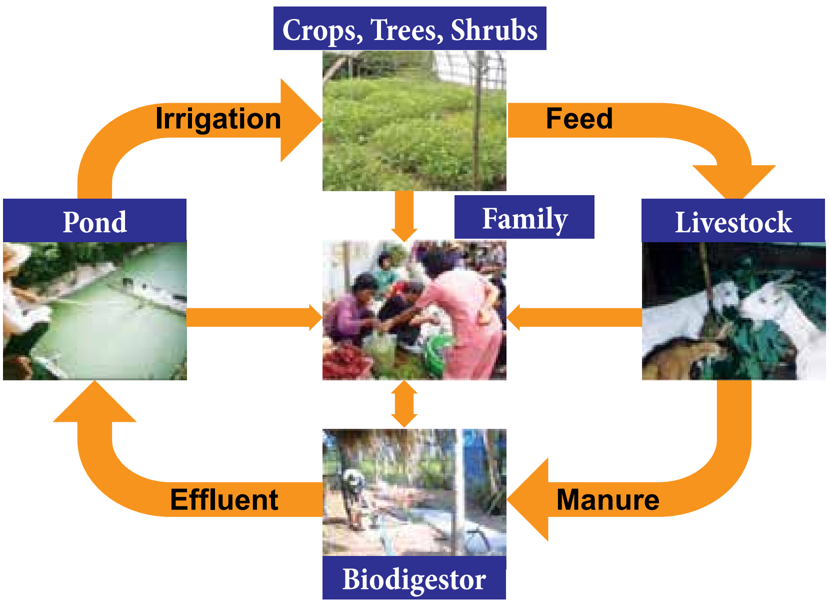
- Food crop should find a place
- Family food requirement should be planned
- Fodder production to meet the demand of animal component
- Specific enterprise based crops; e.g. mulberry/sunflower linked to honey
- Infrastructure based cropping
- Sufficient employment to family members

Animal component in IFS

- One or more animal components or combination of animal component may be planned
- Complimentary enterprises should be identified
- Composting should be the interface between animal and crop enterprises
- Market should be considered before hand
- Need based demand driven enterprises should be prioritized

Allocation of resource in IFS

- List the resources available and required
- Prioritize the resources based on scarcity
- Resource demand will be prioritized based on economic impact and sustainability
- Scares resource on the farm should be allocated for the most important activity
- Recycling of resources should be planned
- Resource based contingent plan should be prepared in advance. This will serve as a security and sustainable alternative in case of crisis



1.9. Mechanization

Modernization of agriculture requires appropriate machinery for ensuring timely field operations, effective application of agricultural inputs and reducing drudgery in agriculture.

Advantages of mechanization

- Increase cropping intensity
- Ensure large area coverage and timeliness
- Increasing farm labour productivity
- Increases crop productivity and profitability

First step in mechanization

- Get good hands on training
- Read manufacturer information
- Give attention to maintenance
- Understand do's and don'ts with respect to equipments and machinery used
- Take utmost care in following safety tips given in the manufacture information booklet.

Selection of farm machinery

- Select based on holding size
- Economic feasibility
- Availability of skilled labour to operate
- Workout the feasibility of hiring v/s owning
- Decide between universal equipment v/s crop specific equipment when multiple crops are grown
- If the initial investment is huge, think of community ownership/custom hire centres, etc.

Benefits of Agricultural Mechanization

Benefits	Value, %
Saving in seed	15-20
Saving in fertilizer	15-20
Saving in time	20-30
Reduction in labours	20-30
Increase in cropping intensity	5-20
Higher productivity	10-15
Substantial reduction in drudgery of farm workers especially that of women	

Farm Mechanisation Potential

Land Preparation


Wooden Plunk


Bullock Drawn Country Plough


Laser Guided Land Leveller


Field Operation of Tractor Drawn Disc Plough

Seeding and Planting Machinery


CRIDA 2 Row Planter

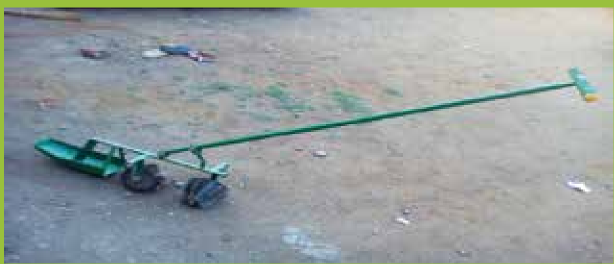

Seed Treating Drum


Field of Operation of Yanji Transplanter for SRI


Tractor Drawn CRIDA 9 Row Planter

Inter-Cultivation Equipments


Grubber Weeder
Cost savings of up to 60% are possible at the early stages of crop growth.


Cono Weeder
Weeding under wetland paddy cultivation


Tractor – Operated Cotton Weeder


Wheel Hoe
Reduces the cost of weeding up to 50%


B.D. 3 Tyne Cultivator

Plant Protection Equipments

Knapsack Power Sprayer

Tree Sprayer

Blower Sprayer

Power Tiller Mounted Sprayer

Harvesting Equipments

Coconut Tree Climber

- Used for picking of coconuts
- Average time taken for climbing up and down is about 6.30 min for a 13 m tree and time for fixing and removing the device on the tree is 4 minutes.

Groundnut Digger

Banana Clump Remover

Austoft Chopper Harvester

Cotton Stalk Puller

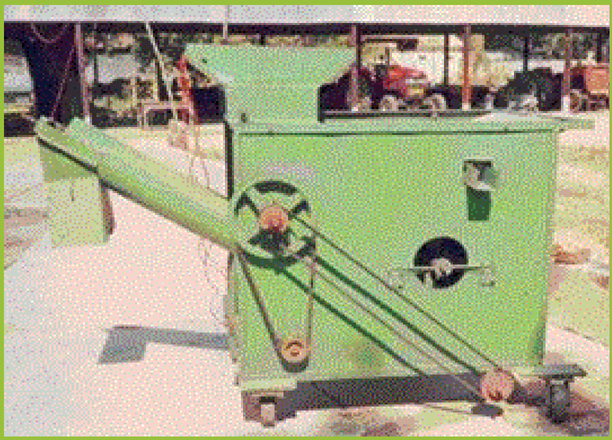
Threshing Equipments

Groundnut Pod Stripper

Castor Sheller

Winnowing and Clearing Equipments

Winnowing Fan



Seed Cleaner

1.10. Good Agricultural Practices (GAP)

Good Agricultural Practices (GAP) are *"practices that address environmental, economic and social sustainability for on-farm processes, and which result in safe and quality food and non-food agricultural products"*.

What are GAP codes, standards and regulations?

Good Agricultural Practices (GAP) codes, standards and regulations are guidelines which have been developed in recent years by the food industry, producers' organizations, governments and NGOs aiming to codify agricultural practices at farm level for a range of commodities.

Why do GAP codes, standards and regulations exist?

These GAP codes, programmes or standards exist because of:

- Growing concerns about food quality and safety worldwide.
- Fulfilment of trade and government regulatory requirements.
- Specific requirements especially for niche markets.

Objectives

- Ensuring safety and quality of produce in the food chain.
- Capturing new market advantages by modifying supply chain governance.
- Improving natural resources used, workers' health and working conditions to creating new market opportunities for farmers and exporters in developing countries.

The benefits of GAP codes

- Standards and regulations are numerous, including food quality and safety improvement.
- Facilitation of market access.
- Reduction in non-compliance risks regarding permitted pesticides, Maximum Residue Limits (MRLs) and other contamination hazards.

GAP related to crop protection

- Use resistant cultivars and varieties.
- Crop sequences, associations and cultural practices.
- Biological prevention of pests and diseases.
- Maintain regular and quantitative assessment of the balance status between pests and diseases and beneficial organisms of all crops.
- Adopt organic control practices where and when applicable.
- Apply pest and disease forecasting techniques where available.
- Determine interventions following consideration of all possible methods and their short and long-term effects on farm productivity and environmental implications. This will allow the minimizing of agrochemicals, in particular, to promote Integrated Pest Management (IPM).
- Store and use agrochemicals according to legal requirements of registration for individual crops, rates, timings, and pre-harvest intervals
- Ensure that agrochemicals are only applied by specially trained and knowledgeable persons.
- Ensure that equipment used for the handling and application of agrochemicals complies with established safety and maintenance standards.
- Maintain accurate records of agrochemical use.
- Identify the GAP in each protection method.

Crop rotation systems

- Sequence crops by selecting pest host relation.
- Selected crop for rotation in order to break the life cycle of pest (Jowar should be rotated with pulses to combat striga weed).
- The selected crop for rotation should not be the food of previous crop pest.
- To select appropriate crops for rotation:
  - Analyze the pest habitat
  - Follow forecasts
  - Monitor pest and natural enemies

Privilege resistant species

- Cultivate plant varieties which are less prone to pest attack.
- The resistant varieties reduce production cost.
- Pest resistant transgenic crops developed for specific pest can be used. This is new avenue for reducing pesticide load.

Seeding techniques

- Depth of placement
- Method of placement
- Time of placement
- Seed treatments
- Managing the above based on pest nature will give good results

Promote useful animals

- Keep good predator population.
- Promote growth of beneficial insects.
- Create an environment congenial for predators; e.g. keeping bird perch in the field.
- Identify the useful animals and study their habitat for providing the required environment.

Observe and control populations

- Follow forecast-short term and long term.
- Study habitat of pest and congenial weather.
- Accordingly take necessary precautions to manage pest.

Give priority to mechanical and biological measures (instead of chemical)

- Get the full knowledge about botanical pesticides.
- Get the knowledge on available parasites and predator/friendly insects and pests.
- Accordingly develop action plan for mechanical and biological measures.

- **Use non cash inputs:** Saves money.
- **Use information on plant protection:** Analyze spatial and temporal distribution and trend analysis.

Monitoring of performance through taking notes each year/season.

- Keep the pest management record along with season, weather and other agriculture activity.
- Document the pest load and control achieved
- Use this experience for future planning.

Precision farming: Use precision farming modules and apply Information Technology (IT) to economize and for effective monitoring.

Good Agriculture Practices help the farmers to make use of the opportunities available in International Markets for selling their products and realising better farm profits.

1.10. Lessons Learnt

1. Critical factors to be considered while deciding the crops and cropping pattern are climatic factors, soil conservation, water, cropping system options, past and present experiences of farmers, expected profit and risk, economic conditions of farmers including land holding, labour availability, mechanization potential technology availability and suitability, demand and availability of market policies and schemes, public and private extension influence, availability of required agricultural inputs including agricultural credit and post harvest storage and processing technologies.
2. Soil, water and wind erosion may be managed through various recommended practices.
3. Method of irrigation has to be decided considering the quantity of water available and crop to be grown.
4. Recommended certified seeds may be used.
5. Mechanisation enhances quality of agricultural operations and minimises the cost and dependence on labour.
6. Good Agricultural Practices (GAP) may be considered essentials to enhance the price and market competitiveness of the produce.

2. Soil and Plant Nutrition

2.1. Objectives of the session

- To increase the awareness and understanding about the soil, its structure, physical, chemical and biological properties and soil fertility.
- To strengthen the farmer's knowledge to manage the soil fertility in an economically and environmentally sustainable manner.

2.2. What we know at the end of the session

- Soil composition
- Physical, chemical and biological characteristics of soil
- Soil testing
- Plant nutrition requirement
- Organic and inorganic fertilizers
- Integrated Nutrient Management (INM) for efficient, economic and sustainable production

Know your Soil

2.3. What is Soil?

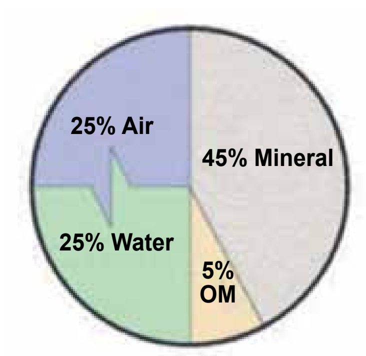
Soil is a thin layer of earth's crust, which serves as natural medium for the growth of plants. Rocks are the important sources for the parent materials over which soils are developed.

Soil Constituents

Rocks, the source of parent matererial

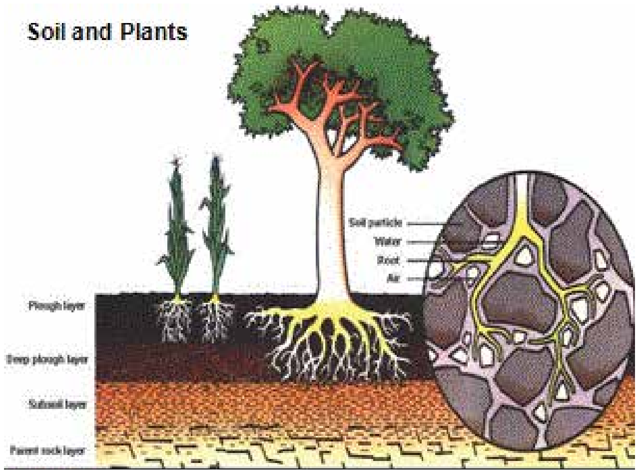
Soil is a dynamic medium made up of minerals, organic matter, water, air and living creatures including bacteria and earthworms.

It was formed and is forever changing due to 5 major physical factors: parent material, time, climate, organisms present and topography. The way in which we manage soil is another major factor influencing the character of the soil.



Soil Constituents

Soil features, properties and their importance

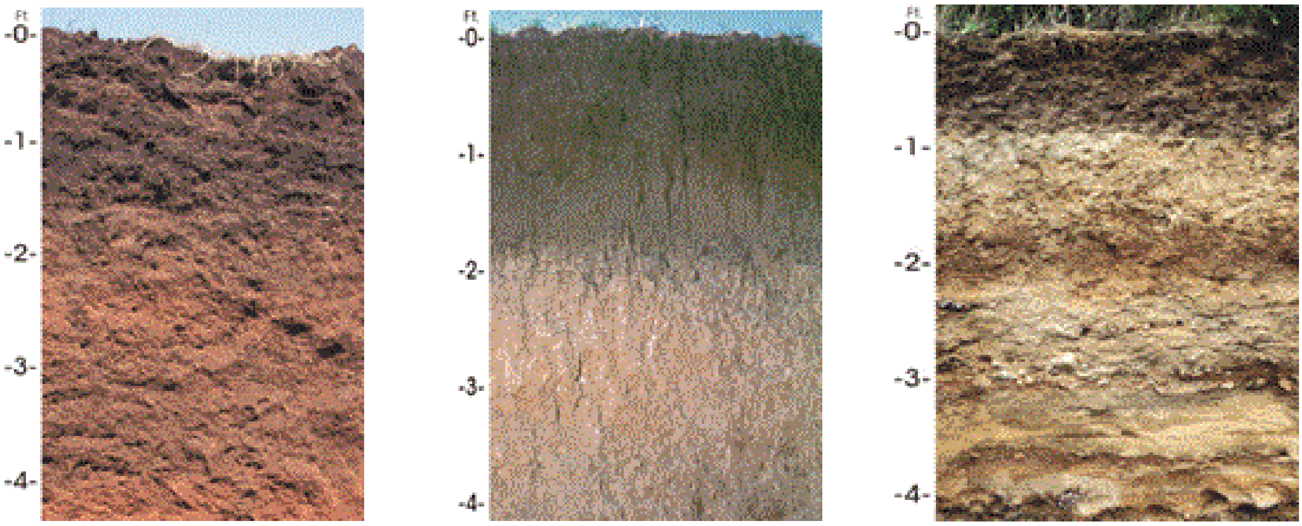


Soil features and properties

Soil colour

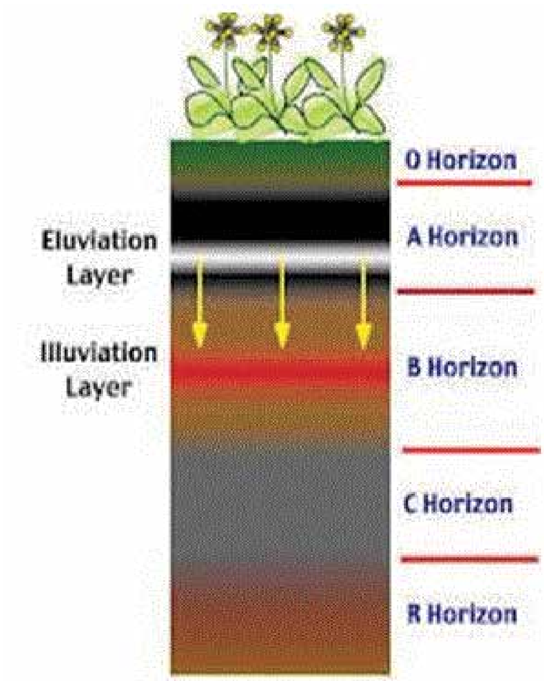
- Dark colour indicates usually medium to high fertility due to high amount of organic matter. These soils have usually high amount of nutrients, good water holding capacity and structure and are well aerated.
- Light colour indicates medium to low fertility. These soils may have leaching issue (water makes organic matter and other nutrients move downward faster).

Soil



Soil depth

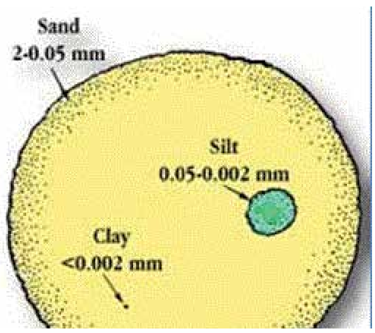
- The depth of soil to which the roots of a plant can readily penetrate to in order to reach water and nutrients.
- Minimum of 3-5 feet is desirable, deeper soils are better because they can hold more nutrients and water.



Soil Depth

Soil texture

- Texture refers to relative proportion of mineral particles (sand, silt, and clay) in soil. Many properties of soils; e.g. drainage, water holding capacity, aeration and the nutrient availability; depend largely on soil texture.
- **Sandy:** Low fertility and water holding capacity but good aeration.
- **Loamy:** Medium fertility and good aeration.
- **Clayey:** High fertility and poor aeration, hard to plough.

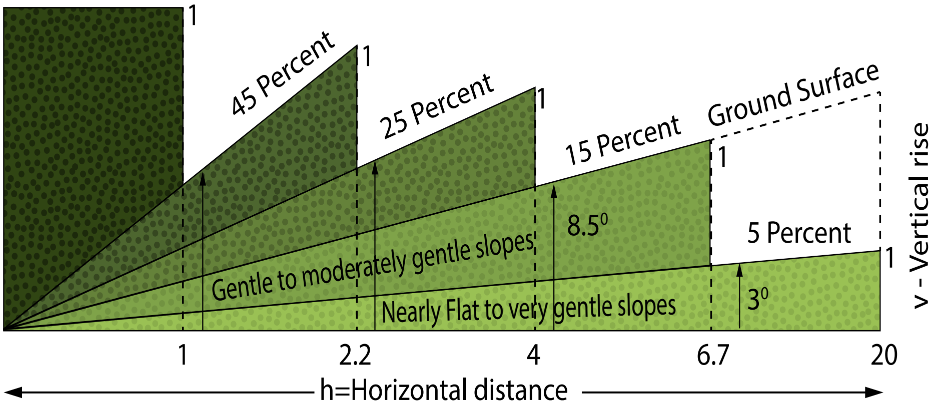


Soil Constituents

- Farmers may refer to their soil as heavy or light, illustrating the ease of working. The heavy soils are usually hard to plough and require much more effort than light soils. Organic matter may be added to improve the soil texture.

Land slope

- Soil gradient is the angle of inclination of the soil surface from the soil.
- It is expressed in percentage, which is the number of feet raise or fall in 100 feet from the horizontal distance.
- Mild gradient up to 1% is desirable.
- Higher gradients are not desirable as it leads to soil and water erosion.
- Perfect levelling is required only for paddy crop.

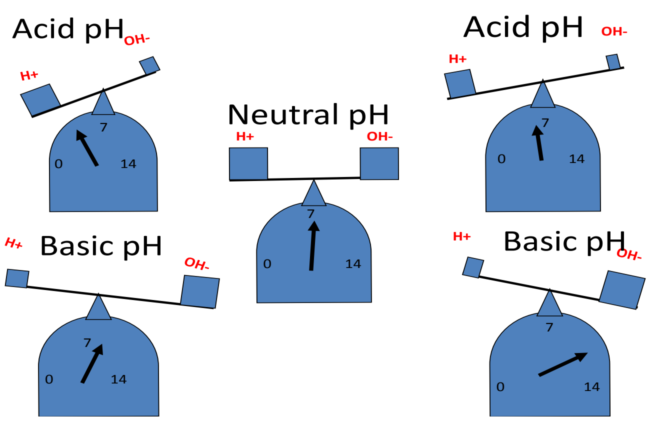


Land Slope

Soil pH

- Soil pH is of utmost importance in plant growth as it influences nutrient availability, toxicities and the activity of soil organisms.

PH Range	Soil Reaction Rating
<4.6	Extremely acid
4.6-5.5	Strongly acid
5.6-6.5	Moderately acid
6.6-6.9	Slightly acid
7.0	Neutral
7.1-8.5	Moderately alkaline
>8.5	Strongly alkaline



Tips for soil pH management

- Acid soils are to be corrected by using lime, quantity of lime application is as per soil test report.
- Alkali soils are to be corrected by gypsum/sulphur, quantity of application is as per soil test report.
- Saline – alkali soils should be treated with gypsum and improved drainage.

Soil organic matter

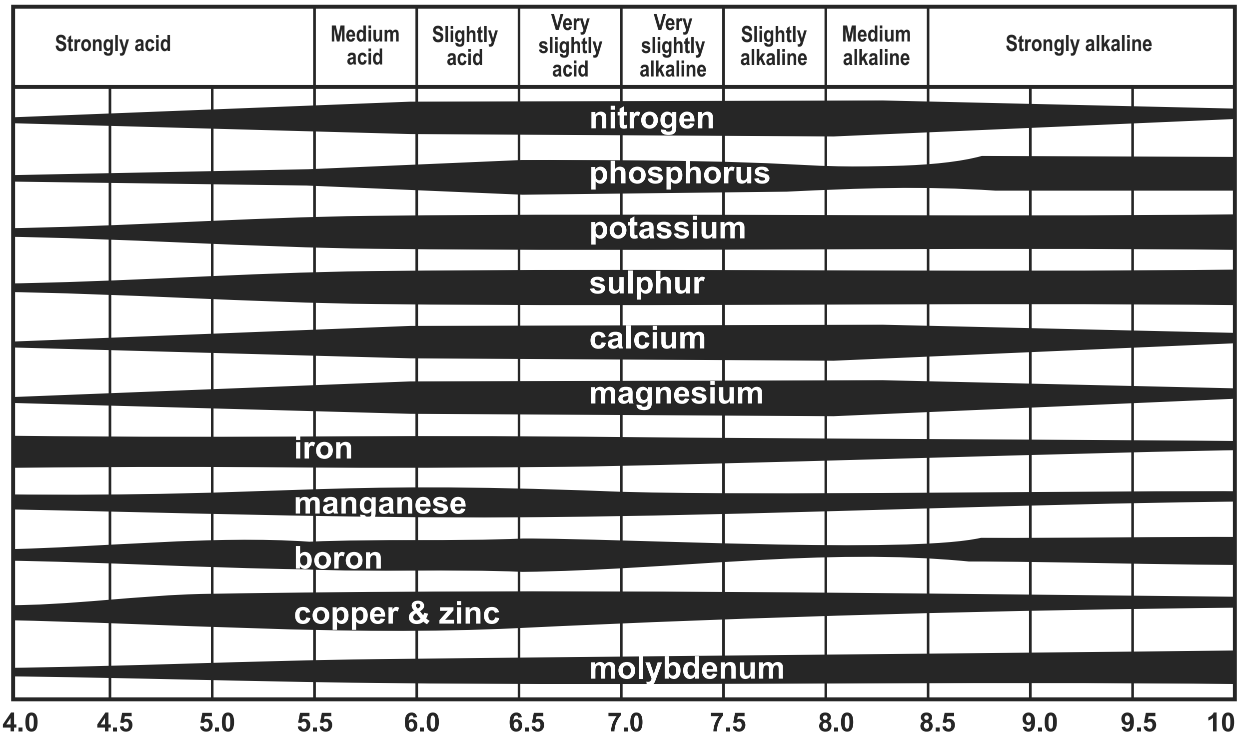
- Soil organic matter is the mix of plant and animal matter in different stages of decay.
- Soil organic matter plays a key role in biological, physical, and chemical function in soil.

Soil organic matter helps by:

- Providing nutrients for soil organisms
- Acting as major reservoir of plant nutrients
- Making nutrient exchange between soil and root of the plants easier
- Improving soil structure
- Influencing soil temperature
- Reducing the risk of soil erosion
- Increasing water holding capacity

Soil organic matter can be improved through:

- Recycling the crop residue back to field without wasting and burning
- Applying compost



Effect of Soil pH on Nutrient Availability

- Applying various organic manures
- Mulching organic wastes
- Using green manures and cover crops
- Suitable crop rotation
- Reducing soil tillage
- Avoiding soil erosion

Carbon - Nitrogen Ratio (C:N Ratio)

There are chemical elements in the organic matter, which are extremely important, especially in their relation or proportion to each other. They are Carbon and Nitrogen. The relationship is called Carbon - Nitrogen Ratio (C:N Ratio). For example, composed manure has 20:1 and sawdust has 400:1 of carbon and nitrogen. Generally speaking, the legumes are highest in nitrogen and have low C:N Ratio, which is highly desirable. Farmers can use blood meal, bone meal, poultry manure, cottonseed meal and soybean meal and other nitrogen rich material as organic matter, which enhance the decomposition.

Electrical Conductivity (EC): EC is normally considered to be a measurement of the dissolved salts in a solution.

General interpretation of EC values

Through application of zypsum, the saline/sodic soils can be amended. The quantity of zypsum to be applied is decided by EC value. Farmers having problem of saline/sodic soils can go for soil testing and approach extension officials for further guidance.

Soil	EC(mS/cm)	Crop reaction
Salt free	0 - 2	Salinity effect negligible, except for more sensitive crops
Slightly saline	4 - 8	Yield of many crops restricted
Moderately saline	8 - 15	Only tolerant crops yield satisfactorily
Highly saline	> 15	Only very tolerant crops yield satisfactorily

Soil fertility

- Soil fertility is generally defined as "ability of soil to supply plant nutrients". Soil structure, soil texture, temperature, water, light and air also play an important role in maintaining soil fertility.
- Plant nutrients which are often scarce in soil are nitrogen, potassium and phosphorus since plants use large amounts for their growth and survival.
- Important nutrients, their function and deficiency symptoms are described below.

2.4. Deficiency Symptoms of Nutrients in Plants


Nitrogen (N) – deficiency symptoms

Nitrogen (N) – deficiency symptoms

1. Stunted growth.
2. Appearance of light green to pale-yellow colour on the older leaves, starting from the tips. This is followed by death and/or dropping of the older leaves depending upon the degree of deficiency.
3. In acute deficiency, flowering is greatly reduced.
4. Lower protein content.


Phosphorous (P) – deficiency symptoms

Phosphorous (P) – deficiency symptoms

1. Overall stunted appearance, the mature leaves have characteristic dark to blue-green colouration, restricted root development.
2. In acute deficiency, occasional purpling of leaves and stems; spindly growth.
3. Delayed maturity and poor seed and fruit development.


Potassium (K) – deficiency symptoms

Potassium (K) – deficiency symptoms

1. Chlorosis along the leaf margins followed by scorching and browning of tips of older leaves. These symptoms then gradually progress inwards.
2. Slow and stunted growth of plants.
3. Stalks weaken and plant lodge easily.
4. Shrivelled seeds of fruits.

Calcium (Ca) – deficiency symptoms

Calcium (Ca) – deficiency symptoms

1. Calcium deficiencies are not seen in the field because secondary effects associated with high acidity limit growth.
2. The young leaves of new plants are affected first. These are often distorted, small and abnormally dark green.
3. Leaves may be cup-shaped and crinkled and the terminal buds deteriorate with some breakdown of petioles.
4. Root growth is markedly impaired; rooting of roots occurs.
5. Dessication of growing points (terminal buds) of plants under severe deficiency.
6. Buds and blossoms shed prematurely.
7. Stem structure weakened.

Magnesium (Mg) – deficiency symptoms

Magnesium (Mg) – deficiency symptoms

1. Interveinal chlorosis, mainly of older leaves, producing a streaked or patchy effect; with acute deficiency, the affected tissue may dry up and die.
2. Leaves usually small, brittle in final stages and curve upwards at margin.
3. In some vegetable plants, chlorotic spot between veins, with tints of orange, red and purple.
4. Twigs weak and prone to fungus attack, usually premature, leaf drop.

Sulphur (S) deficiency symptoms

Sulphur (S) deficiency symptoms

1. Younger leaves turn uniformly yellowish green or chlorotic.
2. Root growth is restricted, flower production often indeterminate.
3. Stems are stiff, woody and small in diameter.

Zinc (Zn) deficiency symptoms

Zinc (Zn) deficiency symptoms

1. Deficiency symptoms mostly appear on the 2nd or 3rd fully mature leaves from the top of plants.
2. In maize, from light yellow striping to a broad band of white or yellow tissue with reddish purple veins between the midrib and edges of the leaf, occurring mainly in the lower half of the leaf.
3. In wheat, a longitudinal band of white or yellow leaf tissue, followed by interveinal chlorotic mottling and white to brown necrotic lesions in the middle of the leaf blade; eventual collapse of the affected leaves near the middle.
4. In rice, after 15-20 days of transplanting, small scattered light yellow spots appear on the older leaves which later enlarge, coalesce and turn deep brown, the entire leaf becomes rust-brown in colour and dries out within a month.
5. In citrus, irregular interveinal chlorosis; terminal leaves become small and barrowed (little-leaf); fruit-bud formation is severely reduced, twigs die back plants.

Copper (Cu) - deficiency symptoms

Copper (Cu) - deficiency symptoms

1. In cereals, yellowing and curling of the leaf blades, restricted ear production and poor grain set, indeterminate tillering.
2. In citrus, die back of new growth; exanthema pockets of gum develop between the bark and the weed; the fruit shows brown spots.

Iron (Fe) - deficiency symptoms

Iron (Fe) - deficiency symptoms

1. Typical interveinal chlorosls; youngest leaves first affected, points and margins of leaves keep their green colour longest.
2. In severe case, the entire leaf, veins and interveinal area turn yellow and may eventually become bleached.


Boron (B) - deficiency symptoms

Boron (B) - deficiency symptoms

1. Death of growing plants (shoot tips).
2. The leaves have a thick texture, sometimes curling and becoming brittle.
3. Flowers do not form and root growth is stunted.
4. "Brown heart" in root crops characterized by dark spots on the thickest part of the root or splitting at centre.

Manganese (Mn) - deficiency symptoms

Manganese (Mn) - deficiency symptoms

1. Chlorosis between the veins of young leaves, characterized by the appearance of chlorotic and necrotic spots in the interveinal areas.
2. Greyish areas appear near the base of the younger leaves and become yellowish to yellow orange.
3. Symptoms of deficiency popularly known in sugarcane as "streak" disease.


Molybdenum (Mo) - deficiency symptoms

Molybdenum (Mo) - deficiency symptoms

1. Chlorotic interveinal mottling of the lower leaves, followed by marginal necrosis and in folding of the leaves.
2. In cauliflower, the leaf tissues wither leaving only the midrib and a few small pieces of leaf blade ("whiptail").
3. Molybdenum deficiency is markedly evident in leguminous plants.

Some common deficiency symptoms are:

Chlorosis - It is the loss of chlorophyll leading to yellowing in leaves. It is caused by the deficiency of elements like K, Mg, N, S, Fe, Mn, Zn and Mo.

Necrosis are death of tissues, particularly leaf tissue is caused by deficiency of K, Ca, Mg

Inhibition of cell division is caused due to lack or deficiency of N, K, S and Mo.

Premature fall of leaves and buds

- deficiency of K and P.

	**Stunted/Retarded plant growth** caused by the deficiency of N, P, K, Zn, Ca.
	**Delay in flowering** due to deficiency of N, S and Mo.

In case the farmers observe the above symptoms, farmers are advised to consult the local extension worker for remedies.

2.5. Different fertilizers and their nutrient content

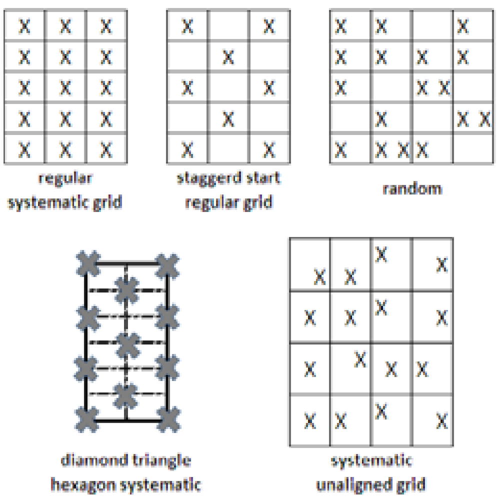
Important chemical fertilizers are the source of major nutrients. Different fertilizers and their nutrient content are illustrated in the table below:

Fertilizer	Nutrient content (%)		
	N	P*	K
Single nutrient fertilizers			
Ammonium sulphate	20	0	0
Urea	46	0	0
Calcium ammonium nitrate[1]	28	0	0
Single super phosphate	0	7	0
Triple Super Phosphate	0	20	0
Potassium sulphate	0	0	40
Muriate of Potash[2]	0	0	48
Double fertilizers			
Ammonium Phosphate	11	23	0
Diammonium Phosphate[3]	18	20	0
Complete Fertilizers			
Sampurna[4]	19	19	19
Vijaya Complex[5]	17	17	17
IFFCO Grade I[6]	10	26	26

A. Soil Analysis: Key to a Successful Nutrient Management Plan

Higher crop yields and quality of the crops depend largely on the efficient supply of nutrients. Soil provides not only the medium but also functions as the source of these nutrients for the plants. Soil resources get depleted with every harvest and need to be replenished for every crop. However, one must know which nutrients are to what extent depleted and what addition of fertilizers should be planned accordingly. Soil analysis, in this regard, helps in determining the level of nutrients and in deciding the required amount of fertilizer application.

Accuracy of soil analysis is directly related to the quality of the soil sample taken. Application of appropriate fertilizers with the proper nutrient mix will help not only to increase the productivity and farm income but also provide a more realistic chance to obtain the desired yield. There are various methods for taking a soil sample from a field. The right method of sampling may be decided by consulting local extension officer or your DESAI-trainer.



Soil Sampling Methods

[1]*__Calcium ammonium nitrate__- To be used with caution.*

[2]*__Muriate of Potash__- To be used with caution and protective equipment (Respirator)*

[3]*__Diammonium Phosphate__- To be used with caution.*

[4]*__Sampurna- Compilation Source__- Manufacturer (Zuari Agro Chemicals).*
[5]*__Vijaya Complex__- Indigenous Complex Fertilizer.*
[6]*__IFFCO Grade I__- Compilation Source- Manufacturer*
(To be used with caution.)

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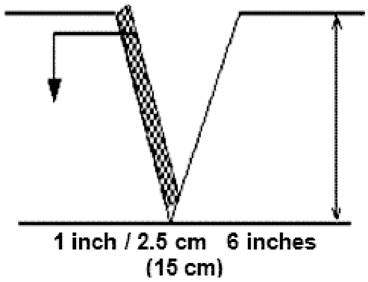
Soil sampling: an illustration

Selecting sampling spot

Remove the surface litter at the sampling spot

Make a 'V' shaped cut to a depth of 15 cm in the sampling spot



Collect soils - V shaped cut:
Remove thick slices of soil from top to bottom of exposed face of the 'V' shaped cut and place in a clean container

Mix the samples thoroughly

Quartering is done by dividing the thoroughly mixed sample into four equal parts

Two opposite quarters are discarded and the remaining is mixed

Collect the sample in a clean cloth or polythene bag

Process of collection

Label with required information:

- Name of the farmer
- Location of the farm
- Survey number
- Previous crop grown, present crop
- Crop to be grown in the next season
- Date of collection
- Name of the sampler, etc

Places for soil testing:

- Krishi Vigyan Kendra (KVK)
- State/district agricultural labs
- Agriculture University, Research Stations
- ATMA Officials/Agripreneurs

B. Plant Analysis

Plant Analysis is the second tool, after soil testing that is critical to improve crop nutrition and yield. Only plant analysis can identify the actual nutrient status of a plant or crop. While soil testing identifies the nutrients offered to the crop or plants, Plant analysis identifies how well the plants utilizes the soil nutrients and applied nutrients. Plant analysis allows the plant to tell us what nutrients it needs.

It is critical that the correct plant part and stage of growth be sampled. The normal nutrient concentration differs between the various plant parts. Also, the normal nutrient concentration of each plant part changes as the plant matures. It is important to keep these factors in mind to assess the nutrients status of plants. The plant parts required to be collected for different crops are as follows.

Crop	Growth stage	Plant part to be sampled	Quantity
Cotton	(a) Seedling, 6" to 12" tall	Entire above ground portion of plant.	15 –20 plants
	(b) Prior to or at first bloom (c) When first squares appear	Youngest fully mature leaves from the main stem of plant. Discard the petioles.	15 – 20 leaves
Sugar Cane	2 Months to mature	Second fully mature leaf without sheath.	15 to 25 leaves
Sunflower	(a) Seedling stage	Entire above ground portion of plant.	15 to 20 plants
	(b) Vegetative to full bloom	Youngest fully mature leaf. No petiole.	15 to 20 leaves

C. Irrigation Water Analysis

Irrigation water, irrespective of its source, always contains some soluble salts. The suitability of waters for a specific purpose depends upon the types and amounts of dissolved salts. Some of the dissolved salts or other constituents may be useful for crops but the quality or suitability of water for irrigation purposes is assessed in terms of the presence of undesirable constituents. Some of the dissolved ions such as NO3 are useful for crops.

The most important characteristic that determine the quality of irrigation water are:

- pH
- Total concentration of soluble salts are judged through Electrical Conductivity (EC).
- Relative proportion of sodium to other cations such as Ca and Mg are referred to as Sodium Adsorption Ratio (SAR).
- Concentration of boron or other elements that may be toxic to plants.
- Concentration of carbonates and bi-carbonate as related to the concentration of calcium plus magnesium are referred to as Residual Sodium Carbonate (RSC).
- Content of anions such as chloride, sulphate and nitrate.

Methods of water sample collection

As water quality and suitability plays an important role in deciding production and productivity of crops, farmers are advised consult extension worker to test the water.

Nutrient requirement and productivity

Keeping soil fertility and soil nutrients at optimum level helps increasing the productivity of the soil. Although the ability of the soil to supply the plants with the required nutrients depends also on the soil condition like i.e. soil texture, soil structure and soil organic matter. Therefore, soil fertility management does not only include nutrient management but also soil condition management.

Apart from other soil management practices, soil fertility focuses on:

- Maintain a balance between nutrient uptake and nutrient application.
- Adequate fertility for the plants at the specific growth stages.
- Soil fertility and organic matter maintenance
- Minimizing the nutrient loss by avoiding excess application.

The soil nutrient demand usually is based on:

- Soil nutrient level
- Crop variety and yielding ability
- Soil moisture
- Targeted yield

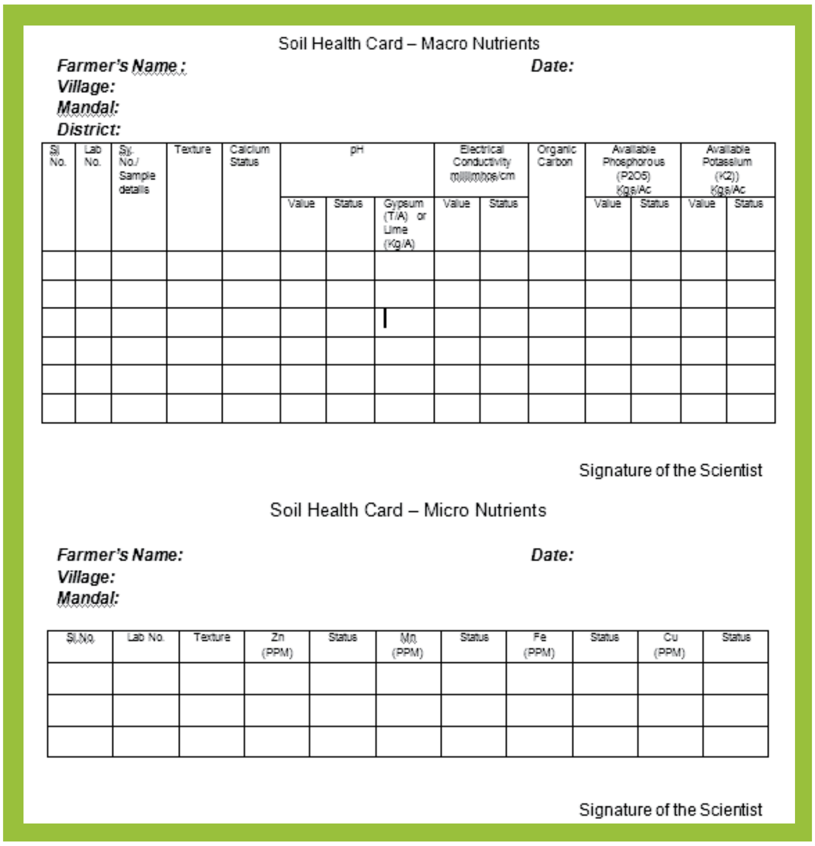
Soil analysis direct the farmer on quantity and quality of fertilisers to be used. Farmers can get the crop specific nutrient (fertilizer) recommendations after consulting the local extension service or their DESAI-trainer and discussing with them the soil analysis report.

C. Recommended Fertilizer Dose for Important Crops

General considerations before fertilizer application:

- Nutrients and not fertilizers should be bought. (Think nutrients, not fertilizers).
- Each nutrient applied as fertilizer should give a desired production response.
- The cost of fertilization must be calculated on the basis of applied plant nutrients per unit area of land.
- Calculate one nutrient at a time, considering available sources, prices and feasibility of using.
- A sample Soil Health Card, issued to the farmers based on the soil test results, should be used by the farmer to calculate the quantities of fertilizers required:

	N (kgs/ha)	P2 O5 (kgs/ha	K2O (kgs/ha	Remarks
Banana	110	35	330	Apply 50% extra fertilizers at 2nd , 4th, 6th & 8th months after planting for tissue culture banana
Cotton	120	60	60	(TCHB – 213)
Citrus (sweet orange)	0.6 kgs	0.2 kgs	0.3 kgs	From 6th year onwards
Mango	1.0 kg	1.0 kg	1.5 kg	Kg of NPK/tree for 6th year onwards
Sugar cane	275	-	112.5	
Sun flower	60	90	60	Irrigated Hybrid
	40	50	40	Rainfed/Varieties



Soil Health Card – Macro Nutrients

Farmer's Name : *Date:*
Village:
Mandal:
District:

Sl No.	Lab No.	Sy. No./ Sample details	Texture	Calcium Status	pH			Electrical Conductivity millimhos/cm		Organic Carbon	Available Phosphorous (P2O5) Kgs/Ac		Available Potassium (K2)) Kgs/Ac	
					Value	Status	Gypsum (T/A) or Lime (Kg/A)	Value	Status		Value	Status	Value	Status

Signature of the Scientist

Soil Health Card – Micro Nutrients

Farmer's Name: *Date:*
Village:
Mandal:

Sl.No.	Lab No.	Texture	Zn (PPM)	Status	Mn (PPM)	Status	Fe (PPM)	Status	Cu (PPM)	Status

Signature of the Scientist

If the samples are all above the critical level, there is no deficiency of any element.

Nutrient	SL, SCL	CL
Zn	0.65	0.70
Cu	0.20	0.30
Fe	4.00	6.00
Mn	2.00	3.00

Fertilizers: Chemical or Organic

Chemical fertilizers

Advantages:

- Nutrients are immediately available for plant uptake.
- Price is lower as compared to organic fertilizer.
- Small quantities are required because they are nutrient rich.

Disadvantages:

- Over application usually results in economic and environmental losses.
- Over supply makes plant tissues soft and vulnerable to diseases and pathogens.
- Increased rate of soil organic matter decomposition resulting in soil degradation.
- Many nutrients applied are easily lost through different chemical reactions.

Organic fertilizers

Advantages:

- Balanced nutrient supply.
- Enhance the soil biological activity.
- Help in improving soil structure.
- Increase the organic matter content.
- Slow release of nutrients makes soil on the long run fertile.
- Help in combating plant diseases.

Disadvantages

- Low nutrient content.
- Only effective in the long run.
- It may not supply all the nutrients required for plant growth.
- High cost.
- Bulkiness.

Efficient fertilizer use

Good knowledge and management practices can improve the fertilizer use efficiency.

- Select the crops and varieties that suit the locality and have best fertilizer response.
- Select right kind of fertilizer according to crop and soil.
- The fertilization should be planned for the cropping pattern and not for single crops.
- Fertilization application rate should be decided only after discussing your soil analysis report with your local extension officer or Desai-Trainer.
- Balanced fertilization should be practiced.
- Crops should only be sown at the locally recommended periods.
- Maintain optimum plant population and proper plant spacing.
- Effective control of pests and diseases will help in maximizing the fertilizer efficiency.
- To maximize the yield increase through fertilizer, all other growth critical factors must also be optimum e.g. crop must be irrigated at critical growth stages.

Fertilizer application methods

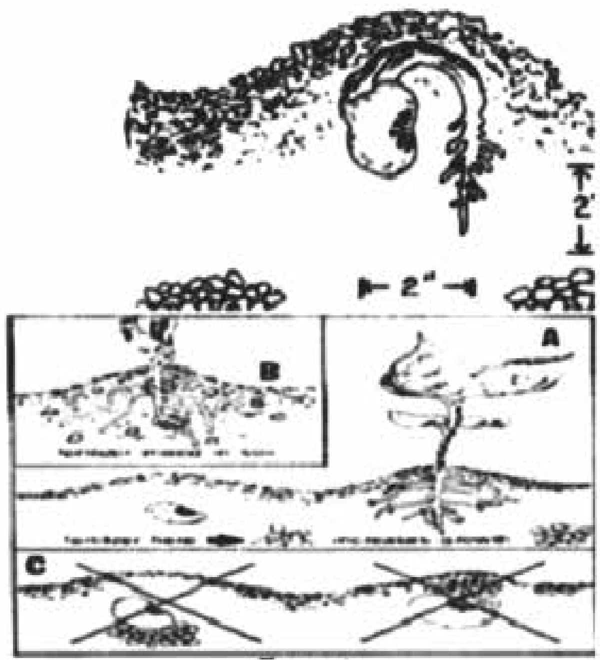
Broadcasting: fertilizer is distributed manually over the cropped field.

- The most common fertilizer application method.
- Highly inefficient method
- High economic and nutrient losses

Broadcasting

Placement: application in band or packets near the plants.

- Two sub-types:
  - i. Band application
  - ii. Spot Application
- The fertilizer use efficiency is high.
- Labour intensive.
- Efficient method but with high labour input.

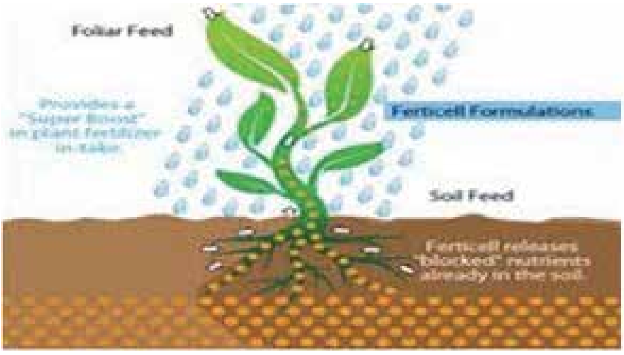


Placement

Ring application: Spread the fertilizer around the tree at a distance of about one meter.

Ring Application

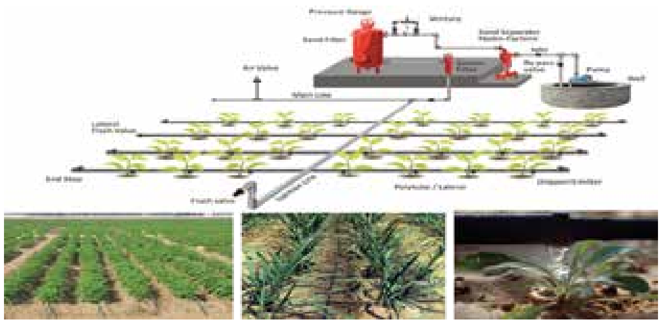
Foliar application: liquid fertilizers are sprayed on the crops.



Foliar Application

- Highly efficient
- Special equipment required
- High cost
- Only selected fertilizers can be applied

Fertigation along with irrigation



Fertigation

Method of application and nutrients

The method of application should be chosen according to the nutrient, crop, soil and cultivation method.

- **Nitrogen** application should be applied in splits and slow release mode to minimize loss.
- Frequent application in small quantity through foliar application is most efficient and results in quick recovery of crops.
- Use slow release nitrogen for plantation crops and long duration crops.
- Under puddle condition, use coated urea-neem oil. Coal tar sulphur coating will make urea to release nitrogen slow to match the uptake pattern.
- Urea can be cured with soil (1 part urea with 5-10 parts soil) to reduce the losses.
- Phosphate should be placed 4 to 6 cm below and 4 to 6 cm away from the seeds to ensure maximum availability.
- Phosphatic fertilizers give better response when placed in bands near the plant rows.
- Potassic fertilizers can be applied in one dose as basal application but for long duration crops the fertilizer application may be done in 2 to 3 splits.

Fertilizer calculations

Here is an example for the application of fertilizers based on the soil test recommendations. Suppose the recommendations are 120 kg N, 60 Kg P and 40 Kg K per ha. Calculate the quantity of urea, super-phosphate and muriate of potash fertilizers needed to supply the recommended doses!

Urea content is 46%, so to supply 46 kg N/ha 100 kg urea is required. To supply 120 kg N/ha 100/46*120=260.9 or 261 kg urea is required. Similarly super phosphate content is 16% P2O5 and the recommendation is given in the form of **Phosphorous.**

- % P = %P2O5 X 0.44
- Kg P = kg P2O5 X 0.44
- Kg P = 16 X 0.44
- Kg P = 7.04

So super phosphate contains 7.04 P and the recommendation is given as 60 kg P.

To supply 60 kg P=100/7.04*60=852.27 or 852 kg single super phosphate is required. Similarly, we have to calculate the dose of potash through Muriate of Potash (MOP) as MOP contains 60% K2O.

So MOP contains 49.8 K and the recommendation given is 40 kg K/ha.

To supply 40 kg K=100/49.8*40=80.3 kg or 80 kg MOP is required. So based on fertilizer analysis and soil test information the fertilizer application rates are calculated as above.

The formula for calculating the fertilizer to be applied (kg/ha) =

$$\frac{100}{\text{Nutrient content in the fertilizer material (\%)}} \times \text{recommended dose (kg/ha)}$$

Nutrient cost comparison

Example - 1

1. Urea with 46% N costs Rs.562.20 per 100 kg.
2. Ammonium sulphate 20% N costs Rs.1029 per 100 kg.

Urea has 46% N i.e. 46 kg N in every 100 kg urea. Therefore unit value of N in urea: 562.2/46= Rs.12.22 per kg N. Ammonium sulphate has 20.6% N i.e. 20 kg N in every 100 kg fertilizer and 24% sulphur. Therefore unit value of N in ammonium sulphate: 1029/20.6= Rs.49.95 per kg N. Thus, the nitrogen is cheaper in urea. Yet, ammonium sulphate also has 24% sulphur in it.

Therefore, for soils with sulphur deficiency, ammonium sulphate is a better choice and for soils with normal sulphur levels, urea presents a better N source.

Example - 2

1. Single Super Phosphate (SSP) with 7 % P costs Rs.480 per 100 kg.
2. Di - Ammonium Phosphate (DAP) 20% P and 18% N costs Rs.1596 per 100 kg.

SSP has 7% P i.e. 7 kg P in every 100 kg SSP. Therefore unit value of P in SSP: 480/7= Rs.68.57 per kg P.

Whereas, DAP has 20% P and 18% N, i.e. 20 kg P and 18 kg of N in every 100 kg of DAP.

Cost of Nitrogen in 100 kg of DAP = (18 X 12.22) = Rs.219.96. Therefore unit value of P in DAP: (1596 – 219.96 = 1376.00); i.e. 1376/20= Rs.68.80 per kg P

Thus, the unit cost of P is the same in both the fertilisers. Yet, DAP also has 18% N in it. Therefore, for soils with nitrogen requirement, DAP is the better choice.

GUIDE FOR MIXING FERTILIZERS

1. Muriate of Potash	2. Sulphate of Potash	3. Sulphate of ammonia	4. Calcium ammonium nitrate	5. Sodium Nitrate	6. Calcium cynanamide	7. Urea	8. Superphosphate single or triple	9. Ammonium phosphate	10. Basic slag	11. Calcium carbonate	
□	□	□	⊠	⊠	⊠	⊠	□	□	□	□	1. Muriate of Potash
□	□	□	⊠	⊠	□	⊠	□	□	□	□	2. Sulphate of Potash
□	□	□	□	⊠	■	⊠	□	□	■	■	3. Sulphate of ammonia
⊠	⊠	□	□	⊠	■	⊠	⊠	⊠	■	□	4. Calcium ammonium nitrate
⊠	□	□	□	□	⊠	⊠	□	□	□	□	5. Sodium Nitrate
⊠	□	■	■	⊠	□	⊠	■	■	□	□	6. Calcium cynanamide
⊠	⊠	⊠	⊠	⊠	⊠	⊠	⊠	⊠	⊠	⊠	7. Urea
□	□	□	⊠	⊠	■	⊠	□	□	■	■	8. Superphosphate single or triple
□	□	□	⊠	⊠	■	⊠	□	□	■	■	9. Ammonium phosphate
□	□	■	■	□	□	⊠	■	■	□	□	10. Basic slag
□	□	■	□	□	□	⊠	■	■	□	□	11. Calcium carbonate

□ Fertilizers which can be mixed

⊠ Fertilizers which may be mixed shortly before use

■ Fertilizers which can not be mixed

Note: The crossing point of the required vertical column and horizontal colum indicates the possibility of mixing or otherwise of the fertilizer

Compatibility of Fertilizers

Organic fertilizers

Organic manures are natural products used to provide the nutrients to crops. Examples of organic manure are cow dung, farmyard manure, green manure, compost from crop residues, vermicompost and other biological waste.

Organic manures increase the organic matter in the soil. Organic matter in turn releases the plant food in available form for the use of crops. However, organic manures should not be seen only as carriers of plant food. These manures also enable a soil to hold more water and also help to improve the drainage in clay soils. They provide organic acids that help to dissolve soil nutrients and make them available for the plants.

Additionally, organic manures have low nutrient content and, therefore, need to be applied in large quantities. For example, to get a standard mixture of 25 kg NPK, almost 1000-2000 kg organic manure is required, whereas the same amount of NPK can be easily obtained with a 50 kg bag of NPK fertilizer.

Compost: Compost is well decomposed organic wastes like plant residues, animal dung and urine earth from cattle sheds, waste fodder, etc.

Tips for FYM preparation

- Provide shade to compost site
- Make compost pit in a well drained elevated place
- Smaller heaps of manageable shape are better
- Keep the pit free from weeds
- Pit should be near to cattle shed and water source
- Allowing for full maturity
- Also use urine
- If ash is added do not heap in the pit but spread uniformly

How to use

- Manure should not be kept exposed in the field before application
- Preferably apply in bands into the soil instead of broadcast. In case broadcast, work it into the soil immediately

Advantage

- Complete plant nutrient
- Improved soil structure increases soil aeration and drainability
- Organic matter acts as biological clay and increase nutrient and water holding capacity of soil
- It reduces the bulk density to desirable level (1.3g/cc)
- It provides enough food for micro organisms in the soil and helps to build up microorganisms in the soil
- Nutrients slow release nature is very useful for long duration and plantation crops

Disadvantage

- It is bulky
- Cost of handling, transportation and storage are high
- High labour input
- High cost per kg of nutrients

Vermi-Compost: Vermi-composting is a process by which earthworms convert organic waste into fertile manure. Important spp of earthworm used for vermicomposting in Indian conditions are Epigeic Eugeniae, Eisenia Foetida, and Perionyx Excavatus.

Earthworms

- Feed on soil and soil organic matter and convert it into compost
- Encourage growth of useful micro organisms
- Aerate and pulverize the soil
- Make soil porous and improving drainage
- Increasing water holding capacity of soil
- Give strength to plant immunity system

Advantages of vermi-compost

- Easy to use
- Low cost to produce
- Convert organic matter in to good manure in short time

How to make it

- Enrich vermi-compost with bio-agents
- Earthworms are our friends. Protect them and nurture in the field

Green manures

- Select important species
- Find a season to fit the green manuring crop in to cropping system
- Incorporate when 45 days old
- Allow 15-20 days before next planting for decomposition

Example: Sunhemp and Daincha

Green leaf manure

- Grow leguminous trees on road side, on bunds and waste land

- Loppings of trees can be incorporated 15 days before planting
- Provide approximately 5 to 10 tons

Example: Pongamia and Neem

Oilcakes

- Concentrated organic matter
- Mix with chemical fertilizer is useful as they make fertilizer slow release
- Use to enrich compost/organics
- Use preferably non edible oil cakes

Liquid Fertilisers

- Liquid form of fertilizers are applied with irrigation water or for direct application through foliar spray to augment yield and improve quality of a variety of crops like fruits, vegetables, oil seeds, pulses, cereals, cotton, tobacco, sugarcane, tea, etc.
- It will ease handling, less labour requirement as well as the possibility of mixing with herbicides have made the liquid fertilisers more acceptable to farmers

Fertigation

- Fertigation is the judicious application of fertilizers by combining it with irrigation water
- Fertigation can be achieved through fertilizer tank, venturi system, injector pump, Non-Electric Proportional Liquid Dispenser (NEPLD) and automated system

Advantages of Fertigation

- Ensures a regular flow of water as well as , resulting in increased growth rates for higher yields
- Offers greater versatility in the timing of the nutrient application to meet specific crop demands
- Improves availability of nutrients and their uptake by the roots
- Safer application method which eliminates the danger of burning the plant root system
- Offers simpler and more convenient application than soil application of fertilizer, thus, saving time, labour, equipment and energy
- Improves fertilizer use efficiency
- Reduction of soil compaction and mechanical damage to the crops
- Potential reduction of environmental contamination
- Convenient use of compound and ready-mix nutrient solutions which also contain a small concentration of micronutrients

Nutrient content of important organic manures

Nutrient content of important organic manures

Organic Manure	Percentage of Nutrients		
	Nitrogen	Phosphorus (P2O5)	Potassium (K2O)
Poultry Manure	1.2-1.5	-	-
Sheep Manure	0.8-1.6	-	-
Farmyard Manure	0.4	0. 3	0.2
Compost	0.5	0.25	0.5
Bone Meal	3.5	21.0	-

Bio-Fertilizers

These are products of microbial origin containing live cells of micro organisms multiplied in a laboratory and mixed with a carrier material like finely powdered coal, lignite or humus and supplied in a solid form.

Basically two types of bio fertilizer distinguished based on their capacity to supply nitrogen or phosphorus. Some bio-fertilizers have the capacity to supply nitrogen because they have the capacity to absorb nitrogen gas from the atmosphere in association with plants and use the nutrient for their cell synthesis. The nitrogen fixed becomes available to crop plants after the plant material is incorporated in to soil. This process is known as symbiotic nitrogen fixation. Other nitrogen fixing organisms can live freely in soil and fix nitrogen. There is one more type where the organism establishes a weak or associative symbiosis and fixes nitrogen.

In addition to nitrogen fixers there are some bacteria and fungi, which are used as bio-fertilizers to enhance phosphorus supply to crops. They solubilize insoluble or difficulty soluble phosphorus through their capacity to produce organic acids.

Plant Nutrient	Microorganism	Crops Benefited
NITROGEN	a) Symbiotic Rhizobium Azolla	All leguminous crops; Rice
	b) Associative Symbiosis Azospirillum	All cereal crops; Sugar cane
	c) Non Symbiotic Azotobacter Blue- green Algae	All crops; Rice

PHOSPHORUS	Microorganism	Crops Benefited
a) Solubilizing Effect	Bacteria: (Bacillus, Pseudomonas)	All Crops
	Fungi: (Aspergillus Penicillium)	All Crops
b) Absorbing Effect	Ecto Mycorrhizal	Tree crops
	Endomycorhizae	All Crops

All bio-fertilizers are microorganisms belonging to group of bacteria or fungi or blue-green algae.

The capacity of N-Fixers to supply N to crops varies from 10 kg – 100 kg/ha and symbiotic nitrogen fixers particularly rhizobium bacteria are very efficient. The P-Solubilizers and P-Absorbers can mobilize insoluble phosphorus to the extent of 10 Kg to 50 kg P/ha. Mycorrihizae are symbiotic fungi, which get associated with plant roots. Azolla is a water fern which establishes a very active and beneficial association with blue-green algae - Anabaena

Algal inoculants are cultivated on submerged soils in tanks, dried and supplied as soil based culture (Inoculants). Endomycorrhizal inoculants are supplied as pieces of roots of grass plants infected with mycorrhizae. Ectomycorrihizal inoculants are cultivated in the laboratory and mixed with carrier material for supply. Azolla are supplied as fresh fronds (leaf). The bio-fertilizers are inoculated to seeds or applied directly to soil after suspending them in water. Azolla is applied directly to soil, which on multiplication establishes a very active symbiotic association with Anabaena, a blue-green alga.

- The best nutrient management method is Integrated Nutrient Management.
- Use bulk quantity of organic matter to provide good physical and chemical properties to soil.
- Smaller quantities of chemical fertilizer to provide quick release, which matches the uptake pattern of crops.

I. Package of Practices – Banana

Banana is one of the most commonly grown fruit crop of the country. India produces about 26.217 MT of banana from an area of 0.709 Mha with an average productivity of 37.0 mt/ha. Major producing states are Tamil Nadu, Maharashtra, Karnataka, Gujarat, Andhra Pradesh, Assam and Madhya Pradesh. Tamil Nadu has 0.1244 mha under banana and the total production during 2008-09 is 6.667 Mmt with 53.6 mt/ha productivity. In Gujarat, banana crop is cultivated in 11 districts covering an area of about 60900 ha. Gujarat is ranking 3rd among the states of India with an average productivity of 58.7t/ha. Unripe fruits are used for making chips, vegetable flour, etc. Ripened fruits are used for preparing drink, jam, beer, salad, etc. Banana fruits are having numerous medicinal uses (as per Ayurveda). From psedostem threads are obtained and used for making rope and cloths.

Prevailing Varieties: Basarai, Lokhandi, Robasta, Shreemanti and Grand Naine.

Climate: Banana grows well in warm and humid climate with an average temperature of 27oC and rainfall of 2,000 to 2,500 mm. However, it requires assured irrigation facility.

Soil type suitability: Loamy and salty clay loam soils with good fertility status are best suited for banana cultivation. However, extremely clayey and sandy soils are not suitable for banana crop.

Land preparation: The land should be ploughed, harrowed and planked to achieve levelled fields. For planting banana, dugout a pit of 30 x 30 x 30 cm size at a spacing of 1.5 x 1.5 m.

Soil sterilization: After land preparation, the pits should be exposed to the sunlight for 10 to 15 days. In the case of heavy infestation of soil born pest and diseases, the soil solarization should be done using transparent plastic for a period of 15 to 20 days during summer season (preferably during May).

Time of plating: Optimum time for planting is 15th June to 15th July. Banana planting is done either by suckers or tissue culture plant.

Suckers: Suckers should be selected from a healthy field of banana. Sufficient care should be taken that suckers should not be damaged during digging and transport. Fresh sucker, weighing about 500 to 1,500 g, should be selected. For control of fungal diseases, suckers should be dipped in a solution of aurofugine[1] (10 g / 100 litres of water) for 1.5 hour prior to planting.

Tissue culture plant: Select healthy and uniform plants.

Irrigation and scheduling: The water requirement of banana crop by surface method of irrigation (25-30 irrigations) is 1,500-1,800 mm. By drip irrigation method, water requirement is 900-1,080 mm. The drip system should be operated on alternate day for a period of 1.05 to 2.25 hours during winter and 2.5 to 2.75 hrs during summer at a pressure of 1.2 kg/cm2.

[1]***Aurofugine***- *To be used with caution.*

Details of Drip System

Lateral spacing	1.5 m
Dripper per plant	2
Spacing between two drippers	30 cm away on either side of stem
Dripper discharge rate	4.0 lph

Water requirement for banana under drip irrigation

Months after planting	Liters / day / plant
1-3	5
3-5	9
5-8	11
8-11	10

Application of Fertilizers

Basal

FYM should be applied @ 20 to 25 t/ha at the time of land preparation. Top dressing of fertilizer should be done as given below

For using water-soluble fertilizers, the following schedule of fertigation may be followed:

Nutrient	Fertilizer application schedule
N (180g/plant)	7 to 8 splits at an interval of 15 to 20 days.
P (72g/plant)	7 to 8 splits at an interval of 15 to 20 days.
K (180g/plant)	7 to 8 splits at an interval of 15 to 20 days.

Time of application: **The total quantities of water-soluble fertilizers should be applied in 7 to 8 splits at an interval of 15 to 20 days. The first split should be applied at the time of planting before monsoon and the rest should be applied after the cessation of monsoon.**

Weeding: **Banana fields should be kept weed free either by hand weeding/interculturing or by weedicide (diuron @ 1.2 kg/ha as pre emergence) application. Mulching with black plastic (50 micron) or sugarcane trash (@ 10 t/ha) should be done. If required, hand weeding should be done prior to mulching.**

Desuckering: **First desuckering should be done manually. To minimize the regeneration of suckers, 3 ml of diesel or kerosene should be injected into the cut portion of the suckers.**

Plant protection measures

Diseases	Control Measures
Bunchy top	Aphid should be controlled by applying systemic insecticide viz; Acetamiprid: (0.2gm/litre).
Premature fruit ripening	Sucker should be dipped in the solution of Aurofugin[1] 10 g in 100 litres of water for 1.5 hrs.
Pests	**Control Measures**
Rhizome weevil and nematode	Neem based products such as Neem oil, Neem cake, Neem seed Kernal extract (NSKE) can be applied.

Bunch coverage: **After the complete formation of the bunch, it should be covered by LLDP film bag (blue, white or black colour). This improves quality as well as yield of banana.**

Time of harvesting: **Maturity varies with variety but usually the crop takes about 12-14 months to mature.**

Yield: **By adopting the above practices, the banana yields about 70 to 80 t/ha. Generally, size and colour based grading is done.**

Post harvest handling and storage: **Bananas can be stored for up to a week in a cool place but unripe bananas should not be stored in the refrigerator as this may irreversibly interrupt the ripening process. If the banana is no longer green, then it is ripe and can be stored for a maximum of one week. For storage, banana should be stored at 13o to 140C. Bunches should be kept out of light after harvest since this hastens ripening and softening. For export, hands are cut into units of 4-16 fingers, graded for both length and girths and carefully placed in poly-lined boxes to hold 12 to 18 kg depending on export requirement. Prior to packaging fruits are cleaned in water or dilute sodium hypochlorite solution to remove the latex and treated with thiobendazole.**

Plastic packaging: **Keeping quality of banana can be increased when packed in 400 gauge LDPE (Low-Density Poly Ethylene) bags with or without ventilation either under ambient temperature or in a zero energy cool chamber (13.5°C). Storing of banana fruits in unvented polybags at low temperature could extend the shelf life of the fruits up to 19.33 days.**

Cost economics

Annual system cost (Rs /ha)	17,500
Cost of cultivation (Rs/ha)	1, 08,000
Total cost (Rs/ha)	1, 25,500
Total income (Rs/ha)	2, 10,000 to 2, 40,000
Net income (Rs/ha)	84,500 to 1, 14,500
C:B ratio	1:1.67

[1] ***Aurofugin***- *To be used with caution.*

II. Packages of Practices – MANGO

Mango is the most important fruit crop of India. India produces about 12.750 Mt of Mango from an area of 2.309 Mha with an average productivity of 5.5 mt/ha. Major producing states are Andhra Pradesh, Uttar Pradesh, Bihar, Karnataka, Tamil Nadu, West Bengal, Orissa & Maharashtra. Uttar Pradesh has 0.2712 mha under mango and the total production during 2008-09 is 3.465 Mt with 12.8 mt/ha productivity. India ranks first in the world for mango production and area under cultivation. Mango is a rich source of vitamin A and has a fairly good content of vitamin C. Mango fruits are used for preparation of pickle, chatani, amchur, jam, squash, nectar and many other delicious products.

Climate and soil: Mango can be grown from alluvial to lateritic soils except black cotton soil which has poor drainage. The temperature between 24 and 270C is ideal for mango cultivation.

Variety: Dashehari, Langra, Chausa, Bombay Green, Lucknow Safeda, Mallika and Amrapali.

Multiplication of genuine planting material: Mango can be propagated by veneer, wedge and soft wood grafting. The protected nurseries in polyhouses and use of sprinkler and drip is becoming common for raising humidity level, which is required for higher grafting success rate.

Preparation of land: The land should be prepared one month before planting. The pits of 1m x 1m x 1m size are dug. The pits are exposed for 2 to 4 weeks to kill harmful soil organisms.

Soil sterilization: Soil sterilization can be achieved through both physical and chemical means. Physical control measures include steam and solar energy. Chemical control methods include herbicides and fumigants. Soil sterilization can also be achieved by using transparent plastic mulch film (25 micron thickness) termed as soil solarisation.

Planting: Square and rectangular systems are popular. Before planting, pits are filled with FYM at the rate of 15-20 kg/plant. The grafts should be planted during July to September.

Planting density: High density (3m x 6m or 5m x 5m) planting helps increase the yield/unit area. Normal planting distance of mango is 8m x 8m.

Canopy Management: Training should be done after 6th month of planting. It is essential to space the branches properly and to help in intercultural operation. At initial branching height between 60 to 70 cm is appropriate.

Water requirement of the crop

Age of the plant	Water requirement of the crop in litres/day/tree
Young plant (up to 3 years)	9-12 lts
3-6 years	30-35 lts
6-10 years	50-60 lts
9-12 years	80-90 lts
Fully grown trees	120 lts

A young tree requires 2 drippers at a distance of 1m on lateral lines, while fully-grown tees require 2 drippers with double lateral lines at 1-1.5 m distance.

Application of fertilizers: Mango should be manured with phosphorus twice in a year i.e. the beginning of the monsoon (June-July) and during the period of post-monsoon (September-October). Usually fertilizers (N and K) are applied in split doses in the month of June-July, September-October, January-February and March-April. For adult trees (10 years or above) 1,000g N, 75g P2O5, 75g K2O and 100 kg FYM per year should be applied. Application of micronutrients such as Zinc and Boron help in cell elongation process.

Malformation: Deblossoming at bud stage (1 cm long) alone or in combination with spray of 200 ppm NAA lowers the number of malformed panicle.

Alternate bearing management: Use of paclobutrazol (5-10g/m canopy diameter), 3 months before budburst applied through soil drenching can be used for obtaining regular bearing.

Weed management: Black plastic mulch (100 micron) restricts the germination of weed seeds and suppresses the weed growth. The size of the film requirement for young plant is 1 m x 1 m, and for 8 years onwards film requirement is 2.5 m x 2.5 m around the tree.

Intercropping: In the interspaces of mango orchard, certain vegetable can be intercropped viz. onion, tomato, radish, carrot, ginger, turmeric, methi, cabbage, etc. Moreover, fruit crops can also be grown viz. papaya, pineapple, etc. for the initial 4-5 years.

Mulching: Soil drenching with paclobutrazol (5g and 10g/tree) coupled with black polythene mulch (100 micron) results into minimum outbreak of September to October vegetative flushing, giving an early and profuse flowering and a higher annual yield.

Plant Protection Measures

Insect pests	Symptoms	Control measures
Mango hopper	Pest starts attacking during flowering season.	Spraying of acetamiprid or thiamethoxam (0.2gm/litre of water).
Mealy bug	Nymphs suck juice from young shoots, panicles and flower pedicels.	Raking of soil around the trunk and mixing with neem cake around tree trunk is effective.
Stem borer	Pest makes tunnel through the main trunk and branches.	Clearing tunnels with hard wire, pouring, Emamectin benzoate (50 gm / litre) and plugged with mud.
Fruit fly	Pest makes the fruits rot by laying its eggs in clusters, just before the ripening, under the peel of fruits.	Application of acetamiprid or thiamethoxam (0.2gm/litre of water).

Disease	Symptoms	Control measures
Anthracnose	It attacks leaves, flowering panicles and fruits.	Spraying of Copper Oxychloride (0.03%)can control this disease.
Powdery mildew	Whitish powdery growth on the leaves.	Wettable sulphur (0.02%) and Bayleton (0.05%).
Bacterial Canker	Unattractive fruits because affected parts of fruits show longitudinal crack and oozing of bacterial exudate and leading to fruit drop.	StreptocycIine(100-200ppm), agrimycin (17% streptomycin) -100 (100ppm) and copper oxychloride[1] (0.03%).

Yield and quality control:

From a well grown up tree one can expect an average yield of 50-225 marketable fruits (50 kg) per plant per year.

Harvesting and post harvest management:

From the 4th year onwards the mango fruits can be harvested at the mature green stage during morning hours. After harvesting fruits are graded according to their size, weight, colour and maturity. Packaging of fruits should be done in corrugated fibreboard (CFB) boxes. Tissue paper and polythene foam paper are used for wrapping high-value fresh mangoes. Polyethylene lining has been found beneficial as it maintains humidity, which results in lesser shrinkage during storage. Dashahari treated with calcium chloride solution (4%) at sub-atmospheric pressure of 500 mm Hg for 5 minutes can be stored at 12 oC for 27 days.

Cost economics of drip irrigated mango (one ha)

Rate of interest	10.5%
Life of system	7.5 years
Expected yield	19 t/ha
Planting distance	5m x 5m
Cost of cultivation	Rs.24, 000
Fixed cost	Rs.30, 298
Annual cost of drip system	Rs.8,713
Expected cost benefit ratio	1: 6.0

III. Package of Practices - Sugarcane

Time Schedule	Recommended Operations
Before planting	Plough the land up to 45 cm depth. Apply 25 tons per hectare of well-decomposed Farm Yard Manure (FYM) or decomposed molasses or compost and deep plough the field with tractor. Make ridges and furrows with 80 cm spacing having 20 cm height and up to 10 m length. Apply 375 kg super phosphate in the furrows.
Planting day	Select about 75,000 two-budded setts per hectare from 6-8 months old nursery or raise seedlings in poly bags with single-budded setts. Treat the setts for 10 minutes by dipping in a solution prepared with Thiamine, 2.5 kg urea and 2.5 kg lime in 250 litres of water. Plant the setts 2 cm deep with buds on the sides. For every 10 furrows, plant setts on 2 rows in one furrow for gap filing purpose.
3rd day after planting	To control weeds, spray atrataf @ 2.5 kg/ha in 500 litres of water with hand sprayer.
5th day after planting	Spread sugarcane trash up to a height of 15 cm on the ridge.
25th day after planting	Gap-fill with seedlings raised in polybags or the plants taken from the 2 rows planted- furrows in every 10th furrow.
30th day after planting	Mix 5 kg Azospirillum and 5 kg phosphobacterium per ha mixed with 250 kg powdered FYM. Apply at the bottom of the plants and irrigate immediately.
From 35th day after planting	From 35 to 100 days after planting Irrigate once in 7-10 days. To prevent attack by early shoot borer, apply Sulphur on the setts and cover with soil. If 25-30% of the shoots are affected, then for every 100 metre length of furrow, mix Sulphur and apply using a hand sprayer on the tips and bottom of the shoots.

45th day after planting	Do hand weeding. Apply in pits a mixture of 110 kg nitrogen[1], 60 kg of potash and 35 kg of neem cake per hectare.
60th, 90th and 120th days after planting	Spray a mixture of urea 2.5%, potassium chloride 2.5% during periods of drought. On the 60th day, apply a mixture of 5kg azospirillum, 5kg phosphobacterium and 250 kg of decomposed FYM in powder form at the bottom of the plants and irrigate immediately. On the 90th day, do hand weeding; apply in pits (after earthing up) a mixture of 110 kg of nitrogen[1], 60 kg of potassium and 35 kg of neem cake per hectare.
120th day after planting	Under drought conditions, apply 60 kg of potassium and irrigate immediately.
150 days to 225 days after planting	Carry out de-trashing at 150 days after planting. If inter-node borer exists, release parasites 6 times @ 5 cc per hectare once in 15 days. 101-210 days – irrigate once in 7 days. 210th day- detrash and tie lodged canes. 225th day- spray acetamiprid or thiamethoxam (2ml/litre of water) to control mealy bugs, white fly and scales.
260th day	Spray Emamectin benzoate (50 gm / litre) of water under the leaves (if required) to control pyrilla and all sucking pests.
270 to 360 days	Irrigate once in 15 days. Stop irrigation 15 days before harvesting.
Harvest	Cut canes at the bottom close to the ground with sickles or sharp knife. Remove trash, roots, water shoots and cane tops and send clean canes to the factory.

[1]***Nitrogen***- *To be used with caution and preventive measures (Gloves, etc).*

SCHEDULE OF OPERATIONS: RATOON CROP

Time Schedule	Recommended Operations
	Remove trash, stubble shave uniformly under correct moisture conditions with sharp spades. Mix 15 tons of FYM or 25 tons of compost or 25 tons of decayed molasses with 375 kg Superphosphate (75 kg of P2O5), 135 kg of nitrogen and 35 kg of neem cake per hectare in pits. Irrigate immediately, cutting the sides of the ridges and ensuring mixing of applied manures well with the soil. Control weeds by spraying atrataf @2.5 kg in 500 litres of water with hand spray.
9th -10th day	Mix 5kg azospirillum and 5 kg phosphobacteria per hectare with 250 kg of powdered FYM and apply at the bottom of the plants and irrigate immediately. Spread trash obtained from plant canes on the furrows.
25th – 30th day	Gap-fill with grown up plants. To prevent attack by early shoot borer, apply Sulphur on the setts and cover with soil.
35th day	Mix 5kg azospirillum per hectare with 250 kg of powdered FYM and apply at the bottom of the plants and irrigate immediately.
From 1-35 days	Irrigate once in 7 days.
From 35-90 days	Irrigate once in 10 days.
60th day	Do hand weeding. Apply in pits a mixture of 110 kg nitrogen, 60 kg of potash and 35 kg of neem cake per hectare and follow light earthing-up.
90th day	Apply 60 kg of potash additionally in drought situations.
91- 250 days	Irrigate once in 7 days.

30, 60 and 90 days	Spray a mixture of urea 2.5%, and potassium chloride 2.5%on the leaves in drought situations.
120th day	Detrash and earth up well.
121 to 210 days	Release Trichogramma parasites (when required) once in 15 days.
180 days:	Detrash a second time.
210 days	Spray acetamiprid or thiamethoxam (0.2gm/litre of water) to control mealy bugs, white fly and scale insects.
251-360 days	Irrigate once in 15 days and stop irrigation 15 days before harvest.
Harvest	Cut canes at the bottom close to the ground with sickles or sharp knife. Remove trash, roots, water shoots and cane tops and send clean canes to the factory.

Note: Water Management – Irrigation gap has to be adjusted depending up on quantity of rain fall Reduce gap between irrigation in sandy soils and increase it in block soils.

IV. Package of Practice for Nagpur Mandarin Cultivation

Selection of site

Orchard: Soil should be well drained and of shallow or medium depth. Deep heavy soils having more than 60% clay contents are not suitable for citrus plantation. **Nursery:** Nursery should be located at least 500 meters away preferably on western side of the orchard to minimize incidence of insect pests and diseases.

Raising of citrus nursery

Sowing of Rootstock Seeds

Potting mixture of soil, sand and FYM or compost should be used in equal proportion (1:1:1) for filling of trays in primary nursery and polythene bags in secondary nursery. Before it is used for filling the bags/trays, the potting mixture should be solarised. For solarisation, it is spread on the concrete platform in 4" thick layer in the month of April – May, sufficiently moistened with water, then covered fully with the white polythene sheet, sealing its edges with soil. Then it should be left undisturbed for 11/2 to 2 months in the hot sun for solarisation.

- Only certified seeds of rough lemon or Rangpur lime rootstocks should be used.
- Shade dried medium size bold seeds of rootstocks should be treated with vitavax or thiram (@ 3g/kg seed) and sown on the raised beds or in plastic trays during September – October.
- Stagnation of water in beds should be avoided otherwise roots of young plants may start rooting.
- Uniform seedlings of medium height only be selected discarding either vigorous or the weak and dwarf ones while transferring to secondary nursery. Plants having hooked or bent roots should be discarded.
- Disease free budgrafts of nagpur mandarin

Plant Protection Measures

- Phytophthora infected plants must be eliminated. In case of phytophthora infection drenching of plants with either metalaxyl MZ72 @ 2.75 g/l water or fosetyl Al @ 2.5 g/l water should be done. Second spray should be given after 40 days.
- To prevent infestation of insect pests like citrus leaf miner and thrips plants should be sprayed either with acetamiprid or thiamethoxam (0.2gm/litre of water) at 10 days intervals.
- The growth of plants is also affected adversely due to mite attack which can be controlled by spraying plans with Fenazaquin 10% EC @ 4ml/litre and wettable sulfur @ 3 g/l water, alternatively at 15-20 days interval.
- In containerized nursery irrigation, fertilizer application, weed control, insect pest and disease control as well as cultural operations can be performed at ease.

Budding

- Budsticks should be used from the authorized and certified source only.
- Budstick should be drawn from the last years flush. The stick should have pencil thickness, be roundish and have whitish longitudinal streaks.
- Budstick should not be drawn from rubbery wood or kikarpani plants.
- Budding should be performed at 10" – 12" height on the rootstock seedling.

Disease Free Bud Grafts of Nagpur Mandarin

Orchard Establishment

Pits size

- Pits for planting should be 2'6" x 2'6" x 2'6" (75 x 75 x 75 cm) size and spaced at 6 x 6 m distance.
- To avoid soil borne fungi or nematodes soil of roots should be removed.

Pre-planting treatment for budlings

- Roots of budling should be dipped in the solution of metalaxy1MZ72 2.75 g for 10-15 minutes before planting.

Planting of budlings

- While planting care should be taken that rootstock union remains at least 6" above ground.

Manure and fertilizer application

- Nitrogen containing fertilizers should be applied in three equal splits in January, July and November months; phosphorus containing fertilizers in two splits in January and July months and Potassium containing fertilizers may be applied as singly dose in January.
- Surveys conducted by NRCC in Kalmeshwar, Katol, Narkhed, Saoner, Hingna and Ramtek tehsils of Nagpur district have revealed N deficiency in most of the orchards and P deficiency in leaf and soil of 50% orchards. Similarly, leaf and soil K was either at desired levels or even more than it. Citrus trees are nitrogen loving plants. They respond well to the applied nitrogenous fertilizers. During fruit development K may also be applied as it may fell deficient. Supplementary doses of P and K at 200 and 100 g/tree, respectively, may be included in the fertilizer package recommended for bearing orchard.

Fertilizer doses

Fertilizers /Age of tree	I Year	II Year	III Year	IV Year and Above
Nitrogen	150	300	450	600
Phosphorus	50	100	150	200
Potassium	25	50	75	100

As far as possible 1/3rd of the dose of N may be given through farm yard manure/compost, oil cakes etc.

Nagpur Mandarin

Leaf sampling

For correct diagnosis of nutritional status use of correct sampling technique is very important. For this, it is important to know as to how many leaves, when, from which part of the plant and from how many trees should be sampled. In case of ambia bahar, 5-6 month old leaves in August-October and for mrig bahar 6-8 months old leaves in December and February should be sampled. As far as possible, the 2nd, 3rd of 4th leaf should be picked from the tip of the non-bearing shoot, preferably at 1.5 – 2 m above the ground and sampled.

Drip irrigation

With the help of a drip system of irrigation, the required quantity of water can be provided right at the feeder root system. Similarly, water-soluble fertilizers and micronutrients also can be given through the drip system. Water requirement of irrigation depends upon age of the tree and season of the year (Table 1). Mulching with drip irrigation maintains moisture in soil for a longer period.

Water requirement of the nagpur mandarin (litres / day / tree)

Month	Age of the three (years)									
	1	2	3	4	5	6	7	8	9	>10
January	7	15	22	30	44	62	72	82	92	102
February	9	20	30	40	60	82	96	101	121	137
March	12	26	40	53	78	109	127	145	163	181
April	14	29	43	63	87	123	143	163	183	204
May	17	34	52	74	102	143	166	188	211	235
June	11	22	34	48	67	95	110	126	142	157
July	8	18	26	41	56	79	92	105	118	131
August	7	14	23	34	42	60	70	80	90	100
September	8	15	25	36	45	65	76	87	98	108
October	9	17	27	40	52	79	92	105	118	131
November	8	15	25	36	45	63	74	85	96	150
December	6	11	19	24	35	49	57	65	73	82

Weed control

For effective and economic control of mono and dicotyledonous weeds, pre-emergence weedicides, diuron 3 kg at the end of May and 120 days thereafter should be done. For post-emergence weed control, glyphosate @ 4 l/ha should be sprayed on weeds before flowering.

Fruit drop

Fruit drop in citrus is of serious nature which occurs at least twice; i.e. the first time when the fruits are little more than the marble size and the second time when the fruits are fully developed or at the time of colour break. This drop is very serious in the ambia bahar crop pre-harvest fruit drop in nagpur mandarin, which is called as pre-harvest fruit drop and is important from economic point of view to the orchardists. To control the fruit drop that occurs after fruit set, two foliar sprays of either 2,4-D or GA3 at 15 ppm + urea 1% and copper oxychloride[2] (0.3%) at monthly intervals in April – May are recommended. Same spray concentration is recommended for controlling pre-harvest drop in the months of September and October. 2,4-D and GA3 may be dissolved earlier in little quantity (30 – 40 ml for 1g) of some organic solvent such as alcohol or acetone before making the spray solution.

Control of Insect Pests

Blackfly (Kolshi): Nymphs of citrus blackfly attack the young flush, suck the sap and excrete sweet and sticky liquid, which favours rapid development of black sooty mould that covers entire plant surface. The process of photosynthesis is hampered greatly resulting in stunted growth of plants, low intensity of flowering, scarce fruiting which are insipid in taste and decline of citrus orchard sets. For control of blackfly two sprays of insecticides viz. acetamiprid 0.2 g/litre water should be given at 50% egg hatching stage that normally occurs in the II weed of July and the I week of December and April. One additional spray which targets the adult blackfly population when it is as its peak helps tremendously in controlling the pest.

Method of spraying: Spraying should be directed at the underside of the leaf, ensuring the complete drenching of the tree. For proper coverage and penetration of canopy, use of power sprayers for spraying operation should be envisaged. Insecticides should be used alternatively for better results.

Citrus psylla: Numerous young brownish nymphs of psylla are seen crawling on the young flush. Several dirty gray colour adults can be seen sitting in line with tails upwards. Voluminous desapping by the nymphs results into the drop of flush, flowers and berries. Affected branches dry and die-back sets in. The nymphs also excrete white crystalline powder, which invites fungal infestation. Psylla can be controlled by spraying acetamiprid 0.2 g/litre water twice with 10 days interval during the initial days of flushing. Bark eating caterpillar (Larva) are incidence of mites on fruits (Lalya).

Leaf miner: Serpentine mines are seen on the new leaves and also young stems are mined when the incidence is severe. At times death of young shoots may occur. The problem is quite serious in nursery and in young orchard. To control citrus leaf miners, spray either imidacloprid 5 ml (To be used in proper doses under guidance of an agriculture expert) and thiamethoxam 0.2g/litre water.

Bark eating caterpillar: The pest is noticed predominantly in the older and ignored orchards. The hanging wooden frass and tunnel at the joint of two branches during October – April indicates the presence of the pest. Larva remains hidden inside the tunnel during daytime and becomes active in the night and feeds on the bark nearby the tunnel. This results into snapping of food supply, ultimately yellowing of leaves on the branch and its slow decline. For control of the pest the wooden frass should be cleaned and each tunnel should be administered with Emamectin benzoate/spinosad @0.4 ml/litre water.

Fruit sucking moth: The moth attacks the ripening fruits during late hours in the evening. The moth punctures it making a hole in the ripening fruit to suck the juice through which an infection may take place. Soon rottening starts, leading to fruit drop. Collection of dropped fruits and their destruction followed by smoking of orchard late evening hours is suggested. For fruit fly control, hanging of methl eugenol (feromone) traps is an effective method to check the pest.

Mites: Citrus rust mites attack in mrig bahar fruits especially during September – November. The fruit surface, particularly the side exposed to sun, is brushed and develops a big patch of dark brown colour – called 'Lalya' – only after 1 to 11/2 months. To check this, two alternating prophylactic sprays at 15-20 days interval with Fenazaquin 10% EC @ 4ml/litre and wettable sulphur @ 3 g/l water are recommended in September – October. Similarly, a phytophthora caused gummosis spray of acetamiprid/thiamethoxam @0.2g/litre water at the berry stage of the ambai fruits is required to protect the fruits from unpleasant scars.

Precautionary measures: Avoid water stagnation in orchard by providing channels along the slope for proper drainage. Moreover, pruning of intermingling branches to allow aeration and sunlight to prevent dampness in orchard are suggested. Presence of guava, pomegranate, chiku, mango trees, etc. near the orchard act as the alternate hosts for blackfly, therefore, such trees should also be covered with insecticidal sprays.

Control of Diseases

Twig Blight: Drying of fruit bearing branches after harvest is a common phenomenon. Removal of such dried shoots along with the 2 cm lower live part, followed by a fungicidal spray i.e. copper oxychloride or bordeaux paste application is recommended to check twig blight.

Gummosis, root rot and collar rot

- Proper diagnosis of the disease is must.
- Affected trees should be treated with metalaxyl MZ 72 @ 2.75 g or fosetyl AL @ 2.5 g/l water till drenching once in May – June. The second spray should be given after 40 days. The tree trunk and soil of the tree basin should also be sprayed/drenched.
- Removal of the rotten roots, cleaning the wound on the gum-oozing trunk with sharp knife and then pasting with metalaxyl[3] MZ 72 should be done.
- Apply bordeaux paste (1 kg $CuSO_4$ + 1 kg CaOH + 10 l water) on tree trunk upto 2 to 21/2 ft from ground twice i.e. before monsoon (May) and after monsoon (October).

Precautionary measures

- Avoid flood irrigation
- Follow the double ring system of irrigation so that water does not come in contact with the tree trunk.
- Avoid deep ploughing under the tree to prevent damage/injury of the root system. Declining nagpur mandarin orchard rejuvenated orchard.

Pre-harvest spray of fungicide: Three pre-harvest sprays of fungicides like difenoconazole @2ml/litre water at 15 day intervals till drenching prevent pre-harvest fruit drop by 54%. It also controls post harvest diseases upto 70%. This spray remains effective for 3 weeks at normal temperatures.

Post-harvest fungicidal treatment: Fruits dipped in fungicidal solution difenoconazole for 5 minutes are safe to minimize rotting upto 70% and can safely be stored for 3 weeks under normal condition.

Precautionary measures

- Follow plant protection measures to keep bearing tree disease free and healthy.
- Post-harvest handling of fruits must be followed carefully to avoid any sort of injury of fruits.
- Avoid the use of copper fungicides like b,ordeaux mixture, blitox, phytolan etc. at this stage.

Rejuvenation of declining citrus orchards: Declining citrus orchards can be rejuvenated with the use of developed technologies and brought into productive stage.

Harvesting

- Traditionally fruits are harvested by twisting and pulling forcefully which may lead to a hole in the neck of the fruit or injury to the stem end. The fruits should be selectively harvested when ¾th of the skin turns yellow. TSS: Acidity ratio should not be less than 14 in both ambia and mrig bahar fruits. TSS should be at least 10%. Once this stage is reached, harvesting should not be delayed for colour development since other fruits may turn loose.
- Fruit should not be allowed to come in contact with soil and straw and also should not be exposed to hot sun.
- Packing should be done immediately after harvesting.
- For waxing on a large scale packing line can be used. Washing and waxing (stay fresh high shine was 2.5 g + Tebuconazole) and grading of fruits is also done automatically at the end.
- Graded fruits are packed in corrugated fibre board boxes (50 x 30 x 30 cm) which are telescopic with holes on both the sides covering 5% of total side portion. To keep the fruits safe from moisture in the store house, boxes should be externally laminated with plastic.
- For small scale (1 – 1.5 tonne) and short duration (20 – 25 days) storage an evaporative cool chamber is recommended, costing Rs.10,000/- to 12,000/-.
- For delaying or postponing the harvest of the fruits, two sprays of GA3 (100 mg / 10 litre of water) in 15 day intervals at the point of the colour break are recommended.
- For long duration cold storage, a temperature of 6-7 0 C and humidity 90-95% is desirable. In such a situation, citrus fruits should not be stored for more than 45 days. To avoid chilling injury, care should be taken that the temperature does not go down upto 40 C.

2.6. Lessons Learnt

1. Dark coloured soils are more fertile compared to light coloured soils. Neutral pH is more suitable for majority of crops. Lime is used for amending acid soils and gypsum for alkaline soil.
2. Soil testing is essential for judicious application of fertilisers based on nutrients status in the soils.
3. Integrated Nutrient Management (INM) practices are useful for efficient, economic and sustainable production.
4. Package and practices recommended by research stations may be adopted in totality

3. Plant Protection

3.1. Objectives of the session

- To increase the awareness and understanding about the crop pest, diseases and weeds.
- To strengthen the farmer's knowledge on effective management of insects, diseases and weeds in crops through Integrated Pest Management.
- To sensitise farmers on safe handling of chemicals.

3.2. What we know at the end of the session

- Insects and their life cycles
- Methods of insect control including Integrated Pest Management (IPM)
- Plant protection equipments
- Symptoms of major diseases
- Integrated disease management
- Major weeds
- Methods of controlling weeds
- Safe handling of chemicals

I. Pest Management

3.3. Crop pest and their importance

Pest is any organism which is detrimental to crop production. Pest cause damage to the plant to the extent of 30 - 90 per cent, sometimes it even causes total loss. Pest includes insects, diseases and weeds. Non insect pest includes nematodes, snails and rodents.

Insect is any of many small invertebrate animals having a segmented body and three pairs of legs and usually two pairs of wings. Some insects are beneficial and some are harmful to agriculture.

What is the difference between complete and incomplete metamorphosis in insects?

Incomplete and complete metamorphosis differs in the number of life cycle stages that insects go through during their transformation from egg to adult. The complete metamorphosis has 4 life cycle stages and an incomplete metamorphosis has 3 life cycle stages.

Complete metamorphosis

Complete metamorphosis has four distinct life cycle stages: **egg, larva, pupa and adult.**

Examples of insects that go through complete metamorphosis are butterflies, silkworms, mealworms and ladybugs. The larva can be worm-like even though the six legs are still visible. The larva form moths and butterflies are called caterpillars. Maggots are the larval stage of flies. The larvae eat constantly and grow rapidly. A hard, protective case forms around the larva at pupa stage. The pupa stage for a butterfly is called a chrysalis. The pupa stage for moth is called cocoon.

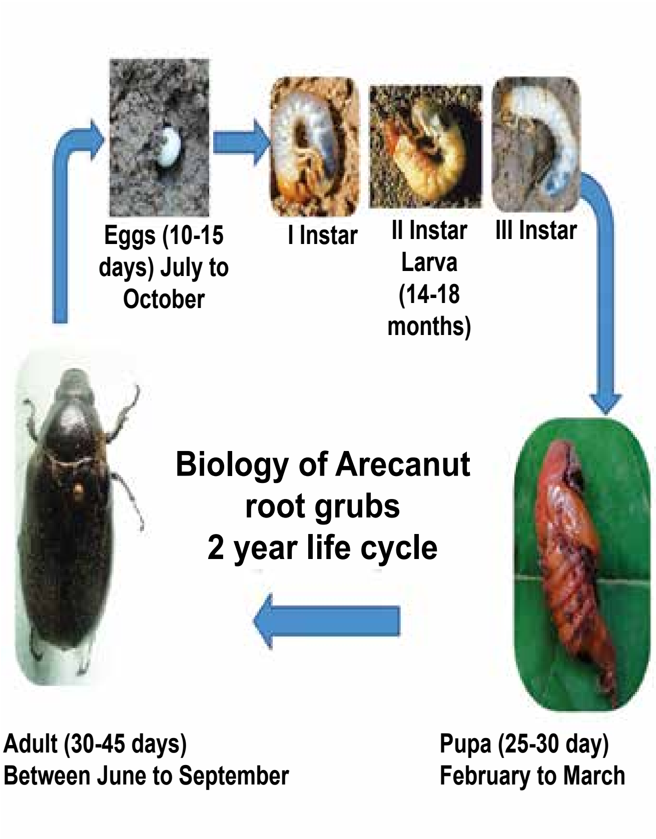
Incomplete metamorphosis

Incomplete metamorphosis has three life cycle stages: **egg, nymph and adult**. The nymph looks similar too but is a smaller version of the adult. The nymph is also wingless. Examples are grasshoppers, cockroaches, ants and praying mantids.

3.4. Life stages of insects

- **EGG** is the initial stages of the insect. Normally an insect lays at least 30 to 300.
- Egg hatch into **LARVA** or worms. Larva is the

damaging stage of insect to any crop. This stage is normally seen in the field.



Life stages of insects

- **PUPA** is an inactive stage preparing itself to develop to an adult.
- Mostly **ADULTS** insects are harmless but many bugs and beetles are harmful to plants.

Mite damage in coconut buttons

Larva damaging leaves

Bollworm damaging cotton

Semilooper feeding on caster

How the damage is visible?

- The larva eats the leaf, fruits or the whole plant parts. Hence the damage is visible.
- Some insects scrape the plant tissues. They also cut the growing parts. Beetles, bugs, thrips and hoppers usually suck the sap from the plants and growing parts, affecting the healthy plant.

Grub of green lace wing

Ladybird beetle

- Not at all. HONEY BEES are also insects which help in pollination (brings pollen from one plant to another), thereby, increasing the yield.

They also provide us valuable HONEY and other products. Another example of "friendly" insects are SILK WORMS.

- NATURAL ENEMIES are insects that are beneficial to man since they feed from the egg or larvae and pupae of crop pest. They are called biological control agents. Examples are the trichogramma parasite for sugarcane borers, crub of green lacewing and ladybird beetle.

3.5. Insect classification

Insects can be classified into 3 groups, depending on their behaviour in the farm such as

- Pests
- Beneficial insects
- Neutral insects

Pest: Whether an insect species is a pest or not depends on the situation. This means that a certain insect could be a pest in one situation but not in another situation. For example, the caterpillars of diamondback moth feed on cabbage and other plants of the cruciferae family. A farmer who grows cauliflower will therefore consider it a pest. Yet, for a farmer who grows potatoes or bananas, the diamondback moth is a neutral insect. When there is nothing to feed it will not even occur. In a paddy crop, black gram is a weed/pest since it is growing unnecessarily. In a black gram field, cow pea is a weed since it can grow. Pest will occur only if there is a host.

Beneficial insects

Some insects are beneficial to the farmer, because they are the natural enemies of harmful insects. Predators feed on other insects and hence control the pest. For example, the assassin bug kills caterpillars and ladybird beetles feed on aphids. Some other insects are beneficial as they help in pollination of plants, e.g. honey bees. There are commercially beneficial insects such as silkworm, which produces silk.

Neutral insects

A neutral insect is neither a pest nor beneficial. Yet again, it really depends on the context. A mosquito in the rice field can be considered as neutral insect.

3.6. How to control pest

Natural control

Understanding insect life cycle: Insects multiply in large numbers. However the survival rate is very less as the nature maintains insect population. Factors like extreme temperature, heavy rains, heavy wind, water stagnation, birds, lizards, spiders, animals, other insects and diseases control the pest. Natural barriers like sea, river, lake, mountains, etc. prevent movement and spread of pest.

Mechanical control: Some of the recommended practices are

- Removal of affected parts
- Collection and destruction of insects
- Drying of seeds
- Tar coating of trees to protect from termites
- Provision of barriers to prevent the entry of pests like green house / screen house, covering of pomegranate fruits with butter paper, etc.
- Clipping off the withered shoots

Agronomical methods: Recommended cultivation practices are as follows.

- Summer ploughing: Opens up the soil and exposes pest to hot sun and predators.
- Trap crop: Is growing the most favoured crop of the insect along with the main crop. The insect feeds from the trap crop and the main crop remains not unaffected. Growing of resistant varieties also prevents the pest attack.

Main Crop	Trap Crop
Tomato	Marigold
Cotton	Okra, Castor, Onion, Garlic
Maize	Sorghum

- **Mixed cropping:** Is growing more than one crop.
- **Inter cropping:** Growing another crop along with main crop, which increases the population of natural enemy.

Main Crop	Intercrop
Cabbage	Tomato, mustard
Cotton	Black gram, green gram
Maize	Sorghum

- **Crop Rotation:** Is growing of different crops in sequence instead of one single crop.

- **Keeping the fields clean:** Managing the weeds which provides home for pests, treating the seeds with pesticides, growing the crops in seasons where pest incidence is less, application of correct dose of fertilizer at correct time, optimum use of water are some of the agronomical methods to control pests.

Ladybird beetle

Pest load and monitoring

Pest monitoring: Is the practice of examining the crop to know whether the pest has affected the crop and the extent of damage to decide whether to spray the pesticide or not.

Economic Threshold Level (ETL): ETL is the pest population density at which the control measure has to be taken up to prevent the pest from reaching economic injury level.

ETL for some of the crops are as follows

- Brown plant hopper in paddy - nymph or adult hopper 5-10 / hill
- Leaf miner in groundnut: 2 larvae / 10 plant or 20-30% plant infestation
- Whitefly in cotton: 5-10 adults / leaf or 20 nymphs / leaf

Biological control methods: There are several biological agents that controls pest such as

- The insect that kills another insect by living inside its body is called Parasite. There is egg, larval and pupal parasites. Trichogramma is an egg parasite used commercially on sugarcane borers.
- Predators are insects that eat other insects. For example, the Ladybird Beetle, dragonfly, damselfly, etc. Insects are also affected by many diseases. Some are fungal, bacterial and viral. Verticillium, a fungal agent, bacillus, a bacterial agent, and NPV (Virus) are commercially used.

Physical Control

- An age old practice is to mix pulses with red earth to protect pulses from pulse beetle.
- Drying of seeds in hot sun and using radiation are some of the physical control methods.

Traps

- **Light Trap:** Most of the insects are attracted towards light. This principle is used to monitor and control pests.
- **Pheromone Trap:** Insects are trapped using the scent of one sex. This is commonly used for the control of cotton bollworms.
- **Yellow Sticky Traps:** Some of the small sucking insects are attracted by yellow colour. Hence yellow coloured containers smeared with sticky materials are kept inside the field to attract sucking insect.

Chemical control

Manual pesticide spraying

- Pesticide application is the last resort to control pests.
- Pesticides to be selected carefully to control the pests.
- Dose and spraying equipments should be selected carefully.
- Get the correct recommendation from the extension worker.
- Understand the difference between chemical name and trade name. Trade name is the trader given name but the chemical name is originated based on the chemical ingredient.

Botanicals: are plant origin pesticides like neem based formulations. We can also use neem seed kernal extract or neem oil along with soap as pesticides. Commercial botanicals are available in the market and botanicals can be prepared, which cost less and are eco friendly.

Trap

For example, Neem Seed Kernal Extract (NSKE) 5%, Gronim, Achook and Neemazal.

3.7. Pesticides

Pesticides are the chemicals used to control pests. The pesticides can be broadly classified into three groups based on how they act on insects.

Contact and stomach poison: When insect comes into contact or when the insect eats the pesticide sprayed parts, it gets killed. Contact and stomach poison is used for controlling larvae that feed on leaves. Examples include Flubendamide. Some of the pesticides derived from plants also have contact action, for example pyrethrum, sabadilla, etc.

Manual Pesticide Spraying

Systemic poison: When the chemical is sprayed on the plant, it is absorbed by the plant and transferred to the entire plant system. Sucking type of pest like aphids, leafhoppers thrips suck the sap from plants. This kind of sucking pests can be controlled by systemic poison like Imidacloprid (To be used with caution), flubendamide, acetamiprid, thiamethoxam, etc.

1) **Fumigant:** Forms vapour and acts on breathing system of the insects. When insects breathe it, they get killed. Examples include, hydrogen cyanamide, sulfuryl fluoride, etc.

Insecticides are poisons which cannot be used directly. They need to be used as per directions given on the container or as per the recommendation of extension workers. Insecticides are available in different forms of which some are:

i) **Dust (D):** Poisons are mixed with gypsum, talc or clay so that it can be used in powder form. There is no need to mix with water.

ii) **Wettable Powders (WP/WDP):**Mixed with dry fillers with sticking agents but these dry powders can be mixed with water.

iii) **Granules (G):** Dry formulations in which poisons are mixed with calcium or gypsum in granulated form. Main merit of this formulation is handling will be easy, it is not carried away by the wind and it is less toxic to plants.

iv) **Liquid forms, Soluble Liquids (SL), Emulsifiable Concentrates (EC), and Soluble Concentrates (SC):** Liquid forms of pesticides. Since the pesticides are directly soluble in water, they are mixed with organic solvents, EC formulations disperse in water easily as they are mixed with emulsifiers.

v) **Fumigants:** are the poisons in gas form. Normally it is used to fumigate godowns, grains, storage rooms and ships even for rat control.

vi) **Poison baits:** Poisons mixed with food material. The latter acts as attractants. Baits are exclusively used to control rats. As the rats are more sensitive, pre baiting is necessary.

3.8. Types of chemicals

- Organochlorines, Organophosphates, Carbamates, Synthetic Pyrethroids are types of chemicals which were used before, but are not recommended now.
- New Molecules: Invention is a continuous process to identify more effective and less hazardous chemical search for better insect management. Examples include Nicotinamides, spinozids, triazoles (Hexaconozole, Propiconozole).

3.9. Sprayers/Dusters:

Types of applicators

Based on what to apply

- Sprayers are used to sprinkle the soluble chemicals.
- Dusters are used to spread the dust formulation of the pesticide.

Based on power source

- Manually operated sprayers/duster
- Power operated sprayers/duster

- Fuel operated
- Battery operated
- Solar panel powered operated
- Self propelled; i.e. they have their own power source for movement
- Normally 50 -100 litres of spray fluid are required for an acre. If we use manually operated sprayers 200-250 litres of spray fluid are required to cover an acre.
- There are a number of models with a variety of features available in the market. One can choose the model depending upon the requirement. Each instrument has got its own merits and demerits.
- Nozzle is the terminal part of the sprayer, which delivers the spray liquid to the plants. Different types of nozzles are available for different purpose.

Types of sprayer

Battery Operated Sprayer

Capacity 16 litres
Battery 12V/8AH (Fuse:6A)
Maintenance free Power Battery
Charger: Input: 220V/50HZ
Output: 12V/1000MA
Nozzle: Single, Double

Rocker Sprayer:

A minimum of 3 persons are necessary. One person to use the lever to create pressure and other persons to spray.

Motorized Power Sprayer is used for spraying of vegetable and ornamental crops in a large area.

Battery Operated Power Sprayer meant for use in the field.

3.10. Integrated Pest Management (IPM)

"IPM is a sustainable approach for managing pests by combining biological, cultural, physical and chemical tools in a way that minimizes economic, health, and environmental risks."

3.11. Some insecticidal materials for common household use

Kerosene emulsion

This is a contact insecticide useful against many sucking insects. Finely divide 500 g of ordinary bar soap and dissolve it in 4.5 litres of boiling water. Cool and add 9 litres of kerosene. The mixture is then vigorously agitated until the oil is completely emulsified. The stock solution can be diluted with 15-20 times of water before spraying.

Tobacco decoction

This is very effective for controlling aphids infesting vegetable crops. Tobacco decoction can be prepared by steaming 500 gm of tobacco in 4.5 litres of water for 24 hours. Then, 320 gm of ordinary sliced bar soap is dissolved separately in another vessel. The soap solution is added to tobacco decoction and the stock solution is diluted 6 - 7 times.

Neem seed suspension

This is very effective as a repellent against locusts and grasshoppers. Kernels of mature neem fruits should be crushed into a coarse powder. For obtaining 0.1% concentration, 1 g of powdered neem seed is required for a litre of water. The required quantity of the coarse powder should be put in a small bag of muslin cloth and dipped in water contained in a bucket and squeezed till the water becomes light brownish. This has to be sprayed on crops.

The latest registered chemicals are available in www.cibrc.nic.in or please verify with extension officer.

II. Disease Management

3.12. Disease

Disease is an impairment of the normal state of a plant that interrupts or modifies its vital functions. All species of plants, wild and cultivated alike are subject to disease. Although each species is susceptible to characteristic diseases, these are, in each case, relatively few in number. The occurrence and prevalence of plant diseases vary from season to season, depending on the presence of the pathogen, environmental conditions, and the crops and varieties grown. Some plant varieties are particularly subject to outbreaks of diseases.

Disease of crops and their importance

- Plant become diseased when it is continuously disturbed by some causal agent including an abnormal process that disrupts the plants.
- There are more than 80,000 plant diseases.
- Diseases reduce the yield of the crops and sometimes lead to disaster e.g. late blight of potato, Panama of banana, etc.
- Managing outbreak of diseases is challenge to the farmer.

Causes of plant disease

Infectious diseases: caused due to fungi, bacteria, viruses, nematodes, etc.

Non-infectious diseases: caused due to unfavourable extraneous condition such as scorching sunlight, high temperature, moisture stress or deficiency of micronutrients, pH, heavy metal toxicity, atmospheric pollution, etc.

3.13. The disease cycle

The main events of stages comprising the disease cycle include the following: production and dissemination of the primary inoculum, primary infection, growth and development of the pathogen, secondary infection and over wintering.

The primary inoculum is the part of the pathogen (that is, bacterial or fungal spores or fungal mycelium) that over winters (over-seasons) and causes the first infection of the season, known as primary infection. In general, the greater the amount of inoculum and the nearer it is to its host, the greater the potential for a disease epidemic.

Dissemination refers to the spread or dispersal of the pathogen from an inoculum source to a host. Dissemination can occur by wind, splashing rain, insects, infested pruning tools, infected or infested transplants, and other means. Spread can occur over short distances within the tree canopy or from distant sources.

Primary infection occurs when the pathogen comes into contact with a susceptible host under favourable environmental conditions. Pathogens penetrate the surface of a plant directly or enter through wounds or natural openings.

Growth and development of a pathogen usually occurs on or within infected plant tissue.

Secondary infection results from spores or cells produced following primary infection or from other secondary infections. The secondary infection cycle can be repeated many times during the growing season. The number of cycles is dependent on the biology of the pathogen and its host and the duration of environmental conditions needed for infection.

Over wintering or over seasoning is the ability of a pathogen to survive from one growing season to the next. Pathogens of apple survive the winter in a number of different ways.

For a plant disease to occur, a susceptible host, a pathogen (casual agent), and favourable environmental conditions must be present and interact with one another. If any one of these requirements is not met, a plant disease will not occur. At present, our ability to manipulate the environment is limited to only a few practices such as pruning to promote drying, bedding to improve soil drainage, and scheduling of irrigation. Severe disease outbreaks can be prevented by manipulating the host - through the use of resistant cultivars - and the pathogen -- through cultural practices and fungicidal or bactericidal sprays.

3.14. Disease control measures of important crops

Crops	Disease	Symptoms	Control Measures
Banana	1. Sigatoka leaf spot	On leaves small light yellow or brownish green narrow streaks appear. They enlarge in size becomes linear, oblong, brown to black spots with dark brown brand and yellow halo. Black specks of fungal fructification appear in the affected leaves. Rapid drying and defoliation of the leaves.	Remove affected leaves and burn. Spray any one of the following fungicides commencing from November at monthly interval-difenoconazole (2ml/litre), azoxystrobin (2ml/litre), Copper oxychloride 2.5 g/lit. Alternation of fungicides for every spray prevents fungicidal resistance. Always add 5 ml of wetting agent like Sandovit, Triton AE, Teepol etc. per 10 lit of spray fluid.
	2. Anthracnose	The skin at the distal ends of the fingers turn black, shrivels. The fungus produces masses of conidia which form a pinkish coat. The entire fruit and bunch is affected in severe cases. Sometimes main stalk of bunch diseased. The bunch becomes black and rotten. Acervuli produces cylindrical conidiophores, hyaline, septate, branched. Conidia hyaline, non-septate, oval to elliptical.	Spray copper oxychloride 0.25% or Bordeaux mixture 1%. Post harvest dipping of fruits in difenoconazole (2ml/litre).
	3. Bunchy-top	Dark broken bands of green tissues on the veins, leaves and petioles. Plants are extremely stunted. Leaves are reduced in size marginal chlorosis and curling. Leaves upright and become brittle. Many leaves are crowded at the top. Branches size will very small. If infected earlier no bunch will be produced. The disease is transmitted primarily by infected suckers. Secondary spread is through the aphid vector.	Spray Methyl Demeton 2 ml/lit to control it. The sprays may be directed towards crown and pseudostem base upto ground level at 21 days interval atleast thrice.

Crops	Disease	Symptoms	Control Measures
Banana cont...	4. Panama disease	Yellowing of the lower most leaves starting from margin to midrib of the leaves. Yellowing extends upwards and finally heartleaf alone remains green for some time and it is also affected. The leaves break near the base and hang down around pseudostem. Longitudinal splitting of pseudostem. Discolouration of vascular vessels as red or brown streaks. The fungus spreads through use of infected rhizomes. Continuous cultivation results in build up of inoculum.	Uproot and destroy severely affected plants. Apply lime at 1 – 2 kg in the pits after removal of the affected plants. In the field, Panama wilt disease can be prevented by corm injection methods. A small portion of soil is removed to expose the upper portion of the corm. An oblique hole at 45° angle is made to a depth of 10 cm. Capsule application for 50 mg of Pseudomonas fluoresces is injected into the hole with the help of 'corm injector' on 2nd, 4th and 6th month after planting.
Mango	1. Powdery mildew	It attacks the leaves, flowers, stalks of panicle and fruits. Shedding of infected leaves occurs when the disease is severe. The affected fruits do not grow in size & may drop before attaining pea size.	Application of Sulphur dust (350 mesh) in the early morning will protect new flush or spraying Wettable sulphur 0.2% will control powdery mildew.
	2. Anthracnose and stalk end-rot	Produces leaf spots, blossom blight, wither tip, twigs blight and fruit rot. Small blister like spots develop on the leaves and twigs. Young leaves wither and dry Tender twigs wither and die back symptom appears. Affected branches ultimately dry up. Black spots appear on fruits. The fruit pulp becomes hard, crack and decay at ripening. Infected fruits drop.	Pre-harvest spraying of Thiophanate methyl 1g/lit 3 times at 15 days interval will control anthracnose and stalk end-rot.
	3. Sooty mould	The fungi produce mycelium which is superficial and dark. They row on sugary secretions of the plant hoppers. Black encrustation is formed which affect the photosynthetic activity. The fungus grows on the leaf surface on the sugary substances secreted by jassids, aphids and scale insects.	Spraying thiamethoxam @ 2 ml/ litre + Maida 5% (1 kg Maida or starch) boiled with 1 lit of water and diluted to 20 litres will control the incidence of sooty mould. Avoid spraying during cloudy weather.

Crops	Disease	Symptoms	Control Measures
Mango cont...	4. Mango malformation	The dark epicarp around the base of the pedicel. In the initial stage the affected area enlarges to form a circular, black patch. Under humid atmosphere extends rapidly and turns the whole fruit completely black within two or three days. The pulp becomes brown and somewhat softer. Dead twigs and bark of the trees, spread by rains.	Apply plant growth regulators (NAA/GA/Ethephon @ 50-200ppm) at Bud Inception stage. Harvest mangoes on clear dry day. Injury should be avoided to fruits at all stages of handling.
Cotton	1. Seedling diseases (seed-rot, rootrot, and damping off	Seed-rot, root-rot, pre emergence and-post emergence damping-off.	Fungicide seed treatments help control seed rots and some pre emergence damping off. However, an additional soil treatment of fungicide must be used to control root-rots and most damping-off. In addition, producers must follow all other recommended cotton production practices to decrease seedling diseases. Some of these practices include use of correct planting equipment and date of planting, good seed bed preparation, correct use of herbicides and insecticides and use of high germinating seed.
	2. Fusarium wilt	Plants become stunted, yellowed, followed by defoliation. Yellowing first occurs around leaf edges and advances inward. Cross sections of infected stems usually reveals a brown Discoloration which is more intense in outer layers of tissue. Infected plants fruit earlier and produce smaller boll.	Reduce nematode population. Crop rotations. Use resistant varieties.
	3. Boll rot	Boll rots usually first appear as water soaked spots. Later, as infection-spreads, bolls turn black and may be-covered with a moldy fungus growth. Badly infected bolls may drop from-plant.	Avoid excessive rates of nitrogen. Practice skip-row planting. Timely defoliation will reduce boll rots. Reduce insects which injure bolls. Growth regulators such as Pix can be used effectively to reduce boll rots.

Crops	Disease	Symptoms	Control Measures
Cotton cont...	4. Leaf spot	Various types of leaf spots and blights. Many spots occur on leaves toward maturity, but these are not usually damaging to the plant at this stage of growth.	Use fungicide seed treatments. Destroy crop residues. Use crop rotations and plant resistant varieties when available (esp. when Bacterial Blight is severe). Keep potash levels at least medium to high.
	5. Verticillium wilt	Seedlings may become infected and turn yellow, dry out and die. Plants that become infected later in the season are stunted and exhibit a yellow condition along leaf margins and between the major vein. Severely affected plants will shed their leaves. A brown Discoloration of the interior of the stem can usually be found later in the season. This discoloration is. Distributed evenly across the inside of the stem.	Plant resistant varieties when Verticillium Wilt is severe. A variety that matures very early may in some years escape injury from Verticillium Wilt.
Citrus	1. Root rot, foot rot and gummosis	Rotted roots, cracked bark, accompanied by gumming Water-soaked, reddish-brown to black bark at the soil line Discoloured tissue in the lower trunk; yellowing, sparse foliage and death of the tree.	Two sprays with drenching either by Fosetyl-Al (2.5g/L)or Metalaxyl MZ-72(2.75g/1 water covering the whole plant canopy and basin of affected plant at 40 days interval after onset of monsoon provided significant control .For the control of gummosis, scraping of the affected parts followed by application of Metalaxyl[2] MZ-72 paste.
	2. Citrus canker	Disease affecting citrus species that is caused by the bacterial Infection causes lesions on the leaves, stems, and fruit of citrus trees. While not harmful to humans, canker significantly affects the vitality of citrus trees, causing leaves and fruit to drop prematurely.	Pruning and destruction of infected twigs followed by three to four sprays with copper oxychloride (COC) 0.3% + streptocycline 100 ppm at monthly intervals after the onset of monsoon.

Crops	Disease	Symptoms	Control Measures
Citrus cont...	3. Citrus decline	Symptoms vary with the cause of the malady. The affected trees do not always die completely, but remain in a state for decandance and unproductive for a number of years. Some-times they may suddenly wilt and die in a day or two. In early stages, Symptoms are restricted to a few limbs, but eventually the whole tree is involved. Trees show sparse mottling leaves, stunted growth, sickly appearance. Midrib and lateral veins of old, mature leaves turn yellow with interveinal areas along the veins showing diffuse yellowing. Leaves may turn yellow and are shed with the onset of summer or autumn and the die-back of twigs starts. Dead shoots stand out prominently and may be found dead right down to the main trunk. The entire tree bears short twigs carrying narrow small leaves on their lower portion. Subsequent secondary growth consists of short, upright small, weak shoots showing a variety of discolouration of leaves. Often these leaves have green veins of green blotches. Occasionally, small, circular, green spots appear on yellow tissue on leaves. The die-back of weak shoots continues. There is excessive flowering, but the fruits are not carried to maturity. The fruits show distinct sun-blotching. The feeder root system becomes depleted, roots turn black and sometimes are covered with rotting bark. Either only a few trees or entire orchard may be affected.	Good cultural practices, improvement in soil fertility and drainage, control of insect pests, nematodes, etc. may be useful to minimize the incidence of decline. Use of resistant rootstocks and certified budwood for propagation is also useful.

3.15 Concept of Seed Treatment

The concept of seed treatment is the use and application of biological and chemical agents that control or contain primary soil and seed borne infestation of insects and diseases which pose devastating consequences to crop production and improving crop safety leading to good establishment of healthy and vigorous plants resulting better yields.

The benefits of seed treatment are as follows:

- Increased germination
- Ensures uniform seedling emergence.
- Protect seeds or seedlings from early season diseases and insect pests improving crop emergence and its growth.
- Use of plant growth hormones may enhance crop performance during the growing season.
- Rhizobium inoculation enhances the nitrogen fixing capability of legume crops, and their productivity.
- Improved plant population and thus higher productivity.

Pest/Disease	Seed Treatment	Remarks
Root rot, wilt	Trichoderma spp. 4-6 gm/kg. seed.	For seed dressing metalseed dresser / earthern pots or polythene bags are used.
Root rot disease	Trichoderma 5-10 gm/kg. seed (before transplanting).	-do-
Bacterial sheath blight	Pseudomonas flourescens 0.5% W.P. 10 gm/kg.	
White tip nematode	Seed soaking in 0.2% solution.	-do-
Anthracnose spp. **Damping off**	Seed treatment with Trichoderma viride4g/kg.	-do-
Soil borne infection of fungal disease	Trichoderma viride @ 2 gm/kg. seed and Pseudomonas flourescens,@10gm/kg. Captan 75 WS @ 1.5 to 2.5 gm a.i./litre for soil drenching.	-do-
Jassid, aphid, thrips	Imidacloprid 70 WS @ 10-15 gm a.i./kg seed (To be used in proper doses under guidance of an agriculture expert).	
Wilt, blight and root brot	Trichoderma spp. @ 4 gm/kg. seed.	For seed dressing metal seed dresser/earthern pots or polythene bags are used.
Root knot nematode	Paecilomyces lilacinusand Pseudomonas fluorescens @ 10 gm/kg as seed dresser.	- do-
Soil borne infection of fungal disease **Early blight** **Damping off** **Wilt**	T. viride @ 2 gm/100gmseed. Captan 75 WS @ 1.5 to 2.0 gm a.i./litre for soil drenching. Pseudomonas fluorescens and V. clamydosporium @ 10gm/kg as seeddresser	For seed dressing metal seed dresser/earthern pots or polythene bags are used.

Bacterial wilt	Pseudomonas fluorescens @ 10gm/kg.	-do-
Pest/Disease	**Seed Treatment**	**Remarks**
Seed rot	Trichoderma viride @ 6 gm/kg seed.	-do-
Jassids, whitefly	Thiamethoxam (0.2gm/litre of water)	
Termite	Treat the seed before sowing with any one of the following insecticides. i) Chlorpyriphos @ 4 ml/kg seed (Best available option and use with caution).	For seed dressing metal seed dresser / earthern pots or polythene bags are used.
Bunt/false smut/ loose smut/covered smut	Carboxin 75 % WP. Tebuconazole 2 DS @ 1.5 to 1.87 gm a.i. per kg seed. T. viride 1.15 % WP @ 4 gm/kg.	
Wilt and damping off	Seed treatment with Trichoderma viridi 1% WP @ 9 gm/kg seeds.	
Soil and tuber borne diseases	Seed treatment with boric acid 3% for 20 minuts before storage.	

3.16. Nematode management

Nematodes are thread-like roundworms invisible to the naked eye. Species parasitic on plants attack roots and other plant parts, causing stunting and yield reduction. Nematode-infected plants are not only weakened but their root systems are more susceptible to secondary infections by fungi or bacteria.

Correct identification is the first step when a nematode problem is suspected. The second step is to determine whether populations are high enough to threaten the crop. Root knot nematodes, the most common pathogenic nematodes in vegetables, cannot penetrate roots when soil temperatures are below 50 degrees F, and will not reproduce when soil temperatures are below 58 degrees F. Their reproductive rate is slower at cooler temperatures, so populations build up more slowly. Thus, cool season crops are less likely to be damaged. E.g., early spring potatoes, are rarely damaged by nematodes.

Nematode management practices

Isolation: Once a nematode problem is confirmed, affected areas and plants should be isolated because transplants, machinery and irrigation water can all spread nematode infections. From initially small-infested areas, nematodes can spread across a field at a rate of 3 feet per year.

Crop rotation and cover crops: Crops susceptible to root knot nematodes include all cole crop species, beans, cucumber, muskmelon, watermelon, bendi, potato, sweet potato and tomato. All potatoes are susceptible to nematodes except for a few cultivars resistant to the golden nematode.

Rotation to non-host crops such as corn, cucurbits, potatoes and tomatoes is an effective control for the cyst nematode. But is less likely to control the root knot nematode because of its wider host range. All species of Meloidogyne are called 'root knot' nematode but each species has a different host range, causing confusion over which crops or cultivars are resistant or tolerant to which species of root knot nematode. Rotations to non-host crops for more than a year reduce populations below damaging levels but will not eliminate them.

Cover crops: significantly reduce subsequent damage to crop.

Increasing Soil Organic Matter: Higher soil organic matter content protects plants against nematodes by increasing soil water-holding capacity and enhancing the activity of naturally-occurring biological organisms that compete with nematodes in the soil.

Fallow Period: A fallow period of two years with no susceptible plants in the field decreases nematode populations. *Marigold* as a rotation crop suppress nematodes.

Plant resistance: Nematode resistant cultivars may be used to reduce the incidence of nematodes

Symptoms

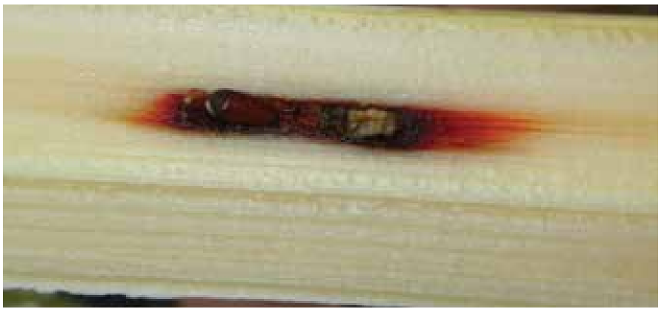
- Understanding symptom and description of disease will help in identification at field level.


Blast infected leaf of paddy


Bacterial blight of paddy


Altenaria leaf spot of redgram


Wilt affected redgram plant


Red rot in sugarcane


Ratoon stunting in sugarcane


Anthracnose in cotton


Gray mould in cotton

3.17. Control measures - Tips for the farmers

- Correct identification of disease in your farm is essential for effective control of disease.
- With little experience, you can identify the disease. However you can contact agricultural officers of your area along with disease specimen and seek their help in identifying the disease.
- You can also give disease-affected plant parts to the Agri clinics for clinical test before undertaking control measures.
- Follow Integrated Diseases Management such as host plant resistance, agronomic practices, judicious use of fungicides, pesticides for vector control, bio-pesticides for pathogen control etc., as indicated below.

3.18. Integrated disease management practices in the field

- Select varieties and hybrids resistant to the most common or economically important diseases in consultation with agricultural officers of your area.

For example:

Diseases of Cotton	Resistant Variety/Hybrid
Verticillium Wilt	MCU 5 VT, Surabhi, Savitha (Hybrid)
Bacterial Leaf Blight	MCU 10, L 604, L 389

- Plant only good quality, disease-free seed having good germination. E.g. Cotton seed having above 80% germination will have vigorous growth and will not suffer from infection of soil borne diseases
- Use seed-treatment with fungicides to control diseases.
- Plant when soil temperature and moisture are most favourable for specific crop. E.g: If the farmers take up sowing of cotton during the warmer temperature (>65 F), there will be better germination and growth.
- Avoid planting the same crop in a field year after year. E.g.:
  I) Growing paddy in the Veriticillium wilt infested field will reduce the incidence of microbial population in the soil.
  II) Growing Chrysanthemum will inhibit the growth of the Veriticillium in the soil.
- Incorporate the crop residues of the previous crop by tilling well before planting season. E.g. Disease affected plants should be burnt immediately.
- Choose right sowing time and maintain appropriate plant population by adopting recommended spacing.
- Apply a balanced fertilizers based on a soil test. E.g.: Potassium deficiency leads to susceptibility of Alternaria leaf spot.
- Apply recommended amount of Farmyard Manures/compost at regular intervals and maintain soil-organic matter content. E.g.: Over dose of chemical fertilisers lead to more vegetative growth and more disease.
- Enrich soil with beneficial micro-organisms like Trichoderma.
- Keep the land weed free. Weeds can serve as alternate hosts for pathogens and helps disease development.
- Timing and duration of irrigation should match the crop and water requirement without allowing for excess water. E.g.: Excessive irrigation favour soil borne pathogen
- Maintain insect population below ETL to reduce the incident of disease transfer by insects.
- Regular crop monitoring is essential for effective disease management.
- Use bio-pesticide as far as possible to control disease. Chemical may be used only as last resort.

III. Weed Management

3.19. Weed and its relevance in crop production

- Weeds are the plants, which grow where they are not wanted
- Weeds compete with crops for water, soil nutrients, light and space
- Weeds reduce crop yields to the extent of up to 50 percent
- Critical period of weed competition is approximately 1/3rd of the duration of the crop

Characteristics of weeds: "One year seeding, seven years weeding"

- Produces larger number of seeds compared to crops. E.g.: Amaranthus retroflexus produces 1,96,405 seeds/plant, whereas wheat & rice produces only 90 to 100 seeds/plant
- Most of the weed seeds are small in size
- Easy and diverse means of seed dispersion
- Seeds germinate earlier and grow faster
- Flower earlier and mature ahead of the crop
- Germinate under tough conditions, season bound
- Seeds are dormant for long period and germinate during suitable season
- Good viability for years
- Tolerate moisture stress
- Possess stronger and deeper root system

Effect of weed competition on crop growth and yield

- Crop suffers from nutritional deficiency
- Growth is reduced
- Water requirement will be more
- Lowers the input response
- Pest and disease incidence will be more
- Yield is affected
- Cost of production will increase

3.20. Critical period of weed competition for important crops

Crops	Days from sowing
Rice (lowland)	35
Rice (upland)	60
Sorghum	30
Maize	30
Cotton	35
Sugarcane	90
Groundnut	45
Soybean	45
Onion	60
Tomato	30

3.21. Different types of common weeds

GRASSY WEEDS

Acrachne racemosa

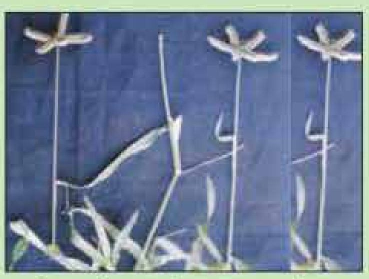
Aristida funiculata

Avena ludoviciana

Brachiaria reptans

Chloris inflata
Syn: *Chloris barbata*

Cynodon dactylon



Dactyloctenium aegyptium

Digitaria bicornis

Digitaria longiflora

Dinebra arabica

Dinebra retroflexa

Dinebra

Echinochloa colona

E. crus-galli

Echinocloa glabrescens

BROAD LEAVED WEEDS

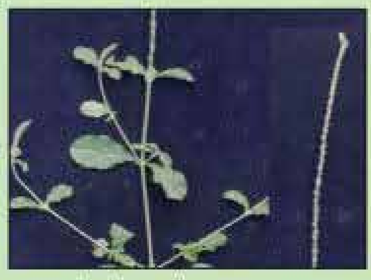

Abutilon hirtum


Abutilon indicum


Acalypha ciliata


Acalypha indica


Acanthospermum hispidum


Achyranthes aspera


Aerva lanata


Aerva tomentosa


Aeschynomene indica


Ageratum conyzoides


Alternanthera


Alternanthera paranychioides


Alternanthera sessilis


Alternanthera pungens


Allmania nodiflora

SEDGES

Cyperus castaneus

Cyperus comprestris

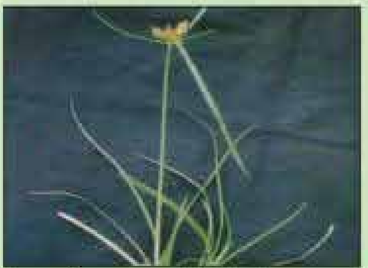
Cyperus difformis

Cyperus digitatus

Cyperus halpan

Cyperus iria

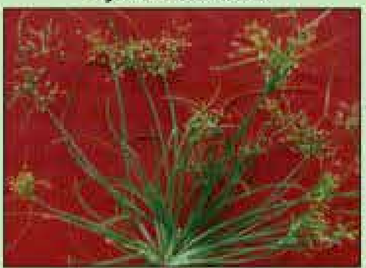
Cyperus nutans



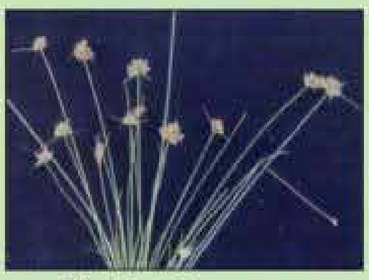
Cyperus polystachyos
Syn: P. odoratus

Cyperus rotundus

Cyperus tenuculmis
Syn: C. zollingeri



Fimbristylis aestivalis



Fimbristylis argentea

Fimbristylis complanata

Fimbristylis milliacea

Fimbristylis woodrowii

Source: Directorate of Weed Science Research, Jabalpur

3.22. Control measures of weeds

Principles of weed control

- Prevention
- Eradication
- Control
- Management

a) Preventive weed control

- Avoid using crop that are infested with weed seeds for sowing
- Avoid adding weeds to the manure pits
- Nursery or planting material should be free from weeds
- Keep irrigation channels, fence-lines and uncropped areas clean
- Constantly look for weed, destroy the weeds then and there
- Use good quality certified seeds, which are free from weed seeds
- Use pre-emergence herbicides to prevent germination of weeds

b) Eradication (complete removal): Weeds are killed or completely removed from a given area, will not reappear unless it is introduced again. However,

- It is very difficult
- Involves high cost
- Can be used in green houses and nurseries

c) Control: Weeds growth is restricted and killed when necessary so that it does not affect crop growth.

d) Weed management: Managing the population of weeds using all possible methods.

(i) Mechanical method:

- **Tillage:** Using plough or disc, weeds are removed from soil and exposed to sunlight.
- **Hoeing:** Using hand hoe, annual and biennials are completely removed.
- **Hand weeding:** Either by physical removal or pulling out of weeds by hand or using some implements.
- **Digging:** Advisable in the case of perennial weeds.
- **Using sickle**: Top portion of weeds are removed using sickle, thereby weeds seed production is controlled.
- **Burning:** Burning is often an economical and practical means of controlling weeds but not always possible in crop production field.
- **Flooding:** Kills weeds by reducing oxygen to plant growth. This is possible only under garden land or wetland condition.

Merits of mechanical method

- Oldest and effective method
- Safe method for environment
- High skill is not necessary
- Weeding is possible in between plants
- Deep rooted weeds can be controlled effectively

Demerits of mechanical method

- Labour and time consuming
- Possibility of crop damage
- Requires ideal moisture
- Costly

ii) Cultural weed control

- **Summer ploughing:** Is done immediately after summer showers. This exposes weeds to hot sun.
- **Field preparation:** Makes the field weed free by constant removal.
- **Select crop** that can compete better with weeds like cowpea, sudan gross, sorghum are good competitors. Fast growing crops suppresses the weed effectively.
- **Maintenance of optimum plant population:** Adequate plant population covers the land and, hence, growth of the weed will be difficult. Close row crops are better than wide row.
- **Crop rotation:** Minimize the dominance of particular weed in the cropping system.
- **Growing of intercrops:** Inter cropping covers the land quickly and reduce growth of the weeds. E.g,: Growing crop such as cowpea/ soybean, etc. in wide spaced crops like maize/pigeon pea/sugar cane, etc.
- **Mulching:** Mulch is a protective covering of material maintained on soil surface. It has smothering effect and reduces the weed growth. Mulching can be done through degradable farm waste or through plastic sheets. In case of plastic, black is the most popular colour used in commercial horticulture crop production, especially for weed control.
- **Solarisation:** Done by covering the pre soaked field with transparent polythene cover, which increases the temperature by 5 – 10 °C.

- **Stale seedbed:** Weeds are allowed to germinate and non-residual herbicide is sprayed to kill the young weed seedlings.
- **Blind tillage:** Ploughing after sowing of the crop and before plants emerge. Normally seed drills are used under rainfed condition for this purpose.
- **Crop management practices:** Vigorous and fast growing crop varieties for better competition with weeds.

Merits of cultural method

- Low cost
- Easy to adopt
- Less technical skill is sufficient
- No damage to crops
- Effective weed control

Demerits of cultural method

- Time taking and difficult
- Perennial and problematic weeds can not be controlled

iii) Biological weed control: Natural enemy of a weed plant is used to control the weed

Examples:

- Zygogramma biolorata for control of Parthenium
- Hirsch – Manniella spinicaudata is a rice root nematode to control most upland rice weeds
- Azolla in rice

Advantages

- Eco friendly
- Easy
- Low cost

Disadvantages

- They may have alternate host or switching over to alternate host
- Multiplication of bio agent in many cases is difficult

iv) Chemical Weed Control: Herbicides are chemicals used to control weeds.

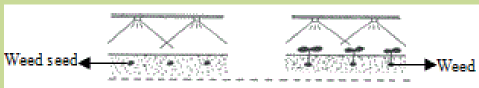
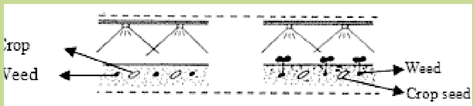
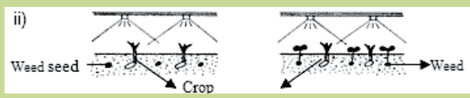
Merits

- Recommended for adverse soil and climatic conditions
- Applied even before weeds emerge and make the environment weed free
- Suitable for all types of crops
- Controls the targeted weeds only
- Controls many perennial weed species
- **Cost effective** compared to labour

Demerits

- Pollutes the environment
- Affects the soil
- Herbicide drift affects adjoining field
- Requires minimum technical knowledge
- Leaves residual effects
- Some herbicides are costly
- No suitable herbicides are available for mixed and inter-cropping system

3.23. Classification of herbicides

Method of application	**Soil herbicides**; e.g: Fluchloralin **Foliar herbicide:** e.g. Glyphosate
Mode of action	**Selective herbicide:** Kills only weeds. **Non selective herbicide:** Kills the entire vegetation.
Mobility	**Contact herbicide;** Kills when comes in contact with plant. **Translocated herbicide:** poison moves from treated parts to untreated part: Eg. Glyphosate
Time of application	**Pre-plant:** Before sowing or along sowing Eg. Glyphosate for Hariyali, Basalin for groundnut  **Pre emergence:** Before weeds germinate;Eg. Thiobencarb  **Post emergence**: applied after weeds germinate i) Eg. Bispyribac Sodium. 

Formulations	**Emulsifiable Concentrate (EC):** Liquid form. **Wettable powders:** Poisons mixed with inert carrier. **Granules (G)** poisons are mixed with granular forms. **Water Soluble Concentrates (WSC)** forms are also available

3.24. Methods of application

- Spraying
- Broadcasting

Foliar application

Blanket spray: Both crop and weed is sprayed with weedicide.

Blanket Spray

Directed spray: Application of herbicides on weeds only avoiding crop using hood.

Eg: Spraying glyphosate[1] in between rows of tapioca using hood to control Hariyali.

Protected spray: Crops are covered and herbicides are sprayed on weeds. This method is expensive.

Spot treatment: Applied only where weeds are present.

3.25. Control of parthenium (perennial weed)

- Manual removal and destruction of Parthenium plants before flowering using hand glouse/ machineries (or)
- Uniform spraying of sodium chloride 200g + 2 ml soap oil/litre of water (or)
- Spraying of 2,4-D sodium salt 8 g or glyphosate10 ml + 20g ammonium sulphate + 2 ml soap solution/litre of water before flowering (or)
- Post-emergence application of metribuzin 3 g / litre of water under non-crop situation.
- Raising competitive plants like Cassia serecea and Abutilon indicum on fallow lands to replace Parthenium (or)
- Biological control by Mexican beetle, fungal pathogen and nematodes

3.26. Control of perennial weeds in orchards

Perennial weeds like Cyperus rotundus, cynodon dactylon, etc. in orchards can be controlled effectively by spraying glyphosate[1] at 2.5 to 5.0 L ha-1 dissolved in 500 liters of water. Falling of the spray fluid on young fruit plant foliage should be avoided. Second spray is required when there is re-growth of weed. (Cost Rs.700/- to 1400 ha).

3.27. Precautions while spraying the herbicides

- Select right kind of herbicide for right kind of crop and spray. Any mistake in choosing the herbicide may result in loss of total crop.
- Dosage should be accurate and good quality of water should be used.
- Use always correct nozzle for spraying. Spraying should be done from front to backwards. (We should not step into the sprayed field for a minimum of 3 days).
- The soil should have sufficient moisture for effective control.
- For paddy, a thin film of water should be maintained for 3 days and it should not be drained.

3.28. List of Herbicides for different crops

* There may be slight variation in doses. Please consult local Agricultural Extension Officer/Scientists from KVK or read carefully the leaflet attached along with the herbicide:

Crop	Herbicide	Dose (kg ai/ha)	Trade name and formulation	Time of application
Rice	Thiobencarb	1.25	Machete 50% EC Delchlor 50% EC	Pre-emergence
	Anilophos	0.40	Thunder 50% EC Saturn 50% EC	Pre-emergence
	Pendimethalin	0.90	Arozin 30% EC Aniloguard 30% EC	Pre-emergence
Finger millet	Pendimethalin	0.90	Stomp 30% EC	Pre-emergence
	2,4-D Na salt	1.00	Fernoxone 80% SS	Post-emergence
Maize	Pendimethalin	0.75	Stomp 30% EC	Pre-emergence
Cotton	Metolachlor	1.00	Dual 50% EC	Pre-emergence
	Pendimethalin	1.00	Stomp 30% EC	Pre-emergence
Groundnut	Metolachlor	1.00	Dual 50% EC	Pre-emergence
	Pendimethalin	0.90	Stomp 30% EC	Pre-emergence
Vegetables	Pendimethalin	1.00	Stomp 30% EC	Pre-emergence
Pulses	Pendimethalin	0.60	Stomp 30% EC	Pre-emergence
Wheat	Isoproturon	0.60	Arelon 75% WP	Pre-emergence
Citrus	Glyphosate 4	4 Kg/ha		Post-emergence

3.29. Herbicide mixtures

Sometimes involves mixing of two or more herbicides used for effective and economical weed control at reduced dosage.

Two types of mixtures available are:

1. Tank mixtures made with the desired proportion of herbicides before application. eg: Anilophos + Bispyribac Sodium – Rice
2. Ready mix – formulated by the manufacturer. Ready mix available in the world market eg: Bispyribac Sodium+Glyphosate.

Mixing Ammonium sulphate with glyphosate increases the efficiency where nitrogen increases the translocation.

3.30. Integrated Weed Management (IWM)

Combination of two or more weed control methods at low input levels to reduce weed composition in a given cropping system below the economic threshold level. It:

- Aims to minimize the residue problems
- Minimize the effect on the ecosystem

3.31. Beneficial Effects of Weeds or Economic Uses of Weeds

Several weeds have been put to certain economic uses since ages. Some of the examples are:

- Typha and Saccharum sp is used for making ropes and thatch boards.
- Chicory Cichorium intybus roots are used for adding flavor to coffee powder.
- Amaranthus viridis, Chenopodium album and Portulaca sp. are used as leafy vegetable.
- Hariyali grass (Cynodon dactylon) and Cenchrus Ciliaris, Dichanthium Annulatum andEclipta alba weeds of grass land serve as food for animals.
- Weeds act as alternate host for predators and parasites of insect pests, which feed on the weeds. For example, Trichogramma chilonis feed upon eggs of caster semi looper, which damage the castor plants. E.g. Commelina sp (Copper), Eichornia crassipes (Copper Zinc, lead and cadmium in water bodies.
- Several species of weeds like Tephorsia purpurea and Croton sparsiflora in South India are used as green manures, whereas Eichornia crassipes and Pistia stratiotes are used for composting.
- Argemone Mexicana is used for reclamation of alkali soils.
- Some weeds have medicinal properties and used to cure snake bite (Leucas aspera), gastric troubles (Calotropis procera), skin disorders (Argemone mexicana) and jaundice (Phyllanthus nirur) and Striga Orobanchioides to control diabetes.
- Agarbathis (Cyperus rotundus), aromatic oils, (Andropogan sp & Simbopogon sp) are prepared from weeds.
- Air pollution determined by wild mustard and chickweed respectively.
- Aquatic weeds are useful in paper, pulp and fiber industry.
- Chenopodium album is used as mulch to reduce evaporation losses, whereas Agropyron repens (quack grass) is used to control soil erosion because of its prolific root system.
- Weeds like Lantana camara, Amaranthus viridis, Chenopodium Albu and Eichhornia crassipes are used for beautification.
- Agropyron repense are used for soil conservation, whereas Dicanthium Annulatum are used as stabilizing field bunds.
- Opuntia Dellini is used as biological fence.

3.34. Lessons Learnt

1. Not all insects are harmful. Farmers need to identify beneficial and harmful insects.
2. Summer ploughing, growing trap crops, adopting mixed cropping, intercropping, crop rotation and keeping the field clean are the important agronomical practices farmers needs to follow for effective pest management.
3. Farmers need to understand the Economic Threshold level of major pest of the crops grown.
4. Biological control methods results in sustainable pest control.
5. Farmers are advised to harvest the fruits and vegetables after the waiting period to minimize the residual effect.
6. Seed treatment prevents soil and seed borne infestation of insects and diseases
7. Select varieties and hybrids resistant to most common and economically important diseases.
8. One year seeding, seven years weeding. Hence, prevent spreading of seeds of weeds.

Categories of Pesticides and Precautions

Insecticide	Group	Label of Toxicity	Precaution
Flubendiamide	Diamide	Category D	To be used with Caution.
Emamectin Benzoate	Macrocyclic Lactone - Avermectin	Category D	To be used with Caution.
Spinosad	Macrocyclic Lactone – Spinosyn	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Acetamiprid	Neonicotinoids	Category D	To be used with Caution.
Imidacloprid	Neonicotinoids	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Thiamethoxam	Neonicotinoids	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Chlorpyriphos	Organothiophosphate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Profenophos	Organothiophosphate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Fenazaquin	Unclassified	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.

Fungicide	Group	Label of Toxicity	Precaution
Thiophanate methyl	Benzimidazole precursor	Category B	To be used with great caution (since highly hazardous for human health and/or environment) and under the guidance of an Agriculture Expert.
Carboxin	Carboxamide/ oxathin	Category D	To be used with Caution.
Copper oxychloride	Inorganic copper	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Fosetyl	organophosphorus	Category D	To be used with Caution.
Captan	Phthalimide	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Metalaxyl	phenylamide	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Azoxystrobin	Strobilurins	Category D	To be used with Caution.

Difenoconazole	Triazoles	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Triadimefon	Triazoles	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Tebuconazole	Triazoles	Category C	To be used with caution and under the guidance of an Agriculture Expert.

Herbicide	Group	Label of Toxicity	Precaution
2,4-D	Chlorophenoxy acid or ester	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Butachlor	Chloroacetanilide	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Metolachlor	Chloroacetanilide	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Fluchloralin	Dinitroaniline	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Pendimethalin	Dinitroaniline	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Anilophos	Organophosphorus	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Diuron	Phenylurea	Category B	To be used with great caution, since probably carcinogenic, and under the guidance of an Agriculture Expert.
Isoproturon	Phenylurea	Category C	To be used with caution and under the guidance of an Agriculture Expert.
Glyphosate	Phosphonoglycine	Category B	To be used with great caution, since probably carcinogenic, and under the guidance of an Agriculture Expert.
Atrazine	Triazines	Category B	To be used with great caution and under the guidance of an Agriculture Expert.
Thiobencarb	Thiocarbamate	Category B	To be used with great caution (since toxic to bees etc.) and under the guidance of an Agriculture Expert.
Metribuzin	Triazinone	Category B	To be used with great caution and under the guidance of an Agriculture Expert.

4. Farm Management

4.1. Objectives of the session

- To equip the farmer to take advantage of improved technologies and market opportunities to increase income.
- To assist the farmer to make proper plan and adopts his production to assure food security for the family.
- To empower the farmer for professional negotiations with buyers, input dealers and credit institutions.
- To educate the farmer to make profitable decisions considering available resources and anticipate risks including market fluctuations.

4.2. What we know at the end of the session

- Importance of farm management
- Basic information which support better farm management decision
- Market driven enterprises
- Matching resources with calendar of activities
- Selection of cropping pattern
- Understanding of cost benefit analysis
- Risk analysis in agriculture

Module 1: Is farming a business?

What is a farm?

Farm is a socio economic unit. It is composed of farm family, farm enterprises and structures. Farmer is a grower cum manager. The farmer has to decide how much land, labour, capital and type of technology to use to produce in a given season. Ultimately, the farmer has to earn a profit to support his livelihood.

What will the farmer do in farm management?

The farmer would efficiently use the available resources to increase profits through deciding among the best alternatives available.

Some basic functions of farm management

The farmer performs following basic functions to effectively manage the farm:

Diagnosis: Analysis of past performance of farm, its weakness/strengths.

Planning: Planning for the future crops/animals considering the opportunities and threats.

Implementation: Efficient implementation with least cost.

Monitoring: Reduce the losses and increase the profits by reducing the costs and choosing better technologies based on the observed opportunities.

Evaluation: Evaluating the actions for repeating the succeses in future.

What will happen if the farmer does not do farm management?

In the absence of good farm management, the farmer may experience losses in farming for the following the reasons:

- There is continuous changes in supply and price of agri inputs like seeds, fertilizers, irrigation, power etc.
- There is continuous change in prices of the produce (outputs) in the market due to demand and supply changes.
- There is continuous change in the farm technologies.

Therefore the objectives of farming as a business are:

- How to choose best variety/crop/cropping pattern
- How to minimise input cost by judicious use
- How to increase the production and productivity
- How to enhance the quality
- How to plan market driven production
- Choosing the better source of finance and better avenues for investment
- Efficient risk management

For better farm management, the farmer should have thorough knowledge of the following aspects:

- Farm map
- Soil slope and topography
- Soil type (physical and chemical properties)
- Soil colour such as red soil or black soil
- Weather parameter such as rainfall, temperature, relative humidity, etc.
- Vegetative cover such as trees, weeds, etc.
- Irrigation potential from borewell/tubewell/nala/channels
- Drainage facilities - whether water gets logged or not
- Technology available and whether the farmer can access them easily
- Risk factors like hand loans and high rate of interest
- Market facilities - whether they are near to his farm or far off
- Communication facilities like cell phone and internet connectivity
- Physical and infrastructure facilities such as godowns, roads for transport, vehicles, custom hiring centers, etc.
- Whether a farmer can afford the crop/animal he/she wishes to grow or rear considering the above conditions
- Supporting programmes and schemes/subsidies

Farmers should maintain farm records to have a holistic knowledge of their production system

For example: If a farmer maintains a record for all the cost of production such as inputs, labour, etc. for the entire crop cycle along with yield and income obtained from selling the produce, he/she can compare with the next crop cycle to understand whether his/her profit increased or decreased. The records also provides information on activities which contributed for his/her profit or loss so that, the farmer can take alternative decisions to enhance his/her net income.

Farm Resources: To make good farm management decisions, farmers need some basic knowledge on farm resources such as the extent of land available for cultivation, source of irrigation, family labour, availability of labour, skill level of labours, livestock, availability of fodder, availability of farm machinery, availability of inputs such as seeds and fertilizers, credit requirement and availability, source of credit, market demand for produce, infrastructure such as cold storage and godowns. etc. For example, regarding manpower and livestock the farmers have to understand following issue:

Man Power

- **Skill:** Is the labour employed is skilled, e.g. cotton picking skill?
- **Knowledge:** Does the farmer/labour have a thorough knowledge of the package of practices of the crop?
- **Attitude:** Does he/she have a positive attitude towards the technology?

Live Stock

- **Breed:** Selection of a suitable breed
- **Production capacity:** For instance, milk production, meat production and egg laying capacity.
- **Adaptability:** Does the selected breed adapt to the local situation?
- **Drafting capacity:** Knowledge on the draught capacity of animal.
- **Resistance:** Is the foreign breeds resistant to local Indian conditions. E.g. the Holstein Fresian is highly sensitive to high temperatures.
- **Feeding habit:** Are the upgrade breeds or imported breeds capable of feeding on locally available feed materials

Module 2: Know your farm resources

Inputs	Tools and equipment	Labour	Money	Land
Seeds Fertiliser Insecticide Fungicide	Plough, hoe, sprayer, thresher, Tractors	Family and Paid workers	Self finance and credit	Own / Rented land Share-cropping

What does one need to know about the market if one wants to do good business?

The market for agricultural produce	The market for inputs and equipment
The location of the market	The locations of sale
Who is the buyer?	Who sells the inputs and equipments?
The quality of product that is demanded by the market	The quality of inputs and equipment
The price of the product compared to other markets	The price of sale of the inputs and equipments
When to sell	When to buy

How does the price of agricultural products change?

The price of agriculture products change according to the season of the year	The price of agriculture products change between years
At times of abundance, the prices are lowest	The price of a product that is needed by more and more people will rise from one year to the next
At times of scarcity, the prices are highest	The price a product that is produced in greater abundance will fall from one year to the next
The quality of inputs and equipment	
The price of sale of the inputs and equipments	
When to buy	

Important Lesson

To DO successful farm business, the farmer must be well aware of prices (of inputs and produce) at different markets. This allows the farmer to plan production as well as make decisions on the purchase of inputs and the sale of produce.

Module 3: Manage your farm for enough income to sustain yourself

FARM PLANNING: Farm planning is to help the farmers to move to a higher level of production and income, starting from where he/she is now with the resource available to him/her. In this process, the farmer has to consider different types of enterprises like:

- Land based (agril. production activities, pisciculture, plantation, seed production, etc.)
- Animal component based (diary, poultry, goatery, piggery, duckery, etc.)
- Nursery/orchard
- Non-land based (mushroom, apiculture, vermiculture, etc.)

In the present example, three crops namely paddy, cotton and maize have been taken into comparison for one season. With these three crops, farmers can grow the following combinations in a year:

- Paddy – Paddy (Kharif followed by Rabi)
- Paddy – Maize (Kharif followed by Rabi)
- Cotton – Maize (Kharif followed by Rabi)
- Maize – Maize (Kharif followed by Rabi)

Exercise 1: Agricultural Calendar of Operations to Plan the Production of Paddy.

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
The tasks of the farmer		**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calendar of operations throughout the year

Here we will see how to determine if farm business was good or bad. We will calculate the “Income” and “Expenditure” from different produce. Master Trainers may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet 2: Paddy

Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line
- Add the money spent (“Expenditure”) on inputs and labour
- Multiply the yield by the price of sale (“Income”)
- Subtract the sum of “money-out” from the “Income”
- Determine if there was a gain or a loss

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
a) Cost of organic & Green Manuring (In-situ plouging)				
b) Application cost				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Litre			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/profenophos)	Litres			

Activity	Unit	Quantity	Price	Total (Rs.)
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				
Irrigation cost if any	Power	month		
Sub-Total				
Cost of harvest				
a) Combined harvester	Hours			
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 2 – Paddy

Cost of Cultivation of Paddy (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	8	800	6400
b) Animal / labour	Days	6	600	3600
Sub Total				**10000**
Seeds and sowing				
a) Cost of seed	Kgs	50 kgs	20	1000
b) Cost of seed treatment				50
c) Cost of sowing (Human Labour)	Days	25	200	5000
d) Cost of thinning/gap filling	Days	5	200	1000
Sub-Total				**7050**
Manures and Fertilizers				
a) Cost of organic & Green Manuring (In-situ plouging)				350
b) Application cost				100

Activity	Unit	Quantity	Price	Total (Rs.)
c) Cost of fertilizer	Kgs	120 N	12	1440
		60 P	50	3000
		40 K	28	1120
d) Application cost (Human Labour Male)	Days	3	200	600
Sub-Total				**6610**
Weed control				
a) Cost of Manual weeding	Labour	30	200	6000
b) Cost of herbicide if any (butachlor)	Litre	2.5	200	500
Sub-Total				**6500**
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/profenophos)	Litres	5	350	1750
Furadon-3G	Kg	15	60	900
c) Cost of Application	Labour	6	200	1200
Sub-Total				**3850**
Irrigation cost if any	Power	5 months	500	2500
Sub-Total				**2500**
Cost of harvest				
a) Combined harvester	Hours	4.5	1700	7650
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days	10	200	2000
Sub total				**9650**
Total cost of cultivation				**46160**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	50	1500	75000
b) Gross returns received per ha (Rs.)	Cart load	8	800	6400
c) Total cost involved per Ha (Rs.)				**81400**
d) Net returns per Ha (Rs.)				**46160**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**35240**
e) Cost benefit ratio				**1:1.76**

Exercise 1: Agricultural Calendar of Operations to Plan the Production of Maize

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
	The tasks of the farmer	**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calender of operations throughout the year.

Here we will see how to determine if farm business was good or bad. We will calculate the "Income" and "Expenditure" from different produce. Master Trainer may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet - 2 (Maize – Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line
- Add the money spent ("Expenditure") on inputs and labour
- Multiply the yield by the price of sale ("Income")
- Subtract the sum of "money-out" from the "Income"
- Determine if there was a gain or a loss

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Thiamethoxam/profenophos)	Litres			
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				

Activity	Unit	Quantity	Price	Total (Rs.)
Irrigation cost if any	Power	month		
Sub-Total				
Cost of harvest				
a) Combined harvester	Hours			
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 2 - Maize

Cost of Cultivation of Maize (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	5	650	3250
b) Animal / labour	Days	5	500	2500
Sub Total				**3750**
Seeds and sowing				
a) Cost of seed	Kgs	20	81.25	1625
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days	5	200	1000
d) Cost of thinning/gap filling	Days	5	200	1000
Sub-Total				**2625**
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	150 N	12	1800
		60 P	50	3000
		50 K	28	1400
d) Application cost (Human Labour Male)	Days	10	150	1500
Sub-Total				**7700**
Weed control				

Activity	Unit	Quantity	Price	Total (Rs.)
a) Cost of Manual weeding	Labour	20	150	3000
b) Cost of herbicide if any (butachlor)	Days	4	500	2000
Sub-Total				**5000**
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides	Litres	5	300	1500
Furadon-3G	Kg	10	60	600
c) Cost of Application	Labour	4	200	8
Sub-Total				**2900**
Irrigation cost if any	Power	4	500	2000
Sub-Total				**2000**
Cost of harvest				
a) human labour	days	20	200	4000
b) Threshing (machine)	quintals	40	60	2400
Post harvest charges				
b) Cleaning and bagging (Human Labor)	days	15	200	**3000**
Sub total				**7400**
Total cost of cultivation				**31375**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	40	1000	40000
b) Gross returns received per ha (Rs.)	Cart load	4	500	2000
c) Total cost involved per Ha (Rs.)				42000
d) Net returns per Ha (Rs.)				**31375**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**10625**
e) Cost benefit ratio				**1:1.34**

Exercise 1: Agricultural Calendar - Operations to Plan the Production of Cotton

The times of work…													
Of the main season are shown by a square													
Of the off-season are shown by a circle													
The tasks of the farmer		**January**	**Feb.**	**March**	**April**	**May**	**June**	**July**	**August**	**Sept.**	**October**	**Nov.**	**Dec.**
	Prepare the field												
	Plough the field												
	Purchase seeds												
	Sow												
	Fertilizer application												
	Weeding												
	Apply insecticide												
	Harvest and store												

Important Lesson:

For a good yield, the farmer plans to do the necessary work in the field and apply the inputs at the right time based on the calender of operations through out the year.

Here we will see how to determine if firm business was good or bad. We will calculate the "Income" and "Expenditure" from different produce. Master Trainer may give the following exercise sheet to the farmers to work out the details for the respective crops by changing the relevant package of practices as applicable to local situations.

Exercise Sheet - 2 (Cotton - Bt Cotton) – Blank sheet to be filled by Farmer based on worked solution given below.

Steps:

- Multiply the quantity with the price in each line.
- Add the money spent (“Expenditure”) on inputs and labour
- Multiply the yield by the price of sale (“Income”)
- Subtract the sum of “money-out” from the “Income”
- Determine if there was a gain or a loss

Cost of Cultivation Cotton (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours			
b) Animal / labour	Days			
Sub Total				
Seeds and sowing				
a) Cost of seed	Kgs			
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days			
d) Cost of thinning/gap filling	Days			
Sub-Total				
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	N		
		P		
		K		
d) Application cost (Human Labour Male)	Days			
Sub-Total				
Weed control				
a) Cost of Manual weeding	Labour			
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Imidacloprid/Thiamethoxam/profenophos)	Litres			
Furadon-3G	Kg			
c) Cost of Application	Labour			
Sub-Total				

Activity	Unit	Quantity	Price	Total (Rs.)
Irrigation cost if any	Power			
Sub-Total				
Cost of harvest				
a) Picking	Kgs			
Sub total				
Total cost of cultivation				
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls			
b) Gross returns received per ha (Rs.)				
c) Total cost involved per Ha (Rs.)				
d) Net returns per Ha (Rs.)				
e) Cost benefit ratio (Gross Returns divided by Total Cost)				

Solution to exercise 3 – Cotton

Cost of Cultivation of Cotton (per hectare)

Activity	Unit	Quantity	Price	Total (Rs.)
Preparatory cultivation				
a) Machine / labour	No of hours	5	650	3250
b) Animal / labour	Days	5	500	2500
Sub Total				5750
Seeds and sowing				
a) Cost of seed	Kgs	0.9	1860	1674
b) Cost of seed treatment				
c) Cost of sowing (Human Labour)	Days	8	200	1600
d) Cost of thinning/gap filling	Days	2	200	400
Sub-Total				**3674**
Manures and Fertilizers				
c) Cost of fertilizer	Kgs	150 N	12	1800
		60 P	50	3000
		60 K	28	1680
d) Application cost (Human Labour Male)	Days	30	150	4500
Sub-Total				10,980
Weed control				
a) Cost of Manual weeding	Labour	75	150	11250

Activity	Unit	Quantity	Price	Total (Rs.)
b) Cost of herbicide if any (butachlor)	Days			
Sub-Total				11250
Plant Protection				
a) Cost of bio-agents				
b) Cost of pesticides (Imidacloprid/ Thiamethoxam/profenophos)	Litres	10	350	3500
Furadon-3G	Kg			
c) Cost of Application	Labour	20	200	4000
Sub-Total				**7500**
Irrigation cost if any	Power	**6**	**500**	**3000**
Sub-Total				**3000**
Cost of harvest				
a) Picking	Kgs	2500	6	15000
Sub total				**15000**
Total cost of cultivation				**57154**
Yield Kgs/Ha. and returns				
a) Qty. produced Qtls. per ha	qtls	25		
b) Gross returns received per ha (Rs.)			3600	90000
c) Total cost involved per Ha (Rs.)				57154
d) Net returns per Ha (Rs.)				**32846**
e) Cost benefit ratio (Gross Returns divided by Total Cost)				**1:1.57**

Note: The fixed cost and recurring cost like the interest, depreciation, opportunity cost, etc have not been taken into calculation in the three crops above. Therefore, other things remaining constant, Only the variable cost and returns have been used for the exercise.

Module 4: Income and expenditure statement

Comparing results to know whether you are doing successful farm business

Please tell what is good and what is bad business and indicate the reasons.

Crop	Unit	1 ha of Paddy	1 ha maize	1 ha Cotton
Production	Quintal	50 q	Main product = 40 q By Product (stover/ Straw) = 4 cart load	25 q
Income	Rs/ha	81400	42000	90000
Expenditure	Rs/ha	46160	31375	57154
Profit or Loss?	Rs/ha	+ 35240	+ 10625	+ 32846
		Profit	Profit	Profit
Rank		I	III	II

* It proves that Paddy after Paddy realises the highest profit throughout the year followed by Cotton – Maize. Paddy – Cotton is not feasible.
Remember: Even if Paddy after Paddy promises the highest profit throughout the year, farmer should avoid continuous monocropping to ensure soil fertility through crop diversification and rotation to realize sustained profitability of their farm.

Main Lessons

- To know if you are doing successful business with a crop, you need to know the "Income" and "Expenditure" accurately.
- The farmer records the inputs & labour used in a field, and calculates the "Income" and "Expenditure"
- From the "Income" the farmer subtracts the Expenditure. The result indicates whether farm is making profit or loss.
- The farmer makes a **PROFIT** or **GOOD BUSINESS** if the "Income" is greater than the "Expenditure."
- It is a **LOSS**, if the "Expenditure" is greater than the "Income." In that case it is **BAD BUSINESS.**
- A loss is illustrated by the the minus (dash) and a profit by the plus in front of the number.
- A good farmer will abandon loss making crop or use a better technique to make a profit.
- To ensure a profit, the farmer needs to visualise Income and Expenditure before production.
- The Difference between Income and Expenditure indicates whether we are making a loss or profit from the use of the land.
- The Unit Cost of a crop indicates if it can compete with the same crop produced elsewhere. In the case of food crops, the Unit Cost indicates if it is preferable to buy the produce in the market.
- The good farmer calculates well ahead of the season to decide what to produce and which techniques to use.
- During the production season the good farmer keeps records on money spent for farm operations and inputs.
- After the harvest, the good farmer evaluates the profit and identifies what changes are needed to improve the planning and profit for the next production season

What are Fixed Costs?

Certain costs are called fixed costs. These are costs for equipment and tools that the farmer owns and are used for various crops over several years, such as sprayers, irrigation pumps, buildings etc. The Fixed Costs do not vary with the size of the field.

Blank Exercise sheet of Income and Expenditure Statement: Comparing Results to know whether you are doing successful Farm Business

After, all the calculations farmers will determine the opportunities to increase revenues. By looking at the money on this page farmers will learn

- How to make investment decisions and determine the best opportunities by using Gross Margin, Labour Productivity and Capital Productivity.
- Rank crops based on Profit or Loss
- What crops and techniques to choose?
- Make a choice based on this ranking

	Unit	Paddy	1 ha maize	1 ha Cotton
Surface Area	Ha	1	1	1
1. Money-Out (Variable Costs) Rs/ha	Rs/ha			
Cost of Inputs	Rs/ha			
Labour Costs	Rs/ha			
2. Income (Gross revenue)	Rs/ha			
Production	Kg			
Price	Rs/kg			
Yield x Price of Sale	Rs/ha			
Value of Produce (Gross Income)	Rs/ha			
Expenditure	Rs/ha			
Profit or Loss?				
Rank				

Module 5: Manage your Money throughout the Year

Bad management of money

- How does one know if the money is managed badly?
- What are the causes?
- What must one do to manage money well during the year?

One must plan! The person, who fails to plan, plans to fail!

First Step: Please look at Income and Expenditure for different crops on the farm and also look at household expenditures. Below are the expenditures of a Household of 5 persons (1 child not yet in school, 1 child in primary school, 1 old parent along with husband and wife). We discuss if we can predict all these expenditures.

Money Needs	Expenditures (Rs)	Period	Can be foreseen
Provisions (food, fuel and household items for an average family size of 5 members)	36,000 (3000 X12)	Each month	Yes
School fees Free education for primary school children			
Clothing			
School uniforms for 1 child	1,000	June	Yes
Clothing /year	5,000	April	Yes
Happy events			
Diwali /dussehra	1000	November	Yes
Baisakhi/Ugadi	1000	January	Yes
Unexpected events	3600	Each month @ Rs 300	No
Health expenditure	2400	Per year	No
Total expenditure	50000		

Second Step:

- Let us put these numbers into a financial calendar. On the next page you will see the numbers calculated in Module 5.
- How much money is left at the end of each month?
- How much money is left at the end of the year?

The trainer explains how to do it.

Case 1: Paddy – Paddy financial calendar based on a farm using current practices (Rs) – Exercise

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Paddy (1 ha) - Kharif												
Cost of Inputs including labour						6550	18610	7400	3450	8150	2000	
Paddy(1 ha) - Summer												
Cost of Inputs including labour	18610	7400	3450	8150	2000							6550
Household	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000
School fees and material												
Happy events				1000							1000	
Clothing				5000		1000						
Unexpected events	300	300	300	300	300	300	300	300	300	300	300	300
Total per month	**21910**	**10700**	**6750**	**17450**	**5300**	**14850**	**21910**	**10700**	**6750**	**11450**	**6300**	**9850**

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Paddy (81 Paddy (81400) Sum-mer					81400							
Paddy (81400) - Kharif											81400	
Total per month					81400						81400	

Expenditure	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Expenditure per Month	21910	10700	6750	17450	5300	14850	21910	10700	6750	11450	6300	9850
Total Expenditure for the Year	142320 (Total expenditure per year = 46160 (Cost of Cultivation for Paddy per season) X2 seasons + 50000 (Family Expenditure per year											
Total balance per year	Income = (81400 X 2 = 162800 -142320 = 20480 (Net balance)											

* As the income is realised only during May and November, the monthly balance is shown as negative which indicates that the farmers raises a hand loan/personal loan or shall keep the gross income (Rs 81,400 X 2 = 162,800) and spend through out the year based on the need.

Note: In this example all produce from the farm is sold as per the prevailing market rates. The profit of the farmer depends on the price for the produce i.e., whether the Minimum Support Price (MSP) or market rates are prevailing - which are both fluctuating.

Third Step:

Fill the second financial calendar. The expenditures for inputs and labour are those from the exercise Sheets in Module 5 – using improved practices.

Fourth Step:

Discuss the differences and which situation is preferable. What changes are necessary?

Main Lessons

- In the agricultural enterprise, expenditure for the farm and the household are incurred every month. But the revenue comes only after sale of produce. Therefore there are months of the year where the expenditures are greater than the revenues. These months are called "deficit months."
- For this reason, the good farmer makes a financial calendar. The farmer plans expenditures for production and household needs in consultation with his spouse.
- To be able to cover the expenditures in deficit months, the good farmer saves money from the sales of produce ("surplus months").
- Improved techniques may contribute to improve the revenues of the farmer.
- The needs for inputs can be identified with calculations of gross margin and the financial calendar. This information can be used to make savings in a targeted way or to solicit credit for production.
- The above calculations are done for a farmer with one hectare for Paddy – Paddy combination in Kharif and Rabi for one year. The same exercise should be repeated for taking farm management decisions for the combinations such as
  - Cotton – Maize (Kharif – Rabi)
  - Paddy – Maize (Kharif – Rabi)
  - Maize – Maize (Kharif – Rabi)

Risk in Agriculture

The farmer does not like risks because they are difficult to predict. However, he/she can determine the risk during the planning so that the impact on revenues could be minimised. For example, Module 4 indicates income and expenditure of paddy, maize and cotton. With available data, Paddy - Paddy is the best preferred cropping pattern. However, the farmer can make a judicious decision taking prevailing and anticipated risks into considerations. The anticipated risks and possible decisions are as follows:

Anticipated Risks	Remarks
Reduction in Paddy Market price by 10%	Cotton – Maize is preferred
Inadequate water due to drought for Paddy - Paddy	Cotton – Maize is preferred followed by Maize - Maize. Quantity of water available decides the cropping pattern.

Anticipated Risks	Remarks
Lack of capital for investment for Paddy – Paddy or Cotton - Maize	Paddy and Maize is preferred followed by Maize - Maize
Serious pest problems in Cotton	Paddy – Paddy is preferred

Main Lessons

- Comparing the gross margins of different crops and the production techniques, helps to make decisions on using the land to maximise revenue. This comparison is important to all agricultural entrepreneurs.
- Comparing the labour productivity helps to identify the crops and techniques that make best use of labour (family or wage labour). It also indicates if it is profitable to work on your own farm.
- Comparing the capital productivity indicates which crops or production techniques make best use of money invested.
- Production decisions are based on these comparisons.
- The good agricultural entrepreneur knows that a fluctuation in prices constitutes a risk and revenues. Risks are concern for traditional as well as improved varieties and techniques.
- To evaluate the impact of this market risk, the entrepreneur calculates estimated the gross margin with a much lower price (pessimistic) than the current price (or last season's price). If the pessimistic gross margin estimate can still satisfy the revenue objectives, then the risk is acceptable.
- To evaluate the impact of production risks, the agricultural entrepreneurs calculates a gross margin using a yield lower (pessimistic) than expected. If the pessimistic gross margin estimate can still satisfy the revenue objectives, then the risk is acceptable.

5. Occupational Health and Safety of Farmers

5.1. Objectives of the session

- To create an awareness about causes of health hazards, risks and fatalities in agriculture.
- To impart the knowledge on preventive measures of health hazards in agriculture.
- To enlighten the farmers on use of first aid in emergencies.

5.2. What we know at the end of the session

- Important occupational health hazards in agriculture
- Factors responsible for increasing risk of injury or illness for farmers
- Safe handling of agro chemicals
- Colour coding of pesticides
- First aid measures for pesticide poisoning
- Care in use of pesticides by farmers
- Safety tips to reduce the risk of injuries and fatalities while handling machineries
- First aid for accidents

In agriculture, farmers work under open condition in natural environment, which expose them to various occupational hazards, especially more caused due to handling of chemicals and machineries. Besides, natural hazards are caused due to snakebite, wild animal attack, etc. Knowledge on preventive and curative aspects of these occupational hazards would reduce risks and ensure the safety to the farmers.

5.3. Important occupational health hazards in agriculture

Exposure	Health Effect	Specificity to Agriculture
Hot Weather	Dehydration, heat cramps, heat exhaustion, heat stroke, skin cancer	Most agricultural operations are performed outdoors
Snakes, insects	Fatal or injurious bites and stings	Close proximity results in high incidence
Sharp tools	Injuries ranging from cuts to fatalities	Most farm situations require a wide variety of skill levels for which workers have little knowledge
Physical labour, carrying loads	Back pain and body pain	Agricultural work involves uncomfortable conditions and sustained carrying of excessive loads
Pesticides	Acute poisoning or chronic poisoning	Pesticides can be hazardous and must be used with Personal Protective Equipment (PPE)
Dusts, fumes	Irritation of the eyes and respiratory tract	Agricultural workers are exposed to a wide range of dusts and gases during plant protection with few exposure controls and limited use of PPE

Gases, pathogen	• Skin diseases such as fungal infections and allergic reactions • Parasitic diseases such as malaria, sleeping sickness and hookworm • Animal related diseases such as anthrax, bovine tuberculosis and rabies (at least 40 of the 250 animal related diseases are occupational diseases in agriculture) • Cancers	• Workers are in direct contact with environmental pathogens, fungi, infected animals, and allergenic plants • Workers have intimate contact with parasites in soil, waste water/sewage, dirty tools and unhygienic housing • Workers have ongoing, close contact with animals through raising and sheltering • Agricultural workers are exposed to a mix of biological agents, pesticides, and diesel fumes, all linked with cancer
Others	• Electricity shocks, fire, road accidents, livestock and wild animal attacks, falling into wells, lightening, psychological depression, suicides, etc.	• Loss of life and injuries and suffering to the dependents

Adapted from IFPRI, 2006

5.5. Factors that may increase risk of injury or illness for farm workers

Age: Injury rates are highest under 15 and over 65 years of age.

Equipment and machinery: Most farm accidents and fatalities involve machinery. Proper machine guarding and regular equipment maintenance according to manufacturers' recommendations can help prevent accidents.

Non-availability of protective equipments: Using protective equipment, such as seat belts on tractors and personal protective equipment (such as safety gloves, cover alls, boots, hats, aprons, goggles and face shields) could significantly reduce farming injuries.

Lack of medical care: Hospitals and emergency medical care are typically not readily accessible in rural areas near farms. However, the farmers may utilize for example 108 mobile medical emergency services during emergencies.

Socio - economic risks

- There are dangers due to indiscriminate pesticide/weedicide application to soil, water, air, food chain, natural enemies of pest, resistance of pest, etc.
- Pesticides are hazardous, so there will be associated risk and impact of risk on social system including economic damage. So there should be a proper understanding on hazards of pesticides.

Effects on human beings

- Pesticides accumulate in fatty tissues and reproductive cells lead to birth defects, abnormalities, abortions, premature deliveries, etc.
- Farm workers who regularly spray pesticides are susceptible to impaired eyesight.
- The liver is particularly susceptible to damage by chlorinated hydrocarbons, which can lead to higher risk of serious infection.

Environmental exposures lead to

- Air pollution
- Soil and water pollution
- Food contamination

Looking into the implications of pesticides on socio-economic dimensions, use of pesticides should be judiciously planned only in critical situations with utmost care. Management of pest below Economic Threshold Level (ETL) is most desirable.

5.6. Safe handling of agro chemicals

Agro chemicals are widely used in agriculture, which are major source of health hazards to farmers. Important tips to farmers for safe use of agro chemicals are as follows:

Buying pesticides

Before buying think over this

- Which pest is to be controlled?
- Is damage crossing threshold level requiring pesticide use?
- Have you observed any predators that check pest attack?
- What is the recommendation for the observed problem?
- Which is the least toxic and low persistent chemical among recommendations?

While buying

- Buy from a reputed and licensed store.
- Buy only required quantity and not to go for bulk purchase.
- Do not buy leaky containers.
- See the expiry date.
- Buy only ISI / BIS marked products.
- Don't buy banned and restricted chemicals.

Transportation

- Do not transport/carry pesticides along with food products.
- Do not spill or allow leakage while transporting.
- If leakage is observed clean the vehicle.

Storage

- Don't keep chemicals in kitchen or with animal feed.
- Keep out of reach of children.
- Keep under lock and key.
- Keep in well ventilated separate room.
- Avoid cross contamination of different chemicals.
- Do not keep with human or animal medicine.
- Properly reseal before storing again.

Application

- Read the label and follow the instructions.
- Don't work alone while handling and applying.
- Don't allow children and animals near mixing and application site.
- Use long wooden stick for mixing.
- Mix only the quantity that you need for the next application.
- Avoid excessive spraying and unintended site application.
- Read the label and instructions carefully before opening the pack.
- Never eat and drink while applying.
- Use Personal Protective Equipment (PPE), particularly long shirt and pants, closed shoes as well as protection mask and gloves.
- Avoid application during rainy period.
- Never blow out clogged nozzles or hoses with your mouth, use pin or fine wire for cleaning.
- Avoid windy conditions and do not spray against wind direction.

After application

- Immediately after application take bath and change cloths.
- All clothes must be washed after spraying/dusting and wash them separately.
- Never leave residues of pesticide in sprayers and dusters.
- While cleaning see that water used for cleaning should not enter the drinking water stream/wells.
- Do not go into the treated field until the recommended safety period has passed.
- Do not harvest produce before safe period.

Safe container disposal

- Burry the containers in the field after use.
- Don't use for food and feed storage.
- Do not sell empty container.
- Do not wash the used containers in community water sources.

Colour Coding of pesticides:

The colour band on the pesticide container indicates the hazard level of pesticides. While handling pesticides farmers should take colour band into consideration.

- Red – Very toxic to toxic
- Yellow – Harmful
- Blue - Moderately hazardous
- Green - Acute hazard unlikely in normal use

5.7. First aid measures for pesticide poisoning

- In case of skin contact, remove contaminant contacts and wash with clean water.
- In case of inhalation, remove from site and provide good clean air site, keep the head and shoulder upright.
- In case of unconscious and breathing stops, provide artificial respiration.
- If pesticide is swallowed, induce vomiting by giving 2-3 liters salt water. Give milk after that.

- Take the patient to doctor at the earliest.
- Take the container along with patient to consult doctor.

Snake and other animal bites or attacks and precautions

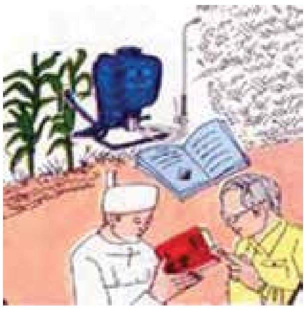
Snakebite is a routinely occurring life threatening emergency in India. The mortality and morbidity associated with the diverse presentation of snakebites can be decreased if a proper history of the patient's background and habits combined with a thorough knowledge of the specific features of the regional snakes are kept in mind.

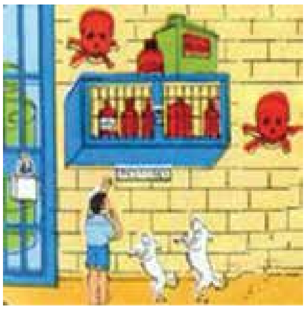
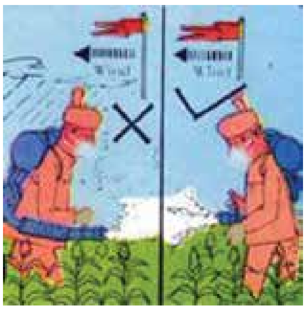
Physical and mental drudgery

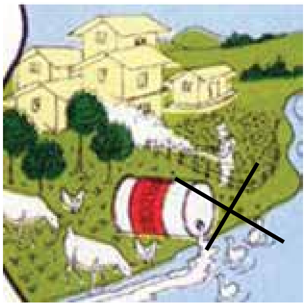
To reduce drudgery related to hard work in different agricultural operations, several technologies for land preparation, weeding, pesticide application and various other farm works have been developed including farm machinery which need to be used by the farmers to keep themselves fit and healthy. Farmers are advised to contact local extension functionaries and scientists to obtain information on such drudgery reducing farm machineries.

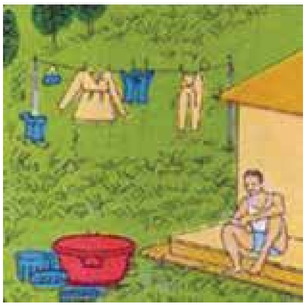
5.8. Care in use of pesticides by farmers

Useful tips for safe use of pesticides by farmers are:

	• Identify the pest and ascertain the damage done. • Use pesticide only if it has exceeded the economical injury level. • Use only the recommended pesticide, which is the least toxic.
	• Read instructions manual of the pesticide and equipment. • Check the spraying equipment and accessories, which are to be used.
	• Ascertain that all components are clean, especially filling and suction, strainer, sprayer tank, cut off device and nozzle • Replace worn out parts such as 'O' ring, seal, gasket, worn out nozzle tip, hose clamps and valves. • Test the sprayer and ascertain whether it pumps the required output at rated pressure. Check the nozzle spray pattern and discharge rate.
	• Calibrate the sprayer. Set sprayer speed and nozzle swath by adjusting spray height and nozzle spacing.

	• Make sure that appropriate protective clothing is available and is used • Train all concerned with the application and also understand the recommendations. Ensure that soap, towel and plenty of water is available.
	• Pesticide should be kept in dry and locked store.
During spraying	
	• Take only sufficient pesticide for the day's application from the store to the site. • DO NOT transfer pesticides from original container and packing into the containers. • Recheck the use instructions of pesticide and equipment. • Make sure pesticides are mixed in the correct quantities.
	• Wear appropriate clothing. • Avoid contamination of the skin especially eyes and mouth. • Liquid formulation should be poured carefully to avoid splashing.
	• Never eat, drink or smoke when mixing or applying pesticides. NEVER blow out clogged nozzles or hoses with your mouth.
	• Follow correct spray technique. Spray plant crop thoroughly by operating sprayer at correct speed and correct pressure.

	• Follow correct spray technique. Spray plant crop thoroughly by operating sprayer at correct speed and correct pressure.
	• Never allow children or their unauthorized persons to be nearby during mixing. Never leave pesticides unattended in the field. • Never spray if the wind is blowing towards grazing livestock or pastures regularly used. • Spraying should be done in the direction of Wind.
After spraying	
	• Remaining pesticides left in the tank after spraying should be emptied and disposed off in pits dug on wasteland. • Never empty the tank into irrigation canals or ponds.
	• Never leave unused pesticides in sprayers. Always clean equipment properly. After use, oil it and then keep away in storeroom.
	• Do not use empty pesticide containers for any purpose. • Crush and bury the containers preferably in a land filled dump.
	• Clean buckets, sticks, measuring jars etc used in preparing the spray solution.

	• Remove and wash protective clothing and footwear. • Wash yourself well and put on clean clothing.
	• Keep an accurate record of pesticide usage. • Prevent persons from entering treated areas until it is safe to do so. • Mark the sprayed plots with a flag.
	Source : tnauagriportal

5.9 Basic Measures to Manage Weeds, Diseases and Pest Problems

- Prevention is better than cure, hence follow the preventive measures discussed above
- Keep pest below ETL using preventive techniques
- Give more attention to pest warnings by Agriculture Department officials and experts
- Be watch full on weather forecast by ICAR and IMD
- Consult KVK or Scientist in early stages
- Discuss the problems with fellow farmers, you may get many traditional tips
- Understand the scientific rationale behind each pest management ITKs
- Have regular contact with local Research and Development wings of agriculture and allied sectors to get a recent development.

5.10. Safety tips to reduce the risk of injuries and fatalities while handling machineries

Contrary to the popular image of fresh air and peaceful surroundings, a farm is not a hazard free work setting. Every year, thousands of farm workers are injured and hundreds die in farming accidents. Safety in agriculture is one of the main concern, especially when handling with farm tools and machinery. Many accidents in agriculture go unnoticed because they will not be reported. So learning from mistakes will be less. Safety tips to reduce the risk of injuries and fatalities while handling machineries are as follows:

- Safety can be improved on farm by increasing awareness of farming hazards and making a conscious effort to prepare for emergency situations including fires, vehicle accidents, electrical shocks from equipment and wires and chemical exposures.
- Be especially alert to hazards that may affect children and the elderly.
- Minimize hazards by carefully selecting the products to ensure safety.
- Always use seat belts when operating tractors.
- Read and follow instructions in equipment operator's manuals and on product labels.
- Inspect equipment routinely for problems that may cause accidents.
- Discuss safety hazards and emergency procedures with all concerned.
- Take precautions to prevent entrapment and suffocation caused by unstable surfaces of grain storage bins and silos. Never "walk on the grain."
- Be aware that methane gas, carbon dioxide, ammonia and hydrogen sulfide can form in unventilated grain silos and manure pits and can suffocate or poison farmers or explode.
- Wear clothing that fits well and is not loose fitting to avoid being caught in pinch points.

- Never reach over or work near unguarded rotating parts.
- Turn off machinery to attend to repairs.
- Always replace shields that were removed for maintenance.
- Plug unused or failed borewell pits
- Check all equipment for potential w.rap points (e.g. where clothing or hair could be wrapped around a shaft) and if possible, shield those points.
- Replace any damaged manufacturer installed warning labels and place warnings on equipment parts not previously labeled (consider painting them with a bright color, perhaps with a wide stripe).
- Stay alert and warn others when working with shear and cutting points (e.g. objects with blades or hard edges used to cut). Some shear and cutting points cannot be guarded, which can result in severe cuts, lost limbs or injuries from objects thrown by the cutting type equipment.
- Wait until a tractor has stopped completely before stepping into the hitching position.
- Never touch free-wheeling parts (i.e. parts that continue to spin after the power is shut off) until they have stopped moving. This could take 2-2½ minutes.
- Be aware of burn points: mufflers, manifolds and even gear cases.
- Hydraulic systems contain fluid under extreme pressure. Before loosening, tightening, removing or otherwise working with any fittings or parts, relieve this pressure (shut off the hydraulic pump, lower implements and follow instructions in the operator's manual).
- Keep equipment in good repair and safety features up to date.
- "Proper machine inspection and maintenance can help prevent accidents".
- When it comes to machinery maintenance, a shield and guard to cover spinning parts or blades should be kept in place.
- Follow the one seat, one rider rule. If there is only one seat on the equipment, there should only be one rider – an adult.
- Don't allow children to play or ride on equipment or in areas where machinery is used or stored.
- Under aged children should not operate 2, 3, and 4 wheeled vehicles.
- Do not allow riders or passengers in the back of pickup trucks.
- Before starting machinery, all operators should know where kids are located. You may be unable to hear or see children, especially behind large wheels or in blind spots.
- All equipment should be parked and locked with the keys removed when not in use.
- Keep hand tools out of reach of children, especially those with sharp or hot parts.

Better safety and health practices reduce farmer fatalities, injuries and illnesses as well as associated costs such as workers' compensation insurance premiums, lost production and medical expenses. A safer and more healthy workplace improves morale and productivity.

First aid for accidents: The farmers injured should be given first aid at the earliest in order to reduce the damage caused by the injuries. Important tips are as follows:

- The treatment should be given from a person trained in basic first aid, using supplies from a first-aid kit.
- Medical treatment and care given at the site of any medical emergency or while transporting any victim to a medical facility.
- Make sure that first-aid trained personnel are available to provide quick and effective first-aid. Alternatively, one of the family members should be trained.
- Make sure first-aid supplies at your workplace are appropriate to your occupational setting. The response time of your emergency medical services is very important.
- Good knowledge about locally available antidotes or medicinal plants, which can be used.
- Keep the first aid kit(s) in the work place.
- Keep clean water in the work place.
- Keep mobile numbers of trained person on first aid, ambulances, local hospital, local doctors and vehicle owners in the near by vicinity.

5.11. Suggested items for your first-aid kit

- Sterile adhesive bandages
- Small roll of absorbent cotton pads of different sizes
- Adhesive tape

- Triangular and roller bandages
- Cotton (1 roll)
- Band-aids (Plasters)
- Scissors
- Pen torch
- Latex gloves (2 pair)
- Tweezers
- Needle
- Moistened towels and clean dry cloth pieces.
- Antiseptic (Savlon or dettol)

First Aid Kit

- Thermometer
- Tube of petroleum jelly or other lubricant
- Assorted sizes of safety pins
- Cleansing agent/soap

Non-prescription drugs

- Aspirin or paracetamol pain relievers
- Antidiarrhea medication
- Antihistamine cream for Bee Stings.
- Antacid (for stomach upset)
- Laxative

Kits should be checked at least weekly to ensure adequate number of needed items is available. Kits may be kept in the work place.

Make sure that first-aid supplies are:

- Easily accessible to all farmers.
- Stored in containers that protect them from damage, deterioration or contamination.
- Containers must be clearly marked, not locked, and may be sealed.
- Able to be moved to the location of an injured or acutely ill worker.
- Make sure emergency washing facilities are functional and readily accessible.

5.12. Lessons Learnt

1. Awareness on occupational health and safety issues is must for every farmer.
2. Majority of the health hazards in Agriculture are preventive in nature provided farmers are aware about.
3. Farmers should handle the agro chemicals safely. Little negligence may cost the life of human beings and livestock.
4. Dispose the pesticide containers safely.
5. First aid knowledge and skills saves the lives.
6. Keep away children from farm machineries.

6. Farmers' Access to Services

6.1. Objectives of the session

- To enhance awareness about source of extension, information and services among farmers.
- To expose farmers to public and private extension services.
- To encourage farmers to avail extension services through ICT means.
- To enhance farmers knowledge on agricultural credit, insurance and legal aspects.

6.2. What we know at the end of the session

Sources of extension and nature of services provided by following extension service providers:

- Public extension services
- Private extension services
- Institutional sources
- ICT sources
- Agricultural credit
- Agricultural insurance
- Legal aspects

Farmers' Access to Services

6.3. Important services required for farmers

- Information
- Inputs (seed, fertilizer, pesticide, machinery, etc.)
- Infrastructure (cold storage, godown, feed mixing unit, etc.)
- Market (market yard, market intelligence, transport, etc.)
- Developmental schemes/programmes
- Credit and insurance, etc.

The above services are needed with dimensions of accessibility, quality, cost effectiveness and timeliness.

Information is the critical input required for the farmers to bring about changes starting from selection of crops till the marketing.

Past experience: The most important source of information to the farmers is his/her past experience itself. However before acting upon information based on past experience, he/she needs to cross check the relevance to the present context.

Progressive farmers: are the small segment of rural life who are socially, economically and technologically advanced compared to other farmers. They go in search of advanced technologies proactively, adopt and harvest the benefit of technologies. They are the nearest and easiest source of agricultural information to other farmers. For example, contact farmers, award winning farmers, Block Farmers Advisory Committee (BFAC) members, District Farmers Advisory Committee (DFAC) members, State Farmers Advisory Committee (SFAC) members, farmers running farm school, etc.

Input Dealers: are mostly village level businessmen who sell seeds, fertilizers, pesticides and machineries to the farmers. Also provides extension advisories by their strength of proximity to the farmers. However maximum precautions has to be exercised for accessing the advisory services as they are not professionally qualified extension functionaries. However, some input dealers are trained through various programme like Diploma in Agricultural Extension Service for Input Dealers (DAESI) who can provide quality agricultural information. In other cases, advisory from local public extension functionaries and inputs based on the advisory from input dealers can be accessed.

Cooperative Societies: District Central Cooperative banks and Primary Societies

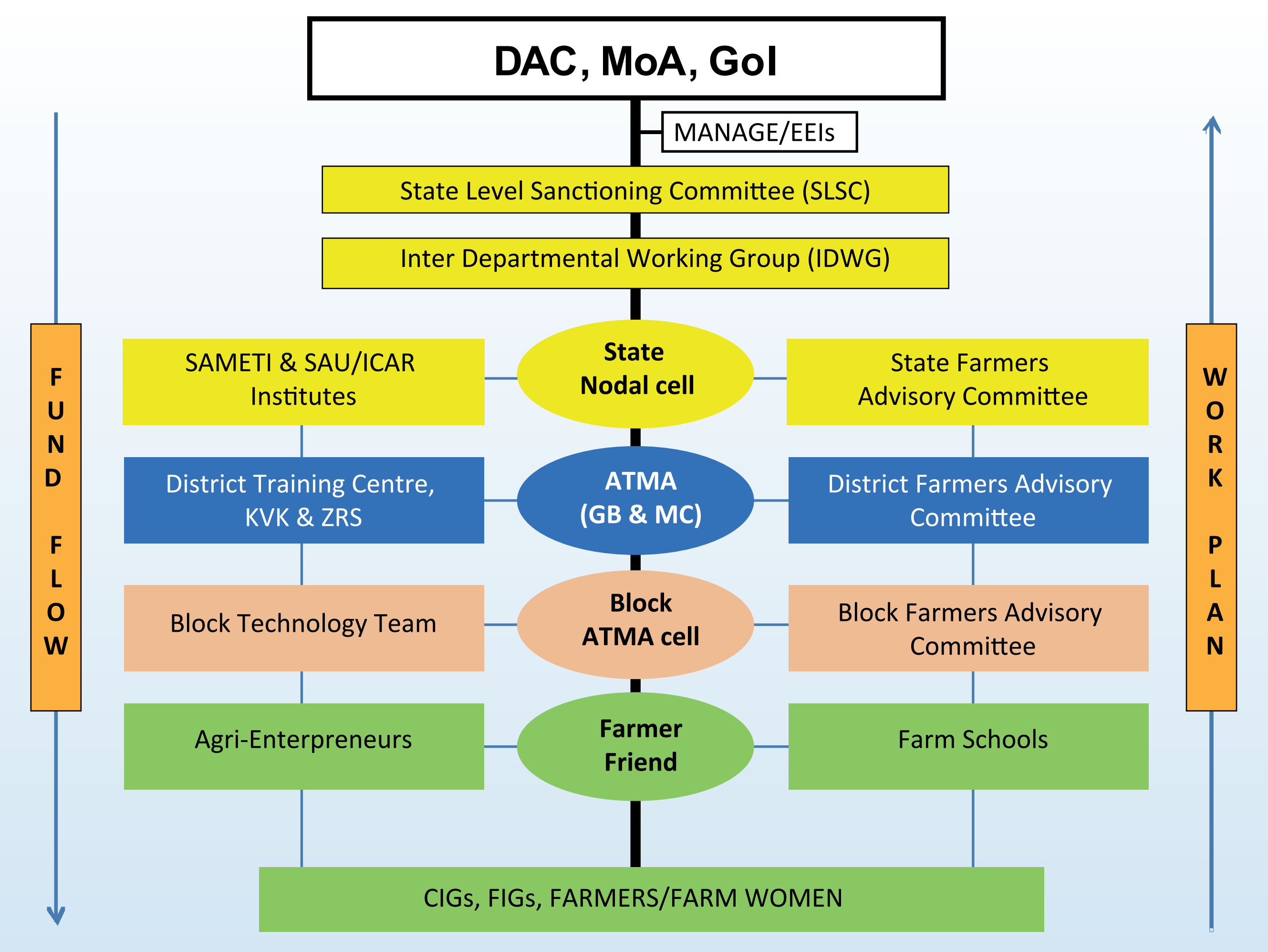
- Extend agriculture credits under priority sector
- Implement NABARD schemes
- Implement Government schemes
- Implement social security schemes
- Extends crop loans

Land (Agriculture & Rural) Development Banks

- Extend medium and long term credit to agriculture
- Implement government schemes

Public sector extension: Represented mainly by the State Agriculture and allied departments continues to be the most important source of information for the majority of farmers. Each department such as Agriculture, Horticulture, Sericulture, Animal Husbandry, Marketing, etc. have their own extension manpower and delivery mechanism to reach the farmers. The focus is on transfer of technology and undertaking agricultural development programmes of state and central Governments.

Agricultural Technology Management Agency (ATMA): An agency undertaking agricultural development programmes at district level in which agriculture and allied departments along with private sectors work together. At village level farmer friend, at block level Block Technology Manager and Subject Matter Specialists provide agricultural information and benefits of schemes/ programmes to the farmers. ATMA organize various extension activities like field visits, trainings, demonstrations, field days, farmers – Scientists interactions, exhibitions, exposure visits, campaign, etc. Publish extension information through print and electronic media, Provide alerts to farmers on agriculture operations and precautions.



Commodity Interest Groups of Farmers (CIGs): A group of farmers growing same crop/ enterprise, share common problems and benefits. Matured CIGs can collectively access information, inputs, infrastructure, credit and market linkages. Some CIGs have independent setup for research, extension, credit and marketing purpose. E.g. Amul for dairy farmers, Maha Grapes for Grape farmers.

Kisan Call Center (KCC): An online agricultural advisory service provided by Government to farmers. Farmers can access information on crops, livestock, fisheries, inputs, credit, government scheme benefits, through toll free number 1800 180 1551 OR 1551 from 6 A.M. to 10 P.M. except on Sundays and Gazetted holidays. Beyond these hours the calls are attended in the IVRS mode. Qualified professional in local language provides the advisory.

Krishi Vigyan Kendra (KVK): A district scientific organizations that works on technology generation, refinement and dissemination. KVK consists of qualified multidisciplinary experts focusing mainly on locally relevant agricultural issues. KVKs organize front line demonstrations, exposure visits, training programmes, exhibitions, field days and provide agricultural literature to the farmers. Some of the KVKs do provide input support for farmers.

Agriculture university extension system

- Maintains supportive extension service to line departments.
- Supplements efforts of line departments for service through their extension units, research stations and through teaching campuses.
- Develop innovative extension strategies.
- Provide technical knowledge to line departments.
- Organize front line demonstration of their technologies.
- Disseminate technologies through public- private partnerships.

Indian Council of Agriculture Research (ICAR)

- Generates agricultural technologies.
- Provides extension support through its research institutes and KVKs.
- Develops innovative extension strategies.
- Provides technical knowledge to line departments.
- Organizes front line demonstration of their technologies.
- Disseminate technologies through public-private partnerships.

Commodity Boards and National Institutes: Commodity Boards like Coffee Board, Rubber Board, Spice Board, Tea Board, Coconut Development Board, Tobacco Board, Silk Board, Cotton Corporation of India, National institutes like National Institute of Agricultural Extension Management (MANAGE), Central Food and Technology Research Institute (CFTRI), Defence Research and Development Organisation (DRDO), National Institute of Plant Health Management (NIPHM), National Institute of Agricultural Marketing (NIAM), National Horticulture Mission (NHM), National Horticulture Board (NHB). etc., provide extension advisory services to their respective clientele group.

International institution: International institution namely International Center for Research in Semi Arid and Tropics (ICRISAT) in Hyderabad is also serving farmers on crops in semiarid and tropics.

Agriclinics and agribusiness centers

- Agriclinics and Agribusiness Centers are advisory and business centers managed by agricultural professionals in rural areas.
- They provide client specific advisory services free/payment basis.
- Details on Agripreneurs are available at www.agriclinics.net.

Agribusiness companies

- Almost all the Agribusiness Companies provide extension advisory services to farmers in their specialized crops/inputs.
- Contract farming assures extension, input, management and market access to farmers by agribusiness companies. Credit is also provided in few cases.
- Few agribusiness companies are providing different services to farmers through farmers one stop shop concept.

Non-Governmental Organisation (NGOs)

Many NGOs are working in various aspects of agricultural development such as farm advisory, input supply, infrastructures, processing, marketing, community mobilization, micro finance, livelihood development, etc. Farmers can take the advantages of such NGOs wherever available.

Mass Media

- Rapidly expanding mass media ensures easy access to information to farmers on real time basis.
- Agricultural magazines, newspapers provide updated information on agriculture to farmers regularly.
- Community radio, radio, and television provides updated information to the farmers.
- Mobile is also widely used as channel for reaching farmers by many organizations.

Internet opens worldwide information to the doorstep of farmers. Some of the important agri portals useful to the farmers along with key information available are as follows:

- www.icar.org: Research institutes and major technologies
- www.indiaagristat.com: Agricultural related statistics
- www.isapindia.org: Query Redress Services (QRS)
- www.indiaagronet.com: Agricultural jobs, buy and sell and exhibitions
- www.agriwatch.com: Market prices of agricultural commodities
- www.indiancommodities.com: Forecasts of prices of commodities, online trade and warehousing
- \www.krishiworld.com: Multilingual portal, organic farming, crop specific information, disease and pest, market watch and home gardening
- www.agriculture-industry-india.com: Export – import directory, agro trade leads, and agro Trade events
- www.agricoop.nic.in: central sector schemes and policies
- www.apeda.com: Export and import procedure and schemes
- www.fert.nic.in: Fertiliser details
- www.mofpi.nic.in: Food processing technologies and schemes
- www.agmarknet.nic.in: Prices of commodities and trends
- www.icrisat.org: Technologies for semi arid tropics
- www.ikisan.com: Agro informatics, soft ware services and education
- www.uttamkrishi.com: Hindi website and toll free help line
- www.nafed-india.com: Cooperative marketing of agricultural produce
- www.agritech.tnau.ac.in: Technologies, special technologies, schemes and services
- www.nhm.nic.in: Horticultural technologies and schemes

6.4. Accessing financial services – sources

- Self Help Groups (Micro Finance Institutions)
- Nationalized and other private banks
- Cooperative banks and societies
- Subsidy schemes of State/Central Governments

Important types of credits and savings

Kisan Credit Card Scheme (KCC) aims at providing adequate and timely support from the banking system to the farmers for their short-term credit needs for cultivation of crops. This mainly helps farmers for purchase of inputs during the cropping season. Credit card scheme proposed to introduce flexibility to the system and improve cost efficiency.

Benefits of KCC

- Simplifies disbursement procedures.
- Removes rigidity regarding cash and kind.
- No need to apply for a loan for every crop and every season.
- Assured availability of credit at any time enabling reduced interest burden for the farmer.
- Helps to buy seeds, fertilizers at farmer's convenience and choice.
- Helps to buy on cash-avail discount from dealers.
- Credit facility for 3 years – no need for seasonal appraisal.
- Maximum credit limit based on agriculture income.
- Any number of withdrawals permitted subject to credit limit.
- Repayment only after harvest.
- Rate of interest as applicable to agriculture advance.
- Security, margin and documentation norms as applicable to agricultural advance.

How to get Kisan credit cards

- Approach nearest public sector bank and get the details.
- Eligible farmers will get a Kisan Credit Card and a passbook. It contains details like name,

address, particulars of land holding, borrowing limit, validity period, a passport size photograph of holder which may serve both as an identity card and facilitate recording of transactions on an ongoing basis.

- Borrower is required to produce the card cum pass book whenever he/she operates the account.

Banks implementing KCC

- Allahabad Bank - Kisan Credit Card (KCC)
- Andhra Bank - AB Kisan Green Card
- Bank of Baroda – BKCC
- Bank of India - Kisan Samadhan Card
- Canara Bank – KCC
- Corporation Bank – KCC
- Dena Bank - Kisan Gold Credit Card
- Oriental Bank of Commerce -Oriental Green Card (OGC)
- Punjab National bank - PNB Krishi Card
- State Bank of Hyderabad –KCC
- State Bank of India –KCC
- Syndicate Bank –SKCC
- Vijaya Bank -Vijaya Kisan Card
- Personal Accident Insurance Package" is provided to the Kisan Credit Card (KCC) holders.

Salient features of the scheme

- This scheme covers all the Kisan Credit Card Holders against death or permanent disability within the country.
- All KCC holders up to the age of 70 years are eligible.

The benefits under the scheme are as under

- Death due to accident caused by outward, violent and visible means: Rs.50,000/-
- Permanent total disability: Rs.50,000/-
- Loss of two limbs or two eyes or one limb and one eye: Rs.50,000/-
- Loss of one limb or one eye: Rs.25,000/-
- Period of Master Policy - Valid for a period of 3 years.
- **Period of Insurance** - Insurance cover will be in force for a period of one year from the date of receipt of premium from the participating Banks in cases where annual premium is paid. In case of three year cover, the period of insurance would be for three years from the date of receipt of premium.
- **Premium** - Out of the Annual premium of Rs.15/- per KCC holder, Bank has to pay Rs.10/- and Rs.5/- has to be recovered from KCC holder.
- **Claims Procedure** - In case of death, disablement claims & death due to drowning: Claim administration will be done by the designated office of the Insurance Companies. Separate procedure is to be followed.

Credit support from nationalized banks

- Extend agriculture credits under priority sector, implement NABARD Schemes, Government schemes and social security schemes
- Extend crop and agriculture investment/term loans

Extension of Bank-SHG Linkage Programme to Agricultural Purposes

- The loans extended under both the SHG linkage programme and Primary Agriculture Co - operative Societies (PACS) originate from joint efforts of people and financial institutions.
- The microfinance (through SHGs- Bank linkage) meets both consumption and production needs of the people compared to the conventional system that caters only to the production needs.

Commodity Boards Finance

- Give crop loans and subsidies
- Crop pledge loan
- Market loan

6.5. Agricultural insurance

Insurance coverage is given to farmers mainly by Agricultural Insurance Corporation Ltd (AIC) and many other private agencies. All loanee farmers automatically eligible for agricultural insurance coverage. Non loanee farmers can also avail this benefit by payment of nominal premium. Some of the important agricultural insurance schemes available at present are as follows.

- National Agricultural Insurance Scheme
- WBCIS - Weather Based Crop Insurance Scheme
- MNAIS - Modified National Agricultural Insurance Scheme
- RISC - Rainfall Insurance Scheme for Coffee

(Coffee Insurance)
- Rubber Insurance
- Coconut Insurance
- Varsha Bima/Rainfall Insurance
- Rabi Weather Insurance
- Wheat Insurance (Weather & Biomass)
- Potato Insurance
- Bio-Fuel Tree/Plant Insurance
- Pulpwood Tree Insurance
- Cardamom Plant & Yield Insurance

6.6 . Legal aspects

It is important for farmers to know the legal aspects of important inputs namely seeds, fertilizers, pesticides, etc.

The Seeds Act 1966 deals with regulations related to production, certification, quality control, sales, seed analysis, seed inspection, export and import, penalty, exemptions and amendment. The details are available under **http://agricoop.nic.in/seedsact.htm**

The Fertiliser (Control) order 1985 deals with regulations related to fertilizer Price control, control on distribution, registration of dealers, manufacture of fertilizers, mixtures, restrictions on manufacturing, import, sale, enforcement authorities, analysis of samples, specifications and penalty. The details are available under **www.agricoop.nic.in/sublegi/FertilizerControlOrder.htm**

Insecticides Act, 1968: An act to regulate the import, manufacture, sale, transport, distribution and use of insecticides with a view to prevent risk to human beings or animals and for matters connected. The details are available under **http://cibrc.nic.in/insecticides_act.htm**

For more information, farmers may contact nearest banks, agricultural department or Kisan Call Centers.

5.12. Lessons Learnt

1. Important extension sources to farmers are progressive farmers, input dealers, Cooperative Society, ATMA, Kissan Call Center (Toll Free No. 1800 180 1551), Krishi Vigyan Kendra, Agricultural Universities, ICAR Institutions, Commodity Boards, National Institutes, Interantional Institutes, Agriclinics and Agribusiness Centers, NGOs, Radio, TV, Newspapers, Agricultural Magazines and agricultural related websites.
2. Important sources of finance to farmers are banks, cooperatives and SHGs.
3. Kissan Credit Cards provides adequate and timely financial support from the banking system to the farmers.
4. Non loanee farmers are advised to take the benefits of agricultural insurance coverage by paying nominal premium, maintain close liaison with extension, credit and insurance providers regularly.

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Food and Agriculture Organization of the United Nations

Crop production manual

A guide to fruit and vegetable production in the Federated States of Micronesia



Crop production manual

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Compiled by:
Sayed Mohammad Naim Khalid

This manual was produced under TCP/MIC/3601, "Strengthening the capacity of Farmers Associations to increase production and marketing of root crops, fruits and vegetables in FSM" project.

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Contents

Preface

This manual is intended to be a guide to the agriculture extension personnel, horticultural producers as well as agricultural institutions in their endeavor to improve productivity in the horticultural sub-sector. This manual will raise awareness and provide decision support information about opportunities at farm and how to do basic farming for subsistence and market and ultimately increase the income of farmers and improve their livelihoods.

Pohnpei and Yap States currently produce limited amount of food locally and export is also limited. The imports of substantial quantities of vegetables, fruits and root crops is amounting to millions of dollars annually. This is partly owing to the fact that the necessary information on crop production locally is not readily available to assist producers in their production.

This manual is designed to provide guidelines on seedling production and management, plant spacing, cropping program, soil fertility and crop protection. It is hoped that this manual will contribute modestly to the goal of improving vegetable production in the country.

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- Mr. Jackson Phillip – Program Coordinator, College of Micronesia FSM
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- Dr. Leon Numaco – Horticulture Consultant, FAO
- Mr. Gumercindo Tumbali – Farm Management & Marketing Consultant, FAO
- Mr. Kadalino Lorens – Agriculturist, Division of Agriculture, Department of R&D Pohnpei State, FSM
- Mr. Santos, Gibson – Soil Conservation Technician, NRCS/USDA, FSM
- Mr. Nicholson Solomon – Director, Department of Resources & Development, Pohnpei, FSM

Farming basics

Farming basics

Introduction
With so much emphasis today on eating natural, organic foods – not to mention the rising costs associated with buying them, many people are considering growing their own vegetables in a home garden. It's very easy to grow your own vegetables, and at harvest time, vegetable gardening is a very rewarding pastime.
All your garden really needs is sunshine (6–8 hours of sun each day, in the summer), some soil, fertilizer, and a little attention to watering, weeding and pest management.
As a gardening beginner, your first vegetable garden will require the largest amount of work, but don't let that dissuade you – the work you put in up front won't have to be repeated next year, and the rewards speak for themselves.

Why start your own farm?
Across the world, many people are having little farms. In FSM most people have access to land and the climate is suitable for crop production. And for the cost of seeds, tools, and water, they're enjoying food that's as fresh and local as it can possibly be, all while spending less time shopping for groceries and more time at home, connecting with their family and friends.

As you can see, there are many great reasons to start your own farm. Apart from allowing you to grow your own food, conserving energy and saving money, own farm will also help to control runoff from rainfall and guard against flooding.
A farm can act as a sound buffer, lowering noise levels in your neighborhood. Your farm will attract a variety of local and migratory wildlife like birds and butterflies, not to mention helping to conserve important creatures like bees – who are vital to our ecosystem – and helping to reduce CO_2 in our atmosphere, replacing it with oxygen. Additionally, a farm is a great space for both children and adults to experience the outdoors, and get off the sofa. A garden can create ground shade, helping with cooling, and can work as a filter for dust and pollutants.

Picking a good spot for your garden
Now that you've decided to start your farm, the first step is to find the perfect location. There are several things you should keep in mind:
Start Small. As we mentioned, your first farm is typically the most work to set up, so starting small is a good idea. Set up a manageable area for your garden bed(s), whether they are in the ground, raised beds, or in containers, and know what you're

going to plant (and the space requirements of those plants) ahead of time.

Sunlight. Sunlight is the most important thing for any farm. Vegetables require at least six hours of sun each day – and if you can get 8 hours, that's better. Don't worry too much about afternoon or morning sun, as long as the garden gets at least six hours total, you're golden.
Good Soil. As farmers, we know that good soil means good plants! When looking at the soil in your own ground, darker tends to be better, but just about any soil is workable – as long as it's not full of rocks, roots, or other obstructions. Even mediocre soil can be improved with minerals, manure, and sand to make it excellent. And, if all else fails, a raised bed with formulated soil is always an option.

Water source. Apart from sun and soil, water is the third piece of the puzzle to make a healthy garden. Choose a location close to a water source – like river, rain water collection, a hose, well, or other hydrant where you can easily make sure the garden stays watered – and while we're at it, plants don't grow in totally soaked soil, so avoid areas that tend to collect rainwater. Higher ground is best!

Wind shelter. This one is especially important on a rooftop, or out in an open plain. Find natural windbreakers in the surrounding area or consider building a fence to protect your plants from the wind to minimize damage from wind exposure.

Choosing what crops to plant

Now that you have a location, it's time to decide what to plant. You'll want to maximize your space and grow plants suited to your local area, as well as considering the time vs. reward – corn, for example, is delicious, but takes up lots of space and takes months to get a single harvest, while pole beans are quire space-efficient and produce beans for weeks.

Easy to grow. For the beginning, having vegetables that are basically

Figure 1: seedling production in trays

foolproof is a great plan. Plants like Chinese cabbage, tomatoes, gourds like squash and eggplant, greens such as lettuce or Swiss chard, and root vegetables like taro tend to be very simple.

Suitable for your region. When planning your farm, pay a visit to your farmer's market and find out what's already grown locally. This will help you to ensure you aren't fighting an uphill battle in getting a plant that isn't exactly suited to your area to thrive – plus, by growing what other farmers are growing, you can always go to those folks for help if needed.

Companion planting. Companion planting can help ensure that all your plants thrive by assisting with pest control. By planting companions next to each other, the idea is that we want all the natural pests attracted to the first plant to be repelled by the second, and vice-versa.
For example, onion pairs well with tomatoes and leafy greens like lettuce, but if planted next to beans or peas you may run into pest troubles.

Finding seeds/seedlings
Buying/receiving seeds can be pretty overwhelming for farmers because of the sheer volume of different companies and varieties out there. The best advice is, "Read the label" or, "Ask your extension officer". This will let you know how difficult the plant is to grow, when to plant, and what's required.

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Figure 1: Seeding production in trays

Planning your farm
Farmers usually make mistake here. First time farms can easily get out of hand, or unwieldy, or fall victim to neglect, wind, waterlogging, or pests. To prevent your farm experience from being a disaster, take some time to plan the farm.

Layout
No matter what kind of layout you end up using for your farm, the first thing to think about is location. Choose a basically level area in higher ground if possible, for good drainage. Take in mind that sun is a concern (6–8 hours) as well as protection from wind.

Raised beds

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Figure 2: An example of raised bed

Raised bed gardening has become one of the most popular way of growing vegetables today. The way they allow you to control space, soil, water levels, and location make them ideal for many gardeners.
If you go with a raised bed layout, construct beds that are no more than 3–4 feet wide, with space for walking paths in between. This will ensure

that you're able to properly attend to – and harvest – every plant.

Traditional rows
For those who have the space, traditional row gardening can be a great option. Planting vegetables in long rows allows you to increase the space between plants, improving air flow around them and giving them more space to grow.

Here's a rule of thumb for row gardening – try to plant the taller plants toward the west, and shorter toward the east. That way your shorter plants don't spend those vital sunshine hours in the shade of the taller ones.

Square foot gardening
For beginner farmers, square foot gardens often tend to be a great way to begin. Larger gardens are prone to weeds and can be overwhelming for the novice. Square foot gardening is similar to raised bed gardening, but it is further subdivided into individual square-foot plots.

Mulching
For farmers, the purpose of mulch is fairly straightforward: mulch is like a barrier keeping sunlight and ambient air out of the soil. Mulching allows soil to stay cooler, meaning less stress on your plant roots from heat, and in cooler months.

On the other hand, it does create cool, dark spaces that are the favored environments of Giant African snails, slugs, earwigs, and other undesirables – so only use a thin layer. Mulch should not touch the stem of plants.

When to plant
When considering the question of when to plant, you'll have to go back to the question of *what* to plant. Remember, reading all those labels on your seed packets will ensure that you're planting things at the right times.
Choose seeds that make good

© FAO/ S M. Naim Khalid

Figure 3: Production in furrows and rows

companions, and thrive at similar intervals for the best beginner gardening experience.

Figure 4: An example of mulch with straw

If you want to think ahead, consider growing something that thrives in heat and can be harvested early – like Chinese cabbage, and then replacing them as the weather cools with a cool-weather plant like broccoli. Of course, if you'd rather simply do one planting and one harvest, especially as a beginner, that's just fine too.

Improving the soil

Most soils are entirely capable of growing a vegetable garden – and even mediocre soil can be improved. For a really successful garden, some soil improvements is a must. Let's take a look at some of the ways you can improve your soil.

Testing the soil

Now, brace yourself, because we're going to talk a little chemistry. Don't panic, you don't need to be a scientist. So, as you may remember from chemistry or biology class, pH is a measure of the acidity of a substance – remember, 7 is neutral, and a pH lower than 7 is acidic, while a pH above 7 is basic.

Soil is no exception, and it has a pH value, and unsurprisingly, plants tend to like neutral soil. For more information and help please contact the extension agents and local agriculture department.

Clearing weeds

The amount of time you'll spend worrying about weeds, in a sense, depends on which style garden you use. While square-foot gardens and raised beds using controlled soils can remain fairly weed-free, a traditional row garden can inherit weeds from plant life that inhabited the soil before it was tilled.

A good way to prevent weeds from starting is to smother them. After clearing the ground for your garden, just leave the clippings of the grass and weeds that were there previously, and cover the area with newspaper.

Keep the newspaper damp by watering it each day, and walk over it to keep everything pressed down. This will discourage the more stubborn weeds from re-growing.

As the plant matter from your clippings begins to decay, earthworms will be attracted, and they will aerate the soil and add nutrients – a bonus!

Compost

Compost is a great way to create extremely nutrient-rich soil, and it couldn't be easier to make. Just create some layers of organic material – dry leaves, kitchen veggie scraps, shredded paper, plant clippings, and just a little bit of soil.
Over time, this will turn into something called humus, and it's excellent for building soil. Further details on how to make compost is presented in ***Annex 1.***

Manure

In addition to composting, plenty of fertilizers and other synthetic soil enrichment methods are around to help improve soil. However, for many gardeners, nothing is better than the tried-and-true method of manure.
In comparison, manure contains a little less nutrient content than the sparkly synthetic fertilizers, but nothing is better at providing carbon and carbon compounds. This is organic material that helps build and fortify the structure of soil, and that's something no fertilizer can do.

Planting the crops

Now that you have your garden laid out, your soil prepared, and your seeds at the ready, it's time to get to the good stuff: planting your garden. Planting time is the second most exciting time to be a gardener (the first is, of course, harvest time) – this is when you can finally put all the hard (but worthwhile) work of building your first garden behind you, and really begin the enjoyable process of growing your own food. Plants can be cultivated by direct seeding or transplanting.

How to plant the seeds

The first step to actually utilizing your garden to grow delicious vegetables is to plant your seeds. If you ask ten gardeners how they plant their seeds, you'll likely get ten different answers, as everyone develops their own routine. But, as a general guide, here's our preferred method of starting seeds for beginners. Seeds can be directly planted in soil or grown in nursery and then transplanted to prepared beds.
Begin by marking out rows with stakes on either side of the garden, and stretching string between them. Alternatively, you can use a long, straight piece of wood as a guide. Whichever method you use, use a hoe or trowel along your guide to cut a furrow into the soil. It should look like a V-shaped indentation a few inches deep – refer to your seed packets for more specific depth requirements. Now that you have your row defined, you can distribute the seeds along the row, taking care to distribute them

more or less evenly in small groups, or individually for larger seeds. Don't be afraid to use additional seeds. This will help cut down on loss should some of your seeds fail to germinate. Finally, once your row has been populated with seeds, use your trowel or hoe to pull fine soil over the seeds – take care to avoid large clods of soil and rocks. Finally, give the soil some taps with your tool of choice to firm it and ensure that moisture in the soil has good contact with the seeds, and to help the soil to retain moisture.

Once you've planted and covered your seeds, use a gentle shower from your garden hose to water them – be gentle here, you don't want to disturb the seeds or the soil. Keep up the practice of making sure these seeds are watered to ensure healthy germination until you see sprouts.

Once the seeds have sprouted, look for them to develop into seedlings. They will start to sprout leaves, and in the case of multiple plants, wait for one of the plants to develop three well defined sets of leaves.

Once the third set has developed, then do a little thinning – you'll want to thin out the smaller, weaker plants so that only the stronger ones are left. This ensures that the smaller plants aren't stealing moisture and nutrients from the champions of the batch.

It's best, and easiest, to thin while these seedling are small, so that pulling them up doesn't disturb the roots of the stronger plants.

Succession planting

As a gardener, you may wish to simply garden for a single season: plant, tend, harvest, and then leave the garden beds be until next summer. That's totally fine, but if you'd like a garden that really works for you year-round, consider succession planting. Succession planting isn't difficult, and it will truly maximize the rewards you get from your home vegetable garden. By using succession planting, you could double or triple the volume of fresh vegetables you harvest from your garden.

By continuously planting and harvesting each season in spring, Summer, and Fall, with weather-appropriate plants, you can turn your garden into a very efficient food source.

Crop rotation

Those of us who have been gardening for years all have a story about a bumper crop one year leading to a minuscule harvest the next.

Bountiful plants followed by punier versions is a very common tale among gardeners, and the culprit isn't the validity of your (or our) green thumb – the answer to fighting this boom-and-slump cycle is crop rotation.

Crop rotation is, essentially, an extension of the work you did in initially laying out your garden, and deciding where to plant what.

Moving the plant groups that you have placed together to new locations – and new soil – every year is something that some gardeners don't do, but it can be a huge improvement to nearly any garden, in several ways.
First of all, rotating crops year after year helps to keep that soil that we spent so much time worrying about healthy and fertile. By planting the same plants in the same soil each year, those same plants are going to drain the same nutrients – in the same way – year after year, and over time that soil will become less effective.
Additionally, rotating your plant groups helps to mitigate soil diseases and pests mainly nematodes.
What are those? These are conditions like verticillium wilt and rootworms that prefer certain types of plants – so planting the same thing in the same location each year helps them set up a permanent foothold.

It is recommended that crop locations are rotated to a different plant family after each planting over two years.
Some examples are as follow:
Night Shade: Tomato, Eggplant, Pepper
Morning Glory: Sweet Potato
Legumes-Bean: Long Beans, Wing Beans

Maintaining the garden

Once you've planted your seeds, it's time to get to maintaining those plants. Garden maintenance isn't difficult, it just requires a little attention every day. This is what, for many of us, makes gardening so enjoyable.
It gets us out of the house and into the warm sun each day, carefully watching and tending to the plants as we see the fruits – sometimes literally – of our labor thrive.

Watering

When, how often, and how much water to give a garden is one of the most common questions beginner gardeners ask. Watering your garden can be as simple as watching rain falls, or as complex as installing automatic irrigation systems. Finding the happy place somewhere in between is where most beginner should flourish.
In terms of how much water your garden needs, that really depends on many factors. The plants, your location, the type of soil you have – all of these are factors in how much you'll need to water your garden.
In terms of soil, sandy soils tend to (of course) hold less water than heavier dirt or clay soils. So, consider how long it takes that soil to dry out.
This is another reason that soil building with compost or manure is highly recommended – this creates healthy soil that drains rainwater sufficiently while retaining just enough water for plants.
As a rule of thumb, your garden soil should be just moist. If the weather is rainy, you may not need to add any

water at all. If it’s hot and dry, then you’ll want to bring out the hose. Watering in the morning tends to be best, so that evaporation is reduced, but watering in the afternoon is okay, too. If you water later in the day, try not to let the plants get too wet – this can encourage fungal growth if the dampness persists into cool evenings.

Weed control

Weeds compete for sunlight, water and nutrient.

If you’ve taken steps to prevent weeds when you created your garden, weed control on an ongoing basis should be fairly manageable – particularly if you’re using raised beds or square-foot gardening methods, which tend to not inherit weeds.

Weeds can surprise you – even in a weed-free patch of soil, you may find that once you plant, you start seeing weed growth, too.

This is because the seeds for those weeds can remain deep under the soil, where they don’t get enough sun or moisture to grow, until they’re dredged up toward the surface by your tiller, hoe, or trowel. For this reason, you should keep an eye on your garden for weed growth.

When you do see weeds popping up, don’t wait for them to become a bigger problem. Young weeds are much easier to deal with than older ones. Once you’ve pulled them, just toss them on the ground – as they decay, they will help add nutrients to your soil.

Pest control

The best thing you can do for pest control is companion planting, as discussed earlier. Another good practice is to plant flowers in addition to your vegetables – this will attract bees and other insects, which will act as natural pest deterrents.

Still, even with plenty of preparation, sometimes pests like aphids, corn earworm, leaf footed bug, mites, scale and others will appear. The best pest control method depends on your plants, and the type of pests you have. Few natural pesticides that you can make at home are presented in ***Annex 2.***

For advice on pest management contact your Extension Agent

Fertilizer

So, you’ve composted and mulched and manured – your soil is in tip-top shape. But, as we’ve mentioned, even manure doesn’t have the nutrient content of a good fertilizer, so it may be good to consider using some (especially if you don’t use manure). There are 3 major kinds of fertilizer:

- **Organic fertilizers** are just what they sound like: they’re made from natural materials, and have to be broken down naturally by microorganisms in your soil, resulting in a slow, gradual release of nutrients.

- **Water-soluble fertilizers** are mixed with water and **sprayed** onto soil and plants as a liquid, rapidly feeding nutrients to the plants.
- **Synthetic fertilizers** are nutrient-rich chemical compounds that attempt to emulate the effects of organic fertilizers on a more rapid, and more nutrient-rich, basis. Plants get food from the ground, sunlight, air and water. For best growth, plants often need more food than is in the soil, especially after the first crop. They will need another source of food from fertilizer or compost.

Harvesting the crops

This is the time you've been waiting for! All throughout the season you've weeded, you've watered, you've nurtured and you've fertilized. Now those veggies are ripe, and it's harvest time. Here are some tips for harvest time to help you make the most of your home garden.

First of all, resist the urge to harvest all in one day. Vegetables ripen at different times, and some plants produce harvestable vegetables continuously for weeks.

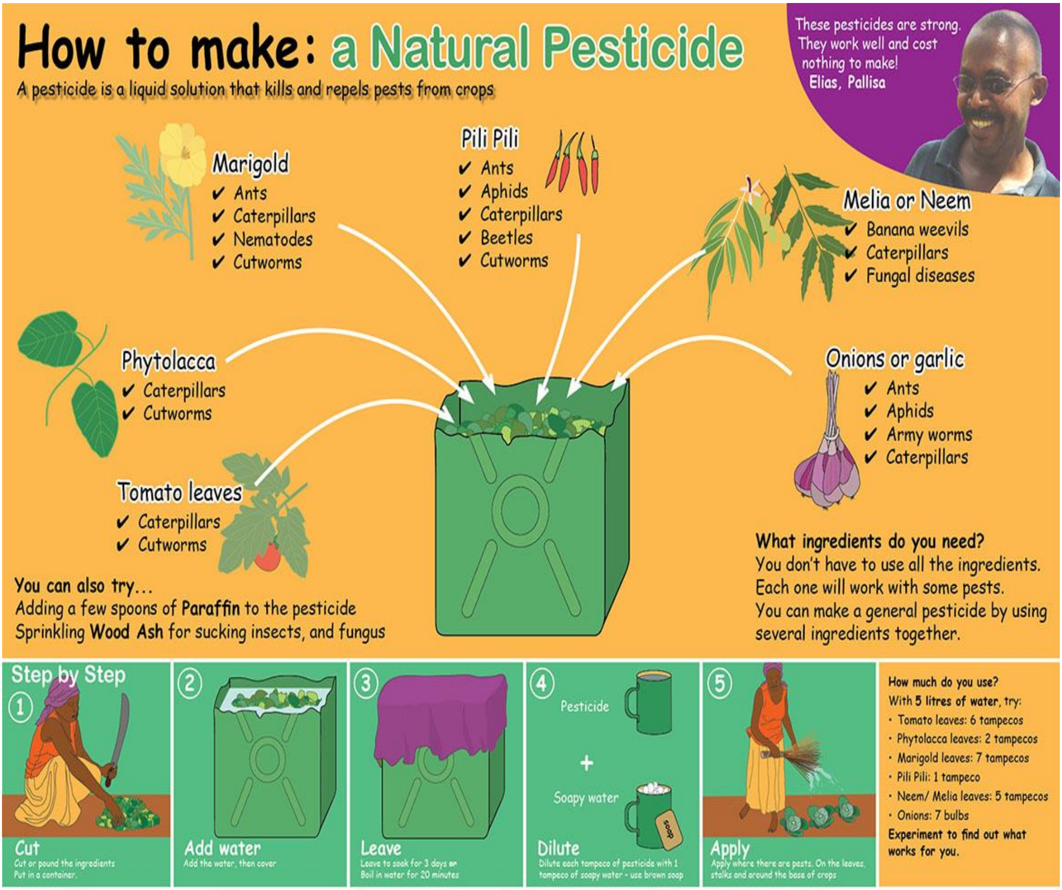
Make harvesting part of your daily garden routine – harvest veggies as they ripen. This ensures that you get a constant supply of fresh, ripe veggies, and it encourages production in the plant (since it isn't wasting energy and nutrients on a vegetable that's already ripe anymore).

Always harvest fruits and vegetable when they are ripe do not allow over ripping or rotting, over ripe fruits and vegetable can become breeding places for pests (e.g. fruit flies) Do not allow rotten or bad produce on the ground again this will great breeding places for pests.

For leafy vegetables, cut through the whole stalk at an angle – harvest these in the cool morning time if you can. For summer squash, you can harvest by holding the vine in one hand while holding the squash in your other hand. A gentle pull is all it takes. In the case of winter squash, cut the vines about 3 inches from the squash.

For root vegetables like onions, carrots, and potatoes, loosen the soil around them with a garden fork before pulling – you don't want to damage the veggies by trying to pull them through solid soil! If you can, try to leverage the veggies out with the fork after loosening the soil, rather than by pulling alone

Organic pest control



Root crops production

Sweet potato (*Ipomoea batatas*)

© FAO/ S. Sherzard

Soil:

- Can be grown in loamy soil with a pH range of 5.6–6.6

Season and planting:

- All year round
- Plant the terminal vine cuttings (80 000/ha) at 20 cm spacing
- The cuttings should be 10–15 cm in length with 2–3 nodes and to be collected from matured vines aged 3 months and above

Preparation of field:

- Plough the field to fine tilth. The soil depth should be at least 30 cm. Form ridges and furrows 60 cm apart or beds

Irrigation:

- Irrigate before planting, on 3rd day and then after once a week
- Stop irrigation one week before harvest
- No irrigation needed if soil is moist

After cultivation:

- The field should be kept clean by hand weeding till vines are fully developed
- Earth up the field on 25th, 50th and 75th day after planting

Pest management

- Remove previous sweet potato crop residues and alternate host i.e., Ipomoea sp. And destroy them
- Use pest free planting materials

Harvest

- Immediately after maturity and destroy the crop residues

Yield:

- 20–25 t/ha of tubers in 110–120 days

Swamp taro (*Cyrtosperma merkusii*)

© FAO

Value:

- The Swamp Taro is one of the few subsistence crops that grows well on atolls
- The main product is the corm
- The young leaves and inflorescences can be eaten as vegetables and the petioles yield a fibre suitable for weaving
- The big leaves are used as a food wrapper and also to cover the earth oven

Propagation

- This species is propagated using setts, which are suckers, the top of the corm with about 30 cm
- (12 in) of petiole, or cormlets, which are young, immature corms produced by a more mature plant

Cultivation

- Grown in fresh water and coastal swamps
- Grown in purpose-built swamp pits in low-lying coral atolls

Climate

- It needs a constant supply of fresh water & organic material e.g. leaves and can be grown in slightly brackish water, thrives in freshwater swamps and can even be found in swiftly flowing rivers and streams
- Mean annual temperature ranges from 230 °C to 310 °C

Food value

- High concentrations of iron, zinc, and calcium and the yellow corm varieties are usually higher in Beta-carotene

Cassava (*Manihot esculenta*)

© FAO/ K. Hardfield

Recommended varieties

- Beqa
- Yapia Damu
- Vulatolu

Seed rate

- Mechanised: 20 000 cuttings/ha.
- Traditional: 20 000–30 000 cuttings per ha

Spacing

- Ridges-1 m between rows
- Plants within rows 50 cm
- Cutting 30 cm in length
- Mounds: 0.5 m/in diameter
- Traditional-1 m x 1 m

Weed management

- Hand Weeding

Harvest yield

- Early Varieties: Mature in 8–10 months.
- Late Varieties: Mature in 12 months

Yield

- 20–30 tonnes/ha

Food value

- Source of Vitamin A & Vitamin C

Taro (*Colocasia esculenta*)

© FAO/ K. Hadfield

Recommended varieties:

- Sawa Toantoal, Pasdora, Sawhn korsae, Sawa Pwetepwet, kuat, Sawa mwahng, Sawa Alahl, Sawa likodopw

Seed rate:

- Traditional Farming System: 10 000 suckers/ha
- Mechanize System: 16 660 suckers/ha

Planting time:

- July to January
- Off Season: March to June
- Wet Zone: Throughout the year
- Intermediate Zone: Sept to March

Spacing:

Traditional System:

- Between rows: 1 m
- Plants within rows: 1 m

Mechanize Systems:

- Between rows: 1 m
- Plants within rows: 60 cm

Fertilizer:

- Poultry Manure: 10 tonnes/ha. Broadcast and mix well with soil before planting

Weed management:

- Hand weeding

Disease management:

- Corm Rot: Improve drainage & Have soil test for Nutrient status
- Shot Hole Spot: A seasonal disease; which disappears when the weather changes
- Remove infected leaves & burn.
- Good husbandry: - regular weeding and timely fertilizer application minimize loss

Harvest yield

- Harvesting at 6–7 months for Hybrid
- Varieties whilst traditional varieties are ready at 9 months for harvest

Yield:

- 20–25 tonnes/ha

Food value:

- Contains large amount of
- Vitamin A, Vitamin B1, Vitamin B2 and Vitamin

Arrowleaf elephant (*Xanthosoma saggitifolium*)

© FAO

Cropping season:

- All year around

Recommended varieties

- Vula
- Dravuloa

Seed rate:

- 10 000 plants/ha

Planting time:

- January to December

Spacing:

- Between rows: 1 m
- Plants within rows: 1 m
- Planting depth: 30 cm

Weed management:

- Hand Weed.

Harvest

- Harvest 12 months after planting

Yield

- 15–20 tonnes/ha.

Food value:

- Source of Vitamin B, Vitamin C, Starch and Protein

Ginger (*Zingiber officinale*)

Recommended varieties:

- White Ginger
- Red Ginger

Seed rate:

- Immature: 7 500 kg
- Mature: 5 000 kg
- Planting Time: September

Spacing:

Slope Land:

- Immature: 60 cm between rows & 15 cm within rows
- Mature: 60 cm between rows & 20 cm within rows

Flatland:

- Immature: 90 cm between rows, 15 cm within rows
- Mature: 90 cm between rows & 20 cm within rows

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 4 weeks before planting

Disease management:

- Remove infected plants
- ensure good drainage system

Insect management:

- Hot water treatment of Planting materials at 51 °C for 10 minutes
- Crop rotation with Cassava & Dalo
- Sanitation
- Proper selection of seed materials

Harvest

- Immature: Harvest at 5 months from planting
- Mature: 10 months from planting

Yield:

- Immature: 20–25 tonnes/ha
- Mature: 25–30 tonnes/ha

Food value:

- Good source of Energy, Potassium,
- Calcium & Sodium

Turmeric (*Curcuma longa*)

Cropping season:

- August to November

Recommended varieties:

- White
- Yellow

Seed rate:

- 10 to 12 tonnes/ha

Planting time:

- September to October

Spacing:

- Between rows: 60 cm
- Plants within rows: 40 cm

Germination:

- Require free drainage

Weed management:

- Hand weed or hoe regularly

Disease management:

- Hot water treatment of planting material at 51°C for 10 min
- Crop rotation with cassava & Dalo
- Sanitation-remove all rhizomes from the field after harvesting
- Proper selection of seed-choose healthy seeds

Harvest

- 10 months after planting

Yield

- 15 to 25 tonnes

Food value

- (Powder) Dietary fibre, Potassium, Iron, (very high), Calcium, Calories, Magnesium, Vitamin C, Thiamin, Riboflavin, Niacin
- White turmeric has medicinal value

Yam (*Dioscorea alata*)

© FAO/ K. Hadfield

Recommended varieties:

- Abya, Gabrach, Wonbey, Rowal, Nagbchey, Surney, Pohnape, Alien, Gamed, Tenmen, Dugyeb, Defrow, Yu Ban, Chugum

Seed rate:

- Ridges: 4.2 tonnes/ha (16 670 setts/ha)
- Mounds: 3.1 tonnes/ha (12 500 mounds/ha)

Planting time:

- Early Varieties: Between October–December
- Late Varieties: Between Jan–March

Spacing:

- Plant Spacing: Ridges–1 m between ridges
- & 0.6 m within ridges
- Mounds: 1 m between mounds and 0.8 m within mounds

Fertilizer:

- Poultry Manure: 10 tonnes/ha.
- Broadcast and mix well with soil before planting

Weed management:

- Hand weeding

Disease management

- Rotate with non-host plants like Cassava, Vegetables (Cabbage, Lettuce)

Insect management

- Use clean material
- Practice Crop rotation

Harvest

- October to March Harvest Index is when leaves start to senescence & are falling off

Yield:

- 15 to 20 tonnes/ha

Food value:

- Fibre, Potassium, Modest amount of Vitamin B1, Vitamin C & fair amount of Iron

Vegetable production

Capsicum (*Capsicum grossum*)

Recommended varieties:

- Yolo Wonder A
- Yolo Wonder B
- Blue Star

Seed rate:

- 300 grams/ha

Planting time:

- Cool season (April to Sept). Can be grown all year around under green house

Land preparation:

- Field should be well prepared, 2 ploughing & 2 harrowing is recommended

Spacing:

- Between rows: 65 cm–75 cm
- Plants within rows: 30–40 cm

Germination:

- 6 to 10 days after sowing

Transplanting

- Transplanting during cloudy days or late in the afternoon
- Seedlings raised in seed trays can be planted any time of the day
- Water the plants after transplanting and continue afterwards

Fertilizer

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with 2 weeks before planting

Weed management:

- Practice inter-row cultivation, hoeing or hand weeding

Disease management

- Use resistant varieties,
- Uproot affected plants & pack in bags, bury and burn. Practice crop rotation of non-host plants
- Destroy affected plants and control Mites & Aphids. Use clean planting materials.
- Avoid planting during wet weather. Remove all infected plants. Avoid damaging the crop during weeding. Use disease free seedling

Harvest

- Fruits are ready for harvest at 3 months after planting and picking continues for 2–3 months

Yield:

- 8–12 tonnes/ha

Food value:

- A rich source of Vitamin A and Vitamin C

Chillies (*Capsicum annuum*)

Recommended varieties:

- Red Fire - Long Red Cayenne
- Bird's Eye
- Bongo Chilly

Seed rate:

- 300 grams/ha

Planting time:

- Best to plant during hot weather season from
- September to February. Can be planted all year around

Planting methods:

- Plant seeds in seedbeds or seedling trays
- transplant to field at 3 leaf stage

Spacing:

- Between rows: 75 cm
- Plants within rows: 30 cm

Germination:

- 5 to 10 days after sowing

Transplanting:

- Transplanting during cloudy days or late in the afternoon
- Water the plants after transplanting and continue afterwards

Fertilizer:

- Poultry Manure: 10 tonnes/ha Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Practice inter-row cultivation

Harvest

- Fruits appear 90–120 days after planting and harvest weekly for one year

Yield:

- Fresh 16 tonnes/ha, Dried 4–6 tonnes/ha

Food value:

- Dried - Dietary Fibre, (Very High)
- Calcium; Vitamin A, Riboflavin and Niaci

Chinese cabbage (*Brassica chinensis*)

© FAO/ K. Hadfield

Recommended varieties:

- Pak Choy
- Kwang Moon
- Wong Bok

Seed rate:

- 300 grams/ha

Planting time:

- Cool season but can be grown throughout the year

Planting methods:

- Sow seeds in prepared seedbeds or seedling trays
- Transplant at 3 leaf stage

Spacing:

- Row to row: 50–75 cm
- within rows: 30 cm

Germination:

- 4 to 6 days after sowing

Fertilizer:

- Poultry Manure: 5 tonnes/ha Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Practice manual weed control

Disease management:

- Crop rotation
- Remove and destroy diseased plants by either burning or burying, as soon as symptoms appear
- Select only healthy planting material

Harvest

- Usually takes 30–60 days to get ready depending on variety

Yield:

- Fresh 12 tonnes/ha

Food value:

- Source of Vitamin A, Vitamin B & Vitamin C

Corriander (Coriandrum sativum)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Recommended varieties:

- Small round seeded
- Large oblong seeded

Seed rate:

- 10 kg/ha

Planting time:

- All year round but better in April to August

Spacing:

- Between rows: 22.5 to 30 cm
- Plants within rows: 4–6 cm

Germination:

- 6 to 10 days after sowing

Fertilizer/manure:

- Poultry manure: 5 tonnes/ha
- Broadcast, mix well into the soil before planting

Weed management:

- Hand weed or hoe when necessary.
- Carry out inter-row cultivation

Disease management:

- Maintain good drainage to prevent root rot

Insect management:

- Generally it is pest free

Harvest

- Regular harvest when the plants are 15–20 cm about the ground

Yield:

- 6 to 8 tonnes/ ha. Use in flavoring, curries and soup

Food value:

- (Leaves) Calories, Protein,
- Iron, Vitamin A, Thiamin, Riboflavin, Niacin, very high in Vitamin C, Potassium, Calcium, Magnesium. Nutritionally a good source but the quantities eaten are too small to be significant

Cucumber (*Cucumis sativus*)

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Recommended varieties:

- Early Set
- Cascade
- Bountiful No. 2
- Space Master
- Early Perfection

Seed rate:

- 2 kg/ha

Planting time:

- All year around, fruits best during cool season

Planting methods:

- Seeds are sown directly into well cultivated soil

Spacing:

- Between rows : 1 m
- Plants within rows: 30 cm (trellising) – 50 cm (ground creeping)

Germination:

- 5 to 7 days after sowing

Fertilizer:

- Poultry Manure: 5 tonnes/ha Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Hand weeding or hoeing is necessary
- Weeds are removed when plants are still standing
- Inter-row cultivation using horse drawn scarifies can be used

Disease management:

- Use healthy seeds of resistant varieties
- Avoid planting at high density

Harvest:

- Harvest at 50–60 days after planting, continue picking of fruits for 3 weeks

Yield:

- Fresh 12–15 tonnes per hectare

Food value:

- Vitamin C

Eggplant (*Solanum melongena*)

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Recommended varieties:

- Chahat
- Pritam/Long Purple

Seed rate:

- 300 grams/ha

Planting time:

- All year around but best the hot and wet season

Spacing:

- Between rows: 1.5 m
- Plants within rows: 50 cm

Germination:

- 5 to 10 days after sowing

Transplanting:

- Transplant at 3 leaf stage
- Seedlings raised in seed trays can be planted any time of the day

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Inter row cultivation.
- Practice of hoeing in between the rows and within the plants

Disease management:

- crop rotation
- Uproot infected plants & burn.
- Plant on well-drained soil.
- Field sanitation
- Keep field weed free

Harvest

- Harvest 60–90 days after planting
- For the local market, harvesting can continue for over a year

Yield:

- 20–25 tonnes/ha

Food value

- Dietary Fibre, Vitamin C

French bean (*Phaseolus vulgaris*)

Recommended varieties
- Contender
- Butter Bean
- Labrador

Seed rate
- 45 kg–50 kg/ha

Planting time
- Best yields from April to September

Spacing
- Between rows: 50 cm
- Plants within rows: 15 cm–20 cm

Germination:
- 3 to 6 days after sowing

Fertilizer
- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management
- Practice of hoeing
- Inter row cultivation

Disease management
- Plough plant remnants after harvest & rotate with Mildew resistant vegetable like
- Tomato, Cabbage & Eggplant

Harvest
- Harvest tender pods 40–60 days after planting
- Picking continues for 4–6 weeks

Yield:
- 7–10 tonnes/ha

Food value
- Dietary Fibre, Vitamin C

Lettuce (*Lactusa sativa*)

Recommended varieties:

Head Type:

- Great Lakes
- Boxhill

Leafy Type:

- Green Mignonette
- Butter crunch
- Rapid

Seed rate:

- 300 grams/ha

Planting time:

Head Type:

- March to October

Leafy type:

- All year round

Spacing:

- Between rows: 75 cm
- within rows: 30–40 cm

Germination:

- 5 to 8 days after sowing
- Raised the seedlings in seed trays/seed bed and transplant at 3 to 4 leaf stage
- Seedlings raised in seed trays

Fertilizer:

- Poultry Manure: 5 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Manual weed control
- Inter row cultivation

Disease management:

- Practice crop rotation
- Avoid planting with high density
- Remove all left over crops

Harvest

- Leafy Lettuce matures in 50–80 days
- Head Type matures in 12–15 weeks

Yield:

- 8–10 tonnes/ha

Food value:

- Dietary Fibre, Source of Vitamin A,
- Vitamin B and C

Long bean (*Vigna sesquipedalis*)

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Recommended varieties:

- Local White
- Yard Long (Dark Green)

Seed rate:

- 7 kg/ha

Planting time:

- Performs best during hot and wet season

Spacing:

- Between rows: 65–75 cm
- Plants within rows: 15–20 cm

Germination:

- 3 to 6 days after sowing

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Hand weed or hoe when necessary
- Inter row cultivation

Disease management:

- Plough plant remnants thoroughly after harvesting, rotate with Vegetables like Cabbage, Eggplant and Tomatoes

Harvest

- Harvest at 50–60 days from planting, pick pods when still tender and harvesting continues for about 2 – 3 weeks

Yield:

- 7–10 tonnes /ha

Food value:

- Dietary Fibre, Vitamin C, Niacin, Vitamin B Complex, Iron and Zinc

Okra (*Abelmoschus esculentus*)

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Recommended varieties:

- Clemson Spineless
- Local Long White
- Dwarf long Pod

Seed rate:

- 8 kg/ha

Planting time:

- All year around but better during hot months

Spacing:

- Between rows: 75–1 m
- within rows: 20–30 cm

Germination:

- 3 to 6 days after sowing

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Hand weed or hoe when necessary and
- practice inter row cultivation

Disease management:

- Rotate with crops
- Plant with other vegetables of different families

Harvest

- Harvesting of tender pods at 60–90 days from planting; and is carried out 2 to 3 times a week and continue for 1 year

Yield:

- 15 tonnes

Food value:

- Dietary Fibre, Potassium, Calcium, Magnesium, Vitamin C

Pumpkin (*Cucurbita maxima*)

© FAO/ K. Hadfield

Recommended varieties:

- Queensland Blue
- Butternut
- Local Selection

Seed rate:

- 1.5 kg/ha

Planting time:

- All year around

Spacing:

- Between rows: 1.5 m
- Plants within rows: 1 m

Germination:

- 3 to 6 days after sowing

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Broadcast and mix well with soil 2 weeks before planting

Weed management:

- Hand weed or hoe when necessary but do not damage the stem
- Inter row cultivation is also recommended

Disease management:

- Rotate with crops of different family such as Eggplants

Harvest

- Harvest at 12–15 weeks from planting

Yield:

- 10–15 tonnes

Food value:

- Dietary Fibre, Potassium, Vitamin C & Vitamin A

Radish (*Raphanus sativus*)

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Recommended varieties:

- Long Whiteside
- Awa Cross
- Everest

Seed rate:

- 10 kg/ha

Planting time:

- Can be planted all year round

Spacing:

- Between rows: 50 cm
- Plants within rows: 5 cm
- Broadcast then thin out to 5 cm apart 2–3 weeks after germination

Fertilizer:

- Poultry Manure: 5 tonnes/ha
- Broadcast and mix well with soil before planting

Weed management:

- Pull out weeds or use a hoe

Harvest

- Harvest at 30–40 days from planting

Yield:

- 8–15 tonnes/ha

Food value:

- Dietary Fibre & Vitamin C

Spring onion (*Allium cepa*)

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Recommended varieties:

- White Lisbon
- Yellow Bermuda

Seed rate:

- 4 kg/ha

Planting time:

- At the start of the cool season.

Spacing:

- Between rows: 50 cm
- Plants within rows: 8 cm

Germination:

- 6 to 10 days after sowing

Fertilizer/manure:

- Poultry Manure: 5 tonnes/ha
- Mix well in the soil 2 weeks before planting

Weed management:

- Mulch with straw materials.
- Hand weed or hoe when necessary; spray pre emergence herbicide

Harvest

- 8 to 12 weeks after planting

Yield:

- 10 to 12 tonnes/ha

Food value:

- Source of Iron, Zinc, Vitamin C and Thiamin

Tomato (*Lycopersicon esculentum*)

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Recommended varieties:

- Alton
- Redland
- Summer taste
- Alafua Large

Cropping season:

- May to October

Seed rate:

- 300 grams/ha

Planting time:

- Main season in the cool months

Spacing:

- Between rows: 0.75 m to 1.0 m Plants within rows 30 to
- 40 cm for staked varieties
- Between rows: 1.5 m
- Plants within rows
- 30 cm for indeterminate varieties grown in open fields

Fertilizer manure:

- Poultry Manure: 12 tonnes/ha
- Broadcast 2–3 weeks after planting. Soil analysis should be done before fertilizer application

Weed management:

- Hand weed
- Inter row cultivation
- Mulching

Disease management:

- Dig, remove and destroy infected plant
- Improve drainage
- Use a two-year rotation
- Use resistant varieties

Harvest

- 10 to 12 weeks after transplanting and picking continues for 5 weeks

Yield:

- 10 to 15 tonnes

Food value:

- Source of Potassium, Calcium, Sodium, Dietary fibre and Protein

Zucchini (*Cucurbita pepo*)

Cropping season:
- Cool dry but can be grown all year round

Recommended varieties:
- Marrow
- Black jack

Seed rate:
- 3 kg/ha

Planting time:
- Cool season

Spacing:
- Between rows: 1 m
- Plants within rows: 30 cm

Germination:
- 5 to 10 days after sowing

Fertilizer/manure:
- Poultry Manure: 12 tonnes/ha Broadcast, mix well before planting

Weed management:
- Hand weed or hoe when necessary
- Practice mulching to control weeds and retain soil moisture

Harvest
- 6 to 8 weeks after planting

Yield:
- 8 to 10 tonnes/ha

Food value:
- Dietary fibre, and Vitamin C

Fruit production

Acid lime (*Citrus aurantifolia*)

Varieties:

- PKM1, Vikram

Soil and climate:

- Tropical and sub-tropical
- Deep well drained loamy soils are the best

Season:

- December–February and June–September

Planting:

- Space: 5 m x 6 m
- Pit: 75 x 75 x 75 cm pits

Irrigation:

- Irrigate copiously after planting
- After establishment at 7–10 days interval
- Avoid water stagnation

Manures and fertilizers

- Farm yard manure
- compost

Intercropping:

- Legumes and vegetable crops can be raised during pre-bearing age

Insect/pest management

- Aphids: Spray neem oil at 3 ml/lit
- Shoot borer: Prune the withered shoots 4 cm below the dried portions
- Citrus Butterfly: Spray two rounds of Bacillus thuringiensis at 1g/lit or neem oil at 10 ml/lit during new flush formation

Disease management

- Twig blight: Prune dried twigs
- Scab: Spray 1 percent Bordeaux mixture

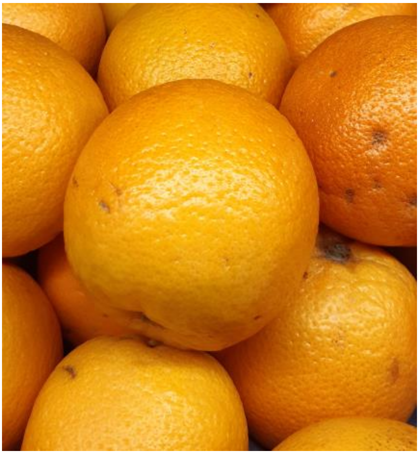
Harvest:

- Starts bearing from 3rd year after planting

Yield:

- 25 t /ha /year

Sweet orange (*Citrus sinensis*)



Varieties:

- Sathugudi

Soil and climate:

- Deep well drained loamy soils
- pH - 6.5 to 7.5
- A dry climate with about 50–75 cm of rainfall from June–September and with well-defined summer and winter season is ideal

Season:

- July to September

Planting material:

- Budded plants (Root Stock: lime is best, now Rough lemon is also preferred)

Preparation of field:

- Dig pits at 75 x 75 x 75 cm in size at 7 x 7 m spacing
- Fill up the pits with top soil and 10 kg of FYM. Plant the budded plants in the centre of the pits and stake it

Irrigation:

- Immediately after planting irrigate copiously
- Once in 10 days

Manures and fertilizers

- Farm yard manure
- Compost

Intercropping:

- Legumes and vegetable crops can be raised during pre-bearing age

Harvest:

- Starts bearing from 5th year after planting

Yield:

- 30 t/ha

Mandarin orange (*Citrus reticulate*)

Varieties:

- Coorg Orange and Kodai Orange

Soil and climate:

- Sub-tropical climate
- 150 cm–250 cm rainfall is required.
- Deep well drained loamy soils are the best
- Soil pH should be between 5.5–6.5

Season:

- November–December
- Planting: Seedlings and budded plants
- Spacing: 6 x 6 m, pit size 75 x 75 x 75 cm. planting during May–June and September–October

Manures and fertilizers:

- Manures are to be applied in the basin 70 cm away from the trunk and incorporated

After cultivation:

- Remove water shoots, root stock sprouts, dead and diseased shoots
- Remove laterals of the main stem up to 45 cm from ground level
- Basins should be provided for each tree with gradient slope

Pest management

- Aphids : Spray neem oil 3 ml/lit or Fish oil rosin soap at 25 g/lit
- Bag the fruits with polythene bags punctured at the bottom
- Apply smoke and set up light traps or food lures (pieces of citrus fruits)
- Prune the withered shoots 4 cm below the dried portions

Harvest:

- Budded plants start bearing from 3–5 years after planting while seedlings take 5–7 years

Yield:

- 15–20 t/ha/year

Sapota (*Manilkhara achras*)

Varieties:

- Oval, Cricket Ball, Kirtibarti, Guthi, CO 1, CO 2, CO 3, P KM 1, PKM 2, PKM 3, PKM 4, PKM (Sa) 5 and Kalipatti

Soil and climate:

- It can be grown in all types of soils

Planting materials:

Grafted on *Manilkhara hexandra* (Pala) rootstock

Season of planting:

- June–December

Spacing:

- 8 x 8 m (156 plants / ha) for conventional planting.
- Adopt high density planting at 8 x 4 m (312 plants/ha) for high productivity

Planting:

- Dig pits of 1 x 1 x 1 m in size.
- Fill up with top soil mixed with 10 kg of FYM

Irrigation:

- Irrigate copiously immediately after planting and on the third day and once in 10 days afterwards till the graft establishes

Manures and Fertilizers

- Manures may be applied 45 cm away from the trunk
- FYM – 10 kg
- After cultivation: Remove the root stock sprouts, water shoots, criss-cross and lower branches

Inter cropping:

- Legumes and short duration vegetable crops may be raised as inter crop during pre-bearing stage

Harvest:

- Collect when fruit brown in colour. Immature fruits it is green
- The mature fruits are harvested by hand picking

Yield:

- 20–25 t/ha year

Avocado (*Persia americana*)

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Fruiting season:
- March to July

Recommended varieties:
Local Selection

Seed Rate:
- 123 plants/ha

Planting time:
- All year round

Spacing:
- Plant 9 x 9 m apart and away from other trees and buildings

Germination:
- Seeds are quick to germinate

Fertilizer/manure:
- Heavy mulch around the base of the plant ensures steady growth
- Soil analysis should be done before fertilizer application

Disease management:
- Root Rot: Maintain good drainage. Prune regularly

Insect management:
- No significant insect pests of concern

Harvest
- Normally fruits appear after 6 to 7 years from planting but grafted/budded plants come into bearing earlier at about 4 to 5 years

Yield:
- About 10 to 15 tonnes/ha from an orchard of about 10 to 12 years.
- Prune regularly

Food value:
- Dietary fibre, Potassium & Vitamin C

Passionfruit (*Passiflora edulis*)

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Recommended varieties

- Local Yellow

Seed rate

- 1111 Seedlings/ha

Planting time:

- cool and dry months

Spacing:

- Between Rows : 3 m
- Plants within rows : 3 m
- Posts: 3 m within rows 6.2 m within rows
- Seeds germinate in 10 days after sowing, ready for transplanting within 6-8 weeks

Fertilizer:

Use compost

Weed management:

- Ring weeding during early stages of growth.
- Hand pollination to be practiced to obtain high yield, to be carried out in the afternoon; after 2 pm
- Integrate apiculture farming with passionfruit for pollination purpose

Disease management:

- Collar Rot: The disease can be controlled by good site selection & planting on raised bed

Insect management

- Red Spider Mite: Controlled by spraying

Yield:

- 1st Year: 12–18 tonnes/ha
- 2nd Year: 20–25 tonnes/ha
- 3rd Year: 10–12 tonnes/ha

Food value:

- Good Source of Iron and Vitamin C

Pineapple (*Ananas comosus*)

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Fruiting season:

- Main Season November to April
- Off Season: February to October

Recommended varieties:

- Smooth Canyenne (Large juicy fruit)
- Ripley Queen (Small sweeter fruit - thorny leaves)

Seed rate:

- Sloppy Land (37 037 suckers/ha)
- Flatland – 48 000 suckers/ha

Planting time:

- Dry season
- Use raised beds on flat land
- Practice phase planting for all year round production

Spacing:

- Sloppy Land: 1.2 m between ridges, 0.6 m between rows per ridge & 0.3 m between plants (Double rows)
- Flat Land: 1.4 m between ridges, 0.4 m between rows per ridge and 0.2 m between plants (Double rows)
- Best planting material are suckers weigh 250–300 or 25–30 cm in height

Fertilizer

- Use compost

Weed management:

- Manual Weeding
- inter row cultivation

Disease management:

- Planting during dry season with good field drainage

Yield:

- Planted Crop: 40–70 tonnes/ha at 1–1.5 kg/fruit
- 1st Ratoon: 30–40 tonnes/ha with ~ weight of 1.0–1.5 kg
- 2nd Ratoon: 20–25 tonnes/ha with ~ weight of 1–1.2 kg

Food value:

- Good source of Vitamin C and Vitamin B1
- & Fibre

Watermelon (*Citullus lanatus*)

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Recommended varieties:

- Charleston Grey
- Sugar Babe
- Farmers Giant

Seed rate:

- 1.5–2 kg/ha

Planting time:

- April to September during the cool season but can be grown all year round

Spacing:

- Between Rows: 3 m
- Plants within Rows: 1 m
- Germination: 6–10 days after sowing

Fertilizer:

- Poultry Manure: 10 tonnes/ha
- Mix well in the soil 2 weeks before planting

Weed management:

- Hand weeding of hoeing as necessary
- Practice of mulching to retain moisture and control weeds

Disease management:

- Have soil analyzed for level of K, and Ca
- High level of N and low level of K & Ca causes blossom end rot
- Keep your crop free from insects to avoid spread of viral diseases

Harvest

- Harvest at 70–120 days from planting

Yield:

- 15–20 tonnes/ha

Food Value:

- Vitamin C

Banana (*Musa sapientum*)

Recommended varieties:
- Kudud
- Mocao
- Mangat Kingit
- Peleu
- Utin wai
- Utin Lihli
- Others (etc)

Seed rate:
- 1 666 suckers/ha

Planting time:
- Recommended from October to March otherwise all year round

Spacing:
- Between Rows: 3 m
- Plants within Rows: 2 m
- Planting Materials: Select healthy and disease free as planting materials

Fertilizer:
- Apply compost around the plant based on canopy diameter

Weed management:
- Ring weeding

Disease management:
- Remove infected plants & bury.
- Field sanitation
- Good planting materials,
- Hot water treatment for Nematodes

Insect management:
- Keep plantation clear of any plant debris & weeds
- Use suckers from non-infected areas
- Practices good husbandry practices

Harvest
- Fruits appear after 9–10 months from planting & ripens about 3 months from fruit set

Yield (dry):
- 1666 bunches in 1st year, 2500 bunches in 2nd Year if two suckers are maintained per stool

Food Value:
- Potassium, Vitamin A & Vitamin C

Cocoa (*Theobroma cacao*)

Recommended varieties:

- Amelonado
- Trinitario
- Keravat

Seed rate:

- 2500 plants/ha

Planting time:

- Can be planted all year
- Dry Zone: Mid September to December
- Wet Zone: October to December

Spacing:

- Between Rows: 2 m
- Plants within Rows: 2 m

Planting materials:

- Select healthy and disease free as planting materials

Fertilizer:

- Compost
- Apply fertilizer around the plant based on canopy diameter

Weed management:

- Ring weeding

Disease management

- Canker: Remove & destroy diseased plants
- Sanitation: Remove disease parts away from the Cocoa field, burn & bury. Plant the recommended variety
- Remove black pods regularly & bury or burn outside the plantation
- Prune shade trees & overgrown Cocoa branches

Harvest:

- Harvest at 3 years after planting

Yield:

- 2.5 tonnes/ha Wet beans.
- Or 2.0 tonnes/ha Dry

Food value:

- Source of Thiamin, Niacin & Vitamin B12

Coconut (*Cocos nucifera*)

Recommended varieties:

- Rotuman Tall
- Niu Leka
- Niu Magimagi
- Niu Drau
- Niu Kitu
- Niu Yabia
- Malayan Red Dwarf x
- Rotuman Tall
- Malayan Yellow Dwarf x
- Rotuman Tall

Seed rate:

- 123 plants/ha
- 6 to 7 months old seedlings are used as planting materials

Planting time:

- Best time for planting is at the onset of the rainy season - October to April

Spacing:

- Triangular : 9 x 9 m
- Square : 10 x10 m

Fertilizer:

- Compost

Weed management:

- Ring weeding

Disease management:

- Remove and destroy diseased seedlings and destroy diseased plants

Insect management:

- Rhinoceros Beetle
  - Biological control using virus, fungus & pheromone traps
- Stick Insects - Cultural method, Good field sanitation

Harvest

- Bearing
- Tall: 5–7 years to bear nuts
- Dwarf: 3–4 years to bear nuts
- Hybrid: 3–4 years to bear nuts

Yield: in dried copra

- Tall : 0.7 1.3 tonnes/ha Dwarf: 0.7–0.8 tonnes/ha
- Hybrid 2–3 tonnes/ha

Food value:

- Vitamin C, Vitamin B1, B2 & Iron.

Kura (Noni) (*Morinda citrifolia*)

Cropping season:
- All year round

Recommended varieties:
- Local Selections
- Small fruit varieties are preferred

Seed rate:
- 625 seedlings/ha

Planting time:
- Can be planted all year round but best during October to March enhance plant growth

Spacing:
- Between rows: 4 m
- Plants within rows: 4 m

Germination:
- Seeds germinate in 25–30 days and within 12–16 weeks, plants are ready for transplanting

Fertilizer/manure:
- Kura is grown naturally (organically) to be planted in new areas
- Soil analysis should be done before fertilizer application

Weed management:
- Ring weed round and in between the plants during establishment stage
- Pruning of main shoots to dwarf the trees for ease of harvest

Disease management:
- No economic diseases

Insect management:
- No economic pests

Harvest
- Fruiting will start at 13–15 months after transplanting

Yield:
- 1.0 to 1.5 kg/tree/week under optimum management practices

Food value:
- Vitamin C

Mango (*Mangifera indica*)

Recommended varieties
- Kensington
- Tommy Atkins
- Mexican Kent
- Parrot
- Mango Dina

Seed rate:
- 125 plants/ha

Planting time:
- Planting is recommended during the Wet
- Season (November to March)

Spacing:
- Between Rows: 9 m
- Plants within Rows: 9 m

Germination:
- Grafted seedlings enhance early flowering & fruiting
- Grafted plants can be produced from reliable nursery few days before transplanting

Fertilizer:
- Compost

Bearing trees:
- 2 –3 kg annually

Weed management:
- Manually control weeds

Insect management:
- Set up protein bait traps
- Good field sanitation
- Bury all fallen fruits to prevent pests population build up

Harvest
- Grafted plants start to fruit within 3 years
- Yields vary depending on the varieties age of tree & weather conditions

Yield:
- Improved Varieties: 25–80 kg/tree in 5th–7th year
- 70–150 kg/tree in 8th to 15th Year

Food value:
- Rich in Vitamin A as well as Vitamin C

Papaya (*Carica papaya*)


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Recommended varieties
- Sunrise Solo (Export Variety)
- Waimanalo (Local markets)

Seed rate
- 1 667 plants/ha

Planting time:
- Can be planted all year round.

Spacing:
- Between Rows: 3 m
- Plants within Rows: 2 m

Germination:
- The seeds germinate in 10 to 12 days after sowing. In cooler seasons it takes longer 18 to 21 days
- The seedlings are grown in plastic pots for 6 to 7 weeks after sowing before transplanted in field

Fertilizer:
- Compost

Weed management
- Plastic mulching controls weeds

Disease management
- Plant papaya in well drained fields

Insect management
- Good field sanitation, remove and bury fallen fruits

Harvest
- Fruit ripens at 8 to 10 weeks after flowering. Maximum yield of
- 60 to 80 tonnes/ha

Food value:
- Excellent source of Vitamin A and Vitamin C

Soursop (*Annona muricata*)

Fruiting season:
- October to April

Recommended varieties:
- Local Selection

Seed rate:
- 500 plants/ha

Planting time:
- Can be planted all year round
- Planted during November to March enhance plant establishment

Spacing:
- Between rows: 4.5 m
- Plants within rows: 4.5 m
- Germination: Propagated by seeds, cuttings or grafted on same rootstock
- Seedlings are grown in nursery and transplanted in to the field at 8 to 10 leaf stage

Fertilizer/manure:
- Compost & farm yard manure

Weed management:
- Ring weed the plants - hand weeding

Disease management:
- Root rot can be a problem if grown in waterlogged areas
- Avoid planting in poorly drained sites

Insect management:
- Birds and Bats eat ripe fruits on the tree
- Harvest fruits before full ripeness

Harvest
- Fruiting starts in 2 to 4 years after planting

Yields:
- 8 to 10 tonnes/ha/year after 3 years from planting
- Economic life 10 to 12 years

Food value:
- Fair source of Protein, Dietary fibre, Potassium and Calcium

Vanilla (*Vanilla fragrans*)

© Apolima edge

Recommended varieties:

- Bourbon vanilla

Seed rate:

- 1 111 plants/ha

Planting time:

- Can be planted all year round

Spacing:

- Between Rows: 3 m
- Plants within Rows: 3 m
- Cutting: 1.5 m long sprout in 15–20 days after planting

Fertilizer:

- Require heavy mulching: 20–30 cm around base. (Coconut husk, dry leaves & rotten decaying timber can be safely used as mulch)
- Poor Soils: Apply 20–30 g of Nitrogen &Phosphorus, 60–100 g Potash per Vine per year beside the organic mulch
- Soil analysis should be done before fertilizer application

Weed management:

- Hand weeding or use of brush cutter at least four times a year

Disease management:

- Remove & destroy infected plants.
- Use healthy and disease free planting materials
- Wash hands after handling infected plants
- Wait for a month for the virus to die, then replant the area

Insect management:

- Slugs & Snails: Control by hand picking and use Snail bait
- Keep area clean

Harvest

- Harvest at 3 years

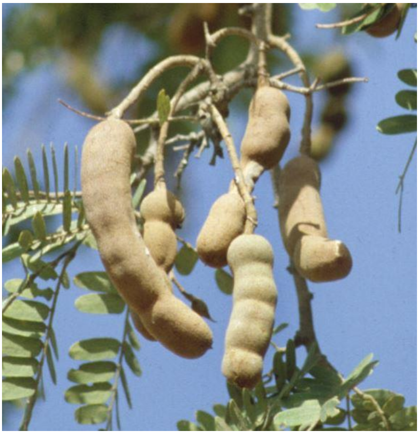
Yield:

- 300–600 kg cured beans

Food value:

- Food flavor

Tamarind (*Tamarindus indica* L)



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Varieties
- Tamarindo

Propagation
- Seed, vegetative and tissue culture propagation methods

Soil and climate
- Often known as the hurricane-resistant tree
- sodic and saline soils
- It can grow well between 21°C and 37°C

Season
- evergreen

Planting
- Plant in rainy season
- Plant deeper than 1.5 cm.
- Transplant the seedling when 10 cm to a pit of 1 x 1 x 1 m
- Spacing 4 x 4 m or 5 x 5 m.

Irrigation
- Annual rainfall of 500–1500 mm.

Intercropping:
- Legumes and vegetable

Manure and fertilizers
- None

Insects
- Shot hole borers, leaf feeding caterpillars, mealy bugs and scale insects

Diseases
- Tree rots, stony fruit disease, bark parasite and bacterial leaf spots

Post-harvest
- Fresh fruits are often dried using small-scale dehydrators, however in most countries rural households dry pods in the sun

Yield
- Young tree yields 20–30 kg fruits per year
- Yield 150–200 kg/tree/year

Products
- Tamarind juice, concentrate, tamarind pulp and pickles

Dragon fruit (*Hylocereus undatus)*

© FAO/ N.Tohovaka

Varieties
- Yellow dragon, Purple haze, Costa Rican sunset

Propagation
- Seed, vegetative and tissue culture propagation methods

Soil and climate
- Well drained red yellow podzolic, lateritic soil and reddish brown earth
- It can grow well between 20°C–30°C. pH 5.5–6.5

Planting
- 60 cm cutting.
- Open areas with more sun light is preferred
  - Spacing: 3 X 3 m
- Pit: 30 cm deep and 20 cm wide

Irrigation
- Annual rainfall of 1700–2500 mm
- Mulching is necessary to reduce moisture loss

Trellising:
- trellis and train vine

Manure and fertilizers
- Organic manure 100 g/plant of (compost)/each 4 months

Insects
- Generally insect free.

Weeds
- Manually remove creeping weeds
- Intercropping

Harvest
- Selective harvesting at full maturity
- Pink red color is indicator of maturity

Post-harvest
- Shelf life is up to 10 days.
- Store at 15–20°C at 85–90 percent relative humidity.

Yield
- Around 12 000 kg/ha

Nutritional value
- Vitamins and fiber

Star apple *(C. cainito L.)*

© Apolima edge

Soil and climate

- Grows on any type of soil.
- pH 5.5–6.0 is desirable

Seed preparation

- Select seeds from healthy, sound and ripe fruits
- Plant 1 cm deep & 2–3 cm apart
- Cultivate in shaded area

Transplant

- When 3–5 leaves have developed, transplant the seedlings

Propagation

- Sexually by seeds and asexually by marcotting, inarching, grafting and budding

Land Preparation and planting

- Clear land & plow deeply
- Spacing: 10 X 12 m row x hill
- Put 500 g to 1 kg organic fertilizer or animal manure in each hole

Intercropping

- pineapple, root crops, perennial leguminous crops and vegetable crops

Irrigation

- Water newly cultivated plants
- Water plant while fruiting to keep their juiciness

Fertilizer

- Organic fertilizer.
- 0.5 –1 kg of chicken or cow manure

Pest management

- Twig borers, carpenter moth, mealy bugs, scales, fruit flies, ants and bats

Harvesting

- It bear fruits at the age of 3–5 years
- grown tree bear 1 000 fruits
- Ripe fruit change color and are soft

Food value

- Used as ingredient of ice-cream and sherbet

Mangosteen (*Garcinia mangostana*)

© FAO/ A. Pierdomenico

Soil and climate

- humid tropical lowlands, RH 80–100 percent
- deep soil, permeable soils with high moisture and organic matter
- Need dry seasons
- Temperatures below 5°C and above 38°C can be lethal

Propagation

- Choose large-sized seed and early ripening fruits, to avoid selecting for an inherited late-fruiting
- Soak seed for 24 hrs in water then cultivate
- Transplant at 2-leaf stage
- Grafting, air layering, budding, cutting, inarching and saddle grafting are other methods of cultivation

Planting

- Tree spacing: 8–10 x 8–10 m
- holes measuring 1 x 1 x 1 m deep
- Fill holes with compost, well decayed manure and topsoil
- Training and pruning

Fertilizer

- Limited information available

Irrigation

- Limited information available

Weed management

- Mix cropping
- Use cover crops and intercropping
- Use mulch

Harvesting

- Hand harvest is preferred
- Yields of between 200 and 2000 fruits per tree, or 4–8 t/ha
- Fruit on 5–8 year after transplanting

Food value

- Rich in carbohydrate. The fruit rind contains tannin and dye

Lanzones (Lansium domesticum C)

© Apolima edge

Soil and climate

- sandy loam to clay loam soil, well drained, slightly acidic (5.3 to 6.5)
- needs ample amount of water

Seed

- Seeds from ripe fruits are harvested and extracted after soaking in water for 1–2 days to soften the aril

Transplanting

- produce 2–3 seedlings per seed
- transplant in 7 X 11 polybags

Propagation

- The seedlings can be asexually propagated within 7–12 months from transplanting.
- cleft grafting, marcotting, cutting, inarching and top working

Planting

- Plant in rainy season
- Use compost in the hole and then add top soil

Irrigation

- No water, no fruit
- Irrigate frequently
- Use well drainage system

Fertilizers

- Use organic manure

Training and pruning

- Do not cut top of the erect seedling
- Remove water sprouts of the grafted plants
- Paint all wounds

Pest management

- Ants, mealy bugs, aphids, mites, borers, fruit flies scale insects

Harvesting

- Collect fruit when cluster of fruit is ripe
- Fruit stalk color changes from green to brown when fruit ripen

Food value

- dessert fruit, high in phosphorus and potassium

Rambutan *(Nephelium lappaceum Linn.)*

© Apolima edge

Soil and climate

- Deep, clay-loam or rich well-drained sandy loam rich in organic matter
- pH 4.5–6.5

Seed

- Select well-developed seeds from mature ripe fruits for rootstock

Propagation

- Rootstocks are ready for asexual propagation or cleft grafting in 6–8 months
- Cleft grafting

Planting

- Spacing: 10 X 10 m row X hill, 100 trees/ha
- Hole size 30 X 30 X 30 cm.
- Put 1 kg organic matter in each hole before planting

Weed management

- Shading, intercropping, cover cropping and mulching

Irrigation

- When no water, there is no flower and fruit
- Irrigate frequently
- Use well drainage system

Fertilizers

- Use organic manure at 10–30 kg/year/tree

Training and pruning

- yearly pruning of the dominant or apical shoots and lateral branches

Pest management

- Fruit borer, leaf eating loofer, thrips, mealy bugs, mites

Disease management

- Powdery mildew, vein necrosis, sooty mold

Harvesting

- Tree bears fruit on 3–6 year old
- When color changes from green yellow or red, fruit is ripe
- Cut clusters, use sharp knives

Food value

- Source of vitamin A, C and fiber

Barbados cherry (*Malpighia punicifolia* L.)


© Apolima edge

Soil and climate

- The tree does well on limestone, marl and clay, as long as they are well drained
- The pH should be at least 5.5.
- Mature trees can survive brief exposure to 28 F (-2.22 °C)
- It can tolerate long periods of drought, though it may not fruit until the coming of rain

Seed

- Select well-developed seeds from desirable clones

Propagation

- Air layering, cutting, seed and grafting

Pruning

- It is bushy shrub or small tree (to 15 feet)
- Multiple or single trunks which can be trained
- Occasionally, bushes appear to be composed of canes
- Branches are brittle, and easily broken

Pest management

- Caribbean fruit fly, fruit worm

Disease management

- Root-knot nematode , burrowing nematode, leaf spotting

Harvesting

- Manual picking of fruit
- 13.5–28 kg/tree or 10–15 tonnes/ha

Food value

- source of calcium, phosphorus and vitamin A

Annex 1: A guide to backyard composting

What is Compost?

Compost is a natural fertilizer and soil conditioner. You can make it at home from organic materials such as kitchen scraps and garden waste. When put into a pile, these materials naturally decompose, turning into a rich, soil-like material called compost or humus. Composting is basically a way of speeding up the natural process of decomposition.

The Keys to Good Compost

Balanced diet: For optimal decomposition, the carbon–nitrogen ratio in a compost pile should be about 30:1. Carbon-rich (“brown”) materials include dry leaves, corn stalks, and sawdust. Nitrogen-rich (“green”) materials include food scraps, coffee grounds, and grass clippings.

Temperature: Compost piles are most active at temperatures of 44 to 52 °C. Decomposition drops with the ambient temperature, and stops altogether if the pile freezes.

Oxygen: Compost depends on the production of aerobic (oxygen-loving) bacteria, which do the work of decomposition.

Moisture: Compost should be moist, but not wet-excess water will decrease oxygen levels, slowing down decomposition.

Ten Easy Steps to Making Compost

1. **Select a site:** In a sunny, well-drained location, measure out an area to site your bin. Three square feet is an ideal bin size, and is the minimum size necessary to generate the required heat in the shortest possible time.
2. **Purchase a bin:** Contact your municipality, a local store, or build your own rodent-proof compost bin.
3. **Form base layer:** In the bottom of the bin, arrange a six-inch layer of coarse materials such as sticks, pruning, and bark pieces. This will allow air to filter into the center of the heap without smothering the soil surface.
4. **Alternate layers:** After the base layer is formed, you can start using your compost bin daily. As you accumulate kitchen or yard waste, add it to the bin in layers, starting with 2 to 4 inches of “green” organic matter. Follow this with more carbon-rich “brown” matter, and continue to alternate between green and brown, ensuring that no organic layer is ever more than 15 inches deep.

5. **Moisten:** Lightly water the pile if necessary-compost ingredients should be damp, not soaking.
6. **Cover:** The compost pile should always be topped by a thick carbon (brown) layer. Using a lid will discourage rodents and other animals.
7. **Monitor:** Each time you add material to the bin, give it a look and a sniff. If the pile has an unpleasant odor, or does not appear to be gradually shrinking, this indicates a problem with the pile.
8. **Add more layers:** The pile will shrink as its contents decompose; continue adding material.
9. **Check:** Compost is generally ready to use after about 2–3 months. This can vary depending on things like temperature and the materials used. Once your bin starts to get full, check to see if the bottom portion of the pile is ready to harvest in order to make room at the top.
10. **Harvest:** Begin harvesting when the compost at the bottom and center is decomposed. Dig out the compost with a shovel, using the door at the bottom of a commercial bin. If you have built your own bin, remove the top new layers and dig the compost from the center.

Further information:
A guide to backyard composting.
https://www.evergreen.ca/downloads/pdfs/Backyard-Composting-Guide.pdf

Annex 2: Natural and homemade pesticides

Name of insecticide:	**Oil spray insecticide**
What pests can be controlled by it?	aphids, mites, thrips
How to make it?	Mix 1 cup of vegetable oil with 1 tablespoon of soap (cover and shake thoroughly), and then when ready to apply, add 2 teaspoons of the oil spray mix with 1 quart of water, shake thoroughly.
How to use it?	Spray directly on the surfaces of the plants which are being affected by the little pests. The oil coats the bodies of the insects, effectively suffocating them, as it blocks the pores through which they breathe.

Name of insecticide:	**Soap spray insecticide**
What pests can be controlled by it?	mites, aphids, earwigs, leafhopper, spider mites, whiteflies, beetles, and other hungry little insects
How to make it?	mix 1 1/2 teaspoons of a mild liquid soap (such as castile soap) with 1 quart of water
How to use it?	Spray the mixture directly on the infected surfaces of the plants. It is always recommended to NOT apply it during the hot sunny part of the day, but rather in the evenings or early mornings

Name of insecticide:	**Neem oil insecticide, fungicide**
What pests can be controlled by it?	powdery mildew, aphids, thrips and whiteflies
How to make it?	Start out with a basic mixture of 2 teaspoons neem oil and 1 teaspoon of mild liquid soap shaken thoroughly with 1 quart of water. Neem oil can be extracted from the seeds of the neem tree.
How to use it?	Spray on the affected plant foliage. It is capable of disrupting the life cycle of insects at all stages (adult, larvae, and egg). Neem oil acts as a hormone disruptor

	and as an "antifeedant" for insects that feed on leaves and other plant parts.

Name of insecticide:	**Diatomaceous earth as a natural pesticide**
What pests can be controlled by it?	snails and slugs as well as other crawling insects
How to make it?	Fossilized algae (diatoms), buy from shops/supermarkets.
How to use it?	Simply dust the ground around your plants, or even sprinkle it on the foliage. This material works not by poisoning or smothering the insects, but instead by virtue of its abrasive qualities and its affinity for absorbing the lipids (a waxy substance) from insects' exoskeleton, which then dehydrates them to death. In order to be an effective, diatomaceous earth needs to be reapplied after every rain.

Name of insecticide:	**Garlic insecticide spray**
What pests can be controlled by it?	Different insects aphids, ants, borers, caterpillar, slugs, whiteflies,
How to make it?	Take 2 whole bulbs (not just 2 cloves) and puree them in a blender or food processor with a small amount of water. quart of water. Let the mixture sit overnight, then strain it into a quart jar, adding 1/2 cup of vegetable oil (optional), 1 teaspoon of mild liquid soap, and enough water to fill the jar.
How to use it?	Use 1 cup of mixture with 1 quart of water and spray liberally on infested plants. Deterrent and killer of insects.

Name of insecticide:	**Pyrethrins**
What pests can be controlled by it?	Toxic to a broad spectrum of pests. Flea, potato, and bean beetles.
How to make it?	Derived from the painted daisy, *Chrysanthemum cinerariifolium*.
How to use it?	Apply dust during cloudy weather or early evening. A nerve toxin, often combined or DE. Degrades rapidly.

Name of insecticide:	**Chile pepper insecticide spray**
What pests can be controlled by it?	Different pests.
How to make it?	Chile spray can be made from either fresh hot peppers or chile pepper powder. To make a basic chile spray from pepper powder, mix 1 tablespoon of chile powder with 1 quart of water and several drops of mild liquid soap. To make chile spray from fresh chile peppers, blend or puree 1/2 cup of peppers with 1 cup of water, then add 1 quart of water and bring to a boil. Let sit until cooled, then strain out the chile material, add several drops of liquid soap to it and spray as desired.
How to use it?	Spray on the infested plant. It is insect repellent. [Caution: Hot chile peppers can be very potent on humans as well, so be sure to wear gloves when handling them, and keep any sprays made from them away from eyes, nose, and mouth.]

Name of insecticide:	**Tomato leaf as a natural insecticide**
What pests can be controlled by it?	aphids
How to make it?	chop 2 cups of fresh tomato leaves (which can be taken from the bottom part of the plant) into 1 quart of water, and let steep overnight. Strain out the plant material
How to use it?	spray onto plant foliage

Name of insecticide:	***Bacillus thuringiensis –or* Bt.**
What pests can be controlled by it?	Toxic primarily to caterpillars, Cabbageworms and cutworms, Colorado potato beetle.
How to make it?	Purchase from an agricultural shop/store.
How to use it?	Available as spray or dust. Apply late afternoon and reapply after rain. Mix with insecticidal soap for better coverage. Bacterial toxin; causes caterpillar death usually within 24 hours. Dissipates in 2 days or less.

Annex 3: Synthetic fertilizers

There are 16 main nutrients or elements that plants require for normal growth and development. These nutrients are divided into two main groups, the macro-elements that are required in relatively large quantities and the micro-elements or trace elements, which are required in very small quantities.

The macro-elements are carbon (C), hydrogen (H), oxygen (O), Nitrogen (N), potassium (K), phosphorous (P), calcium (Ca), magnesium (Mg) and sulphur (S).

The trace elements are iron (Fe), copper (Cu), boron (B), molybdenum (Mo), chlorine (Cl), manganese (Mn) and zinc (Zn).

N, P and K are the most well-known elements that plants require, the remaining macro elements (phosphorous (P), calcium (Ca), magnesium (Mg) and sulphur (S)) are also important.

What follows is a short discussion of the main reasons why each nutrient is important. There is also a discussion on how to identify their deficit.

What fertilizer?	**Nitrogen (N)**
Why does a plant need N?	• Nitrogen is essential for the synthesis of proteins in plants. • N is necessary as a building block for genetic material. • N is an essential part of the green pigment chlorophyll. • N is good for leafy vegetables or as a general tonic to boost plant growth.
Where do we find N?	• Plants use nitrogen in two forms, these are ammonium and nitrate • Ammonium will stimulate leafy growth. • Applying nitrate or urea as a foliar spray very quickly stimulates crop growth. • Nitrogen is also found in organic matter, such as lawn clippings, compost, manure as well as blood or bone meal.
N deficiency diagnosis?	• Plants are stunted • Leaves become pale green or yellow (chlorosis) • Yellowing is normally seen on older leaves first

	• Yellowing starts at the tip of the leaf progressing down the middle of the leaf to the leaf base, spreading across the leaf blade as a whole.

What fertilizer?	**Phosphorus (P)**
Why does a plant need P?	• Plants require phosphorous all the time • There is a strong relationship between phosphorous and nitrogen requirements • If there is no N, the plant cannot take up P from the growth medium • P is essential for growth and development of stems, roots, seeds, flowers and seedlings • In crops P improves crop quality, increases root growth and leads to earlier crop maturity.
Where do we find P?	• Phosphorous deficiency can be corrected by adding phosphorus to irrigation water in the form of e.g. potassium phosphate, or a foliar application of ammonium phosphate. As with nitrogen scorching of leaves could occur • A more long-term source of phosphorous is super-phosphate which is applied to the soil
P deficiency diagnosis?	• Plants are stunted x Leaves take on a purplish color. • The undersides of the leaves become characteristically purple especially on the veins. • Fruits mature late and seeds do not develop properly • The change in color usually develops on older leaves first.

What fertilizer?	**Potassium (K)**
Why does a plant need K?	• Makes plant structure • Improve plant growth. • Aids in the plants overall vigor, strength, water uptake and disease resistance. • K plays a role in maintaining plant water balance, controls transpiration, and activates enzymes. • K improves the plants’ flower, fruit and seed quality.

Where do we find K?	• Potassium deficiency can be overcome by foliar application of potassium Sulphate or potassium nitrate. A more long-term source of potassium is potash worked into the soil.
What does a plant that is deficient in K look like?	• The first sign of potassium deficiency is that the leaves turn dark green x In time leaves become a purple brown colour. • This discoloration is followed by yellowing of leaf edges leading to a browning dying off (necrosis) of the tissue. • Weak stems, with yellowing or browning around the edges and tips of older leaves are a tell-tail sign of K deficiency.

What fertilizer?	Calcium (Ca)
Why does a plant need Ca?	• Calcium is a major constituent of cell walls. • Ca is involved in nitrogen metabolism and activates enzymes. • Ca helps to build strong stems.
Where do we find Ca?	• The effects of can generally be reversed. • This is done by a foliar application of compounds such as calcium nitrate. Calcium is also found in agricultural lime, super-phosphate and gypsum.
What does a plant that is deficient in Ca look like?	• Common symptoms of calcium deficits are stunting, wilting and dark green discoloration. • Leaf margins become scorched x Roots are poorly developed and the root tips die off. • In fruit crops like tomatoes, cucumbers and peppers calcium deficiency causes blossom end rot. This condition is irreversible.

What fertilizer?	Magnesium (Mg)
Why does a plant need Mg?	• Magnesium is essential a part of the green pigment chlorophyll and is thus an extremely important element.
Where do we find Mg?	• Magnesium deficiency is common and can be corrected by foliar application of Epson salts. Magnesium is found in a number of commercial fertilizers.
What does a plant that is	• A plant is deficient in magnesium develops yellow leaves • Usually the older leaves of the plant, rather than the new young leave develop this symptom.

deficient in Mg look like?	The margins of leaves turn yellow, spreading to the leaf blade as a whole.

What fertilizer?	**Sulphur (S)**
Why does a plant need S?	• Sulphur is important for the production of chlorophyll • S is also important as a protein constituent.
Where do we find S?	• Mg is found in super phosphate and gypsum.
What does a plant that is deficient in S look like?	• A sulphur deficiency affects the quality and flavour of fruit and vegetables. • It is seen as a light purple discoloration of petioles, stems and veins, with the leaves turning pale yellow. • Dead spots and patches may develop on leaves.

What fertilizer?	**Iron (Fe)**
Why does a plant need Fe?	• Iron plays a vital role in the formation of chlorophyll during photosynthesis.
Where do we find Fe?	• Iron can be applied as iron chelates and iron salts. • Deficiency develops if the growth medium pH is too high or if anaerobic conditions develop in soil or too much magnesium is found in the in the rooting medium.
What does a plant that is deficient in Fe look like?	• Iron deficiency symptoms are similar to those of magnesium. • The major symptoms are yellowing of young developing leaves. • The veins remain green but the rest of the tissue becomes yellow, causing a mottled leaf.

What fertilizer?	**Manganese (Mn)**
Why does a plant need Mn?	• Mn is Essential for the manufacturing of “sugars” • Mn is required for nitrogen metabolism.

Where do we find Mn?	• Manganese can be applied as manganese sulphate • Care must be taken however as this element is toxic at high concentrations.
What does a plant that is deficient in Mn look like?	• Develops first on young tissues and can easily be confused with iron deficiency. • The distinguishing factor is that Mn deficits cause more overall leaf discoloration and may also cause necrotic spots and lesions. • In severe cases leaves become distorted.

What fertilizer?	**Copper (Cu)**
Why does a plant need Cu?	• Plays a role in the activation of several enzymes, effects cell wall formation • Plants require very little Cu to be present.
Where do we find Cu?	• Copper deficits can be remedied by the application of copper sulphate.
What does a plant that is deficient in Cu look like?	• Cu deficiency causes stunting of the plants leading to shortened inter-nodes and small leaves. • Chlorite blotches develop on older leaves, gradually spreading to younger leaves. • Affected leaves turn dull green to bronze with the edges curling upwards.

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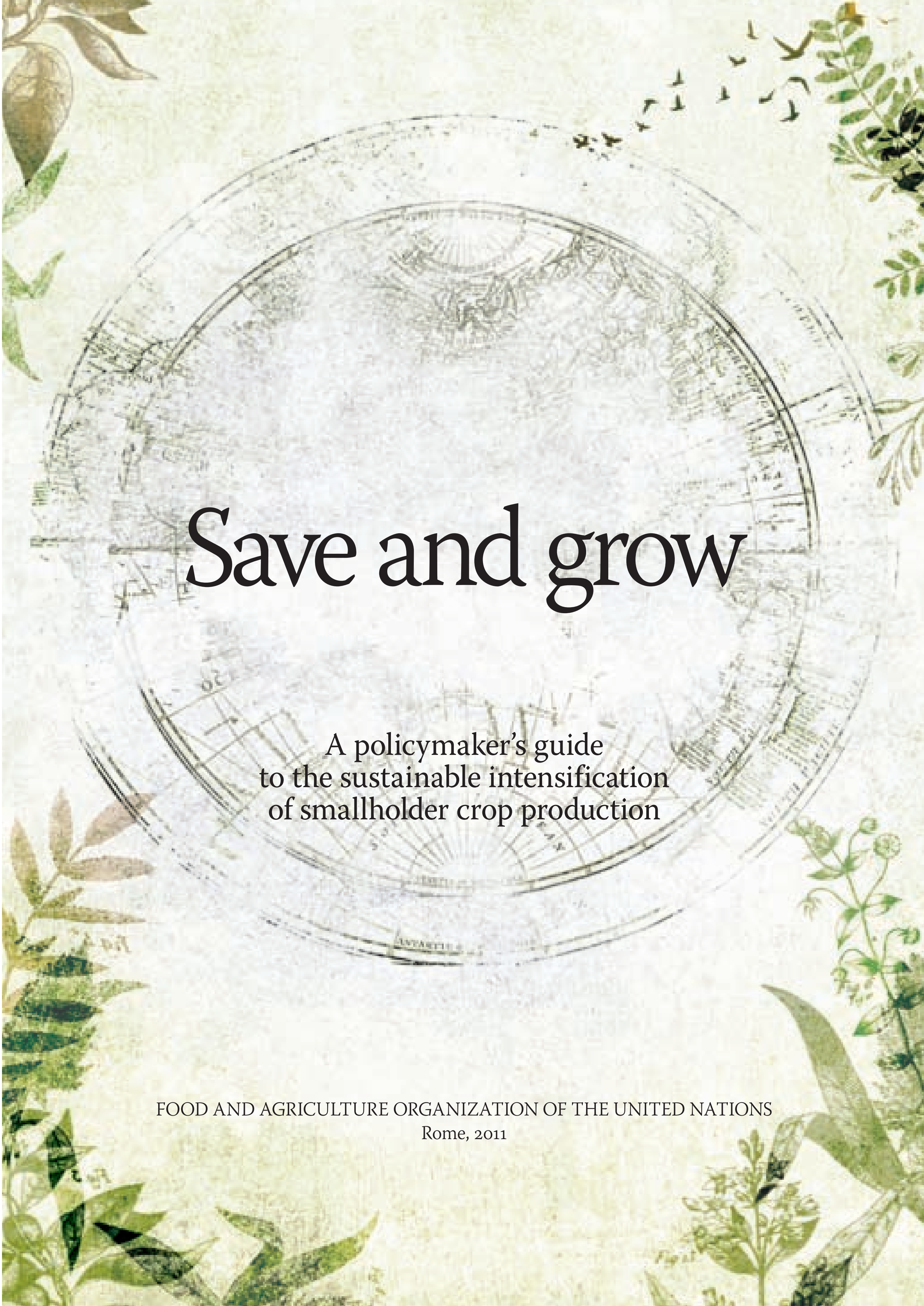

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Foreword

With the publication of *Save and Grow* in 2011, FAO proposed a new paradigm of intensive crop production, one that is both highly productive and environmentally sustainable. FAO recognized that, over the past half-century, agriculture based on the intensive use of inputs has increased global food production and average per capita food consumption. In the process, however, it has depleted the natural resources of many agro-ecosystems, jeopardizing future productivity, and added to the greenhouse gases responsible for climate change. Moreover, it has not significantly reduced the number of chronically hungry, which is currently estimated at 870 million people.

The challenge is to place food production and consumption on a truly sustainable footing. Between now and 2050, the global population is projected to rise from about 7 billion to 9.2 billion, demanding – if current trends continue – a 60 percent increase in global food production. Given the diminishing area of unused land with good agricultural potential, meeting that demand will require ever higher crop yields. Those increases, in turn, need to be achieved in the face of heightened competition for land and water, rising fuel and fertilizer prices, and the impact of climate change.

***Save and Grow* addresses** the crop production dimension of sustainable food management. In essence, it calls for "greening" the Green Revolution through an ecosystem approach that draws on nature's contributions to crop growth, such as soil organic matter, water flow regulation, pollination and bio-control of insect pests and diseases. It offers a rich toolkit of relevant, adoptable and adaptable ecosystem-based practices that can help the world's 500 million smallholder farm families to achieve higher productivity, profitability and resource use efficiency, while enhancing natural capital.

This eco-friendly farming often combines traditional knowledge with modern technologies that are adapted to the needs of small-scale producers. It also encourages the use of conservation agriculture, which boosts yields while restoring soil health. It controls insect pests by protecting their natural enemies rather than by spraying crops indiscriminately with pesticides. Through judicious use of mineral fertilizer, it avoids "collateral damage" to water quality. It uses precision irrigation to deliver the right amount of water when and where it is needed. The *Save and Grow* approach is fully consistent with the principles of climate-smart agriculture – it builds resilience to climate change and reduces greenhouse gas emissions through, for example, increased sequestration of carbon in soil.

For such a holistic approach to be adopted, environmental virtue alone is not enough: farmers must see tangible advantages in terms of higher incomes, reduced costs and sustainable livelihoods, as well as compensation for the environmental benefits they generate. Policymakers need to provide incentives, such as rewarding good management of agro-ecosystems and expanding the scale of publicly funded and managed research. Action is needed to establish and protect rights to resources, especially for the most vulnerable. Developed countries can support sustainable intensification with relevant external assistance to the developing world. And there are huge opportunities for sharing experiences among developing countries through South-South Cooperation.

We also need to recognize that producing food sustainably is only part of the challenge. On the consumption side, there needs to be a shift to nutritious diets with a smaller environmental footprint, and a reduction in food losses and waste, currently estimated at almost 1.3 billion tonnes annually. Ultimately, success in ending hunger and making the transition to sustainable patterns of production and consumption requires transparent, participatory, results-focused and accountable systems of governance of food and agriculture, from global to local levels.

This third reprint of *Save and Grow* comes following the Rio+20 Conference in June 2012 and the launch of the Zero Hunger Challenge by the United Nations Secretary-General, Ban Ki-moon. The challenge has five elements: guarantee year-round access to adequate food, end stunting in children, double small farmer productivity, foster sustainable food production systems, and reduce food waste and loss to zero. In assisting countries to adopt *Save and Grow* policies and approaches, FAO is responding to that challenge and helping to build the hunger-free world we all want.

José Graziano da Silva
Director-General
Food and Agriculture Organization
of the United Nations

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Acknowledgements

This book was produced under the direction of Shivaji Pandey, Director of FAO's Plant Production and Protection Division. Guidance was provided by a steering committee and a technical advisory group. Final technical editing was done by Mangala Rai (President of the National Academy of Agricultural Sciences, India), Timothy Reeves (former Director-General of the International Maize and Wheat Improvement Center), and Shivaji Pandey.

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Overview

1. The challenge

To feed a growing world population, we have no option but to intensify crop production. But farmers face unprecedented constraints. In order to grow, agriculture must learn to save.

The Green Revolution led to a quantum leap in food production and bolstered world food security. In many countries, however, intensive crop production has depleted agriculture's natural resource base, jeopardizing future productivity. In order to meet projected demand over the next 40 years, farmers in the developing world must double food production, a challenge made even more daunting by the combined effects of climate change and growing competition for land, water and energy. This book presents a new paradigm: sustainable crop production intensification (SCPI), which produces more from the same area of land while conserving resources, reducing negative impacts on the environment and enhancing natural capital and the flow of ecosystem services.

2. Farming systems

Crop production intensification will be built on farming systems that offer a range of productivity, socio-economic and environmental benefits to producers and to society at large.

The ecosystem approach to crop production regenerates and sustains the health of farmland. Farming systems for SCPI will be based on conservation agriculture practices, the use of good seed of high-yielding adapted varieties, integrated pest management, plant nutrition based on healthy soils, efficient water management, and the integration of crops, pastures, trees and livestock. The very nature of sustainable production systems is dynamic: they should offer farmers many possible combinations of practices to choose from and adapt, according to their local production conditions and constraints. Such systems are knowledge-intensive. Policies for SCPI should build capacity through extension approaches such as farmer field schools, and facilitate local production of specialized farm tools.

3. Soil health

Agriculture must, literally, return to its roots by rediscovering the importance of healthy soil, drawing on natural sources of plant nutrition, and using mineral fertilizer wisely.

Soils rich in biota and organic matter are the foundation of increased crop productivity. The best yields are achieved when nutrients come from a mix of mineral fertilizers and natural sources, such as manure and nitrogen-fixing crops and trees. Judicious use of mineral fertilizers saves money and ensures that nutrients reach the plant and do not pollute air, soil and waterways. Policies to promote soil health should encourage conservation agriculture and mixed crop-livestock and agro-forestry systems that enhance soil fertility. They should remove incentives that encourage mechanical tillage and the wasteful use of fertilizers, and transfer to farmers precision approaches such as urea deep placement and site-specific nutrient management.

4. Crops and varieties

Farmers will need a genetically diverse portfolio of improved crop varieties that are suited to a range of agro-ecosystems and farming practices, and resilient to climate change.

Genetically improved cereal varieties accounted for some 50 percent of the increase in yields over the past few decades. Plant breeders must achieve similar results in the future. However, timely delivery to farmers of high-yielding varieties requires big improvements in the system that connects plant germplasm collections, plant breeding and seed delivery. Over the past century, about 75 percent of plant genetic resources (PGR) has been lost and a third of today's diversity could disappear by 2050. Increased support to PGR collection, conservation and utilization is crucial. Funding is also needed to revitalize public plant breeding programmes. Policies should help to link formal and farmer-saved seed systems, and foster the emergence of local seed enterprises.

5. Water management

Sustainable intensification requires smarter, precision technologies for irrigation and farming practices that use ecosystem approaches to conserve water.

Cities and industries are competing intensely with agriculture for the use of water. Despite its high productivity, irrigation is under growing pressure to reduce its environmental impact, including soil salinization and nitrate contamination of aquifers. Knowledge-based precision irrigation that provides reliable and flexible water application, along with deficit irrigation and wastewater-reuse, will be a major platform for sustainable intensification. Policies will need to eliminate perverse subsidies that encourage farmers to waste water. In rainfed areas, climate change threatens millions of small farms. Increasing rainfed productivity will depend on the use of improved, drought tolerant varieties and management practices that save water.

6. Plant protection

Pesticides kill pests, but also pests' natural enemies, and their overuse can harm farmers, consumers and the environment. The first line of defence is a healthy agro-ecosystem.

In well managed farming systems, crop losses to insects can often be kept to an acceptable minimum by deploying resistant varieties, conserving predators and managing crop nutrient levels to reduce insect reproduction. Recommended measures against diseases include use of clean planting material, crop rotations to suppress pathogens, and eliminating infected host plants. Effective weed management entails timely manual weeding, minimized tillage and the use of surface residues. When necessary, lower risk synthetic pesticides should be used for targeted control, in the right quantity and at the right time. Integrated pest management can be promoted through farmer field schools, local production of biocontrol agents, strict pesticide regulations, and removal of pesticide subsidies.

7. Policies and institutions

To encourage smallholders to adopt sustainable crop production intensification, fundamental changes are needed in agricultural development policies and institutions.

First, farming needs to be profitable: smallholders must be able to afford inputs and be sure of earning a reasonable price for their crops. Some countries protect income by fixing minimum prices for commodities; others are exploring "smart subsidies" on inputs, targeted to low-income producers. Policymakers also need to devise incentives for small-scale farmers to use natural resources wisely – for example, through payments for environmental services and land tenure that entitles them to benefit from increases in the value of natural capital – and reduce the transaction costs of access to credit, which is urgently needed for investment. In many countries, regulations are needed to protect farmers from unscrupulous dealers selling bogus seed and other inputs. Major investment will be needed to rebuild research and technology transfer capacity in developing countries in order to provide farmers with appropriate technologies and to enhance their skills through farmer field schools.

CROP PRODUCTION GUIDE
AGRICULTURE
2020

Directorate of Agriculture
Chepauk, Chennai - 600 005
&
Tamil Nadu Agricultural University
Coimbatore - 641 003

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முகவுரை

வேளாண்மையில் பயிரிடும் பரப்பு குறைந்து வரும் இக்காலத்தில் பயிரின் உற்பத்தியை பெருக்கி உழவரின் இலாபத்தை அதிகரிப்பது என்பது மத்திய மற்றும் மாநில அரசுகளின் முக்கியமான நோக்கமாக உள்ளது. இதனை அடைய புதிய தொழில் நுட்பங்கள் முக்கிய பங்கு வகிக்கின்றன. பயிர்களின் உற்பத்தியை பெருக்குவதில் தமிழகம் முதன்மை பெற்று "கிரிஷி கருமன் விருதை" தொடர்ந்து ஐந்து முறை பெற்று மாநிலத்திற்கு பெருமை சேர்த்து தமிழ்நாடு வேளாண்மைப்பல்கலைக் கழகமும், தமிழக வேளாண்மைத் துறையும் சாதனை படைத்துள்ளது. இத்தகைய தொழில் நுட்பங்களை உழவர்களுக்கு கொண்டு செல்வது ஒரு முக்கியமான வளர்ச்சிப் பாதையாகும். இவ்வகையில் **பயிர் உற்பத்திக் கையேடு – வேளாண்மை 2020** ஒரு உடனடி தீர்வுக்கான பார்வைக் கையேடாக உள்ளது. இதில் தொகுக்கப்பட்டுள்ள தமிழ்நாடு வேளாண்மைப் பல்கலைக்கழக வேளாண் பயிர்களின் ஆராய்ச்சி முடிவுகளை கடைப்பிடிக்கப்படும் போது பல்வேறு பயிர்களின் உற்பத்தித்திறன் அதிகரிக்கும் என்பது திண்ணமாகும்.

முக்கிய பயிரான நெல் முதற்கொண்டு அனைத்து வேளாண் பயிர்களுக்கான காலநிலை, இரகங்கள், பயிர் மேலாண்மை, பயிர் பாதுகாப்பு மற்றும் விதை உற்பத்தி குறிப்புகள் இக்கையேட்டில் குறிப்பிடப்பட்டுள்ளன. இவை தவிர காளாண் உற்பத்தி, வேளாண்மை கழிவில் உர உற்பத்தி, கழிவு நீரை சுத்திகரித்து உபயோகித்தல், பட்டுப்புழு வளர்ப்பு, வேளாண் காடுகள், ஒருங்கிணைந்த பண்ணைய முறைகள், பிரச்சனைக்குரிய களை மேலாண்மை, இறுகிய நில மேலாண்மை, நீர்நுட்பங்கள், பண்ணைக்கருவி மற்றும் உபகரணங்கள், வேளாண் பதனிடும் தொழில் நுட்பங்கள் ஆகியவை இக்கையேட்டில் இடம் பெற்றுள்ளன.

இக்கையேடு வேளாண்மைத்துறை கள அலுவலர்களுக்கு பெரிதும் உதவியாக இருப்பது மட்டுமன்றி நிர்வாகத்தினர்களுக்கும், ஆராய்ச்சியாளர்களுக்கும், வேளாண் மாணாக்கர்களுக்கும் மிகவும் உதவியாக இருக்கும். இக்கையேட்டினை உருவாக்க உறுதுணையாக இருந்த அனைத்து தமிழ்நாடு வேளாண் பல்கலைக் கழக விஞ்ஞானிகளுக்கும் தொகுப்பாகிய ஆராய்ச்சி இயக்குனர் அவர்களுக்கும், வழிநடத்திய துணை வேந்தர் அவர்களுக்கும் எனது மனமார்ந்த மகிழ்ச்சியையும், நன்றியையும் தெரிவித்துக் கொள்கிறேன்.

(இரா. துரைக்கண்ணு)

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PREFACE

The Indian agriculture has taken a paradigm shift from conventional farming to precision agriculture with an intend to overcome a bundle of constraints such as shrinking arable lands, frequent droughts, declining organic matter, multi-nutrient deficiencies besides exodus of people from farming. Despite the fact that the country faces extreme weather conditions, the State of Tamil Nadu retained its glory by getting the **Krishi Karman Award 2019** for the fifth time in a row for continued enhancement of food grain and oilseed production. The data vividly indicate the successful accomplishments of both scientists of TNAU and officials in the Department of Agriculture in the past several years.

The Tamil Nadu Agricultural University, Coimbatore, is playing a pivotal role in design and development of crop varieties, technologies and farm implements to suit the prevailing situations and adapt to the climate change scenarios. In this context, there was an urgent need for standard operational protocols and technology package of practices to fit various agro-climatic zones of the state of Tamil Nadu. In this context, the TNAU under the dynamic leadership of the Vice Chancellor, Director of Research and other technical directors, the "**Crop Production Guide (CPG) – Agriculture 2020**" was revised, updated and put together as a technology capsule prescribed for adoption by the Department of Agriculture and in turn to the farmers of the State. While doing the revision, the Director of Agriculture and entire team of Department officials ensured that the field level problems are addressed.

The **CPG - Agriculture 2020** carries recommended crops and varieties, seed to seed crop management technologies, seed invigoration techniques, soil-test crop response based fertilizer prescription, drip fertigation, integrated weed management practices, technology capsule for pests and diseases management, labour saving farm machineries and post-harvest management practices.

I take this opportunity to thank and the Vice Chancellor, Director of Agriculture, Director of Research, other Technical Directors and Department officials for their assiduous efforts to orchestrate the document that serve as the base for the agricultural growth and development in the State of Tamil Nadu.

(Gagandeep Singh Bedi)

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PREFACE

The Tamil Nadu Agricultural University, Coimbatore, is one of the best State Agricultural Universities in the country, evolving several crop varieties, technologies and farm implements to promote productivity and profitability of farms while ensuring environmental safety and rural livelihood. The TNAU has a well articulated and structured research framework to develop varieties and technologies to enable adoption by the farmers of the State.

The Research Council of TNAU identifies the field problems that are developed as research projects, monitored and evaluated during the annual crop scientists meets and the research outcomes are presented to the State level Scientific Workers Conference to take the final decision on the varieties or technologies to be recommended for adoption. Such a meticulous planning and execution help us in assembling basket of varieties and technologies that are packaged as the Crop Production Guide.

The TNAU has taken efforts to revise and update the "**Crop Production Guide (CPG) – Agriculture 2020**" involving the Directors and Deans in the university besides Department of Agriculture and Department officials. In the past one year, there was a close coordination between the Department and University in undertaking joint efforts to resolve the unresolved field problems. Such network helped us to improve the CPG as a technology package suitable for the farmers of the Tamil Nadu State. The CPG carries recommended crop varieties, improved management technologies, precision farming practices, technology capsule for the management of pests and diseases besides post-harvest management practices.

I take this opportunity to thank the Agricultural Production Commissioner & Principal Secretary to the Government, Director of Agriculture, Director of Research, other Technical Directors, Deans and Department officials for their contribution towards the publication of the CPG - Agriculture 2020.

(N. KUMAR)

1. RICE (*Oryza sativa* L.)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
36 - 38	10 - 12	30 - 32	1000 - 4500	up to 2000

Tropical and sub tropical hot and humid climate. Minimum temperature required for germination, flowering and grain formation is 10, 23 and 20°C, respectively. Optimum temperature for growth, flowering and grain formation is 21 - 36, 25 - 29 and 20 - 25, respectively. Short day plant.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Varieties
I. Cauvery Delta Zone		
a. Thanjavur/ Tiruvarur		
Kuruvai	(Jun -Jul)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba	(Aug)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009, TRY 3
Late Samba / Thaladi	(Sep -Oct)	VGD 1, TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, CO 43 sub.1, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4
Navarai	(Dec -Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
b. Nagapattinam		
Late Kuruvai	(July)	ADT(R)48, MDU 5, CO 51, ADT 53, ADT(R)45, ADT 37, ADT 36
Samba	(Aug – Sep)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009,
		CO(R)50, CO 52, ADT (R)46, TKM 13, TNAU Rice TRY 3*, ADT 39, ADT 38, CO 43 sub 1, CO 43
Semi dry cultivation in Thanjavur, Thiruvarur and Nagapattinam	August	CR 1009, CR 1009 sub 1, ADT 51,TRY 3
	September	ADT 38, ADT 39, ADT 46, Co 50, Co 52, TKM 13, TRY 3
c. Tiruchirapalli		
Kuruvai	(Jun -Jul)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9, TRY 2*
Samba / Thaladi	(Aug -Sep)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009

		TNAU Rice TRY 3*, TRY 1*, CO 43, TKM 13, VGD 1, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, Imp.White Ponni, ADT (R) 46, TNAU Rice Hybrid CO 4
Summer	(Dec.)	ADT 53, CO 51, ADT (R) 45, TPS 5, ADT 36, ADT 37, CORH 3, ASD 16,
d. Flood affected areas	(Aug-Sep)	CR 1009 sub1
e.Salt affected areas		TRY 1, TRY 2, TRY 3 and Co 43
II. North Eastern Zone		
a. Kanchipuram/Tiruvallur		
Sornavari	(April -May)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba/ Late Samba	(Aug- Sep)	VGD 1, TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4
Navarai	(Dec -Jan)	ADT 53, CO 51, ADT (R) 45, MDU 6, ADT 36, ADT 37, CORH 3, TKM 9
Rainfed direct seeded and Semi-dry	(July - Aug)	Anna (R) 4, ADT 36, ADT 39, TKM 9, TKM 11
b. Vellore/Tiruvannamalai		
Sornavari	(April-May)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba	(Aug)	VGD 1, TKM 13, CO 52, Imp.White Ponni, ASD 19, TNAU Rice ADT 49, CO (R) 50, ADT 39, ADT 38, CO 43, ADT (R) 46, TNAU Rice Hybrid CO 4
Navarai	(Dec -Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
c. Cuddalore/ Villupuram		
Sornavari	(April -May)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5, MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba	(Aug)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009,
		VGD 1,TKM 13, CO 52, Imp.White Ponni,, TRY 3*, TNAU Rice Hybrid CO 4,CO (R) 50, TNAU Rice ADT 49, CO 43, TRY 1, ADT(R) 46, ADT 38
Navarai	(Dec-Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Delta regions of Cuddalore		
Samba	(Aug)	CR 1009, CR 1009 Sub 1, ADT 51
Late samba/Thaladi	(Sep-Oct)	ADT 38, ADT 39, ADT 46, Co 50, Co 52, TKM 13, Improved white ponni, TRY 3

Salt affected areas (Cuddalore)		TRY 1, TRY 2, TRY 3 and Co 43
III. Western zone		
a. Coimbatore/Tiruppur/Erode		
Kar	(May - Jun)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16
Samba /Late Samba	(Aug - Sep)	CO 52, TKM 13, VGD 1, CO 43, Imp.White Ponni, TNAU Rice Hybrid CO 4, CO (R) 50, ADT(R) 46, TNAU Rice ADT 49, ADT 39,
Navarai	(Dec -Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16
b. Karur/Perambalur/Ariyalur		
Kuruvai	(Jun -Jul)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba	(Aug)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009
Late Samba / Thaladi	(Sep -Oct)	VGD 1,TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4, TRY 1*
Navarai	(Dec -Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Delta regions of Karur		
Late Samba	(Sep -Oct)	ADT 38, ADT 39, ADT 46, Co 50, Co 52, TKM 13, Improved white ponni, TRY 3
Delta regions of Ariyalur		
Samba	(Aug)	CR 1009, CR 1009 Sub 1, ADT 51
Late samba/Thaladi	(Sep-Oct)	ADT 38, ADT 39, ADT 46, Co 50, Co 52, TKM 13, Improved white ponni, TRY 3
IV. North Western Zone		
a. Salem/Namakkal		
Kar	(May - Jun)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba	(Aug)	VGD 1,TKM 13, CO 52, Imp.White Ponni, CO 43, TRY 1*, TNAU Rice TRY 3*, CO (R) 50, TNAU Rice ADT 49, TNAU Rice Hybrid CO 4
Navarai	(Dec - Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
b. Dharmapuri/ Krishnagiri		
Kar	(May -Jun)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9
Samba/Late Samba	(Aug - Oct)	VGD 1,TKM 13, CO 52, Paiyur 1, Imp.White Ponni, TNAU Rice ADT 49, ADT 39, ASD 19, CO 43
Navarai	(Dec- Jan)	ADT 53, CO 51, ADT (R) 45, TPS 5 , MDU 6, ADT 36, ADT 37, CORH 3, ASD 16, TKM 9

V. High Altitude zone		
a. The Nilgiris		
Samba	(Jul -Aug)	CO(R)50, CO 52, ADT 39
VI. Southern zone		
a. Pudukottai		
Kuruvai	(Jun -Jul)	ADT 53, CO 51, ADT 43, ADT (R) 45, TPS 5 , ADT 36, ADT 37, CORH 3, ASD 16
Samba	(Aug)	CR 1009 *Sub*1, TNAU Rice ADT 50, ADT 51, CR 1009
Late Samba/Thaladi	(Sep - Oct)	VGD 1,TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4
Rainfed direct seeded/ Semi-dry	(Jul -Aug)	CR 1009 *Sub1,* CR 1009, ADT 39
Delta regions	(Sep -Oct)	ADT 38, ADT 39, ADT 46, Co 50, Co 52, TKM 13, Improved white ponni, TRY 3
Salt affected areas		TRY 1, TRY 2, TRY 3 and Co 43
b. Madurai/Dindigul/Theni		
Kar	(May -Jun)	MDU 6, CO 51, ADT 53, ASD 16, MDU 5, ADT43, ADT(R)45
Samba/ Late Samba	(Aug- Sep)	VGD 1,TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4
Navarai	(Dec -Jan)	MDU 6, CO 51, ADT 53, ADT 36, ADT 37, ADT(R)45, ASD 16
Semi-dry	(Jul -Aug)	Anna (R) 4, MDU 5, PMK (R) 3
c. Ramanathapuram		
Samba	(Aug)	Imp.White Ponni, TNAU Rice TRY 3, TRY 1*, VGD 1,TKM 13, CO 52, TNAU Rice Hybrid CO 4, CO (R) 50, CO 43, ADT 39
Rainfed direct seeded & Semidry	(Jul -Aug)	Anna (R) 4, MDU 5, MDU 6, PMK (R) 3, ADT 36, ADT 53, CO 51
d. Virudhunagar		
Samba	(Sep-Oct)	VGD 1,TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White Ponni, ADT (R) 46, TNAU Rice TRY 3*, TNAU Rice Hybrid CO 4
Rainfed direct seeded	(Jul -Aug)	Anna (R) 4, ADT 36, MDU 6, PMK (R) 3, ADT (R) 45
e. Sivaganga		
Semi-dry	(Jul –Aug)	ADT 36, MDU 6, PMK (R) 3, Anna (R) 4, ADT 53, CO 51, ADT 39, TKM 13, VGD 1
f. Tirunelveli, Thoothukudi		
Early kar	(Apr - May)	TPS 5 , ASD 16, ASD 18, ADT 53, CO 51, ADT 43, ADT (R) 45, ADT 36, ADT 37, CORH 3, TKM 9
Kar	(May -Jun)	
Pishanam/Late Pishanam	(Sep-Oct.)	TPS 3, ASD 19, VGD 1, TKM 13, CO 52, CO (R) 50, ADT 39, ADT 38, TNAU Rice ADT 49, CO 43, Imp.White

		Ponni, ADT (R) 46, TRY 1, TNAU Rice Hybrid CO 4
Semi dry	(July- Aug)	Anna (R) 4, PMK (R) 3
Drought affected areas (Ramanathapuram, Virudhunagar, Tiruvallur& parts of Madurai)		Anna (R) 4, PMK 3
VII. High Rainfall zone		
a. Kanyakumari		
Kar	(May –Jun)	TPS 5, ASD 16, ADT 36, ASD 18, ADT 43,ADT(R) 45, CORH 3, ADT 53, CO 51,
Pishanam / Late samba	(Sep – Oct)	ASD 19, TPS 3, CR 1009 *Sub1*, CR 1009, CO 43, TRY 1*, TNAU Rice TRY 3, VGD 1,TKM 13, ADT 39, CO 52, ADT (R) 46, TNAU Rice Hybrid CO 4, CO(R) 50, Imp.White Ponni, CO 43
Semi-dry	(Jul – Aug)	ADT 36, TKM 9

* suitable for salt affected soils

***Note of Caution of the varieties**: ADT43 is recommended for Kar, Sornavari and Kuruvai seasons and should not be grown during cold weather period. Improved white ponni is also susceptible to blast and care should be taken on plant protection measures. All samba/late samba season varieties are likely to get infected with false smut and hence prophylactic spraying has to beadopted.*

Kuruvai/Navarai/Sornavari : Short duration
late samba/thaladi : Medium duration
Samba : long duration

II. PARTICULARS OF RICE VARIETIES
SHORT DURATION VARIETIES

PARTICULARS	**CO 51**	**MDU 6**	**TPS 5**
Year of Release	2013	2015	2014
Year of Notification	SO.268(E)/28.1.2015 (SVRC) SO.1007(E)/30.3.2017(CVRC)	SO.1379(E)/27.03.2018	SO.1556(E)/11.06.2015
Parentage	ADT 43 / RR 272-1745	MDU 5 / ACM 96136	ASD 16 / ADT 37
Duration (Days)	105-110	115-120	118
Average Yield (kg/ha)	6641	6118	6301
1000 grain wt (g)	16.0	17.3	22.7
Grain L/B ratio	3.0	3.09	2.3
Grain type	Medium Slender	Long Slender	Short bold
Morphological Characters			
Habit	Semi dwarf, erect	Erect ,good tillering	Erect
Leaf sheath	Green	Green	Green
Septum	-	Green	Green
Ligule	-	Pale green	Light green

Auricle	Pale Green	Pale green	Light green
Panicle	Intermediate, droopy	Intermediate, droopy	Well exserted panicle
Husk colour	Straw	Straw	Straw
Rice colour	White	White	White
Abdominal white	Occasionally present	Occasionally present	Occasionally present
Grain size (mm)			
Length	5.5	6.8	6.1
Breadth	1.8	2.2	2.7
Thickness	--	--	
Seed source	Seed centre,TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore	Seed centre, TNAU, Coimbatore
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.

PARTICULARS	**ADT 36**	**ADT 37**	**ADT 43**
Year of Release	1980	1987	1998
Year of Notification	SO 19(E)/ 14.01.1982	SO.280(E)/ 13.04.1989	SO.425(E)/ 8.6.1999
Parentage	Triveni/ IR 20	BG 280-1 2/ PTB 33	IR 50/ Imp. White Ponni
Duration (Days)	110	105	110
Average Yield (kg/ha)	5500	6200	5900
1000 grain wt (g)	20.6	23.4	15.5
Grain L/B ratio	3.1	1.79	2.81
Grain type	Medium	Short bold	Medium slender
Morphological Characters			
Habit	Semi dwarf,Erect	Semi dwarf, Erect	Semi dwarf, slightly open
Leaf sheath	Green	Green	Light green
Septum	Green	White	Cream
Ligule	Colourless	White	White
Auricle	Colourless	White	-
Panicle	Long compact	Compact	Moderately long, Intermediate type,

			drooping
Husk colour	Straw	Straw	Straw
Rice colour	White	White	White
Abdominal white	Absent	White, Present	Very occasionally present
Grain size (mm)			
Length	7.8	5	5.46
Breadth	2.5	2.8	1.94
Thickness	2.0	1.88	1.63
Seed source	Seed centre, TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.

PARTICULARS	**ADT (R) 45**	**ASD 16**	**ASD 18**
Year of Release	2001	1986	1991
Year of Notification	SO.1134(E)/15.11.2001	SO.867(E)/26.11.1986	SO.615(E)/17.8.1993
Parentage	IR50 / ADT 37	ADT 31/CO 39	ADT 31/IR 50
Duration (Days)	110	110 - 115	105 - 110
Average Yield (kg/ha)	5400	5600	5900
1000 grain wt (g)	17.5	24.2	21.8
Grain L/B ratio	2.98	2.6	3.2
Grain type	Medium slender	Short Bold	Medium slender
Morphological Characters			
Habit	Semi dwarf, erect	Semi dwarf,erect	Semi dwarf
Leaf sheath	Green	Green	Pale Green
Septum	Cream	Green	Light green
Ligule	White	White	White clefted
Auricle	-	Colourless	Pale green
Panicle	Compact	Long Compact	Medium, compact exerted

Husk colour	Straw	Straw	Straw
Rice colour	White	White	White
Abdominal white	Absent	Present	Slightly present
Grain size (mm)			
Length	8.00	7.86	8.64
Breadth	2.16	3.02	2.7
Thickness	1.97	1.96	2.2
Seed source	Seed centre, TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.

PARTICULARS	**Anna (R) 4**	**CORH 3 (hybrid)**	**ADT 53**
Year of Release	2009	2006	2019
Year of Notification	SO.2137(E)/ 31.08.2010	SO.1178(E)/ 20.7.2007	SO.3220(E)/ 5.9.2019
Parentage	Pantdhan 10 x IET 9911	TNAU CMS 2A/CB 87R	ADT 43 / JGL 384
Duration (Days)	105-110	110-115	110-115
Average Yield (kg/ha)	3700	7500	6334
1000 grain wt (g)	25.7	22.0	14.5
Grain L/B ratio	3.45	2.95	3.1
Grain type	Long slender	Medium slender	Medium Slender
Morphological Characters			
Habit	Semidwarf erect	Semi dwarf	Medium tall, erect
Leaf sheath	Green	Green	Green
Septum	-	-	Cream
Ligule	-	-	White, Split shape
Auricle	Pale green	Pale green	Light green
Panicle	Intermediate	Long, compact, drooping	Intermediate Compact
Husk colour	Straw	Straw	Straw
Rice colour	White	White	White
Abdominal white	Absent	Occasionally present	Absent

Grain size (mm)			
Length	6.90	6.2	5.8
Breadth	2.00	2.1	1.9
Thickness	-	1.2	1.02
Seed source	Seed centre,TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders.

PARTICULARS	**TKM 11**	**TRY 2**
Year of Release	1998	2001
Year of Notification	SO.425(E)/8.6.1999	SO.1134(E)/15.11.2001
Parentage	C 22/BJ 1	IET6238/IR36
Duration (Days)	110-120	115-120
Average Yield (kg/ha)	3000	5362
1000 grain wt (g)	21.4	22.8
Grain L/B ratio	3.2	3.5
Grain type	Long slender	Long slender
Morphological Characters		
Habit	Erect	Semi dwarf,erect
Leaf sheath	Green	Green
Septum	cream	Light green
Ligule	Colourless	Distinct
Auricle	Light green	Hairy light brown
Panicle	Long, compact, drooping	Compact
Husk colour	-	Straw
Rice colour	White	White
Abdominal white	-	Absent
Grain size (mm)		
Length	9.3	9.1

Breadth	2.3	2.6
Thickness	1.6	1.7
Seed source	Seed centre,TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

PARTICULARS	**ADT (R) 48**	**MDU 5**	**PMK (R) 3**	**TKM 9**
Year of Release	2005	*1996*	2003	1978
Year of Notification	SO.599(E)/ 25.04.2006	SO.662(E)/ 17.09.1997	SO.1177(E)/ 25.08.2005	SO.19(E)/ 14.01.1982
Parentage	IET 11412/IR 64	*O.glaberrima*/ Pokkali	UPLRI 7/CO 43	TKM 7 / IR 8
Duration(Days)	94-99	95 - 100	110-115	100-105
Average Yield (Kg / ha)	4800	4500	3025	5019
1000 grain wt(g)	22.0	21.1	26.10	25.13
GrainL/B ratio	3.25	3.12	2.64	2.71
Grain type	Long slender	Medium slender	Long bold	Short bold
Morphological characters				
Habit	Semidwarf erect	Erect	Erect	Dwarf
Leaf sheath	Green	Green	Green	-
Septum	Cream	-	-	Light blue
Ligule	Acute, prominent	Colourless	Pale green	-
Auricle	-	Colourless	-	-
Panicle	Intermediate	Intermediate	Intermediate	Compact
Husk colour	Straw	Straw	Gold yellow with brown streaks	Straw
Rice colour	White	White	White	Red
Abdominal white	Occassionally present	-	-	Present
Grain size(mm)				
Length	9.15	8.45	6.75	8.12
Breadth	2.54	2.7	2.38	2.99
Thickness	1.90	-	2.08	2.01
Seed source	Seed centre, TNAU,	Seed centre,TNAU,	Seed centre, TNAU,	Seed centre, TNAU,

	Coimbatore-3	Coimbatore-3	Coimbatore-3	Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

MEDIUM DURATION VARIETIES

PARTICULARS	Rice CO 52	TKM 13	TNAU Rice TRY 3	VGD 1
Year of Release	2017	2015	2010	2019
Year of Notification	SO.1379(E)/27.03.2018	SO.3540(E)/ 22.11.2016	SO.1708(E)/26.07.2012	SO.3220(E)/5.9.2019
Parentage	BPT 5204 / CO(R) 50	WGL 32100 / Swarna	ADT 43 / *Jeeraga Samba*	ADT 43/ Seeragasamba
Duration (Days)	130-135	130	135	130 - 135
Average Yield kg/ha	6191	5938	5833	5859
1000 grain wt (g)	14.10	13.8	23.0	8.8 to 8.9
Grain L/B ratio	3.0	2.83	2.58	2.1
Grain type	Medium Slender	Medium Slender	Medium	Short bold
Morphological Characters				
Habit	Erect, Medium Tall	Semi dwarf, erect, non- lodging	Intermediate erect	Semi dwarf, erect
Leaf sheath	Green	Green	Green	Green
Septum	--	Cream	--	Cream
Ligule	--	Split, White	Cleft, White	White
Auricle	White	Present, Colourless	Light Green	light green
Panicle	Long, compact, Droopy	Well exerted, Compact	Intermediate, Compact	Compact and drooping at maturity
Husk colour	Straw	Straw	Straw	Straw
Rice colour	White	White	White	White
Abdominal white	Occasionally present	Occasionally present	Occasionally present	Absent
Grain size (mm)				
Length	5.5	5.44	6.2	3.7

Breadth	1.8	1.92	2.4	1.8
Thickness	--	--	1.5	1.25
Seed source	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders
PARTICULARS	**TNAU Rice ADT 49**	**CO (R) 50**	**ADT 39**	**ADT 38**
Year of Release	2011	2010	1988	1987
Year of Notification	SO.1708(E)/ 26.07.2012	SO.1708(E)/ 26.07.2012	SO.280(E)/ 13.04.1989	SO.280(E)/ 13.04.1989
Parentage	CR1009/ Jeeraga Samba	CO 43 / ADT 38	IR 8/IR 20	IR 1529-680-3-2/ IR 4432-52-6-4/ IR 7963-30-2
Duration (Days)	130- 135	130-135	120 - 125	130 - 135
Average Yield kg/ha	6173	6338	5000	6200
1000 grain wt (g)	14.0	20.5	18	21
Grain L/B ratio	2.77	2.90	2.9	3.2
Grain type	Medium Slender	Medium	Medium slender	Long Slender
Morphological Characters				
Habit	Semi dwarf, Erect	Medium tall with New plant type	Semi dwarf	Semi dwarf, erect
Leaf sheath	Green	Green	Green	Green
Septum	Cream	-	Light Cream	White
Ligule	Split , white	-	Papery white	White Non-prominent
Auricle	Colourless	Pale green	Non-pigmented	White
Panicle	Compact	Long compact droopy	Medium, Modera-tely dense	Long moderately
Husk colour	Straw	Straw	Straw	dense
Rice colour	White	White	White	Straw
Abdominal white	Occasionally present	Occasionally present	Absent	White

Grain size (mm)				Absent
Length	7.36	6.10	7.6	6.9
Breadth	2.24	2.10	2.3	2.4
Thickness	1.69	-	1.9	2
Seed source	Seed centre, TNAU, Coimbatore	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

PARTICULARS	**Imp. White Ponni**	**ADT (R) 46**	**CO 43**	**Paiyur 1**
Year of Release	1986	2002	1982	1982
Year of Notification	SO.280(E)/ 13.04.1989	SO.1177(E)/ 25.08.2005	SO.596(E)/ 13.8.1984	SO.596(E)/ 13.8.1984
Parentage	Taichung 65/2 MayangEbos*80	ADT38 / CO 45	Dasal x IR 20	IR 1721-14/IR 1330-3-3-2
Duration (Days)	135 - 140	135	135 - 140	135-140
Average Yield kg/ha	4500	6656	5200	5900
1000 grain wt (g)	16.4	23.8	20	-
Grain L/B ratio	3.22	3.12	3.5	-
Grain type	Medium slender	Long Slender	Medium slender	Medium slender
Morphological Characters				
Habit	Medium tall	Erect, semi-dwarf	Erect	Medium tall
Leaf sheath	Green	Green	Green	-
Septum	Green	Cream	Green	-
Ligule	White	Long white	White, longer	-
Auricle	Colourless	Pale green	Colourless	-
Panicle	Long drooping	Intermediate	Long drooping	-
Husk colour	Straw	Straw	Straw	Straw
Rice colour	White	White	White	White
Abdominal white	Absent	Absent	Absent	Absent
Grain size (mm)				

Length	8	9.58	8.1	-
Breadth	3	2.46	2.3	-
Thickness	2	1.95	1.8	-
Seed source	Seed centre,TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

PARTICULARS	**TPS 3**	**TNAU Rice hybrid CO 4**	**TRY 1**	**ASD 19**
Year of Release	1993	2011	1995	1995
Year of Notification	SO.360(E)/ 1.5.1997	SO.1708(E)/ 26.07.2012	SO.92(E)/ 2.2.2001	SO.360(E)/ 1.5.1997
Parentage	RP31-492/LMN	TNAU CMS 23 A / CB 174 R	IR578-172-2-2/ BR-1-2-B-1	Lalnakanda/ IR 30
Duration (Days)	135-140	130 - 135	135-140	127 (120-132)
Average Yield (kg/ha)	5253	7348	5255	5800
1000 grain wt (g)	23.2	20.40	24	18.39
Grain L/B ratio	2.06	2.96	2.6	3.06
Grain type	Short bold	Medium slender	Medium	Short, slender
Morphological Characters				
Habit	Semi dwarf/erect	Semi dwarf	Erect	Semi-dwarf, erect
Leaf sheath	Green	Green	Green	Light green
Septum	Cream	-	White	Cream
Ligule	-	-	White	White
Auricle	-	Pale green	White	Palegreen
Panicle	Long	Long compact droopy	Long, moderately compact	Compact, dense drooping & well exerted
Husk colour	Straw	Straw	Straw	Straw
Rice colour	White	White	White	White
Abdominal white	Present	Occasionally present	Absent	Absent
Grain size (mm)				
Length	7.96	5.67	6.2	8.28
Breadth	3.0	1.91	2.4	2.32

Thickness	2.0	-	1.8	1.72
Seed source	Seed centre, TNAU, Coimbatore-3	Seed centre, TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3	Seed centre,TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

LONG DURATION VARIETIES

PARTICULARS	**Rice ADT 51**	**CR 1009 *Sub* 1**	**TNAU Rice ADT 50**	**CR 1009**
Year of Release	2017	2015	2012	1982
Year of Notification	S.O. 6318(E) /26.12.2018	SO.3540(E)/22. 11.2016	SO.268(E)/28.01.2 015	SO.499(E)/08.07. 1983
Parentage	BPT 5204 / I.W.Ponni	CR 1009 / FR 13 A (MAB)	BPT 5204 / CR 1009	Pankaj/Jagannat h
Duration (Days)	154	150-155	149	155 - 160
Average Yield (kg/ha)	6587	5759	5945	5300
1000 grain wt (g)	23.9	23.0	15.9	23.5
Grain L/B ratio	2.74	2.05	2.56	2.2
Grain type	Medium	Short bold	Medium Slender	Short bold
Morphological Characters				
Habit	Erect Semi dwarf	Semi dwarf tolerance to submergence	Medium tall	Erect
Leaf sheath	Green	Green	Green	Green
Septum	Cream	-	Cream	Green
Ligule	White	-	Split , white	White
Auricle	Present, Light green	Pale Green	Absent	Colourless
Panicle	Well exerted, compact panicle	Intermediate	Compact	Medium drooping
Husk colour	Straw	Straw	Straw	Straw
Rice colour	White	White	White	White
Abdominal white	Absent	Occasionally present	Occasionally present	Absent
Grain size (mm)				
Length	6.3	5.06	7.24	6.9
Breadth	2.3	2.46	3.50	3.1
Thickness	1.56	--	1.65	2.1
Seed source	Seed centre, TNAU,	Seed centre, TNAU,	Seed centre,	Seed centre,

	Coimbatore-3	Coimbatore-3	TNAU, Coimbatore	TNAU, Coimbatore-3
Marketability	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders	Bold variety : Direct procurement centre Fine variety : Direct procurement centre/private traders

CROP MANAGEMENTSYSTEMS OF RICE CULTIVATION IN TAMIL NADU

Rice is cultivated under **puddled** and **un-puddled lowland** situations in Tamil Nadu. 'Transplanting' and 'direct wet seeding' are the two environments under puddled lowland. Whereas, un-puddled lowland cultivation undergoes different environments like, dry seeding exclusively with rainfall, locally called as 'rainfed rice', with supplemental irrigation during peak vegetative and reproductive phases by the rain water collected / harvested in tanks ('semi-dry rice') and also assured irrigation from canal after 30-45 days of dry situation (also called semi-dry rice) . They are grouped as follows:

1. **Transplanted puddled lowland rice**
2. **Direct seeded lowland rice**
   a. Wet seeded rice in puddled soil
   b. Dry seeded rice in un-puddled soil
      i) Rainfed
      ii) Semi dry - supplemental irrigation
      iii) Semi dry - canal irrigation* (contingent crop)

3. **Dry seeded upland rice** – This system of rice cultivation is there in areas with high rainfall (like Assam and NE frontiers of India) where the land is slopy and terraced and there is no possibility for bunding to stagnate the water. Grain yield is poor due to loss of nutrients and soil mainly caused by water erosion. Moisture availability is mostly at saturation or at wet range. There is very limited area in Dharmapuri district, Tamil Nadu.

4. **Deep water rice** cultivation exists in certain pockets of Nagapattinam and Tiruvarur districts particularly during NE monsoon with heavy downpour.

1. TRANSPLANTED PUDDLED LOWLAND RICE

TRANSPLANTED RICE

1.1 Nursery management

1.1.1. Wet nursery Nursery area

Select 20 cents (800 m^2) of land area near to water source for raising seedlings for one hectare.

Seed rate

30 kg for long duration
40 kg for medium duration
60 kg for short duration varieties and 20 kg for hybrids

Seed treatment

a. Treat the seeds in Carbendazim or Pyroquilon or Tricyclozole solution at 2 g/l of water for 1 kg of seeds. Soak the seeds in water for 10 hrs and drain excess water.
b. This wet seed treatment gives protection to the seedlings up to 40 days from seedling disease such as blast and this method is better than dry seed treatment.
c. If the seeds are required for sowing immediately, keep the soaked seed in gunny in dark and cover with extra gunnies and leave for 24hrs for sprouting.
d. **Seed treatment with *Pseudomonas fluorescens***: Treat the seeds with talc based formulation of *Pseudomonas fluorescens* 10g/kg of seed and soak in 1lit of water overnight. Decant the excess water and allow the seeds to sprout for 24hrs and then sow.
e. **Seed treatment with biofertilizers :** Five packets (1kg/ha) each of *Azospirillum* and Phosphobacteria or five packets (1kg/ha) of Azophos bioinoculants are mixed with sufficient water wherein the seeds are soaked overnight before sowing in the nursery bed (The bacterial suspension after decanting may be poured over the nursery area itself).

   Carrier based formulation: Treat one hectare of seeds with 1 kg each of biofertilizers *viz., Azsopirillum,* Phosphobacteria, (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) using rice gruel, shade dry for 30 minutes before sowing.

   Liquid formulation : Treat one hectare of seeds with 125 ml of each biofertilizers *viz., Azsopirillum,* Phosphobacteria (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) shade dry for 30 minutes before sowing.

   - Biocontrol agents are compatible with biofertilizers
   - Biofertilizers and biocontrol agents can be mixed together for seed soaking
   - Fungicides and biocontrol agents are incompatible

Forming Seedbeds

- Mark plots of 2.5m breadth with channels 30cm wide all around the seedbeds.
- Length of the seed bed may vary from 8 to 10m according to soil and slope of the land.
- Collect the puddled soil from the channel and spread on the seedbeds or drag a heavy stone along the channel to lower it, so that the seed bed is at a higher level.
- Level the surface of the seedbed, so that the water drains into the channel.

Sowing

- Sow the sprouted seeds uniformly on the seedbed having thin film of water in the surface.

Water Management

- Drain the water 18 to 24 hrs after sowing
- Care must be taken to avoid stagnation of water on the seedbed.
- Allow enough water to saturate the soil from 3rd to 5th day. From 5th day onwards, increase the water depth to 1.5 cm depending on the height of the seedlings.
- Thereafter maintain 2.5 cm depth of water.

Weed Management

- Apply any one of the pre-emergence herbicides *viz*., Pyrazosulfuron ethyl @ 20 g/ha on 3rd or 4th day after sowing to control weeds in the lowland nursery. Keep a thin film of water and allow it to disappear. Avoid drainage of water. This will control germinating weeds.
- Pre-emergence herbicide Butachlor 1.0 l/ha (or) Pendimethalin 1.0 l/ha. Herbicides should be applied on 8 DAS with thin layer of water in the field.

Nutrient management

- Apply 1 tonne of fully decomposed FYM or compost to 20 cents nursery and spread the manure uniformly on dry soil.
- Basal application of DAP is recommended when the seedlings are to be pulled out in 20-25 days after sowing in less fertile nursery soils.
- For that situation, before the last puddling, apply 40 kg of DAP and if not readily available, apply straight fertilizers 16 kg of urea and 120 kg of super phosphate.
- If seedlings are to be pulled out after 25 days, application of DAP is to be done 10 days prior to pulling out.
- For clayey soils where root snapping is a problem, 4 kg of gypsum and 1 kg of DAP/cent can be applied at 10 days after sowing.
- Soil application of 100 g $ZnSO_4$ /cent can be followed.

1.1.2. Dry nursery

- Dry ploughed field with fine tilth is required.
- Nursery area with sand and loamy soil status is more suitable for this type of nursery.

- Area 20 cents.
- Plots of 1 to 1.5 m width of beds and channels may be formed. Length may be according to the slope and soil. Raised beds are more ideal if the soil is clayey in nature.
- Seed rate and seed treatment as that of wet nursery.
- Sowing may be dry seeding. Seeds may be covered with sand and finely powdered well decomposed farm yard manure.
- Irrigation may be done to wet the soil to saturation.
- Optimum age for transplanting – 4^{th} leaf stage
- This type of nursery is handy in times of delayed receipt of canal water.
- During transplanting seedlings may be dipped in 2% $ZnSO_4$ or ZnO for 30 min and then transplanted.

1.2. Main Field Management

1.2.1. Land preparation

- Plough the land during summer to economize the water requirement for initial preparation of land.
- Flood the field 1 or 2 days before ploughing and allow water to soak in. Keep the surface of the field covered with water.
- Keep water to a depth of 2.5cm at the time of puddling.
- Special technologies for problem soils:
  a) For fluffy paddy soils: compact the soil by passing 400kg stone roller or oil-drum with stones inside, eight times at proper moisture level (moisture level at friable condition of soil which is approximately 13 to18%) once in three years, to prevent the sinking of draught animals and workers during puddling.
  b) For sodic soils with pH values of more than 8.5, plough at optimum moisture regime, apply gypsum at 50% gypsum requirement uniformly, impound water, provide drainage for leaching out soluble salts and apply green leaf manure at 5 t/ha, 10 to 15 days before transplanting. Mix 37.5 kg of zinc sulphate per ha with sand to make a total quantity of 75 kg and spread the mixture uniformly on the leveled field. Do not incorporate the mixture in the soil. Rice under sodic soil responds well to these practices.
  c) For saline soils with EC values of more than 4 dS/m, provide lateral and main drainage channels (60cm deep and 45cm wide), apply green leaf manure at 5 t/ha at 10 to 15 days before transplanting and 25% extra dose of nitrogen in addition to recommended P and K and $ZnSO_4$
  d) For acid soils apply lime based on the soil analysis for obtaining normal rice yields. Lime is applied 2.5 t/ha before last ploughing. Apply lime at this rate to each crop up to the 5^{th} crop.

1.2.2.Stand Establishment

Optimum age of seedlings for quick establishment

- Optimum age of the seedlings is 18-22 days for short, 25-30 days for medium and 35-40 days for long duration varieties.

Pulling out the seedlings

- Pull out the seedlings at the appropriate time (4^{th} leaf stage).
- Pulling at 3^{rd} leaf stage is also possible. These seedlings can produce more tillers, provided enough care taken during the establishment phase (See section 1.8 Integrated Crop Management (ICM) - Rice-SRI) through thin film of water management and perfect leveling of main field.
- Transplanting after 5^{th} and higher order leaf numbers will affect the performance of the crop and grain yield. Then they are called as 'aged seedlings'. Special package is needed to minimize the grain yield loss while planting those aged seedlings.

Root dipping

- Prepare the slurry with 5 packets (1 kg/ha) each of *Azospirillum* and Phosphobacteria or 5 packets of (1 kg/ha) Azophos inoculant in 40 lit. of water and dip the root portion of the seedlings for 15 - 30 minutes in bacterial suspension and transplant.

Planting seedlings in the main field

Soil	**Medium and low fertility**			**High fertility**		
Duration	Short	Medium	Long	Short	Medium	Long
Spacing (cm) Hills / m^2	15x10 66	20x10 50	20x15 33	20x10 50	20x15 33	20x20 25

- Transplant 2-3 seedlings/hill for short duration and 2 seedlings/hill for medium and long duration varieties
- Shallow planting (3 cm) ensures quick establishment and more tillers.
- Deeper planting (> 5cm) leads to delayed establishment and reduced tillers.
- Line planting permits rotary weeding and its associated benefits.
- Allow a minimum row spacing of 20 cm to use rotary weeder.
- Fill up the gaps between 7^{th} and 10^{th} DAT.

Management of Aged seedlings*

* **Which developed tillers / underwent node elongation in the nursery itself and About half of its leaf producing capacity may be already over.**

- Follow the spacing recommended to medium and low fertility soil
- Plant two to three seedlings per hill
- Avoid cluster planting of aged seedlings, which are hindering the formation of new tillers.
- New tillers alone are capable of producing normal harvestable panicle. Weak panicle may appear in the mother culm within three weeks after transplanting and vanishes well before harvest.
- To encourage the tiller production, enhance the basal N application by 50% from the recommended and thereafter follow the normal schedule recommended for other stages.

Gap filling

- Fill the gaps if any within 7 - 10 days after planting.Nutrient management Application of organic manures
- Apply 12.5 t of FYM or compost or green leaf manure @ 6.25 t/ha.

- If green manure is raised @ 50 kg seeds/ha *in situ,* incorporate it to a depth of 15 cm using a green manure trampler or tractor.
- In the place of green manure, press-mud / composted coir-pith can also be used.

1.2.3. Nutrient Management

Stubble incorporation

- Apply 10 kg N/ha (22 kg urea) at the time of first puddling while incorporating the stubbles of previous crop to compensate immobilization of N by the stubbles.
- This may be done at least 10 days prior to planting of subsequent crop. This recommendation is more suitable for double crop wetlands, wherein, the second crop is transplanted in succession with short turn around period.

Biofertilizer application

- Broadcast 10 kg of soil based powdered BGA flakes at 10 DAT for the dry season crop. Maintain a thin film of water for multiplication.
- Raise Azolla as a dual crop by inoculating 250 kg/ha 3 to 5 DAT and then incorporate during weeding for the wet season crop.
- Mix 2 kg each of biofertilizers *viz., Azsopirillum,* Phosphobacteria (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) with 25 kg of FYM and 25 kg of sand and broad cast uniformly before transplanting and
- *Pseudomonas fluorescens* (Pf 1) at 2.5 kg/ha mixed with 50 kg FYM and 25 kg of soil and broadcast the mixture uniformly before transplanting.

Application of inorganic fertilizers

- Apply fertilizer nutrients as per STCR-IPNS recommendations for desired yield target (Appendix I) (or)
- N dose may be through Leaf Color Chart (**LCC**)*
- P & K may be through Site Specific Nutrition Management by Omission plot technique**
- If the above recommendation are not able to be followed, adopt blanket recommendation a follows:

Nutrients	**N**	**P_2O_5**	**K_2O**
	(kg/ha)		
Short duration varieties (**dry season**)			
a) Cauvery delta & Coimbatore tract	150	50	50
b) For other tracts	120	40	40
Medium and long duration varieties (**wet season**)	150	50	50
Hybrid rice	175	60	60
Low N responsive cultivars (like Improved White Ponni)	75*	50	50

* For Ponni, N should be applied in three splits at AT, PI and H stages** in addition to GLM or FYM application.
**Phenological stages of rice (days after sowing)

Stages	Short (105)	Medium (135)	Long (150)
Active Tillering (AT)	35-40	50-55	55-60
Panicle Initiation (PI)	45-50	70-75	85-90
Heading (H)	70-75	100-105	115-120

N management through LCC

- Time of application is decided by LCC score
- Take observations from 14 DAT in transplanted rice or 21 DAS in direct seeded rice.
- Repeat the observations at weekly intervals up to heading
- Observe the leaf colour in the fully opened third leaf from the top as index leaf.
- Match the leaf color with the colours in the chart during morning hours (8-10 am).
- Take observation in 10 places.
- LCC critical value is 3.0 in low N response cultures like White Ponni and 4.0 in other cultivars and hybrids
- When 6/10 observations show less than the critical colour value, N can be applied as per the following recommendation : Application of 25 kg N ha^{-1} (1 bag urea) at 7 DAT followed by N @ 40 kg ha^{-1} each time for kuruvai/ short duration rice / 30 kg ha^{-1} each time for medium & long duration rice as and when the leaf colour value falls below the critical value of 4 for varieties and hybrids and critical value of 3 for white ponni, monitored from 14 DAT.
- For aged seedlings : Basal application of 35 kg N per ha is recommended to avoid yield loss when seedlings aged 35 - 45 days are used for transplanting and the LCC based N management can be followed from 14 DAT.

****Recommendation of P&K fertilizer rates based on SSNM approach for rice growing tracts of Tamil Nadu (other than Cauvery Delta)**

Sl. No.	Location	Calibrated SSNM fertilizer dose (kg/ha)*	
		P_2O_5	K_2O
1	Cauvery delta		
	(i) Old delta	**35**	**50**
	(ii) New delta	**35**	**80**
2	Coimbatore District		
	(i) General	30	40
	(ii) Annamalai block	30	80
3	Killikulam	30	50
4	Trichy	35	50
5	Ambasamudram	40	50
6	Bhavanisagar	20	25
7	Paiyur	25	45
8	Yethapur	30	45

9	Aruppukottai	20	30
10	Cuddalore	30	50

** The above SSNM based fertilizer P and K arrived based on yield response are recommended for specific soil series prevailing in different rice growing areas for adoption by farmers

Split application of N and K

- Apply N and K in four equal splits viz., basal, tillering, panicle initiation and heading stages.
- Tillering and Panicle initiation periods are crucial and should not be reduced with the recommended quantity.
- N management through **LCC** may be adopted wherever chart is available

Application of P fertilizer

- P may be applied as basal and incorporated.
- When the green manure is applied, rock phosphate can be used as a cheap source of P fertilizer. If rock phosphate is applied, the succeeding rice crop need not be supplied with P. Application of rock phosphate + single super phosphate or DAP mixed in different proportions (75:25 or 50:50) is equally effective as SSP or DAP alone.

Application of micronutrients

Soil Application

- Soil application of 25 kg zinc sulphate/ha mixed with 50 kg dry sand or apply 25 kg of TNAU Wetland rice MN mixture/ha enriched in FYM at 1:10 ratio incubated for 30 days at friable moisture, just before transplanting.
- It is enough to apply 12.5 kg zinc sulphate /ha, if green manure (6.25 t/ha) or enriched FYM, is applied.
- For saline and sodic acid 37.5 kg $ZnSO_4$ can be applied.
- Apply 500 kg of gypsum/ha (as source of Ca and S nutrients) at last ploughing. Application of 50 kg $FeSO_4$ + 12.5 t FYM /ha, 40 kg S as gypsum can be followed, if the soils are deficient in respective elements.
- For Cauvery delta zone, application of 5 kg $CuSO_4$ can be recommended.

Foliar nutrition

- Foliar spray of 1% urea + 2% MAP + 1% KCl at Panicle Initiation (PI) and 10 days after first spray to inprove grain filling rate and yield in all varieties.
- If deficiency symptom appears in the standing crop (15 days after transplanting) foliar application of 0.5% zinc sulphate + 1.0% urea can be given at 7-10 days for short duration and 15 days interval for medium and copy duration crop until the Zn deficiency symptoms disappear.
- Biofortification strategies; For biofortification of Zn in rice, the efficient cultivars viz., CO51, CO47, ADT 47, ADT 37 may be grown with the basal soil application of 50 kg $ZnSO_4$ alongwith foliar spraying of 0.50% $ZnSO_4$ thrice at 50% flowering, milky and dough stages to enrich the grain Zn content.

Nutrient deficiency / toxicity symptoms

- **Nitrogen deficiency**: Plants become stunted and yellow in appearance first on lower leaves. In case of severe deficiency the leaves will turn brown and die. Deficiency symptoms first appear at the leaf-tip and progress along the midrib until the entire leaf is dead.
- **Potassium deficiency**: Bluish green leaves - when young, older leaves irregular.

Chlorotic and necrotic areas - grain formation is poor - weakening of the straw which results in lodging.

- **Magnesium deficiency:** Leaves are chlorotic with white tips.
- **Zinc deficiency:** Lower leaves have chlorotic particularly towards the base. Deficient plants give a brown rusty appearance.
- **Copper deficiency:** Leaves develop chlorotic streaks on either side of the midrib and appearance of dark brown necrotic lesions on leaf tips. Unfolding of the new leaves will also be seen.
- **Iron toxicity:** Brown spots on the lower leaves starting from tips and proceeding to the leaf base and turns into green or orange purple leaves and spreading to the next above leaves.

Neem treated urea and coal-tar treated urea

Blend the urea with crushed neem seed or neem cake 20% by weight. Powder neem cake to pass through 2mm sieve before mixing with urea. Keep it overnight before use (or) urea can be mixed with gypsum in 1:3 ratios, or urea can be mixed with gypsum and neem cake at 5:4:1 ratio to increase the nitrogen use efficiency. For treating 100 kg urea, take one kg coal-tar and 1.5 litres of kerosene. Melt coal-tar over a low flame and dissolve it in kerosene. Mix urea with the solution thoroughly in a plastic container, using a stick. Allow it to dry in shade on a polythene sheet. This can be stored for a month and applied basally.

N management through LCC For sodic soil

In the case of sodic soils, LCC critical value is 4.0 for varieties and 5.0 for the hybrids.

Other special cultural practices (Contingent Plan)

Application of Pink Pigmented Facultative Methylotroph (*Methylobacterium* sp.) as seed treatment (@ 200 g / 10 kg seeds), soil application (@ 2 kg / ha) and foliar spray (@ 500 ml / ha) at panicle initiation and flag leaf stages for alleviation of water stress effects in both SRI and transplanted system of rice cultivation.

1.2.4. Weed management

- Use of rotary weeder from 15 DAT at 10 days interval. It saves labour for weeding, aerates the soil and root zone, prolongs the root activity, and improves the grain filling though efficient translocation and ultimately the grain yield.
- Cultural practices like dual cropping of rice-azolla, and rice-green manure (described in wet seeded rice section 2.5 & 2.6 of this chapter) reduces the weed infestation to a greater extent.
- Summer ploughing and cultivation of irrigated dry crops during post-rainy periods reduces the weed infestation.

Pre-emergence herbicides

- Use Butachlor 1.25kg/ha or Anilophos 0.4kg/ha as pre-emergence application. Alternatively, pre-emergence application of herbicide mixture viz., Butachlor 0.6kg + 2,4 DEE 0.75kg/ha, or Anilophos + 2, 4 DEE 'ready-mix' at 0.4kg/ha followed by one hand weeding on 30 - 35 DAT will have a broad spectrum of weed control.

- Any herbicide has to be mixed with 50kg of dry sand on the day of application (3 - 4 DAT) and applied uniformly to the field with thin film water on the 3rd DAT. Water should not be drained for next 2 days from the field (or) fresh irrigation should not be given.
- Pre-emergence application of pretilachor at 1.0 kg ha^{-1} on 3 DAT + weeding with Twin row rotary weeder at 40 DAT
- PE Pyrazosulfuron ethyl @ 20 g ha^{-1} on 3 DAT + hand weeding (HW) on 45 DAT.
- PE butachlor 0.75 kg ha^{-1} + bensulfuron methyl 50 g ha^{-1} on 3 DAT + HW on 45 DAT
- PE Oxadiazon 87.5 g ha^{-1} followed by Post emergence (POE) 2,4-D 1 kg ha^{-1} along with hand weeding on 35 DAT.
- PE butachlor 0.75 kg per hectare + bensulfuron methyl 50 g ha^{-1} on 3 DAT followed by mechanical weeding on 45 DAT is effective for broad spectrum weed control.
- Crop growth and yield were enhanced by butachlor 1.2 + 2,4-DEE 1.5 lit ha^{-1} with 100% inorganic nitrogen.
- Conventional tillage of one dry ploughing and two passes of cage wheel puddling combined with pre-emergence application of butachlor at 1.25 kg ha^{-1} under lowland situation.
- Stale bed preparation by pre-puddling minimum tillage with glyphosate combine with post- plant pre emergence butachlor 1.25 kg ha^{-1} resulted in increased rice grain yield, net income and B: C ratio in rice-rice cropping.
- If pre-emergence herbicide application is not done, hand weeding has to be done on 15th DAT.
- 2,4-D sodium salt (Fernoxone 80% WP) 1.25 kg/ha dissolved in 625 litres with a high volume sprayer, three weeks after transplanting or when the weeds are in 3 - 4 leaf stage.
- Early post emergence application of Bispyripac sodium 40 g ha^{-1} (2-3 leaf stage of weeds) + Hand weeding on 45 DAT
- Pre emergence application of Pretilachlor @ 750 g/ha at 3 DAT followed by post emergence application of Chlorimuron methyl + Metsulfuron methyl @ 4 g/ha on 25 DAT had higher weed contol efficiency and net return.
- Pre emergence application of Butachlor @ 1.0 kg ai/ha on 3 DAT + Finger type single row or double row rotary weeders weeding on 45 DAT. If pre emergence application is avoided, then finger type single row/double row rotary weeders weeding on 20 and 40 DAT.

1.2.5. Water management

- Puddling and leveling minimizes the water requirement
- Plough with tractor drawn cage wheel to reduce percolation losses and to save water requirement up to 20%.
- Maintain 2.5cm of water over the puddle and allow the green manure to decompose for a minimum of 7 days in the case of less fibrous plants like sunnhemp and 15 days for more fibrous green manure plants like Kolinchi (*Tephrosia purpurea*).
- At the time of transplanting, a shallow depth of 2cm of water is adequate since high depth of water will lead to deep planting resulting in reduction of tillering.

- Maintain 2 cm of water up to seven days of transplanting.
- After the establishment stage, cyclic submergence of water (as in table) is the best practice for rice crop. This cyclic 5cm submergence has to be continued throughout the crop period.

  Days after disappearance of ponded water at which irrigation is to be given

Soil type	Summer	Winter
Loamy	1 day	3 days
Clay	Just before/immediately after disappearance	1 - 2 days

- Moisture stress due to inadequate water at rooting and tillering stage causes poor root growth leading to reduction in tillering, poor stand and low yield.
- Critical stages of water requirement in rice are a) panicle initiation, b) booting, c) heading and
  d) flowering. During these stages, the irrigation interval should not exceed the stipulated time so as to cause the depletion of moisture below the saturation level.
- During booting and maturity stages continuous inundation of 5cm and above leads to advancement in root decay and leaf senescence, delay in heading and reduction in the number of filled grains per panicle and poor harvest index.
- Provide adequate drainage facilities to drain excess water or strictly follow irrigation schedule of one day after disappearance of ponded water. Last irrigation may be 15 days ahead of harvest.

Precautions for irrigation

- The field plot size can be 25 to 50 cents depending on the source of irrigation.
- Field to field irrigation should be avoided. Field should be irrigated individually from a channel.
- Small bund may be formed parallel to the main bund of the field at a distance of 30 to 45 cm within the field to avoid leakages of water through main bund crevices.
- To minimize percolation loss, the depth of stagnated water should be 5 cm or less.
- In water logged condition, form open drains, about 60 cm in depth and 45 cm width across the field.
- Care should be taken not to allow development of cracks.
- In canal command area, conjunctive use of surface and ground water may be resorted to for judicious use of water.

Alternate Wetting and Drying Irrigation (AWDI)

- Safe Alternate Wetting and Drying Irrigation (AWDI) is to monitor the depth of ponded water on the field using ‘Field Water Tube’ (FWT) which is made of 40 cm long plastic pipe with a diameter of 15 cm so that water table is easily visible.
- Tube is perforated with 0.5 cm diameter holes in the bottom and the top 15 cm portion is non-perforated.

- Above the perforated portion, markings are made for 5 cm so that irrigation at 5 cm depth could be done.
- One Field Water Tube is required for adopting the AWDI in an area of 1 acre. The FWT is installed in the field using mallet and it is inserted upto the perforated portion buried inside the soil. The soil inside the tube is to be removed.
- FWT to be installed near the field levies so that the water level inside the FWT could be monitored easily.
- Safe AWDI of 10 cm depletion in light soils and 15 cm depletion in heavy soils was found to improve the water use efficiency in rice.

Non-Puddled machine Transplanted Rice (NPTR)

- Traditional transplanted rice cultivation requires 1200-1400 mm of water of which puddling consumes 250 mm of water.
- In NPTR, puddling is replaced with dry ploughing (using cultivator and rotavator) followed by laser leveling and wetting.
- Soil is allowed to settle for 12-24 hrs before transplanting very light irrigation is given again to maintain a uniform depth of 1 cm standing water.
- Machine transplanting is adopted in the wetted soil.
- Alternate Wetting and Drying Irrigation method is followed for water management.
- Though there was a yield reduction, considerable water saving under NPTR from 120 to 245 mm.

1.3. **Insect management:** See Crop Protection Chapter

1.4. **Disease management:** See Crop Protection Chapter

1.5. Harvesting

- Taking the average duration of the crop as an indication, drain the water from the field 7 to 10 days before the expected harvest date as draining hastens maturity and improves harvesting conditions.
- When 80% of the panicles turn straw colour, the crop is ready for harvest. Even at this stage, the leaves of some of the varieties may remain green.
- Confirm maturity by selecting the most mature tiller and dehusk a few grains. If the rice is clear and firm, it is in hard dough stage.
- When most of the grains at the base of the panicle in the selected tiller are in a hard dough stage, the crop is ready for harvest. At this stage harvest the crop, thresh and winnow the grains.
- Dry the grains to 12% moisture level for storage. Grain yield in rice is estimated only at 14% moisture for any comparison.
- Maturity may be hastened by 3-4 days by spraying 20% NaCl a week before harvest to escape monsoon rains.

1.6. Seedling throwing method of stand establishment

- 20 days old seedlings of short duration rice varieties
- Requirement of seedlings will be approximately 20% more than the line planting or equal to random planting.
- The seedlings are thrown into the puddled leveled field by labour without using force.
- Suitable for all seasons except *Thaladi* or heavy rain season.
- 50% labour shaving as compared to line planting and 35% to random planting.

- Up to 7-10 days of seedling throwing care should be taken to maintain thin film of water (similar to wet seeded rice).
- Other cultural operations are same as in transplanted rice
- Grain yield will be equal to line planted crop and 10-12% higher than random planted crop.

1.7. Transplanted hybrid rice

Seed rate	20 kg per hectare
Nursery	Basal application of DAP at 2 kg/cent of nursery area. Sparse sowing of seeds at one kg/cent of nursery area will give robust seedlings with 1-2 tillers per seedling at the time of planting. If the soil is heavy, apply 4 kg gypsum/cent of nursery area, 10 days before pulling of seedlings.
Age of seedling	20 to 25 days
Spacing (cm)	20 x 10 (50 hills/m^2) or 25 x 10 (40 hills/m^2) according to soil fertility
Seedlings/ hill	One (along with tillers if already produced)
Fertilizer	175:60:60 kg N, P_2O_5 and K_2O/ha

Other package of practices: same as in transplanted rice varieties.

1.8. INTEGRATED CROP MANAGEMENT (ICM) - RICE (SRI - System of Rice Intensification)

1.8.1.Season

- Dry season with assured irrigation is more suitable.
- Difficulty in crop establishment may be seen in areas with heavy downpour (NE Monsoon periods of Tamil Nadu

1.8.2.Varieties

- Hybrids and varieties with heavy tillering feature

1.8.3.Nursery

1.8.3.1.Seed rate

- 5-7 kg/ha for single seedling per hill
- 12 -15 kg/ha for two seedlings per hill wherever difficulty in establishment of rice is seen

1.8.3.2. Mat nursery preparation

- *Preparation of nursery area:* Prepare 100 m^2 nursery to plant 1 ha. Select a level area near the water source. Spread a plastic sheet or used polythene gunny bags on the shallow raised bed to prevent roots growing deep into soil.
- *Preparation of soil mixture:* Four (4) m^3 of soil mix is needed for each 100 m^2 of nursery. Mix 70% soil + 20% well-decomposed pressmud / bio-gas slurry / FYM + 10% rice hull. Incorporate 1.5 kg of powdered DAP or 2 kg 17-17-17 NPK fertilizer in the soil mixture.

- *Filling in soil mixture:* Place a wooden frame of 0.5 m long, 1 m wide and 4 cm deep divided into 4 equal segments on the plastic sheet or banana leaves. Fill the frame almost to the top with the soil mixture.
- *Seed Treatment with biofertilizers* : Five packets (1 kg/ha) of *Azospirillum* and five packets (1 kg/ha) of Phosphobacteria or five packets (1 kg/ha) of Azophos. Biofertilizers are mixed with water used for soaking and kept for 4 hrs. The bacterial suspension after draining may be sprinkled in the nursery before sowing the treated seeds
- *Pre-germinating the seeds 2 days before sowing:* Soak the seeds for 24 hr, drain and incubate the soaked seeds for 24 hr, sow when the seeds sprout and radicle (seed root) grows to 2-3 mm long.
- *Soil application of biofertilizers :* Application of *Azospirillum* @ 2 kg and Arbuscular mycorrhizal fungi @ 5 kg for 100 m^2 nursery area
- *Sowing:* Sow the pre-germinated seeds weighing 90 -100 g / m^{-2} (100g dry seed may weigh 130g after sprouting) uniformly and cover them with dry soil to a thickness of 5mm. Sprinkle water immediately using rose can to soak the bed and remove the wooden frame and continue the process until the required area is completed.
- *Watering:* Water the nursery with rose can as and when needed (twice or thrice a day) to keep the soil moist. Protect the nursery from heavy rains for the first 5 DAS. At 6 DAS, maintain thin film of water all around the seedling mats. Drain the water 2 days before removing the seedling mats for transplanting.
- *Spraying fertilizer solution* (optional): If seedling growth is slow, sprinkle 0.5% urea + 0.5% zinc sulphate solution at 8-10 DAS.
- *Lifting seedling mats:* Seedlings reach sufficient height for planting at 15 days. Lift the seedling mats and transport them to main field.
- For elite seedling production under modified mat nursery : seed fortification with 1.0% KCl mixed with native soil and powdered DAP @ 2.0 kg per cent along with *Pseudomonas* 240 g/ cent followed by drenching with 0.5 % urea solution on 9 DAS

1.8.4. Main field preparation

- Puddled lowland prepared as described in transplanted section
- Perfect leveling is a pre-requisite for the water management proposed hereunder

1.8.5. Transplanting

- 1-2 seedlings of 14-15 days old
- Square planting of 25 x 25 cm (10 x 10 inch)
- Fill up the gaps between 7^{th} and 10^{th} DAT.
- Transplant within 30 minutes of pulling out of seedlings.
- There may be difficulty in crop establishment in areas with heavy downpour (North East Monsoon periods of Tamil Nadu)

1.8.6. Irrigation management

- Irrigation to be done so as to moist the soil during early period upto 10 days
- Restoring irrigation to a maximum depth of 2.5 cm after development of hairline

cracks in the soil until panicle initiation (PI)

- Increasing irrigation depth to 5.0 cm after PI one day after disappearance of ponded water till completion of flowering stage.
- Placing of water pipe as safe alternate wetting and drying irrigation (AWDI) reduces the total number of irrigation given to rice crop (Perforated water pipe is placed 10 – 15 cm below the soil surface and the water lever moderation observed for time of irrigation)

1.8.7. Weed management

- Using rotary weeder / Cono weeder / power operated two row weeder
- Moving the weeder with forward and backward motion to burry the weeds and as well as to aerate the soil at 7-10 days interval from 10-15 days after planting on either direction of the row and column.
- Manual weeding is also essential to remove the weeds closer to rice root zone.

1.8.8. Nutrient management

- As per transplanted rice.
- Use of LCC has more advantage in N management.
- Green manure and farm yard manure application will enhance the growth and yield of rice in this system approach.
- Under sodic soils, during rotary weeding, apply Azophosmet @ 2.2 kg/ha and PPFM as foliar spray @ 500 ml/ha

1.8.9. Other package of practices as recommended to transplanted rice

- STCR based fertilizer recommendation for transplanted rice (for some selected districts) is given in the **Appendix I.**

2. WET SEEDED PUDDLED LOWLAND RICE

WET SEEDED RICE

2.1 Area

- Direct wet seeding can be followed in all the areas wherein transplanting is in vogue.

2.2. Season

- As that of translated rice

2.3.Field preparation

- On receipt of showers during the months of May - July repeated ploughing should be carried out so as to conserve the moisture, destroy the weeds and break the clods.
- After inundation puddling is to be done as per transplanting. More care should be taken to level the field to zero level.
- Stagnation of water in patches during germination and early establishment of the crop leads to uneven crop stand.
- Land leveling has say over efficient weed and water management practices.
- Provision of shallow trenches (15 cm width) at an interval of 3m all along the field will facilitate the draining of excess water at the early growth stage.

2.4. Varieties

All the varieties recommended for transplanting can do well under direct wet seeded conditions also. However, the following varieties are more suited.

Varieties	Duration (days)	Time of sowing
Ponmani	160 to 165	1st to 30th August for *Samba*
CO 43, IR20, ADT 38 ADT 39, Ponni, Improved White Ponni	125 to135	1st to 30th September for *Thaladi*
ADT 36, ADT 37	105 to 110	1st to 10th June for *Kuruvai* 1st to 10th October for late *Thaladi*

2.5. Sowing

- Follow a seed rate of 60 kg / ha
- Pre-germinate the seeds as for wet nursery
- Seed treatments as adopted for transplanted rice
- Sow the seeds by **drum seeder** or broadcast uniformly with thin film of water.
- Dual cropping of rice-green manure is economic for nutrient budget and efficient for grain production. For this method use 'TNAU Rice-Green manure seeder'.

TNAU Rice cum Green manure seeder

- Manually drawn seeder developed at TNAU to sow pre-germinated paddy and green manure daincha crop (*Sesbania aculeata*) in alternate rows in puddled soil.
- On attaining a height of 40 cm after about one month of sowing the daincha crop was trampled by using long handled IRRI design cono weeder.
- Seeder sows four paddy rows and four daincha rows in a single pass.
- Using one (male) operator and two women labourers half of ha can be sown with the seeder in a day of 8 hours.
- Paddy was sown at 60 kg/ha seed rate and green manure crop at 20 kg/ha seed rate. The distance between the adjacent rows is 12.5 cm. When compared to sole wet seeded rice, weeds are better controlled in the wet seeded rice inter-cropped with green manure.
- Also intercropping of rice with green manure dhaincha and incorporation at 7.0 t/ha enhanced the growth and yield of rice and beneficial in terms of N addition (40 kg N /ha).
- There is greater possibility of intercropping green manures during early stage of rice crop with increased grain yield by one tones / ha.

2.6. After cultivation

- Thinning and gap filling should be done 14 - 21 days after sowing, taking advantage of the immediate rain.
- If dual cropped with green manure, incorporate the green manure when grown to 40 cm height or at 30 days after sowing, whichever is earlier, using Cono-weeder.
- Green manure incorporated fields may be operated again with rotary weeder a week later in order to aerate the soil and to exploit organic acids formed if any.

2.7. Manures and fertilizer application

- For direct wet seeded lowland rice, the recommendation is same at that of transplanted rice.
- Apply N and K as 25% each at 21 DAS, at active tillering, PI and heading stages.
- If N applied through LCC, use the critical value 4 for line sown drill seeded rice.
- Entire P as basal applied in the last plough or at the time of incorporation of green manure/ compost.
- Biofertilizers as recommended to transplanted rice may be followed wherever feasible and moisture available.
- Micro nutrient, foliar application and biofertlizers as recommended to transplanted rice.

2.8. Weed management

- In wet seeded rice, pre-emergence application of Pretilachlor 0.75 kg/ha on 8 DAS on 3-4 DAS followed by one hand weeding on 40 DAS in direct drum seeded rice
- In wet seeded rice, sowing with drum seeder and cono weeding (manual / power weeder) is done at 10, 20 and 30 DAS
- In wet seeded rice, hand weeding twice on 15 - 20 DAT and 45 DAT will control the weeds effectively (or) Pendimethalin 1.0 lit/ha at 8 DAT with optimum moisture condition and one hand weeding on 45 DAT.
- In rice -rice -fallow system intercropping of *Sesbania rostrata* control the weeds of rice field along with incorporation of *Sesbania rostrata* in to the field and one hand weeding on 35 DAS.
- Apply PE Pretilachlor 0.45 kg ha^{-1} on 3 DAS + Roto cylindrical weeder + weeding on 45 DAS in wet seeded rice have good control of weeds like *Echinochloa crusgalli, Panicum repens, Eclipta alba* and *Monochoria vaginalis.*
- Pre-emergence application of Pendimethalin 1.0 kg/ha at 3 DAS followed by post emergence application of bispyribac sodium 25 g/ha at 25 DAS along with one hand weeding 45 DAS effectively reduced weed density in wet seeded rice.
- Pre emergence application of pyrazosulfuron ethyl at 20 g a.i /ha on 3 DAS followed by cono weeding on 25 DAS had higher weed control efficiency in drum seeded rice.
- Combination of drum seeded rice intercropped with green manure (dhaincha) along with pre-emergence herbicide application of Pretilachlor (30.7 EC) @ 0.45 kg ha^{-1} + safener on 5 DAS is the best weed control method in drum seeded rice.

2.9. Water management

- During first one week irrigate the soil with thin film of water.
- Depth of irrigation may be increased to 2.5 cm progressively as per the crop age.
- Follow schedule as given in transplanted rice.

2.10. Insect management: See Crop Protection Chapter

2.11. Disease management: See Crop Protection Chapter

Other package of practices

- As recommended in transplanted rice

3. DRY SEEDED RAINFED UN-PUDDLED LOWLAND RICE

RAINFED RICE

The crop establishment, growth and maturity depend up on the rainfall received. There will be standing water after crop establishment for a minimum period of few days to a maximum up to grain filling, depending up on the rainfall. This type of cultivation in Tamil Nadu is called as **'rainfed rice'**, with the assumption that the soil moisture will be under unsaturated (dry) condition during establishment or entire growth period, with reference to tropical climate.

3.1. Area

- Coastal districts of Tamil Nadu like Kanchipuram, Tiruvallur, Pudukottai, Ramanathapuram, Virudhunagar, Sivagangai and Kanyakumari.

3.2. Season

- June – July – (Coastal northern districts)
- September – October (Coastal southern districts)

3.3. Field preparation

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

3.4. Varieties

- Short duration varieties as mentioned in season and varieties including local land races suitable for those tracts.

3.5. Sowing

- Seed rate: 75kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early moisture stress.
- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 1 kg or *Azospirillum and Phosphobacteria* @ 1 kg each per ha seed, whichever is available.
- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.

3.6. After cultivation

- 10 packets (2 kg/ha) each of *Azospirillum* inoculant and Phosphobacteria or 10 packets

  (2 kg/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14 - 21days after sowing, taking advantage of the immediate rain
- Foliar spray of Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice.

3.7. Manures and fertilizer application

- Blanket recommendation : 50:25:25 kg $N:P_2O_5:K_2O$ /ha
- Apply a basal dose of 750 kg of FYM enriched with fertilizer phosphorus (P at 25 kg/ha)
- Apply N and K in two equal splits at 20 - 25 and 40 - 45 days after germination.
- If the moisture availability from the tillering phase is substantial, three splits (25 kg N and 12.5 kg K at 20-25, 40-45 and 60-65 DAG) can be adopted.
- N at PI may be enhanced to 40 kg, if the tiller production is high (may be when the estimated LAI is greater than 5.0) and moisture availability ensured by standing water for 10 days.
- Basal application of $FeSO_4$ at 50 kg/ha + 12.5 t FYM is desirable for iron deficient soil (or) apply TNAU Rainfed rice MN mixture @12.5 kg/ha as EFYM at 1:10 ratio incubated for 30 days at friable moisture.
- Need based foliar application of 0.5% $ZnSO_4$ and 1% $FeSO_4$ + 0.1% citric acid may be taken up at tillering and PI stages.
- Foliar spray of 1% urea + 2% MAP + 1% KCl at PI and 10 days after may be taken up for enhancing the rice yield if sufficient soil moisture is ensured
- Apply 25 kg $ZnSO_4$ if the soil is Zn deficient.

3.8. Weed management

- First weeding can be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply pendimethalin 1.0 kg/ha on 5 days after sowing on the day of receipt of soaking rain followed by one hand weeding on 30 to 35 days after sowing.

3.9. Insect management: See Crop Protection Chapter

3.10. Disease management: See Crop Protection Chapter

3.11. Harvesting

Same as that for wet rice cultivation

4. DRY SEEDED RAINFED UN-PUDDLED LOWLAND RICE WITH SUPPLEMENTAL IRRIGATION

Semi dry rice

It is called as **semi-dry rice**. Crop establishment is as that of rainfed rice but the rain water collected in village tank (Kanmai) is supplemented to protect the crop during peak vegetative and reproductive phases. Interaction between applied nutrients and crop is positive here due to better moisture availability than rainfed rice and hence varieties may be improved ones and nutrient levels may be higher than the previous system.

4.1. Area

- Kanchipuram, Tiruvallur, Ramanathapuram,Sivaganga, Kanyakumari, Nagapattinam, Tiruvarur and Pudukottai.

4.2. Seasons

- July to August - Kanchipuram/Tiruvallur, Kanyakumari
- August - Nagapattinam/Tiruvarur, Pudukottai
- September to October - Ramanathapuram, Sivaganga

4.3. Field preparation

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

4.4. Varieties

- Short duration varieties as mentioned in season and vanities including local land races suitable for those tracts.
- Since there is supplemental irrigation high yielding improved short duration varieties can yield more yield than the land races.

4.5. Sowing

- Seed rate: 75 kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early moisture stress.
- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 1 kg/ha or *Azospirillum and Phosphobacteria @ 1 kg/ha* each per ha seed, whichever is available.
- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.

- Sowing of seed by multi crop planter (Happy Seeder) under dry condition @ 40 kg/ha

4.6. After cultivation

- 10 packets (2kg/ha) each of *Azospirillum* inoculants and Phosphobacteria or 10 packets (2 kg/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14-21days after sowing, taking advantage of the immediate rain
- Foliar spray of Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice. .

4.7. Manures and fertilizer application

- Blanket recommendation : 75:25:37.5 kg $N:P_2O_5:K_2O$ /ha
- Apply a basal dose of 750 kg of FYM enriched with fertilizer phosphorus (P at 25 kg/ha)
- Apply N & K in three splits at 20-25, 40-45 and 60-65 days after germination.
- Each split may follow 25kg N and 12.5 kg K_2O.
- If the moisture availability is substantial, the split at 40-45 DAS (panicle initiation) may be applied up to 40kg N and 12.5kg K_2O to enhance the growth and the grain yield.
- Basal application of $ZnSO_4$ at 25kg/ha and $FeSO_4$ at 50 kg/ha + 12.5 t FYM is desirable wherever zinc and iron deficiency were noted (or) apply TNAU Rainfed rice MN mixture @12.5 kg/ha as EFYM at 1:10 ratio incubated for 30 days at friable moisture.
- Need based foliar application of 0.5% $ZnSO_4$ and 1% $FeSO_4$ + 0.1% citric acid at tillering and PI stages.
- Foliar spray of 1% urea + 2% MAP + 1% KCl at PI and 10 days after may be taken up for enhancing the rice yield if sufficient soil moisture is ensured

4.8. Weed management

- First weeding should be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply Pendimethalin 1.0 kg/ha on 5 days after sowing followed by one hand weeding on 30 to 35 days after sowing.
- PE butachlor 1.0 kg ha^{-1} followed by weeding using finger type single row and double row rotary weeders resulted in higher grain yield and net profit.
- Application of Pretilachlor@0.45 l/ha on 5 DAS and two machine weeding (Power weeder) on 30 and 45 DAS, if sowing is done by using Happy seeder.

4.9. Water management

- The crop to be irrigated from 30-35 days onwards, utilizing water impounded in the tanks.
- Irrigation to be given to a depth of 2.5 - 5.0 cm only. The schedule of irrigating

one day after disappearance of ponded water to be followed in order to save water and to bring additional area under rice cultivation.

4.10. Insect management: See Crop Protection Chapter
4.11.Disease management: See Crop Protection Chapter

4.12.Harvest

- It is same as that of transplanted rice.
- These areas are more suitable for combine-harvester

5. DRY SEEDED IRRIGATED UN-PUDDLED LOWLAND RICE

Also be called 'semi-dry rice'

It is a contingent plan to command areas, anticipating the release of water; rice crop can be established under rainfed condition up to a maximum of 45 days as that of previous two situations. Filed is converted to wet condition on receipt of canal water. Conversion depends up on receipt of canal water and nutrient management is decided according to the period of irrigation.

5.1.Area

- Tiruvarur and Nagapattinam districts

5.2.Season

- Samba / Thaladi seasons command areas.

5.3. Field preparation

- Dry plough to get fine tilth taking advantage of rains and soil moisture availability.
- Apply gypsum at 1 t/ha basally wherever soil crusting and soil hardening problem exist.
- Perfect land leveling for efficient weed and water management.
- Provide shallow trenches (15 cm width) at an interval of 3m all along the field to facilitate draining excess water at the early growth stage.

5.4. Varieties

- Medium duration varieties, if sown in August and short duration varieties beyond September, as mentioned in season and vanities.
- Since there is assured irrigation from canal, high yielding improved short or medium duration varieties can be cultivated depending up on the situation (month of sowing, nearness to canal, depth of standing water during NEM etc).

5.5. Sowing

- Seed rate: 75kg/ha dry seed for any recommended variety.
- Seed hardening with 1% KCl for 16 hours (seed and KCl solution 1:1) and shade dried to bring to storable moisture. This will enable the crop to withstand early

moisture stress.

- On the day of sowing, treat the hardened seeds first with *Pseudomonas fluorescens* 10g/kg of seed and then with *Azophos* 1 kg/ha or *Azospirillum and Phosphobacteria* @ 1 kg/ha each per ha seed, whichever is available.
- Drill sow with 20 cm inter row spacing using seed drill.
- The seeds can also be sown behind the country plough
- Depth of sowing should be 3 - 5 cm and the top soil can be made compact with leveling board.
- Pre-monsoon sowing is advocated for uniform germination.
- Pre-monsoon sowing with medium duration variety is an advantage for higher grain yield and as well to manage the heavy rainy season.

5.6. After cultivation

- 10 packets (2 kg/ha) each of *Azospirillum* inoculant and Phosphobacteria or 10 packets (2 kg/ha) of Azophos mixed with 25 kg of FYM may be broadcasted uniformly over the field just after the receipt soaking rain / moisture.
- Thinning and gap filling should be done 14 - 21 days after sowing, taking advantage of the immediate rain.

5.7. Manures and fertilizer application

- Apply FYM/compost at 12.5 t/ha or 750 kg of FYM enriched with 50 kg P_2O_5 as basal dose in clay soils of Nagapattinam / Tiruvarur district.
- Blanket recommendation : 75:50:37.5 kg $N:P_2O_5:K_2O$ /ha
- N and K in three splits at around 20-25, 40-45 and 60-65 days for short duration varieties or four splits for medium duration varieties at around 20-25, 40-45, 60-65 and 80-85 days after germination is suitable.
- Each split may follow 25kg N and 12.5 kg K_2O.
- If the moisture availability is substantial and canal water received from tillering phases itself, the split at panicle initiation (40-45 DAS in short duration and 60-65 DAS in medium duration) may be applied up to 40kg N and 12.5kg K_2O to enhance the growth and the grain yield.
- To induce tolerance under short and prolonged drought situation in Kuruvai season, apart from seed treatment, foliar spray with 1% KCl + CCC at 500ppm during vegetative stage is effective in mitigating the drought and in increasing the yield.
- Basal application of $ZnSO_4$ at 25 kg/ha and $FeSO_4$ at 50 kg/ha + 12.5t FYM is desirable wherever zinc and iron deficiency were noted (or) apply TNAU Rainfed rice MN mixture @12.5 kg/ha as EFYM at 1:10 ratio incubated for 30 days at friable moisture.
- Need based foliar application of 0.5% $ZnSO_4$ and 1% $FeSO_4$ + 0.1% citiric acid at tillering and PI stages
- Foliar spray of 1% urea + 2% MAP + 1% KCl at PI and 10 days later may be taken up for enhancing the rice yield if sufficient soil moisture is ensured

5.8. Weed management

- First weeding should be done between 15 and 21 days after germination.
- Second weeding may be done 30 - 45 days after first weeding.
- Apply pendimethalin 1.0 kg/ha on 5 days after sowing followed by one hand weeding on 30 to 35 days after sowing.
- Application of Pretilachlor@0.45 l/ha on 5 DAS and two machine weeding (Power weeder) on 30 and 45 DAS, if sowing is done by using Happy seeder.

5.9. Other special cultural practices

- Foliar spray of Cycocel 1000 ppm (1 ml of commercial product in one lit. of water) under water deficit situations to mitigate ill-effects.
- Foliar spray of Kaolin 3% or KCl 1% to overcome moisture stress at different physiological stages of rice.
- For delayed water release in LBP area, irrigating rice to 5cm depth three days after disappearance of pounded water and growing ADT 38 rice can be resorted to if the release of water is delayed up to September.
- First top dressing should be applied immediately after the receipt of sufficient rain or canal water.
- Hand weeding, thinning and gap filling should be done before N-fertilizer application.
- Subsequent top dressings in two or three splits should be done before heading.

5.10. Water management

- As that of irrigated rice when canal water is used for irrigation
- Possibility of subsequent conversion to deep water situation as seen in this tract, specfic variety should be chosen.

5.11. Insect management: See Crop Protection Chapter

5.12. **Disease management:** See Crop Protection Chapter

5.13. Harvest

5.13.1. As that of transplanted rice. This area is more suitable to combine harvester.

5.14. DEEP WATER RICE

5.14.1. Cultivation is like the methods described in this section except the harvest. Harvest may some times restricted only to panicle because of the standing water even after maturity.

DRY SEEDED UPLAND RICE

Establishment

5.14.2. As that of section 3 to 5.

Area

5.14.3. There are small batches in and around Dharmapuri district. Rainfall availability in these tract is better than the rainfed rice cultivated in other parts of Tamil Nadu. There is no bund to stagnate the water. Moisture availability is there but crop growth depends on the nutrient status.

Other Cultural practices

5.14.4. As recommended to semi-dry rice (sec. 4)

5.14.5. Nutrient may be split applied depending upon the growth.

5.14.6. LCC based N application is more suitable for this tract.

5.14.7. Use of PPFM-Pink Pigmented Facultative Microbes (seed treatment @ 0.2 kg / 5 kg seeds, soil application basal @ 2.0 kg/ha and foliar spray@ 500 ml/ha at PI & flag leaf stages)for mitigation of terminal drought is recommended.

Intercropping

5.14.8. Blackgram for every four rows of rice.

Grain Yield

5.14.9. Grain yield depends up on the moisture availability and nutrient status.

AEROBIC RICE

5.14.10. Suitable variety PMK (R) 3

5.14.11. Optimum plant population : 50 hills per m^2 (20 x 10 cm)

5.14.12. Green manure intercrop in aerobic rice : Daincha intercropping and incorporation at 25 DAS

5.14.13. Ridges and furrows

5.14.14. Weed management : Pre emergence application of pendimethalin at 0.75 kg/ha followed by two hand weeding or mechanical weeding on 25 and 45 DAS

5.14.15. PE pendimethalin 1.0 kg ha^{-1} along with single tyne sweep weeding on 45 DAS which was comparable with PE along with hand weeding.

5.14.16. Fertilizer dose : 150 : 50 : 50 kg NPK/ha.

5.14.17. N in four splits : 20 % at 15 DAS, 30 % at tillering and PI and 20% at flowering or Nitrogen management at LCC value of 4

5.14.18. Basal application of $ZnSO_4$ at 25 kg/ha and $FeSO_4$ + 12.5t FYM at 50 kg/ha is desirable wherever zinc and iron deficiency were noted (or) apply TNAU Rainfed rice MN mixture @12.5 kg/ha as EFYM at 1:10 ratio incubated for 30 days at friable moisture.

5.14.19. Need based foliar application of 0.5% $ZnSO_4$ and 1% $FeSO_4$ + 0.1% citric acid may be taken up at tillering and PI stages

5.14.20. Irrigation : IW/CPE ratio of 1.0 with 3 cm depth of water – total water requirement of 650 mm.

5.14.21. **Surface drip fertigation:** Under aerobic rice conditions, schedule surface drip irrigation (with the lateral distance of 80 cm) at 125 % Open Pan Evaporation (PE) for clay soil / 150 % PE for sandy soil along with fertigation of 500 ml / ha of *Azophosmet* (composite biofertilizer) as seed treatment (@ 200 g / 10 kg seeds) and fertigation through drip system @ 500 ml / ha to be given during panicle initiation and flag leaf stages

5.14.22. **Sub-surface drip biogation:** Under aerobic rice conditions, schedule sub-surface drip fertigation (laterals concealed at 10 cm soil depth at a distance of 80 cm) scheduled at 125 % Open Pan Evaporation (PE) for clay soil / 150 % PE for sandy soil along with fertigation of *Azophosmet* as seed treatment @ 200 g 10 kg / seeds and fertigation @ 500 ml / ha and along with biogation of seaweed extract @ 500 ml / ha to be given during panicle initiation and flag leaf stages

POST HARVEST TECHNOLOGY OF RICE PROCESSING OF RICE

Parboiling

5.14.23. Parboiling is a hydrothermal treatment followed by drying before milling for the production of milled parboiled grain. Parboiling of paddy has been known in the orient for centuries. Nearly 50 per cent of the paddy produced in India at present is parboiled.

5.14.24. In general, the three major steps in parboiling, i.e. soaking, steaming and drying and have a great influence on the final characteristics and quality of parboiled rice.

5.14.25. Parboiling is the latest premilling treatment which improves the quality of rice. The traditional parboiling process in India is carried out in different ways.

- Paddy
- Cleaning
- Soaking in water (8 h)
- Draining
- Steaming (20 minutes)
- Aerating (3h) and heaping (3h)
- Tempering (1h)
- Sun drying (2-4 h)
- Dried paddy (14% moisture)

Improved parboiling method of CFTRI, Mysore, India (Batch)

- Paddy
- Cleaning
- Soaking in hot water for 3 h (70 ^{0} C)
- Draining the water
- Steaming
- Shade drying (2-4 h)
- Dried paddy (15 % moisture)

MILLING OF PADDY

- Milling of dried paddy (raw and parboiled) Destoner (remove dust, dirt, chaff and stones)
- Sheller

? Husk Brown rice and unshelled paddy (aspirated through fan box)
? Huller (primary polishing)
? Bran Polished rice
? Cone polishing
? Bran Head rice
? Packaging

PROCESSED PRODUCTS

Parched rice *

It is prepared by throwing rice in sand heated to a high temperature in an iron or mud pan. On stirring, rice begins to crackle and swell. Then the content of the pan are removed and sieved to separate the parched rice from sand. Parboiled rice is used for making grayish to brilliant white colour parched rice and sold either salted or unsalted. It is eaten as such or mixed with butter milk or milk.

Expanded cereals Expanded rice (Pori) *

5.14.26. Expanded rice (murmura, pori, muri) is a traditional convenience food widely consumed in India either as such or with Jaggery, roasted Bengal gram and shredded vegetables and spices. The product is mostly produced in home or cottage sector by skilled artisans.

5.14.27. In the traditional process, the paddy is soaked in water preferably over night until saturation, drained and then either steamed or dry roasted in sand for parboiling. The parboiled paddy is milled, salted and again roasted in sand for expansion.

Flow chart

Paddy
↓
Soaking in water (30% m.c)
↓
Parboiled rice (14% m.c)
↓
Milling (15% m.c)
↓
Dry heat parboiled rice in pans (110^{o}C)
↓
Salting (Sodium chloride 10g/100ml water per kg of rice)
↓
Roasting in hot sand (250^{o}C)
↓
Expanded rice (0.5% m.c)

Puffing / Popping * Puffed rice : (using rice)

This popular ready-to-eat snack product is obtained by puffing milled parboiled rice. In the traditional process rice is gently heated on the furnace without sand to reduce the moisture content slightly. It is then mixed with salt solution and again roasted on furnace in small batches with sand on a strong fire for a few seconds to produce the expanded rice. Rice expands about 8 times retaining the grain shape and is highly porous and crisp.

Parched paddy or puffed rice: (using paddy)

Sun dried paddy is filled in mud jars and is moistened with hot water. After 2-3 min. the water is decanted and the jars are kept in an inverted position for 8-10 hours. Next the paddy is exposed to the sun for a short time and then parched in hot sand as in the preparation of parched rice. Puffed rice is prepared by throwing pretreated paddy into sand heated to a high temperature in an iron pan. During parching the grain swell and burst into a soft white product. The parched grains are sieved to remove sand and winnowed to separate the husk.

Puffed rice from parboiled rice

The rice is soaked in salt water to increase the moisture to about 20%. The moist rice is introduced into a hot vessel at about 250-275^{o}C for 30-40 seconds. The rice puffs suddenly.

Popped rice

This is yet another traditional value added product prepared from raw paddy. The paddy at a moisture content of 12-14% is directly roasted in iron pans using sand as a medium at a temperature of 150-200^{o}C. The production of popped rice is comparatively less and the product is mainly used in religious functions and ceremonies.

Flaking *

Flaked rice is another important value added product prepared from paddy. Traditionally, it is prepared from soaked paddy, after heat treatment and immediate flattening using a flaking machine (an edge runner)

Paddy
↓
Soaking (12-18 hours)
↓
Roasting (200 – 250^{0}C for 20-30 sec)
↓
Tempering (5 min)
↓
Flaking (using an edge runner) Flaking machine
↓
Flaked rice

Flaked rice is made from parboiled rice. Paddy is soaked in water for 2 -3 days to soften the kernel followed by boiling water for a few minutes and the water is drained off. The paddy is heated in a shallow earthen vessel or sand in iron pan till the husks break open. It is pounded by a wooden pestle which flattens the kernel and removes the husk. The husk is separated by winnowing. Flaked rice is thin and papery and of white colour.

Quick cooking rice is made by steeping polished rice in water to a moisture content of 35 per cent, cooking under pressure and drying. Alternatively the rice may be subjected to freezing, thawing and dehydration.

Derived products

Polished rice may be precooked and canned as rice pudding and also used to make dry breakfast cereals.

RICE AND RICE PRODUCTS

Modernization of rice milling Industry also results in production of quality by-products viz., broken rice, husk and rice bran. Technology is now available for the production of value-added products from these by-products.

Byproducts of Rice Broken rice

The broken rice is widely used in the food preparations and in the industries for making flour and in the manufacture of baby foods. The starch extracted from broken rice finds wider application in the pharmaceutical, textile and other industries.

Rice husk

Rice husk that contains about 38% cellulose and 32% lignin and is one of the most abundant renewable agriculture based fuel materials. The production of rice husk is about 80 million tonnes per year, equivalent in energy to about 170 million barrels of oil. Paddy husk contains about 22 per cent ash of which 95 per cent is silica. Because of its high silica content, it is used as an abrasive. Large quantities of husk are used in India as fuel for boilers, kilns and household purposes.

Rice bran

Commercially rice bran is the most valuable by-product, which is characterized by its high fat (15 to 20%) and protein content. It also contains vitamins, minerals and many other useful chemicals. It is a potential source of edible oil. Because of its nutritional value, it is being used as feed for poultry and livestock. More stable defatted bran containing higher percentage of protein, vitamins and minerals is an excellent ingredient for both food and feed. The bran is the most nutritious byproduct of rice milling and is used almost exclusively as a feedstuff. It is generally contaminated with husk, which lowers its nutritive value. Rice bran contains about 12 per cent protein and 15 per cent fat.

Rice bran oil

Bran oil is obtained by the extraction of rice bran with solvents. Bran oil is also obtained in the solvent extraction milling of rice. The oil contains a high percentage of unsaturated fatty acids, yet it is quite stable because of the presence of natural antioxidants. When refined, bleached and deodorized, it is used for salad dressing and as cooking oil. Bran after solvent extraction has a higher percentage of protein that the original material. With its low fat content it keeps well.

Importance

Rice bran oil is the oil extracted from the germ and inner husk of rice. Rice bran oil is rich in vitamin E, γ-oryzanol (an antioxidant that may help prevent heart attacks) and phytosterols (compounds believed to help lower cholesterol absorption) which may provide associated health benefits. It has a mild taste and is popular in Asian cuisine because of its suitability for high- temperature cooking methods such as deep-frying and stir-frying. Rice bran oil is mostly monounsaturated - a tablespoon contains 7 grams of monounsaturated

fat, three of saturated fat and five of polyunsaturated fat.
Rice bran oil also contains components of vitamin E that may benefit health. The unique components, such as oryzanol or tocotrienol, have been drawing people's attention. Numerous studies show rice bran oil reduces the harmful cholesterol (LDL) without reducing good cholesterol (HDL). In those studies, Oryzanol is reported as the key element responsible for that function. Tocotrienol, on the other hand, is highlighted as the most precious and powerful vitamin E existing in nature and is said to have an anti-cancer effect, too. As a Vitamin-E source, rice bran oil is rich not only in alpha Tocopherol but also has the highest amount of Tocotrienol in liquid form vegetable oils.

Uses

Rice bran oil is ideal oil for margarine and shortening. The flavor gives the good palatability and the desired prime form crystal provides smooth plasticity and spreading qualities. When processed to retain high levels of tocols, rice bran oil may be used as a natural antioxidant source for topically coating a wide range of products such as crackers, nuts, and similar snacks to extend shelf life.

Rice polishing

Rice polishing is also rich in nutrients. They are not recovered in sizeable quantity in India. They are mostly used as animal feed.

Uses of defatted bran and bran

Defatted bran can be successfully used as an ingredient in the bakery products such as bread, cake, biscuits etc. After finer grinding, it can be incorporated into maida flour up to 20 per cent for the preparation of bakery products.

Appendix – I

1. Cereals

Rice (1)

Soil :	River alluvium (Noyyal series)	FN = 4.39 T – 0.52 SN – 0.80 ON
Season:	Kharif	FP_2O_5 = 2.22 T – 3.63 SP–0.98 OP
Target :	7 t ha^{-1}	FK_2O = 2.44 T – 0.39 SK– 0.72 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	18	300	150	67	25*	148	65	25*
220	20	350	140	60	25*	138	58	25*
240	22	400	130	53	25*	128	51	25*
260	24	450	119	45	25*	117	43	25*
280	26	500	109	38	25*	107	36	25*

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

Rice (2)

Soil : River alluvium (Noyyal series)
Season: Rabi
Target : 7 t ha^{-1}

FN = 4.63 T – 0.56 SN – 0.90 ON
FP_2O_5 = 1.98 T – 3.18 SP– 0.99 OP
FK_2O = 2.57 T – 0.42 SK – 0.67 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	18	300	159	58	25*	157	56	25*
220	20	350	148	52	25*	146	50	25*
240	22	400	137	46	25*	135	44	25*
260	24	450	126	39	25*	124	37	25*
280	26	500	114	33	25*	112	31	25*

*Maintenance dose

Rice - SRI (3)

Soil : River alluvium (Noyyal series)
Season: Kharif
Target : 8 t ha^{-1}

FN = 4.33 T – 0.53 SN – 0.68 ON
FP_2O_5 = 2.08 T – 3.18 SP – 0.70 OP
FK_2O = 2.78 T – 0.30 SK – 0.63 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	18	300	187	75**	75**	183	75**	75**
220	20	350	177	75**	75**	173	71	75**
240	22	400	166	73	69	162	64	68
260	24	450	156	67	54	152	58	53
280	26	500	145	61	39	141	52	38

** Maximum dose

Rice - SRI (4)

Soil : River alluvium (Noyyal series)
Season: Rabi
Target : 8 t ha^{-1}

FN = 4.20 T – 0.45 SN – 0.68 ON
FP_2O_5 = 2.05 T – 2.65 SP – 0.66 OP
FK_2O = 2.85 T – 0.29 SK – 0.59 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	18	300	193	75**	75**	191	75**	75**
220	20	350	184	75**	75**	182	75**	75**
240	22	400	175	75**	75**	173	76	75**
260	24	450	166	75**	65	164	70	66
280	26	500	157	72	50	155	65	51

** Maximum dose

Rice - SRI – White Ponni (5)

Soil : River alluvium (Noyyal series)
Season: Rabi
Target : 6 t ha^{-1}

FN = 3.43 T – 0.34 SN – 0.64 ON
FP_2O_5 = 1.83 T – 3.24 SP – 0.61 OP
FK_2O = 1.98 T – 0.18 SK – 0.37 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	18	300	85	25*	32	86	25*	35
220	20	350	78	25*	25*	79	25*	26
240	22	400	71	25*	25*	72	25*	25*
260	24	450	64	25*	25*	65	25*	25*
280	26	500	58	25*	25*	59	25*	25*

* Maintenance dose

Rice - SRI (6)

Soil : River alluvium (Ambasamudram series)
Season: Rabi
Target : 7 t ha^{-1}

FN = 3.54T– 0.30 SN – 0.94 ON
FP_2O_5 = 1.37T– 0.41 SP – 0.80 OP
FK_2O = 2.61T – 0.64 SK –0.61 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP^a**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	20	150	135	65	54	135	60	57
225	30	175	127	61	38	127	56	41
250	40	200	120	57	25*	120	52	25*
275	50	225	112	52	25*	112	47	25*
300	60	250	105	48	25*	105	43	25*

* Maintenance dose; SP^a- Bray P

Rice - SRI (7)

Soil : Red non calcareous (Vannapatti series)
Season : Rabi
Target : 7 t ha^{-1}

FN = 3.49 T - 0.36 SN - 0.74 ON
FP_2O_5 = 1.66 T - 2.76 SP - 0.69 OP
FK_2O = 2.19 T - 0.66 SK - 0.52 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	12	100	119	60**	54**	115	51**	57**
220	14	120	112	55**	41**	108	46	44**
240	16	140	105	49**	28**	101	40	31**
260	18	160	98	44	25	94	35	25*
280	20	180	91	38	25	87	29	25*

** Maintenance dose

Rice (8)

Soil : Red sandy loam (Irugur series) — FN = 5.19 T - 0.89 SN - 0.98 ON

Season : Kharif — FP_2O_5 = 2.27 T - 4.50 SP - 1.09 OP

Target : 7 t ha^{-1} — FK_2O = 3.11 T - 0.59 SK - 1.02 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	160	168	60**	60**	166	60**	60**
180	14	180	150	60**	60**	148	60**	60**
200	16	200	132	60**	60**	130	57	60**
220	18	220	115	55	55	113	48	58
240	20	240	97	46	43	95	39	46

** Maximum dose

Rice (9)

Soil : Red -Sandy loam (Irugur series) — FN = 4.88 T - 0.68 SN - 0.72 ON

Season : Rabi — FP_2O_5 = 2.06 T - 2.91 SP - 2.27 OP

Target : 7 t ha^{-1} — FK_2O = 2.89 T - 0.47 SK - 0.59 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	12	200	166	75**	75	164	75**	75**
200	14	220	153	75**	66	151	73	69
220	16	240	139	75**	57	137	68	60
240	18	260	125	69	47	123	62	50
260	20	280	112	63	38	110	56	41

** Maximum dose

Rice (10)

Soil : Black alluvium (Adanur series) — FN = 2.80 T - 0.29 SN - 0.89 ON

Season: Rabi — FP_2O_5 = 1.35 T - 1.28 SP - 1.78 OP

Target : 8 t ha^{-1} — FK_2O = 2.50 T - 0.42 SK - 1.14 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	16	240	119	65	66	117	58	69
200	18	260	113	62	58	111	55	61
220	20	280	107	59	49	105	52	52
240	22	300	101	57	41	99	50	44
260	24	320	96	54	33	94	47	36

Rice (11)

Soil : Black alluvium (Kalathur series)
Season: Kharif (Kuruvai)
Target : 7t ha^{-1}

FN = 5.29 T - 0.75 SN - 0.89 ON
FP_2O_5 = 1.65 T - 1.76 SP - 0.78 OP
FK_2O = 2.73 T - 0.37 SK- 0.82 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	18	260	167	61	62	165	54	65
220	20	280	152	57	55	150	50	58
240	22	300	137	54	47	135	47	50
260	24	320	122	50	40	120	43	43
280	26	340	107	47	32	105	40	35

Rice(12)

Soil : Black alluvium (Kalathur series)
Season: Rabi
Target : 7 t ha^{-1}

FN = 5.34 T - 0.67 SN - 0.73 ON
FP_2O_5 = 1.90 T - 1.86 SP - 0.70 OP
FK_2O = 2.81 T - 0.33 SK - 0.80 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	18	260	187	75**	75**	185	70	75**
220	20	280	173	73	71	171	66	74
240	22	300	160	69	65	158	62	68
260	24	320	147	65	58	145	58	61
280	26	340	133	62	52	131	55	55

** Maximum dose

Rice (13)

Soil : River alluvium (Manakkarai series)
Season: Kharif (Kuruvai)
Target : 7 t ha^{-1}

FN = 4.25 T - 0.60 SN - 0.79 ON
FP_2O_5 = 2.71 T - 4.39 SP - 0.89 OP
FK_2O = 3.83 T - 0.60 SK - 0.82 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
180	14	240	137	75**	75**	135	75**	75**
200	16	260	125	75**	75**	123	75**	75**
220	18	280	113	75**	67	111	75**	70
240	20	300	101	75**	55	99	72	58
260	22	320	89	70	43	87	63	46

** Maximum dose

Rice (14)

Soil : River alluvium (Manakkarai series)
Season: Rabi (Pishanam)
Target : 7 t ha^{-1}

FN = 4.47 T - 0.58 SN - 0.79 ON
FP_2O_5 = 2.66 T - 3.68 SP - 0.89 OP
FK_2O = 4.08 T - 0.65 SK - 0.82 OK

Initial soil test values (kg ha^{-1})			NPK (kg ha^{-1}) + GM @ 6.25 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	14	240	156	75**	75**	154	75**	75**
200	16	260	144	75**	75**	142	75**	75**
220	18	280	132	75**	71	130	75**	74
240	20	300	121	75**	58	119	75**	61
260	22	320	109	75**	45	107	75**	48

** Maximum dose

12 Points for SRI

Use of quality certified / hybrid seed Seed rate 2 kg per acre
40 m^2 nursery for one acre Raised bed nursery / Tray nursery
14 days old seedling (3-4 leaf stage) Levelling with lazer leveller
Marker for square planting
Square planting with 25 cm x 25 cm Single seedling per hill
Alternate wetting and drying method of irrigation
Cono weeding four times from 10 DAT at an interval of 10-15 days Use of leaf colour chart (LCC) for N management

CROP PROTECTION

A) Pest management

Nursery

Seed treatment with Imidacloprid 48FS @ 2.5 g/kg seed.
An area of 800 sq.m. (20 cents) nursery is required for planting one ha of main field. Forty litres of spray fluid is required for spraying the nursery area.

Pests	Management strategies
Thrips *Stenchaetothrips biformis*	• Sampling: Wet your palm with water and pass over the foliage in 12 places at random in the nursery and count the number of thrips. • ETYL: If thrips population exceeds 60 numbers in 12 passes or if rolling of half of leaf area of first and 2nd leaves in 10% of seedlings is noticed. • Spray any one of the following insecticides: Monocrotophos 36% SL 40 ml

	Thiamethoxam 25% WG 4 g
Green leafhopper *Nephotettix nigropictus* *N. cincticeps* *N. virescens*	• Sampling: Take 25 net sweepings in the nursery area. If the population exceeds 60 for 25 sweepings or 20/m^2 by actual counting, • Maintain 2.5 cm of water in the nursery and broadcast carbofuran 3% CG 3.5 kg in 20 cents
Caseworm *Parapoynx stagnalis*	• Mix 250 ml of kerosene with sand and apply to the standing water • Dislodge the cases by passing a rope and drain water • Collect the cases and destroy • Spray any one of the following insecticides: Monocrotophos 36% SL 40 ml Quinalphos 25% EC 80 ml
Army worm *Spodoptera mauritia*	• Drain water from the nursery • Spray chlorpyriphos 20% EC 80 ml during evening hours.

ii) Main field

- Remove/destroy stubbles after harvest
- Keep the fields free from weeds
- Trim field bunds
- Provide effective drainage
- Avoid use of excessive 'N' fertilizers.
- Avoid close planting, especially in BPH and leaffolder prone areas/seasons
- Leave 30 cm space at every 2.5 m
- Use irrigation water judiciously
- Use light traps (1/ha) to monitor pest incidence
- Use pheromone traps (12/ha) to monitor stem borer and leaffolder incidence
- Remove and destroy egg masses of stem borer
- In BPH prone areas/seasons, avoid use of resurgence causing chemicals like synthetic pyrethroids and quinalphos
- Use suggested insecticides at recommended doses based on ETL
- Avoid repeated use of same insecticide
- Dose recommended are per ha, unless otherwise specified

Economic threshold level (ETL) for important pests

Pest	ETL
Stem borer	2 egg masses/m^2 or 10% dead hearts or 2% white ear
Leaffolder	10% leaf damage at vegetative phase and 5% of flag leaf damage at flowering

Gall midge	10% silver shoots
Whorl maggot	25% damaged leaves
Thrips	60 numbers in 12 passes or rolling of the first and second leaves in 10% of seedlings.
Brown planthopper	1 hopper/ tiller in the absence of predatory spider and 2 hoppers / tiller when spider is present at 1/hill.
Green leafhopper	60/25 net sweeps or 5/hill at vegetative stage or 10/hill at flowering or 2/hill in tungro endemic area
Earhead bug	5 bugs/100 earheads at flowering and 16 bugs/100 ear heads from milky stage to grain maturity

Pests	Management strategies
Stem borer *Scirpophaga incertulas*	• Release of the egg parasitoid, *Trichogramma japonicum* thrice (at weekly interval from 37 DAT) @ 1,00,000/ha each release (when moth activity is noticed) • *Bacillus thuringiensis* var. *kurstaki* @ 1.50 kg/ha • Spray any one of the following insecticides: (per ha) Acephate 75 % SP 670-1000 g Acephate 95 % SG 590 g Azadirachtin 0.03% 1000 ml Carbofuran 3% CG 25 kg Carbosulfan 6% G 16.7 kg Carbosulfan 25% EC 800-1000 ml Cartap hydrochloride 50 % SP 1000 g Chlorantraniliprole 18.5% SC 150 ml Chlorantraniliprole 0.4% G 10 kg Chlorpyriphos 20% EC 1250 ml Fipronil 5% SC 1000-1500 g Fipronil 80%WG 50- 62.5 kg Flubendiamide 20% WG 125 g Flubendiamide 39.35% M/M SC 50 g Thiacloprid 21.7% SC 500 g Thiamethoxam 25% WG 100 g

Leaffolder *Cnaphalocrocis medinalis*	• Release *Trichogramma chilonis* thrice (at weekly interval from 30 DAT) @ 1,00,000/ha each (when moth activity is noticed) • Spray *Bacillus thuringiensis* var. *kurstaki* 1.50 kg/ha • Apply *Beauveria bassiana* 1.15 WP 2.5 kg/ha • Spray any one of the following insecticides per ha: Acephate 75 % SP 666-1000 ml Acephate 95 % SG 590 g Azadirachtin 0.03% 1000 ml Chlorpyriphos 20% EC 1250 ml Carbosulfan 6% G 16.7 kg Cartaphydrochloride 50 % SP 1000 g Chlorantraniliprole 18.5% SC 150 g Chlorantraniliprole 0.4% G 10 kg Fipronil 80%WG 50-62.5 g Flubendiamide 20% WG 125-250 g Flubendiamide 39.35% M/M SC 50 g Indoxacarb 15.8% EC 200 g Thiamethoxam 25% WG 100 g
Gall midge *Orseolia oryzae*	• Distribute *Platygaster oryzae* parasitised galls at 1 per 10 m^2 on 10 days after transplanting (DAT), when natural parasitisation is noticed in abundance. • Spray any one of the following insecticides per ha: Carbosulfan 25% EC 800-1000 ml Chlorpyriphos 20% EC 1250 ml Fipronil 5% SC 1000-1500 g Fipronil 0.3% G 16.67 - 25 kg Quinalphos 5% G 5 kg Thiamethoxam 25% WG 100 g
Whorl maggot *Hydrellia sasakii*	• Spray any one of the following insecticides per ha: Cartap hydrochloride 4% G 18.75 – 25 kg Chlorpyriphos 20% EC 1250 ml Fipronil 5% SC 1000-1500 g Fipronil 0.3% GR 16.67- 25 kg
Case worm *Parapoynx stagnalis*	Spray phenthoate 50% EC 1000 ml
Hispa/ spiny beetle *Dicladispa armigera*	• Spray any one of the following insecticides per ha: Carbofuran 3% CG 25 kg Chlorpyriphos 20% EC 1250 ml Malathion 5% DP 25 kg

	Malathion 50%EC 1150 ml
Grasshopper	Dust chlorpyriphos 1.5% DP 25 kg/ha
Thrips *Stenchaetothrips biformis*	• Spray any one of the following insecticides per ha: Azadirachtin 0.15% W/W 1.5 – 2.5 kg Thiamethoxam 25% WG 100 g
Brown planthopper *Nilaparvata lugens*	• Avoid excessive use of nitrogen • Control irrigation by intermittent draining • Set up light traps during night or yellow pan traps during day time • Drain water before use of insecticides • Direct spray towards the base of the plants. • Spray any one of the following insecticides per ha: Acephate 75 % SP 666-1000 g Acephate 95 % SG 590 g Acetamiprid 20% SP 50-100 g Azadirachtin 0.03% 1000 ml Neem oil 3% 15 lit Buprofezin 25% SC 800 ml Carbosulfan 25% EC 800-1000 ml Clothianidin 50% WG 20-24 g Chlorantraniliprole 18.5% SC 150 g Chlorantraniliprole 0.4% G 10 kg Chlorpyriphos 1.5% DP 25 kg Chlorpyriphos 20% EC 1250 ml Dinotefuran 20% SG 150-200g Fenobucarb 50% EC 500-1500 ml Fipronil 5% SC 1000-1500 ml Fipronil 0.3% GR 16.67-25 kg Imidacloprid 70% WG 30-35 kg Imidacloprid 17.8 SL 100-125 ml Pymetrozine 50% WG 300g
White backed planthopper *Sogatella furcifera*	• Spray any one of the following insecticides per ha: Phosphamidon 40% SL 1000 ml Azadirachtin 0.03% 1000 ml Buprofezin 25% SC 800 ml Carbosulfan 25% EC 800-1000 ml Chlorantraniliprole 18.5% SC 150 g Chlorantraniliprole 0.4% G 10 kg

	Fipronil 5% SC 1000-1500 ml Fipronil 0.3% GR 16.67-25 kg Imidacloprid 70% WG 30-35 kg Imidacloprid 17.8% SL 100-125 ml
Green leafhopper *Nephotettix nigropictus* *N. cincticeps* *N. virescens*	• Spray any one of the following insecticides twice, 15 and 30 days after transplanting per ha: Phosphamidon 40% SL 1000 ml Carbofuran 3% CG 25 kg Buprofezin 25% SC 800 g Carbosulfan 25% EC 800-1000 ml Fipronil 5% SC 1000-1500 g Fipronil 0.3% G 16.67-25 kg Imidacloprid 17.8% SL 100 -125 ml Thiamethoxam 25% WG 100 g • The vegetation on the bunds should also be sprayed with the insecticides • Set up light traps to attract and control the leafhoppers as well as to monitor the vector population. • Destroy/ kill the leafhoppers attracted to light trap
Mealybug *Brevennia rehi*	• Spray methyl demeton 25% EC 1000 ml/ha
Blue leafhopper/ white leafhopper	• Spray methyl demeton 25% EC 500-1000 ml/ha
Black bug *Scotinophara lurida*	• Spray neem seed kernel extract 5% (25 kg kernel/ha)
Earhead bug *Leptocorisa acuta and L. oratorius*	• Dust/ spray any one of the following, the first during flowering and second a week later (per ha): Quinalphos 1.5% D 25 kg Malathion 50% EC 500 ml Neem seed kernel extract 5% (25 kg kernel/ha) Notchi or *Ipomoea* or *Prosopis* leaf extract 10% KKM 10% D 25 kg
Termite *Anacanthotermus viarum*	• Apply chopped paddy straw treated with chlorpyriphos 1.5% DP 25 kg/ha
Mite *Oligonychus oryzae*	Spray any one of the following insecticides per ha: • Dicofol 18.5% EC 1250 ml Azadirachtin 0.03% 1000 ml

Rat *Rattus rattus rufuscens, Rattus meltada*	• Poison bait at 1 part zinc phosphide with 49 parts popped corn/rice/dry fish or warfarin 0.5% 1 part with 19 parts of popped corn/rice/dry fish or bromodialone 0.25 w/w (1:49) at 0.005%. Mix one part of bromodialone + 49 parts of bait and keep inside the field. • Mechanical collection and destruction • Narrow bund maintenance (45 x 30 cms) • Setting up of owl perches • Setting up of Thanjavur bow trap @ 100/ha

IPM module

- Seed treatment with Imidacloprid 48%FS @ 2.5 g/kg
- *Pseudomonas fluorescens* – Seed treatment (10 g/kg), seedling dip (2.5 kg/ha), main field application (2.5 kg/ha)
- Pest and disease management in nursery (preferably neem seed kernel extract 5% or Neem oil 2%)
- Integrated Nutrient Management
- Use of neem cake coated urea (5 : 1)
- Incorporation of green manures / biofertilizers
- 'N' management by Leaf Colour Chart (LCC)
- 'K' application – basal (50%) + one top dressing (50%)
- Adoption of cultural practices
- Variety selection
- Spacing based on season, variety and location (endemic / hot spot)
- Rogueing space (1' for every 8')
- Water management – alternate wetting and drying and submergence of recommended level during critical periods only
- Release of biocontrol agents (*Trichogramma japonicum* for stem borer and *Trichogramma chilonis* for leaffolder), when the moth activity is noticed
- Set up bird (owl) perches at 40 to 50 /ha
- Application of botanicals especially Neem seed kernel extract 5% and Neem oil 2%
- ETL based insecticide / fungicide application (No synthetic pyrethroids)
- Integrated rodent management
  Narrow bund maintenance (45 x 30 cms)
  Zinc phosphide baiting (49: 1)
  Trapping with Thanjavur bow trap (100 nos./ha)
  Baiting with bromodialone

Resurgence

Repeated application of the following insecticides can cause resurgence of insect pests Avoid spraying of synthetic pyrethroids and the following insecticides

- BPH, *Nilaparvata lugens* **:** acephate, carbofuran, chlorpyriphos, deltamethrin, ethopenprox, fenthion, fenvalerate, methomyl, methylparathion, monocrotophos, permethrin, perthane, phosalone, quinalphos, thiometon, vamidothion
- GLH, *Nephotettix virescens :* deltamethrin, phorate
- WBPH, *Sogatella furcifera :* cypermethrin, deltamethrin, fenvalerate
- Leaf folder, *Cnaphalocrocis medinalis :* Carbofuran

B. Disease Management

I. Disease management in nursery

Dry seed treatment	• Treat the seeds with thiram or captan or carboxin or carbendazim @ 2 g/kg of seeds • Treat the seeds at least 24 hours prior to soaking for sprouting • The treated seeds can be stored for 30 days without any loss in viability
Wet seed treatment	• Treat the seeds with carbendazim or tricyclazole @ 2 g/l/kg of seeds • Soak the seeds in the solution for 2 hours • Drain the solution, sprout the seeds and sow in the nursery bed • This wet seed treatment gives protection to the seedlings up to 40 days from seedling diseases such as blast and this method is better than dry seed treatment or • Treat the seeds with talc based formulation of *Pseudomonas fluorescens* @ 10g/kg of seed and soak in 1lit of water overnight • Decant the excess water and allow to sprout the seeds for 24 h and then sow **CIB Recommendation** • Treat the seeds with carbendazim 25% + mancozeb 50% WS @ 3-3.5 g/kg of seeds
Seedling dip with *Pseudomonas fluorescens*	• Stagnate water to a depth of 2.5cm over an area of $25m^2$ in the main field • Sprinkle 2.5 kg of the talc based formulation of *P. fluorescens* and mix with stagnated water • The seedlings pulled out from the nursery are to be soaked for 30 min. in the stagnated water and then transplanted
❖ Biocontrol agents are compatible with biofertilizers ❖ Biofertilizers and biocontrol agents can be mixed together for seed soaking ❖ Fungicides and biocontrol agents are incompatible	

II. Disease management in main field

Name of the Disease	Recommendations
Blast : ***Pyricularia grisea*** **(*Magnaporthe grisea)***	**1. Cultural methods** • Remove collateral weed hosts from bunds and channels • Use only disease free seedlings • Avoid excess nitrogen • Apply N in three split doses (50% as basal, 25% at tillering phase and 25% at panicle initiation stage) • Use resistant varieties like CO 47, CO 52 and TNAU rice hybrid CO 4 and moderately resistant varieties like CO 50 and CO 51 in endemic areas **2. Chemical method** • Spray carbendazim 50WP @ 500 g/ha or tricyclozole 75 WP @ 500 g/ha or metominostrobin 20 SC @ 500 ml/ha or Azoxystrobin 25 SC @ 500 ml/ha after observing initial infection of the disease **CIB Recommendation** • Spray isoprothiolane 40 % EC @ 750 ml/ha or kasugamycin 3% S.L @ 1000 -1500 ml/ha or kasugamycin 5% + copper oxychloride 45% WP @ 700gm/ha or picoxystrobin 22.52% SC @ 600 ml/ha or tebuconazole 25% WG @ 750 gm/ha or mancozeb 75 % WP @ 1.5 - 2.0 kg/ha or aureofungin 46.15% SP @ 1% after observing initial infection of the disease and repeat after 15 days, if required. **3. Biological control** • Treat the seeds with *Pseudomonas fluorescens* TNAU liquid formulation @ 10 ml/kg of seeds • Seedling root dipping with *P. fluorescens* TNAU liquid formulation @ 500 ml for one hectare seedlings • Soil application with *P. fluorescens* TNAU liquid formulation @ 500 ml/ha • Foliar spray with *P. fluorescens* TNAU liquid formulation @ 5 ml/l
Brown spot: *Drechslera oryzae* (*Cochliobolus miyabeanus*)	• Spray metominostrobin @ 500 ml/ha after observing initial infection of the disease **CIB Recommendation** • Spray propineb 70% WP @ 1500 – 2000 gm/ha or propineb 54.2% + tricyclazole15 % WP @ 1250 gm/ha or carbendazim 5% GR @ 12.5 kg/ha **r combined infection of blast and brown spot** • Spray propineb 54.2% + tricyclazole 15 % WP @ 1250 gm/ha after

<table>
<tr><td></td><td>observing initial infection of the disease.
For combined infection of blast, sheath blight and brown spot
• Spray azoxystrobin 16.7 % + tricyclazole 33.3% SC @ 500 ml/ha after observing initial infection of the disease</td></tr>
<tr><td>Sheath rot:
Sarocladium oryzae</td><td>Botanicals
• Spray neem oil 3% or Ipomoea leaf powder extract @ 25 kg/ha or Prosopis leaf powder extract @ 25 kg/ha. First spray at boot leaf stage and second at 15 days later
Chemical method
• Spray carbendazim @ 500 g/ha or metominostrobin @ 500 ml/ha or hexaconazole 75% WG @ 100 mg/ lit. First spray at the time of disease appearance and second spray at 15 days later
Biological control
• Treat the seeds with Pseudomonas fluorescens TNAU liquid formulation @ 10 ml/kg of seeds
• Seedling root dipping with P. fluorescens TNAU liquid formulation @ 500 ml for one hectare seedlings
• Soil application with P. fluorescens TNAU liquid formulation @ 500ml/ha
• Foliar spray with P. fluorescens TNAU liquid formulation @ 5ml/l</td></tr>
<tr><td>Sheath blight:
Rhizoctonia solani (Thanatephorus cucumeris)</td><td>Cultural method
• Apply neem cake @ 150 kg/ha to soil
Using botanical
• Foliar spray with neem oil 3% @ 15 l/ha starting from disease appearance
Chemical method
• Spray carbendazim 50 WP @ 500 g/ha or azoxystrobin @ 500 ml/ha or hexaconazole 75% WG @ 100 mg/l. First spray at the time of disease appearance and second spray at 15 days later.
CIB Recommendation
• Spray azoxystrobin 11% + tebuconazole 18.3% w/w SC @ 750 ml/ha or azoxystrobin 7.1% + propiconazole 11.9 % W/W SE @ 500 ml/ha or flusilazole 40% EC @ 300 ml/ha or iprodione 50 % W.P @ 2.25 kg/ha or pencycuron 22.9 % SC @ 600-750 ml/ha or propiconazole 25% E.C @ 500 ml/ha or thifluzamide 24 % SC @ 375 ml/ha or carbendazim 25 % + flusilazole 12.5% SE @ 800-960 ml/ha after observing initial infection of the disease
For combined infection of blast and sheath blight
• Spray hexaconazole 4% + carbendazim 16% SC @ 750 gm/ha or hexaconazole 5% EC @ 1000 ml/ha or iprodione 25% + carbendazim 25% WP@ 500 gm/ha or carpropamid 27.8% SC @ 100 ml/ha or iprobenphos 48 % EC @ 200 ml/ha or kresoxim-methyl 44.3% SC @ 500 gm/ha or tebuconazole 25.9% E.C. @ 750 ml/ha or tricyclazole 45% + hexaconazole 10% WG @ 500 gm/ha or carbendazim 1.92% + mancozeb 10.08% GR @ 12.5 kg/ha after observing initial infection of</td></tr>
</table>

	the disease. Repeat the applications as per severity of diseases. Broadcast the granules under standing water condition **Biological control** • Treat the seeds with *Pseudomonas fluorescens* TNAU liquid formulation @ 10 ml/kg of seeds • Seedling root dipping with *P. fluorescens* TNAU liquid formulation @ 500 ml for one hectare seedlings • Soil application with *P. fluorescens* TNAU liquid formulation @ 500 ml/ha • Foliar spray with *P. fluorescens* TNAU liquid formulation @ 5 ml/l
Rice grain discoloration: *Helminthosporium oryzae, Alternaria tenuis, Fusarium Moniliforme, Sarocladium oryzae*	• Spray carbendazim + thiram + mancozeb (1:1:1) @ 0.2% at 50% flowering stage **CIB Recommendation** • Spray tebuconazole 50% + trifloxystrobin 25% WG @ 200 gm/ha at 50% flowering stage **For combined infection of sheath blight, leaf blast and neck blast and grain discolouration** • Spray tebuconazole 50% + trifloxystrobin 25% @ 200 gm/ha For combined infection of blast, brown spot and grain discolouration, spray tricyclazole 18% + mancozeb 62% WP @ 1000 – 1250 g/ha
Bacterial leaf blight: *Xanthomonas oryzae* pv. *oryzae* and **Bacterial leaf streak:** *Xanthomonas oryzae* pv. *oryzicola*	• Spray 20% fresh cow dung extract twice (starting from initial appearance of the disease and another at fortnightly interval) or spray twice copper hydroxide 77 WP @1.25 kg/ha 30 and 45 days after planting or spray streptomycin sulphate + tetracycline combination @ 300 g + copper oxychloride @ 1.25 kg/ha. If necessary repeat 15 days later • Application of bleaching powder @ 5 kg/ha in the irrigation water is recommended at the kresek stage • Spray neem oil 60 EC @ 3% or NSKE @ 5% for the control of sheath rot, sheath blight, grain discolouration and bacterial blight
False smut: *Ustiloginoidea virens*	• Two sprays with propiconazole 25 EC @ 500 ml/ha or copper hydroxide 77 WP @ 1.25 kg/ha at boot leaf and 50% flowering stages **CIB Recommendation** Two sprays with copper hydroxide 77 WP @ 2.0 kg/ha at boot leaf and 50% flowering stages
Rice tungro disease: *Rice tungro Bacilliform virus* and *Rice tungro Spherical virus* (Vectors: *Nephotettix*	**Physical methods** • Set up light traps to attract and control the leaf hopper vectors as well as to monitor the population. • In the early morning, the population of leafhopper alighting near the light trap should be killed by spraying / dusting the insecticides. This should be practiced every day **Chemical method** • Spray phosphamidon 40% SL 1000 ml/ha or carbofuran 3% CG 25 kg/ha or buprofezin 25% SC 800 g/ha or carbosulfan 25% EC 800-

virescens *N. nigropictus* *N. parvus* *N. malayanus* *Recilia dorsalis*)	1000 ml/ha or fipronil 5% SC 1000-1500 g/ha or fipronil 0.3% G 16.67-25 kg/ha or imidacloprid 17.8% SL 100 -125 ml/ha or thiamethoxam 25% WG 100 g/ha twice at 15 and 30 days after transplanting • The vegetation on the bunds should also be sprayed with the insecticides
Rice Orange leaf: ***Candidatus Phytoplasma*** (Vector: *Nephotettix virescens,* *N. nigropictus*)	**Cultural method** • Plough the stubbles as soon as the crop is harvested to prevent the survival of orange leaf pathogen during offseason **Chemical method** • Spray phosphamidon 40% SL 1000 ml/ha or carbofuran 3% CG 25 kg/ha or buprofezin 25% SC 800 g/ha or carbosulfan 25% EC 800-1000 ml/ha or fipronil 5% SC 1000-1500 g/ha or fipronil 0.3% G 16.67-25 kg/ha or imidacloprid 17.8% SL 100 -125 ml/ha or thiamethoxam 25% WG 100 g/ha twice at 15 and 30 days after transplanting • The vegetation on the bunds should also be sprayed with the insecticides

C) Nematode management

❖ Application of carbofuran 3G @ 1 kg a.i./ha both in nursery and in main field at 45days after planting reduces of rice root-knot nematode, *Meloidogyne graminicola*.

RICE - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety, if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, a distance of 3 m all around the field from the same and other varieties of the crop.

Pre-sowing seed management

- In dormant cultivars, break the dormancy by soaking the seeds in equal volume of 0.1 N conc. HNO_3 or 0.5 % KNO_3 for 12 - 16 h.
- Upgrade the seeds adapting specific gravity grading with salt solution prepared by dissolving 1.5 kg of common salt in 10 lit of water. Remove the floaters and sinkers should be used for sowing after repeated washing with water.
- Harden the seeds for rainfed or direct sowing by soaking in equal volume of 1% KCl solution for 16 h and dry back the seeds to original moisture content.
- Soak the seed in 4% *Pseudomonas fluorescens* for 12 h at the ratio of 1 : 1 and dry back the seeds to original seed moisture content under shade.
- Soak the seeds in equal volume of 80 µM concentration of sodium nitroprusside for 16 hrs to raise the nursery in saline / sodic soils.

Method of planting

- SRI method can be adapted.

Recommendation for planting under saline soil condition

- Incorporation of green manure like daincha in soil.
- Shallow planting at 3 - 4 seedlings / hill.
- Basal application of gypsum @ 500 kg / ha.
- Foliar spray with 0.5 % $FeSO_4$ and $ZnSO_4$ at tillering stage.

Recommendation for Zinc deficient soils

- Apply $ZnSO_4$ @ 25 kg / ha.

Fertilizer recommendation for different duration varieties

- Short duration : NPK @ 120:40:40 kg / ha
- Medium duration : NPK @ 150:50:60 kg / ha
- Long duration : NPK @ 150:50:80 kg / ha

Roguing space

- Leave a roguing space of 30 cm for every 150 cm.

Foliar application

- Foliar spray of 2 % DAP at boot leaf stage and at 5 - 10% flowering.

Harvesting

- When 90 % of the panicle are in golden yellow colour with the moisture content of 20% for short and medium duration varieties and 17% moisture for long duration varieties.

Threshing

- Thresh either manually or using mechanical threshers at a seed moisture content of 16 - 17%.

Drying

- Dry the seeds to 12 - 13 % moisture content for short term storage and 8-9 % moisture for long term storage.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat the seeds with halogen mixture @ 3 g / kg ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1) as eco-friendly treatment.
- Seeds of poor storable paddy varieties (ADT 38, ADT 46) at 10 per cent moisture content are to be treated with halopolymer @ 4 g / kg + Bavistin @ 2 g / kg + Imidacloprid @ 1 ml / kg and stored in super grain bag for extending the storability.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8-9 months) with a seed moisture content of 12 - 13 %.
- Store the seeds in polylined gunny bag for medium term storage (12- 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

RICE - HYBRID SEED PRODUCTION TECHNIQUES

Land requirement

- Select fertile land with good drainage and irrigation facilities.
- Previous crop should not be different varieties / hybrids of paddy

Isolation

- Space isolation : 100 m
- Time isolation : 25 days (later)
- Barrier isolation : Either a distance of 30 m with vegetative barrier or plastic sheet with 2 m height.

Staggered sowing

- Male parent should be sown in three to four staggered sowings based on the duration of parental lines for continuous availability of pollen till the completion of flowering in the female parent.

Main field management

Spacing

- Between 'A' lines 10 cm; between 'R' lines 30 cm; between A and 'R' line 20 cm : within rows 15 cm.

Planting design

- Two paired row @ 2 - 3 seedlings / hill.

Fertilizer application

- Apply NPK @ 150 : 60 : 60 kg / ha.
- Apply N and K in 3 split doses during the basal, active tillering and panicle initiation stages.

Foliar Application

- Foliar spray of 2 % DAP at boot leaf stage and another at 5 - 10 % flowering stage.

Special operations

Foliar spray of GA_3 @ 75 g/ha for Panicle exertion

- First foliar spray of 40 g of GA_3 at 5-10 % panicle emergence stage followed by 35 g of GA_3 at 24 h after first spray.

Note: GA_3 should be dissolved in 70 % ethyl alcohol.

Supplementary pollination

- Rope pulling or shaking the pollen parent (R line) with the help of two bamboo sticks at 30 - 40% of spikelets opening stage is followed as supplementary pollination technique. This process is repeated for 3 to 4 times during the day time (10 am to 1 pm) at an interval of 30 min and continued for 7 to 10 days during flowering period.

Harvesting

- Harvest the male parent (R line) first and remove completely from the field.
- Then harvest the seed parent (A line).

Grading

- For getting better seed quality, size grade the seeds using 1.3 mm x 19 mm oblong sieve.
- Size graded seeds may be upgraded by density grading using specific gravity separator. Heavy and medium fractions with 90 - 92% recovery are selected for seed purpose.

Drying

- Sundry the seeds to reduce the moisture content to 12 - 13 %.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg.
- Treat the seeds with halogen mixture @ 3 g / kg ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 as eco-friendly treatment.

2. MILLETS

(i) SORGHUM (*Sorghum bicolor*)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
40	7 - 8	27 - 35	400 - 600	up to 2300

Tropical crop. It can tolerate drought conditions as well as water logging condition. Short day plant

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Sl. No.	Agro ecological zones	Districts	Season	Varieties/ Hybrids
1	North Eastern Zone	Vellore, Thiruvannamalai, Cuddalore, Villupuram, Thiruvallur and Kancheepuram	Jan-Feb (Thaipattam) April - May (Chithiraipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	CO 30
2	North Western Zone	Salem, Namakkal, Dharmapuri and Krishnagiri		
3	Western Zone	Coimbatore, Erode, Karur, Tiruppur, Theni and Dindigul		
4	Cauvery Delta Zone	Trichy, Thanjavur, Thiruvarur, Nagapattinam, Pudukkottai Perambalur and Ariyalur	Jan-Feb (Thaipattam) April - May (Chithiraipattam) June-July (Adipattam)	
5	Southern Zone	Madurai, Sivagangai, Virudhunagar, Ramanathapuram Tirunelveli and Thoothukudi	Jan-Feb (Thaipattam) April - May (Chithiraipattam) Sep-Oct (Puratassipattam)	CO 30 and K 12

II. PARTICULARS OF SORGHUM VARIETIES

PARTICULARS	K 12	CO 30
Year of Release	2015	2010
Year of Notification	SO.1379(E)/ 27.03.2018	SO.1708(E)/26.07.2012
Parentage	Derivative of SPV 772 × S 35-29	Derivative of APK 1 x TNS 291
Duration (Days)	95-100	95-105
Area (Districts)	Southern districts of Tamil Nadu	All districts
Season (Pattam)		
Rainfed	Puratasi	Adi,Puratasi
Irrigated	Chithirai	Thai, Chithirai

Grain yield (kg/ha)		
Rainfed	3123	2400
Irrigated	-	3360
Fodder yield (kg/ha)		
Rainfed	11900	7000
Irrigated	-	9200
Stalk	Juicy	Juicy
Plant height (cm)	225-240	220-240
Sheath Colour	Reddish purple	Tan Green
Midrib	White	Dull white
Earhead shape	Elongated	Cylindrical
Compactness	Semi Compact	Semi Compact
Grain Colour	Creamy white	White
Special features	Tolerant to drought, photo insensitive, moderately resistant to shoot fly and stem borer, resistant to downy mildew. Suitable for rainfed situation	High dry Matter digestibility, tolerance to shoot fly, grain mould and downy mildew

CROP MANAGEMENT

I. SELECTION OF SEEDS

Good quality seeds are to be collected from disease and pest-free fields.

Quantity of seed required

Irrigated Transplanted - 7.5 kg/ha; Direct sown - 10 kg/ha Rainfed Direct sown - 15 kg/ha Sorghum under irrigated condition is raised both as a direct sown and transplanted crop. Transplanted crop has the following advantages:

a. Main field duration is reduced by 10 days.
b. Shoot fly, which attacks direct sown crops during the first 3 weeks and which is difficult to control, can be effectively and economically controlled in the nursery itself.
c. Seedlings which show chlorotic and downy mildew symptoms can be eliminated, thereby incidence of downy mildew in the main field can be minimised.
d. Optimum population can be maintained as only healthy seedlings are used for transplanting.
e. Seed rate can also be reduced by 2.5 kg/ha.

II. NURSERY PRACTICES

1. NURSERY PREPARATION

For raising seedlings to plant one hectare, select 7.5 cents (300 m^2) near a water source where water will not stagnate.

2. APPLICATION OF FYM TO THE NURSERY

i. Apply 750 kg of FYM or compost for 7.5 cents nursery and apply another 500 kg of compost or FYM for covering the seeds after sowing.
ii. Spread the manure evenly on the unploughed soil and incorporate by ploughing or apply just before last ploughing.

3. LAYING THE NURSERY

i. Provide three separate units of size 2 m x 1.5 m with 30 cm space in between the plots and all around the unit for irrigation.
ii. Excavate the soil from the inter-space and all around to a depth of 15 cm to form channels and spread the soil removed on the bed and level.

4. PRE-TREATMENT OF SEEDS

i. Treat the seeds 24 hours prior to sowing with Carbendazim or Captan or Thiram at 2g/kg of seed.
ii. Carrier based formulation: Treat one hectare of seeds with 1 kg each of biofertilizers *viz., Azsopirillum,* Phosphobacteria (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) and 25 g of powder formulation of AM fungi using binder (polymer), shade dry for 30 minutes before sowing.
iii. Liquid formulation: Treat one hectare of seeds with 125 ml of each biofertilizers *viz., Azsopirillum,* Phosphobacteria (or) Azophos and Silicate solubilizing bacteria (SSB) shade dry for 30 minutes before sowing

5. SOWING AND COVERING THE SEEDS

1. Make shallow rills, not deeper than 1 cm on the bed by passing the fingers vertically over it.
2. Broadcast 7.5 kg of treated seeds evenly on the beds.
3. Cover by leveling the rills by passing the hand lightly over the soil.

6. WATER MANAGEMENT

i. Provide one inlet to each nursery unit.
ii. Allow water to enter through the inlet and cover all the channels till the raised beds are covered with water and then cut off.
iii. Adjust the frequency of irrigation according to the soil types as follows:

Number of irrigations	Red soil	Heavy soil
First irrigation	Immediately after sowing	Immediately after sowing
Second irrigation	3rd day after sowing	4th day after sowing
Third irrigation	7th day after sowing	9th day after sowing
Fourth irrigation	12th day after sowing	16th day after sowing

NOTE: Do not keep the seedlings in the nursery for more than 18 days. If older seedlings are used, establishment and yield are adversely affected. Do not allow cracks to develop in the nursery by properly adjusting the quantity of irrigation water.

MAIN FIELD PREPARATION FOR IRRIGATED CROP

1. PLOUGHING

Plough the field with an iron plough once (or) twice. Sorghum does not require fine tilth since it adversely affects germination and yield in the case of direct sown crop. To overcome the subsoil hard pan in Alfisols (deep red soils) chiselling the field at 0.5 m intervals to a depth of 40 cm on both the directions of the field followed by disc ploughing once and cultivator ploughing twice help to increase the yield of sorghum and the succeeding crops.

Application of FYM and 100% of recommended N can also be followed. In soils with sub-soil hard pan, chiselling should be done every year at the start of the cropping sequence to create a favourable physical environment.

2. APPLICATION OF FYM

Spread 12.5 t/ha FYM or any compost along with 2 kg each *Azospirillum,* Phosphobacteria or 2 kg of Azophos on the unploughed field and incorporate the manure in the soil. Apply well decomposed poultry manure @ 5 t/ha to improve the grain yield as well as physical properties of soils.

3. FORMATION OF RIDGES AND FURROWS

i. Form ridges and furrows of 6 m length and 45 cm apart
ii. Form irrigation channels across the furrows
iii. Alternatively form beds of sixe 10 m^2 and 20 m^2 depending on the availability of water.

4. APPLICATION OF FERTILIZERS Transplanted crop

If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 P_2O_5 45 K_2O kg/ha. Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of P_2O_5 and K_2O basally before planting. Apply 30 kg S basally for sulphur deficient soils. Soil test crop response based integrated plant nutrition system (STCR-IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets (ready reckoners are furnished).

Sorghum - Hybrid (1)

Soil : Red sandy loam (Irugur series)
Target : 4.0 - 5.0 t ha^{-1}

$FN = 4.86T - 0.53\ SN - 0.98\ ON$
$FP_2O_5 = 1.63T - 0.87\ SP - 0.90\ OP$
$FK_2O = 4.56T - 0.59SK - 0.76\ OK$

Initial soil test values (kg ha^{-1})			Yield target – 4 t ha^{-1}			Yield target – 5 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
160	12	160	58	23	48	106	39	68**
180	14	180	47	23*	36	96	37	68**
200	16	200	36	23*	24	85	36	68**
220	18	220	26	23*	23*	74	34	58
240	20	240	45*	23*	23*	64	32	46

* Maintenance dose; ** Maximum dose

Sorghum -Varieties (2)

Soil :	Mixed black calcareous (Perianaickenpalayam series)	FN = 6.06T-0.81SN-0.53 ON
Target :	4.0- 5.0 t ha^{-1}	FP_2O_5 = 2.06T-3.14 SP-0.72 OP
		FK_2O = 5.03T-0.47SK-0.66 OK

Initial soil test values (kg ha^{-1})			**Yield target – 4 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 5 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	12	300	45*	23*	23*	105	33	71
200	14	340	45*	23*	23*	89	27	52
220	16	380	45*	23*	23*	73	23*	33
240	18	420	45*	23*	23*	57	23*	23*
260	20	460	45*	23*	23*	45*	23*	23*

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}.

i. In the case of ridge planted crop, open a furrow 5 cm deep on the side of the ridge at two thirds the distance from the top of the ridge and place the fertilizer mixture along the furrow and cover with soil upto 2 cm.
ii. Soil application of Azospirillum at 10 packets (2 kg/ha) and 10 packets (2000g/ha) of phosphobacteria or 20 packets of Azophos (4000g/ha) after mixing with 25 kg of FYM + 25 kg of soil may be carried out before sowing/planting.

Direct sown crop

i. Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 P_2 O_5, 45 2 5 K_2O kg/ha.
ii. Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of $P_{25}O_5$ and K_2 O basally before sowing and if basal application is not possible the same could be top dressed within 24 hours.
iii. In the case of bed planted crop, mark lines to a depth of 5 cm and 45 cm apart. Place the fertilizer mixture at the depth of 5 cm along the lines. Cover the lines upto 2 cm from the top before sowing.
iv. In the case of sorghum raised as a mixed crop with a pulse crop (Blackgram, Greengram or Cowpea) open furrows 30 cm apart to a depth of 5 cm.
v. Apply fertilizer mixture in two lines in which sorghum is to be raised and cover upto 2 cm.
vi. Skip the third row in which the pulse crop is to be raised and place fertilizer mixture in the next two rows and cover upto 2 cm with soil.

vii. Application of bio-fertilizers: When Azospirillum is used apply only 75% of recommended N for irrigated sorghum.

5. APPLICATION OF MICRONUTRIENTS Transplanted Crop

i. Mix 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg and apply the mixture over the furrows and on top one third of the ridges (or) TNAU MN mixture @ 12.5 kg ha^{-1} for irrigated; 7.5 kg for rainfed crop as enriched FYM. Prepare enriched FYM @ 1:10 ratio of MN Mixture & FYM at friable moisture and incubate for one month in shade.

ii. If micronutrient mixture is not available, mix 25 kg of zinc sulphate with sand to make a total quantity of 50 kg and apply on the furrows and on the top one third of the ridges.

Direct Sown Crop

i. Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg.

ii. Spread the mixture evenly on the beds.

iii. Basal application of 25 kg $ZnSO_4/ha^{-1}$ (or) 12.5 kg kg Zn SO_4 + 12.5 t/ha FYM for deficient soils.

iv. Basal application of $FeSO_4$ @ 50 kg/ha along with 12.5 t/ha FYM for iron deficient soils.

v. Foliar spraying of 0.5% $ZnSO_4$, 1.0% $FeSO_4$+0.1% citric acid thrice on 30, 40 and 50 DAS if deciciency observed in plants.

III. MANAGEMENT OF MAIN FIELD Spacing: 45 x 15 cm Population: $15/m^2$

1. TRANSPLANTED CROP

i. Pull out the seedlings when they are 15 to 18 days old.

ii. Prepare slurry with 5 packets of Azospirillum (1000g/ha) and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of Azophos (2000 g/ha) in 40 lit. of water and dip the root portion of the seedlings in the solution for 15-30 minutes and transplant.

iii. Let in water through the furrows

iv. Plant one seedling per hill

v. Plant the seedlings at a depth of 3 to 5 cm.

vi. Plant the seedlings on the side of the ridge, half the distance from the top of the ridge and the bottom.

2. DIRECT SOWN CROP

i. In the case of pure crop of sorghum, maintain the seed rate at 10kg/ha.

ii. In the case of inter crop of sorghum with pulse crop, maintain the seed rate of sorghum at 10 kg/ha and pulse crop at 10 kg/ha.

iii. In the case of pure crop of sorghum, sow the seeds with a spacing of 15 cm between seeds in the rows which are 45 cm apart.

iv. Maintain one plant per hill.

v. If shootfly attack is there, remove the side shots and retain one healthy shoot.

vi. Sow the seeds over the lines where fertilizers are placed.

vii. Sow the seeds at a depth of 2 cm and cover with soil.

viii. In the case of sorghum intercropped with pulses sow one paired row of sorghum alternated with a single row of pulses. The spacing between the row of sorghum and pulse crop is 30 cm.

Forage cowpea CO 1 can be intercropped in sorghum at two rows of fodder cowpea in between paired rows of sorghum.

3. WEED MANAGEMENT

i. Apply PE Atrazine @ 0.25 kg/ha on 3-5 DAS followed by 2,4-D @ 1 kg/ha on 20-25 DAS on the soil surface, using Backpack/Knapsack/Rocker sprayer fitted with a flat fan nozzle using 500 litres of water/ha (or) if herbicides are not used, hand weeding twice on 10-15 DAS and 30-35 DAS.
ii. Apply PE Atrazine@0.25 kg/ha on 3-5 DAS followed by one hand weeding on 30-35 DAS.
iii. In line sown crop, apply PE Atrazine @ 0.25 kg/ha on 3-5 DAS followed by Twin Wheel hoe weeder weeding on 30-35 DAS.
iv. In transplanted crop, apply PE Atrazine @ 0.25 kg/ha on 3-5 DAT followed by 2,4-D @ 1 kg/ha on 20-25 DAT.
v. If pulse crop is to be raised as an intercrop in sorghum do not use Atrazine, spray PE Pendimethalin @ 0.75 kg/ha on 3-5 DAS

4. THINNING OF THE SEEDLINGS AND GAP FILLING Direct sown crop

Thin the seedlings and gap fill with the seedlings thinned out. Maintain a spacing of 15 cm between plants after the first hand weeding. Thin the pulse crop to a spacing of 10 cm between plants for all pulse crop except cowpea, for which spacing is maintained at 20 cm between plants.

5. DEFICIENCY SYMPTOMS

Zinc: Deficiency symptoms first appear in the newly formed leaves at 20 to 30 days age. Older leaves have yellow streaks or chlorotic striping between veins.

Iron:Interveinal chlorosis will be observed. If the deficiency continues the entire leaf including the veins may exhibit chlorotic symptoms. Newly formed leaves exhibit chlorotic symptoms. The entire crop may exhibit bleached appearance, dry and may die.

Direct sown crop

i. Foliar spraying of 1% $FeSO_4$+0.1% citric acid thrice if deficieny symptom appeared.
ii. Recommendation given in transplanted crop may be followed.

NOTE:

a. Spray only if micronutrient mixture is not applied.
b. Apply in case of iron deficiency.
c. If soil is calcareous

IV. WATER MANAGEMENT

Regulate irrigation according to the following growth phase of the crop.

	Transplanted crop	Direct sown crop
Growth phase	1 to 40 days	1 to 33 days
Flowering phase	41 to 70 days	34 to 65 days
Maturity phase	71 to 95 days	66 to 95 days

Stages	No. of Irrigation	Days of Transplanting / Sowing of Crop	
		Transplanted	Direct Sown
Light Soils			
i. Irrigate for germination	1	1st day	1st day
or establishment	2	4th day	4th day
Regulate irrigation	1	15th day	15th day
during vegetative phase	2	28th day	28th day
iii. Flowering phase (copious irrigation)	1	40th day	40th day
	2	52nd day	52nd day
	3	..	64th day
iv. Maturity phase (Control irrigation)	1	65th day	76th day
	2	..	88th day
v. Stop irrigation thereafter	..	..	
Heavy soils			
i. Irrigate for germination	1	1st day	1st day
or establishment	2	4th day	4th day
ii. Regulate irrigation	1	17th day	17th day
during vegetative phase	2	30th day	30th day
iii. Flowering phase (give	1	40th day	45th day
copious irrigation)	2	52nd day	60th day
	3	.. day	75th
iv. Maturity phase (Control irrigation)	1	72nd day day	90th
v. Stop irrigation thereafter			

NOTE : Adjust irrigation schedule according to the weather conditions and depending upon the receipt of rains. Contingent Plans to be done before 75% of soil moisture is lost from available water. Foliar Spray of 3% Kaolin (30 g in one litre of water) during period of stress will mitigate the ill effects.

V. **HARVESTING AND PROCESSING**

i. Consider the average duration of the crop and observe the crop. When the crop matures the leaves turn yellow and present a dried up appearance.
ii. The grains are hard and firm.
iii. At this stage, harvest the crop by cutting the earheads separately.
iv. Cut the stalk after a week, allow it to dry and then stack.
v. In the case of tall varieties, cut the stem at 10 to 15 cm above ground level and afterwards separate the earheads and stack the stalk.
vi. Dry the earheads.
vii. Thresh using a mechanical thresher or by drawing a stone roller over the earheads or by using cattle and dry the produce and store.

RATOON SORGHUM CROP

1. RATOONING TECHNIQUE

i. Harvest the main crop leaving 15 cm stubbles.
ii. Remove the first formed two sprouts from the main crop and allow only the later formed two sprouts to grow. Allow two tillers per hill.

2. HOEING AND WEEDING

i. Remove the weeds immediately after harvest of the main crop.
ii. Hoe and weed twice on 15^{th} and 30^{th} day after cutting.

3. APPLICATION OF FERTILIZERS

i. Apply 100 kg N/ha in two split doses.
ii. Apply the first dose on 15^{th} day after cutting and the second on 45^{th} day after cutting.
iii. Apply 50 kg P_2O_5 /ha along with the application of N on 45^{th} day.

4. WATER MANAGEMENT

i. Irrigate immediately after cutting the main crop.
ii. Irrigation should not be delayed for more than 24 hours after cutting.
iii. Irrigate on 3^{rd} or 4^{th} day after cutting.
iv. Subsequently irrigate once in 7 - 10 days.
v. Stop irrigation on 70 - 80 days after ratooning.

5. HARVEST

Harvest the crop when the grains turn yellow.

NOTE: The duration of the ratoon crop is about 15 days less than the main crop.

RAINFED SORGHUM

1. RAINFALL

Average and well distributed rainfall of 250-300 mm is optimum for rainfed sorghum.

2. DISTRIBUTION

Madurai, Dindigul, Theni, Ramanathapuram, Tirunelveli, Thoothukudi, Virudhunagar, Sivagangai, Tiruchirapalli, Erode, Salem, Namakkal, Coimbatore and Dharmapuri Districts.

3. SEASON

The crop can be grown in South West and North East monsoon seasons provided the rainfall is evenly distributed.

4. FIELD PREPARATION

i. Field has to be prepared well in advance taking advantage of early showers. FYM application should be done @ 12.5 t / ha and well incorporated at the time of ploughing.

ii. Chiseling for soils with hard pan
Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5 m interval, first in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5 t FYM or composted Coir pith/ha besides chiseling to get an additional yield of about 30% over control.

iii. To conserve the soil moisture sow the seeds in flat beds and form furrows between crop rows during inter cultivation or during third week after sowing.

5. SEED RATE

15 kg/ha

6. SEED TREATMENT Direct sown crop

Seed hardening ensures high germination. The seeds are pre-soaked in 2% potassium dihydrogen phosphate solution for 6 hours in equal volume and then dried back to its original moisture content in shade and are used for sowing. (or)

i) Harden the seeds with 1% aquous fresh leaf extract of *Prosopis juliflora* and pungam, (*Pongamia pinnata*) mixed in 1:1 for 16 hrs at 1:0.6 ratio (Seed and solution) followed by drying and subsequently pelleting the seeds with Pungam leaf powder @300 g/kg with gruel.

ii) Halogenise the seeds containing CaOCl, $CaCO_3$ and arappu leaf powder @ 5:4:1 ratio or iodine based (containing 2 mg of Iodine in 3 g of $CaCO_3$) formulation @ 3g/kg packed in polylined cloth bag to maintain seed viability for more than 10 month.

iii) Treat the seeds with three packets of azospirillum (600 g) and 3 packets of phosphobacteria or 6 packets of Azophos (1200 g/ha). In the main field, apply 10 packets of azospirillum 2000g/ha and 10 packets (2000g/ha) of phosphobacteria or 20 packets of Azophos (4000 g/ha) with phosphobacteria 2 kg with 25 kg FYM + 25 kg soil.

iv) The seed is pelletised with 15 g of Chloropyriphos in 150 ml of gum and shade dried.

7. SOWING

Sow the seeds well before the onset of monsoon at 5 cm depth (by seed drill or by country plough).

Pre-monsoon sowing

Sow the hardened seeds at 5 cm depth with seed cum fertilizer drill to ensure uniform depth of sowing and fertilizer application before the onset of monsoon as detailed below:

	District	Optimum period
1.	Coimbatore	37-38th week (II to III week of Sep.)
2.	Erode	38th week (III week of Sep.)
3.	Sivaganga	40th week (I week of Oct.)
4.	Ramanathapuram	40th week (I week of Oct.)
5.	Thoothukudi	39-40th week (Last week of Sep. to I week of Oct)
6.	Vellore, Tiruvannamalai	37th-38th week (Sep. II week to Sep. III week)

i. Sow the sorghum seeds over the line where the fertilizers are placed.
ii. Sow the seeds at a depth of 5 cm and cover with the soil.
iii. Sow the seeds with the spacings of 15 cm in the paired rows spaced 60 cm apart.
iv. Sow the pulse seeds to fall 10 cm apart in the furrows between the paired rows of sorghum.

8. SPACING

45 x 15 cm or 45 x 10 cm.

9. FERTILIZER

Apply 12.5 t/ha of Composted Coir pith + NPK at 40:20:0; Apply enriched FYM @ 750 kg/ha. The recommended dose of 40 kg N and 20 kg P_2O_5 /ha for rainfed sorghum can be halved if FYM @ 5 t/ha is applied.

10. WEED MANAGEMENT

Keep sorghum field free of weeds from second week after germination till 5th week. If sufficient moisture is available spray Atrazine 0.25 kg/ha as pre-emergence application within 3 days after the receipt of the soaking rainfall for sole sorghum and for sorghum based intercropping system with pulses, use Pendimethalin at 0.75 kg/ha.

Under rainfed sorghum intercropped with cowpea as a pre-plant incorporation of isoproturan @ 0.5 kg ha^{-1} gave good control of weed with applied after 1st and 2nd spell of rainfall pendimethalin 1.0 kg ha^{-1} will be safer for both the crops.

11. CROPPING SYSTEM

- The most profitable and remunerative sorghum based cropping system adopted is sorghum with cowpea, redgram, lab-lab, blackgram.
- In rainfed Vertisol, adopt paired row planting in sorghum and sow one row of blackgram/ cowpea in between paired rows of sorghum to have 100% population of sorghum plus 33% population of blackgram/cowpea.
- Intercropping of sunflower CO 1, with the main crop of sorghum CO 26 in 4:2 ratio is recommended under rainfed conditions during North-East monsoon for black soils of CBE.
- Intercropping of soybean with sorghum in the ratio 4:2 is recommended for kharif seasons.
- For sorghum - blackgram intercropping system as well as sole cropping, application of 20 kg N and 20 kg P O /ha through enriched FYM and treating the seeds with Azospirillum is 25 recommended for Aruppukottai region.
- For sorghum (CO 25) + Fodder cowpea (CO 1) intercropping system, application of 20 kg N and 20 kg P O /ha with enriched FYM is recommended for Coimbatore region 25
- The intercropping system, fodder sorghum (K 7) + Fodder cowpea (CO 5) at 3:2 ratio is found profitable for rainfed Vertisols of Aruppukottai.
- Tamarind and Neem trees upto 3-4 years from date of planting form an ideal tree component for agroforestry in black cotton soils of Kovilpatti. Sorghum and blackgram gave higher yield even at 50 per cent of the recommended level of fertilizer application.

CROP PROTECTION

A. Pest management

- Protect nursery by applying any one of the following insecticides (in 6 litres of water) on 7th and 14th day of sowing
  - Methyl demeton 25EC 12 ml
  - Dimethoate 30EC 12 ml
- Plough soon after harvest, remove and destroy the stubbles.
- Treat seeds with chlorpyriphos 20EC or phosalone 35 EC (4 ml/kg) or imidacloprid 48FS or imidacloprid 70WS or thiamethoxam 30FS (10 g/kg) before sowing.
- Avoid repeated application of insecticides which may induce resurgence
- The sowing of sorghum should be completed in as short a time as possible to avoid continuous flowering which favours grain midge and earhead bug multiplication in an area.
- Set up light traps till mid night to monitor, attract and kill adults of stem borer, grain midge and earhead caterpillars.

Economic threshold level (ETL) for important pests

Insect pest	ETL
Shoot fly	1 egg/plant in 10% of plants in the first two weeks of sowing or 10 % dead hearts
Mite	5 mites/cm^2 of leaf area

Stem borer	10 % damage
Grain midge	5 / earhead
Earhead caterpillar	2 / earhead
Earhead bug	10 / earhead
Shoot fly *Atherigona soccata*	• Take up early sowing of sorghum immediately after the receipt of South West or North East monsoon to minimise shoot fly incidence • In case of direct seeding, use increased seed rate up to 12.5 kg/ha and remove shoot fly damaged seedlings at the time of thinning • In case of trasnsplanting, transplant only healthy seedlings • Spray dimethoate 30EC 12 ml for an area of 120 m^2 nursery • Set up fish meal trap @ 12/ha till the crop is 30 days old • Plough soon after harvest, remove and destroy the stubbles **Apply any one of the following/ha** • Carbofuran 3CG 33.3 kg (at the time of sowing) • Dimethoate 30EC 500 ml • Neem seed kernel extract 5% • Quinalphos 25EC 1500 ml
Mite, *Oligonychus indicus*	Spray quinalphos 25EC 1500 ml/ha Direct the spray fluid towards the under surface of the leaves
Aphids *Rhopalosiphum maidis Melanaphis sacchari*	Spray dimethoate 30EC 500 ml/ha
Stem borer, *Chilo partellus, Sesamia inferens*	• Sowing lab lab / cowpea as an intercrop to minimize stem borer damage (Sorghum: Lab lab /cowpea 4:1) • Apply carbofuran 3CG 17 kg/ha (with sand) to make up a total quantity of 50 kg/ha and apply in leaf whorls
Grain midge, *Contarinia sorgicola*	• **Apply any one of the following/ha on 3rd and 18th day after panicle emergence** • Dimethoate 30EC 1650ml • Malathion 50EC 1600 ml • Malathion 5D 25 kg • Neem seed kernel extract 5% • Phosalone 35EC 1150 ml • Phosalone 4D 25 kg
Earhead bug, *Calocoris angustatus*	**Apply any one of the following/ha on 3rd and 18th day after panicle emergence** • Malathion 50EC 1000 ml • Malathion 5D 25 kg • Neem seed kernel extract 5% • Quinalphos 1.5DP 25 kg
Earhead caterpillar, *Helicoverpa armigera*	• Set up sex pheromone traps at 12 nos./ha to attract males of *Helicoverpa armigera* from flowering to grain hardening • Apply NPV at 1.5 X10^{12} POB along with crude sugar 2.5 kg + cotton seed kernel powder 250 g on the earheads twice at 10 days interval (preferably during early morning or evening) **Apply any one of the following/ha on 3rd and 18th day after panicle emergenc**

	• Malathion 5D 25 kg • Phosalone 4D 25 kg
Rice weevil, *Sitophilus oryzae*	• Treat seeds with chlorpyriphos 20EC 4 ml/kg

Disease Management

Nursery practices

Seed treatment: Treat the seeds 24h prior to sowing with carbendazim or captan or thiram @ 2 g/kg of seeds or metalaxyl @ 6 g/ kg of seeds.

Name of the Disease	Recommendations
Rust: *Puccinia purpurea*	• Spray mancozeb @ 1 kg/ha. Repeat fungicidal application after 10 days
Ergot or Sugary disease: *Sphacelia sorghi*	• Adjust the sowing period to prevent flowering during rainy and winter seasons • Spray mancozeb @ 1000 g/ha or propiconazole @ 500 ml/ha at 5 - 10% flowering and at 50% flowering stages. Repeat the spray after a week, if necessary
Head Mould: Fungal complex *Fusarium, Curvularia, Alternaria, Aspergillus* and *Phoma*sp.	• Spray mancozeb or captan @ 1000 g + aureofungin sol 100 g/ha in case of intermittent rainfall during earhead emergence and repeat, if necessary a week later
Downy Mildew: *Peronosclerospora sorghi*	• Rogue out infected plants up to 45 days of sowing • Spray metalaxyl + mancozeb @ 500 g or mancozeb @ 1000 g/ha after symptom development **CIB recommendation** • Seed treatment with metalaxyl-M 31.8% ES @ 2 ml/kg of seed or slurry seed treatment with metalaxyl35%WS @ 2 g/ kg seed
Charcoal Rot: *Macrophomina phaseolina*	• Treat the seeds with *Pseudomonas fluorescens* @10 g/kg or *Trichoderma viride* @ 4 g/kg of seed
Grain smut : *Sphacelotheca sorghi*	**CIB recommendation** • Treat the seeds with sulphur 80% WP @ 3-4 g/kg seed

SORGHUM - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety, if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 100 m all around the field from the same and other varieties of the crop.
- The distance may be extended to 400 m for the presence of Johnson grass.

Season

- June - July and October - November.

Pre-sowing seed treatment

- Soak the seeds in KH_2PO_4 2 % for 16 h at of 1:0.6 ratio and dry back the seeds to original seed moisture content (8 - 9 %) under shade. This can be adopted both for the garden and dry land ecosystem.
- Soak the seed in 4% *Pseudomonas fluorescens* for 12 h at 1 : 1 ratio and dry back the seeds to original seed moisture content under shade.

Fertilizer requirement

- As basal application NPK @ 100 : 50 : 50 kg / ha.

Spacing

- 45 x 10 cm.

Pre-harvest sanitation spray

- Spray 2 % carbendazim at ten days before harvest against black mould.

Harvesting

- Seeds attain physiological maturity 40 - 45 days after 50 % flowering.
- Harvest the earheads as once over harvest, when the seeds have attained the characteristic yellow colour.

Threshing

- Thresh the earheads either manually or mechanically at a moisture content of 15 - 18 %.

Seed grading

- Size grade the seeds either with 9 / 64" or depending upon the variety.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg.
- Treat the seeds with halogen mixture @ 3 g / kg ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8-9 months) with a seed moisture content of 10 - 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12- 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

SORGHUM - HYBRID SEED PRODUCTION

Land requirement

- Fertile land with good drainage and irrigation facility.

- Field should not have volunteer plants. Hence, the previous crop should not be the same or different variety / hybrid of sorghum.

Isolation

- For certified / quality seed production leave a distance of 200 m all around the field from same and other varieties of sorghum.

Season

- For increased seed set and effective synchronization, sow the crop during October - November.

Planting ratio

- Sow the female and male parents in the ratio of 4:2 for foundation seed production and 5:2 for certified seed production

Border rows

- Sow the male parent in four rows around the field for the availability of adequate pollen.

Fertilizer requirement

- NPK @ 100: 50: 50 kg / ha.
- Apply NPK @ 50:50:50 kg / ha as basal; 25 kg of nitrogen after first weeding and during boot leaf stage as top dressing.

Foliar Application

- Foliar spray of 0.5% $FeSO_4$ at primordial initiation stage and there after two sprays at ten days interval to enhance the seed set.

Synchronization techniques (Adopt any one of the following)

- Apply 1 % urea at flower initiation to the delayed parent.
- Withhold one irrigation to the advanced parent.
- Staggering the sowing of male and female parents depending upon the hybrid and location.
- Foliar spray of cycocel (CCC) @ 300 ppm to delay the flower formation
- Foliar spray of growth retardant, MH @ 500 ppm at 45 DAS to the advanced parent.

Harvesting

- Harvest the male parent (R line) first and remove from field.
- Harvest the hybrid crop when 90 % of seeds in the earhead have attained the characteristic yellow colour.

Other management practices

The techniques recommended for varieties can be adapted.

(i) CUMBU (*Pennisetum glaucum* (L) R. Br.)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
40	8 - 10	27 - 35	250 - 350	up to 1800

Tropical warm weather crop. Grow in a wide range of ecological conditions and can still yield well even under unfavourable conditions of drought stress and high temperatures. Best suited temperature for crop growth is between 27 - 30°C. Short day plant.

CROP IMPROVEMENT

I SEASON AND VARIETIES

Sl.No.	Agro ecological zones	Districts	Season	Varieties/ Hybrids
1	North Eastern Zone	Vellore, Thiruvannamalai, Cuddalore, Villupuram, Thiruvallur, Kancheepuram	Jan-Feb (Thaipattam) April - May (Chithiraipattam) June-July (Adipattam) Sep-Oct Puratassipattam)	CO 10 and Hybrid CO 9
2	North Western Zone	Salem, Namakkal, Dharmapuri, Krishnagiri		
3	Western Zone	Coimbatore, Erode, Karur, Tiruppur, Theni and Dindigul		
4	Cauvery Delta Zone	Trichy, Thanjavur, Thiruvarur, Nagapattinam, Pudukkottai Perambalur and Ariyalur	Jan-Feb (Thaipattam) April - May (Chithiraipattam) June-July (Adipattam)	
5	Southern Zone	Madurai, Sivagangai, Virudhunagar, Ramanathapuram Tirunelveli and Thoothukudi	Jan-Feb (Thaipattam) April - May (Chithiraipattam) Sep-Oct (Puratassipattam)	

II. PARTICULARS OF CUMBU HYBRID AND COMPOSITE

PARTICULARS	Hybrid CO 9	CO 10
Year of Release	2011	2016
Year of Notification	SO.1708(E)/26.07.2012	SO. 2238 (E) /29.06.2016
Parentage	ICMA 93111A x PT 6029-30	Composite of five elite inbred lines
Season-irrigated/ rainfed	Both	Both
Duration (Days)	75-80	85-90
Grain yield (kg/ha)		
Rainfed	2707	2923
Irrigated	3728	3526
Plant height (cm)	160-180	160-180
Tillers (No.)	4-6	4-6

Pigmentation	-	-
Hairiness	Absent	Absent
Days to 50% bloom	45-50	47-50
Shape of earhead	Candle to Cylindrical	Spindle
Bristles	Absent	Absent
Length of earhead (cm)	25-35	25-30
Earhead girth diameter (cm)	3.1-3.6	3.1-3.6
Grain Colour	Grey yellow	Grey brown
1000 grains weight (gm)	13-14	12-13
Special features	Short duration, High Fe content (8mg/100g) Resistant to downy Mildew	High protein Content (12.07%) and Resistant to downy mildew

CROP MANAGEMENT

II NURSERY

1. PREPARATION OF LAND

i. For raising seedlings to plant one ha select 7.5 cents near a water source. Water should not stagnate.

ii. Plough the land and bring it to the fine tilth.

2. APPLICATION OF FYM

Apply 750 kg of FYM or compost and incorporate by ploughing. Cover the seeds with 500 kg of FYM.

3. FORMING RAISED BED

i. In each cent mark 6 plots of the size 3 m x 1.5 m with 30 cm channel in between the plots and all around.

ii. Form the channel to a depth of 15 cm.

iii. Spread the earth excavated from the channel on the beds and level.

NOTE: The Unit of 6 plots in one cent will form one unit for irrigation.

4. REMOVAL OF ERGOT AFFECTED SEEDS AND SCLEROTIA TO PREVENT PRIMARY INFECTION

i. Dissolve one kg of common salt in 10 litres of water.

ii. Drop the seeds into the salt solution

iii. Remove the ergot and sclerotia affected seeds which will float.

iv. Wash seeds in fresh water 2 or 3 times to remove the salt on the seeds.

v. Dry the seeds in shade.

vi. Treat the seeds with three packets (600g) of the Azospirillum inoculant and 3 packets (600g) of phosphobacteria or 6 packets (1200g) of azophos.

5. TREATMENT OF THE NURSERY BED WITH INSECTICIDES

Apply phorate 10 G 180 g or Carbofuran 3 G 600 g mixed with 2 kg of moist sand, spread on the beds and work into the top 2 cm of soil to protect the seedlings from shootfly infestation.

6. **SOWING AND COVERING THE SEEDS**

i. Open small rills not deeper than 1 cm on the bed by passing the fingers over it.

ii. Sow 3.75 kg of seeds in 7.5 cents (0.5 kg / cent) and use increased seed rate upto 12.5 kg per ha in shootfly endemic area and transplant only healthy seedlings.

iii. Cover the seeds by smoothening out the rills with hand. Sprinkle 500 kg of FYM or compost evenly and cover the seeds completely with hands.

NOTE: Do not sow the seeds deep as germination will be affected.

7. **IRRIGATION TO THE SEED BED**

i. Provide one inlet to each unit so as to allow water in the channels.

ii. Allow water to enter the channel and turn off the water when the raised bed is completely wet.

iii. Irrigate as per the following schedule.

	Light Soil	Heavy Soil
1^{st}	immediately after sowing	Immediately after sowing
2^{nd}	on 3^{rd} day after sowing	On 3^{rd} day after sowing
3^{rd}	on 7^{th} day after sowing	On 9^{th} day after sowing
4^{th}	on 12^{th} day after sowing	On 16^{th} day after sowing
5^{th}	on 17^{th} day after sowing	

8. **PROTECTION OF SEEDLINGS IN THE NURSERY FROM PEST ATTACK**

If seed bed is not treated before sowing, protect the nursery by applying any one of the insecticides given below on the 7^{th} and 14^{th} day of sowing by mixing in 6 litres of water. Endosulfan 35 EC 12ml ; Methyl demeton 25 EC 12 ml, Dimethoat 30 EC 12 ml.

Note:

1. The seedlings should not be kept in nursery for more than 18 days. Otherwise the establishment and yield will be affected adversely.
2. Ensure that cracks should not develop in the nursery. This can be avoided by properly adjusting the quantity of irrigation water.

I. PREPARATION OF MAIN FIELD

1. **FIELD PREPARATION**

i. Plough with an iron plough twice and with country plough twice. Bring the soil into fine tilth.

ii. CHISELING FOR SOILS WITH HARD PAN: Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5m interval, first in one direction then in the direction perpendicular to the previous one, once in three years.

2. APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost or composted coir pith uniformly on unploughed soil. Incorporate the manure by working the country plough and apply Azospirillum to the soil @ 10 packets per ha (2000 g) and 10 packets (2000g) of phosphobacteria (or) 20 packets (4000g) of azophos with 25kg of soil and 25 kg of FYM.

3. FORMING RIDGES AND FURROWS/BEDS

i. Form ridges and furrows (using 3 ridges) 6 m long and 45 cm apart. If pulses is intercropped, form ridges and furrows 6 m long and 30 cm apart.
ii. If ridge planting is not followed, form beds of the size 10 m^2 or 30 m^2 depending upon water availability.
iii. Form irrigation channels.
iv. To conserve soil moisture under rainfed condition, sow the seeds in flat and form furrows between crop rows during intercultivation on third week after sowing

4. APPLICATION OF FERTILIZERS

Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendation is not available follow the blanket recommendation of 70:35:35 kg N, $P_2 O_5$, K_2O/ha for all varieties. For hybrids, apply 80 kg N, 40 kg P_2O_5 and 40 kg K_2O per ha. Apply the recommended N in three splits as 25:50:25 per cent at 0.15 and 30 DAS and full dose of phosphorus and potassium basally. Combined application of azospirillum and phosphobacteria or azophos along with 75 per cent of the recommended level of N and P is recommended for rainfed conditions. Apply 30 kg S basally for S deficient soils.

Method of application: For transplanted crop, open a furrow more than 5 cm deep on the side of the ridge (1/3 distance from the bottom), place the fertilizer and cover. For the direct sown crop, mark the lines more than 5 cm deep 45 cm apart in the beds. Place the fertilizer below 5 cm depth and cover upto 2 cm from the top before sowing. In the case of intercropping with pulses, mark lines more than 5 cm deep 30 cm apart in the beds. Apply fertilizer only in the rows in which cumbu is to be sown and cover upto 2 cm. When azospirillum inoculant is used for seeds, seedlings use only 50 kg N/ha for variety, 60 kg N/ha for hybrid, as soil application in other words, reduce 25% N of soil test recommendations.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yieldtargets. (ready reckoners are furnished)

Pearl millet- Hybrid

Soil :	Mixed black calcareous (Perianaickenpalayam series)	FN = 6.04 T - 0.49 SN - 0.80 ON FP_2O_5 = 2.78 T - 1.65 SP - 0.97 OP
Target :	3.0 – 4.0 t ha^{-1}	FK_2O = 3.29 T - 0.17 SK - 0.58 OK

Initial soil test value (kg ha^{-1})			Yield target – 3 t ha^{-1}			Yield target – 4 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	15	300	40*	25	20*	98	52	53
200	20	325	40*	20*	20*	89	44	48
220	25	350	40*	20*	20*	79	36	44
240	30	375	40*	20*	20*	69	28	40
260	35	400	40*	20*	20*	59	19	36

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

5. APPLICATION OF MICRONUTRIENT MIXTURE

Apply 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture. Mix the mixture with enough sand to make 50 kg and apply on the surface just before planting/after sowing and cover the seeds. Broadcast the mixture on the surface of seed line (or) Apply TNAU MN mixture @ 12.5t/ha for irrigated and 7.5 kg/ha for rainfed crops as enriched FYM (prepare enriched FYM at 1:10 ratio of MN mixture and FYM at friable moisture and incubate for one month in shade). If micronutrient mixture is not available apply 25 kg of zinc sulphate per ha. Mix the chemical with enough sand to make 50 kg and apply as above. For Mn deficiency apply 12.5 kg $MnSO_4ha^{-1}$ basally or foliar spraying of 0.2% $MnSO_4$ thrice can be followed.

II. MANAGEMENT OF MAIN FIELD

1. TRANSPLANTING SEEDLINGS OR SOWING PRE-TREATED SEEDS Transplanted Crop

i. Pull out the seedlings when they are 15 to 18 days old.
ii. Adopt the spacing 45 x 15 cm for all the varieties / hybrids.
iii. Plant seedlings on the side of ridge, half way from the bottom. Depth of planting should be 3 to 5 cm.
iv. Root dipping with bio-fertilizers: Prepare the slurry with 5 packets (1000 g)/ha of Azospirillum inoculant and 5 packets (1000g/ha) of phosphobacteria or 10 packets of azophos (2000g/ha) in 40 lit. of water and dip the roots of the seedlings 15 - 30 minutes before planting.

Direct sown crop

Soaking of cumbu seeds either in 2% Potassium chloride (KCl) or 3% Sodium Chloride (NaCl) for 16 hours followed by 5 hours shade drying improves germination and stand.

i. Adopt the spacing of 45 x 15 cm for all varieties / hybrids. If pulse is intercropped, adopt a spacing of 30 x 15 cm for cumbu and 30 x 10 cm for pulses. One pair row of cumbu is alternated with a single row of pulse crop.
ii. In the furrows in which fertilizers have been applied, place 5 kg of seed, allowing them to fall 4 - 5 cm apart (Use higher seed rate of 5 kg to offset mortality). The optimum population should be 1,45,000 per ha. Use increased seed rate upto 12.5 kg per hectare in shoot fly endemic area and remove the shootfly damaged seedlings at the time of thinning.
iii. Where pulse seeds are to be sown, drop pulse seeds to fall 5 cm apart and cover.

2. WEED MANAGEMENT Transplanted crop

Spray PE Atrazine 0.25 kg/ha on 3 DAT followed by one hand weeding on 30 - 35 DAT. If herbicide is not used hand weeding twice on 15 DAT and 30 - 35 DAT.

Direct Sown crop

i. Apply the PE Atrazine 0.25 kg/ha on 3 DAS as spray on the soil surface using Back-pack/Knapsack/Rocker sprayer fitted with flat type nozzle using 500 litres of water/ha.
ii. Apply herbicide when there is sufficient moisture in the soil.
iii. Hand weeding 30 - 35 DAT if pre-emergence herbicide is applied.
iv. If pre-emergence herbicide is not applied hand weeding twice on 15 and 30 DAT.

3. THINNING AND GAP FILLING

In direct sown crop after 1st weeding at the time of irrigation, gap fill and thin the crop to a spacing of 15 cm between plants; cowpea crop to 20 cm between plants and other pulses crops to 10 cm between plants.

4. TOP DRESSING OF FERTILIZERS

i. Top dress the nitrogen at 15 and 30 days after transplanting or direct sowing.
ii. In transplanted crop, open a furrow 5 cm deep with a stick or hoe at the bottom of the furrow, place the fertilizer and cover.
iii. In the case of direct sown crop apply the fertilizer in band. If intercropped with pulses apply the fertilizer to cumbu crop only.
iv. After the application of fertilizer, irrigate the crop.

III. WATER MANAGEMENT

STAGES	Days after transplantation/sowing — Transplanted Crop	Direct Sown Crop
Light Soils		
i. Germination	1^{st} day after transplanting	1^{st} day after sowing
	4^{th} day	4^{th} day
ii. Vegetative phase	15^{th} Day	17^{th} day
	28^{th} day	30^{th} day
iii. Flowering phase	40^{th} day	42^{nd} day
	52^{nd} day	55^{th} day
	65^{th} day	68^{th} day
iv. Maturity phase	77^{th} day	79^{th} day
Total	8 irrigations	8 irrigations
Heavy Soils		
i. Germination	1^{st} day after planting	1^{st} day after sowing
	4^{th} day	5^{th} day
ii. Vegetative phase	15^{th} day	15^{th} day
	28^{th} day	30^{th} day
iii. Flowering phase	42^{nd} day	45^{th} day
	54^{th} day	57^{th} day
iv. Maturity Phase	66^{th} day	70^{th} day
Total	7 irrigations	7 irrigations

NOTE: This is only a guideline and the irrigation schedule is to be adjusted depending upon the prevailing weather conditions.

IV. HARVESTING THE CROP

1. **SYMPTOMS OF MATURITY**
   i. Leaves will turn yellow and present a dried appearance.
   ii. Grains will be hardened.

2. **HARVESTING**
   i. Cut the earheads separately.
   ii. Cut the straw after a week, allowing it to dry and stack it in the field till it can be transported.

3. **THRESHING, CLEANING, DRYING AND STORING**
   i. Dry the earheads
   ii. Thresh in a mechanical thresher or
   iii. Spread it and drag a stone roller over it or

iv. Cattle thresh.
v. Dry the seeds below 10 per cent and mix 100 kg of grains with 1kg of activated kaolin to reduce the rice weevil and rice moth incidence.
vi. Spray Malathion 50EC 10 ml/ lit @ 3 lit of spray fluid/100 m^2 over the bags during storage godowns,
vii. For grain purpose the grain should be dried well below 10% moisture and stored in gunny bags.

CROP PROTECTION

Protection of seedlings in the nursery from pest attack

If seed bed is not treated before sowing, protect the nursery by applying any one of the insecticides given below on the 7^{th} and 14^{th} day of sowing by mixing in 6 litres of water; Methyl demeton 25 EC 12 ml, Dimethoate 30 EC 12 ml.

Note:

1. The seedlings should not be kept in nursery for more than 18 days. Otherwise the establishment and yield will be affected adversely.
2. Ensure that cracks should not develop in the nursery. This can be avoided by properly adjusting the quantity of irrigation water.

A. PEST MANAGEMENT

Pest management strategies

Pest	Management strategies
Shoot fly *Atherigona approximata*	▪ Use seeds pelleted with insecticides (see sorghum) ▪ Seed treatment with imidacloprid 70 WS 10 g/kg of seeds ▪ Plough soon after harvest, remove and destroy the stubbles. ▪ Set up the TNAU low cost fish meal trap 12/ha till the crop is 30 days old. ▪ **Spray any one of the following :** Methyl demeton 25 EC 500 ml/ha Dimethoate 30 EC 500 ml/ha Neem seed kernel extract 5%
Ear midge *Geromyia pennisetti*	▪ **Apply any one of the following at 50 % flowering :** Carbaryl 10 D 25 kg/ha Malathion 5 D 25 kg/ha Carbaryl 50 WP 750 g/ha or dimethoate 30 EC 600 ml/ha (500 l of spray fluid/ ha).

B. Disease Management

Seed treatment:

- **For removal of ergot / sclerotia to prevent primary infection:**
  Dissolve 1 kg of common salt in 10 litres of water and add the seeds into the salt solution. Remove the floating ergot and sclerotia affected seeds. Wash the seeds in fresh water for 2 to 3 times to remove the salt, shade dry the seeds and treat the seeds with thiram @ 2 g /kg of seed.
- Treat the seeds with metalaxyl @ 6 g/kg for the management of downy mildew in endemic areas

Name of the Disease	Recommendations
Sugary or Ergot disease: *Claviceps fusiformis*	• Spray carbendazim @ 500 g or mancozeb @1000 g /ha during 5 - 10% flowering and repeat at 50% flowering stage
Rust: *Puccinia substriata*	• Sow during December – May to reduce the level of incidence • Spray wettable sulphur@ 2500 g / ha or mancozeb @ 1000 g/ha during initiation of disease symptom and repeat after 10 days, if necessary
Downy Mildew: *Sclerospora graminicola*	• Grow downy mildew resistant varieties CO (Cu) 9 and TNAU-Cumbu Hybrid-CO 9 and CO 10 • Transplant the seedlings to reduce the disease incidence • Remove the infected seedlings in both transplanted and direct sown crop up to 45days • Spray metalaxyl + mancozeb @ 500 g or mancozeb @ 1000 g/ha

Integrated management strategies for major pest and diseases of pearl millet

Treat the seeds with metalaxyl @ 6 g/kg of seed + imidacloprid @ 5 g/kg of seeds + remove the downy mildew infected plants up to 45 days of sowing + spray mancozeb @ 1000 g/ha + spray NSKE 5% at 50% flowering to manage downy mildew, rust and shoot fly.

CUMBU (PEARL MILLET) - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety, if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 200 m all around the field from the same and other varieties of pearl millet.

Season

- October - December and June - September.

Pre-sowing seed treatment

- Soak the seeds in 2 % KCl for 16 h at 1:1 ratio and dry back the seeds to original seed moisture content (8 - 9 %) under shade. This can be adapted both for the garden and dry land ecosystem.

Fertilizer requirement

- Apply NPK @ 100 : 50 : 50 kg / ha.
- Apply NPK @ 50 :50:50 kg / ha as basal and 50 kg N on 30 days after sowing as top dressing.

Spacing

- 45 x 20 cm.

Foliar spray

- Spray 1 % DAP at peak tillering stage to increase seed filling.

Harvesting

- Seeds attain physiological maturity at 27 - 30 days after 50 % flowering.
- Harvest the earheads when the seed attained the characteristic pale green colour, as once over harvest at 20 - 25 % moisture content.
- Harvest the crop two times when the tillers number is more.
- Earheads from late-formed tillers (after 7 earheads from first formed tillers) should not be selected for seed purpose.

Threshing

- Thresh the earheads either manually or mechanically at moisture content of 15 - 20%.

Drying

- Dry the seeds either under sun or using mechanical hot air driers to reduce the moisture content to 10%.

Seed grading

- Grade the seeds with 4 / 64” (or) 5 / 64” round perforated metal sieve for grading.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed (or)
- Treat the seeds with halogen mixture @ 3 g / kg ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1) as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 10 - 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

CUMBU (PEARL MILLET) - HYBRID SEED PRODUCTION

Land requirement

- Select fertile land with good drainage and irrigation facilities.
- Field should not have volunteer plants. Hence, the previous crop should not be the same or different variety / hybrid of pearl millet.

Isolation

- All around the field, leave 200 m distance from same and other varieties / hybrids of pearl millet.

Season

- October - November and June - July.

Spacing

- 45 x 20 cm.

Planting ratio

- Sow the female and male lines in the ratio of 8 : 2 to 12 : 2 depending upon the hybrids.

Fertilizer requirement

- Apply NPK @ 120:60:60 kg / ha as basal application.

Foliar spray

- Spray 2 % DAP at peak tillering stage for enhanced seed set.

Synchronization techniques

- Stagger the sowing of male and female parents depending upon the hybrid and location.

Harvesting

- Harvest the male parent (R line) first and remove from the field.
- Harvest the hybrid crop when 90 % seeds on the ear head have attained the characteristic pale green colour.

(ii) **RAGI (*Eleusinecoracana*)**

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
40	8 - 10	25 - 35	500 - 1000	up to 2100

Tropical and sub tropical. It is a heat loving plant and requires minimum of 8 - 10°C for germination, 26 - 29°C for the growth. Does not tolerate heavy rainfall and requires a dry spell during grain ripening. Short day plant.

CROP IMPROVEMENT
SEASONS AND VARIETIES

Sl. No.	Agro ecological zones	Districts	Season	Varieties
1	North Eastern Zone	Vellore, Thiruvannamalai, Cuddalore, Villupuram, Thiruvallur and Kancheepuram	Dec-Jan (Marghazipattam) April-May (Chithiraipattam) June-July (Adipattam)	CO (Ra) 14 CO 15
2	North Western Zone	Salem, Namakkal, Dharmapuri and Krishnagiri	Dec-Jan (Marghazipattam) April-May (Chithiraipattam) June-July (Adipattam)	CO (Ra) 14 CO 15 Paiyur 2
3	Western Zone	Coimbatore, Erode, Karur, Tiruppur and Dindigul	Dec-Jan (Marghazipattam) April-May (Chithiraipattam) June-July (Adipattam)	CO (Ra) 14 CO 15
4	Cauvery Delta Zone	Trichy, Thanjavur, Thiruvarur, Nagapattinam, Pudukkottai, Perambalur and Ariyalur	Dec-Jan (Marghazipattam) April-May (Chithiraipattam) June-July (Adipattam)	
5	Southern Zone	Madurai and Theni	Dec-Jan (Marghazipattam) April-May (Chithiraipattam) Sep-Oct (Puratassipattam)	CO (Ra) 14 CO 15
		Sivagangai, Virudhunagar, Ramanathapuram, Tirunelveli and Thoothukudi	Sep-Oct (Puratassipattam)	
6	Hilly and High Altitude Zone	Ooty	Dec-Jan (Marghazipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	

I. PARTICULARS OF RAGI VARIETIES

PARTICULARS	Paiyur 2	CO (Ra) 14	CO 15
Year of Release	2008	2013	2015
Year of Notification	SO.2187(E)/27.08.2009	SO.1177(E)/25.08.2005	SO.2805(E)/25.08.2017
Parentage	VL 145 x Selection 10	Malawi 1305 x CO 13	CO 11 x PR 202
Duration (days)	115	105-110	120 – 125

Rainfed/ Irrigated	Rainfed	Both	Both
Grain yield (kg/ha)			
Irrigated	--	2892	3461
Rainfed	2527	2794	2950
Straw yield (kg/ha)			
Irrigated	--	8113	6698
Rainfed	4200	8503	5030
Stem	Erect	Erect	Erect
Height (cm)	90	115-120	95-100
Tillers	3-4	8-9	5-7
Days to 50% flowering	81	72	84-88
Ear size and shape	Incurved	Top curved	Large, Compact fingers top curved
Fingers	7-8	9-12	8-11
Ear length (cm)	7.0	10-12	9-12
Grain colour	Brown	Brown	Copper brown
1000 grain wt (g)	2.9	3.1	3.2

CROP MANAGEMENT

I. PREPARATION OF NURSERY (IRRIGATED TRANSPLANTED CROP)

1. PREPARATION OF LAND

i. For raising seedlings to plant one ha of main field, select 12.5 cents (500 m^2) of nursery area near a water source, where water does not stagnate.

ii. Mix 37.5 kg of super phosphate with 500 kg of FYM or compost and spread the mixture evenly on the nursery area.

iii. Plough two or three times with a mould board plough or five times with a country plough.

2. FORMING RAISED BED

i. Mark units of 6 plots each of size 3 m x 1.5 m. Provide 30 cm space between plots for irrigation.

ii. Excavate the soil from the interspace and all around to a depth of 15 cm to form channels and spread the soil removed from the channels on the bed and level.

3. PRE-TREATMENT OF THE SEEDS WITH FUNGICIDES

i. Seed treatment with Azospirillum may be done @ 3 packets/ha (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha).

ii. Mix the seeds in a polythene bag to ensure a uniform coating of seeds with Thiram 4 g/ kg or Captan 4 g/kg or Carbendazim 2 g/kg of seeds.

4. SOWING AND COVERING THE SEEDS

i. Make shallow rills not deeper than one cm on the beds by passing the fingers vertically over them.

ii. Broadcast 5 kg of treated hand seeds evenly on the beds.
iii. Cover the seeds by the hand lightly over the soil.
iv. Sprinkle 500 kg of powdered FYM over the beds evenly to cover the seeds which are exposed and compact the surface lightly.

NOTE: Do not sow the seeds deep as germination will be adversely affected.

5. WATER MANAGEMENT

i. Provide one inlet to each nursery unit.
ii. Allow water to enter so as to cover all the channels around the bed. Allow the water in the channel to raise till the raised beds are fully wet and then cut off water.
iii. Adjust the frequency of irrigation according to the soil type.

No. of irrigations	RED SOILS	HEAVY SOILS
1st	Immediately after sowing	Immediately after sowing
2nd	3rd day after sowing	4th day after sowing
3rd	7th day after sowing	9th day after sowing
4th	12th day after sowing	16th day after sowing
5th	17th day after sowing	

NOTE:

1. One irrigation is given on the 3rd day in the case of red soil to soften the hard crust formed on the soil surface and also to facilitate seedlings to emerge out.
2. Do not allow cracks to develop in the nursery bed by properly adjusting the quantity of irrigation water.

6. PULLING OUT THE SEEDLINGS FOR PLANTING

Pull out seedlings on the 17th to 20th day of sowing for planting.

II. PREPARATION OF MAIN FIELD

1. PLOUGHING THE FIELD

Plough twice with mould board plough or thrice with wooden plough till a good tilth is obtained.

2. APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field and then plough and incorporate in the soil. NOTE: Do not spread and leave the manure uncovered in the field as nutrients will be lost.

3. APPLICATION OF FERTILIZERS

In soils having high intensive cropping system viz., Ragi-Maize-Cowpea, having high soil available K (310 kg/ha) potassium need not be applied.

- If soil test recommendation is not available, adopt a blanket recommendation of 60 kg N, 30 kg P_2O_5 and 30 kg K_2O per ha.
- Apply half the dose of N and full dose of P_2O_5 basally before sowing and the remaining 50% in two equal splits at 25-30 and 40-45 days after sowing is recommended.
- Broadcast the fertilizer mixture over the field before the last ploughing and incorporate into the soil by working a country plough.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Ragi (1)

Soil : Mixed black calcareous (Perianaickenpalayam series)

Target : 3.5 - 4.0 t ha^{-1}

FN = 4.35T-0.37 SN-0.98 ON

FP_2O_5 = 1.18T-1.03 SP-0.80 OP

FK_2O = 2.68T-0.14SK-0.40 OK

Initial soil test values (kg ha^{-1})			Yield target – 3.5 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 4.0t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	12	300	34	15*	15*	55	15*	25
200	14	340	30*	15*	15*	48	15*	20
220	16	380	30*	15*	15*	41	15*	15*
240	18	420	30*	15*	15*	33	15*	15*
260	20	460	30*	15*	15*	30*	15*	15*

* Maintenance dose

Ragi (2)

Soil : Red sandy loam (Somayanur series)

Target : 3.5 – 4.0t ha^{-1}

FN =4.94T-0.55 SN

FP_2O_5=1.36T-0.96 SP

FK_2O=4.20T-0.46 SK

Initial soil test value (kg ha^{-1})			Yield target – 3.5 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 4.0 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	160	33	15*	33	58	15*	54
180	14	180	30*	15*	24	47	15*	45
200	16	200	30*	15*	15*	36	15*	36
220	18	220	30*	15*	15*	30*	15*	27
240	20	240	30*	15*	15*	30*	15*	18

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

Apply 10 packets/ha (2000 g) of Azospirillum and 10 packets (2000 g/ha) of Phosphorous solubilizing bacteria or 20 packets of Azophos (4000 g/ha) after mixing with 25 kg of soil and 25 kg FYM before transplanting. Apply TNAU MN mixture @12.5 kg/ha for irrigated and 7.5 kg/ha for rainfed crops as enriched FYM (prepare enriched FYM at 1:10 ratio of MN mixture and FYM at friable moisture and incubate for one month in shade.

4. FORMING BEDS AND CHANNELS

i. Form beds of size 10 m^2 to 20 m^2 according to topography of the field.
ii. Provide suitable irrigation channels.

5. APPLICATION OF MICRONUTRIENTS

Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg/ha (or) Apply the mixture evenly on the beds. (or) For alleviating Zn deficiency in plants, spray 0.5% $ZnSO_4$ on 30, 40 and 50 days after sowing. For specific micronutrient deficiencies, apply 25 kg $ZnSO_4$, 10 kg $CuSO_4$ and 50 kg $FeSO_4$ + 12.5 t FYM /ha can be followed.

III. MANAGEMENT OF MAIN FIELD

1. TRANSPLANTING THE SEEDLINGS

i. Let water into the bed, level the bed, if it is not levelled.
ii. Plant 2 seedlings per hill.
iii. Plant the seedlings at a depth of 3 cm.
iv. Plant 18 to 20 days old seedlings.
v. Adopt a spacing of 30x10 cm for planting.
vi. Adopt 22.5 x 10 cm spacing for direct sowing.
vii. Root dipping with Azospirillum: Prepare slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of Azophos (2000 g/ha) in 40 litres of water and dip the root portion of the seedlings in the solution for 15-30 minutes and transplant.

2. WEED MANAGEMENT

i. Apply PE Oxyfluorfen @ 0.05 kg /ha on 3 DAS using Backpack Knapsack/Rocker sprayer fitted with flat fan type of nozzle with 500 litre of water/ha followed by one hand weeding on 20 DAS.
ii. Apply the herbicides when there is sufficient moisture in the soil or irrigate immediately after application of herbicide.
iii. If pre-emergence herbicide is not applied, hand weed twice on 10th and 20th DAT.
iv. For rainfed direct seeded crop, apply post emergence herbicide; 2,4-DEE or 2,4-D Na salt at
0.5 kg/ha on 10 DAS depending on the moisture availability.

3. HOEING AND HAND WEEDING

i. Hoe and hand weed on the 15th day of planting in light soils and 17th day of planting in heavy soils and subsequently on 30th and 32nd days, respectively.

ii. Allow the weeds to dry for 2 or 3 days after hand weeding before giving irrigation. NOTE: Do not adopt hoeing and hand weeding if herbicide is applied.

IV. WATER MANAGEMENT

Regulate irrigation according to the following growth phases of the crop

Stages/ Phase	No. of irrigations	Crop duration days		
		80	100	120
Vegetative phase (Nursery)	As per soil type	1 to 16	1 to 18	1 to 20
Vegetative phase (main field)		1 to18	1 to 20	1 to 22
Flowering phase		19 to 40	21 to 55	23 to 69
Maturity phase		Beyond 40 days	Beyond 55 days	Beyond 69 days
Heavy soils				
Establishment	1	1st day	1st day	1st day
(1-7 days)	2	5th day	5th day	5th day
Vegetative phase	1	18th day	20th day	20th day
(8-20 days)	2	31st day	33rd day	30th day
Flowering phase	1	41st day	42nd day	37th day
(21-55 days)	2	51st day	52nd day	44th day
	3	--	--	63rd day
Maturity phase	1	61st day	62nd day	78th day
(56-120 days)	2	--	--	93rd day
Stop irrigation thereafter				
Light soils				
Establishment	1	1st day	1st day	1st day
(1 – 7 days)	2	5th day	5th day	5th day
Vegetative phase	1	15th day	16th day	16th day
(8 - 20 days)	2	26th day	28th day	28th day
Flowering phase	1	36th day	36th day	36th day
(21 - 55 days)	2	45th day	45th day	45th day
	3	--	54th day	54th day
Maturity phase	1	58th day	69th day	78th day
(56 - 120 days)	2	70th day	85th day	93rd day

NOTE: The irrigation schedule is given only as a general guideline. Regulate irrigation depending upon the prevailing weather conditions and receipt of rain.

V. HARVESTING

1. DECIDE WHEN TO HARVEST

i. Ragi crop does not mature uniformly and hence the harvest is to be taken up in two stages.

ii. When the earhead on the main shoot and 50% of the earheads on the crop turn brown, the crop is ready for the first harvest.

2. HARVEST OF THE CROP First harvest

i. Cut all earheads which have turned brown.
ii. Dry, thresh and clean the grains by winnowing.

Second Harvest

i. Seven days after the first harvest, cut all the earheads including the green ones.
ii. Cure the grains to obtain maturity by heaping the harvested earheads in shade for one day without drying, so that the humidity and temperature increase and the grains get cured.
iii. Dry, thresh and clean the grains by winnowing and store the grains in gunnies.

Threshing

Green earheads if harvested will contaminate the seeds with immature seeds and interfere cleaning, drying and grading. Dry earheads until seed moisture content reaches 15% and separate manually by threshing with bamboo stick or machine thresher.

Precleaning and drying

Threshed seeds should be precleaned before sundrying, seeds must be dried to 12% moisture content before grading.

Protection from storage pests

1. Grain purpose: Dry the seeds adequately to reduce the moisture level to 10%.
2. Seed purpose: Admix one kg of Activated kaolin or Malathion 5% D for every 100 kg of seed. Pack in gunny or polythene lined gunny bags for storage.

Special problems

i. Root Aphids: Mix Dimethoate 3 ml in one litre of water and drench the rhizosphere of the infested and surrounding plants with the insecticidal solution.
ii. Rainfed ragi: Azospirillum mixed with FYM and applied to field saves the cost of nitrogen by 50% with a comparable yield obtained with 40 kg N/ha.
iii. Management of aged seedlings of ragi under rainfed conditions: When planting **ragi** seedlings beyond 21 days, increase the number of seedlings to 3/hill and increase N level by 25% to minimise yield loss.
iv. Apply VAM culture (*Glomus fasciculatum*) at 100 g/m^2 in the nursery and also treat with Azospirillum and Phosphobacterium as seed treatment, seedling dip and field application to reduce the reniform nematode population in ragi.

RAGI : RAINFED

Rainfall

Average and well distributed rainfall of 450-500 mm is optimum for rainfed ragi

Season

Finger millet is grown in different seasons in different parts of the country. As a rainfed crop, it

is normally sown in June- July in Tamil Nadu. It also grown in winter season (rabi) by planting in September – October in Tamil Nadu and as a summer irrigated crop by planting January – February.

Tillage

Fall ploughing is advantageous for moisture conservation. In the month of April or May, one deep ploughing with mould board plough followed by ploughing with wooden plough twice is necessary. Before sowing secondary tillage with cultivator and multiple tooth hoe to prepare smooth seed bed is necessary.

Seed rate and planting

A plant population of 4 – 5 lakhs per ha is optimum for getting higher yields and higher or lower population than the optimum will reduce the yield. Line sowing is ideal and seed drills giving spacing of 22.5 – 30 cm between rows should be used. Finger millet seeds are very small (400 seeds/g) and the recommended seed rate is 10 kg/ha. Therefore, even when seed drill is used thinning within the row leaving a spacing of 7.5 – 10 cm between plants, must be followed.

Sowing by seed-cum-fertilizer drill is advantageous for line sowing besides efficient utilization of applied nutrients.

Maintenance of optimum plant population is an important prerequisite for getting higher yield under rainfed conditions. Poor germination, often, is the result of inadequate moisture after sowing in low rainfall areas. Under these conditions, the adoption of a simple technique like seed hardening will not only improve germination and subsequent plant stand but also impart early seedling vigour and tolerance to drought.

The procedure of seed hardening technique is as follows.

1. Sole seeds in water for 6 hours. Use one litre water for every kg seed for soaking.
2. Drain the water and keep the seeds in wet cloth bag tightly tied for two days.
3. At this stage, the seeds will show initial signs of germination.
4. Remove seeds from the wet cloth bag and dry them in shade on a dry cloth for 2 days.
5. Use the above hardened seeds for sowing.

Manuring and fertilization

Finger millet responds well to fertilizer application especially to N and P. The recommended doses of fertilizers vary from state to state for rainfed crop. Recommended dose of 40:20:20 kg/ha N:P:K was applied. With judicious application of farmyard manure inorganic fertilizer efficiency is enhanced. Entire P_2O_5 and K_2O are to be applied at sowing, whereas nitrogen is to be applied in two or three split doses depending upon moisture availability. In areas of good rainfall and moisture availability, 50% of recommended nitrogen is to be applied at sowing and the remaining 50% in two equal splits at 25-30 and 40-45 days after sowing. In areas of uncertain rainfall, 50% at sowing and the remaining 50% around 35 days after sowing is recommended.

Bio-fertilizers

Treating seeds with *Azospirillum brasilense* (N fixing bacterium) and *Aspergillus awamori* (P solubilizing fungs) @ 25 g/kg seed is beneficial. In case seeds are to be treated with seed dressing chemicals, treat the seeds first with seed dressing chemicals and then with bio-fertilizers at the time of sowing.

Procedures for inoculating seeds with biofertilizers

1. Bio-fertilizer culture specific to the crop is to be used @ 25 g per kg of seed.
2. Sticker solution is necessary for effective seed inoculation. This can be prepared by dissolving 25 g jaggery or sugar in 250 ml water and boiling for 5 minutes. The solution thus prepared is cooled.
3. Smear the seeds well using the required quantity of sticker solution. Then add culture to the seeds and mix thoroughly so as to get a fine coating of culture on the seed.
4. The culture-coated seeds is to be dried well in shade to avoid clumping of seeds.
5. Use the inoculated seeds for sowing.

Weed control

i. In line sown crop 2-3 inter-cultivations are necessary. In assured rainfall and irrigated areas spraying 2,4-D sodium salt @ 0.75 kg.a.i./ha as post-emergent spray on 20-25 days after sowing effectively controls weeds.
ii. Apply, Isoproturon @ 0.5 a.i/ha as pre-emergence on 3 DAS is also effective in control of weeds. In broadcast crop two effective hand weedings will minimize weeds as inter cultivations is not possible.
iii. For direct sown rainfed ragi post-emergence application of 2, 4 D Na salt (or) EE formulation at 0.5 kg ha^{-1} applied on 10 days after sowing and at 0.75 kg ha^{-1} applied on 15 days after sowing will give effective weed control as well as higher grain yield.

Cropping systems Crop rotation

Rotation with legumes like greengram / blackgram / field bean / soybean / horse gram or ground nut in southern state will minimize inorganic fertilizer application and also sustain higher yields.

Intercropping

Finger millet based inter cropping system with pigeon pea at 4:1 ratio is recommended for rainfed situation to obtain high grain yield

CROP PROTECTION

A. Pest Management

Aphids *Schizaphis graminum, Rhopalosiphum maidis*	Spray dimethoate 30EC 20 ml per 5 cent nursery
Stem borer *Sesamia inferens*	Apply carbofuran 3CG 50 kg/ha in leaf whorls
Root aphid *Tetraneura nigriabdominalis*	Drench dimethoate 30EC 1:1 (with water) in the rhizosphere of infested and surrounding plants
Ear head bug *Calocoris angustatus*	**Apply any one of the following/ha on 3rd and 18th day after panicle emergence** • Malathion 5D 25 kg • Neem seed kernel extract 5% • Malathion 50EC 500 ml/ha (twice at 10% heading and 9 days after)

B. DISEASE MANAGEMENT

Nursery:

Treat the seeds with thiram or captan @ 4 g or carbendazim @ 2 g/kg or *Pseudomonas fluorescens* @ 10 g/kg of seed.

Main field:

Name of the Disease	Recommendations
Blast: *Pyricularia grisea*	• Spray carbendazim @ 500 g or iprobenphos(IBP) @ 500 ml/ha immediately after symptom development and repeat at flowering stage and 15 days later to manage neck and finger blast • Spray aureofungin sol 100 ppm (100 mg/l) at 50% ear head emergence followed by spray with mancozeb 1000 g/ha or *Pseudomonas fluorescens* @ 0.2% ten days later • Two sprays of tricyclazole 75% WP @ 500 g/ha at maximum tillering and heading stages
Virus diseases **Mosaic and Mottle streak**	• Rogue out the virus infected plants • Spray methyl demeton 25EC @ 500 ml/ha on noticing symptoms and repeat twice at 20 days interval, if necessary

RAGI (FINGER MILLET) - VARIETAL SEED PRODUCTION

Land Requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 3 m all around the field from the same and other varieties of the finger millet.

Pre-sowing seed treatment

- Soak the seeds in 0.5 % $CaCl_2$ for 6 h at 1:1 ratio and dry back the seeds to original seed moisture content (8 - 9 %) under shade. This can be adapted both for the garden and dry land ecosystem.
- Treat the graded seed with carbendazim @ 2 g / kg of seed.

Nursery sowing

- In raised bed, sow the seeds not deeper than 1 cm and sprinkle with 200 kg of powdered FYM. Lightly level and compact the surface of nursery.

Harvesting

- Harvest the crop in 2 harvests.
- First harvest should be taken up when 50 % of seeds in the ear-heads have attained the characteristic brown colour.
- Second harvest should be taken up a week to ten days after first havest, when all the remaining earheads turned brown (spikelets are non-shattering).

Threshing

- Dry the earheads until the seed moisture content is reduced to 15 % and seeds are separated manually by threshing with pliable bamboo stick or machine thresher.
- Pre-clean the threshed seeds before sun drying.
- Dry the seeds to 12 % moisture content before grading.

Seed grading

- Grade the seeds either with BSS 10 x 10 or BSS 12 x 12 depending upon the variety.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 10 to 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 to 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(i) MAIZE (*Zea mays* L.)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
40 - 44	6 - 7	21 - 32	500 - 750	up to 3000

Tropical and sub tropical. Minimum temperature for germination is 6 - 7°C, suitable temperature for germination and growth is 21 - 23 and 30 - 32°C, respectively. Day neutral plant.

CROP IMPROVEMENT
SEASON AND VARIETIES

Sl. No.	Agro ecological zones	Districts	Season	Hybrids
1	North Eastern Zone	Vellore, Thiruvannamalai, Cuddalore, Villupuram, Thiruvallur, Kancheepuram	Jan-Feb (Thaipattam) April - May Chithiraipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Hybrid CO 6 and COH(M) 8
2	North Western Zone	Salem, Namakkal, Dharmapuri, Krishnagiri		
3	Western Zone	Coimbatore, Erode, Karur, Tiruppur, Theni and Dindigul		
4	Cauvery Delta Zone	Trichy, Thanjavur, Thiruvarur, Nagapattinam, Pudukkottai Perambalur and Ariyalur	Jan-Feb (Thaipattam) April - May Chithiraipattam) June-July (Adipattam)	
5	Southern Zone	Madurai and Virudhunagar	Jan-Feb (Thaipattam) June-July (Adipattam) Sep-Oct(Puratassipattam)	
		Sivagangai, Ramanathapuram, Tirunelveli and Thoothukudi	Sep-Oct(Puratassipattam)	
6	High rainfall zone	Kanyakumari	Jan-Feb (Thaipattam) June-July (Adipattam) Sep-Oct(Puratassipattam)	
7	Hilly and High Altitude Zone	Ooty	Jan-Feb (Thaipattam) June-July (Adipattam) Sep-Oct(Puratassipattam)	

II. PARTICULARS OF MAIZE HYBRIDS

PARTICULARS	CO 6	COH(M) 8
Year of Release	2012	2014
Year of Notification	SO.1708(E)/26.07.2012	SO.1919(E)/30.07.2014
Parentage	UMI 1200 x UMI 1230	UMI 1201 x UMI1230
Duration (days)	110	85 – 95
Area of adaption	All maize growing areas	All maize growing areas
Rainfed/Irrigated	Both	Both
Grain yield (kg/ha)		
Irrigated		7600
Rainfed	5500	5500
Special features	High shelling (81%) with high test weight (40 g /100 seeds). Multiple disease resistance to sorghum downy mildew, *Maydis* leaf blight, *Turcicum* leaf blight, Post flowering stock rot and Banded leaf and sheath blight. Moderately resistant to stem borer.	Medium maturity hybrid, grains are bold, orange yellow in colour and semi dent type. Single cross normal corn. Multiple disease resistance *viz*. MLB, TLB, RDM, DM and moderately resistant to PFSR and *polysora* rust under artificial Conditions. Moderately resistant to stem borer *(chilo partellus*) and resistant to cyst nematode *(Heterodera zeae).*
Stem Colour	Green	Green
Leaf: Anthocyanin Colouration of sheath	Present	Present
Ear: Anthocyanin Colouration of silk	Present	Present
Cob size	Big	Big
Ear: Husk Coverage	Fully covered	Fully covered
Colour of top of Grains	Orange Yellow	Orange yellow
Type of kernels	Semi dent	Semi dent

CROP MANAGEMENT

i. IRRIGATED MAIZE

1. APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field along with 10 packets of Azospirillum (2000 g/ha) and incorporate in the soil.

2. FIELD PREPARATION

Plough the field with disc plough once followed by cultivator ploughing twice, after spreading FYM or compost till a fine tilth is obtained.

3. FORMING RIDGES AND FURROWS OR BEDS

Form ridges and furrows providing sufficient irrigation channels. The ridges should

be 6 m long and 60 cm apart.
If ridges and furrows are not made, form beds of size 10 m^2 or 20 m^2 depending on the availability of water.
Use a bund former or ridge plough to economise cost of production.

4. APPLICATION OF FERTILIZERS

i. Soil test crop response based integrated plant nutrition system (STCR- IPNS) I If soil test recommendation is not available adopt a blanket recommendation of 135:62.5:50 Kg/NPK ha^{-1} for varieties and 250:75:75 kg NPK ha^{-1} for hybrid maize
ii. Apply quarter of the dose of N; full dose of P_2O_5 and K_2O basally before sowing.
iii. In the case of ridge planted crop, open a furrow 6 cm deep on the side of the ridge, at two thirds the distance from the top of the ridge.
iv. Apply the fertilizer mixture along the furrows evenly and cover to a depth of 4 cm with soil.
v. If bed system of planting is followed, open furrows 6 cm deep at a distance of 60 cm apart.
vi. Place the fertilizer mixture along the furrows evenly and cover to a depth of 4 cm with soil.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Maize - Hybrid (1)

Soil : Red sandy loam (Palaviduthi series)
Target : 9 – 10 t ha^{-1}

F N = 3.96 T – 0.62 SN – 0.69 ON
FP_2O_5= 1.56 T – 1.93 SP – 0.60 OP
FK_2O = 1.66 T – 0.27 SK – 0.49 OK

Initial soil test values (kg ha^{-1})			**Yield target – 9 t ha^{-1}**			**Yield target – 10 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	14	200	177	83	65	217	99	82
220	16	220	165	80	60	205	95	77
240	18	240	153	76	55	192	91	71
260	20	260	140	72	49	180	87	66
280	22	280	128	68	44	167	84	60

Maize - Hybrid (2)

Soil : Mixed black calcareous (Perianaickenpalayam series)

Target : 9 – 10 t ha^{-1}

F N = 4.01 T – 0.76 SN – 0.83 ON

FP_2O_5= 1.57 T – 2.71 SP – 0.61 OP

FK_2O = 2.09 T – 0.26 SK – 0.65 OK

Initial soil test values (kg ha^{-1})			Yield target – 9 t ha^{-1}			Yield target – 10 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	14	300	154	73	80	194	89	101
220	16	350	139	68	67	179	84	88
240	18	400	125*	63	54	164	78	75
260	20	450	125*	57	41	148	73	62
280	22	500	125*	52	38*	133	67	49

* Maintenance dose

Maize- Hybrid (3)

Soil : Black calcareous (Pilamedu series)

Target : 10 – 11t ha^{-1}

F N = 3.78 T – 0.78 SN – 0.89 ON

FP_2O_5= 1.47 T – 2.02 SP – 0.91 OP

FK_2O = 1.79 T – 0.14 SK – 0.62 OK

Initial soil test values (kg ha^{-1})			Yield target – 10 t ha^{-1}			Yield target – 11 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	12	400	178	91	91	215	105	109
200	14	450	162	87	84	200	101	102
220	16	500	146	83	77	184	97	95
240	18	550	131	79	70	169	93	88
260	20	600	125*	75	63	153	89	81

*****maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

When Azospirillum is used as seed and soil application, apply 100 kg of N/ha (25% reduction on the total N recommended by soil test).

Defieciency symptoms

Nitrogen deficiency	:	Leaves become yellow, older leaves show drying at the tips which progress along mid veins, stalks become slender.
Phosphorus deficiency	:	Leaves are purplish green during early growth. Growth spindly, slow maturity, irregular ear formation.
Potassium deficiency	:	Leaves show yellow or yellowish green streaks, become corrugated. Tips and marginal scorch. Tips end in ears are poorly filled. Stalks have short internode. Plants become weak and may fall down.
Magnesium deficiency	:	Older leaves are the first to become chlorotic at margins and between veins. Streaked appearance of leaves. Necrotic or chlorotic spots seen in leaves.
Zinc deficiency	:	Older leaves have yellow streaks or chlorotic striping between veins. In several cases, unfolding of young leaves, which may be white or yellow.
Iron deficiency	:	Interveinal chlorosis. The entire crop may exhibit bleached appearance.

5. APPLICATION OF MICRONUTRIENT

i. 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu, mixed with sand to make a total quantity of 50 kg/ha is to be applied. (or)

ii. Apply TNAU MN mixture @ 30 kg/ha as enriched FYM (Prepare enriched FYM at 1:10 ratio MN mixture and FYM; mix at friable moisture and incubate for one month in shade.

iii. Apply zinc sulphate @ 37.5 kg ha^{-1} for hybrid maize and 25 kg ha^{-1} for varieties can be followed in Zn deficient soils.

iv. Apply 50 kg $FeSO_4$ + 12.5 t FYM ha^{-1} along with 40 kg S as elemental sulphur for calcareous soils. Apply the mixture over the furrows and two thirds in the top of ridges, if ridge planting is followed. If bed system of sowing is followed, apply the micronutrient mixture over the beds.

v. Apply 40 kg S, 10 kg borax and 50 kg $FeSO_4$ + 12.5 t FYM for specific respective nutrient deficiency in soils.

vi. For zinc and iron deficiencies in plants foliar spraying 0.5% $ZnSO_4$, 1% $FeSO_4$ + 0.1% citric acid thrice on 30, 40 and 50 days after sowing can be followed.

6. SEED RATE

Select good quality seeds and adopt the seed rate of 20 kg/ha for CO 1 and TNAU Maize Hybrid CO 6 and 25 kg /ha for COBC 1.

7. SPACING

Adopt a spacing of 25 cm between plants in the rows which are 60 cm apart.
Population : For varieties and hybrids 6 – 7 plants / sq. m. and
For baby corn, 8 – 9 plants / sq. m.

8. SEED TREATMENT

Step 1: Use pelleted seeds with insecticides (treat one kg of seeds with Chlorpyriphos 20EC or Monocrotophos 36 WSC or Phosalone 35 EC @ 4 ml + 0.5 gram gum in 20 ml of water) for the control of stem borer or seed treatment with imidacloprid 70 WS 10 g/kg of seeds.

Step 2: Seed treatment with Metalaxyl or Thiram @ 2 g/kg of seed for the control of downy mildew and crazy top

Step 3: Seeds treated with fungicides may be treated with three packets (600 g/ha) of Azospirillum before sowing.

9. SOWING

i. Dibble the seeds at a depth of 4 cm along the furrow in which fertilizers are placed and cover with soil.
ii. Put one seed per hole if the germination is assured otherwise put two seeds per hole

10. WEED MANAGEMENT

i. Apply Atrazine @ 0.50 – 0.75 kg/ha as pre-emergence on 3-5 DAS using Backpack/ Knapsack/ Rocker sprayer fitted with a flat fan nozzle using 500 litres of water/ha followed by one hand weeding on 30-35 DAS. (or)
ii. Apply Atrazine @ 0.50 kg/ha as pre-emergence on 3-5 DAS followed by 2,4-D @ 1 kg/ha on 20-25 DAS, using Backpack/Knapsack/Rocker sprayer fitted with a flat fan nozzle using 500 litres of water/ha.
iii. In line sown crop, apply PE Atrazine @ 0.50 kg/ha on 3-5 DAS followed by Twin Wheel hoe weeder weeding on 30-35 DAS.
iv. Apply herbicide when there is sufficient moisture in the soil.
v. Do not disturb the soil after herbicide application.
vi. If pulse crop is to be raised as intercrop, do not use Atrazine. Spray Pendimethalin @0.75 kg/ha as pre emergence on 3-5 DAS.

11. THINNING AND GAP FILLING

i. If two seeds were sown, leave only one healthy and vigorous seedling per hole and remove the other on the 12-15 days after sowing.
ii. Where seedlings have not germinated, dibble presoaked seeds at the rate of 2 seeds per hole and immediately irrigate.

12. HOEING, HAND-WEEDING AND EARTHING UP

i. Hoe and hand-weed on the 30th day of sowing.
ii. Earth up and form new ridges so that the plants come directly on the top of the ridges. This will provide additional anchorage to the plants.

13. TOP DRESSING WITH NITROGEN

i. Place half of the dose of N on the 25^{th} day of sowing along the furrows evenly and cover it with soil.
ii. Place the remaining quarter of N on the 45^{th} day of sowing

14. WATER MANAGEMENT

Maize crop is sensitive to both moisture stress and excessive moisture, hence regulate irrigation according to the requirement. Ensure optimum moisture availability during the most critical phase (45 to 65 days after sowing); otherwise yield will be reduced by a considerable extent.

Regulate irrigation according to the following growth phase of the crop.

Germination & establishment phase	1 to 14 days
Vegetative phase	15 to 39 days
Flowering phase	40 to 65 days
Maturity phase	66 to 95 days

Heavy soils		
Stage	**No. of irrigation**	**Days after sowing**
Germination & establishment	3	After sowing, Life irrigation -4th,12th day
Vegetative	2	25th, 36th day
Flowering(Irrigate copiously)	2	48th, 60th day
Maturity phase (Control irrigation)	2	72nd, 85th day
Light soils		
Germination & establishment	3	After sowing, Life irrigation -4th ,12thday
Vegetative Phase	3	22nd ,32nd & 40th day
Flowering phase (Irrigate copiously)	3	50th ,60th & 72nd day
Maturity phase (Controlled irrigation)	2	85th , 95th day

DRIP IRRIGATION TO MAIZE

Irrigation once in 2 days

Irrigation based on climatological approach Irrigation volume:

= (Pe x Kp x Kc x A x Wp) – Re

Pe – Pan evaporation rate (mm/day) Kp – Pan co-efficient (0.75 to 0.80)

Kc – Crop co-efficient (0.4 – Vegetative stage; 0.75 – Flowering stage; 1.05 – Grain formation stage) A – Area (75 x 30 cm)

Wp – Wetted percentage (80% for maize) Re – Effective rainfall (mm)

$$\text{Irrigation duration} = \frac{\text{Water requirement per plant once in 2 days}}{\text{No. of dripper / plant x Discharge rate (lph)}}$$

DRIP FERTIGATION TECHNOLOGY

- Method of planting : paired row planting
  (60/90 × 30 cm)
- Fertilizer dose = 150:75:75 kg NPK per ha
- Drip fertigation with Water soluble fertilizer (WSF)
  - N Polyfeed 19-19-19
  - P MAP 12-61-00
  - K KNO_3 13-00-45

Fertigation Device : Ventury assembly (3/4") with injector pump (0.5 HP)

Fertigation schedule for Hybrid maize with Water Soluble Fertilizers at (75 % RDF)

Stage (days)	Duration (days)	Fertilizer form	Fertilizer grade			Dose / ha/ day	Total Qty (Kg/ha)	Nutrients kg/ha		
			N	P	K			N	P	K
6 to 25	20	MAP	12	61	0	2.813	56.25	6.75	34.31	0.00
	20	Urea	46	0	0	0.938	18.75	8.63	0.00	0.00
26-60	35	PolyFeed	19	19	19	2.143	75.00	14.25	14.25	14.25
	35	Multi-K	13	0	45	1.500	52.50	6.83	0.00	23.63
	35	Urea	46	0	0	2.143	75.00	34.50	0.00	0.00
61-75	15	PolyFeed	19	19	19	2.750	41.25	7.84	7.84	7.84
	15	Multi-K	13	0	45	1.600	24.00	3.12	0.00	10.80
	15	Urea	46	0	0	4.500	67.50	31.05	0.00	0.00
								112.96	56.40	56.51

Fertigation schedule for Hybrid maize with Normal Fertilizers (100% RDF)

Stage (days)	Duration (days)	Fertilizer form	Fertilizer grade			Dose / ha/ day	Total Qty (Kg/ha)	Nutrients kg/ha		
			N	P	K			N	P	K
6 to 25	20	DAP	18	46	0	5.00	100	18.0	46.0	0.0
	20	Urea	46	0	0	2.50	50	23.0	0.0	0.0
26-60	35	DAP	18	46	0	1.86	65	11.7	29.9	0.0
	35	Urea	46	0	0	4.29	150	69.0	0.0	0.0
	35	MOP	0	0	60	2.14	75	0.0	0.0	45.0
61-75	15	Urea	46	0	0	4.13	62	28.5	0.0	0.0
	15	MOP	0	0	60	3.33	50	0.0	0.0	30.0
								150.2	75.9	75.0

HARVESTING STAGE OF HARVEST

Observe the following symptoms, taking into consideration the average duration of the crop.

i. The sheath covering the cob will turn yellow and dry at maturity.

ii. The seeds become fairly hard and dry. At this stage the crop is ready for harvest.

HARVESTING THE CROP

i. Tear off the cob sheath by using the gunny needle and remove the cobs from the plant.

ii. Carry out harvest operations at a single stage for easy transportation.

THRESHING THE COBS

i. Dry the cobs under the sun till the grains are dry.

ii. Use mechanical threshers or by running the tractor over dried cobs to separate the grains from the shank.

iii. Clean the seeds by winnowing

iv. Collect and store the dry grains in gunnies.

STACKING THE STRAW FOR FEEDING CATTLE

i. Maize straw can also be used as a good cattle feed when it is green.

ii. Harvest the crop and cut the green straw into bits with a chaff cutter or chopping knife and feed the cattle.

I. RAINFED MAIZE

1. FIELD PREPARATION

Chisel the soil having hard pan formation at shallow depths with chisel plough at 0.5 m interval first in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5 t/ha of FYM or compost or composted coir pith besides chiselling, to get an additional yield of about 30% over control.

2. APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field along with 10 packets of Azospirillum (2000 g/ha) and incorporate in the soil.

3. APPLICATION OF FERTILIZER

i. Apply NPK as per soil test recommendation as far as possible. If soil test recommendation is not available, adopt a blanket recommendation of 60 : 30 : 30 NPK kg/ha for Alfisols and 40 : 20 : 0 NPK kg/ha for Vertisols.

ii. Apply half of N and full dose of $P_2 O_5$ and $K_2 O$ with enriched FYM as basal along with Azospirillum (10 packets/ha).

iii. Top dress remaining half of N at tasseling.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yieldtargets. (ready reckoners are furnished)

Rainfed Maize

Soil :	Red sandy loam (Irugur series)	FN = 3.23 T - 0.42 SN - 0.52 ON
Target :	4 - 5 t ha^{-1}	FP_2O_5 = 1.51T - 1.98 SP - 0.94 OP
		FK_2O = 1.73T - 0.21 SK - 0.48 OK

Initial soil test values (kg ha^{-1})			**Yield target – 4 t ha^{-1}**			**Yield target – 5 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	10	200	30*	16	15*	62	31	20
180	12	220	30*	15*	15*	54	27	15*
200	14	240	30*	15*	15*	46	23	15*
220	16	260	30*	15*	15*	37	19	15*
240	18	280	30*	15*	15*	30*	15*	15*

* Maintenance dose ** Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}.

Apply TNAU MN mixture @ 7.5 kg /ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).

4. SEED RATE

Select good quality seeds. Adopt the seed rate @ 20 kg/ha for hybrids and 25 kg/ha for varieties

5. SPACING

Adopt a spacing of 45 cm between rows and 20 cm between plants in the row. Population : 10 - 11plants/m^2

6. PRE-TREATMENT OF SEEDS WITH BIOFERTILIZER

Seeds treated with fungicides may be treated with three packets (600 g/ha) of Azospirillum

7. SOWING

Dibble or drill the seeds at a depth of 4 cm.

8. CROPPING SYSTEMS

i. Intercropping system of maize + cowpea or maize + blackgram is recommended for higher net returns in the red lateritic soils of Southern districts.
ii. For Vertisols of Southern district, maize + redgram intercropping systems is ideal.

CROP PHYSIOLOGY

Foliar spray of TNAU Maize Maxim @ 3 kg/acre in 200 litres of water at tassel initiation and at grain filling stages improves grain filling, grain yield and drought tolerance.

CROP PROTECTION

A. PEST MANAGEMENT

Shoot fly, *Atherigona orientalis*	• Set up TNAU low cost fish meal trap 12 nos./ha till the crop is 30 days old • **Apply any one of the following insecticides/ ha** Carbofuran 3CG 33.3 kg Dimethoate 30EC 1.2 lit Methyl demeton 25EC 1.0 lit Monocrotophos 36SL 625 ml
Stem borer, *Chilo partellus*	• Release egg parasitoid, *Trichogramma chilonis* @ 12 cc/ha coinciding with egg laying period thrice at weekly interval. • Conserve larval parasitoid, *Cotesia flavipes* • Apply carbofuran 3CG 33.3 kg/ha • If granular insecticides are not used, spray dimethoate 30EC 660 ml/ha
Aphids *Rhopalosiphum maidis*	• Spray dimethoate 30EC 1.2 lit/ha
Cob borer *Helicoverpa armigera*	Spray azadirachtin 1% (10000 ppm) 1500 ml/ha
Thrips	Apply carbofuran 3CG 33.3 kg/ha
Fall armyworm (FAW), *Spodoptera frugiperda* (invasive pest)	• Apply neem cake @ 250 kg/ha during last ploughing and treat seeds with thiamethoxam 30 FS or *Beauveria bassiana* @ 10 g/ kg • Adopt spacing of 60 x 25 cm for irrigated and 45 x 20 cm for rainfed maize and rogue spacing of 75 cm for every 10 rows • Raise border crop of cowpea, sunflower or gingelly, and intercrop with black gram or green gram to attract and conserve natural enemies • Raise border crop of Bajra Napier for irrigated maize or grain sorghum variety for rainfed maize to attract FAW adults on border crops • Use solar light trap @ one /ha and sex pheromone traps @ 50/ha for mass trapping of adults from 10-15 DAS **Apply any one of the following/ ha** • Early whorl stage (15 – 20 DAS) • Azadirachtin 1% EC 20 ml/10 l • Thiodicarb 75 WP 20 g/10 l • Emamectin benzoate 5 SG 4g/10 l • Late whorl stages (40-45 DAS) • *Metarhizium anisopliae* 80 g/10 l with 1 x 10^8 cfu/g • Spinetoram 12 SC 5 ml/10 l • Novaluron 10 EC 15 ml/10 l • Tasseling and cob formation stage (60 – 65 DAS)

	• Flubendiamide 480 SC 4 ml/10 l • Chlorantraniliprole 18.5 SC 4 ml/10 l

B. Disease Management

Seed treatment: Treat the seeds with carbendazim @ 2 g/kg or thiram @ 4 g/kg or metalaxyl @ 3 g/kg of seed

Name of the Disease	Recommendations
Rust: *Puccinia sorghi*	**CIB recommendation** • Spray kresoxim-methyl 44.3% SC @ 1 ml/l of water
Downy mildew or Crazy top: *Peronosclerospora sorghi*	• Sow resistant hybrid TNAU maize hybrid CO-6 and COH (M) 8 • Rogue out downy mildew infected plants • Spray metalaxyl + mancozeb @ 1000 g or mancozeb 1000 g/ha at 20 days after sowing **CIB recommendation** • Treat the seeds with metalaxyl-M 31.8% ES @ 2.4 ml/kg seed or with metalaxyl 35% WS @ 700gms with a dilution of 0.75 to 1ltr / 100 kg seeds.
Turcicum leaf blight: *Exserohilum turcicum* **and *Maydis* leaf blight:** *Helminthosporium maydis*	• Spray mancozeb or zineb @ 2-4 g/l on appearance of the disease and repeat at 10 days interval, if necessary • Seed treatment with *Pseudomonas fluorescens* @ 10g/ kg and spray propiconazole 25% EC @ 0.1% on 35 and 50 DAS **CIB recommendation** • Spray kresoxim-methyl 44.3% SC @ 1 ml/ l of water
Post Flowering Stalk rot: *Macrophomina phaseolina*	• Follow crop rotation • Avoid water stress at flowering time to reduce the disease incidence • Avoid nutrient stress and apply potash @ 80 kg/ha in endemic areas • Apply *P. Fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM / sand in soil at 30 days after sowing
	CIB recommendation for combined infections • Spray azoxystrobin 18.2% w/w + cyproconazole 7.3% w/w SC @ 1l/ha to control downy mildew, leaf blights and rust diseases • Spray azoxystrobin 18.2% w/w + difenoconazole 11.4% w/w SC @ 0.1% to control leaf blights and downy mildew diseases

MAIZE - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free from volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified seed production, leave a distance of 200 m all around the field from the same and other varieties of maize.

Pre-sowing seed management

- Soak the seed in 4 % *Pseudomonas fluorescens* for 8 h at 1 : 1 ratio and dry back the seeds to original seed moisture content under shade.

Season

- June - September and November - February.

Spacing

- 45 x 10 cm.

Fertilizer requirement

- The crop requires NPK @ 150:75:75 kg / ha. Apply NPK @ 40:75:40 kg / ha as basal, 50 kg N at 20 days after sowing and 60:0:35 kg NPK at 40 days after sowing.

Harvest

- Harvest the cobs as once over harvest.
- Verify true to type cobs based on kernel and shank colour (cob sorting) variations.
- Remove the diseased cobs.

Shelling

- Shell the cobs either by beating with pliable bamboo stick or using maize sheller with required rpm at a seed moisture content of 15 - 18 %.
- Improper shelling leads to pericarp injury up to 48 % and will promote saprophytic fungal growth.
- Estimate mechanical / pericarp injury through 20 % $FeCl_3$ test or using 0.25 % Tetrazolium chloride solution.

Size grading

- Grade the seeds using 18 / 64” round perforated sieves.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed using 5 ml of water / kg of seed.
- Dry dress the seeds with halogen mixture @ 3 g / kg of seed ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 for grain cum seed

storage.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 10 to 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

MAIZE - HYBRID SEED PRODUCTION

Land requirement

- Select fertile land with good drainage and irrigation.
- Field should not have volunteer plants. Hence, the previous crop should not be the same or different variety / hybrid of maize.

Isolation

- All around the field leave 200 m distance from same and other varieties / hybrids of maize.

Pre-sowing seed treatment

- Coat the seed with polymer @ 6 g / kg + carbendazim @ 2 g / kg + imidachloprid @ 1 ml / kg + micronutrient mixture @ 3 ml / kg of seed.

Spacing

- 60 x 25 cm.

Planting ratio

- Sow the female and male parents in the ratio of 6 : 2.

Border rows

- Sow four rows of male parent all around the field for effective pollination.

Fertilizer requirement

- Apply NPK @ 150: 75: 75 kg / ha in split applications. Three split application 25 kg of N at vegetative,at 5% flowering and 10 days after second application. Apply 18.75 kg of K in two split application at 5% flowering and maturity stages.

Foliar spray

- Spray $ZnSO_4$ 0.5 % + boric acid 0.2 % at 50 % flowering stage to enhance the seed set.

(V) SMALL MILLETS

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Sl. No.	Agro ecological zones	Districts	Season	Crops	Varieties
1	North Eastern Zone	Vellore, Thiruvannamalai, Cuddalore, Villupuram, Thiruvallur and Kancheepuram	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Varagu	CO 3, TNAU 86
				Panivaragu	CO (Pv) 5,TNAU 202
				Kudiraivali	CO (Kv) 2,MDU 1
2	North Western Zone	Salem, Namakkal, Dharmapuri and Krishnagiri	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Varagu	CO 3,TNAU 86
				Panivaragu	CO (Pv) 5,TNAU 202
				Kudiraivali	CO (Kv) 2,MDU 1
3	Western Zone	Coimbatore, Erode, Karur, Tiruppur and Dindigul	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Varagu	CO 3,TNAU 86
				Kudiraivali	CO (Kv) 2,MDU 1
				Panivaragu	CO (Pv) 5,TNAU 202
4	Cauvery Delta Zone	Tiruchy, Thanjavur, Thiruvarur, Nagapattinam, Pudukkottai, Perambalur and Ariyalur	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Varagu	CO 3,TNAU 86
5	Southern Zone	Madurai,Theni, Pudukottai, Sivagangai, Virudhunagar, Ramanathapuram, Tirunelveli and Thoothukudi	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Varagu	CO 3,TNAU 86
				Panivaragu	CO (Pv) 5,TNAU 202
				Kudiraivali	CO (Kv) 2,MDU 1
6	Hilly and High Altitude Zone	Ooty	Feb-March (Masipattam) June-July (Adipattam) Sep-Oct (Puratassipattam)	Tenai	CO (Te) 7
				Samai	ATL 1, CO (Samai) 4, Paiyur 2
				Panivaragu	CO (Pv) 5,TNAU 202

II. PARTICULARS OF SMALL MILLET VARIETIES

TENAI

PARTICULARS	CO (Te) 7
Year of Release	2005
Year of Notification	SO.599(E)/25.04.2006
Parentage	CO 5 x ISE 248
Duration (days)	80-85
Pigmentation	Greenish purple
Tillering ability	High
Panicles	Long, Compact
Grain Character	Yellow
Grain Yield (kg/ha) Straw Yield (t/ha)	1855 5.10
Special features	Non lodging and high yielding

SAMAI

PARTICULARS	Paiyur 2	CO(Samai) 4	ATL 1
Year of Release	2000	2006	2019
Year of Notification	SO.821(E)/ 13.09.2000	SO.1178(E)/ 20.07.2007	SO.3220(E)/ 05.09.2019
Parentage	Pure line selection from PM 295	CO 2 x MS 1684	Pedigree selection from the cross CO (Samai) 4 X TNAU 141
Duration (days)	85	75-80	85 – 90
Pigmentation	Green	Green	Dark Green
Tillering ability	Moderate	High	Moderate
Panicles	long, loose panicle	Open and Loose	semi-compact panicles
Grain Character	Brown	Grayish yellow	Golden Yellow
Grain Yield (kg/ha) Rainfed	850	1567	1587
Special features	Short duration, suitable for little millet – horse gram cropping sequence	Short duration, suitable for the existing double cropped rainfed situation of North, North Western and Western zones of Tamil Nadu	Drought tolerant, Non lodging and uniform maturity, Sturdy culm, Suitable for mechanical harvesting, Highly suitable for the districts in the Eastern Ghats such as Dharmapuri, Tiruvannamalai, Vellore, Salem and Krishnagiri. It is very much adaptive in the plains also under rainfed system.

VARAGU

PARTICULARS	CO 3	TNAU 86
Year of Release	1980	2012
Year of Notification	SO.19(E)/14.01.1982	SO. 2805(E) dt:25.08.2017
Parentage	Selection from Georgia variety	Pure line selection from IPs 85
Duration (days)	120	105 - 110
Pigmentation	Purple stem	Light purple at nodes
Tillering ability	High	Profuse tillering (8-12)
Panicles	Well exposed clusters and spikelets	Long panicles (7 - 10 cm) and grains are arranged in regular rows
Grain Character	Brown & Bold with hard seed Coat	Brown & Bold with oval shape
Grain Yield (kg/ha) Rainfed	1500 - 1800	2700 - 3200
Special features	Tolerant to smut, short duration	Tolerant to head smut, sheath blight and brown spot

PANIVARAGU

PARTICULARS	CO(PV) 5	TNAU 202
Year of Release	2007	2011
Year of Notification	SO. 449(E) / 11.02.2009	SO. 2805(E) dt: 25.08.2017
Parentage	PV 1403 x GPUP 21	PV 1453 x GPUP 16
Duration (days)	70	70 - 75
Pigmentation	Green	Green
Tillering ability	High	High
Panicles	Compact, dense with bold grains	Semi-Compact with bold grains
Grain character	Golden yellow	Yellow
Grain yield (kg/ha) Rainfed	2400	2000
Special features	High tillering, short duration ,fits well in the double cropped rainfed situation	Short duration, no incidence of pest and diseases and nutritive grains suitable for value addition

KUDIRAIVALI

PARTICULARS	CO(KV) 2	MDU 1
Year of Release	2009	2017
Year of Notification	SO.2137(E)/31.08.2010	SO.1379(E)/ 27.03.2018
Parentage	Pure line selection from EF 79	Pure line selection from Arupukkottailocal
Duration (days)	95	95-100
Pigmentation	Green	Green
Tillering ability	High	High (4-9)
Panicles	Compact, Pyramidal	Compact, Pyramidal Shape
Grain Character	Brownish grey	Yellowish grey
Grain Yield (kg/ha)		
Irrigated	-	2200 – 2500

Rainfed	2650	1500 – 1700
Special features	Good grain quality	High milling per cent (70%) and iron content (16 mg/ 100g of grain). Good quality and taste.

CROP MANAGEMENT

Package of practices for Tenai Seeds and sowing

For line planting : 10kg/ha

For sowing : 12.5 kg/ha for use of Gorru or seed drill is recommended.

Seed treatment : Treat 1 kg of seeds with 2 g Thiram or Carbendazim.

Field preparation : Plough the field thoroughly using a small iron plough or country plough to fine tilth. Apply basally FYM/Compost 12.5 t/ha

Fertilizer application :

Nitrogen 44 kg/ha

Phosphorus 22 kg/ha

Spacing : For line planting 22.5 cm x 10 cm (10 cm in between plants)

Weeding : First weeding on 15th DAS and the second on 40th DAS

Thinning : Before 20 DAS

Plant protection : Generally no major problem of pests and diseases

Package of practices for Samai

Seeds and sowing : For line planting 10 kg/ha

For broad casting 12.5 kg/ha

use of Gorru or seed drill is recommended.

Seed treatment : Treat 1 kg of seeds with 2 g Thiram or Carbendazim.

Field preparation : Plough the field thoroughly 2 or 3 times using a small iron plough or country plough to fine tilth.

Fertilizer application: Apply basally FYM/COMPOST : 12.5 t/ha

Nitrogen : 44 kg/ha, Phosphours : 22 kg/ha

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yieldtargets. (ready reckoners are furnished)

Samai

Soil : Red sandy loam (Irugur series)

Target : 1.5 – 2.0 t ha^{-1}

FN = 8.83 T - 0.41 SN - 0.55 ON

FP_2O_5 = 3.75 T - 1.10 SP - 0.62 OP

FK_2O = 4.57 T - 0.15SK - 0.48 OK

Initial soil test values (kg ha^{-1})			**Yield target – 1.5 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 2.0 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	160	22*	14	20	64	33	42
180	14	180	22*	12	17	56	31	39
200	16	200	22*	11*	14	48	28	36
220	18	220	22*	11*	11	39	26	33
240	20	240	22*	11*	8	31	24	30

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

Spacing : For line planting 25 x 10 cm (10 cm in between plants)

Weeding : First weeding is done on the 15th DAS and the second weeding on 40th DAS

Thinning : Thinning is done soon after weeding or before 20 DAS Plant

Protection : Usually no major problem of pests and diseases

Package of practices for Varagu

Seeds and sowing : For line planting 10 kg/ha; For sowing 12.5 kg/ha Use of Gorru or seed drill is recommended.

Seed treatment : Treat 1 kg of seeds with 2 g Thiram or Carbendazim.

Field preparation : Plough the field thoroughly using a small iron plough or country plough to fine tilth.

Fertilizer application: Apply basally FYM/COMPOST : 12.5 t/ha
Nitrogen : 44 kg/ha, Phosphours : 22 kg/ha

Spacing : For line planting 45 x 10 cm (10 cm in between plants)

Weeding : First weeding is done on the 15th DAS and the second weeding on 40th DAS

Thinning : Thinning is done soon after weeding or before 20 DAS Plant

protection : Generally no major problem of pests and diseases

3. **WHEAT (*Triticum aestivum.*)**

CROP IMPROVEMENT

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
31 - 34	3 - 4	20 - 25	600 - 900	up to 3300

Tropical and Sub tropical cool and dry climate. Grown during *rabi* season and has wide adaptability. Wheat can tolerate severe cold and snow and resume growth with the setting in warm weather in spring. Wheat needs cold and dry climate. Long day plant.

I. SEASON AND VARIETY

Suitable Districts

Plains & adjoining areas near to hills and hills in Theni, Dindigul, Karur, Coimbatore, Erode, Salem, Dharmapuri, Vellore, Thiruvannamalai and Kancheepuram Districts

Season

Ideal sowing time is 15th October to 1st week of November. Sowing must be completed within the first fortnight of November.

Variety : COW(W)1, TNAU Samba Wheat COW 2

2. Morphological Description of COW(W) 1

Particulars	**COW (W) 1**	**COW (W) 2**
Parentage	HD2646/HW2002A/CPAN3057	Mutant of NP 200
Duration (days)	85-90	110
Grain yield (Kg /ha)	2364	4040
Stem	Erect	Erect to semi erect
Height (cm)	73 – 78	75-80 cm
Tillers	5-6	10-12
Days to 50% flowering	50 days	73 days
Ear size and shape	Fusiform ears	Long & slightly tappering
Grain colour	Amber	Raddish
1000 grains weight (g)	37	41
Special features	Non lodging, non shattering; tolerance to stem and leaf rust suitable for chappathi and bread making.	Resistant to rust, heat tolerant

3. SEED RATE: 100 kg/ha

CROP MANAGEMENT

1. FIELD PREPARATION

Plough twice with an iron plough and two to three times with cultivator and prepare the land to a fine tilth.

2. APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost on the unploughed field and incorporate in the soil.

3. SEED TREATMENT WITH FUNGICIDES

Treat the seeds with Carbendazim or Thiram at 2 g/kg of seeds 24 hours before sowing

4. FORMING BEDS AND CHANNEL

Form beds size on 10 m^2 or 20 m^2. The irrigation channels are to be provided sufficiently.

5. APPLICATION OF FERTILIZERS

(i) If soil test recommendation is not available, adopt a blanket recommendation of 80:40:40 NPK kg/ha. Apply 37.5 kg $ZnSO_4$, 40 kg S basally for soils having Zn and S deficiencies.

(ii) Apply half of N and full dose of P_2O_5 and K_2O basally before sowing and incorporate in the sowing line.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Wheat - Hills(1)

Soil Laterite (Ooty Series)

Target 3.5 – 4.0 t ha^{-1}

FN = 7.60 T- 0.55 SN - 0.92 ON

FP_2O_5 = 3.59T - 0.26 SP - 0.54 OP

FK_2O = 3.88T - 0.45SK - 0.51 OK

Initial soil test values (kg ha^{-1})			**Yield target – 3.5 t ha^{-1}**			**Yield target – 4.0 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP^a**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	200	175	96	49	27	134	67	45**
225	250	200	82	36	16	120	54	35
250	300	225	69	30*	15*	107	41	24
275	350	250	55	30*	15*	93	30*	15*
300	400	275	50*	30*	15*	79	30*	15*

* Maintenance dose ;** Maximum dose;SP^a- Bray P

Wheat - Plains (2)

Soil :	Mixed black calcareous (Perianaickenpalayam series)	FN	= 8.83 T- 0.71 SN- 0.88 ON
		FP_2O_5	= 4.52T- 1.75 SP- 0.95 OP
Target :	3.5 – 4.0 t ha^{-1}	FK_2O	= 6.05T - 0.20SK - 0.83 OK

Initial soil test values (kg ha^{-1})			**Yield target – 3.5 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 4.0 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	14	300	112	90**	60**	150**	90**	60**
220	16	350	98	90**	60**	142	90**	60**
240	18	400	84	90**	60**	128	90**	60**
260	20	450	69	90**	60**	114	90**	60**
280	22	500	55	90**	60**	99	90**	60**

** Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

6. SOWING

Draw the lines 20 cm apart and sow the seeds continuously after application of fertilizers to a depth of 5 cm. Avoid deep sowing.

7. WEED MANAGEMENT

i. Spray PE Isoproturon 500 g/ha as pre-emergence spray on 3 DAS followed by one hand weeding on 35th DAS.
ii. If herbicide is not applied, give two hand weedings on 20th and 35th DAS.

8. WATER MANAGEMENT

The crop requires 4 - 6 irrigations depending on the soil type and rainfall. Wheat crop requires minimum of 5 irrigations at the following critical stages.

I = Immediately after sowing
II = Crown root intiation : 15-20 DAS III =Active tillering stage : 35-40 DAS
IV= Flowering stage : 50-55 DAS
V= Grain filling stage : 70-75 DAS

Crown root initiation and flowering are the most critical stages. Water stagnation should be avoided at the time of germination.

9. TOP DRESSING

Apply remaining half of N at crown root initiation stage (15-20 DAS).

10. HARVESTING

Harvest the crop when the grains become hard and straw becomes dry and brittle. Thresh and winnow the grains. Use mechanical threshers to reduce the cost of threshing and winnowing.

CROP PROTECTION

Seed treatment: Treat the seed with any one of the following fungicides Carbendazim @ 2 g/kg of seed, Thiram @ 2 g/kg of seed or Carboxin @ 2 g/kg of seed.

4. PULSES

(i) REDGRAM (*Cajanus cajan (L.) Millsp.*)

Climate Requirement

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
30	17	26 - 30	600 - 1400	

Tropical and subtropical legumes, suitable for rainfed in semiarid areas due to its deep taproot, heat tolerance and fast growing habit. During vegetative growth, prefers a fairly moist and warm climate. During flowering and ripening stage, requires bright sunny weather for proper fruit setting. Highly susceptible to frost at the time of flowering, hardy, widely adaptable, and drought resistant. It has low tolerance of soil salinity and waterlogging.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

District/season	Varieties
Vaigasi Pattam (May-June) Krishnagiri, Dharmapuri, Salem, Erode, Coimbatore Dindigul, Theni and Madurai	CO(Rg) 7, CO 8, CO 9
Adi/Avanipattam (June - August) Vellore, Thiruvannamalai, Salem, Namakal, Perumbalur, Ariyalur, Madurai, Dindigul, Theni, Pudukkottai and Sivaganga June 15th to July 15th sowing	CO 8, CO 9
Purattasipattam (September – October) Vellore, Tiruvannamalai, Dharmapuri, Salem, Namakkal, Erode, Coimbatore, Madurai, Dindigul, Theni Pudukkottai, Sivagangai, Perumbalur, Ariyalur	Co (Rg) 7 and VBN(Rg) 3
Markazhipattam (Winter Irrigated) All districts except The Nilgiris and Kanyakumari	Co (Rg) 7 and VBN(Rg) 3
Chithiraipattam (Summer Irrigated) All districts except The Nilgiris and Kanyakumari Wetland bunds	Co (Rg) 7 and VBN(Rg) 3 BSR 1

II.DESCRIPTION OF REDGRAM VARIETIES

PARTICULARS	CO 8	CO 9
Year of Release	2017	2018
Year of Notification	SO.1379(E)/ 27.03.2018	SO.3220(E)/05.09.2019
Parentage	APK 1 x LRG 41	CO6 x IC 525427
50% flowering(days)	120-130	120-130
Maturity(days)	170-180	170-180
Season	Adi pattam	Adi pattam
Grain yield(kg/ha)		
Rainfed	1600 kg/ha	1700 kg/ha
Irrigated	1800 cm	-
Plant height(cm)	165-180	210-240 cm
Plant spread	Erect	Erect

Colour of standard petal	Yellow base with medium pattern of streaks	Yellow base with medium pattern of streaks
Colour of pod	Brown streaks	Green with Brown streaks
Colour of grain	Cremish brown	Brown
100 seed weight(g)	10.22 to 11.44g)	9.9
Pattern of growth	Indeterminate	Indeterminate

Particulars	**Co (Rg) 7**	**VBN(Rg)3**	**BSR 1**
Year of Release	2004	2005	1986
Year of Notification	SO.1177(E)/ 25.08.2005	SO.1572(E)/ 20.09.2006	Not notified
Parentage	Selection from PB 9825	Vamban 1 x Gulbarga	Pureline selection from Mayiladumparai
50% flowering(days)	70-90	65-70	100-110
Duration (days)	120-130	100-105	Perennial
Grain yield(kg/ha)			
Rainfed (kg/ha)	950	880	0.75 to 1.0 kg of green pods/plant
Irrigated (kg/ha)	1168	1530	-
Plant height (cm)	120-130	100-120	150-200
Branches	7-9	3-10	7-10
Plant spread	Semi spreading	Erect, Semi-determinate and open type	Semi spreading
Colour of standard petal	Yellow with light red vein at the base	Yellow with light red vein at the base	Red at dorsal side
Colour of pods	Green with purple streaks	Green with purple streaks	Red with diagonal constriction
Colour of grain	Reddish brown	Reddish brown	Reddish brown
100 grain wt (g)	8.5-11.0	7.0-8.0	12.0
Pattern of growth	Indeterminate	Semi determinate	Indeterminate

CROP MANAGEMENT

SEED RATE

System	**Long duration**		**Short duration**	
	Low Fertility	High Fertility	Low Fertility	High Fertility
Sole Crop	8	5	13	10
Mixed Crop	3	3	5	5
Bund Crop	50 g / 100 meter			

Long duration varieties: CO 6, CO 8, Vamban 2, LRG 41

Short duration varieties : CO (Rg) 7, VBN (Rg) 3

Bund Crop : BSR 1

III. MANAGEMENT OF FIELD OPERATION

1. PREPARATION OF THE LAND

Prepare the land to fine tilth and apply 12.5 t FYM/ha or composted coir pith at the time of last ploughing and form ridges and furrows

2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with rhizobium. Fungicides and biocontrol agents are incompatible.

3. TREATMENT OF THE SEEDS WITH BIOFERTILIZER

a) Fungicide (or) bio control agents treated seeds should be again treated with bacterial culture after 24 hours. Treat the seeds required for sowing 1 ha with 200g each of Rhizobial culture CPR6 / CPR 9, phosphobacteria and PGPR (*Pseudomonas* sp.) using rice gruel, shade dry it before sowing. (or) Treat one hectare of seeds with 25 g each of powder formulation of *Rhizobium* and AM fungi using binder (polymer), shade dry before sowing.

b) If seed treatment is not carried out, apply 2 kg each Rhizobial culture, Phosphobacteria and PGPR (*Pseudomonas* sp.) with 25 kg of FYM and 25 kg of sand, mix uniformly before sowing.

4. APPLICATION OF FERTILIZERS

If soil test is not done, apply fertilizers basally before sowing

a) Apply fertilizers basally before sowing.

Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +10 kg S*/ha Irrigated : 25 kg N + 50 kg P_2O_5 + 25 kg K_2O + 20 kg S*/ha

*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

b) Soil application of 25 kg $ZnSO_4$/ha under irrigated condition

c) Soil application of TNAU micronutrient mixture @ 5 kg/ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).

d) Foliar spraying to mitigate moisture stress

Foliar spraying of 2% KCl + 100 ppm Boric acid during dry spell as mid season management practice in black gram during *Rabi* season is recommended to increase the yield over KCl spray alone .

Nitrogen substitution by organic sources for pulses

50 per cent nitrogen can be substituted through organic source (850 kg of vermicompost per hectare). Lime application is recommended for pulses with soil pH less than 6.0.

SOWING THE SEEDS

Dibble the seeds adopting the following spacing.

Variety	Pure crop		Mixed crop
	Low fertility	High fertility	
CO(Rg) 7	45 cm x 30 cm	60 cm x 30 cm	120 cm x 30 cm
Vamban (Rg) 3, APK 1	45 cm x 20 cm	60 cm x 20 cm	120 cm x 30 cm
CO 6, CO 8, Vamban 2, LRG 41	90 cm x 30 cm	120 - 150 x 30 cm	240 cm x 30 cm
Bund Crop	60 cm for BSR 1 and 30 cm for others.		

5. Season

- Long duration varieties (CO 6, CO 8, Vamban 2, LRG 41) : Second fortnight of July and First fortnight of August months.
- Short duration varieties : January - May and September first fort night.
- Note: Sowing season should be planned in such a way that flowering and pod maturity stage does not coincide with rain.

6. WEED MANAGEMENT

i) Pre emergence application of Pendimethalin 0.75 kg/ha (2.5 litres/ha) on 3 DAS mixed with 500 litres of water using Backpack/Knapsack/Rocker sprayer using flat fan deflector type of nozzle. Then irrigate the field. Following this, one hand weeding may be given on 30-35 DAS

ii) If herbicide is not given, give two hand weedings on 20 and 35 DAS.

iii) In case of labour problem, apply Pendimethalin 0.75 kg (2.5 lit/ha) on 3 DAS followed by early post emergence application of Imazethapyr @ 60 g ai/ha on 15 DAE of weeds (2 - 3 leaves stage of weeds) and quizalofop ethyl @ 50 g ai/ha on 20 DAE of weeds (2 - 3 leaves of weeds) are recommended for controlling broad leaved and grassy weeds, respectively. If both the weeds are present, tank mix application of Imazethapyr @ 60 g ai/ha and quizalofop ethyl @ 50 g ai/ha at 15 - 20 days after emergence of weeds (2 - 3 leaves stage of weeds) is recommended. Apply PE Pendimethalin 30% EC + Imazethapyr 2% EC (Valor 32% EC; Readymix herbicide) @ 1,.0 kg a.i. ha^{-1} at 3 DAS.

iv) Apply PE metalachlor 1.0 kg ha^{-1} on 3 DAS followed by one hand weeding on 40 DAS. Note: At the time of herbicide application, there should be sufficient soil moisture

7. WATER MANAGEMENT

Irrigate immediately after sowing, 3^{rd} day after sowing, bud initiation, 50 % flowering and pod development stages. Water stagnation should be avoided.

8. FOLIAR APPLICATION

a) Foliar spray of NAA 40 mg/l once at pre-flowering and another at 15 days thereafter

b) Foliar spray of DAP 20 g/l or urea 20 g/l once at flowering and another at 15 days there after

c) Foliar spray of salicylic and 100 mg/litre once at preflowering and another at 15 days there after

9. HARVESTING THE CROP

1) Harvest the whole plants when 80% of the pods mature.
2) Heap for 2 – 3 days
3) Dry and process.

10. INTER-CROPPING

a) Raise one row of long duration redgram varieties as inter crop for every six rows of groundnut (6:1) under rainfed situation.

b) Raise one row of short or medium duration redgram as inter crop for every four rows of groundnut (4:1) under rainfed as well as irrigated condition.

c) **Multistoreyed cropping**: For rainfed Vertisols of Virudhunagar, Tirunelveli, Thoothukudi districts recording more than 300 mm of rainfall during the crop growth period, multistoreyed cropping system Agathi + Redgram (Co 5) + Cotton (MCU 10) + Blackgram (Co 5) is highly profitable. (Agathi in I tier with 1 x 1 m spacing - Redgram in II tier with a spacing of 45 x 20 cm - Cotton in the III tier with a spacing of 45 x 15 cm - Blackgram in the IV tier with the spacing of 30 x 10 cm).

For rainfed Vertisols receiving less than 300 mm of rainfall, Agathi + Sorghum (CO 26) + Cotton (MCU 10) + Blackgram (Co 5) system is ideal. (Agathi in I tier with a spacing of 1 x 1 m - sorghum in II tier with a spacing of 45 x 15 cm - cotton in III tier with the spacing of 45 x 15 cm and Blackgram in IV tier with 30 x 10 cm). For both systems, apply 40 kg N and 20 kg P_2O_5/ha.

11. REDGRAM TRANSPLANTING

- Select only long duration redgram varieties
- Transplant within the month of August either under rainfed condition or under irrigated condition
- Select poly bag with a size of 6x4 inches and 200 micron thickness
- Fill the poly bag with native soil: Sand: FYM @1:1:1 and put 3-4 holes in the bottom to avoid water stagnation
- Soak the seeds in 0.2% Calcium chloride for one hour and dry it under shade for 7 hours to harden the seeds
- Treat the hardened seeds with *T. viride* @ 4g/kg and 100 g Rhizobium and 100 g phosphobacterium. Sow the seeds @2/poly bag at 1 cm depth
- Sow the seeds in polybags 30-45 days prior to transplanting
- Plough the field deeply to get fine tilth followed by 2-3 harrowings at 3 weeks prior to transplanting
- In medium to deep soils for raising long duration varieties, dig 15 sqcm pits at 5’ X 3’ for pure crops and 6’ x 3’ for intercropping under irrigated condition. In rainfed condition dig the pits at a spacing of 5’x3’. For short duration varieties dig 15 sq cm pits at 3’ x 2’ spacing.
- Under water logging condition, form furrows before digging pits
- Apply inorganic fertilizers @ 25:50:25 kg NPK /ha at 20-30 days after planting as urea, DAP and potash around the seedlings
- Apply $ZnSO_4$ @ 25 kg/ ha as basal along with FYM or sand

- Nip (removal of top 5 cm) the plants at 20 – 30 days after planting to arrest the terminal growth
- Foliar Spray of Napthalene aceptic acid (NAA) @ 0.5 ml/litre to control flower dropping in red gram.

12. NUTRITIONAL DISORDERS

Redgram / Greengram/Blackgram/Cowpea

Zinc: Symptom appears within a month of sowing. The plants are stripped with yellow or pale green foliage. Veins and mid ribs of the leaves are green although tissue around them becomes yellow and bronzed.

Iron: Reduced concentration of Chlorophyll in leaves - pale leaf colour may be indistinguishable from deficiency of nitrogen or other elements.

CROP PHYSIOLOGY

Foliar spray of TNAU Pulse Wonder @ 2 kg/acre in 200 litres of water at flower initiation stage decreases flower shedding, increases yield and offers moisture stress tolerance

CROP PROTECTION

A. Pest management

Pest	**ETL**
Aphids	20 nos. /2.5 cm shoot length
Pod borers	10% of affected pods
Plume moth	5 larvae /plant
Spotted pod borer	3 larvae /plant

Pest Management strategies

Aphids *Aphis craccivora*	**Spray any one of the following :** Methyl demeton 25 EC 500 ml/ha Dimethoate 30 EC 500 ml/ha
Pod borers **Spotted pod borer** *Maruca vitrata* **Gram pod borer** *Helicoverpa armigera* **Plume moth** *Exelastis atomosa* **Pod fly** *Melanagromyza obtusa*	• For pod borers, raise one row of sunflower as intercrop for every 9 rows of pigeon pea and plant maize as border crop. • Pheromone traps for *Helicoverpa armigera* 12/ha • Bird perches 50/ha • Mechanical collection of grown up larva and blister beetle • Ha NPV 3 x10^{12} POB/ha in 0.1% teepol • *Bacillus thuringiensis* var *kurstaki* 5%WP 1000-1250 g/ha (Note : Insecticide / Ha NPV spray should be made when the larvae are upto third instar) **Apply any one of the following insecticides:** Azadirachtin 0.03 % WSP 2.5kg/ha Benfuracarb 40% EC 2.5l/ha Chlorantraniliprole 18.5% SC 150ml/ha Chlorpyriphos 20 EC 1250 ml / ha Emamectin benzoate 5% SG 220 g/ha Ethion 50% EC 1.0 l/ha Flubendiamide 39.35 % SC 100ml / ha Indoxacarb 14.5% SC 350 ml/ha Indoxacarb 15.8% SC 333 ml/ha Lufenuron 5.4% EC 600ml/ha Methomyl 40%SP 750g/ha Monocrotophos 36%SL 625-1250ml/ha Neem oil 2%

	Quinalphos 1.5%DP 23kg/ha Quinalphos 25 %EC 1400ml/ha Spinosad 45%SC 125 ml/ha Thiodicarb 75 WP 625g / ha
Pod bugs	Dimethoate 30% EC 500ml/ha Methyl demeton 25% EC 500ml/ha

B. Disease management

Seed treatment : Treat the seeds with *Trichoderma asperellum* @ 4 g or *P. fluorescens* @ 10 g or carbendazim 2 g or thiram @ 4 g/kg of seed

Disease	Recommendations
Wilt: *Fusarium udum*	• Apply *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand to soil at 30 days after sowing • Spot drench with carbendazim @ 1 g/ lit
Root rot: *Rhizoctonia bataticola* *(Macrophomina phaseolina)*	
Sterility Mosaic: *Pigeonpea sterility mosaic virus* (Vector : *Aceria cajani*)	• Rogue out the virus infected plants in the early stages of growth • Spray fenazaquin @ 1ml/ l soon after appearance of the disease and if necessary repeat after 15 days

C. Nematode management

Seed treatment with *Pseudomonas fluorescens* and *Trichoderma viride* @ 5+5 g/kg seed manages population of cyst nematode, *Heterodera cajani*.

SEED PRODUCTION
RED GRAM - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified seed production leave a distance of 100 m all around the field from the same and other varieties of red gram.

Pre-sowing seed treatment

- Soak the seeds for 3 h in 100 ppm $ZnSO_4$ (10 g / 100 lit of water) in 1/3 volume before sowing and quickly air dry in shade to their original moisture content.
- Treat the seeds with carbendazim 75 % WP 2 g dissolved in 5 ml of water per kg of seeds and air-dried.
- Pellet the seeds with *Rhizobium* culture (50 g / kg of seed) before sowing.

Foliar application

- Spray 2 % DAP at the time of flowering and a second spray at 15 days after the first spray.
- Spray NAA 40 ppm at the time of flowering and a second spray at 15 days after the first spray.

Pre-harvest sanitation spray

- Spray (0.05 %) malathion 50 EC 3 - 5 days before harvest to minimize the carryover of bruchid infestation from field to storage.

Harvest

- Harvest the pods at physiological maturity stage (approximately 40 days after 50 % flowering).
- Collect the seeds from first and second pickings for quality seeds.

Drying

- Dry the pods to about 15 % moisture content.
- Dry the seeds to 10 % moisture content.

Seed grading

- Size grade the large seeded varieties using BSS 5 x 5 or BSS 6 x 6 wire mesh sieve.
- Discard the discoloured and broken seeds for seed purpose.

Pre-storage seed treatment

- Treat the seeds with carbendazim 2 g using 5 ml of water / kg of seed (or) dry dress the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder at 5:4:1 ratio @ 3 g / kg of seed.

Storage

- Store the seeds with a seed moisture content of 10 - 12 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 8% in 700 gauge polythene bag for long term storage (more than 15 months)

RED GRAM HYBRID SEED PRODUCTION

Land requirement

- Select fertile land with good drainage and irrigation facilities.
- Field should be free from volunteer plants. Hence, the previous crop should not be the same or different variety / hybrid of redgram.

Isolation

- For foundation seed production (parental lines seed production), leave a distance of 200 m all around the field from the same and other varieties / hybrids of redgram.
- For hybrid seed production from the same and other varieties / hybrids of redgram, leave a distance of 100 m all around the field.

Planting ratio

- Sow the female and male lines at a ratio of 4:2.

Border rows

- Sow 2 rows of the male parent all around the field for effective pollination.

Spacing

- 45 x 15 cm.

Fertilizer

- Apply NPK @ 25:50:25 kg / ha^{-1} as basal application.

Roguing

- Pull out all male fertile plants in female rows for genetic purity maintenance.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g using 5 ml of water / kg of seed.
- Dry dress the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizia amara*) leaf powder mixed in 5:4:1 ratio @ 3 g / kg of seed.
- Treat the seeds with turmeric rhizome powder (or) neem leaf powder at 1:50 ratio against bruchid infestation as an eco-friendly seed treatment.

PERENNIAL REDGRAM

Variety	:	BSR 1
Economic uses	:	Tender beans are pinkish green in colour and can be cooked as curry or added to Kurma or Sabji. When the beans mature they can be used as Dhal. Recommended for growing in kitchen gardens, backyards, farm road sides, as border crop in sugarcane, banana and betelvine and as a shade crop in turmeric and as a bund crop in paddy double cropped wetlands.
Season	:	June – July Height of the plant: 150 - 200 cm Number of branches 7 - 10
Flowering	:	Five months from date of sowing
Pit Size	:	Small pits are dug 90 cm apart and the pits are filled with a mixture of well decomposed manure or compost and soil.
Fertilizer application	:	Urea 15 g and superphosphate 30 g / pit.
Planting methods	:	Two to three seeds are dibbled per pit and watered. When they grow six inches height one plant may be retained in each pit.
Irrigation	:	Need based
Harvesting	:	If harvested when the pods are tender the beans will be fit for making curry. Each plant will yield two to three kg of green pods at an average seed yield of 750 g to one kg per plant. After the first harvest the branches are pruned and allowed to grow further. In another 45 - 60 days the plants produce the second flush. For pure crop, about 3 kg of seeds may be required.

(i) BLACKGRAM (*Vigna mungo L.*)

Climate Requirement

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
40	20	27 - 30	400 – 600	1800

Tropical and subtropicalhot and humid growing season. It is generally grown in kharif/rainy and summer season. Heavy rains during flowering stage are harmful to yield of pea crop.

CROP IMROVEMENT
1. SEASON AND VARIETIES

District/Season	Varieties
Adipattam (June-August) All districts except Kanyakumari and Nilgiris	VBN 6, VBN 8
Puratasipattam (September-November) Vellore, Tiruvannamalai Dharmapuri, Salem, Namakkal, Perambalur, Erode, Coimbatore, Madurai, Dindigul, Theni, Pudukottai,, Sivagangai, Ramanathapuram, Virudhunagar, Thoothukudi and Tirunelveli	VBN 6, MDU 1, CO 6, VBN 8, VBN 10
Markazhi – Thaipattam(Winter Irrigated) All districts except Kanyakumari and Nilgiris	VBN 6, CO 6, VBN 8, VBN 10
Rice fallows (January) Thanjavur,Tiruvarur, Nagapattinam, Cuddalore, Villupuram and Kanchipuram	ADT 3, ADT 6, KKM 1, VBN 6, VBN 9
Chithiraipattam (Summer Irrigated) Thanjavur, Tiruvarur, Nagapattinam, Cuddalore, Villupuram, Tiruchirappalli, Perambalur, Thiruvallur, Kancheepuram,	ADT 5, VBN 8

II. DESCRIPTION OF BLACKGRAM VARIETIES

Particulars	VBN 6	VBN 8	VBN 9	VBN 10
Year of Release	2011	2018	2019	2019
Year of Notification	SO.1708(E)/ 26.07.2012	SO.1379(E)/ 27.03.2018	SO.3220(E)/ 05.09.2019	SO.3220(E)/ 05.09.2019
Parentage	Vamban 1 x *Vigna mungo silvestris*	Vamban 3 x VBG 04-008	Mash 114 x Vamban 3	Vamban 1 x UH 04-04
Maturity duration (days)	65-70	65-70	70-75	70-75
Grain yield (kg/ha)	-	-		
Rainfed	850	988	1230	-
Irrigated	890	871	-	1130
Height (cm)	18.6	35-40	35-40	35-40
Hairiness of pods	Hairy	Hairy	Hairy	Hairy
100 grain wt (g)	3.8-4.0	4.5 – 5.0	4.0-5.0	4.0-5.0
Special features	Resistant to Yellow Mosaic, synchronized pod maturity	Resistant to Mungbean Yellow Mosaic Virus (MYMV), leaf crinkle and moderately resistant to powdery mildew diseases	Moderately resistant to Mungbean Yellow Mosaic Virus, Urdben Leaf Crinkle Virus, Leaf Curl Virus and Powdery mildew diseases	Resistant to Mungbean Yellow Mosaic Virus, Urdbean Leaf Crinkle Virus, Leaf Curl Virus diseases

Particulars	CO 6	ADT 3	ADT 5	ADT 6	MDU 1	KKM 1
Year of Release	2010	1982	1988	2017	2014	2017
Year of Notification	SO.632(E)/25.03.11	SO.596(E)/13.08.1984	SO.793(E)/22.11.1991	SO.1379(E)/ 27.03.2018	SO.1556(E)/11.06.2015	SO.1379(E)/ 27.03.2018
Parentage	DU 2 x VB 6	Pure line selection from Thirunelveli local	Pure line selection from Kanpur	Vamban 1 x VBN 04-006	ADB 2003 x VBG 66	COBG 643 X VBN3
Maturity duration (days)	60-65	70-75	65-70	65-70	70-75	65-70
Grain yield (kg/ha)	-	-	-	-	-	
Rainfed	880	720 (Rice fallow)	-	740 (Rice fallow)	-	610 (Rice fallow)
Irrigated	-	-	1545	-	790	-
Height (cm)	30 -35	50	20-25	35-40	30-35	50
Hairiness of pods	Non Hairy	Hairy	Hairy	Hairy	Hairy	
100 grain wt (g)	5.0 - 6.2	3.5- 4.0	3.5-4.5	4.0-5.0	4.5-5.0	
Special features	Moderately resistant to YMV disease. Field tolerance to aphids, pod borer and synchronized maturity	Yellow mosaic incidence will be less during Markazhi and Thai pattam	After 65 days second sett of flowering starts	Moderately resistant to MYMV, LCV and PMD	Moderately resistant to MYMV, Non shattering and Non Lodging. Suitable for *Rabi* season. Good battering quality.	

III. SEED RATE

VARIETIES	Quantity of seed required kg/ha Pure crop	Mixed crop
VBN 6, VBN 8, ADT 5, CO 6, MDU 1	20	10
ADT 3, ADT 6, KKM 1 (Rice fallows)	25	..

Optimum plant population 3,25,000/ha

CROP MANAGEMENT

IV. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

i. Prepare the land to fine tilth and form beds and channels.
ii. Amendments for soil surface crusting: To tide over the soil surface crusting apply lime at the rate of 2t /ha along with FYM at 12.5 t/ha or composted coirpith at 12.5 t/ha to get an additional yield of about 15 - 20%.

2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with Biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

Note: Seed treatment will protect the seedlings from seed borne pathogens, root-rot and seedlings diseases.

3. SEED TREATMENT WITH BIOFERTILIZER

a) Treat the seeds required for sowing 1 ha with 200g each of Rhizobial culture COG 15, Phosphobacteria and PGPR (*Pseudomonas* sp.) using rice gruel, shade dry it before sowing. (or) Treat one hectare of seeds with 25 g each of each of powder formulation of *Rhizobium* and AM fungi using binder (polymer), shade dry before sowing.
b) If seed treatment is not carried out, apply 2 kg each Rhizobial culture, Phosphobacteria and PGPR (*Pseudomonas* sp.) with 25 kg of FYM and 25 kg of sand, mix uniformly before sowing.

4. FERTILIZER APPLICATION

If soil test is not done, apply fertilizers basally before sowing

a) Apply fertilizers basally before sowing.
   Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +20 kg S*/ha Irrigated : 25 kg N + 50 kg P_2O_5 + 25 kg K_2O + 40 kg S*/ha
   *Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

Soil test crop response based integrated plant nutrition system (STCR- IPNS recommendation may be adopted for prescribing fertilizer doses for specified yield targets (ready reckoners are furnished)

Soil :	Mixed black calcareous (Perianackenpalayam series)	FN = 10.84T-0.39 SN FP_2O_5=7.23T-1.00 SP
Target :	0.9 – 1.0 t ha^{-1}	FK_2O=5.20T-0.04 SK

Initial soil test values (kg ha^{-1})			**Yield target – 0.9 t ha^{-1}**			**Yield target – 1.0 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	300	13*	25*	13*	13*	28	13*
180	14	325	13*	25*	13*	13*	26	13*
200	16	350	13*	25*	13*	13*	25*	13*
220	18	375	13*	25*	13*	13*	25*	13*
240	20	400	13*	25*	13*	13*	25*	13*

*maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield targe in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

b) Soil application of 25 kg $ZnSO_4$/ha under irrigated condition (or) 12.5 kg $ZnSO_4$ on EFYM.

c) Soil application of TNAU micronutrient mixture @ 5 kg/ha as Enriched FYM. (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture incubate for one month in shade).

   50 kg $FeSO_4$ as EFYM or 10 kg Fe EDTA per ha.

d) Foliar spray of 1% urea for yield improvement in black gram.

   For yield improvement through increasing the physiological, biochemical attributes, foliar spray of urea 1% on 30 and 45 dayas after sowing is recommended. For rice fallow pulses in Delta area, the present recommendation of foliar spray of 2% DAP may be continued.

   Foliar spraying of 0.5% ZnSO4, 1% FeSO4 + 0.1% citric acid at 30, 45 DAS if the plants shown deficiency symptoms. For yield improvement through increasing the physiological, biochemical attributes, foliar spray of urea 1% on 30, 45 days after sowing is recommended. For rice fallow pulses in Delta area, the present recommendation of foliar spray of 2% DAP may be continued.

e) Foliar spraying to mitigate moisture stress

Foliar spraying of 2% KCl + 100 ppm Boric acid during dry spell as mid season management practice in black gram during *Rabi* season is recommended to increase the yield over KCl spray alone .

Economizing the use of micronutrients through seed treatment for blackgram

Seed coating with biofertilizers and micronutrients viz., Zn, Mo & Co @ 4, 1, 0.5 g/kg of seed is recommended.

Nitrogen substitution by organic sources for pulses

50 per cent nitrogen can be substituted through organic source (850 kg of vermicompost pe hectare). Lime application is recommended for pulses with soil pH less than 6.0.

5. SOWING OF SEEDS

a) For irrigated crop dibble the seeds adopting 30 x 10 cm spacing
b) For rainfed crop dibble the seeds adopting 25 cm x 10 cm spacing

6. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on third day. Irrigate at intervals of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

7. FOLIAR APPLICATION

a. Foliar spray of NAA 40 mg/litre once at pre-flowering and another at 15 days thereafterto reduce flower shedding.
b. i) For rice fallow crops foliar spray of TNAU Pulse wonder @ 5 kg/ha once at flowering to decreases flower shedding.
   ii) For irrigated and rainfed crops, foliar spray of TNAU Pulse wonder @ 5 kg/ha once at flowering
c. Foliar spray of salicylic acid 100 mg/litre once at preflowering and another at 15 days there after to improve translocation efficiency and seed yield.

8. WEED MANAGEMENT

i) Pre emergence application of Pendimethalin 1.0 litres/ha under irrigated condition, PE application of Pendimethalin 0.75 litres/ha under rainfed condition on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litres of water for spraying one ha followed by one hand weeding at 20 DAS (or) EPOE application of quizalofop ethyl @ 50 g ai/ha^{-1} and imazethapyr @ 50 g ai ha^{-1} on 15 – 20 DAS. If herbicides are not applied give two hand weedings on 15 and 30 days after sowing.
ii) Apply PE Pendimethalin 30% EC + Imazethapyr 2% EC (Valor 32% EC; Readymix herbicide) @ 1.0 kg a.i. ha^{-1} at 3 DAS.

9. Multi bloom technology

A special technology being practiced in Pattukottai block of Tanjore district for blackgram and greengram. The soil is alluvial and rich in organic matter and nutrients. The crop is sown during early summer (Jan.-Feb.) as normal crop and fertilizer is applied as per the recommendation for irrigated crop. In addition to that, top dressing of Nitrogen is done with an extra dose of 25 to 30 kg through urea. Since pulses are indeterminate growth habit and continue to produce new flashes, the top dressing will be done on 40-45 days after sowing. The crops complete its first flesh of matured pods during 60-65th day, further their second new flesh within 20-25 days. Therefore two fleshes of pods can be harvested at a time within the duration of 100 days.

CROP PROTECTION

A. Pest management

Economic threshold level for important pests

Pest	**ETL**
Aphids	20nos./2.5 cm shoot length
Pod borers	10% of affected pods
Spotted pod borer	3 larvae/plant
Stem fly	10% of affected plants
Tobacco caterpillar	8 egg masses/100 m

Pest Management strategies

Stem fly *Ophiomyia phaseoli*	Treat seeds with Dimethoate 30% EC 5 ml/kg
Aphids *Aphis craccivora*	**Spray any one of the following** Methyl demeton 25% EC 500 ml/ha Dimethoate 30% EC 500 ml/ha
Whitefly *Bemisia tabaci*	**Spray any one of the following** Methyl demeton 25% EC 500 ml/ha Dimethoate 30% EC 500 ml/ha
Tobacco caterpillar *Spodoptera litura*	➢ Use of light trap to monitor and kill the attracted adult moths. ➢ Set up the sex pheromone trap at 12/ha to monitor the activity of the pest and to synchronize the pesticide application, if need be, at the maximum activity stage. ➢ Growing castor along border and irrigation bunds. ➢ Removal and destruction of egg masses in castor and cotton crops. ➢ Removal and destruction of early stage larvae found in clusters which can be located easily even from a distance. ➢ Collection and destruction of shed materials. ➢ Hand picking and destruction of grownup caterpillars. **Spray any one of the following insecticides** Chlorpyriphos 20 EC 3750 ml/ha Chlorantraniliprole 18.5% SC @150 ml/ha Flubendiamide 39.35% SC 100ml/ha
Blue butterflies *Lampides boeticus* *Euchrysops cnejus*	**Spray any one of the following insecticides** Chlorantraniliprole 18.5% SC 100ml/ha Flubendiamide 39.35% SC 100ml/ha Lufenuron 5.4% EC 600ml/ha Monocrotophos 36% SL 625 ml/ha

Spotted pod borer *Maruca vitrata*	Thiodicarb 75% WP 625g/ha

Pod bugs	Dimethoate 30% EC 500ml/ha Methyl demeton 25% EC 500ml/ha
Storage pests Bruchid- *Callosobruchus chinensis* *C. maculatus*	Dry the seeds adequately to reduce moisture level to 10 %. Use two-in-one or pitfall traps for monitoring the emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce. Mix Malathion 5% D 1 kg with 100kgs of seed Pack in polythene lined gunny bags for storage

B. Disease management

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g or carbendazim @ 2 g or thiram @ 4 g/kg of seed

Disease	Recommendations
Powdery mildew: *Erysiphe polygoni*	• Spray NSKE 5% or neem oil 3% twice at 10 days interval from initial disease symptom appearance • Spray 10% Eucalyptus leaf extract at initiation of the disease and 10 days later • Spray carbendazim @ 500 g or wettable sulphur 1500 g/ha or propiconazole 500 ml/ha at initiation of the disease and 10 days later
Rust: *Uromyces appendiculatus*	Spray mancozeb @ 1000 g or wettable sulphur 1500 g /ha at initiation of the disease and 10 days later
Leaf spot: *Cercospora canescens*	Spray carbendazim @ 500 g/ha or mancozeb @ 1000g /ha at initiation of the disease and 10 days later
Yellow mosaic (Geminivirus) **and** **Leaf crinkle** (Vector: *Bemisia tabaci*) **Leaf curl** (Tospovirus) (Vector: *Frankliniella schultzii,* *Thrips tabaci,* *Scirtothrips dorsalis*)	**Integrated disease management** • Growing resistant varieties such as VBN 4, VBN 6, VBN 7 and VBN8 • Seed treatment with imidacloprid 600FS @ 5 ml/kg of seeds • Installation of yellow sticky traps @ 12 numbers / ha • Rogue out the virus infected plants up to 45 days • Foliar spray of 10% notchi leaf extract at 30 DAS or neem formulation @ 3 ml/l • Spray methyl demeton 25 EC 500 ml/ha or dimethoate 30 EC 500 ml/ha or thiamethoxam 75WG @ 100 g/ha or imidacloprid 17.8 SL @ 250 ml/ha or thiamethoxam 75 WS 1 g /3 lit and repeat after 15 days, if necessary
Root rot: *Rhizoctonia bataticola* *(Macrophomina phaseolina)*	• Seed treatment with *Trichoderma asperellum* @ 4 g/kg or *Pseudomonas fluorescens* @ 10 g/kg • Basal soil application of zinc sulphate 25 kg/ha • Basal soil application of neem cake @ 150 kg/ha • Soil application *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand at 30 days after sowing • Spot drench with carbendazim @ 1 g/ l

Root rot - stem fly complex	Seed treatment with *Beauveria bassiana* + *Pseudomonas fluorescens* @ 5g each/kg of seed

RICE-FALLOWS

VARIETIES AND SEED RATE

VARIETIES	Quantity of seed required kg/ha Sole crop	Mixed crop
ADT 3, ADT 6, KKM 1, VBN 6 (Rice fallows)	25	..

1. **TIME OF SOWING**

   Third week of January –Second week of February

2. **SOWING OF SEEDS**

   a) For relay cropping broadcast the seeds in the standing crop 5 to 10 days before the harvest of the paddy crop uniformly under optimum soil moisture conditions so that the seeds should get embedded in the waxy mire.

   b) For combined harvesting areas, broadcast the seeds before harvesting the paddy crop with machinerie

3. **FOLIAR APPLICATION**

   a. Foliar Spray of NAA 40 mg/litre once at pre-flowering and another at 15 days thereafter

   b. Foliar spray of pulse wonder @ 5 kg/ha once at flowering to decreases flower shedding and improve yield.

   c. Foliar spary of salicylic acid 100 mg/litre once at prefloweing in another and 15 days there after to improve translocation efficiency and seed yield.

4. **HARVESTING**

   i) Picking the matured pods, drying and processing

   ii) Uprooting or cutting the whole plants, heaping, drying and processing

BLACKGRAM - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 5 m all around the field from the same and other varieties of the crop.

Pre-sowing seed treatment

- Remove all discoloured seeds and use only normal coloured seeds for seed purpose.
- If the presence of hard seed percentage exceeds more than 10 %, scarify the seeds

with commercial H_2SO_4 for 2 min.

- Harden blackgram seeds for garden and dry land ecosystem with 100 ppm $ZnSO_4$ for 3 h in 1/3 volume of solution and dry seeds under shade to the original seed moisture content (8 - 9 %)

Fertilizer

- NPK @ 25 : 50 : 25 kg + 5 kg of TNAU micro nutrient mixture / ha.

Foliar application

- Spray 2 % DAP at the time of first appearance of flowers and second spray 15 days after first spray for enhanced seed set.
- Spray NAA 40 ppm at first flowering and at fortnight interval to reduce the flower drop.
- Spray 0.1 % brassinoloid on 35^{th} and 45^{th} day after sowing.

Pre-harvest sanitation spray

- Spray Malathion 50 EC at 0.05 % three to five days before harvest to minimize the bruchid infestation in storage.

Harvest

- Harvest the pods 30 days after 50 per cent flowering. At this stage, the colour of majority of the pods (80 %) will be black. The pod moisture content will be around 17 - 18 %.
- Harvest the pods as pickings, if the flowering period is longer.
- Dry the pods to 13 to 15 per cent moisture content.

Threshing

- Thresh the pods either with pliable bamboo stick or pulse thresher.

Drying

- Dry the seeds to 8 - 9 per cent moisture content.

Seed grading

- Grade the seeds using BSS 7 x 7 wire mesh sieve
- Discard the discoloured and broken seeds for seed purpose.
- Avoid bruchid infected seeds for sowing.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 10 - 12 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 8 % in 700 gauge polythene bag for long term storage (more than15 months).

(ii) GREENGRAM (*Vigna radiata* L.)

Climate requirement

T_Max^oC	T_Min^oC	Optimum ^oC	Rainfall mm	Altitude m MSL
40	20	25 - 32	600 - 800	2000

Tropical and subtropical hot climate. The crop needs a well - distributed rainfall. Heavy rains at flowering are harmful, even moist winds at this stage interfere with fertilization. It can tolerate drought to a great extent. It is considered to be the hardiest pulse among all pulse crops. Day neutral plant.

CROP IMPROVEMENT
i. SEASON AND VARIETIES

District/season	Varieties
Adipattam (June - July) All districts except Kanyakumari and Nilgiris	CO(Gg) 7, VBN(Gg) 2, VBN(Gg) 3, CO 8, VBN 4
Puratasipattam (September - October) Kanchipuram, Tiruvallur, Dharmapuri, Vellore, Tiruvannamalai, Salem, Namakkal, Cuddalore, Villupuram, Thiruchirapalli, Perumbalur, Erode, Coimbatore, Madurai, Dindigul, Theni, Pudukottai, Pudukkottai, Sivagangi, Ramanthapuram, Virudhunagar, Thothukudi and Thirunelveli,	Co(Gg) 7, VBN(Gg) 2, VBN(Gg) 3, CO 8, VBN 4
Margazhi-Thai Pattam (December – January) All districts except Kanyakumari and Nilgiris	VBN(Gg)3, CO(Gg) 7, CO 8
Rice fallows (January - February) Thanjavur, Tiruvarur, Nagapattinam, Cuddalore,	ADT 3
Summer (February - March) Thanjavur, Tiruvarur, Nagapattinam, Cuddalore, Villupuram, Tiruchirapalli, Perambalur, Thiruvallur, Kanchipuram	VBN(Gg) 3, CO(Gg) 7, CO 8, VBN 4

II. DESCRIPTION OF GREENGRAM VARIETIES

Particulars	**Co (Gg) 7**	**CO 8**	**VBN(Gg) 2**	**VBN (Gg) 3**	**ADT 3**	**VBN 4**
Year of Release	2005	2013	2001	2009	1988	2019
Year of Notification	SO.1177(E)/25.08.05	SO.268(E)/28.01.2015	SO.1197(E)/14.11.2002	SO.2137(E)/31.08.2010	SO.793(E)/22.11.1991	SO.3220(E)/05.09.2019
Parentage	MGG336 x CoGG 902	COGG 923 x VC 8040A	VGG 4 x MH 309	CO 1 x Vellore local	H7016 x Rajendran G65	PDM 139 x BB 2664
Maturity duration (days)	60-65	55- 60	65-70	65-75	70-75	65-70

Grain yield (kg/ha)						
Rainfed (kg/ha)	980	-	750	775	500 (rice fallow)	936
Irrigated (kg/ha)	-	845	900	880	-	1251
Height (cm)	30 - 45	55-65	50-60	35-55	40-60	45-55
Pod Colour at maturity	Brown	Brown	Black	Brown	Black	
100 grain wt (g)	3.5 – 4.0	3.5-4	3.6-3.9	2.8-3.5	2.5-3.5	35-40
Special features	High protein content (25.2%), High seed weight and synchronized maturity	Suitable for single/mechanical harvest. Moderately resistant to MYMV and stem necrosis diseases. Moderately resistant to sucking pests like aphids and stem fly	Moderately resistant to Yellow Mosaic, Synchronize pod maturity	Moderately resistant to powdery mildew and Yellow mosaic Indeterminate flowering	Suitable only for Rice fallow and Margazhi pattam	Resident to urbean leaf crinkle virus, moderately resistant to yellow mosaic virus and powdery mildew diseases

I. SEED RATE

Particulars	Quantity of seed required kg/ha	
	Pure crop	Mixed crop
All varieties	20	10
Rice fallows - ADT 3	30	--

II. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

i. Prepare the land to get fine tilth and form beds and channels.

ii Amendments for soil surface crusting: To tide over the soil surface crusting apply lime at the rate of 2 t/ha along with FYM at 12.5 t/ha or composted coir pith at 12.5 t/ha to get an additional yield of about 15 - 20%.

2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram @ 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Bio control agents are compatible with biofertilizers. First treat the seeds with Biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

3. SEED TREATMENT WITH BIOFERTILIZER

a) Treat the seeds required for sowing 1 ha with 200g each of Rhizobial culture COG 15, Phosphobacteria and PGPR (*Pseudomonas* sp.) using rice gruel, shade dry it before sowing. (or) Treat one hectare of seeds with 25 g each of each of powder formulation of *Rhizobium* and AM fungi using binder (polymer), shade dry before sowing.

b) If seed treatment is not carried out, apply 2 kg each Rhizobial culture, Phosphobacteria and PGPR (*Pseudomonas* sp.) with 25 kg of FYM and 25 kg of sand, mix uniformly before sowing.

4. FERTILIZER APPLICATION

If soil test is not done ,

Apply fertilizers basally before sowing.

Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +20 kg S*/ha Irrigated : 25 kg N + 50 kg P_2O_5 + 25 kg K_2O + 40 kg S*/ha

*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a source of phosphorus

Soil test crop response based integrated plant nutrition system (STCR- IPNS recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Soil :	Red sandy loam (Irugur series)	FN = 25.07 T - 0.71 SN
Target	0.8– 0.9 t ha^{-1}	FP_2O_5 = 15.44 T - 5.48 SP
		FK_2O = 11.00 T - 0.19 SK

Initial soil test values (kg ha^{-1})			**Yield target –0. 8 t ha^{-1}**			**Yield target – 0.9 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	160	38**	26	18	38**	41	29
180	14	180	33	25*	14	38**	30	25
200	16	200	19	25*	13*	38**	25*	21
220	18	220	13*	25*	13*	29	25*	17
240	20	240	13*	25*	13*	15	25*	13

* Maintenance dose;** Maximum dose

Note: FN, FP_2O_5 and FK_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{1}.

c) Soil application of 25 kg $ZnSO_4$/ha or 12.5 kg $ZnSO_4$ as EFYM under irrigated condition

d) Soil application of TNAU micronutrient mixture @ 5 kg/ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).
   50 kg $FeSO_4$ as EFYM or 10 by Fe EDTA ha^{-1} for the deficient soils.

Multi-blooming technology for irrigated green gram in new delta region of Thanjavur

For higher yield and income, apply 25:50:25:20 kg NPKS/ha.+25 kg N/ha. in 3 equal splits on 30, 45 and 60 days after sowing + 2% DAP spray on 45 and 60 days after sowing.

Foliar spraying of 0.5% $ZnSO_4$ thrice, 1% $FeSO_4$ + 0.1% citric acid thrice can be followed when plants shows deficiency symptoms at 7-10 days intervals.

e) Foliar spray of 1% urea for yield improvement in green gram

For yield improvement through increasing the physiological, biochemical attributes, foliar spray of urea 1% on 30 and 45 days after sowing is recommended. For rice fallow pulses in Delta area, the present recommendation of foliar spray of 2% DAP may be continued.

Economizing the use of micronutrients through seed treatment for greengram

Seed coating with biofertilizers and micronutrients viz., Zn, Mo & Co @ 4,1,0.5 g/kg of seed is recommended.

5. SOWING

Dibble the seeds adopting a spacing of 30 x 10 cm. For bund crop dibble the seeds at 30 cm spacing.

6. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on third day. Irrigate at interval of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 2.0 per cent as foliar spray during vegetative stage if there is moisture stress.

7. FOLIAR APPLICATION

a. Foliar spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter to reduce flower shedding.
b. i) For rice fallow crops, foliar spray of TNAU Pulse wonder @ 5 kg/ha once at flowering or DAP 20 g/litre once at flowering and another at 15 days thereafter
   ii) For irrigated and rainfed crops foliar spray of TNAU Pulse wonder @ 5 kg/ha once at flowering or DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter.
c. Foliar spray of salicylic acid 100 mg/litre at preflowering and another at 15 days therafter to improve translocation efficiency and seed yield.

8. WEED MANAGEMENT

i) Pre emergence application of Pendimethalin @ 1.0 litres / ha under irrigated condition or PE application of Pendimethalin 0.75 litres per hectare under rainfed condition on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litres of water for spraying one ha. After this, one hand weeding on 30th days after sowing gives weed free environment throughout the crop period (or) EPOE application of quizalofop ethyl @ 50 g ai/ha^{-1} and imazethapyr @ 50 g ai ha^{-1} on 15 – 20 DAS.

ii) If herbicide is not applied give two hand weedings on 15 and 30 days after sowing.

iii) Apply PE Pendimethalin 30% EC + Imazethapyr 2% EC (Valor 32% EC; Readymix herbicide) @ 1.0 kg a.i. ha^{-1} at 3 DAS.

iv) For the irrigated blackgram PE isoproturon @ 0.5 kg ha^{-1} followed by one hand weeding on 30 DAS.

9. MULTI BLOOM TECHNOLOGY

A special technology being practiced in Pattukottai block of Tanjore district for blackgram and greengram. The soil is alluvial and rich in organic matter and nutrients. The crop is sown during early summer (Jan.-Feb.) as normal crop and fertilizer is applied as per the recommendation for irrigated crop. In addition to that, top dressing of Nitrogen is done with an extra dose of 25 to 30 kg through urea. Since pulses are indeterminate in growth habit and continue to produced new flushes, the top dressing will be done on 40-45 days after sowing. The crop complete its first flushes of matured pods during 60-65th day and put further second new flush within 20-25 days. Therefore two flushes of pods can be harvested at a time within the duration of 100 days.

CROP PROTECTION

A. Pest management

Economic threshold level for important pests

Pest	**ETL**
Aphids	20/2.5 cm shoot length
Pod borers	10% of affected pods
Spotted pod borer	3/plant
Stem fly	10%of affected plants
Tobacco cut worm	8 egg masses/100 m

Pests Management strategies

Stem fly *Ophiomyia phaseoli*	Treat seeds with dimethoate 30% EC 5 ml/kg of seed
Aphids *Aphis craccivora*	**Spray any one of the following** Methyl demeton 25% EC 500 ml/ha Dimethoate 30% EC 500 ml/ha
Whitefly *Bemisia tabaci*	**Spray any one of the following** Methyl demeton 25% EC 500 ml/ha

	Dimethoate 30% EC 500 ml/ha
Tobacco cut worm *Spodoptera litura*	➢ Use of light trap to monitor and kill the attracted adult moths. ➢ Set up the sex pheromone trap at 12/ha to monitor the activity of the pest and to synchronize the pesticide application, if need be, at the maximum activity stage. ➢ Growing castor along border and irrigation bunds. ➢ Removal and destruction of egg masses in castor and cotton crops. ➢ Removal and destruction of early stage larvae found in clusters which can be located easily even from a distance. ➢ Collection and destruction of shed materials. ➢ Hand picking and destruction of grownup caterpillars. **Spray any one of the following insecticides** Chlorpyriphos 20 EC 3750 ml/ha Chlorantraniliprole 18.5% SC @150 ml/ha Flubendiamide 39.35% SC 100ml/ha
Blue butterflies *Lampides boeticus* *Euchrysops cnejus*	**Spray any one of the following insecticides** Chlorantraniliprole 18.5% SC 100ml/ha Flubendiamide 39.35% SC 100ml/ha Lufenuron 5.4% EC 600ml/ha Monocrotophos 36% SL 625 ml/ha Thiodicarb 75% WP 625g/ha
Spotted pod borer *Maruca vitrata*	Monocrotophos 36% SL 437 ml/ha Chlorantraniliprole 18.5% SC 100ml/ha
Pod bugs	Dimethoate 30% EC 500ml/ha Methyl demeton 25% EC 500ml/ha
Storage pests Bruchid- *Callosobruchus chinensis* *C. maculatus*	Dry the seeds adequately to reduce moisture level to 10 %. Use pitfall traps or two in one model trap to monitor the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce. Mix Malathion 5 D 1 kg for every 100 kg seed Pack in polythene lined gunny bags for storage

Disease Management

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g or carbendazim @ 2 g or thiram @ 4 g/kg of seed

Disease	Recommendations
Powdery mildew: *Erysiphe polygoni*	• Spray NSKE 5% or neem oil 3% twice at 10 days interval from initial disease symptom appearance • Spray 10% Eucalyptus leaf extract at initiation of the disease and 10 days later • Spray carbendazim @ 500 g or wettable sulphur 1500 g/ha or propiconazole 500 ml/ha at initiation of the disease and 10 days later
Rust: *Uromyces appendiculatus*	Spray mancozeb @ 1000 g or wettable sulphur 1500 g /ha at initiation of the disease and 10 days later

Leaf spot: *Cercospora canescens*	Spray carbendazim @ 500 g/ha or mancozeb @ 1000g /ha at initiation of the disease and 10 days later
Yellow mosaic (Geminivirus) **and** **Leaf crinkle** (Vector: *Bemisia tabaci*) **Leaf curl** (Tospovirus) (Vector: *Frankliniella schultzii, Thrips tabaci, Scirtothrips dorsalis*)	**Integrated disease management** • Seed treatment with imidacloprid 600FS @ 5 ml/kg of seeds • Installation of yellow sticky traps @ 12 numbers / ha • Rogue out the virus infected plants up to 45 days • Foliar spray of 10% notchi leaf extract at 30 DAS or neem formulation @ 3 ml/l • Spray methyl demeton 25 EC 500 ml/ha or dimethoate 30 EC 500 ml/ha or thiamethoxam 75WG @ 100 g/ha or imidacloprid 17.8 SL @ 250 ml/ha or thiamethoxam 75 WS 1 g /3 lit and repeat after 15 days, if necessary
Root rot: *Rhizoctonia bataticola (Macrophomina phaseolina)*	• Seed treatment with *Trichoderma asperellum* @ 4 g/kg or *Pseudomonas fluorescens* @ 10 g/kg • Basal soil application of zinc sulphate 25 kg/ha • Basal soil application of neem cake @ 150 kg/ha • Soil application *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand at 30 days after sowing • Spot drench with carbendazim @ 1 g/ l
Root rot - stem fly complex	Seed treatment with *Beauveria bassiana* + *Pseudomonas fluorescens* @ 5g each/kg of seed

RICE-FALLOWS

VARIETIES AND SEED RATE

Varieties	Quantity of seed required kg/ha	
	Sole crop	Mixed crop
ADT 3	30	-

1. TIME OF SOWING

Third week of January –Second week of February

2. SOWING OF SEEDS

a) For relay cropping broadcast the seeds in the standing crop 5 to 10 days before the harvest of the paddy crop uniformly under optimum soil moisture conditions so that the seeds should get embedded in the waxy mire.

b) For combined harvesting areas, broadcast the seeds before harvesting the paddy crop with machineries

3. FOLIAR APPLICATION

a. Foliar spray of NAA 40 mg/litre once at pre-flowering and another at 15 days thereafter

b. Foliar spray of TNAU pulse wonder @ 5 kg/ha once at flowering or DAP 20 g/lt once at flowering and another at 15 days thereafter

c. Foliar spray of salicylic acid100 mg/litre once at preflowering and anothere at 15 days there after.

4. **HARVESTING**

i) Picking the matured pods, drying and processing

ii) Uprooting or cutting the whole plants, heaping ,drying and processing

SEED PRODUCTION
GREEN GRAM - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free from volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be of the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 5 m all around the field from the same and other varieties of the crop.

Pre-sowing seed treatment

- Remove all discoloured seeds and use only normal coloured seeds (olive green) for seed purpose.
- Avoid bruchid infested seeds for sowing.
- If the presence of hard seed percentage exceeds more than 10 %, scarify the seeds with commercial H_2SO_4 for 2 min.
- Harden the greengram seeds for garden and dry land ecosystem with 100 ppm $MnSO_4$ for 3 h at the ratio of 1:0.3 ratio and dry back to original seed moisture content (8 - 9 %) under shade.

Fertilizer

- NPK @ 25 : 50 : 25 kg + 5 kg TNAU micro nutrient mixture / ha.

Foliar application

- Spray 2 % DAP at the time of first appearance of flowers and second spray 15 days after first spray for enhanced seed set.
- Spray NAA 40 ppm at first flowering and at fortnight interval to reduce the flower drop.
- Spray 0.1 % brassinoloid on 35^{th} and 45^{th} day after sowing.

Pre-harvest sanitation spray

- Spray (0.05 %) Malathion 50 EC three to five days before harvest to minimize the bruchid infestation in storage.

Harvest

- Harvest the pods at 30 days after 50 % flowering when majority of the pods (80 %) are brown in colour.
- Harvest the pods as pickings, if the flowering period is longer.

- Dry the pods to 13 to 15 % moisture content.

Threshing

- Thresh the pods either with pliable bamboo stick or pulse thresher.

ying

- Dry the seeds to 8 - 9 % moisture content.

Seed grading

- Grade the seeds using BSS 7 x 7 wire mesh sieve.
- Discard the discoloured and broken seeds for sowing or storage.

Pre-storage seed treatment

- Treat the seeds with carbendazim 2 g / kg of seed.
- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 10 - 12 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 8 % in 700 gauge polythene bag for long term storage (more than15 months).

(iii) COWPEA (*Vigna unguiculata* (L.) Walp.aggreg.)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
35	15	20 - 30	400 - 600	32

Cowpea is called the "hungry - season crop" because it is the first crop to be harvested before the cereal crops. Cowpea is tolerant of shading and can be combined with tall cereal plants such as sorghum and maize. It is sensitive to waterlogging, though less than other legumes. High moisture may hinder cowpea crops in the sub - humid tropics due to the many diseases. Frost can damage the plant during flowering period.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES
Adipattam (June-August) For all districts except Kanyakumari and Nilgiris	Co(CP) 7
Purattasipattam (September - November) Vellore, Thiruvannamalai, Dharmapuri, Salem, Namakkal, Perembalur, Erode, Coimbatore, Madurai, Dindigul, Theni and Virudhunagar	Co(CP) 7, VBN 3
Margazhi - Thaipattam (December – February) Kanchipuram, Thiruvallur,Vellore, Thiruvannamalai, Dharmapui,Salem,Namakkal, Coimbatore, Erode, Madurai, Dindigul, Theni, Tiruchirappalli, Perambalur, Ariyalur, Karur, Pudukkottai, Tirunelveli and Thoothukudi	Co(CP) 7, VBN 3

I. Description of Cowpea varieties

Particulars	**Co (CP) 7**	**VBN 3**
Year of Release	2002	2017
Year of Notification	SO.1177(E)/25.08.2005	S.O. 6318(E) / 26.12.2018
Parentage	Gamma mutant of Co 4 (20 Kr)	TLS 38 x VCP 16-1
50%flowering(days)	40 – 45	50-55
Duration (days)	70 – 75	75-80
Grain yield(kg/ha)		
Rainfed	1000	1010
Irrigated	1600	-
Plant height (cm)	40 – 55	65 - 70
Stem, branches	Green with purple ring at fruiting nodes, 5 – 8 branches	Determinate plant type, synchronized maturity
Leaves	The terminal leaflet has sub hastate shape	The terminal leaflet has sub globose shape

Colour of pods	Green	Creamy white colour and glabrous pods
Dry	Light brown	Light brown
Colour of grain	Brownish white and square shape.	Light brown and kidney shape
100 grain wt (g)	12 – 14	12.5 – 13.5

III. SEED RATE

Seed rate (pure crop) : 25 kg/ha

CROP MANAGEMENT

IV. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

Prepare the land to fine tilth and form beds and channels.

2. SEED TREATMENT

Treat the seeds with Carbendazim or Thiram 2 g/kg of seed 24 hours before sowing (or) with talc formulation of *Trichoderma viride* @ 4g/kg of seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

3. SEED TREATMENT WITH BIOFERTILIZER

a) Fungicide-treated seeds, should be again treated with a bacterial culture. There should be an interval of atleast 24 hours between fungicidal and biofertilizer treatments.

b) The improved rhizobial strain COC 10 is more effective in increasing the yield. Treat the seeds with one packet (200 g/ha) of Rhizobial culture (COC 10) and one packet (200 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets (2 kg/ha) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

4. FERTILIZER APPLICATION

a) Apply fertilizers basally before sowing.
Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +10 kg S*/ha Irrigated : 25 kg N + 50 kg P_2O_5 + 25 kg K_2O + 20 kg S*/ha
*Note : Applied in the form of gypsum if Single Super Phosphate is not applied as a source of phosphorus

b) Soil application of 25 kg $ZnSO_4$/ha along with 50 kg FYM or sand under irrigated condition

c) Soil application of 10 kg borax, 0.25 kg Ammonium molybdate can be followed if the soil is deficient.

5. **SOWING**

Dibble the seeds adopting the following spacing.

Varieties	Spacing
VBN 3	30 cm X 15 cm
CO(CP) 7	45 cm x 15 cm

6. **WATER MANAGEMENT**

Irrigate immediately after sowing followed by life irrigation on third day. Irrigate at interval of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 2.0 per cent as foliar spray during vegetative stage if there is moisture stress.

7. **FOLIAR APPLICATION**

a. Foliar spray of DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter to enhance flower number and pod setting
b. Foliar spray of NAA 40 mg/litre once at flowering and another at 15 days thereafter to reduce flower drop
c. Foliar spray of salicylic acid 100 mg/litre once at flowering and another at 15 days ther after to improve seed yield.

8. **WEED MANAGEMENT**

i) Pre emergence application of Pendimethalin @ 0.75 litre / ha on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 lit of water for spraying one hectare followed by one hand weeding on 30 days after sowing gives weed free environment throughout the crop period.
ii) If herbicide is not applied, give two hand weeding on 15 and 30 days after sowing.

CROP PROTECTION

A. Pest management

Pests	**ETL**
Aphids	20nos. /2.5 cm shoot length
Spotted pod borer	3larvae /plant
Stem fly	10% of affected plants

Pests Management strategies

Stem fly *Ophiomyia phaseoli*	Seed treatment with dimethoate 30 EC 5 ml/kg of seed
Aphids *Aphis craccivora*	**Spray any one of the following** Methyl demeton 25 EC 500 ml/ha

Whitefly *Bemisia tabaci*	Dimethoate 30 EC 500 ml/ha
Blue butterflies *Lampides boeticus* *Euchrysops cnejus*	Chlorantraniliprole 18.5% SC 100ml/ha
Spotted pod borer *Maruca vitrata*	• Thiodicarb 75% WP 750g/ha
Storage pests Bruchid- *Callosobruchus chinensis* *C. maculatus*	• Dry the seeds adequately to reduce moisture level to 10 %. • Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce. • Mix Malathion 5 D 1 kg for every 100 kg seed • Pack in polythene lined gunny bags for storage
Pod bug	• Dimethoate 30% EC 500ml/ha • Methyl demeton 25% EC 500ml/ha

B. Disease Management

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g/kg or carbendazim @ 2 g/kg or thiram @ 4 g/kg of seeds

Disease	Recommendations
Rust: *Uromyces appendiculatus*	Two sprays of chlorothalonil 0.1% or one spray with 0.1% chlorothalonil followed by 3% neem oil after the appearance of disease
Root rot: *Macrophomina phaseolina* *(Rhizoctonia bataticola)*	• Soil application of *P. fluorescens* or *T. asperellum* @ 2.5 kg/ ha with 50 kg of well decomposed FYM or sand • Spot drenchwith carbendazim @ 1 g /l
Aphid borne mosaic: (Potyvirus) (Vector: *Aphis craccivora, A. fabae, A. gossypii* and *Myzus persicae)*	Roguing out the virus infected plants in the early stage of growth up to 30 days and spraying twice at fortnightly intervals with methyl demeton 25 EC @ 500 ml/ha or dimethoate 30 EC 500 @ ml/ha or imidacloprid 17.8 SL @ 250ml/ha

SEED PRODUCTION
COWPEA - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 5 m all around the field from the same and other varieties of cowpea.

Season

- September - October and June - July.

Intercultural operation

- Clip the tendrils for promotion of flower production.
- Up root and destroy the plants exhibiting severe symptoms of mosaic in the early stages of growth.
- Spray NAA 40 ppm (40 mg in 1 litre) at first flowering to reduce flower drop.
- Spray 2 % DAP at flower initiation and at peak flowering to promote pod setting.

Harvesting

- Seeds attain physiological maturity 27 - 30 days after anthesis. At this stage the seed moisture content will be around 18 per cent.
- Harvest the pods as they turn light straw in colour and the seeds turn brown or mottled.
- Harvest the pods as picking (4 - 5 nos.) at 10 days interval.
- Shade dry the pods for 1 - 2 days and then sundry until they become brittle.
- Beat the pods with pliable bamboo stick or pulse thresher by adjusting the cylinder speed to avoid splitting and cracking of seeds.

Seed grading

- Grade the seeds with 10 / 64" or 12 / 64" round perforated sieves.

Drying

- Remove the broken and immature seeds.
- Dry the seeds to 8 - 10 % moisture content

Pre- storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed along with carbaryl 200 mg / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 10 - 12 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12- 15 months).
- Store the seeds with a seed moisture content less than 8 % in 700 gauge polythene bag for long term storage (more than15 months).

(iv) HORSEGRAM (*Macrotyloma uniflorum*)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
42	20	25 - 32	200 - 700	800

Tropical crops. Extremely drought - resistant crop. Moderately warm, dry climatic conditions are suitable for its optimum growth. It does not grow well on higher altitudes because of cool and wet climate.

I. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES
November (Winter season) (Rainfed) All districts except The Nilgiris and Kanyakumari	Paiyur 2

II. Description of Horsegram varieties

Particulars	Paiyur 2
Year of Release	1998
Year of Notification	SO.425(E)/08.06.1999
Parentage	Gamma irradiation of Co 1
50% flowering(days)	45-50
Maturity Duration (days)	100-105
Grain yield(kg/ha)	
Rainfed	870
Plant height (cm)	40-45
Branches	2 -3 branches
Colour of grain	Pale brown
100 grain wt (g)	3.5

III. SEED RATE

For a pure crop 20 kg/ha is needed.

MANAGEMENT OF FIELD OPERATIONS

1. **FIELD PREPARATION**
Prepare the land to fine tilth.

2. **SEED TREATMENT WITH FUNGICIDES**
Treat the seeds with any one of the following fungicides. Carbendazim or Thiram at 2 g/kg seed.

3. **FERTILIZER RECOMMENDATION**
Apply basally 12.5 t/ha FYM/Compost, 12.5 kg/ha nitrogen, 25 kg/ha phosphorus , 12.5 kg/ha potassium if soil is deficient in NPK status. Application of 25 kg $ZnSO_4$, 25 kg $FeSO_4$ + FYM can be followed if the soil is deficient in Zn and Fe. `

4. **SEED TREATMENT WITH BIOFERTILIZER**
Treat the seeds with one packet (200 g/ha) of Rhizobial culture and one packet (200 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

4. **SOWING**
Dibble the seeds with a spacing of 30 x 10 cm.

5. **WEED MANAGEMENT**
Give one weeding and hoeing on 25-30 days after sowing

6. **HARVESTING**
Harvest the matured whole plant, thresh and extract seeds

CROP PROTECTION

Seed treatment: Treat the seeds with carbendazim @ 2 g/kg or thiram @ 4 g/kg

HORSE GRAM - SEED PRODUCTION
VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 10 m all around the field from the same and other varieties of horsegram.

Season

- October - November.

Pre-sowing seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Priming the seeds with 100 ppm $ZnSO_4$ @ 1:1 seed to solution ratio for 3 hours and drying back to original moisture content.

Crop management

- Foliar spray of 0.5 % $ZnSO_4$ at 50 % flowering for enhancing the productivity
- Spray magnesium chloride against any chlorotic symptom @ 6 g / litre with a power sprayer for 2 - 3 times at 5 days interval.

Intercultural operation

- Clip the tendrils to promote flower production.

Harvesting

- Harvest the pods at once when 75 - 80 % of the pods turned into yellowish brown in colour.
- Timely harvest is essential; care to be taken not to expose the pods to rain or very moist weather which may change the seed coat colour from light brown to dark brown or light black.

Seed grading

- Grade the seeds with round perforated metal sieves having 8 / 64" diameter.
- Remove the discoloured seeds.

Storage

- Store the seeds with a seed moisture content of 8 - 9 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 7 - 8 % in poly lined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 7 % in 700 gauge polythene bag for long term storage (more than15 months).

(v) BENGALGRAM (*Cicer arietinum L.*)

CLIMATE REQUIRMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
35 - 45	6 - 8	20 - 25	500 - 800	2500

Tropical and subtropical winter season crop. The field should have loose tilth and good drainage. Long day plant. Severe cold and frost at the time of flowering causes detrimental effect to gram seed development.

CROP IMPROVEMENT

i. SEASON AND VARIETIES

District /Season	Varieties
November (Winter season) Rainfed Vellore, Tiruvannamalai, Salem, Namakkal, Tiruchirapalli, Perambalur, Karur, Dharmapuri, Pudukottai, Erode, Coimbatore, Madurai, Dindigul, Theni, Virudhunagar, Ramanathapuram, Sivagangai, Tirunelveli, Thoothukudi	CO 4

II. Description of chickpea variety

Particulars	**CO 4**
Year of Release	1998
Year of Notification	SO. 425 (E) / 08.06.1999
Parentage	Cross derivative of ICC 42 x ICC 12237
50% flowering	40
Duration (days)	85
Grain yield (kg/ha)	
Rainfed	1150
Height (cm)	35-40
Branches	3-5
Flower colour	Light pink & veined
Colour of grain	brown
100 grain wt (g)	30-32

(iii) SEED RATE

- CO 3 - 90 kg/ha.
- CO 4 - 75 kg/ha.

CROP MANAGEMENT

ii. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

Prepare the land to fine tilth and apply 12.5 t FYM/ha

2. SEED TREATMENT

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed. Biocontrol agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible. The above seed treatment will protect the seedlings from seed borne pathogens in the early stages.

3. SEED TREATMENT WITH BIOFERTILIZER

Treat the seeds with one packet (200 g/ha) of Rhizobial culture and one packet (200 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10packets of Rhizobium (2 kg/ha) and 10 packets(2 kg) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the biofertilizer treated seeds in shade for 15 minutes before sowing.

4. FERTILIZER APPLICATION

a) Apply fertilizers basally before sowing.
Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +10 kg S*/ha Irrigated : 25 kg N + 5 kg P_2O_5 + 25 kg K_2O + 20 kg S*/ha
*Note : Applied in the form of gypsum, if Single Super Phospate is not applied as a source of phosphorus

5. SOWING

Dibble the seeds by adopting a spacing of 30 cm x 10 cm.

6. WEED MANAGEMENT

i) Pre emergence application of Pendimethalin @ 0.75 litres / ha on 3rd day after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litres of water for spraying one ha followed by one hand weeding on 25 - 30 days after sowing.
ii) If herbicide is not applied give two hand weedings on 15th and 30th day after sowing.

7. INTERCROPPING IN BENGALGRAM

Bengalgram in paired row planting with one or two rows of Coriander as intercrop would give the highest return. Wheat can also be intercropped in deep black cotton soil in Coimbatore, Erode, Salem, Namakkal and Dharmapuri districts.

CROP PROTECTION

A. Pest management

Economic threshold level for important pests

Pest	ETL
Gram caterpillar	2 early instar larvae/plant 5-8 eggs/plant
Aphids	20/2.5 cm shoot length

Pest Management strategies

Aphid *Aphis craccivora*	**Spray any one of the following :** Methyl demeton 25 EC 500 ml/ha Dimethoate 30 EC 500 ml/ha
Gram caterpillar *Helicoverpa armigera*	• Pheromone traps for *Helicoverpa armigera* 12/ha • Bird perches 50/ha • Mechanical collection of grown up larva and blister beetle • Ha NPV 3 x10^{12} POB/ha in 0.1% teepol • *Bacillus thuringiensis* var *kurstaki* 5%WP 1000-1250 g/ha (Note : Insecticide / Ha NPV spray should be made when the larvae are upto third instar) **Apply any one of the following insecticides:** Azadirachtin 0.03 % WSP 2.5kg/ha Benfuracarb 40% EC 2.5l/ha Chlorantraniliprole 18.5% SC 150ml/ha Chlorpyriphos 20 EC 1250 ml / ha Emamectin benzoate 5% SG 220 g/ha Ethion 50% EC 1.0 l/ha Flubendiamide 39.35 % SC 100ml / ha Indoxacarb 14.5% SC 350 ml/ha Indoxacarb 15.8% SC 333 ml/ha Lufenuron 5.4% EC 600ml/ha Methomyl 40%SP 750g/ha Monocrotophos 36%SL 625-1250ml/ha Neem oil 2% Quinalphos 1.5%DP 23kg/ha Quinalphos 25 %EC 1400ml/ha Spinosad 45%SC 125 ml/ha • Thiodicarb 75 WP 625g / ha
Storage pests	• Dry the seeds adequately to reduce moisture level to 10 %. • Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce. • Mix Malathion 5 D 1 kg for every 100 kg of seed • Pack in polythene lined gunny bags for storage

Disease Management

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g/kg or carbendazim @ 2 g/kg or thiram @ 4 g/kg of seeds

Disease	Recommendations
Wilt: *Fusarium oxysporum* f. sp. *ciceri*	Soil application with P. *fluorescens* @ 2.5 kg/ha with 50 kg of well decomposed FYM or sand
Root rot: *Macrophomina phaseolina (Rhizoctonia bataticola)*	• Soil application of *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand • Spot drench with carbendazim @ 1 g/l

BENGAL GRAM - SEED PRODUCTION

Land requirements

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 5 m all around the field from the same and other varieties of bengal gram.

Pre-sowing treatment

- Soak the seeds in 1 % KH_2PO_4 for 3 h in 1/3rd volume of solution and shade dry the seeds to bring back to original seed moisture content.
- Avoid bruchid infested seed for seed purpose.

Harvesting

- Harvest the crop at once when 70 - 80 % of pods are creamy yellow in colour.

Seed grading

- Grade the seeds using 13 / 64” or 18 / 64" sieves depending on the variety.
- Dry the seeds to 8 - 10 % moisture content

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in a ratio of 5:4:1@ 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 9 - 10 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 8 % in 700 gauge polythene bag for long term storage (more than 15 months).

(VII) GARDEN LAB LAB (AVARAI)

(*Lab lab purpureus* (L.) var. *typicus*.)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
42	14	22–28	650 - 3000	2000 - 2400

Tropical and sub tropical crop. Lablab is a summer - growing annual or occasionally short - lived perennial forage legume. Lablab tolerates some flooding but does not withstand poor drainage or prolonged waterlogging. Lablab does better in full sunlight.

CROP IMPROVEMENT

i. SEASON AND VARIETIES

District/season	Varieties
Adipattam (July - Aug) Kanjipuram,Tiruvallur,Dharmapuri,Coimbatore,Madurai,Dindigul,Theni,Vellore,Tiruvannamali,Ramanathapuram,Virudhunagar,Sivagangai, Tirunelveli, Thoothukudi, Salem, Namakkal, Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur,Karur,Pudukkottai,Kanyakumari Erode	CO (Gb) 14
Puratasipattam (September - November) Kanchipuram, Tiruvallur,Tiruchirapalli,Perambalur, Karur,Vellore,Tiruvannamalai, Cuddalore,Villupuram, Dharmapuri, Salem, Namakkal, Pudukkottai, Erode,Coimbatore, Madurai,Dindigul, Theni, Ramanathapuram, Sivagangai, Virudhunagar, Tirunelveli, Thoothukudi, Thanjavur, Tiruvarur, Nagapattinam,	CO (Gb) 14 CO (Gb) 14 CO (Gb) 14
Summer (April) Kanjipuram,Tiruvallur,Vellore,Tiruvannamali,Cuddalore,Villuppuram Dharmapuri, Salem, Namakkal, Thanjavur, Tiruvarur, Nagapattinam, Kanyakumari,Pudukkottai, Erode,Coimbatore,Madurai,Dindigul,Theni,Ramanathapuram, Virudhunagar, Tirunelveli, Thoothukudi, Sivagangai,	CO (Gb) 14 CO (Gb) 14 CO (Gb) 14

II.DESCRIPTION OF AVARAI VARIETY

PARTICULARS	CO (Gb)14
Parentage	Cross derivative of CO 9 X CO 4
Year of release	2007
1 st flowering(days)	35-40
Duration (days)	80-85 days(seed to seed) 70-75 days(vegetable type)
Grain yield(kg/ha)	

Irrigated (kg/ha)	7584 Green pod
Habit	Dwarf ,bushy without tendrils
Height (cm)	56-62
Colour of Flowers	white
Colour of pod	Green
Shape of pod	flat
Colour of grain	Reddish brown
100 seed weight (g)	34-36

I. SEED RATE

Particulars	Quantity of seed required kg/ha Sole crop	Mixed Crop
CO (Gb) 14	25	-

CROP MANAGEMENT

II. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

Prepare the land to fine tilth. Form beds and channels for bushy types.

2. SEED TREATMENT WITH FUNGICIDES

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens*@ 10 g/kg seed. Biocontrol agents are compatible with biofertilizers. First treat the seeds with biocontrol agents and then with Rhizobium. Fungicides and biocontrol agents are incompatible.

3. SEED TREATMENT WITH BACTERIAL CULTURE

Fungicide treated seeds should be again treated with bacterial culture. There should be an interval of atleast 24 hours between fungicidal and bacterial culture treatments. Three packets of bacterial culture are sufficient for treating seeds required for one ha. The bacterial culture slurry may be prepared with rice kanji. Dry the inoculated seeds in shade for 15 minutes before sowing.

4. FERTILIZER APPLICATION

(a) Apply fertilizers basally before sowing.

Rainfed : 12.5 kg N + 25 kg P_2O_5 + 12.5 kg K_2O +10 kg S*/ha Irrigated : 25 kg N + 50 kg P_2O_5 + 25 kg K_2O + 20 kg S*/ha

*Note : Applied in the form of gypsum if Single Super Phospate is not applied as a sourceof phosphorus

(b) Soil application of 25 kg $ZnSO_4$/ha 10 kg borax, 5 kg $CuSO_4$ under irrigated condition if the soil is deficient in respective nutrients.

5. SOWING

Dibble the seeds adopting the following spacing. Varieties

CO (Gb) 1 : 45 cm X 30 cm

6. WEED MANAGEMENT

Pre emergence application of Pendimethalin @ 0.75 litres/ha on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 litres of water for spraying one ha. After this, one hand weeding on 40-45 days after sowing gives weed free environment throughout the crop period.

If herbicide is applied give two hand weedings on 25 and 45days after sowing.

7. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on third day. Irrigate atinterval of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at the rate of 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

8. PRUNING TECHNIQUE

A spacing of about 10 feet between lines and four feet between plants are adopted. Pits are dug and two to three seeds are sown in the middle of the pit. One healthy seedling is allowed to grow and the rest removed. The vine is propped with a stick. When the vine reaches the pandal, the terminal bud is nipped. Allow the branches to trail over the pandal. Each branch may be pruned at three feet length so that the pandal is covered with vines. Branches arising on the main vine below the pandal are removed. When flowering starts, prune the tip of the branches bearing inflorescence having three nodes from the productive axil. Continue this procedure throughout the reproductive phase.

9. HARVESTING

Pick the pods when they are completely dry. Thresh the pods and clean the beans. Pick the tender pods once in a week for vegetable purpose.

CROP PROTECTION

A. Pest management

Economic threshold level for important pests

Pest	**ETL**
Aphids	20 numbers per 2.5 cm shoot length
Spotted pod borer	3 larvae per plant
Gram caterpillar	10% of affected pods

Pest management strategies

Pests	**Management strategies**
Aphid *Aphis craccivora*	**Spray any one of the following :** Methyl demeton 25 EC 500 ml/ha Dimethoate 30 EC 500 ml/ha
Spotted pod borer *Maruca vitrata*	Thiodicarb 75% WP 750g/ha

Gram caterpillar *Helicoverpa armigera*	• Pheromone traps for *Helicoverpa armigera* 12/ha • Bird perches 50/ha • Mechanical collection of grown up larva and blister beetle • Ha NPV 3 x10^{12} POB/ha in 0.1% teepol • *Bacillus thuringiensis* var *kurstaki* 5%WP 1000-1250 g/ha (Note : Insecticide / Ha NPV spray should be made when the larvae are upto third instar) **Apply any one of the following insecticides:** Azadirachtin 0.03 % WSP 2.5kg/ha Benfuracarb 40% EC 2.5l/ha Chlorantraniliprole 18.5% SC 150ml/ha Chlorpyriphos 20 EC 1250 ml / ha Emamectin benzoate 5% SG 220 g/ha Ethion 50% EC 1.0 l/ha Flubendiamide 39.35 % SC 100ml / ha Indoxacarb 14.5% SC 350 ml/ha Indoxacarb 15.8% SC 333 ml/ha Lufenuron 5.4% EC 600ml/ha Methomyl 40%SP 750g/ha Monocrotophos 36%SL 625-1250ml/ha Neem oil 2% Quinalphos 1.5%DP 23kg/ha Quinalphos 25 %EC 1400ml/ha Spinosad 45%SC 125 ml/ha Thiodicarb 75 WP 625g / ha
Pod bug *Riptortus pedestris* *Clavigralla gibbosa*	• Dimethoate 30% EC 500ml/ha • Methyl demeton 25% EC 500ml/ha
Storage pests Bruchid - *Callosobruchus chinensis*	Dry the seeds adequately to reduce moisture level to 10%. Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce. Mix Malathion 5 D 1 kg for every 100 kg seed Pack in polythene lined gunny bags for storage

DISEASE MANAGEMENT

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g/kg or carbendazim @ 2 g/kg or thiram @ 4 g/kg of seeds

Disease	Recommendations
Anthracnose and die-back: *Colletotrichum lindemuthianum*	Spray mancozeb @ 1000g or carbendazim @ 250 g/ha soon after the appearance of the disease and if necessary, spray once again a fortnight later

GARDEN LAB LAB (AVARAI) - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be of the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified seed production leave a distance of 5 m all around the field from the same and other varieties of the crop.

Pre-harvest sanitation spray

- Spray 0.07 % malathion before harvesting the pods to avoid bruchid infestation.

Harvest

- Harvest the pods when they turn straw yellow in colour.
- Discard the terminal pods for seed purpose as they contain immature seeds.
- Dry the pods to 15 - 18 % moisture content.

Drying

- Dry the seeds to 8 - 10 % moisture content.

Seed grading

- Grade the seeds using 18 / 64" round perforated sieves.
- Remove the broken and immature seeds.
- Dry the seeds to 7 to 8 per cent moisture content.

Pre-storage seed treatment

- Treat the seeds with carbendazim at 2 g / kg of seed along with carbaryl at 200 mg / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 9 - 10% in gunny or cloth bags for short term storage (8-9 months).
- Store the seeds with a seed moisture content of 8 - 9 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 8 % in 700 gauge polythene bag for long term storage (more than15 months).

(vi) FIELD LAB-LAB (MOCHAI)

(*Lab lab purpureus* (L.) var. *ignosus*)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
35	4 - 6	18 - 30	800 - 1000	1800 - 3000

Tropical and sub tropical crop. Hot weather and drought stress are damaging to peas during the flowering period. Field peas can be grown as a winter crop in warm and temperate areas because pea seedlings have considerable frost resistance. High humidity is harmful to pea crop due to incidence of disease. Short day plant.

CROP IMPROVEMENT

District /Season	Varieties
All districts except Nilgiris All throughout the year	CO 2

II. Description of mochai variety

PARTICULARS	Co 2
Year of Release	1984
Year of Notification	SO.596(E)/13.08.1984
Parentage	Derivative of Co 8 X Co 1
50% flowering(days)	35-45
Duration (days)	105
Grain yield(kg/ha)	
Rainfed	900
Irrigated	1400
Habit	Erect and bushy determinate photo insensitive
Hight (cms)	60
Colour of Flowers	Purple
Colour of pod	Green
Shape of pod	flat
Colour of grain	Black
100 seed weight (g)	20.0

I. SEED RATE

Particulars	Quantity of seed required kg/ha	
	Sole crop	Mixed crop
CO 1	20	10.0
CO 2	25	12.5

CROP MANAGEMENT
II. MANAGEMENT OF FIELD OPERATIONS

1. FIELD PREPARATION

Prepare the land to fine tilth.

2. SEED TREATMENT WITH FUNGICIDES

Treat the seeds with Carbendezim (or) Thiram @ 2g/kg of seed 24hrs before sowing (or) with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible

3. SEED TREATMENT WITH BACTERIAL CULTURE

Fungicide treated seeds should be again treated with bacterial culture. There should be an interval of atleast 24 hours between fungicidal and bacterial culture treatments. Three packets of bacterial culture are sufficient for treating seeds required for one hectare. The bacterial culture may be prepared with rice kanji. Dry the inoculated seeds in shade for 15 minutes, before sowing.

4. FERTILIZER APPLICATION

Apply 20 kg N and 80 kg P_2O_5 and 40 kg K_2O per ha 40 kg of S as gypsum (220 kg/ha)/ ha as basal dressing. Soil application of 25 kg $ZnSO_4$/ha,10 kg borax, 25 kg $FeSO_4$ + FYM under irrigated condition if the soil is deficient in respective nutrients.

5. FOLIAR APPLICATION

i. Foliar spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter to reduce flower drop and enhance seed set.

ii. Foliar spray of DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter to enhance flower number and pod set

6. SOWING

Dibble the seeds, adopting the following spacing. Strain Sole crop Mixed crop

CO 1	90 cm x 30 cm	200 cm x 30 cm
CO 2	45 cm x 15 cm	200 cm x 15 cm

7. WEED MANAGEMENT

i) Pre emergence application of Pendimethalin @ 2 litres/ha on 3 days after sowing using Backpack/ Knapsack/Rocker sprayer fitted with flat fan nozzle using 500 l of water for spraying one ha. After this, one hand weeding on 40-45 days after sowing gives weed free environment throughout the crop period.

ii) If herbicides are not applied, give two hand weedings on 25^{th} and 45^{th} days after sowing.

8. WATER MANAGEMENT

Irrigate immediately after sowing, followed by life irrigation on third day. Irrigate at interval of 7 to 10 days depending upon soil and climatic conditions. Flowering and pod formation stages are critical periods when irrigation is a must. Avoid water stagnation at all stages. Apply KCl at 0.5 per cent as foliar spray during vegetative stage if there is moisture stress.

9. HARVESTING

Dry pods may be collected for grain purposes. Green mature pods may be collected for vegetable purpose.

CROP PROTECTION

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescens* @ 10 g/kg or carbendazim @ 2 g/kg or thiram @ 4 g/kg seeds

Disease	Recommendations
Anthracnose and die-back: *Colletotrichum lindemuthianum*	Spray mancozeb @ 1000 g or carbendazim @ 250 g/ha soon after the appearance of the disease and if necessary, spray once again a fortnight later

(ix) SOYABEAN (*Glycine max* (L.) Merr.)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
40	10	25 - 32	600 - 750	2000

Tropical and subtropical warm and moist climate. Short day plant. It can withstand short periods of waterlogging and short drought.

CROP IMPROVEMENT

1. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES
Adipattam (June - July) **Purattasipattam** (Sep. - Oct.) **Masipattam** (February - March) **Rice fallows**	Co(Soy)3

II. Description of soybean varieties

Particulars	Co (Soy) 3
Year of Release	2005
Year of Notification	SO.599(E)/25.04.2006
Parentage	Cross derivative of UGM 69 x JS 335
50% flowering	39 – 41 days
Duration (days)	90-100
Grain yield (Kg/ha)	
Rainfed	-
Irrigated	1700
Height (cm)	53.5
Branches	5 - 6
Flower colour	Pink
Colour of grain	Creamy yellow with brown hilum
100 seed weight (g)	10.95 – 11.75

2. SEED TREATMENT WITH FUNGICIDES

a) Treat the seeds with Carbendezim or Thiram @ 2g/kg of seed 24hrs before sowing or with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

b) Coat the seeds with $ZnSO_4$ @ 300 mg/kg using 10% maida solution as adhesive (250 ml/ kg) or gruel and arappu leaf powder (250 g/kg) as carrier to increase the field stand.

3. SEED TREATMENT WITH BIOFERTILIZER

a) Treat the seeds atleast 24 hours before sowing.
Treat the seeds with 3 packets (600 g/ha) of Rhizobial culture (COS-1) and 3 packets (600 g/ha) of Phosphobacteria developed at TNAU using rice kanji as binder. If the seed treatment is not carried out apply 10 packets of Rhizobium (2000 g/ha) and 10 packets (2000 g) of Phosphobacteria with 25 kg of FYM and 25 kg of soil before sowing. Dry the bacterial culture treated seeds in shade for 15 minutes before sowing.

4. FERTILIZER APPLICATION

i) Apply 20 kg N and 80 kg P_2O_5 and 40 kg K_2O per ha 40 kg of S as gypsum (220 kg/ha) as basal dressing. Soil application of 25 kg $ZnSO_4$, 25 kg $MnSO_4$/ha under irrigated condition if the soil is deficient.

ii) Foliar spray of NAA 40 mg/litre and Salicylic acid 100 mg/litre once at pre-flowering and another at 15 days thereafter

iii). Foliar spray of DAP 20 g/litre or urea 20 g/litre once at flowering and another at 15 days thereafter

iv). Foliar spraying of 1% $FeSO_4$ + 0.1% citric acid thrice at 7-10 days interval.

5. SOWING

Dibble the seeds at a depth of 2 - 3 cm adopting a spacing of 30 x 5 cm. In Erode district, Soybean + Castor (60 cm apart) cropping system gives high net return.

6. WATER MANAGEMENT

Irrigate immediately after sowing. Give life irrigation on 3rd day. Further irrigation at intervals of 7 - 10 and 10 - 15 days during summer and winter season respectively to be given depending on soil and weather conditions. Soyabean is very sensitive to excess moisture and the crop is affected, if water stagnates in the fields. The crop should not suffer due to water stress from flowering to maturity. To alleviate moisture stress spray of either Kaolin 3% or liquid paraffin at 1% on the foliage. In Erode district cultivate Soybean + Castor with irrigation at 0.60 IW/CPE ratio (once in 10 to 12 days) is recommended.

7. WEED MANAGEMENT

i) Pendimethalin 1.0 litre /ha after sowing followed by one hand weeding on 30 days after sowing.

ii) If herbicide spray is not given two hand weedings on 20 and 35 days after sowing may be given.

iii) Early Post emergence application of Imazythypur @ 50 g / ha may be applied as post emergence on 20 DAS with one hand weeding on 30 days after sowing.

8 HARVESTING

Yellowing of leaves and shedding, indicate the maturity of the crop. Cut the entire plant when most of the pods have turned yellow, drying and processing.

SOYABEAN IN RICE FALLOWS

Soyabean can be sown in rice fallows from middle of January to middle of March. Seeds can be dibbled at 75 kg/ha.

SPECIAL SITUATIONS

1. Optimum time of sowing Soyabean CO 1 - 2nd fortnight of June in Kharif
2. Intercropping of Soyabean CO 2 in Sugarcane is recommended for North Western Zone.
3. Intercropping of Soyabean in coconut gardens of more than 10 years is recommended.
4. Vermipelleting (50 g/kg) and adopting spacing of 30 x 10 cm and two foliar sprays of 2% DAP during flowering is recommended to achieve higher yield.

RAINFED SOYABEAN

i. VARIETIES

CO 1

ii. SEASON

The crop can be grown in South-West and North-East monsoon seasons. The middle of July is the optimum time of sowing for rainfed Soyabean in North Western Zone.

3. SEED TREATMENT WITH THE FUNGICIDES AND BIOFERTILIZERS

a) Treat the seeds with Carbendezim or Thiram @ 2g/kg of seed 24hrs before sowing or with talc formulation of *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.

- Biocontrol agents are compatible with biofertilizers.
- First treat the seeds with biocontrol agents and then with Rhizobium.
- Fungicides and biocontrol agents are incompatible.

b) Treat the seeds required for ha. with three pockets of Rhizobium and 3 packets of Phosphobacteria

4. FERTILIZER APPLICATION

i) Apply NPK as per soil test recommendation as far as possible. If soil test recommendation is not available adopt blanket recommendation of 20:40:20:20 NPKS kg/ha, if adequate moisture is available.

ii) Apply entire dose of N, P, K and S as basal.

5. SPACING

Adopt a spacing of 30 cm between rows and 5 cm between plants in the row.

6. SOWING

Dibble or drill the seeds.

7. WEED MANAGEMENT

i) If sufficient moisture is available, Pendimethalin 1.0 litre/ha after sowing followed by one hand weeding on 30 days after sowing.

ii) If herbicide spray is not given, two hand weeding on 20 and 35th day after sowing.

iii) Early Post emergence application of Imazythypur @ 40 g ai/ha applied as per amergence on 20 days after sowing with one hand weeding on 30 DAS.

CROP PROTECTION

Spodoptera, Helicoverpa, *Spilosoma*, Semilooper, Leaf miner, Stem fly	*Bacillus thuringiensis* var. *Kurstaki*, Bio-tech. International @ 500-750 g/ha Chlorantraniliprole 18.5% SC @150 ml/ha Ethion 50% EC @1500ml/ha Flubendiamide 39.35% SC 150ml/ha Indoxacarb 15.8%EC 330ml/ha Profenophos 50%EC 1.0 l/ha Spinetorum 11.7% SC 450ml/ha
Girdle beetle	Profenophos 50%EC 1.0 l/ha Thiacloprid 21.7% SC750ml/ha
Leaf weevil	Malathion 50% EC @1500ml/ha Quinalphos 1.5DP 16kg/ha Quinalphos 25%EC 1.0 l/ha

Seed treatment: Treat the seeds with *T. asperellum* @ 4 g or *P. fluorescen s*@ 10 g/kg or carbendazim @ 2 g/kg or thiram @ 4 g/kg of seeds

Disease	Recommendations
Rust: *Phakopspora pachyrhizi*	Spray triadimefon @ 0.1 % or propiconazole @ 0.1% or hexaconazole @ 0.1% at flowering stage or at the onset of disease
Virus diseases **Yellow mosaic** (Gemini virus) (Vector –*Bemisia tabaci*) **Bud blight** (Ilarvirus) (Vector- *Thrips palmi*)	• Rogue out the virus infected plants up to 30 days • Two sprays with thiamethoxam 25 WG @ 100 g/ha or methyl demeton @ 800 ml/ha or imidacloprid 17.8 SL @ 250 ml/ha at 30 and 45 days after sowing to control the vector

SOYABEAN - SEED PRODUCTION
VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be of the same variety or other varieties of the same crop. It can be of the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 3 m all around the field from the same and other varieties of the crop.

Harvest

- Harvest the pods as they turn yellow in colour.

Threshing

- Thresh the pods either manually or mechanically using pliable bamboo sticks.

Seed grading

- Grade the seeds using 14 / 64” or 12 / 64” sieves based on the varieties.

Drying

- Dry the seeds to 7- 8 % moisture content.

Pre-storage seed treatment

- Treat with carbendazim @ 2 g / kg of seed along with carbaryl @ 200 mg / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds with a seed moisture content of 10 - 12 % in gunny or cloth bags for short term storage (8 - 9 months).
- Store the seeds with a seed moisture content of 8 - 10 % in polylined gunny bag for medium term storage (12 - 15 months).
- Store the seeds with a seed moisture content less than 7 % in 700 gauge polythene bag for long term storage (more than15 months).

(x) SWORD BEAN (*Canavalia gladiata L.*)

CLIMATE RQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
38	10	15 - 30	700 - 4200	1500

Tropical and subtropical warm and moist climate. It is widely cultivated in the humid tropics.tolerates salinity and waterlogging. This crop can grow in light shade under trees to serve as a nitrogen - fixing cover crop.

CROP IMPROVEMENT

Sword bean SBS 1 is an introduction and is one of the vegetables with photo-insensitivity. It matures in 110 - 120 days. It can be grown throughout the year and gives good response to irrigation. Tender pods are ready for harvest from 75 days after sowing. As a pure crop it gives an average grain yield of 1356 kg/ha and green pod yield of 7500 kg/ha. This can also be grown as border crop, intercrop and a shade crop.

I. SEASON

June - July (Rainfed), September - October (Rabi), February - March (Summer).

II. DESCRIPTION OF VARIETY - SBS 1

Year of release	1990
Plant habit	Dwarf, erect, bushy
Pigmentation	Green
Branches (No)	4 - 6
Inflorescence	Axillary raceme
Flower	Bold, light purple
Pods	Long, pendulous, green, flat and fleshy (for vegetable use). Becomes very hard on maturity.
100 seed weight (g)	131.6
Seed colour	Milky white
Days to 50% bloom	45 - 50
Salient features	Early duration (110 - 120 days) Vegetable cum grain crop Free from beany odour Highly nutritious and delicious (25.9% protein) No major pests and diseases

III. MANAGEMENT OF FIELD OPERATIONS

- Seed rate (kg/ha) : 110-120 (Pure crop)
- Fertilizers (kg/ha) : 25 N 50 P_2O_5
- Spacing : 45 x 30 cm (irrigated), 30x20 cm rainfed

INTEGRATED PEST MANAGEMENT FOR PULSE PESTS

1. **Stem fly**
   - It attacks blackgram, greengram and cowpea.
   - Adult fly is blackish and lay eggs on the young leaves
   - Affected plants get dried
   - Immature stage will be inside the stem
   - Economic threshold level is 10% damage
2. **Aphids**
   - Attacks blackgram, greengram, lab lab, cowpea and redgram.
   - Congregated on the growing shoots, leaves, flowers and pods.
   - Affected plants will be weak and stunted
   - Because of honeydew ant movements will be there
3. **Whiteflies**
   - Attacks blackgram, greengram, cowpea and soyabean
   - Act as vector for yellow mosaic virus disease
4. **Bugs**
   - Desap the flowers and pods
   - Affected pods show shriveled grains
5. **Pod borers**
   - Gram pod borer, spotted pod borer, blue butterflies, pod fly and blister beetles are the major borers
   - Blister beetles feed on flower buds, flowers and young pods
   - Spotted pod borers web the flowers and young pods
   - Gram pod borer, plume moth and blue butterflies bore into the pods
   - Pod fly feed on the seeds of redgram.

   IPM

   - Take up the sowing of blackgram from September to November with increased seed rate (25 kg/ha) in stem fly endemic areas.
   - Remove alternate hosts
   - Use of pheromone traps @ 12/ha for Gram pod borer
   - Spray insecticides like methyl demeton or dimethoate or monocrotophos @ 500ml/ha to reduce the sucking insects
   - Spray Neem seed kernel extract (25 kg/ha) against pod borers
   - Avoid insecticidal spray when parasitoids and predators activity is high.

6. **Storage pests**
   - Dry the seeds adequately to reduce moisture level to 10 %.
   - Use pitfall traps or two in one model trap to assess the time of emergence of field carried over pulse beetle in storage and accordingly sun-dry the produce.
   - Seed: **Mix any one of the following for every 100 kg** : Activated kaolin 1 kg Malathion 5 D 1 kg TNAU Neem oil 60 EC (C) 1lit. Pungam oil 1lit.
     Monocrotophos 36 SL 400 ml
   - Pack in polythene lined gunny bags for storage

5. OILSEEDS

(i) GROUNDNUT (*Arachis hypogaea*)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
40	15	25 - 35	500 - 700	1160

Tropical crop, wide spectrum adoptable crop which grown in all 3 seasons. Flowering and seed setting affected by cloudy weather. Day neutral plant. Resists drought and tolerate flooding for one week once it establish.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone/ District/Season	Sowing Month	Varieties
I. Western Zone (Irrigated)		
Coimbatore, Tiruppur		
Chithiraipattam	April-May	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Erode,Theni,Dindigul		
Margazhipattam	Dec- Jan	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Western Zone (Rainfed)		
Coimbatore,Tiruppur, Erode, Theni, Dindigul		
Anippattam	June- July	TMVGn 13, VRIGn 7, CO 6, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
II. Southern Zone (Irrigated)		
Ramanathapuram, Thirunelveli		
Thaippattam	Jan- Feb	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Karur, Pudukkottai, Madurai, Virudhunagar		
Margazhipattam	Dec- Jan	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Sivagangai		
Ayppasipattam	Oct- Nov	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Southern Zone (Rainfed)		
Karur, Pudukkottai, Madurai, Sivagangai		
Anippattam	June-July	TMVGn 13, VRIGn 6, VRI Gn 7, VRI 8, CO 6, CO 7, TMV 14, BSR 2
Virudhunagar		
Adippattam	July-Aug	TMVGn 13, VRI 8, CO 7, TMV 14, BSR 2
Ramanathapuram, Thirunelveli		
Purattasipattam	Sep- Oct	TMVGn 13, VRI Gn 7, VRI 8, CO 6, CO 7, TMV 14, BSR 2
Thoothukudi		
Karthigaipattam	Nov- Dec	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
III. North Eastern Zone (Irrigated)		

Villupuram		
Chithiraipattam	April-May	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Thiruvallur, Kancheepuram		
Margazhipattam	Dec- Jan	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Cuddalore		
Ayppasipattam	Oct- Nov	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Vellore, Thiruvannamalai		
Karthigaipattam	Nov- Dec	TMVGn 13, VRIGn 6, VRI 8, CO 7, TMV 14, BSR 2
Thiruvallur, Cuddalore, Vellore		
Anippattam	June-July	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Kancheepuram		
Adippattam	July-Aug	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Thiruvannamalai		
Purattasipattam	Sep- Oct	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Villupuram		
Karthigaipattam	Nov- Dec	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
IV. North Western Zone (Irrigated)		
Perambalur, Ariyalur		
Margazhipattam	Dec- Jan	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Namakkal, Dharmapuri		
Vaigasipattam	May- June	CO 6, VRI GN 7
Salem, Krishnagiri		
Karthigaipattam	Nov- Dec	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
North Western Zone (Rainfed)		
Namakkal		
Vaigasipattam	May- June	CO 6, VRI Gn 7, BSR 2
Salem, Dharmapuri, Krishnagiri		
Anippattam	May- June	TMVGn 13, CO 6, VRI Gn 7, BSR 2
Perambalur, Ariyalur		
Adippattam	July-Aug	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
V. Delta Zone (Irrigated)		
Thiruchirapalli, Thanjavur, Thiruvarur, Nagapattinam		
Margazhipattam	Dec- Jan	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Delta Zone (Rainfed)		
Thiruchirapalli		
Anippattam	June-July	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2
Thanjavur, Nagapattinam		
Margazhipattam	Dec- Jan	TMVGn 13, VRI Gn 6, VRI 8, CO 7, TMV 14, BSR 2

Bunch varieties: :TMV Gn13, TMV 14, VRI Gn 6, VRI 8, CO7, BSR Semi spreding varieties: :VRI Gn 7, CO6

Suitable varieties for irrigated: VRI 8, BSR 2, CO 7

Suitable varieties for rainfed **:** TMV Gn13, TMV 14, BSR 2, CO 6, CO7 and VRI 7

Bold variety (Gujarat) : GG 7

TMV 14 and BSR 2 are alternate varieties for TMV 7

II. DESCRIPTION OF GROUNDNUT VARIETIES

Particulars	TMVGn 13	VRIGn 6
Year of Release	2006	2007
Year of Notification	SO.1178(E)/20.07.2007	SO.449(E)/11.02.2009
Parentage	Selection from Pollachi red	Derivative of ALR 2 X VG 9513
Duration (days)	100-105	120-125
Average Yield of Pods kg/ha		
Rainfed	1613	1916
Irrigated	2580	2403
Shelling %	71.4	75
100-seed weight (g)	44	36
Oil content %	50	50
Special features	Red kernel, high yield and tolerant to terminal drought	Small pods, moderately resistant to late leaf spot, rust and PBND diseases. Resistant to early season drought, high harvest index (34.6%)
Growth habit	Bunch	Bunch
Leaf colour	Green	Light green
Seed colour	Red	Light Rose

Particulars	VRIGn 7	BSR 2
Year of Release	2008	2019
Year of Notification	SO.2187(E)/27.08.2009	SO.3220(E)/05.09.2019
Parentage	Derivative of TMV 1 X JL 24	VR12 x TVG 0004
Duration (days)	120-125	105-110
Average Yield of Pods kg/ha		-
Rainfed	1865	2222
Irrigated	-	2360
Shelling %	72	70.2
100-seed weight (g)	46	40-43
Oil content %	48	45.01
Special features	Moderately resistant to late leaf spot and rust diseases. Moderately resistant to leaf miner	One or two seeded, usually two seeded, medium sized pods, Moderately resistant to late leaf spot and rust diseases
Growth habit	Semi-spreading	Bunch
Leaf colour	Dark green	Green
Seed colour	Rose	Tan

Particulars	VRI 8	TMV 14
Year of Release	2016	2018
Year of Notification	SO.2805(E)/25.08.2017	SO.1498(E)/01.04.2019
Parentage	ALR 3 x AK 303	VRI Gn 6 x R 20012
Duration (days)	105-110 days	95-100 days

Average Yield of Pods kg/ha		
Rainfed	2130	2124
Irrigated	2700	2286
Shelling %	70	70.6
100-seed weight (g)	45-50	38.0
Oil content %	49	48.0
Special features	Moderately resistant to sucking pest and defoliators. Moderately resistant to foliar fungal disease. Medium bold kernel suitable for confectionary/table purpose	Higher dry pod yield than VRI (Gn) 6 & TMV (Gn) 13; Higher shelling percentage than VRI Gn 6 Less incidence of *Spodoptera*, thrips and leaf miner compared to VRI (Gn) 6 and TMV (Gn) 13 under field conditions; Moderately resistant to late leaf spot and rust disease under field conditions
Growth habit	Bunch	Bunch
Leaf colour	Light green	Green
Seed colour	Rose	Rose

Particulars	TNAU CO 6	CO 7
Year of Release	2010	2013
Year of Notification	SO.1708(E)/26.07.2012	SO.2680(E)/01.10.2015
Parentage	Derivative of CS 9 X ICGS 5	Derivative of ICGV 87290 X ICGV 87846
Duration (days)	125-130	100 -105
Average Yield of Pods kg/ha		
Rainfed	1914	2300
Irrigated	-	2806
Shelling %	73.5	71
100-seed weight (g)	48.5	35 - 44
Oil content %	49.5	51
Special features	Dark green foliage, tolerant to foliar diseases	High oil, moderately tolerant to Rust and Late leaf spot , tolerant to Drought
Growth habit	Semi- spreading	Spanish Bunch
Leaf colour	Dark green	Green
Seed colour	Tan testa	Tan testa

CROP MANAGEMENT

I. Rainfed

1. FIELD PREPARATION

i) Plough with tractor using a disc followed by harrow, once or twice with iron plough or 3 - 4 times with country plough till all the clods are broken and a fine tilth is obtained.

ii) <u>Chiselling for soils with hard pan</u>: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t/ha of FYM or composted coir pith besides chiselling.

iii) <u>Amendments for soil surface crusting</u>: a) To tide over the surface crusting, apply lime @ 2 t/ha along with FYM or composted coir pith @ 12.5 t/ha. b) Coir pith at 12.5 t/ha converted to compost by inoculating with *Pleurotus* and applied serves as a good source of nutrients.

2. APPLICATION OF FERTILIZERS

Apply NPK fertilizers as per soil test recommendation. If soil test is not done, follow the blanket recommendation.

N	P	K	
10	10	45	Kg/ha

For rainfed groundnut –castor intercropping system, apply the recommended dose of 10:10: 45 kg NPK ha^{-1} to the main crop of groundnut and for castor apply the recommended dose of 40 kg N

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Rainfed Groundnut

Soil : Red sandy clay loam (Somayanur series)

Target : 1.0-1.2 t ha^{-1}

FN = 7.50 T - 0.33 SN - 0.45 ON

FP_2O_5 =3.50 T - 1.67 SP - 0.55 OP

FK_2O = 6.78 T - 0.31 SK - 0.43OK

Initial soil test values (kg ha^{-1})			Yield target – 1.0 t ha^{-1}			Yield target – 1.2 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	160	5*	5*	23*	15**	8	23*
180	14	180	5*	5*	23*	11	5*	23*
200	16	200	5*	5*	23*	5*	5*	23*
220	18	220	5*	5*	23*	5*	5*	23*
240	20	240	5*	5*	23*	5*	5*	23*

* Maintenance dose;** Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}.

3. FORMING BEDS

i) Form beds of size 10 m^2 to 20 m^2 depending upon the slope of the land and type of soil.

ii) Wherever tractor is engaged, bed former may be used.

Or Ridges and furrows may be laid at 60cm spacing between ridges and sowing taken on both sides of the ridge

Or Raised bed with a width of 60cm and with a furrow of 15cm on either side may be formed and sowing taken on the raised bed

4. APPLICATION OF MICRONUTRIENTS

Apply TNAU MN mixture @ 7.5 kg /ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade). Broadcast evenly on the soil surface immediately after sowing. Do not incorporate micronutrient mixture in to the soil.

5. **NUTRITIONAL DISORDER**

Zinc deficiency: Apply 25 kg $ZnSO_4$/ha as basal.

If soil analysis shows less than 1.2 ppm of zinc, soil application of 25 kg $ZnSO_4$ is recommended. Reduce $ZnSO_4$ application from 25.0 kg ha^{-1} to 12.5 kg ha^{-1} if FYM is applied @ 12.5 t $ha^{-1.}$ For the standing crop, less than 39.4 ppm of zinc in leaves, foliar spray of 0.5% $ZnSo_4$ is recommended.

Iron deficiency: Foliar of spray 1% $FeSO_4$ + 0.1% citric acid thrice on 30, 40 and 50 days after sowing.

Boron deficiency: Application of Borax 10 kg

Sulphur deficiency: Gypsum 400 kg/ha as soil application at 45th day after sowing.

6. **SEED RATE**

Use 120 kg/ha of kernels, 175 kg/ha of kernels for bold seeded varieties.

7. **SPACING**

Adopt a spacing of 30 cm between rows and 10 cm between plants. Wherever groundnut ring mosaic (bud necrosis) is prevalent, adopt a spacing of 15cm x 15 cm.

8. **SEED TREATMENT**

i) Treat the seeds with talc formulation of *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.
Biocontrol agents are compatible with biofertilizers.
Treat the seeds with biocontrol agents first and then with *Rhizobium*.
Fungicides and biocontrol agents are incompatible.

ii) Treat the seeds with *Trichoderma* @ 4g/kg. This can be done just before sowing. It is
compatible with biofertilizers. SUCH SEEDS SHOULD NOT BE TREATED WITH FUNGICIDES (or)

iii) Treat the seeds with Thiram or Mancozeb @ 4 g/kg of seed or Carboxin or Carbendazim at 2 g/kg of seed.

iv) Treat one hectare of seeds with 125 ml of *Rhizobium* (TNAU 14) and 125 ml of Phosphobacteria, shade dry it for 30 minutes before sowing

9. **SOWING**

- Use Kovai seed drill/gorru to sow the seeds in lines.
- Put one seed in each hole. Protect the seeds from crows and squirrels.

10. **INTERCROPPING**

i) Raise one row of cowpea for every five rows of groundnut wherever red hairy caterpillar is endemic.

ii) Raise intercrops like redgram, blackgram, sunflower, gingelly or other pulses.

iii) Cumbu can be raised as intercrop.

iv) Groundnut + Gingelly or Groundnut + Blackgram in the ratio of 4:1 or Groundnut + Cowpea at 6:1 ratio and Groundnut + Sunflower at 6:2 ratio may be raised.

11. WEED MANAGEMENT

i) **Pre-emergence**: Pendimethalin @ 1.0 litre/ha applied through flat fan nozzle with 500 l of water/ha. After 35 - 40 days one hand weeding may be given.

ii) If no herbicide is applied two hand weeding may be given on 20th and 40th day after sowing.

12. EARTHING UP

Accomplish earthing up during second hand weeding/late hand weeding (in herbicide application).

NOTE: i) Earthing up provides medium for the peg development ii) Use the improved hoe with long handle which can be worked more efficiently in a standing position. iii) Do not disturb the soil after 45th day of sowing as it will affect pod formation adversely.

13. APPLICATION OF CALCIUM SULPHATE (GYPSUM)

i) Apply gypsum @ 400 kg/ha by the side of the plants on 40th to 70th day depending upon
soil moisture.

ii) Apply gypsum, hoe and incorporate it in the soil and then earth up.

iii) Avoid gypsum in calciferous soils.

iv) Gypsum is effective in soils deficient in calcium and sulphur.

NOTE: Application of gypsum encourages pod formation and better filling up of the pods.

Application of gypsum at the rate of 50 % basal both in rainfed and irrigated condition reduces Khadhasty malady and pod scab nematode

Combined nutrient spray

Pod filling is a major problem especially in the bold seed varieties. To improve pod filling spraying of nutrient solution is to be given. This can be prepared by soaking DAP 2.5 kg, Ammonium sulphate 1 kg and borax 0.5 kg in 37 lit of water overnight. The next day morning it can be filtered and about 32 litre of mixture can be obtained and it may be diluted with 468 lit of water so as to made up to 500 litre to spray for one ha. Planofix at the rate of 350 ml. can also be mixed while spraying. This can be sprayed on 25th and 35th day after sowing.

14. HARVESTING

i) Observe the crop, considering its average duration. Drying and falling of older leaves and yellowing of the top leaves indicate maturity.

ii) Pull out a few plants at random and shell the pods. If the inner shell is brownish black and not white, then the crop has matured.

iii) Irrigate prior to harvest, if the soil is dry, as this will facilitate easy harvesting. If there is enough moisture in the soil, there is no need for irrigation for harvesting.

iv) If water is not available for irrigating the field prior to harvest, work a mould board plough or work a country plough, so that the plants are uprooted. Engage labour to search pods left out in the soil, if necessary.

NOTE: Do not keep the pulled out plants in heaps when they are wet, especially the bunch varieties, as the pods will start sprouting.

v) Strip off the pods from the plants. Groundnut stripper developed by TNAU can be used.

vi) Dry the pods in the sun for 4 or 5 days. Repeat drying for 2 or 3 more days after an interval of 2 or 3 days to ensure complete drying. When temperature is very high, avoid direct sun drying. Collect the pods in gunnies and store on the ground over a layer of sand to avoid any moisture coming in contact with dry pods.

I. Irrigated

1. FIELD PREPARATION

i) Plough with tractor using a disc followed by harrow, once or twice with iron plough or 3 - 4 times with country plough till all the clods are broken and a fine tilth is obtained.

ii) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t/ha FYM or composted coir pith besides chiselling.

iii) Amendments for soil surface crusting: a) To tide over the surface crusting, apply lime @ 2 t/ha along with FYM or composted coir pith @ 12.5 t/ha. b) When coir pith at 12.5 t/ha is converted into compost by inoculating with *Pleurotus* and applied, it serves as a good source of nutrien

2. APPLICATION OF FERTILIZERS

If soil test is not done, follow the blanket recommendation.

N	P	K
25	50	75 kg/ha

80 kg S as gypsum on 45 DAS

For calcareous soil, application of 40 kg S elemental sulphur along with either 50 kg $FeSO_4$ + 12.5 t FYM or 5 kg Fe EDTA can be used. For sulphur deficient calcareous soil, application of 60 kg S/ha elemental sulphur as basal application is recommended. Growing CO7, ALR3 and CO2 can be recommended in calcareous soils tolerate lime induced iron chlorosis while CO4, TMV2 and ALG320 were highly sensitive to iron deficiency.

N and K in three splits *viz.,* 50 % N & K as basal + 25 % N and K at 20 DAS + 25 % N and K at 45 DAS is recommended.

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Groundnut (1)

Soil : Red sandy loam (Irugur series)

Target : 2.0 - 2.5 t ha^{-1}

FN =6.54T-0.56 SN-0.69 ON

FP_2O_5=3.80T-3.32 SP-0.77 OP

FK_2O=8.35T-0.65SK-0.87 OK

Initial soil test values (kg ha^{-1})			Yield target – 2.0 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 2.5 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
160	12	160	13*	25*	38*	34	35	65
180	14	180	13*	25*	38*	23	29	52
200	16	200	13*	25*	38*	13*	25*	39
220	18	220	13*	25*	38*	13*	25*	38*
240	20	240	13*	25*	38*	13*	25*	38*

* Maintenance dose

Groundnut (2)

Soil : Red sandy clay loam (Somayanur series)

Target : 2.0- 2.5 t ha^{-1}

FN = 6.54T - 0.51SN - 1.10 ON

FP_2O_5 = 4.19 T - 2.95SP - 0.77 OP

FK_2O = 5.47 T - 0.33SK - 0.87 OK

Initial soil test values (kg ha^{-1})			Yield target – 2.0 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 2.5 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
160	12	160	13*	28	38*	38**	49	44
180	14	180	13*	25*	38*	32	43	38*
200	16	200	13*	25*	38*	22	38	38*
220	18	220	13*	25*	38*	13*	32	38*
240	20	240	13*	25*	38*	13*	26	38*

* Maintenance dose;** Maximum dose

Groundnut (3)

Soil : Low level Laterite

Target : 2.0- 2.5 t ha^{-1}

FN = 5.97 T - 0.45 SN

FP_2O_5 = 3.80 T – 3.32 SP

FK_2O = 7.08 T - 0.58 SK

Initial soil test values (kg ha^{-1})			Yield target – 2.0 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 2.5 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
160	12	120	13*	25*	38*	37	35	67
180	14	140	13*	25*	38*	28	29	56
200	16	160	13*	25*	38*	19	25*	44
220	18	180	13*	25*	38*	13*	25*	38*
240	20	200	13*	25*	38*	13*	25*	38*

* Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}.

3. Forming Beds

Form beds of size 10 m land and type of soil to 20 m depending upon the availability of water, slope of the

- Wherever tractor is engaged, bed former may be used. or
- Ridges and furrows may be laid at 60cm spacing between ridges and sowing taken on both sides of the ridge
- Raised bed with a width of 60cm and with a furrow of 15cm on either side may be formed and sowing taken on the raised bed

4. POLYTHENE FILM MULCHING

Broad beds and furrows method of groundnut cultivation is a proven technology from ICRISAT. Considering the favourable environment in the Broad beds and furrows system for the development of groundnut pods, with a little modification in the size, beds are to be formed for the polyethylene film mulched groundnut. Make the beds at a width of 60 cm, leaving 15 cm on the either side for the furrows. In a plot size of 4.5 m x 6.0 m, five beds can be made. After the formation of the bed and fertilizer application, spread black polythene sheet (90 cm width) over the soil surface. The edges of the polyethylene can be sheet Seven micron polythene film sheet @50 kg/ha is required. Holes can be made at required spacing of 30 x10 cm before spreading of the sheets. The seed requirement is similar to normal groundnut cultivation

5. APPLICATION OF MICRONUTRIENTS

- Apply TNAU MN mixture @ 12.5 kg /ha as Enriched FYM . (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).
- Broadcast evenly on the soil surface immediately after sowing. Do not incorporate the micronutrient mixture to the soil.
- To increase flower retention, pod filling and to induce drought tolerance apart from yield improvement, 2 sprays of TNAU groundnut rich @ 5.0 kg/ha (for each spray) at 35 DAS (50 per cent flowering) and 45 DAS (Pod developing stage) in 500 litres of water is recommended

6. NUTRITIONAL DISORDER

Zinc deficiency: Apply 25 kg $ZnSO_4$/ha as basal.

If soil analysis shows less than 1.2 ppm of zinc, soil application of 25 kg $ZnSo_4$ is recommended. Reduce $ZnSO_4$ application from 25.0 kg ha^{-1} to 12.5 kg ha^{-1} if FYM is applied @ 12.5 t $ha^{-1.}$ For the standing crop, less than 39.4 ppm of zinc in leaves, foliar spray of 0.5% $ZnSo_4$ is 4 recommended.

Iron deficiency : Foliar spray of 1% $FeSO_4$ + 0.1% citric acid on 30, 40 and 50 days after sowing. Apply 50 kg $FeSO_4$ + 12.5 t FYM basally.

Boron deficiency: Application of Borax 10 kg basally (or) 0.2% boric acid twice on 40, 50 DAS.

For multinutrient deficiency: Apply 25 kg $ZnSO_4$ + 10 kg borax + 20kg S as Gypsum.

Copper deficiency: Basal application of 10 kg $CuSO_4$ or 0.2% $CuSO_4$. Spray twice after observing plant nutrient deficiencies.

7. SEED RATE

Use 125 kg/ha of kernels. Increase the seed rate by 15% in the case of bold seeded varieties.

8. SPACING

Adopt a spacing of 30 cm between rows and 10 cm between plants. Wherever groundnut ring mosaic (bud necrosis) is prevalent, adopt a spacing of 15cm x 15 cm.

9. SEED TREATMENT

i) Treat the seeds with *Trichoderma viride* @ 4 g/kg seed or *Pseudomonas fluorescens* @ 10 g/kg seed.
Biocontrol agents are compatible with biofertilizers.
First treat the seeds with biocontrol agents and then with *Rhizobium*.
Fungicides and biocontrol agents are incompatible.

ii) Treatment with *Trichoderma* can be done just before sowing. SUCH SEEDS SHOULD
NOT BE TREATED WITH FUNGICIDES. (or)

iii) Treat the seeds with Thiram or Mancozeb @ 4 g/kg of seed or Carboxin or Carbendazim at 2 g/kg of seed.

iv) Treat the seeds with 3 packets (600 g)/ha of Rhizobial culture TNAU14 developed at TNAU using rice kanji as binder. If the seed treatment is not carried out, apply 10 packets/ha (2000 g) with 25 kg of FYM and 25 kg of soil before sowing.

Seed treatment will protect the young seedlings from root-rot and collar rot infection.

10. SOWING

a) Dibble the seeds at 4 cm depth along with fertilizer.

11. WEED MANAGEMENT

i. **Pre-emergence**: Pendimethalin @ 1.0 litre/ha applied on third day after sowing through flat fan nozzle with 500 litres of water/ha followed by irrigation. After 35 - 40 days one hand weeding may be given.

ii. Spray Early post emergence application of Imazethapyr @ 50 ml/ha at 20-30 days after sowing based on weed density as post emergence spray

iii. If no herbicide is applied two hand hoeing may be given on 20th and 40th day after sowing.

iv. Apply, PE Oxyfluorfen @ 200 g/ha on 3rd DAS and followed by one hand weeding on 40-45 DAS

v. Apply, PE Oxadiazon @ 0.8 kg ha^{-1} followed by one earthing up using hoes (or) working star type weeder

12. EARTHING UP:

Accomplish earthing up during second hand weeding/late hand weeding (in herbicide application).

NOTE: i) Earthing up provides medium for the peg development. ii) Use the improved hoe with long handle which can be worked more efficiently in a standing position. iii) Do not disturb the soil after the 45^{th} day of sowing as it will affect pod formation adversely.

13. APPLICATION OF CALCIUM SULPHATE (GYPSUM)

- Apply gypsum @ 400 kg/ha by the side of the plants on the 40^{th} to 45^{th} day of sowing. Apply gypsum, hoe and incorporate in the soil and then earth up.
- Avoid gypsum in calciferous soils.
- Gypsum is effective in soils deficient in calcium and sulphur.

NOTE: Application of gypsum encourages pod formation and better filling up of the pods.

Application of gypsum at the rate of 50 % basal both in rainfed and irrigated condition reduces Khadhasty malady and pod scab nematode

Combined nutrient spray

Pod filling is a major problem especially in the bold seed varieties. To improve pod filling spraying of nutrient solution is to be given. This can be prepared by soaking DAP 2.5 kg, Ammonium sulphate 1 kg and borax 0.5 kg in 37 lit of water overnight. The next day morning it can be filtered and about 32 litre of mixture can be obtained and it may be diluted with 468 lit of water so as to made up to 500 litre to spray for one ha. Planofix at the rate of 350 ml. can also be mixed while spraying. This can be sprayed on 25^{th} and 35^{th} day after sowing. or Spray TNAU Groundnut Rich @ 5.5 kg/ha for 2 sprays (50 per cent flowering and pod developing stage) to increase flower retention and pod filling.

14. WATER MANAGEMENT

Schedule the irrigation at 0.40 and 0.60 IW/CPE ratio during vegetative and reproductive phase respectively. Regulate irrigation as per the growth phase of the crop. Pre-flowering phase : 1 to 25 days Flowering phase : 26 to 60 days Maturity phase: 61 to 105 days Regulate irrigation based on physiological growth phases. Pegging, flowering and pod development phases are critical for irrigation during which period adequate soil moisture is essential. Irrigate as follows:

i) Sowing or pre-sowing
ii) Life irrigation, 4 - 5 days after sowing.
iii) 20 days after sowing
iv) At flowering give two irrigations
v) At pegging stage give one or two irrigations
vi) In pod development stage, 2 - 3 irrigations depending on the soil type

Note: Spraying 0.5% Potassium chloride during flowering and pod development stages *will aid to mitigate the ill effects of water stress. Sprinkler irrigation will save water to the tune of about 30%. Borderstrip irrigation is recommended in command areas in light textured soils. Composted coir pith increases moisture availability and better drainage in heavy textured soil.*

15. HARVESTING

i) Observe the crop, considering its average duration. Drying and falling of older leaves and yellowing of the top leaves indicate maturity.
ii) Pull out a few plants at random and shell the pods. If the inner shell is brownish black and not white, then the crop has matured.
iii) Irrigate prior to harvest, if the soil is dry, as this will facilitate easy harvesting. If there is enough moisture in the soil, there is no need for irrigation for harvesting.
iv) If water is not available for irrigating the field prior to harvest, work a mould board plough or work a country plough, so that the plants are uprooted. Engage labour to search pods left out in the soil, if necessary.

NOTE: Do not keep the pulled out plants in heaps when they are wet, especially the bunch varieties, as the pods will start sprouting.

v) Strip off the pods from the plants. Groundnut stripper developed by TNAU can be used.
vi) Dry the pods in the sun for 4 or 5 days. Repeat drying for 2 or 3 more days after an interval of
2 or 3 days to ensure complete drying. When temperature is very high, avoid direct sun drying. Collect the pods in gunnies and store on the ground over a layer of sand to avoid any moisture coming in contact with dry pods.

CROP PHYSIOLOGY

Foliar spray of TNAU Groundnut Rich @ 2 kg/acre in 200 litres of water at peak flowering and at pod development stages increases flower retention, pod filling and improves moisture stress tolerance and pod yield.

CROP PROTECTION

Economic threshold level for important pests

Pest	**ETL**
Leaf miner	Leaf miner 1 larva / m row
Tobacco caterpillar	8 egg masses/100 m row

Pests Management strategies

Red hairy caterpillar, *Amsacta albistriga*	• Dig out and destroy the pupae from field bunds and shady spots prior to summer rains. • In rain fed crop set up 3 to 4 light traps and bonfires immediately after rains to attract and kill the moths. • Collect and destroy gregarious, early instar larvae on lace-like leaves of intercrops such as redgram

	and cowpea. • Collect and destroy egg masses in the cropped area. • Dig a trench 30 cm deep and 25 cm wide with perpendicular sides around the infested fields to avoid larval migration. • Spray Aa NPV • Apply quinalphos 1.5 DP 25 kg/ha
Virus multiplication Collect medium sized larvae of *Amsacta albistriga* from the field and starve them over night. Make a pure suspension of virus with the nucleus culture in water. Dip *Calotropis* leaves in virus suspension, shade dry and feed them to starved larvae for 1 or 2 days. From third day, normal, untreated leaves can be fed to these larvae. From 5th day, the treated larvae will start dying. Virus infected larvae can be diagnosed by their pinkish ventral surface, their head hanging downwards with white body contents oozing out through ruptured body wall in the late stage. Collect the dying larvae, keep in fresh potable water for a few days, grind the larvae and filter through several layers of fine cloth and collect filtrate (Crude virus suspension). Use virus suspension obtained from 750 medium sized larvae for spraying one hectare along with a sticker 250 ml or Triton in 350 l of water. Use potable water for mixing and spray in the evening hours.	
Tobacco caterpillar**,** *Spodoptera litura*	• Grow castor as border or intercrop in groundnut fields to serve as indicator or trap crop. • Monitor the emergence of adult moths by setting up light trap and pheromone traps. • Collect egg masses and destroy. • Collect the gregarious larvae and destroy them as soon as the early symptoms of lace-like leaves appear on castor, cowpea and groundnut. • Spray Methomyl 40 SP 750ml / ha to control the early instar (1st to 3rd instar) larvae. • Spray NSKE 5% • Apply Nuclear Polyhedrosis Virus 1.5 x 10^{12} POBs/ha with crude sugar 2.5 kg/ha and Teepol 250 ml/ ha.
Leafhopper, *Empoasca kerri*	Intercrop lab lab with groundnut 1:4 ratio **Spray any one of the following insecticides / ha** Imidacloprid 17.8 SL 100 ml Quinalphos 25 EC 1400 ml
Leafminer / Leaf webber, *Aproaerema modicella*	Set up light trap between 8 and 11 pm at ground level **Spray any one of the following insecticides / ha** Methyl demeton 25 EC 1000 ml Quinalphos 25 EC 1400 ml
Podborer (Earwig) *Anisolabis stali*	Apply Carbofuran 3 CG 50 kg/ha to the soil prior to sowing in endemic areas. Repeat soil application of any formulations on the 40th day of sowing and incorporate in the soil during the earthing up.
Whitegrubs *Holotrichia consanguinea, H. serrata*	Apply Carbofuran 3 CG 33.3 kg/ha

Aphid *Aphis craccivora*	**Apply anyone of the following insecticides/ha** Imidacloprid 17.8 SL 100 ml Methyl demeton 25 EC 1000 ml
Thrips *Scirtothrips dorsalis*	Apply Quinalphos 1.5 DP 23.3kg/ha Spray Quinalphos 25 EC 1400 ml/ha
Termites	Apply Thiamethoxam 75 SG@125 g/ha

DISEASE MANAGEMENT

Seed treatment : Treat the seeds with carbendazim @ 2 g/kg or *Trichoderma asperellum* @ 4 g /kg or *Pseudomonas fluorescens* @ 10 g/kg of seeds

Disease	Recommendations
Rust: *Puccinia arachidis*	Spray mancozeb @ 1000 g /ha or chlorothalonil @ 1000 g /ha or wettable sulphur @ 2500 g /ha. If necessary, repeat the spray 15 days later.
Early leaf spot: *Cercopora arachidicola* (*Mycosphaerella arachidis*) **Late leaf spot:** *Phaeoisariopsis personata* (*Mycosphaerella berkeleyii*)	Spray carbendazim @ 500 g/ha or mancozeb @ 1000 g/ha or chlorothalonil @ 1000 g/ha. If necessary, repeat the spray 15 days later. **CIB Recommendation** • Spray hexaconazole 5% EC @ 1500ml/ha or metiram 70% WG 2 kg/ha or propiconazole 25% EC @ 500 ml/ha or pyraclostrobin 20% WG @ 500g/ha or sulphur 40% WP @ 5.65-7.50 kg/ha or sulphur 80% WP @ 2.5-5.0 kg/ha or sulphur 85% DP @ 15-20 kg/ha or carbendazim 12% + mancozeb 63% WP @ 500 g/ha or fluxapyroxad 167 g/l + pyraclostrobin 333 g/l SC @ 300 ml/ha or pyraclostrobin 133g/l + epoxiconaxole 50g/l SE @ 500/ha **For combined infection of leaf spot and rust** • Spray bitertanol 25% WP @ 1 kg/ha or chlorothalonil 75% WP @ 1.5 g/l or mancozeb75% WP @ 1.5 to 2 kg/ha or tebuconazole 25.9% m/m EC @0.50-0.75 l/ha **For combined infection of leaf spot and stem rot** • Spray carbendazim 25% + flusilazole 12.5% SE @ 640-800 g/ha **For collar rot, seed rot, root rot and stem rot** • Treat the seeds with carboxin37.5% + thiram 37.5% DS @ 3g/kg of seeds **For the management of termites, thrips, jassids, root grubs, collar rot and stem rot** • Spray imidacloprid 18.5% + hexaconazole 1.5 % FS @ 200 ml/ha

Combined infection of rust and leaf spot	Spray 10% *Calotropis* leaf extract or spray carbendazim @ 250 g + mancozeb 1000 g/ha or chlorothalonil @ 1000g/ha. If necessary, give the second spray 15 days later.
Root rot: *Macrophomina phaseolina* (*Rhizoctonia bataticola*)	• Soil application of *P. fluorescens* @ 2.5 kg /ha with 50 kg of well decomposed FYM / sand at 30 DAS. • Spot drench with carbendazim @ 1 g / l
Groundnut bud necrosis: (*Groundnut bud nercrosis virus*) (Vector: *Thrips tabaci, Frankliniella schultzeii*)	• Antiviral principles (AVP) from sorghum or coconut leaves. AVPs are extracted as follows: Sorghum or coconut leaves are collected, dried, cut into small bits and powdered. To one kg of leaf powder two litres of water is added and heated to 60°C for one hour. It is then filtered through muslin cloth and diluted to 10 litres and sprayed. To cover one hectare area 500 l of fluid will be required. Two sprays at 10 and 20 days after sowing will be needed. • For vector management, apply quinalphos 1.5 DP 23.3kg/ha or spray quinalphos 25 EC @ 1400 ml/ha

GROUNDNUT – VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 3 m all around the field from the same and other varieties of the crop.

Season

- June - July and December - January.

Spacing

- 25 x 15 cm.

Pre-sowing seed hardening

- Harden the graded seeds by soaking in 0.5 % $CaCl_2$ (50 % seed volume) for 6 h. After 6 h soaking, incubate the seeds in between moist gunny bags for 12 -18 h. Observe the sprouting of radicle periodically at 2 h interval after 12 h of incubation.
- Separate the seeds with sprouted radicle (just visible expression of radicle) and dry under shade.

Fertilizer requirement

- Apply NPK @ 25:50:75 kg / ha as blanket.

- Apply borax as basal application @ 10 kg / ha in boron deficient soils.
- Apply gypsum @ 400 kg / ha at peg formation stage and at earthing.

Foliar application

- NAA@ 200 ppm at 60 days after sowing to arrest late formed flowers and increase the seed yield in groundnut.

Pre-harvest spray to arrest *in situ* germination

- Spray 1250 ppm MH (Maleic Hydrazide) at 60 days after sowing.

Harvest

- Harvest the pods as and when the colour of the inner side of the shell turns black and dry to 10 - 12 per cent moisture.

Drying

- Stake the plants as the pods are exposed outside for easy drying of pods.
- Dry the pods to 15 - 20 % moisture content under sun.

Decortication

- Dry the pods to 16 per cent moisture content and decorticate either manually or using hand operated decorticator with proper adjustment.
- Dry the kernels to 7 to 8 per cent moisture.
- Practice pod verification based on varietal characteristic before grading to remove genetically impure seeds.
- Remove all discoloured pods.
- Reject mechanically injured pods for seed purpose.

Pre-storage seed treatment

- Treat the pods with carbendazim @ 2 g / kg at 6 - 7 % moisture content.

Seed storage

- Store the pods in gunny bags with calcium chloride @ 250 g / 30 kg of pods.
- Store the seeds in gunny for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 6 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 5 %.

(ii) SESAME (*Sesamum indicum*)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
40	20	25 - 35	450 - 500	up to 1600

Tropical crop. It needs fairly high temperature for good growth. Can with stand drought, survive well with winter dew. Short day plant.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone/ District/Season	Sowing Month	Varieties
I. Western Zone (Irrigated)		
Coimbatore, Thiruppur, Erode		
Masipattam	Feb- March	VRI (Sv) 2, TMV 7, VRI 3
Western Zone (Rainfed)		
Coimbatore,Thiruppur, Erode, Dindigul		
Anippattam	June- July	TMV 7
Theni		
Karthigai	Nov- Dec	VRI(Sv) 2, TMV 7
II. Southern Zone (Irrigated)		
Thirunelveli, Karur		
Chithiraipattam	Apr- May	VRI (Sv) 2, TMV 7
Pudukkottai		
Margazhi	Dec- Jan	VRI (Sv) 2, TMV 7, VRI 3
Southern Zone (Rainfed)		
Madurai		
Anipattam	June-July	TMV 7
Virudhunagar, Pudukkottai,		
Adippattam	July-Aug	TMV 7
Karur		
Purattasipattam	Sep- Oct	VRI(Sv) 2, TMV 7
Ramanathapuram, Sivagangai, Thirunelveli, Thoothukudi		
Karthigaipattam	Nov- Dec	VRI(Sv) 2, TMV 7
III. North Eastern Zone (Irrigated)		
Kancheepuram, Cuddalore, Vellore		
Margazhipattam	Dec- Jan	VRI (Sv) 2, TMV 7, VRI 3
Thiruvannamalai		
Masipattam	Feb- March	VRI (Sv) 2, TMV 7, VRI 3
Villupuram		
Chithiraipattam	Apr- May	VRI (Sv) 2, TMV 7
Thiruvallur		
Anipattam	June-July	TMV 7
North Eastern Zone (Rainfed)		

Vellore, Thiruvannamalai		
Anippattam	June-July	TMV 7
Kancheepuram, Cuddalore		
Adippattam	July-Aug	TMV 7
Thiruvallur		
Purattasipattam	Sep- Oct	VRI (Sv) 2, TMV 7
Villupuram		
Karthigaipattam	Nov- Dec	VRI (Sv) 2, TMV 7
IV. North Western Zone (Irrigated)		
Namakkal		
Margazhipattam	Dec- Jan	VRI (Sv) 2, TMV 7, VRI 3
Salem, Perambalur, Ariyalur		
Masipattam	Feb- March	VRI (Sv) 2, TMV 7, VRI 3
North Western Zone (Rainfed)		
Salem,Namakkal, Dharmapuri, Krishnagiri		
Anippattam	June-July	TMV 7
Perambalur, Ariyalur		
Adippattam	July-Aug	TMV 7
V. Delta Zone (Irrigated)		
Thanjavur, Thiruchirapalli		
Masipattam	Feb- March	VRI (Sv) 2, TMV 7, VRI 3
Thiruvarur		
Chithiraipattam	Apr- May	VRI (Sv) 2, TMV 7
Delta Zone (Rainfed)		
Thanjavur, Thiruvarur, Nagapattinam		
Thaippattam	Jan- Feb	TMV 7
Thiruchirapalli		
Purattasipattam	Sep- Oct	VRI (Sv) 2, TMV 7

Suitable Varieties for Irrigated **:** VRI 2, VRI 3, TMV 4, TMV 6, TMV 7
Suitable Varieties for Rainfed :TMV 6, TMV 7
Suitable Varieties for Rice fallow :VRI 1

II Description of sesame varieties

Particulars	VRI(Sv) 2	TMV 7	VRI 3
Year of Release	2005	2009	2017
Year of Notification	SO.599(E)/ 25.04.2006	SO.2137(E)/ 31.08.2010	SO.1379(E)/ 27.03.2018
Parentage	Derivative of VS 9003 X TMV 6	Derivative of SI 250 X ES 22	Derivative of SVPR 1 x TKG 87
Duration (days)	80-85	80-85	75-80
Average Yield (kg/ha)			
Rainfed	650-700	850	-

Irrigated	700-750	920	995 (Margazhipattam) 1055 (Masi pattam)
Oil content %	51.9	50	50
Habit	Profuse branching	Erect, indeterminate, with Profuse	Erect, indeterminate with Profuse branching
Capsules	4 loculed	4 loculed	4 loculed
Seeds	Reddish brown	Brown	white

CROP MANAGEMENT

1. FIELD PREPARATION

a) Plough the field with tractor twice or with mould board plough thrice or five times with a country plough.

b) Break the clods in between ploughings and bring the soil to a fine tilth to facilitate quick germination as the seeds are small.

c) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depth with chisel plough first at 0.5 m interval in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5t FYM/composted coir pith besides chiselling.

d) For irrigated gingelly, form beds of size 10 m^2 or 20 m^2 depending upon the availability, inflow of water and slope of the land. Level the beds perfectly without any depressions to prevent water stagnation, which will affect the germination adversely.

e) In rice fallows, field is ploughed once with optimum moisture, seeds are sown immediately and covered with one more ploughing.

2. APPLICATION OF FERTILIZERS

i) Spread FYM or composted coir pith or compost @ 12.5 t/ha evenly on the unploughed field and plough it in.

ii) If the manure is not applied before commencement of ploughing, spread 12.5 t/ha of FYM or compost evenly on the field before the last ploughing and incorporate in the soil.

iii) If soil tests are not available, follow the blanket recommendations. **Rainfed:** Apply 23:13:13 kg NPK/ha or 17:13:13 kg NPK/ ha + 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha). **Irrigated:** Apply 35:23:23 kg NPK/ha or 21:23:23 kg NPK/ha + 3 packets of Azospirillum (600 g/ha) and 3 packets (600 g/ha) of Phosphobacteria or 6 packets of Azophos (1200 g/ha)

Soil : Black alluvium (Adanur series) $FN = 13.7\,T - 0.46\,SN$
Target : 1.00 - 1.25t ha^{-1} $FP_2O_5 = 6.3T - 1.79\,SP$
$FK_2O = 12.8T - 0.47\,SK$

Initial soil test values (kg ha^{-1})			**Yield target – 1.00t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 1.25 t ha^{-1}** NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	12	180	23	22	13	53**	35**	35**
200	14	200	18*	12*	12*	24	22	26
220	16	220	18*	12*	12*	18*	18	17
240	18	240	18*	12*	12*	18*	15	12*
260	20	260	18*	12*	12*	18*	12*	12*

* Maintenance dose **Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}. Open furrows to a depth of 5 cm and 30 cm apart and place the fertilizer mixture along the furrows and cover to a depth of 3 cm with soil before sowing.

iv) If furrow application is not done, broadcast the fertilizer mixture evenly on the beds before sowing.

v) Apply TNAU MN mixture @ 7.5 kg/ha as enriched FYM for rainfed sesame and TNAU MN mixture @ 12.5 kg/ha as enriched FYM for irrigated sesame. (Prepare enriched FYM at 1:10 ratio of MN mixture and FYM; mix at friable moisture and incubate for one month in shade)

3. APPLICATION OF AZOSPIRILLUM

a) Treat one hectare of seeds with 600 g of *Azospirillum* and 600g of phosphobacteria (or) 600g of Azophos. Apply 2 kg of *Azospirillum* and 2 kg of phosphobacteria (or) 2 kg of Azophos with 25 kg of FYM and 25 kg of sand, mix uniformly before sowing as soil application.

b) Liquid formulation Treat one hectare of seeds with 125 ml of *Azospirillum* and 125 ml of Phosphobacteria, shade dry it for 30 minutes before sowing.

4. NUTRITIONAL DISORDERS

a) **Manganese deficiency** : Leaves develop interveinal chlorosis, chlorotic tissue, later develop light brown or husk coloured necrotic lesions. Mix 10 kg $MnSO_4$ /ha with 45 kg of soil and broadcost evely in the beds after sowing.

b) **Zinc deficiency**: Middle leaves develop chlorosis in the interveinal areas and necrosis along the apical leaf margins. Apply 25 kg Zinc sulphate with 45 kg of soil and broadcast evenly in the beds after sowing.

Note: Do not incorporate the micronutrient in the soil.

5. **SEED RATE**

Adopt a seed rate of 5 kg/ha.

6. **SPACING**

a) Give a spacing of 30 cm between rows and 30 cm between plants. b) For rice fallows, seeds are broadcasted and thinned to maintain 11 plants/m^2

7. **QUALITY OF SEEDS**

Select mature, good quality seeds free from pest and fungal damage.

8. **SEED TREATMENT**

Treat the seed with *Trichoderma*@ 4g/kg. This can be done just before sowing. SUCH SEEDS SHOULD NOT BE TREATED WITH FUNGICIDES or treat the seed with Thiram 4 g or Carbendazim at 2 g/kg of seeds before sowing.

9. **SOWING**

a) Sow the seeds preferably in lines.
b) Mix the seeds with four times its volume of dry sand and drop the mixture evenly along the furrows in which fertilizers are applied.
c) Sow the seeds to a depth of 3 cm and cover with soil.
d) The optimum time of sowing for VRI (SV) 1 sesame is second fortnight of February to first fortnight of March under summer irrigated conditions.

10. **WATER MANAGEMENT**

i) Irrigate at sowing and give life irrigation 7 days after sowing depending on the soil and climatic condition and allow excess water to percolate.
ii) Give one pre-flowering irrigation (25 days): One at flowering and one or two at pod setting. An irrigation at flowering period is critical.

NOTE: The critical stage for moisture requirement is the flowering phase i.e, between 35th to 45th days of sowing. During the maturity phase, moisture status should be low. If more water is given during this phase, maturity of seeds is affected and filling up of the capsules will be poor. Therefore, stop irrigation after 65 days of sowing.

11. **THINNING**

Thin out the seedlings to a spacing of 15 cm between the plants on the 15th day of sowing and 30 cm on 30th day of sowing. This operation is very important for the crop in order to induce basal branches.

12. **WEED MANAGEMENT**

i. Apply, PE application of Pendimethalin 1.0 litre /ha followed by one hand weeding on 25th DAS

13. **HARVESTING**

a) **Decide when to harvest**

i. Observe the crop, considering the average duration of the crop.
ii. Twenty five per cent of the leaves from the bottom are shed and the top leaves

loose their colour and turn yellow at maturity.

iii. The colour of the stem turns yellow.
iv. The colour of the capsules turn yellow upto the middle.
v. Harvest before the bottom capsules turn brown.
vi. Examine the 10^{th} capsule from the bottom by opening. If the seeds attained the full color of the variety harvest may be taken up.
vii. If harvest is delayed/ the capsules will dehisce resulting in yield reduction.

b) Harvest

i. Pull out the plants from the bottom.
ii. Stack in the open, one over the other in a circle with the stems pointing out and the top portion pointing inside.
iii. Cover the top with straw, so that humidity and temperature increases.
iv. Cure like this for 3 days, shake the plants. About 75 per cent of the seeds will fall off.
v. Dry the plants for one more day and again shake the plants. All the mature seeds will fall off.
vi. Winnow the seeds and dry in the sun for 3 days. Stir once in 3 hours to give uniform drying.
vii. Collect the seeds and store in gunnies.

CROP PROTECTION

A. Pest management

Economic threshold level for important pests

Pest	ETL
Shoot & capsule borer	10 larvae/ m^2 in the vegetative stage and 2 larvae / m^2 in the reproductive stage

Pest Management strategies

Shoot & capsule borer**,** *Antigastra catalaunalis*	**Spray any one of the following** Neem seed kernels extract 5% Neem oil 2% Spray Quinalphos 25 EC 2000 ml/ha
Pod borer, *Elasmolomus (= Aphanus) sordidus* **Gall fly,** *Asphondylia ricini* **Whitefly,** *Bemisia tabaci***,** *A. dispersus*	**Spray any one of the following neem based insecticide** NSKE 5% Neem oil 2%
Leafhopper, *Emposca devastans*	**Spray any one of the following** NSKE 5% Neem oil 2% Methyl demeton 25 EC 1200 ml/ha Quinalphos 25 EC 2000 ml/ha
Storage pests *Tribolium castaneum* *Corcyra cephalonica*	Mix one kg of activated clay with 100 kg of seeds after adequate drying of seeds

DISEASE MANAGEMENT

Disease	Recommendations
Powdery mildew: *Erysiphe cichoracearum*	Apply sulphur dust @ 25 kg/ha or spray 0.2% wettable sulphur
Alternaria blight: *Alternaria sesami*	Spray mancozeb @ 1000 g/ha
Cercospora leaf spot: *Cercospora sesami*	Spray mancozeb @ 1000 g/ha
Root rot: *Macrophomina phaseolina* *(Rhizoctonia bataticola)*	• Soil application of *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand at 30 days after sowing. • Spot drench with carbendazim @ 1 g/ l
Phyllody: Phytoplasma (Vector: *Orosius albicinctus*)	• Remove and destroy infected plants. • To control vector, spray NSKE @ 5% or neem oil @ 2% or methyl demeton 25 EC @ 1200 ml/ha or quinalphos 25 EC @ 2000 ml/ha or dimethoate 30 EC 500 ml/ha combined with intercropping of sesamum + redgram (6 : 1)

SESAME - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 200 m all around the field from the same and other varieties of the crop.

Pre- sowing seed treatment

Pellet the seeds with neem leaf powder @ 760 g + 120 g Azotobacter + 120 g phosphobacteria for 1 kg seed to enhance the productivity.

Fertilizer

- Apply NPK @ 50:25:25 kg / ha and Manganese sulphate @ 5 kg / ha as basal application.

Foliar application

- Spray 1 % DAP at the time of first flowering and 10 days after the first spray.

Harvest

- Harvest the crop when 75 - 80 % of the pods started yellowing and bottom 1 or 2 pods have dehisced. At this stage, the pod moisture content will be 50 - 60 % and seed moisture content will be 25 - 30 % and the seeds will be chocolate brown colour.
- Stake the plants in inverted position and cure them for 3 - 4 days.

Threshing

- Beat the staked plants with pliable bamboo stick for removal of seeds.

Seed grading

- Grade the seeds with 4 / 64" round perforated sieve.

Drying

- Dry the seeds to 7- 8 % moisture content.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg along with carbaryl 200 mg / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara)* leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 6 - 7 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with seed moisture content less than 5 %.

(iii) CASTOR (*Ricinus communis*)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
38 – 40	10 - 15	20 - 25	350 - 600	up to 2500

Tropical and requires moderately high temperature 20 to 27°C with low humidity throughout the growing season. It grows best in areas where there are clear warm sunny days. Prolonged cloudy weather with high temperature at the time of flowering resulted in poor seed setting, which is known as sex reversion. High temperature above 41°C at flowering time even for as short period results in blasting of flowers. Very resistant to drought but evenly distributed rainfall is required. Heavy rainfall at flowering reduces the yield. Very susceptible to frost but grow even an altitudes of 1200 to 2100 m, if sown in March- April, perennial varieties are grown at still higher altitude for shade in coffee estates.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES
A. Rainfed	
1. Adipattam (Jun-July)	
All districts	Variety : TMV 5, TMV 6
	Hybrid : YRCH 1, YRCH 2
B. Irrigated	
1. Vaigasi pattam (May - June)	
All districts	Hybrid : YRCH 1, YRCH 2
2. Karthigaipattam (Nov - Dec)	
All districts	Hybrid : YRCH 1, YRCH 2
3. Panguni pattam (March- Apr)	
All districts	Hybrid : YRCH 1, YRCH 2
C. Gardenland (border)	
1. Perennial	
All districts	Variety : YTP 1

DESCRIPTION OF CASTOR VARIETIES

Particulars	CO 1	TMV 5	TMV 6	Hybrid	Hybrid	Variety
				YRCH 1	YRCH 2	YTP 1
Year of Release		1984	1997	2009	2017	2019
Year of Notification		SO.832(E)/ 18.11.1985	SO.647(E)/ 09.09.1997	SO.2137(E)/ 31.08.2010	SO.399(E)/ 24.01.2018	
Parentage	Pureline selection from Anamalai	Derivative of SA 2 X S248/2	Derivative of VP 1 X RC 962	DPC 9 X TMV5	M 619-1 X SKI 215	Cross Derivative of TMV 6 x Salem Local
Duration (days)	perennial	120	160	150-160	170-180	115-120
Yield (kg/ha)						
Rainfed (mixed crop)	-		500			
Rainfed	2.5	850	950	2000	2300	1456 kg/ha
(pure crop)	kg/tree/year					
Irrigated (pure crop)	-	-	-	3000	3500	3kg/plant/ year
Oil content (%)	57	50	51.9	49	49	49
Special features						Suitable for perennial system.
Stem colour	Pinkish green	Rose	Red	Light red	Red	Red
Bloom	No bloom	Triple	Double	Triple	Triple	Triple
(waxy coat)						
Receme/	Bold, sparse	Spiny, non	Medium,	Spiny, non	Semi spiny, non	Long, Conical, Semi compact
capsule	setting, non	dehiscent,	lengthy ,	dehiscent,	dehiscent,	Bold, dehiscent,
	dehiscent	resistant to	spiny	resistant to leaf	resistant to capsule borer, leaf hopper,	resistant to capsule borer, leaf hopper,
		leaf hopper	capsule	hopper	Semilooper, spodoptera	
Suitability	Pure and mixed crop	Pure and mixed crop	Pure and mixed crop	Pure and mixed crop	Pure and mixed crop	Pernennial Pure Border crop & mixed crop

Other features	-	-	-	-	Resistant to wilt, high basal branching and proportion of female flowers more than 95 percent. Resistant to lodging, fertilizer responsive and suitable for rainfed situation and areas of limited irrigation.	Resistant to Semilooper , *Spodoptera*, Thrips and Capsule borer

CROP MANAGEMENT

1. PREPARATION OF THE FIELD

Plough two-three times with country or mould board plough.

2. APPLICATION OF FERTILIZERS

Spread 12.5 t/ha of FYM or compost evenly on the main field before last ploughing and incorporate in to soil by working a country plough. Apply 30 kg sulphur/ ha through gypsum at the time of last ploughing for higher castor yield.

NOTE: Do not leave FYM or compost exposed to sunlight as nutrients will be lost.

3. SEED RATE

Adopt a seed rate of 10 kg/ha for varieties and 5 kg/ha for hybrids.

4. SPACING

Adopt the following spacing.

	Rainfed situation	Irrgiation situation
Varieties	90 cm x 60 cm	90 cm x 90 cm
YTP 1	3mx3m	3mx3m
Hybrids		
YRCH 1	120 cm x 90 cm	150 cm x 120 cm
YRCH2	180 cm x 150 cm	180 cm x 150 cm

Under, irrigated conditions, for clay soils wider spacing of 150 cm x 150 cm for YRCH 1 can also be adopted.

For TMV 5 short duration variety 60 x 30 cm may be adopted.

5. APPLICATION OF FERTILIZERS

Apply NPK fertilizers basally as per soil test recommendations as far as possible. If soil test recommendations are not available, follow the blanket recommendation as follows

Rainfed conditions	**Recommended NPK kg/ ha**
Varieties	45 : 15 : 15 NPK kg / ha
Hybrids – YRCH 1	60 : 30 : 30 NPK kg / ha
YRCH 2	70 : 35 : 35 NPK kg / ha
Irrigated condition	
Varieties	60 : 30 : 30 NPK kg / ha
Hybrids - YRCH1	90 : 45 : 45 NPK kg / ha
YRCH 2	135 : 65 : 65 NPK kg / ha

YRCH 1: In rainfed situations apply 100% P & 50% N&K basally & remaining quantity may be applied in one or two top dressings based on the soil moisture availability.
YRCH 2: In rainfed situations, apply 35 kg P & 37.5 kg N & 17.5 kg K basally and remaining quantity of 37.5 kg N & 17.5 kg K may be applied in one or two top dressings based on the soil moisture availability.

YRCH 1: In irrigated situations, apply 100% P & 50% N&K as basal & remaining quantity N&K may be applied in two equal splits at 30^{th} & 60^{th} DAS.
YRCH 2: under irrigated condition, apply 65 kg P & 67.5 kg N & 32.5 kg K as basal and remaining quantity of 67.5 kg N & 32.5 kg K may be applied in two equal splits at 30^{th} & 60^{th} day after sowing (DAS).

Apply 12.5 kg $ZnSO_4$ ha^{-1} (If the soil available Zn is < 1.2 ppm) and 25 kg $FeSO_4$ ha^{-1} (if the soil available Fe is < 3.7 ppm for non calcareous soil and < 6.3 ppm for calcareous soil). If soil test values are not available.

Rainfed conditions	**Recommended TNAU MN mixture kg/ha**
Varieties	7.5
Hybrids	10.0
Irrigated conditions	
Varieties	12.5
Hybrids	15.0

(Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).

6. PRE TREATMENT OF SEEDS

a) Treat the seeds with Thiram or carbendazim @ 2g / kg of seeds or with *Trichoderma viride* @ 4g/kg of seeds. Treat the seed 24 hours prior to sowing. Soaking the seeds with water for 10 hours enhances the germination.

b) In rainfed situations, seed priming with 1% KCl for 3 hours and sowing a week before onset of monsoon is recommended.

7. SOWING

a) Sow the seeds adopting the recommended spacing. b) Place the seeds at depth 4 - 6 cm. c) Put two seeds in each hole and retain only one healthy seedling by thinning out of weaker seedling/pistillate plant at 20 DAS.

Selective mechanization

Selective mechanization in castor *viz*., sowing with tractor drawn seed drill with a spacing of 120 cm x 90 cm, inter cultivation with power weeder on 20 and 40 DAS, need based plant protection with boom sprayer, harvesting by secateurs and threshing and shelling by castor thresher increased the kernel yield and net return & benefit cost ratio

8. GAP FILLING

Gap fill on the 15th day of sowing and simultaneously thinning may be done leaving one healthy plant.

9. WEED MANAGEMENT

Apply pre emergence herbicide Pendimethalin @ 1 litre/ha or Fluchloralin @ 1 litre/ha on 3 DAS followed by hand weeding twice on 20th & 40th DAS.

10. CASTOR PGR CONSORTIA (CASTOR GOLD)

Foliar application of plant growth regulator consortia @ 0.05 % (0.5 ml/litre of water) on 25 and 60 days after sowing for increasing pistillate flower production, seed setting per cent and seed yield.

11. NIPPING

For perennial castor variety YTP 1, nipping of primary shoot at 10th internode using secateurs is recommended for tripling the productive branches.

12. INTERCROPPING

Raise one row of castor for every six rows of groundnut. In the case of late receipt of monsoon blackgram + castor at 6:1 ratio is recommended. Or Intercropping of castor with Blackgram or Greengram in 1:2 ratio is recommended for rainfed situation.

Intercropping of castor with small onion in 1:2 ratio by adopting 1.5 m x 1.0 m spacing is recommended for irrigated situation.

For hilly areas of Tamil Nadu *viz*., Kalrayan hills, Javadhu hills and Yelagiri hills, Samai (10kg/ha) + Castor 1.0 kg/ha) @ 10:1 ratio-line sowing (25cm x10 cm) with 50 % organic (FYM @ 8.0t/ha) and 50 % inorganic nutrient (22:11 kg N & P/ha) is recommended for realizing maximum profit.

13. HARVESTING THE CROP

Observe the crop considering the average duration of the variety. i) One or more capsules show sign of drying. ii) Cut the matured racemes without damaging the secondaries. iii) Dry the capsule in the sun without heaping it in the shade. iv) Use castor sheller to separate the seeds or beat the dried capsule with wooden planks, winnow and collect the seeds.

For YRCH 1, first harvest has to be done on 90 DAS, subsequently second and third harvest is to be carried out on 120 and 150 DAS, respectively.
For YRCH 2, first harvest has to be done on 110 DAS followed by second and third harvest is to be taken on 140 and 170 DAS, respectively.
The harvested spike should be sun dried and dried capsules can be shelled in the sheller.

CROP PROTECTION

CROP PROTECTION **A. Pest Management**	
Pests	**Management strategies**
Defoliators: **Semiloopers** *Achaea janata* *Paralellia algira*	• Encourage the activity of the larval parasitoid, *Microplitis maculipennis* • Spray Azadirachtin 0.03% 1000 ml • Spray any one of the following insecticides at fifteen days interval Chlorpyrifos 20EC @ 1250 ml/ha Profenophos 50 EC @ 500ml/ha Thiodicarb 75 WP @ 500g/ha Acephate 75SP @ 780 g/ha Flubendiamide 39.35 SC @ 100ml/ha Chlorantraniliprole 18.5 SC @ 150ml/ha
Tobacco caterpillar *Spodoptera litura*	• Use of light trap to monitor and kill the attracted adult moths • Set up the sex pheromone trap @ 12/ha to monitor the activity of the adults and to synchronize the pesticide application • Mechanical collection and destruction of egg masses and early stage larvae found in clusters which can be located easily even from a distance. • Hand picking and destruction of grownup of grown up caterpillars. • Spray NSKE 5 % or Azadirachtin 1 % EC (10000 ppm) 2 ml/ lit. or apply *Bacillus thuringiensis* 2g/lit. during evening hours. • Spraying nuclear polyhedrosis virus at 1.5 x 10^{12} POB per ha and virus in the evening • Apply Poison bait in the soil helps in killing the grown up larvae hide in soil during day time. Poison bait (1 kg carbaryl+10 kg rice bran+ 1 kg jiggery+ 1 litre of water to make the bait in to pellets for one hectare) • Spray any one of the following insecticides at fifteen days interval. Spraying of insecticides should be done either in the early morning or in the evening Chlorpyrifos 20EC @ 1250 ml/ha

	Profenophos 50 EC @ 1000 ml/ha Thiodicarb 75 WP @ 500g /ha Acephate 75SP @ 780 g/ha Flubendiamide 39.35 SC @ 100ml/ha Chlorantraniliprole 18.5 SC @ 150ml/ha

Other defoliators **Hairy caterpillars** *Euproctis fraternal* *Porthesia scintillans* **Slug caterpillar** *Parasa lepida* **Woolly bear** *Pericallia ricini* **Spiny caterpillar** *Ergolis merione* **Tussock caterpillar** *Orgyia postica*	• Spray Azadirachtin 0.03% 1000 ml • Spray *Bacillus thuringiensis* var *kurstaki* 5%WP 1000-1250 g/ha • Spray any one of the following insecticides Profenofos 50EC@ 500ml/ha Chlorpyrifos 20EC @ 1250 ml/ha Flubendiamide 39.35 SC @ 100ml/ha Chlorantraniliprole 18.5 SC @ 150ml/ha
Serpentine leaf miner *Liriomyza trifolii*	• Spray any one of the following plant products Neem seed kernel extract 5% Neem oil 3 % • Spray any one of the following insecticides at fifteen days interval Chlorpyrifos 20EC @ 1250 ml/ha Malathion 50 EC @ 1000 ml/ha
Sucking pests **Green leaf hopper** *Empoasca flavescens*	• Spray any one of the following insecticides at fifteen days interval Dimethoate 30EC @ 825 ml/ha Acetamiprid 20SP @ 100 g/ha Thiamethoxam 25WG @ 200 g/ha Clothianidin 50WDG @ 50 g/ha
White fly *Trialeurodes ricini*	• Monitor the activities of the adult white flies by setting up yellow pan traps and sticky traps smeared with grease or sticky oil @ 25 / acre at 1 foot height above the plant canopy • Collection and removal of white fly infested leaves those which were shed due to severe attack • Spray any one of the following plant products Neem seed kernel extract 5% Neem oil 3 % • Spray Fish oil rosin soap 25g / lit of water • Spray any one of the following insecticides at fifteen days interval Imidacloprid 17.8 SL 125 ml/ha Dimethoate 30 EC @ 825 ml/ha Acetamiprid 20%SP 100g /ha

	Thiamethoxam 25WG @ 200 g/ha Profenophos 50%EC 1000 ml/ha Thiacloprid 21.7%SC 600ml/ha
Flower thrips *Retithrips siriacus* *Scirtothrips dorsalis*	• Spray any one of the following insecticides at fifteen days interval Imidacloprid 17.8 SL 125 ml/ha Dimethoate 30 EC @ 825 ml/ha
Shoot and Capsule borer *Conogethus punctiferalis*	• Spray Neem oil 3% twice at 15 days interval during flowering stage to prevent the adults to lay eggs • Spray any one of the following insecticides from flowering at fifteen days interval Profenofos 50EC@500ml/ha Malathion 50 EC @ 1000ml/ha Indoxacarb 15.8EC @ 500ml/ha Spinosad 45SC@75 ml/ha Thiodicarb 75WP@ 500g/ha

B.Disease Management

Disease	Recommendations
Grey mold: *Botrytis ricini*	• Remove and destroy the infected spikes • During cloudy weather and rainy season, give prophylactic spray with carbendazim @ 2 g/l twice at 15 days interval or prophylactic spray of *P. fluorescens* @ 2g/l and second spray after a fortnight.
***Fusarium* Wilt**	• Seed treatment with carbendazim @ 2gram/ kg of seed • Soil drenching with carbendazim @ 2gram / litre of water.

CASTOR – VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 300 m all around the field from the other varieties / hybrids of the crop.

Season

- June - July and September - October.

Pre-sowing seed treatment

- Seed hardening with 2 % KH_2PO_4 for 16 h (seed to solution ratio 1:1) and dry back to original moisture content.

Fertilizer requirement

- Apply NPK @ 90:70:70 kg / ha as basal.

Spacing

- 90 x 30 cm.

Harvesting

- Harvest the crop as once over harvest when 80 % of the capsules turn into brown colour.

Threshing

- Thresh the capsules either using power operated thresher or manually by trampling or beating with pliable bamboo stick.

Seed grading

- Grade the seeds at 10 % moisture content using 18 / 64" round perforated sieve.
- Discard the broken and immature seeds for seed purpose.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 5 %.

CASTOR - HYBRID SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 300 m all around the field from the other varieties / hybrids of the crop.

Planting ratio

- Sow the female and male parents in the ratio of 3:1 or 4:1 for certified seed production.

Border rows

- Sow four rows of male parents in around the field for the availability of adequate pollen.

Season

- Sow the female line during first fortnight of September for production of more pistillate inflorescence and male line one week later.

Fertilizer requirement

- Apply NPK @ 90:70:70 kg / ha as basal application.

Spacing

- 90 X 30 cm.

Harvesting

- Harvest the racemes as once over harvest when 80 % of the capsules turn to brown colour.
- The seeds from secondary raceme are better than primary and others.

Threshing

- Shell the seeds either using power operated thresher or manually by beating with pliable bamboo stick.
- Avoid hand operated thresher to reduce the mechanical damage.

Seed grading

- Grade the seeds using 18 / 64” round perforated sieve.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1@ 3 g / kg of seed as eco-friendly treatment.
- Mix the seeds with dry sweet flag (or) vasambu (*Achorus calamus)* rhizome powder at the ratio of 1:100 for grain cum seed storage.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than15 months) with a seed moisture content less than 5 %.
- The seeds of female parent are poor storer than male and hybrid.

(iv) SUNFLOWER (*Helianthus annuus*)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
38 - 40	10 - 15	20 - 30	350 - 600	up to2500

Tropical and subtropical climate. During vegetative phase, crop requires cold temperature. Higher temperature (> 38°C) during reproductive stage reduces the oil content. Day neutral plant. Crop gives highest yield of oil per hectare when grown below 1,500m MSL.Cannot tolerate drought and water logging.

CROP IMPROVEMENT

I. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES
A. Rainfed	
1. Adipattam (Jun-July)	
Coimbatore, Erode, Salem, Namakkal,	Variety : COSFV 5
Tirunelveli, Dindigul, Dharmapuri, Tiruchirapalli, Perambalur, Karur	Hybrid : COH 3.
2. Karthigaipattam(Oct- Nov) Cuddalore, Villupuram, Virudhunagar, Sivagangai , Ramanathapuram, Madurai, Dindigul, Theni, Tiruchirapalli, Perambalur, Karur, Tirunelveli	Variety : COSFV 5 Hybrid : COH 3
B. Irrigated	
1. Adipattam (July-August) Coimbatore, Erode, Salem, Namakkal,Tirunelveli, Dindigul, Dharmapuri, Tiruchirapalli, Perambalur, Karur	Variety : COSFV 5 Hybrid : COH 3
2. Karthigaipattam (Nov-Dec) Cuddalore, Villupuram, Virudhunagar, Sivagangai , Ramanathapuram, Madurai, Dindigul,Theni, Tiruchirapalli, Perambalur, Karur, Tirunelveli	Variety : COSFV 5 Hybrid : COH 3
3. Margazhipattam (Dec-Jan) Salem, Namakkal, Dharmapuri, Erode, Coimbatore, Madurai, Dindigul, Theni, Tirunelveli, Thoothukudi	Variety : COSFV 5 Hybrid : COH 3
4. Chithiraipattam (April - May) Coimbatore, Erode, Dharmapuri, Salem, Namakkal, Tiruchirapalli, Perambalur, Karur	Variety : COSFV 5 Hybrid : COH 3

I. DESCRIPTION OF SUNFLOWER VARIETIES

Particulars	COSFV 5	COH 3
Year of Release	2006	2017
Year of Notification	SO.1178(E)/20.07.2007	S.O. 6318(E) / 26.12.2018
Parentage	Cross derivative of *Helianthus annus X H. preacox*	COSF 6A X IR 6
Duration (days)	85-90	90-95
Yield (kg/ha)		
Rainfed	1500	2150
Irrigated	1700	2410
Oil content (%)	40-42	40- 42
Ray floret	Yellow	Yellow
Plant height (cm)	145-165	160-170
Seed size & colour	Dark brown	Black
1000 seed weight (g)	48-50	52
Volume weight (g/100ml)	45-48	47

CROP MANAGEMENT

1. FIELD PREPARATION

Plough once with tractor or twice with iron-plough or three to four times with country-plough till all the clods are broken and a fine tilth is obtained.

2. APPLICATION OF FERTILIZERS

i) Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the field before the last ploughing and incorporate in the soil by working a country plough.

If soil test recommendations are not available, follow the blanket NPK/ha for both irrigated and rainfed crops.

	Season	Blanket recommendation of Nutrients (kg/ha)		
		N	P_2O_5	K_2O
Hybrids	**IRRI**	60	90	60
	RF	*40*	*50*	*40*
Varieties	**IRRI**	60	30	30
	RF	*40*	*50*	*40*

Soil test crop response based integrated plant nutrition system (STCR- IPNS recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (read reckoners are furnished for irrigated sunflower)

Sunflower - Hybrid

Soil : Mixed black calcareous (Perianaickenpalayam series)

Target : 2.0- 2.5 t ha^{-1}

FN =9.60T- 0.49SN-0.68 ON

FP_2O_5=4.20T -1.87SP-0.80 OP

FK_2O=9.24T-0.45SK-0.64 OK

Initial soil test values (kg ha^{-1})			Yield target – 2.0 t ha^{-1}			Yield target – 2.5 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
160	12	300	59	45*	30*	90**	53	56
180	14	325	49	45*	30*	90**	49	45
200	16	350	39	45*	30*	87	45*	34
220	18	375	30*	45*	30*	77	45*	30*
240	20	400	30*	45*	30*	67	45*	30*

* Maintenance dose;** Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

iii) Biofertilizer : Soil application - Mix 10 packets (2000 g/ha) of Azospirillum and 10 packets(2000 g/ha) of Phosphobacteria or 20 packets of Azophos(4000 g/ha) with 25 kg FYM and 25 kg soil and apply before sowing.

3. APPLICATION OF MICRONUTRIENTS

a) Mix 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make total quantity of 50 kg/ha. For **rainfed sunflower** apply TNAU MN mixture @ 7.5 kg ha^{-1} as enriched FYM

b) variety and 10 kg ha^{-1} as enriched FYM for hybrid and for ***Irrigated sunflower*** apply TNAU MN mixture @ 12.5 kg ha^{-1} as enriched FYM for variety and 15 kg ha^{-1} as enriched FYM for hybrid (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mixat friable moisture &incubate for one month in shade).

c) Apply the mixture over the furrows and top two thirds of the ridges before sowing.

d) Do not incorporate the mixture in the soil.

(i) To overcome manganese deficiency foliar spray of 0.5% $MnSO_4$ on 30,40 & 50th day after sowing and

(ii) For zinc deficiency apply 25 kg $ZnSO_4$/ha as basal or 0.5% $ZnSO_4$. Spray on 30,40 & 50th day after sowing.

(iii) For B and S deficient soils, apply 10 kg Borax or 0.2% boric acid twic and 40 kg S as Gypsum/ha.

4. FORMING RIDGES AND FURROWS

i. Form ridges and furrows with 60 cm spacing.

ii. Use bund-former or ridge plough to economise and

iii. Form irrigation channels across and ridges according to the topography of the field.

5. SEED RATE

	Rainfed	Irrigated
Varieties	7 kg/ha	6 kg/ha
Hybrids	5 kg/ha	5 kg/ha

6. SEED TREATMENT

Soaking seeds in 2% for 12 hrs and shade drying is recommended for rainfed $ZnSO_4$ sowing.

7. SEED TREATMENT

i) Treat the seed with *Trichoderma* @4g/kg. This can be done just before sowing. It is compatible with biofertilizers. Such seeds should not be treated with fungicides.

ii) Treat the seeds with Carbendazim or Thiram at 2 g/kg of seed.

iii) Treat the seeds 24 hours prior to sowing.

iv) Treat the seeds required for sowing 1 ha with 600 g of *Azospirillum* and 600g of phosphobacteria (or) 600g of Azophos using rice gruel as binder, shade dry the treated seeds for 30 min and sow immediately.

Liquid formulation Treat one hectare of seeds with 125 ml of *Azospirillum* and 125 ml of Phosphobacteria, shade dry it for 30 minutes before sowing

v) Moist hydration for 24 hours in moist gunny bags followed by drying and seed dressing with Thiram @ 2g/kg to enhance field emergence.

vi) Seeds dried to 8 - 9% moisture content, treated with Thiram @ 2g/kg and packed in polylined (300 guage) cloth bag can store upto 9 months with 70% germination.

8. SOWING

Spacing : Hybrids : 60 cm x 30cm

Varieties : 45 cm x 30cm

i) Place the seeds at a depth of 3 cm along the furrows in which the fertilizer mixture is placed. Put two seeds per hole

9. THINNING

Thin out seedlings leaving only one healthy and vigorous seedling in each hole on the 10^{th} day of sowing.

10. WEED MANAGEMENT

i) Apply Fluchloralin @ 1.0 lit/ha before sowing and incorporate or apply as pre-emergence spray on 3^{rd} day after sowing followed by irrigation or apply Pendimethalin @ 1.0 litre/ha as pre- emergence spray 3 days after sowing. The spray of these herbicides has to be accomplished with Knapsack sprayer fitted with flat fan nozzle using 500 lit water/ha as spray fluid.

All the herbicide application is to be followed by one late hand weeding 30 - 35 days after sowing.

After application of pre emergence herbicide, instead of hand weeding, power weeder can be used if sowing was done with the spacing of 75 x 25 cm.

ii) If pre emergence herbicide was not applied, hand weeding to be done on on 15^{th} and 30^{th} day after sowing and remove the weeds.

11. WATER MANAGEMENT

Irrigate immediately after sowing followed by an irrigation on 4 – 5th day and later at interval of 7 to 8 days according to soil and climatic conditions at seeding, flowering and seed development stage.

12. FOLIAR SPRAY OF NAPHTHALENE ACETIC ACID (NAA)

i) Foliar spray of Napthalene Acetic Acid (NAA) at 20 ppm concentration (280 g NAA in 625 litres of water per ha) on the 30th and 60th day of sowing.

ii) Use a high volume sprayer and give a thorough coverage of the entire plant.

iii) Do not use brackish water.

13. SULPHUR FERTILIZATION

Apply sulphur @ 20 kg/ha through ammonium sulphate or single super phosphate. Or apply gypsum@ 200kg/ha as basal

14. BORIC ACID

Foliar spray of boric acid @ 0.2 % (2g/l of water) to capitulum at ray floret opening stage to improve seed set and seed filling.

15. IMPROVING SEED SET BY MECHANICAL MEANS

a. During the mid flowering phase, improve pollination by :
   i. Mild rubbing of the capitulum with the hand covered with soft cloth or
   ii. Rubbing two flowers face to face gently.
   iii. The mid-flowering phase are: 58 to 60 days of planting for long duration varieties, 45 to 48 days of planting for short duration varieties
   iv. .Do this operation in the morning hours between 9.0 and 11.00 am when pollen shedding is high.

b. Keeping bee hives at the rate of 5/ha improves seed setting.

16. JUDGE WHEN TO HARVEST

Observe the bracts on the backside of the capitula. When they turn lemon yellow, the heads harden and the crop is ready for harvest.

Bird damage: Use of reflective ribbons scares the birds effectively and thus prevents loss of grain.

17. HARVESTING

i. Cut the capitula (flower heads) only

ii. Thresh and clean
   a. Immediately after harvest, dry the heads in the sun for 3 days.
   b. Spread the heads in thin layer and give turning once in 3 hours.

NOTE: Do not heap or store the heads before drying properly as mould fungi will develop and spoil the grain quality.

   c. Thresh using a mechanical thresher, or beat with a stick and separate the grains.
   d. Winnow and clean the seeds
   e. Dry the seeds again in the sun for another two days
   f. Store in gunny bags

CROP PROTECTION

A. Pest management

Weevil *Myllocerus* spp. **Tobacco cutworm** *Spodoptera litura* **Gram podborer** *Helicoverpa armigera*	▪ Hand pick the *Helicoverpa* larvae and destroy. ▪ Spray Azadirachtin 5% W/W 0.5 ml/lit
Leafhopper *Amrasca devastans*	▪ Treat seed with imidacloprid 70 WS at 7 g/kg protection upto 7 weeks. ▪ Spray Imidacloprid 70 WS 490 ml/ha (or) Imidacloprid 17.8 SL 100 ml/ha
Whitefly, *B.tabaci, A.dispersus*	▪ Spray Imidacloprid 70 WS 490 ml/ha (or) Imidacloprid 17.8 SL 100 ml/ha
Thrips	▪ Spray Imidacloprid 17.8 SL 100 ml/ha

B. Disease management

Alternaria leaf spot: *Alternaria helianthi*	• Spray mancozeb @ 1000 g/ha or • Treat the seeds with carbendazim + mancozeb @ 3g/kg + propiconazole 0.1 % sprays at 30 and 45 days after sowing or • Treat the seeds with *Pseudomonas fluorescens* @ 10 g/kg seeds along with foliar spray of hexaconazole or propiconazole @ 0.1% at 45 days after sowing and foliar spray of *P. fluorescens* at 60 days after sowing
Rust: *Puccinia helianthi*	Spray mancozeb @ 1000 g/ha
Charcoal rot: *Macrophomina phaseolina* *(Rhizoctonia bataticola)*	• Soil application of *P. fluorescens* or *T. asperellum* @ 2.5 kg / ha with 50 kg of well decomposed FYM or sand at 30 days after sowing. • Spot drenching with carbendazim @ 1 g/ l
Powdery mildew: *Golovinomyces cichoracearum*	Two sprays of difenoconazole @ 0.05% at 40 and 60 DAS
Head rot: *Rhizopus* sp	Spray mancozeb @ 1000 g/ha in case of intermittent rainfall at the head stage, directing the spray to cover the capitulum. Repeat fungicidal application after 10 days, if humid weather continues
Necrosis virus disease: *Tobacco streak virus* (Ilarvirus) (Vector: Thrips)	• Raise sorghum as border crop (one month prior to sunflower sowing). • Seed treatment with imidacloprid @ 2 g/kg of seeds • Spray imidacloprid 17.8 SL @ 100 ml/ha

SUNFLOWER - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 200 m all around the field from the same and other varieties / hybrids of sunflower.

Spacing

- 45 x 30 cm.

Pre-sowing seed treatment

- Seed soaking in 2 % KNO_3 for 6 hrs to release dormancy.
- Seed hardening with 2 % KH_2PO_4 for 16 h and dry back to original moisture content.
- Seed coating with polymer @ 3 g / kg + imidachloprid @ 2 ml / kg + carbendazim @ 2 g / kg + *Pseudomonas fluorescens* @ 10 g / kg.

Fertilizer

- Apply NPK @ 60:45:45 kg / ha as basal application.

Foliar application

- At the stage of capitulum opening, spray 0.2 % boric acid for increased seed set.

Supplementary pollination

- During flowering, rub the heads with muslin cloth or palm during 8 - 11 am on alternate days till the completion of flowering (7 - 10 days).
- Keep bee hives @ 5 nos. / ha to increase insect activity.

Harvesting

- Harvest the heads when the thalamus drooped and turned to lemon yellow in colour with black coloured seeds.
- Harvest the heads and dry immediately until the seed moisture content reduced to 15 - 16 %.
- Separate the seeds either with mechanical thresher or manually.

Seed grading

- Grade the seeds using 9 / 64" round perforated sieve.
- Upgrade the size graded seed using specific gravity separator.
- Remove the broken and dehulled seeds from the lot.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed along with carbaryl @

200 mg / kg of seed.

- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 7 %.

SUNFLOWER - HYBRID SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 400 m all around the field from the same and other varieties / hybrids of sunflower.

Border rows

- Sow four rows of male parent around the field for the availability of adequate pollen.

Planting Ratio

- Sow female and male plants in a ratio of 4:1 or 6:1

Foliar spray

- Spray 0.2 % boric acid at button opening stage to increase seed set.

Supplementary pollination

- During flowering, collect pollens from male flowers and smear the pollen over the female heads with muslin cloth or palm at the time between 8 - 11 am on alternate days till the completion of flowering.
- Keep bee hives @ 5 nos. / hectare.

Harvesting

- Harvest 'R' lines first and remove from the field before harvesting the hybrid.
- Harvest the earheads of female plants as once over harvest.

(v) SAFFLOWER (*Carthamus tinctorius*)

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
38 - 40	5	22 - 30	350 - 450	up to 1000

Tropical and subtropical semi arid climate. Grown during rabi, primarily as a rainfed crop. Yields are lower under humid or rainy conditions due to reduced seed set and increased disease incidence. Not recommended for areas with >450 mm of annual precipitation. Tolerated very low temperature during the rosette stage, but very sensitive to frost injury after stem elongation until crop maturity. This crop does best in areas with warm temperatures and sunny, dry conditions during the flowering and seed-filling periods. Temperature for seed germination is 15^{o}C.

CROP IMPROVEMENT

DISTRICT/SEASON	VARIETIES
A. Rainfed	
1. Karthigaipattam (Nov-Dec)	
All districts	K 1, CO 1

I. DESCRIPTION OF SAFFLOWER VARIETIES

Particulars	K 1	CO 1
Parentage	Pureline selection from American spiny variety	Pureline selection from CTS 7403 (Non spiny)
Duration (days)	120	125
Yield (kg/ha)		
Rainfed	700	800
Oil content (%)	32	33
Special features	spiny florets, suitable for Southern districts	Non-spiny, tolerant Alternaria, moderately Resistant to wilt

CROP MANAGEMENT

III. PREPARATION OF THE FIELD

1. FIELD PREPARATION

a) Plough with tractor 2-3 times with a mould board plough or 5 times with a country plough.

b) Break the clods in between the ploughings and bring the soil to a fine tilth.

2. APPLICATION OF FYM

a) Spread 12.5 t of FYM or compost or composted coir pith per ha evenly and incorporate in the soil. b) If the manure is not applied before commencement of ploughing, spread the manure evenly before the last ploughing and incorporate in the

soil.
NOTE: Do not leave the organic manure exposed to sunlight as nutrients will be lost.

3. APPLICATION OF FERTILIZERS

Apply N at 20 kg/ha basally.

4. SEED RATE

Adopt a seed rate of 10 kg/ha.

5. SPACING

Adopt a spacing of 45 cm between rows and 15 cm between plants.

6. SELECTION OF GOOD QUALITY SEEDS

Select mature good quality seeds, free from pest damage and fungal attack.

7. PRE-TREATMENT OF SEEDS WITH FUNGICIDES

a) Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. b) Treat the seeds 24 hours prior to sowing.

NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage.

8. SOWING

a. Sow the seeds in line at a depth of 2 to 3 cm and cover with soil. b. Sow using gorru or country plough.

NOTE: First week of November is the best sowing time.

9. THINNING OUT SEEDLINGS

Thin out the seedlings to a spacing of 15 cm between plants on the 15th day of sowing.

10. WEED MANAGEMENT

Weeding with have hoe on 25 and 40 days after of sowing (DAS)

11. HARVESTING

i. Observe the crop considering the average duration of the crop.
ii. The leaves and entire plant loose their colour and turn brown at maturity.
iii. Cut the plants at the bottom.
iv. Keep the plants in the threshing floor and beat the plants (heads) with sticks till the mature seeds are separated.
v. Winnow the seed and dry in the sun.
vi. Collect and store the seeds in gunnies.

CROP PROTECTION

Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. Treat the seeds 24 hours prior to sowing.

NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage.

SAFFLOWER - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 200 m all around the field from the same and other varieties of the crop.

Spacing

- 60 x 20 m.

Fertilizer

- Apply NPK @ 60:60:20 kg / ha as basal application.

Harvest

- Harvest the pods as once over harvest.

Seed grading

- Grade the seeds using BSS 6 x 6 wire mesh sieve.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2g / kgof seed.
- Treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1@ 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 7 %.

(vi) Coconut (*Cocos nucifera*); Palmae

Climate

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
38 - 40	10 - 15	25 - 30	800 - 2500	up to 600

Tropical and subtropical climate. Withstand water logging. Minimum sunshine shou hours per year.

VARIETIES AND HYBRIDS

Varieties

i. East Coast Tall
ii. West Coast Tall
iii. VPM-3 (Selection from Andaman Ordinary Tall)
iv. ALR (CN -1) (Selection from Arasampatty Tall)
v. ALR (CN-2) (Selection from Tiptur Tall)
vi. COD (Dwarf for tender coconut purpose only)
vii. VPM 4 (Selection from WCT)
viii. ALR 3 (Dwarf for tender nut purpose only and Selection from Kenthali Dwarf)

Hybrids

Tall x Dwarf

(To be grown under well managed conditions)

ix. VHC 2 - ECT X MYD
x. VHC 3 - ECT X MOD
xi. VPM 5 – LCT x CCNT

(Besides, the hybrids of ECT x COD, WCT x COD and WCT x MYD are also produced by the State Department of Agriculture. The dwarf x tall type (COD x WCT) which has to be grown under well-managed conditions with assured irrigation is also produced by State Department of Agriculture).

CROP MANAGEMENT

Soil

Red sandy loam, laterite and alluvial soils are suitable. Heavy, imperfectly drained soil is unsuitable.

Seasons

June-July, December - January. The planting can also be taken up in other seasons wherever irrigation and drainage facilities are available.

Spacing

Adopt a spacing of 25' x 25' (7.5 x 7.5 m) with 175 plants/ha. For planting in field border as a single row, adopt 20' spacing between plants.

Planting

Dig pit size of 3’ x 3' x 3'. In the pits, sprinkle Lindane 1.3 % D to prevent white ant damage. Fill the pit to a height of two feet (60 cm) with FYM, red earth and sand mixed in equal proportions. At the center of the pit, remove the soil mixture and plant the seedling after removing all the roots. Press the soil well around the seedling and provide the seedling with shade by using plaited coconut leaves or palmyrah leaves. Keep the pits free from weeds. Remove soil covering the collar region. As the seedlings grow and form stem, fill up the pits gradually by cutting the sides.

Water management

From 5th year onwards, adopt the following irrigation schedule based on pan evaporation for drip irrigation and basin irrigation.

Western region

Months	Normal condition (for best yield)	Moderate water scarcity condition	Severe water scarcity condition
A. Drip irrigation			
February to May	65 lit / day	45 lit/ day	22 lit / day
January, August and September	55 lit / day	35 lit / day	18 lit/day
June and July, October to December	45 lit / day	30 lit/ day	15 lit / day
B. Basin irrigation			
February to May	410 lit / 6 days*		
January, August and September	410 lit /7 days*		
June and July, October to December	410 lit /9 days*		

Eastern region

Months	Normal condition (for best yield)	Moderate water scarcity condition	Severe water scarcity condition
A. Drip irrigation			
March - September	80 lit / day	55 lit / day	27 lit/day
October – February	50 lit / day	35 lit/ day	18 lit /day
B. Basin irrigation			
March – September	410 lit / 5 days*		
October – February	410 lit /8 days*		

* Quantity of water to be applied in the basin. Add 30-40 % of the above quantity of water (135 -165 litres/palm) to meet the conveyance loss.

For drip irrigation, open four pits size of 30 x 30 x 30 cm opposite to each other at one meter distance from the trunk. Place 40 cm long PVC conduit pipe (16 mm) in a slanting position in each pit and place the drippers inside the conduit tube and allow the water to drip 30 cm below the soil surface. Fill the pits with coir pith to prevent evaporation.In the first year, irrigate on alternate days and from the second year to the time of maturity irrigate twice in a week based on the water requirement.

Drought management and soil moisture conservation

Mulching with coconut husks/leaves/coir pith

Apply coconut husks with convex surface facing upwards (100 Nos.) or dried coconut leaves (15 Nos) or coir pith up to a height of 10 cm in the basin of 1.8 m radius around the palms as mulch for soil moisture conservation particularly during summer season.

Burial of coconut husk or coir pith

Husk burial can be done in coconut basins or in the interspaces to overcome drought and button shedding. Bury husks @ 100 Nos. with concave surface facing upwards or 25 kg of coir pith/palm in circular trenches, dug 30 cm width and 60 cm depth at 1.5 metres radius. The husk can be also buried in the trenches at a distance of 3 m from the palm with a size of 45 cm deep and 150 cm width in between two rows of coconut. The soaking of the coconut husk or coir pith as the case may be preserves the monsoon rains.

Manuring

From 5th year onwards, apply 50 kg of FYM or compost or green manure.1.3 kg Urea (560 g N), 2.0 kg Superphosphate (320 g P_2O_5) and 2.0 kg Muriate of potash (1200 g K_2O) in two equal splits during June – July and December – January. Apply manures and fertilizers in circular basins of 1.8 m from the base of the palm, incorporate and irrigate. During 2nd, 3rd and 4th year ¼, ½ and ¾ doses of the above fertilizer schedule should be adopted respectively. Sufficient moisture should be present at the time of manuring. Fertigation may be done at monthly intervals with 75% of the recommended dose of the above fertilizers. Phosphorous may be applied as super phosphate in the basins and incorporated or as DAP through drip when good quality of water is available. TNAU micronutrient mixture is recommended @ 1.0kg/tree/year.

TNAU coconut tonic nutrition

For nut bearing coconut Palm, root feed TNAU coconut tonic @200ml/palm once in six months.

Bio-fertilizer recommendation

At the time of planting, apply 50g of *Azospirillum,* 50 g of Phosphobacteria (or) 50 g of Azophos and 50 g of AM fungi. Mix all the contents with sufficient quantity of FYM or any compost. After planting apply the above biofertilizers once in 6 months/palm near to the feeding roots as that of fertilizer application

Organic recycling

Any one of the green manure crops like sunnhemp, wild indigo, calapagonium or daincha may be sown and ploughed *in situ* at the time of flowering as a substitute of compost to be applied. Sow sunnhemp @ 50 g/palm in the basin and incorporate before flowering. Coir pith compost/vermicompost made from coir pith/ coconut leaves/ other wastes from coconut grove can be applied.

Intercultural operation weed management

The interspace in the coconut garden has to be ploughed twice in a year in June-July and December - January. Intercultural operation is essential to keep weed population under check, to enhance the utilisation of the applied plant nutrients by the coconut trees, to facilitate proper aeration to the roots of coconut, to induce fresh root growth.

Weed management

For the broad-leaved weeds, pre-emergence spraying of atrazine @1.0 kg / ha for the control of grasses and sedges, post emergence spraying of glyphosate @ 10 -15 ml and 20 g Ammonium sulphate + 2 ml soap solution /litre of water.

Inter cropping

Inter/mixed crops may be selected based on the climatic requirement of the inter/mixed crop, irrigation facilities and soil type. The canopy size, age and spacing of the coconut are also to be considered. Market suitability should be taken into consideration before selecting an intercrop.

Below 7 years of age: Any suitable annual crop for particular soil type and climatic condition may be raised as intercrops upto 5 years after planting depending upon the canopy coverage. Groundnut, sesamum, sunflower, tapioca, turmeric and banana can be grown. Avoid crops like paddy and sugarcane etc.

7 – 20 years of age: Green manure crops and fodder crops (Napier grass and guinea grass) alone can be grown.

Above 20 years of age (20 years of age has to be adjusted based on the sunlight transmission of above 50% inside the canopy).

The following crops can be grown depending on the soil and climatic suitability.

(i) **Annuals**	:	Groundnut, bhendi, turmeric, tapioca, sweetpotato, sirukizhangu, elephant foot yam, ginger, pineapple
(ii) **Biennials**	:	Banana varieties, poovan and monthan are suitable.
(iii)**Perennials**	:	Cocoa*, pepper* (Panniyur 1 or Panniyur 2 or Panniyur 5 or Karimunda), nutmeg* and vanilla*

*Suitable areas are Pollachi tract of western region and Kanyakumari district. For vanilla,
use disease free planting material and maintain high vigilance to maintain a disease free crop.

Multiple cropping system

Coconut + banana + sirukizhangu + bhendi is suitable system for the eastern region. Crops like banana, pepper, cocoa, nutmeg, vanilla can be tried under multiple cropping system in suitable areas in the western region. In all the systems, apply recommended quantity of water and manures and fertilizers to the intercrops separately.

Crop physiology

Root feeding of TNAU coconut tonic @ 200 ml / palm twice a year at six months interval decreases button shedding and increases the number and size of nuts.

Crop protection

Pest management

Pests	Management strategies
Rhinoceros beetle *Oryctes rhinoceros*	▪ Remove and burn all dead coconut trees in the garden (which are likely to serve as breeding ground) to maintain good sanitation. ▪ Collect and destroy the various bio-stages of the beetle from the manure pits (breeding ground of the pest) whenever manure is lifted from the pits. ▪ Incorporate the entomopathogen i.e, fungus (*Metarrhizium anisopliae*) in manure pits to check the perpetuation of the pest. ▪ Soak castor cake at 1 kg in 5 l of water in small mud pots and keep them in the coconut gardens to attract and kill the adults. ▪ Treat the longitudinally split tender coconut stem and green petiole of fronds with fresh toddy and keep them in the garden to attract and trap the beetles. ▪ Examine the crowns of tree at every harvest and hook out and kill the adults. ▪ For seedlings, apply 3 naphthalene balls/palm weighing 3.5 g each at the base of inter space in leaf sheath in the 3 inner most leaves of the crown once in 45 days. Set up light traps following the first rains in summer and monsoon period to attract and kill the adult beetles. ▪ Field release of Baculovirus inoculated adult rhinoceros beetle @ 15/ha reduces the leaf and crown damage caused by this beetle. ▪ Apply mixture of either neem seed powder + sand (1:2) @150 g per palm or neem seed kernel powder + sand (1:2) @150 g per palm in the base of the 3 inner most leaves in the crown ▪ Place phorate 10 G 5 g in perforated sachets in two inner most leaf axils for 2 times at 6 months intervals. ▪ Set up rhinolure pheromone trap @ 1/ 2 ha to trap and kill the beetles.
Black headed caterpillar *Opisina arenosella*	▪ The incidence of the pest is noticed from the month of November to May and from August to November after rainfall. The coconut trees of all ages are attacked. • Release the larval (*Bethylid, Braconid* and *Ichneumonid*) and pupal (*Eulophid*) on (chalcid) parasitoids and predators periodically from January, to check the build up of the pest during summer. ▪ Among the larval parasitoids, the bethylid *Goniozus nephantidis* is the most effective in controlling the pest. The optimum level of release is 1:8 of host-parasitoid ratio. The parasitoi should be released @3000/ha under the coconut trees when the pest is in the 2nd or 3rd instar larval stage. Parasitoid release trap may be used to release the parasitoid at the site of feeding. Parasitoids should not be released in the crown region since they will be killed by predators like spiders and reduviid bugs. ▪ Remove and burn all affected leaves/leaflets. ▪ Spray Malathion 50 EC 0.05% (1mi/lit) to cover the undersurface of the leaves thoroughly in case of severe epidemic outbreak of the pest

	in young palms. ▪ **Root feeding for the control of coconut Black headed caterpillar:** Select a fresh and live root, cut sharply at an angle and insert the root in the insecticidal solution containing monocrotophos 36 WSC 10 ml + water 10 ml in a 7 x 10 cm polythene bag. Secure the bag tightly to the root with a cotton thread. Twenty four hours later, check whether there is absorption. If there is no absorption select another root. These methods should not be resorted to as a routine practice and it is suggested only for cases of severe epidemic outbreak of the pest and when the survival of the tree is threatened.
Red palm weevil *Rhynchophorus ferrugineus*	▪ Remove and burn all wilting or damaged palms in coconut gardens to prevent further perpetuation of the pest. ▪ Avoid injuries on stems of palms as the wounds may serve as oviposition sites for the weevil. Fill all holes in the stem with cement. ▪ Avoid the cutting of green leaves. If needed, they should be cut about 120 cm away from the stem. ▪ Fill the crown and the axils of top most three leaves with a mixture of fine sand and neem seed powder or neem seed kernel powder (2:1) once in three months to prevent the attack of rhinoceros beetle damage in which the red palm weevil lays eggs. ▪ Setting up of attractant traps (mud pots) containing sugarcane molasses 2½ kg or toddy 2½ litres + acetic acid 5 ml + yeast 5 g + longitudinally split tender coconut stem/logs of green petiole of leaves of 30 numbers in one acre to trap adult red palm weevils in large numbers. ▪ Install pheromone trap @1/2 ha **Root feeding**: As under black headed caterpillar
Termites *Odontotermes obesus*	▪ Locate termite mounds in or near the coconut nursery or garden and destroy. ▪ Swabbing with neem oil 5% once on the base and upto 2 m height of the trunk for effective control. ▪ Spray copper sulphate 1% or cashew nut shell oil 80% or spray chlorphyriphos @ 3ml/lit of water, neem oil 5% or NSKE 20% to preserve plaited coconut leaves from the termite attack.
Scale insect *Aspidiotus destructor*	▪ Pluck mature nuts and spray monocrotophos 36 WSC 1 ml/ha. ▪ Do not harvest nuts for 45 days after spraying.
Mealy bugs *Pseudococcus longispinus*	▪ Remove leaflets harbouring these insects and destroy them ▪ **Spray any one of the following :** Malathion 50 EC 2 ml/lit Dimethoate 30 EC 1 ml/lit Methyl demeton 25 EC 1 ml/lit Phosphamidon 40 SL 1.25 ml/lit ▪ Monocrotophos 36 WSC 1 ml/lit Neem oil 30ml/lit.

<table>
<tr>
<td>Leaf caterpillars
Turnaca acuta
Nut caterpillar
Nut coreid bug</td>
<td>▪ Collect and destroy the immature stages of the insects by conducting study (or neem compaign) wherever possible and spray carbary 50 WP 2 gm/lit .
▪ Root feeding with monocrotophos 36 WSC @ 10 ml + 10 ml water at 45 days interval for 3 times for control of leaf caterpillar.
▪ Set up light trape to trap and collect adult moths
▪ Spray Dichlorvas 76 WSC 2 ml / lit.</td>
</tr>
<tr>
<td>Slug caterpillar
Contheyla rotunda</td>
<td>Spray any one of the following:
▪ Dichorvos 76 WSC 2 ml/lit
▪ Bacillus thuringiensis 2 g/lit,
▪ Triazophos 40 EC 5 ml
▪ Methyl demeton 25 EC 4 ml/lit
▪ Root feeding with monocrotophos 15 ml + 15 ml of water</td>
</tr>
<tr>
<td>Scolytid bark borer beetles
Xyleborus parvulus</td>
<td>▪ Stem injection through a stove wick soaked in 0.2% dichlorvos and plugging the hole and repeating the treatment using the same wick and
hole a month after.</td>
</tr>
<tr>
<td>Eriophyid mite
Aceria guerreronis</td>
<td>Manurial and fertilizer recommendation (Soil application/tree/year)
Urea 1.3 kg
Superphosphate 2.0 kg
Muriate of potash* 3.5kg
* Increased quantity is recommended to increase the plant resistance to the mite. Neem cake application @ 5 kg Organic manure (well rotten FYM) @ 50 kg
1. Micronutrients (Soil application / tree / year)
Borax 50 g
Gypsum 1.0 kg Magnesim sulphate 500g
Grow sunnhemp as intercrop twice a year (Seed rate 30 kg/ha)

Spot application of ecofriendly botanicals
<table>
<tr><th>Round</th><th>Eco-friendly Botanical</th><th>Quantity / tree</th></tr>
<tr><td>1</td><td>Azadirachtin 1%</td><td>5 ml in one lit. of water</td></tr>
<tr><td>2</td><td>Neem oil + Teepol</td><td>30 ml in one lit. of water</td></tr>
<tr><td>3</td><td>Azadirachtin 1%</td><td>5 ml in one lit. of water</td></tr>
</table>
Method of application
✓ The botanicals should be applied in the sequence indicated above at 45 days interval using a one litre hand sprayer. Rocker or Pedal sprayer can be used for spraying small trees.
✓ The spray should be applied at the crown region by a climber covering only the top six bunches during non rainy season
✓ The bunches must be covered well by the spray fluid and approximately one litre of spray fluid may be required per tree
Precautions and safety measures
❖ Spraying should be avoided during windy season to prevent contamination
❖ At the time of spraying, protective mask and clothing should be used
❖ Wash face and hands cleanly with soap after spraying.</td>
</tr>
</table>

Coconut Rugose spiraling whitefly	The coconut rugose spiraling whitefly was noticed in serious proportion in various district coconut gardens of Tamil Nadu. The insects suck the sap and cause damage in the leaf fronts with copious honey dew secretions on the leaves. It induce development of sooty mould fungus there by leaves become completely black and reduces the photosynthesis rate. The following TNAU technologies can be adopted to manage the spiraling whitefly, ➢ Release of Encarsia guadeloupae @ 100 parasitoids /ac (10 leafbits/ac) ➢ Installation of yellow sticky traps (5 ft. x 1.5 ft.) smeared with castor oil @ 5/ ac ➢ Release of Chrysoperla zastrowi sillemi eggs @ 500/ac in young palms ➢ Pesticide holiday' must be declared to conserve the natural enemies fauna

New invasive pest	
Palm civet *Vivera zibatha*	▪ Poison baiting with ripe banana fruit sandwiched with 0.5 g carbofuran 3 G granules.
Rat *Rattus rattus wroughtoni*	▪ Tree banding with inverted iron cones or Prosophis thorns. Baiting with bromodialone 0.005% at 10 g/tree at crown region twice at an interval of 12 days.

Diseases management

Basal stem rot (*Ganoderma lucidum*)

- Aureofungin-sol @ 2g + copper sulphate @ 1g dissolved in 100 ml water or hexaconazole @ 2 ml with 100 ml of water, applied as root feeding for 3 times at 3 months interval. (The active absorbing root of pencil thickness be selected and a slanting cut is made. The solution is taken in a polythene bag or bottle and the cut end of the root is dipped in the solution)
- Forty liters of 1% Bordeaux mixture should be applied as soil drench around the trunks in a radius of 1.5 meter
- Neem cake @ 5 kg/tree can be applied along with fertilizers and azotobactor @ 200 g/tree

Bud rot (*Phytophthora palmivora*)

- The infected tissues from the crown region should be removed and protected with Bordeaux paste
- Spray 1% bordeaux mixture or copper oxychloride @ 0.25 % on crown region as pre-monsoon spray
- Spray copper oxychloride @ 0.25 % after the onset of monsoon

Stem bleeding (*Ceratocystis paradoxa*)

The bark of the trunk should be removed in the bleeding area and Bordeaux paste should be applied in this area

Preparation of 1% bordeaux mixture

Copper sulphate @ 400g should be dissolved in 20 litres of water and 400 g of lime in another 20 litres of water separately. The copper sulphate solution should be added to the lime solution constantly stirring the mixture. Earthen or wooden vessels alone should be used and metallic containers should not be used. To find out whether the mixture is in correct proportion, a polished knife should be dipped in the mixture for one minute and taken out. If there is reddish brown deposit of copper, additional quantity of lime should be added till there is no deposit in the knife.

Preparation of bordeaux paste:

Take 200 g of copper sulphate and dissolve it in one litre of water and 200 g of lime in one litre of water separately, both are mixed simultaneously in a third vessel and the resultant mixture can be used as a paste.

SPECIAL PROBLEMS IN COCONUT

Rejuvenation of existing garden

The low yield in vast majority of gardens is due to thick population, lack of manuring and irrigation. These gardens could be improved if the following measures are taken.

i. **Thinning of thickly populated gardens**

In the farmer's holdings where thick planting is adopted, many trees give an yield of less than 20 nuts/palm/year. By cutting and removal of these trees, the yield could be increased. Besides, there is saving in the cost of cultivation and increase in net profit. After removal of low yielding trees, the populations should be maintained at 175 palms/ha.

ii. **Ensuring adequate manuring and irrigation**:

The yield can be increased in the existing gardens when manuring + irrigation + cultural practice is adopted as per recommendation.

Pencil point disorder (Micronutrient deficiency)

Because of micronutrient deficiency, the stem will taper towards its tip with lesser number of leaves. The leaf size will be greatly reduced and the leaves will be pale and yellow in colour. Along with the recommended fertilizer dose, 225 g each of Borax, Zinc sulphate, Manganese sulphate, Ferrous sulphate, Copper sulphate and 10 g of Ammonium molybdate may be dissolved in 10 litres of water and poured in the basin of 1.8 m radius. This disorder can be corrected if noticed early. Severely affected palms may be removed and replanted with new seedlings.

Button shedding

Shedding of buttons and premature nuts may be due to any one of the following reasons:

i) Excess acidity or alkalinity
ii) Lack of drainage
iii) Severe drought
iv) Genetic causes
v) Lack of nutrients
vi) Lack of pollination

vii) Hormone deficiency
viii) Pests
ix) Diseases

The following remedial measures are suggested.

Rectification of soil pH

Excess acidity or alkalinity of soil may cause button shedding. If the soil pH is less than 5.5, it is an indication of excess acidity. This could be rectified by adding lime. Increase in alkalinity is indicated by soil pH higher than 8.0. This situation could be rectified by adding gypsum.

Providing adequate drainage facilities

Lack of drainage results in the roots of coconut trees getting suffocated for want of aeration. Shedding of buttons occur under such condition. Drainage channels have to be dug along the contours to drain the excess water during rainy season.

Management of young coconut gardens under waterlogged conditions

(i) A trench between two rows of young coconut palms should be dug during onset of the monsoon rains. The size of the trench is 3 m width, 30 – 45 cm depth to entire length of field. The soil excavated from the trench should be placed along the rows of palms to make a raised bed.
(ii) Form mound around the young palms to a radius of 1.2 m width with height of 30 –45 cm.

Genetic causes

In some trees button shedding may persist even after ensuring adequate manuring, irrigation and crop pest and disease management. This is an indication of inherent defect of the mother palm from which the seed material was obtained. This underlines the need for proper choice of superior mother palm for harvesting seed coconut to ensure uniformly good yielding trees.

Lack of nutrition

Button shedding occurs due to inadequate or lack of manuring. The recommended dose of manurial schedules and proper time of application are important to minimise the button shedding. Apply extra 2 kg of muriate of potash with 200 g of Borax/palm over and above the usual dosage of fertilizer to correct the barren nuts in coconut for period of 3 years.

Boron deficiency or crown choke disorder

Apply 200 g of borax/palm/year in two splits.

Lack of pollination

Button shedding also occurs due to lack of pollination. Setting up of beehives @ 15 units/ha may increase the cross pollination in the garden. Further the additional income obtained through honey, increases the net profit per unit area.

Hormone deficiency

The fertilised female flowers i.e., buttons shed in some cases. By spraying 2, 4- D at 30 ppm or NAA 20 ppm (30 mg per litre of water) on the inflorescence one month after opening of the spathe, the setting percentage could be increased.

Pests

Button shedding may happen due to the attack of bug. Spraying of systemic insecticides like Methyldematon 0.025% (1ml/lit) or Dimethoate 0.03% (1ml/lit) may reduce the occurrence.

Diseases

Button shedding also occurs due to disease incidence such as basal stem rot. Adoption of control measures suggested for the disease reduces not only spread of the disease but also prevents shedding of buttons.

Coconut mother palm selection and nursery management

The need for collecting seed materials from high yielding coconut palms is highly essential in a perennial crop like coconut.
The following points may be remembered.

Mother palm selection

1. Select seed gardens, which contain large proportion of high yielding trees with uniformity in yielding ability. Trees growing closer to households, cattle shed, compost pits and other favorable conditions should be avoided.
2. High yielding mother palms giving not less than 100nuts/palm/annum should be chosen for collecting seednuts. Alternate bearers should be avoided. The age of the palm chosen be middle age i.e., from 25 to 40 years. Even trees with 15 years age can be selected, if it is high yielding and has stabilized yield.
3. The mother palm should have straight trunk, spherical or semi spherical crown, high rate of leaf and spathe production, short and stout petiole, more number of female flowers regular bearing habit, non – buckling bunches, high setting parentage, medium in nut size, high copra outturn and free from pest and diseases. A good regular bearing mother palm produces on an average one leaf and an inflorescence in its axil every month. So, there will be twelve bunches of varying stages of maturity at any one time. Avoid trees producing habitually barren nuts.
4. Harvest seednuts during the months of February - August to get maximum germination and good quality seedlings. Harvest the bunches intended for seednut by lowering them to the ground using a rope to avoid injury to seednuts
5. The seednuts should be round in shape and when tapped by finger should produce metallic sound. Fully ripe nuts develop twelve months after fertilisation.
6. To get more quality seedlings, the seednuts of tall and hybrid are to be air cured for one month followed by sand curing for two months. For dwarf varieties, the air curing should be lesser than one month followed by sand curing for two months.

Nursery management

1. Select nursery area in a well drained plot with coarse texture soil near water source for irrigation. Nursery can be raised in the open space with artificial shade or in the adult coconut garden.
2. Plant seednuts in long and narrow beds at a spacing of 30 x 30 cm either horizontally or vertically in deep trenches with 20-25 cm depth. Five rows of nuts may be planted in each bed accommodating 50 nuts per row.
3. Irrigate the nursery beds once in three days.
4. Keep the nursery free of weeds. To manage the weed problem in coconut nursery, growing sunnhemp 2 times (each harvested at flowering stage) followed by one hand weeding at 6^{th} month was found to be very effective besides yielding green manure for manuring the adult coconut palms.
5. Provide shade to the nursery by raising Sesbania or Leucaena on the sides of beds.
6. The seednuts start germination 6 – 8 weeks after planting and germination continues upto six months. Select seedlings that germinate before 5 months after planting. Remove those nuts which do not germinate 5 months after sowing.
7. Regularly survey for pest and diseases
8. Select seedlings 9 to 12 months after planting. Seedlings, which have germinated earlier, having good girth at collar and early splitting of leaflets, should be selected for planting. Do not select the so called Kakkamukku Pillai i.e., seednuts which have just germinated. Eliminate the seedlings which are deformed or having stunted growth.
9. Remove the seedlings from the nursery by lifting with spade. Do not pull out the seedlings by pulling leaves or stem.
10. Select quality seedlings with a minimum of 6 leaves and girth of 10 cm at collar.

(vii) OILPALM

CLIMATE REQUIREMENT

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
33 - 38	10 - 15	24 - 29	2000 – 4000	up to 900

Tropical and subtropical climate. Cannot tolerate drought and need 80% RH.

INTRODUCTION

Oil palm requires evenly distributed annual rainfall of 2000 mm without a defined dry season. In areas with dry spell, a deep soil with high water holding capacity and a shallow water table augmented with copious irrigation will satisfy the water requirement of the palm. Temperature can be a limiting factor for oil palm production Best oil palm yields are obtained in places where a maximum average temperature of 29-33 C and minimum average temperature of 2-24 C are available. Higher diurnal temperature variation causes floral abortion in regions with a dry season.

The crop requires 1800-2000 sunlight hours annually, more than 300 cal/cm^2 / per day, constant sunlight of atleast 5 hours per day for better oil palm yield.

Moist, deep and well drained medium textured soils rich in humus content are considered ideal. Gravelly and sandy soils, particularly the coastal sands are not ideal for oil palm cultivation. Heavy clay soils with poor drainage properties may pose problems of aeration during rainy seasons.

NURSERY AND ITS MANAGEMENT

Nursery is raised by planting germinated sprouts initially in a pre-nursery bed or in polybags in a primary nursery and transplanting them at five leaf stage to a secondary nursery of large sized polybags. Raising seedlings in large polybags without a pre-nursery stage is also being practiced.

The potting mixture is made by mixing top soil, sand and well decomposed cattle manure in equal proportions. Smaller polybags of 250 guage and 23 x 13 cm size, preferably black are used for raising primary nurseries. These bags are filled with the potting mixture leaving one cm at the top of the bag. A healthy germinated sprout is placed at the centre at 2.5 cm depth. While placing the sprout, care must be taken to keep the plumule of the sprout facing upwards and the radicle downwards in the soil. It is better to plant sprouts soon after the differentiation of radicle and plumule. The seedlings are to be watered daily. Application of a fertilizer mixture containing one part of ammonium sulphate, one part of super phosphate, one part of muriate of potash and two parts of magnesium sulphate is recommended at 15 g at one month stage, 45 g at three months stage and 60 g at six months stage per seedling. This has to be applied 6 - 8 cm away from seedlings during the first application, 10-12 cm away during second and 15-20 cm away during the third application in primary nursery. Surface soil is slightly scratched at the time of fertilizer application.

SINGLE STAGE POLY BAG NURSERY AND SECONDARY NURSERY

The germinated seeds can be directly planted into large black polybags with the advantage of avoiding the pre-nursery stage. At present the single stage polybag nursery is recommended in India. Since the plants are to remain in these polybags for more than one year, good quality polybags of 500 gauge and 40 x 45 cm size are to be used. On the lower half of the bag, perforations are made at an interval of 7.5 cm for drainage. A bag can carry 15 - 18 kg of nursery soil depending on the type of soil mixture used.

The water requirement for different stages of growth of seedlings are as follows: 0 - 2 months at 4 mm/day, 2 - 4 months at 5 mm/day, 4 - 6 months at 7 mm/day and 6 - 8 months at 10 mm/day. It is better to supply if feasible the daily requirement in two halves to prevent overflow and wastage caused by one time application. Application of 9 - 18 lit. of water per seedling per week according to the stage of growth and soil type.

FIELD PLANTING

Prepare the land for oil palm plantings at least 3 months before transplanting the seedlings to the main field. In soils with low permeability, drainage channels are to be constructed to prevent water stagnation in upper layer of soil

AGE OF SEEDLINGS AT TRANSPLANTING

It is advisable to plant well grown seedlings of 12 - 14 months old. At this stage, a well developed tenera seedling will have a height of 1-1.3 m from base and will have more than 13 functional leaves. These seedlings were found to maintain higher leaf production, bear earlier, produce heavy bunches, give higher fruit/bunch ratio and a higher oil to mesocarp in the first year of harvest.

SELECTION OF SEEDLINGS

All deformed, diseased and elongated seedlings are to be discarded. Differences in the height of healthy seedlings ranging from 90 to 159 cm tend to even up after 14 months of transplanting to maintain.

TIME OF TRANSPLANTING

Transplanting to the main field has to be done during the onset of rainy season. In very impermeable soils and where there is chance for the seedlings to suffer severely during rainy season, proper drainage has to be ensured.

SPACING AND METHOD OF PLANTING

The optimum planting density for oil palm is the density of population that gives maximum production from unit area. Triangular system of planting with 9 x 9 x 9 m spacing accommodates 143 palms/ha. is being recommended. For efficient utilization of solar energy the rows are to be oriented in the North-South direction. Equilateral triangular system of planting with 9 m spacing between palms will allow each plant to occupy the centre of a hexagon thus allowing better use of the area.

TRANSPORTING SEEDLINGS AND PREPARING PITS

While transporting seedlings to the planting site one hand is placed at the bottom of the bag while holding the plant collar with the other one. Leather gloves can be used to avoid injury with spines of the leaves.

Pits of 60 cm^3 are taken prior to planting and filled with surrounding top soil and allowed to settle. Rock phosphate is applied at 200 g per planting pit. Nitrogen is not usually applied in the planting pits as the application of fertilizers may damage the root system and affect survival of the plants if there is a dry period soon after planting. Nitrogen and potassium are usually applied 4 – 6 weeks after planting. In Mg deficient soils, magnesium is applied at 100 g as anhydrous $MgSO_4$ or 200 g epsum salt per seedling.

REPLACEMENT AND GAP FILLING

Field inspection is carried out one to two months after planting to gap fill dead plants. Replanting is carried out during the onset of next monsoon. These palms are to be given special care so that they can catch up with the rest of the plantations. Early production of more female inflorescences in the initial 30 months, is an indication of high yielders and all those that fail to produce female bunches will remain as poor yielders. However, replacements are found to be affected to some extent by the vigorous growth of the neighbouring palms which will shade the replanted palms.

FERTILIZER REQUIREMENT

Based on the fertilizer experiments conducted under rainfed conditions in India, the following fertilizer schedule is recommended for oil palm until specific results are derived from multilocational fertilizer trials.

Fertilizer recommendation for oilpalm

Age	**Nutrients (gram/palm/year)**		
	N	**P**	**K**
First year	400	200	400
Second year	800	400	800
Third year onwards	1200	600	2700

METHOD OF FERTILIZER APPLICATION

The fertilizers are preferably applied in two equal split doses during May - June and September -October by uniformly spreading them within a 2 metre circle around the base of the palm and forking to incorporate them into the soil. Supply of sufficient quantity of green leaves or compost is advantageous especially where the soil is poor in organic matter content. Mg deficiency can be corrected through the application of 500 g of MgSO4 palm/year. Borax @ 100 g per palm also recommended.

Urea is found to be the most economic nitrogen source if losses by volatilization and leaching are minimised. Rock phosphate and muriate of potash are the best source for phosphorus and potassium respectively. During the initial years fertilizers may be applied within the area covered by the crown canopy. In the case of older palms, fertilizers are applied depending on the concentration of roots and are usually applied in the weeded circle. Appropriate soil conservation methods such as growing cover crops and platform cutting (on sloppy lands) enhance the efficiency of fertilizers by preventing losses through run off.

NUTRIENTS - FUNCTIONS AND DEFICIENCY SYMPTOMS

The effect of major nutrients on growth and yield of oil palm has been studied in most of the oil palm growing countries in Asia and Africa.

a) Nitrogen: In oil palm, characteristic yellowing symptoms are developed under N deficiency conditions. Nitrogen is found to be essential for rapid growth and fruiting of the palm. It increases the leaf production rate, leaf area, net assimilation rate, number of bunches and bunch weight. Excessive application of nitrogen increases the production of male inflorescence and decreases female inflorescence thereby reducing the sex ratio.

b) Phosphorus: In oil palm seedlings, P deficiency causes the older leaves to become dull and assume a pale olive green colour while in adult palms high incidence of premature desiccation of older leaves occurs. Phosphorus application increases the bunch production rate, bunch weight, number of female inflorescences and thereby the sex ratio. However, lack of response to P due to P fixation in soils is very common in the tropics. Eventhough the main effect of phosphorus on the productivity of the palm has not been significant in most studies, it gives a positive interaction with nitrogen and potassium.

c) Potassium: When potassium is deficient, growth as well as yield is retarded and it is translocated from mature leaves to growing points. Under severe deficiency, the mature leaves become chlorotic and necrotic. Confluent orange spotting is the main K deficiency condition in oil palm in which chlorotic spots, changing from pale green through yellow to orange, develop and enlarge both between and across the leaflet, veins and fuse to form compound lesions of a bright orange colour. Necrosis within spots is common, but irregular. Mid crown yellowing is another prominent K deficiency condition of the palm in which leaves around the 10th position on the phyllotaxy become pale in colour followed by terminal and marginal necrosis. A narrow band along the midrib usually remains green. There is a tendency for later formed leaves to become short and the palm has an unthrifty appearance with much premature withering.

Potassium removal is large compared to the normal exchangeable K content in most top soils. It is mostly required for the production of more number of bunches, maximum number of female inflorescences, increased bunch weight and also for increasing the total dry matter production and yield.

d) Magnesium: In adult oil palm and in seedlings in the field, severe Mg deficiency symptoms are most striking and have been named as ‘orange frond’. While the lower most leaves are dead, those above them show a gradation of colouring from bright orange on the lower leaves to faint yellow on leaves of young and intermediate age. The youngest leaves do not show any discolouration. The most typical Mg-deficiency symptom is the shading effect in which the shaded portion of the leaflet will be dark green while the exposed portion of the same leaflet is chlorotic. Heavy rates of K applications induce Mg-deficiency, particularly on poor acid soils.

Among the secondary nutrients, calcium and sulphur, and probably chlorine, may not pose much problems to oilpalm cultivation in the country.

e) Micronutrients: Micronutrient elements, iron, manganese, copper and zinc are not generally found limiting in the nutrition of oil palm on acid soil conditions. Boron deficiency is occasionally found on young palms in the field showing a reduction of leaf area in certain leaves producing incipient 'little leaf', advanced 'little leaf' with extreme reduction of leaf area and bunching and reduction in the number of leaflets and 'fish-bone' leaf. The 'fish-bone' leaves are abnormally stiff with leaflets reduced to projections. Leaf malformations including 'hook leaf' and corrugated leaflets are some other associated symptoms. Soil application of 50 - 200 g borax, per palm, depending on age, and severity of symptoms is practiced for correcting the malady.

WATER REQUIREMENT

Continuous soil moisture availability encourages vigorous growth and increased yield of oil palm. Adequate supply of water, good soil depth and water holding capacity contribute to water availability. In oil palm as water deficiency increases, stomata will remain closed and the development and opening of spear will be inhibited. Water deficiency adversely affects flower initiation, sex differentiation and therefore, results in low sex ratio due to production of more male inflorescences. It is established that oil palm needs 120 - 150 mm of water to meet its monthly evapo-transpiration needs. In areas where perennial water source is available, basin irrigation is possible. But where the terrain is undulating and water is scarce during summer months, drip irrigation is recommended to keep four drippers per palm in the weeded palm circle to supply atleast 90 litres of water per palm per day during summer months which will vary according to the ETP values in a locality.

FERTIGATION

Drip fertigation with the recommended dose of fertilsers at bimonthly interval was found to increase the yield.

WEED CONTROL

The basin area of oil palm is kept free of weed growth through ring weeding. It is more important for young palms, roots of which are to be kept free from competition from weed. Depending on the extent of weed growth and rainfall, hand weeding is carried out even upto four times in a year during early years of the plantation which is progressively reduced to two rounds a year.

Herbicide application has become common in recent years. Care must be taken in the choice of herbicide and its application to prevent the damage of young palms. It is recommended to preferably apply contact herbicides rather than translocated herbicides. Translocated herbicides like Paraquat which is inactivated when contacted with soil are also used. Herbicides such as 2, 4-D, 2, 4- 5-T, halogenated aliphatic acids Dalapon and TCA are found to produce abnormalities in oil palm seedlings and are to be avoided. Pre-emrergence Atrazine @ 1.0 kg/ha for the control of grasses and sedges and POE Paraquat 10 ml / litre of water.

MAINTENANCE OF PATHS

In young plantation, the maintenance of paths is important for inspection and in later years for harvesting. This is carried out by timely weed control as done in the case of ring weeding.

ABLATION

The bunches produced initially will be very small and have low oil content. Removal of such inflorescences is called ablation or castration. Removal of all inflorescences during the initial three years is found to improve vegetative growth of young palms so that regular harvesting can commence after three and half years of planting. Ablation is done at monthly interval by pulling out the young inflorescence using gloves or with the help of devices such as narrow bladed chisels. Ablation improves drought resistance capacity of young palms by improving shoot and root growth especially in low production areas where dry condition exists.

PRUNING OF LEAVES

In oil palm two leaves are produced per month. Therefore, it becomes necessary to prune excess leaves so as to gain access to bunches for harvest. Severe pruning will adversely affect both growth and yield of palm, cause abortion of female flowers and also reduce the size of the leaves. It was suggested that palms aged 4 - 7 years should retain 6 - 7 leaves per spiral (48 - 56), those aged 8 - 14 years 5 - 6 leaves per spiral (40 - 49) and those above 15 years should have 4 - 5 leaves per spiral (32 - 40). Leaf pruning is carried out in India using chisels so that leaf base that is retained on the palm is as short as possible or otherwise it may catch loose fruits, allow growth of epiphytes and the leaf axils form a potential site for pathogens. The leaf petioles are removed by giving a clear cut at a sufficient distance from the base of the petiole using a sharp chisel for young palms and with the long sickle in taller palms.

Pruning is preferably carried out at the end of the rainy season. It is also better to carry it out during the low crop season when labourers are also available. Pruning is confined to only lower senile leaves during initial harvests but when canopy closes in later years, leaves are cut so as to retain two whorls of fronds below the ripe bunch.

Insect pollination in oil palm

The oil palm, hitherto though to be wind pollinated, has been now proved to be an a insect pollinated species. From West Africa, the original home of oil palm, eight species of pollinating weevils were reported. Occurence of ***Eldeidobius kamerunicus*** in the oil palm plantations of Kerala was introduced during 1985 from where it was introduced and got established in little Andamans during 1986.

The weevils are dark brown in colour. Adult weevils feed on the anther filament. Eggs are deposited inside the male flowers and larva feeds on the spent flowers. Life-cycle is completed within 11 to 13 days. Males live longer than females. The activity of the insects is in accordance with the receptivity of the male and female inflorescences.

It was roughly estimated that 40 palms in a grove might be the minimum to sustain a sufficiently high continuous population of pollinators to pollinate. All are receptive female inflorescences. The weevils carry maximum pollen during the third day of antheses. Antennae, rostrum, thorax, legs etc. are the main sites of pollen land. E.kamerunicushas a fairly good searching ability. It can survive in dry as well as in wet seasons.

Introduction of weevil in India increased the fruit let from 36.8 percent to 56.1 percent resulting in 40 per cent increase in F/B ratio. The maximum attainable pollination potential was as much as to cent percent with 57 percent increase in FFB weight.

For introduction, male flowers cut from palms which have the weevils are transferred to a plantation where one wishes to introduce. In order to make sure that they are not carrying any plant pathogens to other area/countries, we have to breed them under laboratory conditions for seven or eight generations before introduction.

Pest Management

In India, since the import of germplasm is in the form of seeds/sprouts, possibilities for introduction of the pest species from other countries are limited. But many of the pest species of related palm species such as coconut and areca palm, have got adapted to oil palm. Among the 49 species of insects infesting adult oilpalms, 14 species are known pests of coconut and 19 species are known pests of areca palms. Insect pests of oilpalm in India are more or less same as those reported from Malaysia and other South-East Asian countries.

PEST OF ADULT PALMS

The rhinoceros beetle

The rhinoceros beetle is primarily a serious pest of coconut palm, and in recent years has attained the pest status in oilpalm also. The adult beetle which bores through into the spear leaves, resulting in snapping of the fronds at the feeding sites. In oil palm plantations failed female inflorescences, dead palm trunks, persistent leaf axils and empty bunch heaps, act as breeding sites for the pest.

The red palm weevil

Infestation by the red palm weevil Chynchophorus ferrungineus was noticed in majority of oil palm plantations resulting in the death of the palms. Damage is due to the feeding activity of the grubs, usually 12-87 per palm, which bore through and feed on the softer tissues of stem and meristem. Palms infested by R.ferrugineus show gradual wilting and drying of outer whorl of fronds. In some cases roofing of spear was also noticed.

Biological control

In nature, the rhinoceros beetle is suppressed by entomophogens like *Baculovirus oryctusvirus* and *Metarhizium anisopliae*. Release of *Baculovirusoryctes minimise* the pest incidence.

Cultural control

i) Field sanitation and elimination of breeding sites like dead palm trunks, empty bunch heaps etc., within the plantations are essential for the management of both red palm weevil and rhinoceros beetle.
ii) When the infestation by rhinoceros beetle is very high, especially in young plantations, Hand picking of the adult beetles using hooks is very effective.
iii) For red palm weevils, use of attractants incorporating fermented sugarcane juice, acetic acid, yeast etc., to collect and kill the adult weevils is recommended.

Chemical control

i) For rhinoceros beetles, placing 3-4 napthalene balls in the youngest spear axils at weekly intervals is recommended.
ii) For palms with advanced stage of infestation by red palm weevil, stem injection of 5-8 ml of monocrotophos is advised.

Fruit bunch covering against avian pests

Covering the bunches with different materials such as noirenets, reed baskets, plaited coconut leaf baskets and senile oil palm leaf are effective in preventing the fruit damage. But senile oil palm leaf covering is more practical and economical as the material is readily available and involves only the labour charges and cost of rope bits.

Rodent control

Among rats, the burrowing type is more serious which tunnel into the bole of the seedlings. Different baits such as acute poison baits (Zinc phosphide, Aluminium phosphide etc.) anticoagulants (warfarin, fumarin, bromadiolone) and traps such as iron like traps, snap traps, deathfall trap, boro trap etc. may be used as an integrated approach to minimise the rodent damage to the crop.

Disease

Oil palm, a new crop to the country is reported to be affected by a number of diseases and disorders. Among these, bud rot causes considerable economic losses.

Bud rot

- Higher disease incidence is noticed in young plantations. Rotting initiates at the basal portion of the spear closure to the meristem and extends to the whole spear. The spear could be easily pulled off.
- Cleaning the affected tissues and drenching the crown with carbendazim 0.1 percent cures the disease.
- The leaves emerging immediately after the application of fungicides are shorter and successively emerging ones are normal.

Leaf spots

- Leaf spots caused by *Curvularia* noticed on the inner whirl and young leaves. The fungal spots enlarge with a yellow ring around spots. As these spots enlarge the leaf will be scorched.
- *Pestalotiopsis* fungal spots are irregular with grey to brown centre. Numerous black dots, the acervuli of the fungus, are seen on the lesions.

Management

Affected leaves must be cut and burnt Spray Mancozeb @ 0.2%.

HARVESTING

Proper and timely harvesting of fruit bunches is an important operation which determines the quality of oil to a great extent. The yield is expressed as fresh fruit bunches (FFB) in kg per hectare per year or as oil per hectare per year. The bunches usually ripen in six months after anthesis. Unripe fruits contain high water and carbohydrate and very little oil. As the fruit ripens oil content increase to 80 - 85% in mesocarp. Over ripe fruit contains more free fatty acids (FFA) due to decomposition and thus increases the acidity. Usually the ripe fruits, attached to the bunches contain 0.2 to 0.9% FFA and when it comes out of extraction plant the FFA content is above 3%.Ripeness of the fruit is determined by the degree of detachment of the fruit from bunches, change in colour and change in texture of the fruit. Ripening of fruits start from top downwards, nigrescens fruits turning reddish orange and the virescens (green) to reddish brown. Fruits also get detached from tip downward in 11 - 20 days time. Ripeness is faster in young palms than in older palms for the bunches of equal weight. The criteria used in determining the degree of ripeness based on the fruit detachment are as follows:

a) fallen fruits: 10 detached or easily removable fruits for young palms and 5 for adult palms,

b) number of fruits detached after the bunch is cut; 5 or more fruits/kg of bunch weight,

c) quantity of detachment per bunch; fruit detachment on 25% of visible surface of bunch.

These criteria could be applied with flexibility.

FREQUENCY OF HARVESTING

Harvesting rounds should be made as frequent as possible to avoid over ripening of bunches. A bunch which is almost ripe but not ready for harvest for a particular harvesting round should not be over-ripe by next round. In lean period of production, harvesting can be made less frequent and it should be more frequent in peak periods. Harvesting rounds of 7 - 14 days are generally practiced. Other factors determining frequency are, extraction capacity of the mill, transportation facilities, labour availability and skill of the workers. In India, harvesting is usually carried out with a chisel of 6 - 9 cm wide attached to a wooden pole or light hollow aluminium pipe, Bunches are cut without damaging the petiole the leaf that supports it. Use of narrow chisel is usually carried out till the palm reaches two meters above the ground. For taller palms upto 4 meters, a wider chisel of 14 cm is used. The curved knife is attached to a long bamboo or aluminium pole with screws or steel wires to harvest from taller palms. In uneven stands, an adjustable, telescopic type of pole is in use.

Yield of Oilpalm

In well maintained garden the yield of oilpalm will be as furnished below :

Age of oilpalm	Yield Ton/ha/year
3-4 years	5
4-5 years	12
5-6 years	25
6-25 years	30

ECONOMICS

A detailed account of the economics of oilpalm cultivation in India has been furnished. The data furnished therein is modified using current labour charges and oil price and the details on various investments and returns from one hectare adult plantation. This excludes the cost of land as we expect government owned land, leased land, or already owned property will be used for oilpalm cultivation. From the fourth year, the yield of bunches increases upto tenth year, and a stabilized bearing is attained thereafter. The investment during first year under irrigation will be almost three times of that under rainfed conditions mainly on account of the initial expenditure required to install the drip irrigation system. With irrigation the annual returns will exceed the annual expenses from the first harvest itself, i.e, during the fourth year after planting. By the end of sixth year the total returns will be more than total investments including all the expenditure for installing pumpset and the drip irrigation system. A minimum of 22 FFB per hectare can be expected from the tenth year onwards.

TABLE 1 - COST OF PRODUCTION AND (Rs.) PER HECTARE

S. No.	Particulars	Cost of production
1	Labour cost for 200 Nos. @ Rs.120/- per day as casual labour	24,000
2	Fertilizer cost	5000
3	Plant Protection cost	500
	Total cost of production	**29,500**

TABILE 2 : INCOME FROM OILPALM GARDEN DEPENDING UPON THE BUNCH PRODUCTION

S.No.	No. of Bunches/ tree/year	FFB yield t/ha/year	Gross Income Rs./ha/year	Net income (Gross income cost) Rs./ha/year
1	10 bunches @ 10kg/tree/year	14.3	1,02,960	73,460
2	12 bunches @ 15kg/tree/year	25.7	1,85,040	1,55,540
3	12 bunches @ 20kg/tree/year	34.3	2,46,960	2,17,460

(viii) NIGER (Guizotia abyssinica)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
35	10 - 15	15 - 28	650 - 1000	1000 - 1250

Tropical and subtropical dry climate. It can grow in semi-shade or full sun. Flowering was very delayed at day lengths of more than 12 hours.

CROP IMPROVEMENT
I. SEASON AND VARIETIES

DISTRICT/SEASON	VARIETIES

A. Rainfed

1. Adippattam (June- July)
Dharmapuri, Krishnagiri, Hilly regions of Shevroy, Kolli hills, Jawad hills and Thalavadi hills

2. Purattasipattam (Sep-Oct) Dharmapuri, Krishnagiri, Hilly regions of Shevroy, Kolli hills, Jawad hills and Thalavadi hills, Dharmapuri, Krishnagiri, Hilly regions of Shevroy, Kolli hills, Jawad hills and Thalavadi hills

3. Purattasipattam (Sep-Oct) Dharmapuri, Krishnagiri, Hilly regions of Shevroy, Kolli hills, Jawad hills and Thalavadi hills

I. DESCRIPTION OF NIGER VARIETIES

Particulars	Paiyur 1
Parentage	Mass selection from Composite II
Duration (days)	80
Yield (kg/ha)	
Rainfed	260
Oil content (%)	44.6
Plant height (cm)	80-85
Branches	Profuse
Days to 50 % flowering	50
Seed size	Bold
Seed colour	Brown

CROP MANAGEMENT
I. PREPARATION OF THE FIELD

1. FIELD PREPARATION

a) Plough with tractor 2-3 times with a mould board plough or 5 times with a country plough.

b) Break the clods in between the ploughings and bring the soil to a fine tilth.

2. APPLICATION OF FYM

a) Spread 12.5 t of FYM or compost or composted coir pith per ha evenly and incorporate in the soil. b) If the manure is not applied before commencement of ploughing, spread the manure evenly before the last ploughing and incorporate in the soil.

NOTE: Do not leave the organic manure exposed to sunlight as nutrients will be lost.

3. APPLICATION OF FERTILIZERS

Apply N at 20 kg/ha basally.

4. SEED RATE

Adopt a seed rate of 5 kg/ha.

5. SPACING

Adopt a spacing of 30 cm between rows and 10 cm between plants.

6. SELECTION OF GOOD QUALITY SEEDS

Select mature good quality seeds, free from pest damage and fungal attack.

7. PRE-TREATMENT OF SEEDS WITH FUNGICIDES

a) Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. b) Treat the seeds 24 hours prior to sowing.

NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage.

8. SOWING

a. Sow the seeds in line at a depth of 2 to 3 cm and cover with soil. b. Sow using gorru or country plough.

9. THINNING OUT SEEDLINGS

Thin out the seedlings to a spacing of 10 cm between plants on the 15th day of sowing.

10. WEED MANAGEMENT

Hoe and weed on 20th and 35th day of sowing.

11. HARVESTING

i) Observe the crop considering the average duration of the crop.

ii) The leaves and entire plant loose their colour and turn brown at maturity.

iii) Cut the plants at the bottom.

iv) Keep the plants in the threshing floor and beat the plants (heads) with sticks till the mature seeds are separated.

v) Winnow the seed and dry in the sun.

vi) Collect and store the seeds in gunnies.

CROP PROTECTION

Pre-Treatment of Seeds with fungicides

Treat with Carbendazim or Thiram at 4 g/kg of seed in a polythene bag and ensure a uniform coating of the fungicide over the seed. Treat the seeds 24 hours prior to sowing.

NOTE: Seed treatment will protect the young seedlings from root rot disease in the early stage. Varietal production seed

VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production leave a distance of 200 m all around the field from the same and other varieties of niger.

Spacing

- 30 x 30 cm

Fertilizer

- Apply NPK @ 40 : 40 : 20 kg ha^{-1} as basal application.

Harvesting

- Harvest the crop as whole plants at 85 days after sowing, when the seeds attained physiological maturation.

Seed grading

- Grade the seeds using BSS 16 x 16 wire mesh sieve.

Seed treatment

- Slurry treat the seeds with carbendazim @ 2g kg^{-1} of seed (or)
- Slurry treat seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1) @ 3g kg^{-1} of seed as eco – friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8-9 months) with seed moisture content of 8 - 9%.
- Store the seeds in polylined gunny bag for medium term storage (12- 15 months) with seed moisture content of 7 – 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than15 months) with seed moisture content of less than 5%.

Other management practices

- As in crop management techniques.

6. FIBRE CROPS

(i) CLIMATE REQUIREMENT OF COTTON

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
40	12	27 - 32	500 - 700	Up to 1500

Tropical warm season crop. A daily mean temperature of 16°C for seed germination, 21 - 27°C for proper vegetative growth and 27 - 32°C for fruiting phase. Abundant sunshine during boll maturation and harvesting is essential to obtain a good quality crop produce. Heavy showers of rain or heavy irrigation during fruiting period cause shedding of flowers and young bolls.

I. SEASON AND VARIETIES

District/Season	Varieties/Hybrids
Irrigated (Main)	
Winter Irrigated (Aug – Sep)	
Coimbatore, Erode, Madurai, Dindigul, Theni, Salem, Namakkal	MCU 5, MCU 5VT, Suraj, MCU 13, Surabhi, Suvin, CO 14
Dharmapuri	MCU 5, MCU 5VT, Suraj, MCU 13, Surabhi, CO 14
Cuddalore, Villupuram	LRA 5166, SVPR 2, SVPR 4, Surabhi, CO 14
Madurai, Virudhunagar, Tirunelveli, Trichy, Salem, Erode, Dindigul	SVPR 5
Summer – Irrigated (Feb – Mar)	
Erode	MCU 5, MCU 12, MCU 13, Surabhi
Madurai, Virudhunagar, Dindigul, Tirunelveli, Thoothukudi, Theni, Ramanathapuram, Sivagangai,	MCU 5, SVPR 2, SVPR 4, Surabhi, SVPR 5, SVPR 6
Rainfed (Sep – Oct)	
Madurai, Dindigul, Theni, Ramanathapuram, Virudhunagar, Sivagangai, Tirunelveli, Thoothukudi, Dharmapuri, Perambalur, Trichy	LRA 5166, K11, KC 2, SVPR 2,KC 3, SVPR4, K12
Rice Fallow	
Thanjavur, Tiruvarur, Nagapattinam, Karur, Cuddalore and Villupuram	MCU 7, SVPR 3

II. PARTICULARS OF COTTON VARIETIES/HYBRIDS

Varieties/ Hybrids	Year of Release	Year of Notification	Parentage	Season	Irrigated/ Rainfed	Mean yield of seed (kg/ha)	Special features
MCU 5	1968		Multiple cross	Aug-Oct Feb-Mar	Irrigated	1850	Extra long staple (29 mm MHL), Can spun upto 70s, ginning 34%
MCU 7	1972	SO.596(E)/ 13.08.1984	X ray irradiation of X L 1143 EE	Jan-Feb	Rice fallows	1330	Medium staple (23.7 mm MHL), Can spun upto 30s, early maturing with 33.2% ginning outturn. Tolerant to Black arm
MCU 12	1999	SO.821(E)/ 13.09.2000	Derivative from the cross LRA 5166 x MCU 11	Aug-Oct	Irrigated	2000	Shorter in duration than MCU 5, GOT 34.8% Can spun upto 50s
MCU 13	2004	SO.1177(E)/ 25.08.2005	It is a multiple cross derivative involving the parents of [(TCH 665 x LS 149) x (TCH 665 x TCH 21)] x (TCH 21 x EECH) x (TCH 92-7 x EECH)	Aug-Oct Jan-Feb	Irrigated	2200	Early duration Can spun upto 50s
CO 14	2016	SO.2238(E)/ 29.06.2016	(MCU5/TCH92-7) x MCU 5-1	Aug-Oct	Irrigated	1768	Extra long stable cotton (2.5% span length - >35.0mm), Ginning outturn: 34.8%, capable for spinning upto 70s count.
SVPR 2	1996	SO.360(E)/ 01.05.1997	TSDT 22 x JR 36	Feb - Mar Sep-Oct	Irrigated Rainfed	2000	High ginning out turn of 36.4%, medium staple (24.3 mm), can spin 30's, suited to summer irrigated, winter rainfed and tankfed rice fallow tracts of Tamil Nadu.
SVPR 3	2000	SO.821(E)/ 13.09.2000	Selection from L.H 900 x 1301 D.D	Jan-Feb	Rice fallows	1800	Suitable for rice fallow tract, early duration (135-140

							days), Tolerant to drought, leafhopper, alternaria spot, black arm disease.
SVPR 4	2009	SO.2137(E)/ 31.08.2010	Hybrid derivative of MCU 5x S4727	Feb- Mar Sep- Oct	Irrigated Rainfed	1800	Superior medium staple cotton with good fibre strength. suitable for spinning 40's yarn.
SVPR 5	2014	SO.3540(E)/ 22.11.2016	Cross derivative of NDLH 1658 x Surabhi	Aug - Sep Feb- March	Irrigated	1845	Long staple - 29.0mm (UHML), Bundle strength 27.8g/tex(HVI mode), can spin 40 - 50's counts, moderately resistant to Jassid.
SVPR6	2017	SO.1379(E)/ 27.03.2018	Cross derivative of SVPR 2 x BJA 592	Feb- March	Irrigated	2312	Long staple 29.1mm (UHML), Bundle strength 27.3 g/tex(HVI mode), Can spin 40's count, Moderately resistant to Jassid.
KC 2	1997	SO.647(E)/ 09.09.1997	MCU 10 x KC 1	Sep - Oct	Rainfed	1000	High ginning out turn of 37.5%, medium staple cotton - 24.4 mm, Suited for rainfed black cotton soil of Tirunelveli, Thoothukudi and Virudhunagar Districts.
KC 3	2006	SO.1178(E)/ 20.07.2007	Hybrid derivative of TKH 97x KC1	Sep- Oct	Rainfed	1080	Resistant to leaf hopper, medium staple cotton – 26.4 mim, suited to southern districts of Tuticorin, Tirunelveli and Virudhunagar District.
MCU 5 VT			Selection from MCU 5	Aug-Oct	Irrigated	2500 kg/ha	Long staple variety capable of spinning

							upto 60 ‘s count yarn
Suraj			LRA 5166 (CCH 326612 x HLS 329)	Aug-Oct	Irrigated	2500 kg/ha	Long staple and suitable for high density planting system
Varieties/ Hybrids	**Year of Release**	**Year of Notification**	**Parentage**	**Season**	**Irrigated/ Rainfed**	**Mean yield of seed (kg/ha)**	**Special features**
K 11	1993	SO.360(E)/ 01.05.1997	(0794-1-DX H 876) x (0794-1-DX H 450) Multiple Hybrid derivative	Sept – Oct	Rainfed	1100	Better fibre properties with lesser pest incidence than K10
K12	2017	SO.399(E)/ 24.01.2018	K11 x K9	Oct-Nov	Winter rainfed	1193	Early duration: 135-140 days, 2.5% span length 27.7mm, can spun upto 30’s counts. Resistance to drought, leaf hopper
LRA 5166			Laxmi x Reba B.50 x AC 122	Aug-Oct Jan – Feb	Irrigated Rainfed	1800 725	Medium staple (29 mm), can spun upto 40s, ginning 36.2%
Surabhi			MCU 5 VT (MCU 5 x *G.mexicanum*)	Aug-Oct	Irrigated	2200	Extra long staple, *Verticillium* wilt resistant
Suvin			Hybrid derivative from the cross Sujatha x St. Vincent	Aug-Oct	Irrigated	1020	Extra long staple cotton with 28% ginning outturn and 32 mm MHL, spins 100s

Extra long staple : CO 14 and Suvin
Long staple : SPVR 5, SVPR 6, MCU 5, MCU 5 VT, MCU 12 , MCU 13,Surabhi and Suraj
Superior medium staple : LRA 5166, SVPR4, KC 3 and K 12
Medium staple: SVPR 2, SVPR 3, MCU 7 and KC 2
Short staple: K 11
Gossypium hirsutum :CO 14, Surabhi, SPVR 5, SPVR 6, MCU 5, MCU 5 VT, MCU 12 , MCU 13, LRA 5166, SVPR 2, SVPR 3, SVPR 4 , MCU 7, KC 3, KC 2, Suraj
Gossypium barbadense: Suvin
Gossypium arboreum : K 11 and K12
Variety suitable for high density planting : Suraj , CO 15,KC 3, Karunganni Cotton K 12

Definition/terminologies

Staple length

Graders estimate the fibre length by hand stapling and is called staple length and expressed in millimeter.

Microniare

It is the measure of the index of the fibre diameter and is assessed by determining weight per unit length of the fibre. It is expressed in micronaire value (µg/inch).

Fibre strength

Fibre strength is generally considered to be next to fibre length and fineness. It is referred as bundle or tensile strength, essential for high speed spinning. It determines yarn strength. Unit of this parameter was expressed in gms/tex.

Uniformity ratio

It is the ratio of 50% span length to 2.5% span length expressed as percentage.

$$\text{Uniformity ratio} = \frac{50\% \text{ span length x } 100}{2.5\% \text{ span length}}$$

Maturity - Coefficient:

The maturity of cotton fibres is expressed by the maturity ratio parameter, which is calculated on the basis of the fibre circularity coefficient , defined as the ratio of the cell wall area and that of a circle of the same perimeter as the fibre cross-section.

CROP MANAGEMENT
I. PREPARATION OF FIELD FOR IRRIGATED COTTON
CROP FIELD PREPARATION

1.1.1 PREPARATION OF THE FIELD

i) Prepare the field to get a fine tilth.

ii) Chiselling for soils with hard pan: Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5 M interval, first in one direction and then in the direction perpendicular to the previous one, once in three years. Apply 12.5 t farm yard manure or composted coir pith/ha besides chiselling to get increased yield

iii) If intercropping of Greengram/Soyabean is proposed, prepare the main field, so as to provide ridges and furrows to take up sowing 20 days prior to cotton sowing.

1.1.2. APPLICATION OF FYM/COMPOST AND BIOFERTILIZERS

Spread 12.5 t of FYM or compost or 2.5 t of vermicompost per ha if available, uniformly on the unploughed soil.

Apply 2 kg of *Azospirillum* + 2 kg Phosphobacteria (or) 2 kg of Azophos + 2 kg Pink Pigmented Facultative Methylotroph (PPFM) with 25 kg of FYM and 25 kg of sand, mix uniformly before sowing as soil application.

1.1.3. PRE-TREATMENT OF ACID DELINTED SEEDS WITH BIOFERTILIZER

Treat one hectare of seeds with 600g of *Azospirillum,* 600g of Phosphobacteria (or) 600 g of Azophos + 600 g of Silicate Solubilizing Bacteria.

Liquid formulation Treat one hectare of seeds with 125 ml of *Azospirillum,* 125 ml of Phosphobacteria and Silicate solubilizing bacteria (SSB) shade dry for 30 minutes before sowing.

1.1.4. FORMATION OF RIDGES AND FURROWS

i) Form ridges and furrows 10 m long with appropriate spacing depending upon the variety.

ii) Use ridge plough or bund former to form ridges so as to economise on cost of cultivation.

iii) In fields with ragi stubbles, just dibble cotton seeds at the specified spacings.

iv) Adopt the following spacing between ridges for different varieties/hybrids.

Varieties/Hybrids	**Spacing between ridges (cm)**
MCU 5, SVPR 2, LRA 5166, MCU 12, MCU 13	75
Suvin	90
MCU 7	60

NOTE: Adopt higher spacing rows in fertile soils by 15 to 30 cm.

1.1.5. APPLICATION OF INORGANIC FERTILIZERS

v) If soil test recommendations are not available, follow the blanket recommendation for the different varieties.

Varieties / Hybrids	**Quantity of fertilizers (Kg/ha)**		
	N	**P_2O_5**	**K_20**
MCU 7, SVPR 3	60	30	30
MCU 5, MCU 12, MCU 13,Suvin, SVPR 2	80	40	40

ii) If basal application could not be done, apply on the 25th day after sowing.

iii) Apply 50 per cent of N and K full dose of P_2 O_5 as basal and remaining ½ N and K at 40 – 45 DAS for varieties. For hybrids apply N in three splits *viz*., basal, 45 and 65 DAS.

Soil test crop response based integrated plant nutrition system (STCR- IPNS recommendation may be adopted for prescribing fertilizer doses for specified yield targets (ready reckoners are furnished)

Cotton -Variety (1)

Soil : Mixed blackcalcareous (Perianaickenpalayam series)

Target : 2.5 - 3.0 t ha^{-1}

FN = 8.81 T-0.62 SN

FP_2O= 2.53T-1.36SP

FK_2O=4.92T-0.25SK

Initial soil test values (kg ha^{-1})			**Yield target – 2.5 t ha^{-1}**			**Yield target – 3.0 t ha^{-1}**		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	10	300	54	20*	20*	98	30	33
200	12	340	41	20*	20*	85	28	23
220	14	380	40*	20*	20*	73	25	20*
240	16	420	40*	20*	20*	61	22	20*
260	18	460	40*	20*	20*	48	20*	20*

* Maintenance dose

Cotton - Variety (2)

Soil : Red sandy loam (Irugur series)

Target : 2.5 and 3.0 t ha^{-1}

FN = 7.66T-0.43 SN-0.71 ON

FP_2O_5 = 3.22T-3.27 SP-0.87 OP

FK_2O = 5.97T-0.50SK-0.66 OK

Initial soil test values ($kg\ ha^{-1}$)			**Yield target – 2.5 t ha^{-1}** NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 3.0 t ha^{-1}** NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
180	10	200	59	20*	20*	97	32	39
200	12	220	51	20*	20*	89	25	29
220	14	240	42	20*	20*	80	20*	20*
240	16	260	40*	20*	20*	72	20*	20*
260	18	280	40*	20*	20*	63	20*	20*

*Maintenance dose

Cotton under Drip fertigation –Bt Hybrid (3)

Soil : Mixed black calcareous (Perianaickenpalayam series)

Target : 3.5 - 4.0 t ha^{-1}

FN = 8.51 T-0.40 SN-0.73 ON

FP_2O_5 = 4.41T-2.25 SP - 0.75 OP

FK_2O = 6.59T-0.18SK-0.66 OK

Initial soil test values ($kg\ ha^{-1}$)			**Yield target – 3.5 t ha^{-1}** NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**Yield target – 4.0 t ha^{-1}** NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	14	400	163	91	124	205	113	157
220	16	450	155	86	115	197	108	148
240	18	500	147	82	106	189	104	139
260	20	550	139	77	97	181	99	130
280	22	600	131	73	88	173	95	121

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.
Apply 55 kg S as Gysum basally for a sulphur deficient soil.

iv) Foliar application of 2% DAP + 1% KCl or polyfeed and Multi k may be sprayed to improve kapas yield

v) Apply the fertilizers in a band, two-thirds of the distance from the top of the ridge, and incorporate.

1.1.6.APPLICATION OF MICRONUTRIENT MIXTURE

TNAU MN mixture 12.5 kg/ha for variety and 15 kg/ha for hybrid apply as enriched FYM. Enriched FYM is prepared at 1.10 ratio of MN mixture and FYM, mixed at friable moisture and for one month in shade. Need based foliar spray of 2% $Mgso_4$ + 1% urea during boll formation stage.

Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg for one ha.

Yield Maximization and reducing reddening in Bt cotton
Irrigated

Application of TNAU MN mixture (12.5 kg ha^{-1} as EFYM for variety and 15 kg ha^{-1} as EFYM for Bt cotton) along with the recommended NPK to obtain the maximum seed cotton yield with reduced extent of leaf reddening..

1.1.7. NUTRITIONAL DISORDERS

a. In the case of Zinc deficient soils $ZnSO_4$ @ 50 kg/ha as basal or 0.5% $ZnSO_4$ spray thrice at 45, 60 and 75 DAS.
b. When reddening occurs in leaves apply 0.5% $MgSO_4$ + 1.0% urea + 0.1% $ZnSO_4$ foliar spray on 50^{th} and 80^{th} day to correct this malady.
c. Need based foliar spray of 2% $MgSO_4$ + 1.0% urea during boll formation stage to reduce magnesium deficiency.

I. Main Field Operations

Seed Rate

Adopt the following seed rates for different varieties/hybrids

Varieties / Hybrids	Quantity of seed (Kg/ha)		
	With fuzz	Delinted	Naked
MCU 5, MCU 7, MCU 12, MCU 13	15.00	7.50	..
SVPR 2	15.00	..	..
KC 2	20.00	15.00	..
SUVIN	..	..	6.00

1.2.2. SPACING

In a pure crop of cotton, adopt the spacing as below for the different varieties.

Varieties / hybrids	Spacing (cm)	
	Between rows	Between plants
MCU 5, MCU 12, MCU 13	75	30
LRA 5166, SVPR 2		
KC 2	45	15
SUVIN	90	45
MCU 7, SVPR 3	60 or 75 *	30

* Fertile soils

a. If cotton intercropped with other crops, one paired row of cotton is alternated with three rows of intercrop and the total population of cotton crop is maintained at the same level as in the case of pure crop.
b. For intercropping with Greengram / Soyabean, complete the sowing and irrigation 20 days prior to cotton sowing on one side of the ridge.

Varieties/hybrid	Spacing for cotton crop (cm)		
	Within Paired row	Between Paired rows	Between plants
MCU 5, MCU 12,MCU13	60	90	30
SUVIN	80	100	45

Plant two rows of intercrop between each paired row of cotton

Intercrop	Seed rate(kg/ha)	Spacing (cm)	
		Rows	Plants
Blackgram	12.5	30	10
Greengram	12.5	30	10
Cowpea	7.5	30	20
Soyabean	20.0	30	10

For higher returns, advance sowing of either greengram or soyabean 20 days before sowing of cotton in winter season.

1.2.3.ACID-DELINTING OF COTTON SEEDS

i) Choose plastic bucket bucket for acid delinting of seeds.
ii) Do not use earthen wares, metal vessels, porcelain wares or wooden drum for acid delinting as concentrated sulphuric acid will corrode them.
iii) Put the required quantity of seeds in the container and add commercial concentrated sulphuric acid at the rate of 100 ml per kg of fuzzy seed.
iv) Stir vigorously and continuously with a wooden stick for 2 to 3 minutes till the fuzz sticking to the seeds is completely digested and the seed coat attains a dark brown colour of coffee powder.
v) Add water to fill the container. Drain the acid water and repeat the washing 4 or 5 times to remove any trace of acid.
vi) Remove the floating, ill-filled and damaged seeds while retaining the healthy and good seeds which remain at the bottom.
vii) Drain the water completely and dry the delinted seeds in shade.

Advantages of Acid delinting

i) Eliminates some externally seed borne pathogenic organisms.
ii) Kills eggs, larvae and pupae of pink boll worm.
iii) Helps to remove immature, ill-filled, cut and damaged seeds.
iv) Makes seed dressing more effective and easy
v) Facilitates easy sowing and good germination.

1.2.4.1.PRE-TREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES

i) Treat the delinted seeds with talc formulation of Trichoderma viride @ 4g/kg of seed or with Carbendazim (or) Thiram @ 2g/kg of seed.
Biocontrol agents are compatible with biofertilizers.
First treat the seeds with biocontrol agents and then with biofertilizers. Fungicides and biocontrol agents are incompatible.
ii) Treat the delinted fungicide treated seeds with 3 packets (600 g) of Azospirillum and 3 packets of phosphobacteria 600g (or) 6 packets of Azophos (1200 g) and sow immediately.

1.2.4.2.SEED HARDENING

Soak the seeds in equal volume of Pungam leaf extract (1%) for 8 hours and dry back to original moisture to increase germination and vigour. Dry the seeds in shade.

Seed pelleting: Seeds coated with arappu leaf powder (100 g/kg) along with DAP (40 g/kg), micronutrient mixture (15 g/kg) and Azospirillum (200 g/kg) phosphobacteria (200 g/ha) or Azophos (400 g/ha) using 5% maida solution or gruel as adhesive (300 ml/kg) to increase the germination and vigour.

1.2.5.SOWING

i) Dibble the seeds at a depth of 3-5 cm on the side of the ridge 2/3 height from the top and above the band where fertilisers and insecticides are applied, maintaining the correct spacing and then cover seeds with soil.
ii) In the case of intercropping, sow the seeds of the intercrop in between the paired rows of cotton in a row of 5 cm apart and cover the seeds.
iii) Sow the required number of seeds in each hole.

Varieties / hybrids	No. of seeds / hole	
	Fuzzy seeds	Delinted seeds
Hybrids	2	1
Varieties	3	2

1.2.6. WEED MANAGEMENT

i) Apply Pendimethalin @ 1.0 litre/ha three days after sowing or Fluchloralin 1.0 kg a.i /ha on 3DAS + power weeding on 45 DAS followed by earthing up or Trifloxy salfuron @ 10 g/ha on 15 DAS for broad leave weeds and sedges. Or Pre emergence application of Pendimathalin (38.7% CS) 650 ml/ha. Sufficient moisture should be present in the soil at the time of herbicide application. This will ensure weed free condition upto 40 days.
ii) One hand weeding on 45 DAS will keep weed free environment upto 60 DAS.
iii) Hoe and hand weed between 18th to 20th day of sowing, if herbicide is not applied at the time of sowing.

1.2.7.GAP FILLING

a. Take up gap filling on the 10th day of sowing.
i) In the case of TCHB 213, raise seedlings in polythene bags of size 15 x 10 cm.
ii) Fill the polythene bags with a mixture of FYM and soil in the ratio of 1:3.
iii) Dibble one seed per bag on the same day when sowing is taken up in the field.
iv) Pot water and maintain.
v) On the 10th day of sowing, plant seedlings maintained in the polythene bags, one in each of the gaps in the field by cutting open the polythene bag and planting the seedling along with the soil intact and then pot water.

b. In the case of all other varieties, dibble 3 to 4 seeds in each gap and pot water.

1.2.8.THINNING

Thin out the seedlings on the 15th day of sowing. In the case of fertile soils, allow only one seedling per hole, whereas in poor soil allow two seedlings per hole.

1.2.9.TOP DRESSING

iv) Top dress 50% of the recommended dose of N and K on 40 – 45 DAS for varieties.

v) Top dress 1/3rd of recommended dose of N on 40-45 DAS and the remaining 1/3rd on 60- 65th DAS for hybrids.

1.2.10.RECTIFICATION OF RIDGES AND FURROWS

Reform the ridges and furrows after first top dressing in such a way that the plants are on the top of the ridges and well supported by soil.

1.2.11.SPRAYING OF NAPTHALENE ACETIC ACID (NAA)

Spray 40 ppm NAA at 60 and 90 days after sowing on the crop to prevent early shedding of buds and squares and to increase the yield.

NOTE: 40 mg of NAA dissolved in one litre of water will give 40 ppm.

1.2.12.MANAGEMENT STRATEGIES FOR DELAYED SUMMER IRRIGATED COTTON SOWING

KCl 1% spray, twice on 50 and 70 DAS for delayed sowing (first fortnight of March) of summer irrigated cotton in rice-cotton cropping system for Srivilliputhur region.

1.2.13.ARRESTING TERMINAL GROWTH

Nip the terminal portion of the main stem as indicated below:

For varieties having less than 160 days duration nip the terminal portion of the main stem beyond the 15th node (75 to 80 DAS) and for varieties and hybrids having more than 160 days duration beyond the 20th node (85 - 90 DAS).

II. WATER MANAGEMENT

Regulate irrigation according to the following growth phases of the crop.

Stages	No. of Irrigations	Days after dibbling seeds	
		Light soil	**Heavy soil**
Germination Phase(1-15 days)			
Irrigate for germination	1	Immediately after sowing	Immediately after sowing
and establishment	2	Give a life irrigation on	Give a life irrigation on
		5th day of sowing to	5th day of sowing to
		facilitate the seedlings	facilitate the seedlings
		to emerge out	to emerge out
Vegetative phase (16-44 days)			
Regulate	1	Irrigate on the 20th or	Irrigate on the 20th or 21st day
		21st day of sowing, three	of sowing, three days after
		days after hoeing and	hoeing and weeding
		Weeding	
	2	Irrigate again on	Irrigate again on

		the 35th or 36th	the 40th day of
		day of sowing	Sowing
Flowering phase (45-100daysfor hybrids and 87 days for varieties)			
Irrigate copiously	1	48th day	55th day
	2	60th day	70th day
	3	72nd day	85th day
	4	84th day	100th day
	5	96th day	**
Maturity phase (beyond 100 days for hybrids and 88 days for varieties)		For all varieties other than Suvin	
Control irriga	1	108th day	115th day
tion during	2	120th day	130th day
maturity phase	3	130th day	
	4	144th day	
	Stop Irrigation after 150th day		
	For Suvin		
	1	108th day	115th day
	2	120th day	130th day
	3	132nd day	145th day
	4	144th day	160th day
	5	158th day	...
Stop irrigation after 160th day			
NOTE: i.	*If irrigation is given on climatological approach, Schedule the irrigation at 0.40 and 0.60 IW/CPE ratio during vegetative and reproductive phases respectively.*		
ii)	*The irrigation schedule given above is only a guideline and regulate the irrigation*		
	depending upon the prevailing weather condition and receipt of rains.		
iii)	*Adopt alternate furrow or skip furrow irrigation to save irrigation water.*		

The features of the methods are furnished below:

Skip furrow irrigation

a) Suited to heavy soils like clay and loam
b) Alternate furrows should be skipped and may be converted to ridges having a wide bed formation.
c) Short term crops like pulses may be raised in wider bed without exclusive irrigation.
d) Water saving is 50% when compared to control.

Alternate furrow irrigation

a) During any one run of irrigation a particular set of alternate furrows is irrigated.
b) The interval of irrigation should be shortened when compared to the conventional furrows.
c) During the next run, the left over furrows be irrigated.
d) Suited to heavy soils like clay and loam.

III. HARVESTING

a) Harvest at frequent intervals, at less than 7 days interval.
b) Harvest in the morning hours upto 10 to 11 a.m only when there is moisture so that dry leaves and bracts do not stick to the kapas and lower the market value.
c) Pick kapas from well burst bolls only.
d) Remove only the kapas from the bolls and leave the bracts on the plants.
e) After kapas is picked, sort out good puffy ones and keep separately.
f) Keep stained, discoloured and insect attacked kapas separately.

NOTE: Do not mix stained, discoloured and insect damaged kapas with good kapas, as they will spoil the good kapas also and lower the market value of the produce.

IV.POST HARVEST OPERATIONS

1) Immediately after picking, dry the kapas in shade. If it is not dried immediately the colour will change which will lower the market value.
2) Do not dry the kapas under direct sun as the fibre strength and luster will be lost.
3) Grade the kapas into good and second quality ones, if it is not sorted out at the time of picking.
4) Spread a thin layer of dry sand on the ground and keep the kapas over it.

RICE FALLOW COTTON

2.1.PREPARATION OF THE FIELD

i) If the soil is in waxy condition, instead of Zero tillage, the seed rows may be tilled and the seed dibbled in Virudhunagar district.
ii) If the soil is dry and not in condition to take up sowing, let in water and then allow the soil to dry till soil comes to waxy condition.
iii) At the lower level of the field dig a trench 15 cm wide and connect this trench to the outside channel to drain off the excess water.

2.2.PRE-TREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES

iv) Same as for the irrigated crop.
v) Treat the acid delinted and fungicide treated seeds with 3 packets (600g) of Azospirillum and sow immediately.

2.3.SOWING THE SEEDS

Particulars		MCU 7	SVPR 3
a) Seed rate (kg/ha)			
	i) Fuzzy seed	15.0	15
	ii) Acid delinted	7.5	7.5
b) Spacing (cm)			
	i) Between rows	60	60 or 75*
	ii) Between plants	30	30
c) Number of seeds / hole			
	i) Fuzzy seeds	4	4
	ii) Acid delinted	2	2
d) Depth of sowing (cm)		3	3

* In fertile soils

2.4.FILLING UP GAPS

vi) Fill up gaps on the 10th day of sowing.

vii) Dibble 2 to 3 acid delinted seeds or 4 to 5 fuzzy seeds in the gaps in the case of MCU 7 and SVPR 3

2.5.THINNING SEEDLINGS

viii) Thin out seedlings on the 20th day of sowing

ix) Leave only one healthy and vigorous seedling per hill.

2.6. WEED MANAGEMENT

i) Pre-emergence application of Pendimethalin 1.0 litre/ha ensures weed free condition for 40 - 45 days. This should be followed by one hand weeding and earthing up during 40 - 45 days.

ii) Take up hoeing and weeding 20 days after sowing.

iii) Take up this operation when the top soil dries up and comes to proper condition.

2.7.APPLICATION OF FERTILIZERS

a) Apply NPK fertilisers as per soil test recommendations. If soil test is not done follow the blanket recommendation of 60:30:30 kg NPK/ha.

b) Apply half the dose of N and K full dose of P_5O_5 at 35th day in old delta and balance in 55 days the rows of cotton plants. In the case New delta apply full P and 1/3 of N and K at 20 DAS and 2/3 N and K at 40 DAS.

2.8.APPLICATION OF MICRONUTRIENTS

Apply basally12.5 kg/ha micronutrient mixture prepared by Department of Agriculture. Apply $MgSo_4$ basally @ 30 kg/ha to prevent reddening.

2.9.FORMATION OF RIDGES Old delta

a) If soil is in condition, give a hoeing with mammutti and form ridges and incorporate the fertilizer in the soil around the plants between 30th to 35th day of sowing.

b) If soil is not in condition, give one hoeing and weeding and cover the fertilizers.

c) Form long ridges and furrows from one end of the field to the other without forming any separate channels for carrying water to prevent excessive soaking of water.

d) Form ridges and furrows on alternate rows of plants. Skip furrow method of irrigation to prevent excessive irrigation

New delta

a) Give a hoeing with mummutti and form ridges and incorporate the fertiliser in the soil around the plants on the 40th day of sowing.

b) If soil is not in condition give one hoeing and weeding and cover the fertilizers.

c) Form long ridges & furrows on alternate rows of plants to adopt skip furrow irrigation.

Note: In case of zinc deficient soils, apply 50 kg ZnSo4 /ha

2.10. APPLYING OF NAA

Spray 40 ppm of NAA (40 mg of NAA dissolved in one litre of water) at 40/45th day using high volume spray. Repeat the same dose after 15 days of first spraying.

2.11. TOPPING

Arrest terminal growth by nipping the terminal 15th node for controlling excessive vegetative growth. (70-75 DAS

2.12.WATER MANAGEMENT

Regulate irrigation according to the growth phases of the crops.

Stages	No. of	Days after dibbling seeds	
	Irriga		
	tions	Old delta	New delta
1. Vegetative Phase			
Regulate irrigation	1	One wetting on the 30th	One irrigation on the 20th day
during the germination		to 35th day of sowing after	after the application of fertilisers
phase		the application of fertilisers	
	2	...	One irrigation on the 40th day after the application of N
2. Flowering Phase			
Irrigate more frequently	1	45th day of sowing after the	45th day
		application of 2nd dose of N	
	2	55th day	51st day
	3	65th day	56th day
	4	75th day	61st day
	5	85th day	66th day
	6	...	71st day
	7	...	76th day
	8	...	81st day
	9	...	86th day
	10	...	91st day
3. Control	1	99th day	98th day
Irrigation during	2	113th day	105th day
maturity phase	3	...	112th day

Stop irrigation from the 113th day onwards.

Note: 1) The irrigation schedule given above is only a guideline and regulate irrigation depending upon the prevailing weather conditions and receipt of rains.

2) Observe the crop and if the plants show wilting symptoms in the afternoon and in the evening hours, give an additional irrigation.

Harvesting

Post harvest operation

As that of the irrigated cotton.

Pest and disease management

RAINFED COTTON

Follow water harvesting techniques and raise a successful crop of cotton.

3.1. SEASON AND VARIETIES

For Thirumangalam in Madurai district, Sattur in Virudhunagar district and parts of Kovilpatti in Thoothukudi district, where the seasonal rainfall is 375 mm and most of it is received during September or first week of October. Select LRA 5166 (or) SVPR 2 (or) KC 2, KC 3.

In places, where rains are received during October or November, Select K 11 for Ramanathapuram, Virudhunagar, Tirunelveli and Thoothukudi districts.

3.2.PREPARATION OF LAND

3.2.1.PREPARATION OF THE FIELD

i) Start preparation of the land immediately after harvest of the previous crop.
ii) Adopt permanent broad ridges system.

3.2.2.APPLICATION OF FYM OR COMPOST

iii) Spread 12.5 t of FYM or compost or composted coir pith or 2.5 t of vermicompost per ha uniformly on the unploughed soil.
iv) Incorporate the manure in the soil by working the multipurpose implement or country plough.
v) Apply TNAU MN mixture @ 7.5 by as enriched FYM.

1.2.3. APPLICATION OF INORGANIC FERTILIZERS

i) Apply NPK fertilizers as per soil test recommendation as far as possible.
ii) If soil tests are not done, follow the blanket recommendations for the different varieties.

Varieties	Quantity of fertilizers (Kg/ha)		
	N	P_2O_5	K_2O
K 11	20	0	0
SVPR 2	40	20	40
KC 2	40	20	40

Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Rainfed Cotton -Bt Hybrid

Soil : Black (Pilamedu series)

Target : 2.8 - 3.2 t ha^{-1}

$FN = 5.35T-0.24\ SN-0.53\ ON$

$FP_2O_5 = 3.67T-1.99\ SP-0.84\ OP$

$FK_2O = 3.83T-0.13SK-0.55\ OK$

Initial soil test values (kg ha^{-1})			Yield target – 2.8 t ha^{-1}			Yield target – 3.2 t ha^{-1}		
			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
175	14	200	83	45**	45**	90**	45**	45**
200	16	250	77	45**	45**	90**	45**	45**
225	18	300	71	45**	42	90**	45**	45**
250	20	350	65	43	36	86	45**	45**
275	22	400	59	39	29	80	45**	45**

** Maximum dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in q ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure in kg ha^{-1}.

3.2.4. APPLICATION OF MICRONUTRIENT MIXTURE

vi) Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg. (or)
Apply TNAU MN mixture @ 7.5 kg /ha as Enriched FYM (Prepare enriched FYM at 1:10 ratio of MN mixture & FYM ; mix at friable moisture &incubate for one month in shade).

vii) Apply uniformly over the furrows after sowing and cover the seeds. Do not incorporate in the soil.

Yield Maximization and reducing reddening in Bt cotton Rainfed

Application of TNAU MN mixture (7.5 kg ha^{-1} as EFYM for variety and 10 kg ha^{-1} as EFYM for Bt cotton) along with the recommended NPK to obtain maximum seed cotton yield with reduced extent of leaf reddening

SEEDS AND SOWING

viii) Adopt the following seed rates for different varieties/hybrids.

Varieties	Quantity of seeds (kg/ha) Fuzzy seeds	Delinted seeds
K 11	20	..
LRA 5166, SVPR 2	20	15

Note: Delint only LRA 5166 and SVPR 2 seeds. Do not delint seeds of K 11

ix) In the case of mixed crop of cotton, maintaining the same seed rates as for a pure crop and
adopt the following seed rate for the pulses crop. Blackgram/greengram10 kg/ha
Cowpea 7.5 kg/ha

3.2.6.SPACING

x) In the case of pure crop of varieties/hybrids, a spacing of 45 cm between rows and 15 cm between plants may be adopted.

xi) In the case of cotton, intercropped with pulses, one paired row of cotton is alternated with two rows of pulses and the total population of cotton crop is maintained at the same line as that for a pure crop of cotton.

xii) Adopt a spacing of 30 x 10 cm for the pulse crop in between each paired row of cotton. APK 1 Blackgram is best suited for this situation.

3.2.7.ACID DELINTING

Adopt procedure for acid delinting as for an irrigated crop.

3.2.8.PRETREATMENT OF ACID DELINTED SEEDS WITH FUNGICIDES

Same as for the irrigated crop.

3.2.9.SOWING

xiii) Use the multipurpose farming implement to sow the seeds and to apply basal fertilizers simultaneously.

xiv) Fill the hopper in the implement with the fertilizer mixtures and work the implement.

xv) Engage 3 persons for dropping the seeds, 2 for cotton and one for pulses.

In one operation, placement of fertilizer, sowing of seeds and covering will be completed.

NOTE: Cotton and pulses can be sown at a depth of 5 cm in black cotton soil even before the onset of monsoon rains in dry bed sowing. When light rains are received, the moisture will not penetrate deeper and the seeds will not germinate and die away. Only when good rains are received, the moisture level will be sufficient to penetrate to the level of the seed and facilitate germination and proper establishment.

3.2.10. WEED MANAGEMENT

xvi) Pre-emergence application of Pendimethalin (38.7% CS) 650 ml/ha followed by one hand weeding on 40 days after crop emergence. At the time of herbicide application sufficient soil moisture must be there.

xvii) If sufficient soil moisture is not available for applying herbicides hand weeding may be given at 10 - 20 days after crop emergence.

xviii) Integrated weed management in cotton: Post emergence application of pyrithiobac sodium @ 62.5g a.i./ha + quizalofop ethyl @ 50 g a.i./ha at 2 to 4 leaf stage or 45 DAS.

3.2.11. GAP FILLING

Dibble 3 to 4 seeds in each gap if sufficient moisture is available.

3.2.12. THINNING SEEDLINGS

xix) Allow two seedlings per hole and thin out on 15th day of sowing, adopting proper spacing between plants.

xx) Thin the pulse crop on the 20th day of sowing, adopting a spacing of 15 cm between plants for cowpea and 10 cm for other pulse crop.

3.2.13. FOLIAR FERTILIZATION

Spray 0.5% urea and 1% KCl on the 45th and 65th day of sowing if sufficient moisture is available.

In site water harvesting and crop resident addition for rainfed cotton in black soil.

Brand Bed Furrow (BBF) system wth coirpith application @ 5 t/ha for higher soil moisture retension, seed cotton yield and enhanced the carbon storage under vermisols.

3.2.14. INTERCULTIVATION WITH DHANTHULU/BLADE HARROW

Work dhanthulu or blade harrow on the 30th and 45th day of sowing.

NOTE: Other cultivation practices, plant protection measures, harvest etc., are the same as for the irrigated crop.

CROP PHYSIOLOGY

Foliar spray of TNAU Cotton Plus @ 2.5 kg/acre in 200 litres of water at flowering and at boll formation stages reduces flower and square shedding, improves boll bursting, increases seed cotton yield and imparts drought tolerance.

CROP PROTECTION

B. PEST MANAGEMENT

- Remove the cotton crop and dispose off the crop residues as soon as harvest is over.
- Avoid stacking of stalks in the field.
- Avoid ratoon and double cotton crop.
- Adopt proper crop rotation.
- Use optimum irrigation and fertilizers.
- Grow one variety throughout the village as far as possible.
- Treat the seeds with imidacloprid or use designer seed (Delinted seed + polykote @ 3g / kg + carbendazim @ 2 g / kg + imidacloprid @ 7 g / kg + *Pseudomonas fluorescens* 10 g / kg + Azophos 40 g / kg). When the treated seeds are used, it protects against sucking pests upto 45 days after sowing and promotes early vigour of the crop.
- Synchronize the sowing time in the villages and complete the sowing within 10 to 15 days.
- Avoid other malvaceous crops in the vicinity of cotton crop.
- Timely earthing up and other agronomic practices should be done.
- Hand pick and burn periodically egg masses, visible larvae, affected and dropped squares, flowers and fruits and squash pink bollworm in the rosettes.

- Use locally fabricated light traps (modified Robinson type) with 125W mercury lamps to determine the prevalence and insect population fluctuations.
- The magnitude of the activity of the moths of the cotton pink bollworm, the cut worm (*Spodoptera litura*) and the American bollworm can be assessed by setting up the species-specific sex pheromone trap each at the rate of 12 per ha.
- Apply insecticides only where it is absolutely necessary when pest population or damage reaches ET level.
- Intercropping with pulses viz., cowpea, greengram, blackgram, soybean and maize reduces the bollworm incidence and population of sucking pests of cotton, *viz*., aphid and leafhopper with the highest activity of natural enemies *viz*., spiders and predatory lady bird beetles.

Economic threshold level for important pests

Pest	ETL
Thrips	50 nymphs or adults / 50 leaves
Aphids	15% of infested plant
Leafhopper	50 nymphs or adults / 50 leaves
Mite	10 mites / cm^2 leaf area
Boll-worms	
Spotted	10% infested shoots / squares / bolls
Spiny	10% infested shoots / squares / bolls
Pink	10% infested fruiting parts
Helicoverpa	One egg or one larva / plant
Whiteflies	5-10 / leaf
Stem weevil	10% infestation
Tobacco cutworm	8 egg masses / 100 m row

American bollworm *Helicoverpa armigera*	**Monitoring:** Pest monitoring through light traps, pheromone traps and *in situ* assessments by roving and fixed plot surveys has to be intensified at farm, village, block, regional and State levels. For management, an action threshold of one egg per plant or 1 larva per plant may be adopted. **Cultural practices:** • Synchronized sowing of cotton preferably with short duration varieties in each cotton ecosystem. • Avoid continuous cropping of cotton both during winter and summer seasons in the same area as well as rationing. • Avoid monocropping. Growing of less preferred crops like greengram, blackgram, soyabean, castor, sorghum *etc.*, along with the cotton as intercrop or border crop or alternate crop to reduce the pest infestation. • Removal and destruction of crop residues to avoid carry over of the pest to the next season, and avoiding extended period of crop growth by continuous irrigation.

	• Optimizing the use of nitrogenous fertilizers which will not favour the multiplication of the pest. • Judicious water management for the crop to prevent excessive vegetative growth and larval harbourage. **Biological control:** • Application of Nuclear Polyhedrosis Virus (NPV) at 3 x 10^{12} POB / ha in evening hours at 7^{th} and 12^{th} week after sowing. • Conservation and augmentation of natural predators and parasitoids for effective control of the pest. • Inundative release of egg parasitoid, *Trichogramma* spp., at 6.25 cc / ha at 15 days interval 3 times from 45 days after sowing • Egg- larval parasitoid, *Chelonus blackburnii* and predator, *Chrysoperla carnea* at 1,00,000 / ha at 6^{th}, 13^{th} and 14^{th} week after sowing. • ULV spray of NPV at 3 x 10^{12} POB / ha with 10% cotton seed kernel extract, 10% crude sugar, 0.1% each of Tinopal and Teepol for effective control of *Helicoverpa*. **Note:** Dicofol, methyl demeton and monocrotophos are comparatively safer to *Chrysoperla* larva recording low egg mortality. **Chemical control :** • Discourage the indiscriminate use of insecticides, particularly synthetic pyrethroids. • Use of proper insecticides which are comparatively safer to natural enemies at the correct dosage and alternating different groups of insecticides for each round of spray. • Avoid combination of insecticides as tank mix. • Adopt proper delivery system using spraying equipments like hand compression sprayer, knapsack sprayer and mist blower to ensure proper coverage with required quantity of spray fluid and avoid ULV applications or Akela spray applications. • Proper mixing and preparation of spray fluid for each filling of spray fluid tank. **At early stages of square formation apply one of the following insecticides** Acephate 75% SP 780g/ha Azadirachtin 0.03% EC 2500ml/ha Chlorpyriphos 20% EC 1250ml/ha Diflubenzuron 25% WP 300 - 350g/ha Emamectin benzoate 5% SG 190-220g/ha Fipronil 5% SC 2000ml/ha Flubendiamide 20% WG 250g/ha Flubendiamide 39.35% SC 100-125ml/ha Indoxacarb 14.5% SC 500ml/ha Novaluron 10%EC 1000ml/ha NPV of *H. armigera* 0.43%AS 400-600ml/ha

	Profenofos 50% EC 1500-2000ml/ha Pyridalyl 10% EC 750-1000ml/ha Chlorantraniliprole 18.5% SC @150 ml/ha Lufenuron 5.4% EC @600 ml/ha Spinosad 45.0% SC 165-220ml/ha Thiodicarb 75%WP 1000g/ha Monocrotophos 36% SL @1125-2250 ml/ha **Biological control:** *Bacillus thuringiensis*-k750-1000g/ha *Bacillus thuringiensis* var. kurstaki (3a,3b,3c) 5%WP 750-1000g/ha *Beauveria bassiana* 1.15%WP 400g/ha
Spotted bollworm *Erias vitella;* *E. insulana*	**Spraying any one of the following insecticides** Flubendiamide 39.35%SC100-125ml/ha Chlorantraniliprole 18.5% SC @150 ml/ha Indoxacarb 14.5%SC 500ml/ha Diflubenzuron 25%WP 300-350g/ha Profenophos 50%EC 1500-2000ml/ha Fipronil 5% SC 2000ml/ha Spinetoram 11.7 % SC @420-470 ml/ha **Biological control:** *Bacillus thuringiensis* var kurstaki (3a,3b,3c) 5%WP 750-1000g / ha
Pink boll worm *Pectinophora gossypiella*	• Use pheromone traps to monitor the adult moth activity @ 12 / ha • Inundative release of egg parasitoid *Trichogrammatoidea bactrae* @ 40,000 / ha at 15 days interval 3 times from 45 days after sowing with coinciding the incidence of the pest. **Spraying any one of the following insecticides:** Emamectin Benzoate 5% SG 190 – 220 g/ha Chlorpyriphos 50% EC 1000 – 1200 ml / ha Profenofos 50%EC 1500 – 2000ml/ha Diflubenzuron 25%WP 300-350g/ha Thiodicarb 75% WP 1000 g /ha
Tobacco cutworm *Spodoptera litura*	➢ Use of light trap to monitor and kill the attracted adult moths. ➢ Set up the sex pheromone trap at 12/ha to monitor the activity of the pest and to synchronize the pesticide application, if need be, at the maximum activity stage. ➢ Growing castor along border and irrigation bunds. ➢ Removal and destruction of egg masses in castor and cotton crops. ➢ Removal and destruction of early stage larvae found in clusters which can be located easily even from a distance. ➢ Collection and destruction of shed materials. ➢ Hand picking and destruction of grownup caterpillars. **Spray any one of the following insecticides** Chlorpyriphos 20 EC 3750 ml/ha Diflubenzuron 25%WP 300-350g/ha

	Chlorantraniliprole 18.5% SC @150 ml/ha Spinetoram 11.7 % SC @420-470 ml/ha ➢ Spraying of insecticides should be done either in the early morning or in the evening and virus in the evening. ➢ Spraying nuclear polyhedrosis virus at 1.5 x 10^{12} POB per ha.
Stem weevil *Pempherulus affinis*	• Basal application of FYM 25 t/ha and 250 kg/ha of neem cake. • Seed treatment with chlorpyriphos 20EC @ 10ml/kg of seed + drenching collar region with chlorpyriphos 50 EC @ 1200 ml/ha on 15 and 30 days after sowing + Earthing up.
Whitefly *Bemisia tabaci*	• Avoid alternate, cultivated host crops of the whitefly in the vicinity of cotton crop. • Growing cotton only once a year either in winter or summer season in any cotton tract. • Adopting crop rotation with non-preferred hosts such as sorghum, ragi, maize, etc., for the whitefly to check the buildup of the pest. • Removal and destruction of alternate weed hosts like *Abutilon indicum*, *Chrozophore rottlari*, *Solanum nigrum* and *Hibiscus ficulensus* from the fields and neighbouring areas and maintaining field sanitation. • Timely sowing with recommended spacing, preferably wider spacing and judicious application of recommended dose of fertilizers, particularly nitrogenous and irrigation management is essential to arrest the excessive vegetative growth and pest build up. Late sowing may be avoided and the crop growth should not be extended beyond its normal duration. • Field sanitation may be given proper attention. • Cultivation of most preferred alternate host crops like brinjal, bhendi, tomato, tobacco and sunflower may be avoided. In case their cultivation is unavoidable, plant protection measures should be extended to these crops also. • Monitoring the activities of the adult white flies by setting up yellow pan traps and sticky traps at 1 foot height above the plant canopy and also in situ counts. • Collection and removal of white fly infested leaves from the plants and those which were shed due to the attack of the pest and destroying them. **Chemical control:** Acetamiprid 20%SP 100g/ha Azadirachtin 0.15% 2500-5000ml/ha Buprofezin 25% SC 1000ml/ha Chlorpyriphos 20%EC 1250ml/ha Clothianidin 50%WDG 200-250 g/ha(Soil drenching) Clothianidin 50%WDG 40-50 g/ha (Foliar spray) Diafenthiuron 50%WP 600g/ha Dinotefuran 20% SG 150 g/ha Fipronil 5%SC 1500-2000 ml/ha Flonicamid 50% WG @ 150 g/ha

	Imidacloprid 17.8% SL 100-125 ml/ha Profenophos 50%EC 1000 ml/ha Thiacloprid 21.7%SC 500- 600ml/ha Thiamethoxam 30% FS @10 g/Kg seed (Seed dresser) Thiamethoxam 70% WS @430 gm/ha Thiamethoxam 25%WG 200 g/ha Pyriproxyfen 10% EC @ 500- 700 ml/ha Spiromesifen 22.9% SC @600 ml/ha Spray any one of the following plant products alone or in combination with the recommended dose of insecticide Neem seed kernel extract 5% or Neem oil at 5 ml/l of water Fish oil rosin soap 25g / lit of water Notchi leaves 5% extract *Catharanthus rosea* extract 5% Spray any one of the following in early stage (500l) mid and late stages (1000l spray liquid/ha) In the early stages with high volume sprayer, use a goose neck nozzle to cover the under surface of the foliage to get good control of the pest. If high volume sprayers are not available, 375 litres of spray fluid may be used per hectare for application in the low volume motorized knapsack mist blower. • The use of synthetic pyrethroids should be discouraged in cotton to avoid the problem of white fly. Cypermethrin, fenvalerte and deltamethrin cause resurgence of whiteflies. So avoid repeated spraying of pyrethroids. • The plant protection measures should be adopted on a community basis in specified cotton areas. **Biological control:** • *Verticillium lecanii* 1.15%WP 2500g/ha
Thrips *Thrips tabaci*	• Seed treatment with imidacloprid 70WS at 7g / kg protect the crop from aphids, leafhoppers and thrips upto 8 weeks. **Spray any one of the following insecticides** (500l spray fluid/ha) Methyl demeton 25EC 500ml/ha Dimethoate 30EC 500ml/ha Buprofezin 25%SC 1000ml/ha Diafenthiuron 50%WP 600g/ha Clothianidin 50%WDG 200-250 g/ha (Soil drenching) Dinotefuran 20% SG 150 g/ha Fipronil5% SC 1500-2000ml/ha Flonicamid 50% WG @ 150 g/ha Imidacloprid 70%WG 30-35g/ha Imidacloprid 48%FS/100 kg seed 500-900g/ha Imidacloprid 17.8%SL 100-125ml/ha Profenophos 50% EC 1000ml/ha Thiacloprid 21.7% SC 100-125ml/ha Thiamethoxam 70% FS 430 g/ha Thiamethoxam 25%WG 100 g/ha Spinetoram 11.7 % SC @420-470 ml/ha

Aphids *Aphis gossypii*	Seed treatment with imidacloprid 70WS at 7g / kg protect the crop from aphids, leafhoppers and thrips upto 8 weeks. **Spray any one of the following insecticides** Acetamiprid 20%SP 50g/ha Azadirachtin 0.03%EC 2500ml/ha Buprofezin 25%SC 1000ml/ha Clothianidin 50%WDG 200-250 g/ha(Soil drenching) Carbosulfan 25% DS 60g/kg seed Chlorpyriphos 20% EC 1250ml/ha Diafenthiuron 50%WP 600ml/ha Dinotefuran 20% SG 150 g/ha Fipronil 5% SC 1500-2000 ml/ha Flonicamid 50% WG @ 150 g/ha Imidacloprid 70% WG 30-35kg/ha Imidacloprid 17.8% SL 100-125ml/ha Malathion 50% EC 1000ml/ha Profenophos 50% EC 1000ml/ha Thiacloprid 21.7%SC 100-125ml/ha Thiamethoxam 25%WG100 g/ha Thiamethoxam 70% WS @430 gm/ha Thiamethoxam 30% FS @10 g/Kg seed (Seed dresser)
Leaf hopper *Amrasca devastans*	**Spray any one of the following insecticides** Imidacloprid 200SL at 100ml/ha Imidacloprid 70% WG @30-35 g/ha Imidacloprid 17.8% SL @ 100 – 125 ml/ha Acetamiprid 20%SP 50g/ha Azadirachtin 0.03%WSP 2500- 5000g/ha Buprofezin 25%SC 1000ml/ha Clothianidin 50%WDG 30- 40 kg/ha (Foliar spray) Clothianidin 50%WDG 200-250 g/ha(Soil drenching) Diafenthiuron 50%WP 600g/ha Dinotefuran 20% SG 150 g/ha Fipronil 5%SC 1500-2000ml/ha Profenophos 50%EC 1000ml/ha Thiacloprid 21.7%SC 100-125ml/ha Flonicamid 50% WG @ 150 g/ha Thiamethoxam 25%WG 100g/ha Thiamethoxam 30% FS @10 g/Kg seed (Seed dresser) Thiamethoxam 70% WS @430 gm/ha NSKE 5% 25kg/ha • Where the leaf hopper is a big menace apply Neem oil formulation 0.5% or neem oil 3% thrice at fortnightly intervals.
Cotton mealy bug *Phenococcus solenopsis*	• Remove the alternate weeds hosts. • Monitor the incidence regularly and look for crawler emergence. • Take up the management at initial stage to get maximum control. • Use of encyrtid parasitoids, *Acerophagus papayae* @ 100 per village against *Paracoccus marginatus* and *Aenasius*

	bambawaeli against *Phenococcus solenopsis* are recommended. (Consult the specialists for effective chemicals for individual species). • Wherever necessary use botanical insecticides like neem derivatives such as neem oil 2%, NSKE 5% and Fish oil rosin soap 25 g/lit. of water. • Use of profenophos @ 2000 ml / ha may be adopted as an alternative
Yellow mite *Polyphagotarsonemus latus*	Spiromesifen 22.9% SC 600 ml/ha
Red spider mite *Tetranychus cinnabarinus*	Spray Spiromesifen 22.9% SC 600 ml/ha or Dicofol 18.5% EC @ 2700ml/500 lit. of water

Pest management strategies

Resurgence

Repeated application of the following insecticides can cause resurgence of the insect pest of cotton

- *Amrasca devastans :* Deltamethrin
- *Aphis gossypii :* Cypermethrin, deltamethrin, fenvaerate, monocrotophos
- *Bemisia tabaci :* Cypermethrin, deltamethrin, fenvalerate, monocrotophos
- *Ferrisia virgata :* Cypermethrin, deltamethrin, fenvalerate, permethrin
- *Tetranychus urticae :* Acephate, fenvalerate

Disease Management

Name of the disease	Recommendations
Bacterial leaf blight: *Xanthomonas axonopodis* pv. *malvacearum*	• Avoid stacking of infected plants • Spray *streptomycin sulphate* @ 300 ppm + copper oxychloride @ 2.0 kg/ha immediately after the symptom appearance and repeat at 10 days later.
***Alternaria* leaf spot:** *Alternaria macrospora*	Spray any one of the following fungicides / biocontrol agent • Copper oxychloride @ 2 kg or mancozeb @ 1 kg or chlorothalonil @ 500 g/ha or difenaconazole @ 0.05% or krexoxym methyl @ 0.1% or tebuconazole @ 1 ml/l or trifloxystrobin + tebuconazole @ 0.6 g/l or propiconazole @ 1 ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing. • *Bacillus subtilis* (BSC5) @ 0.04% on 60, 90 and 120 days after sowing can also be applied.
Grey mildew: *Ramularia areola*	Spray any one of the following fungicides • Carbendazim @ 250 g/ha or mancozeb @ 1000g or chlorothalonil @ 500 g/ha or difenaconazole @ 0.05% or krexoxym methyl @ 0.1% or tebuconazole @ 1ml/l or propiconazole @ 1ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing.
Boll rot: *Fusarium moniliforme, Colletotrichum capsici,*	Spray any one of the following fungicides • Carbendazim @ 500 g or mancozeb @ 2000 g or copper oxychloride

Aspergillus flavus, A. niger, Rhizopus nigricans, Nematospora, Botryodiplodia	@ 2500 g/ha along with an insecticide recommended for bollworm from 45th day at fortnightly interval.
***Cercospora* leaf spot:** *Cercospora gossypii*	•Spray propiconazole @ 1 ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing.
Damping off and *Fusarium* wilt: *Rhizoctonia solani* and *Fusarium oxysporum* f. sp. *vasinfectum*	•Seed treatment with *Pseudomonas fluorescens* + *Bacillus subtilis* + *Trichoderma asperellum* mixture @ 10 g/kg and soil application of *P. fluorescens* + *B. subtilis* + *T. asperellum* mixture @ 2.5 kg/ha during sowing and at 90 days after sowing.
Root rot: *Macrophomina phaseolina (Rhizoctonia bataticola)*	**Cultural method** •Soil application of neem cake @ 150 kg/ha **Biological control** •Seed treatment with *T. asperellum* @ 10 g/kg followed by basal application of zinc sulphate @ 50 kg/ha. •Seed treatment with *Bacillus* (BSC5) @ 10g/kg followed by soil application @ 2.5 kg/ha with 250 kg of compost at the time of sowing. •Seed treatment with *P. fluorescens* @ 10 g/kg and soil application @ 2.5 kg/ha with 250 kg of compost at the time of sowing. •Seed treatment with *P. fluorescens* + *B. subtilis* + *T.asperellum* mixture @ 10g/kg and soil application of *P. fluorescens* + *B. subtilis* + *T. asperellum* mixture @ 2.5 kg/ha during sowing and at 90 days after sowing. **Chemical control** • Spot drench with carbendazim @ 1 g/l at the base of affected plants as well as surrounding healthy plants • Soil drenching with trifloxystrobin + tebuconazole @ 0.75g/l

Integrated pest and disease management (IPDM) technology for cotton

- Seed treatment with imidacloprid 70 WS @ 10 g/kg seed
- Soil drenching with chlorpyriphos 20 EC @ 1.25 l/ha on 25 days after sowing
- Soil application with *Bacillus subtilis* (BSC5) on 30 days after sowing
- Foliar application of *B. subtilis* (BSC5) @ 10g/l on 60 days after sowing
- Monitoring with yellow sticky traps for whitefly @ 12 numbers / ha
- Monitoring with pheromone trap for *Spodoptera* @ 12 numbers / ha
- Need based application of imidacloprid 17.8SL @ 25 g.a.i./ha
- Need based application of 0.1 per cent trifloxystrobin (25%) + tebuconazole (50%) WG or mancozeb 75 WP @ 0.25 per cent
- Raising of trap crop (castor and maize) along the bunds

Nematode Management

❖ Seed treatment with *Pseudomonas fluorescens* @ 20 g/kg followed by soil application @ 2.5kg/ha reduces reniform nematode, *Rotylenchulus reniformis* in cotton.

RAINFED COTTON

CROP PROTECTION

A. PEST MANAGEMENT

< The control measures recommended for irrigated cotton will hold good.

< **When water is not available, use any one of the following insecticides for the control of bollworms at 25 kg/ha :**

Carbaryl 5 D

Phosalone 4 D

B. DISEASE MANAGEMENT

Name of the disease	Recommendations
Bacterial leaf blight: *Xanthomonas axonopodis* pv. *malvacearum*	• Avoid stacking of infected plants • Spray *streptomycin sulphate* @ 300 ppm + copper oxychloride @ 2.0 kg/ha immediately after the symptom appearance and repeat at 10 days later.
***Alternaria* leaf spot:** *Alternaria macrospora*	Spray any one of the following fungicides / biocontrol agent • Copper oxychloride @ 2 kg or mancozeb @ 1 kg or chlorothalonil @ 500 g/ha or difenaconazole @ 0.05% or krexoxym methyl @ 0.1% or tebuconazole @ 1 ml/l or trifloxystrobin + tebuconazole @ 0.6 g/l or propiconazole @ 1 ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing. • *Bacillus subtilis* (BSC5) @ 0.04% on 60, 90 and 120 days after sowing can also be applied.
Grey mildew: *Ramularia areola*	Spray any one of the following fungicides • Carbendazim @ 250 g/ha or mancozeb @ 1000g or chlorothalonil @ 500 g/ha or difenaconazole @ 0.05% or krexoxym methyl @ 0.1% or tebuconazole @ 1ml/l or propiconazole @ 1ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing.
Boll rot: *Fusarium moniliforme, Colletotrichum capsici, Aspergillus flavus, A. niger, Rhizopus nigricans, Nematospora, Botryodiplodia*	Spray any one of the following fungicides • Carbendazim @ 500 g or mancozeb @ 2000 g or copper oxychloride @ 2500 g/ha along with an insecticide recommended for bollworm from 45th day at fortnightly interval.
***Cercospora* leaf spot:** *Cercospora gossypii*	• Spray propiconazole @ 1 ml/l or metiram 55% + pyraclostrobin 5% WG @ 0.1% at 60, 90 and 120 days after sowing.
Damping off and *Fusarium* wilt: *Rhizoctonia solani* and *Fusarium oxysporum* f. sp. *vasinfectum*	• Seed treatment with *Pseudomonas fluorescens* + *Bacillus subtilis* + *Trichoderma asperellum* mixture @ 10 g/kg and soil application of *P. fluorescens* + *B. subtilis* + *T. asperellum* mixture @ 2.5 kg/ha during sowing and at 90 days after sowing.
Root rot: *Macrophomina phaseolina (Rhizoctonia bataticola)*	**Cultural method** • Soil application of neem cake @ 150 kg/ha **Biological control**

	• Seed treatment with *T. asperellum* @ 10 g/kg followed by basal application of zinc sulphate @ 50 kg/ha. • Seed treatment with *Bacillus* (BSC5) @ 10g/kg followed by soil application @ 2.5 kg/ha with 250 kg of compost at the time of sowing. • Seed treatment with *P. fluorescens* @ 10 g/kg and soil application @ 2.5 kg/ha with 250 kg of compost at the time of sowing. • Seed treatment with *P. fluorescens* + *B. subtilis* + *T.asperellum* mixture @ 10g/kg and soil application of *P. fluorescens* + *B. subtilis* + *T. asperellum* mixture @ 2.5 kg/ha during sowing and at 90 days after sowing. **Chemical control** • Spot drench with carbendazim @ 1 g/l at the base of affected plants as well as surrounding healthy plants • Soil drenching with trifloxystrobin + tebuconazole @ 0.75g/l .

C.Nematode management

Seed treatment with *P.flouresecens* @20g/Kg and soil application @ 2.5 kg/ha
Application of consortia formulation of Pfbv 22 + Bbv 57@ 2.5 Kg/h

COTTON – VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants and designated diseases especially the wilt disease. The previous crop should not be of the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- Leave a distance of 50 m for foundation seeds and 30 m for certified seeds all around the field from the same and other varieties / hybrids of the crop.

Season

- Summer crop : February - March
- Winter crop : August - September

Acid delinting of fuzzy seeds

- Delint the fuzzy seeds with commercial sulphuric acid @ 100 ml / kg of seed for 2 - 5 minutes depending upon the variety (2 minutes for MCU 5 and 5 minutes for MCU 12).
- After acid delinting remove the floaters and insect damaged seeds and separate the brown colour and well filled sinkers.
- Wash the collected seeds thoroughly for 3 to 4 times with fresh water and neutralize with 0.5 % lime solution for removal of traces of acid.

Pre-sowing seed treatment

- Seed hardening with 2 % KCl for 10 hrs in the seed to solution ratio 1:1 and dry back to original moisture content.
- Seed coating with polymer @ 3 g / kg + imidachloprid @ 2 ml / kg + *Pseudomonas fluorescens* @ 10 g / kg + Azophos @ 120 g / kg of seed.
- The above two treatments can be integrated as designer seed treatment.

Foliar application

- Spray 1 % Diammonium phosphate on 70th, 80th and 90th days after sowing.

Roguing

- The crop should be rogued for off-types from vegetative stage to harvesting stage based on plant stature, leaf size, leaf colour, stem colour, flower colour, petal spot, pollen colour, number of sympodia, boll size and shape to maintain genetic purity.

Harvesting

- Pick the fully bursted kapas periodically in six pickings at weekly intervals.
- Consider first five pickings in winter crop and first four pickings in the summer crop for seed purpose, the seed from the subsequent pickings are inferior in quality.
- Do not retain the kapas unpicked in the field for more than a week as it reduces seed quality.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 8 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 6 %.

COTTON - HYBRID SEED PRODUCTION

Land requirement

- Land should be free from volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 30 m all around the field from the same and other varieties / hybrids of cotton.
- Between the parental lines leave an isolation distance of 5 m.

Seeds and sowing

- Male : 2 kg / ha
- Female : 4 kg / ha

Fertilizer requirement

- Compost : 12.5 t / ha, NPK : 20:60:50 kg / ha as basal application

Top dressing

- Top dress @ 12.5 kg N / ha at 60 and 90 days after sowing.
- Earthing up the crop adequately after first top dressing.
- Irrigate the crop immediately after every top dressing.

Foliar application

- Foliar spray of 100 ppm boric acid or 0.5 % zinc sulphate to the male parent at initiation flowering to improve the pollen viability and pollen production.
- Foliar spray of salicylic acid @ 250 ppm at 90 days after sowing for increased seed set.
- Foliar spray of 2 % DAP 4 times at 10 days interval during boll development period (60, 70, 80 and 90 days after sowing) for better development of crossed bolls.

Emasculation and dusting for cross pollination

- Emasculate and dust as far as possible all buds appearing during the first six weeks of reproductive phase to ensure good seed setting and development of bolls.
- Emasculate the female buds on the previous day evening.
- Smear pollen dust to the stigma of all the emasculated flowers for good number of boll formation.
- Restrict emasculation to each day evening to 3 pm to 6 pm and pollination to morning between 10 am to 1 pm to ensure highest purity of hybrid seeds.
- Choose optimum size of bud and avoid too young or too old buds for emasculation.
- Cover the male buds with paper cover during previous day evening for their use on next day.
- Cover emasculated buds with butter paper cover to avoid out crossing.
- Close the crossing programme after 9th week (from commencement of crossing) and remove all buds and flowers appearing subsequently to facilitate the development of crossed bolls.

Topping

- Top the plants either manually or spray Maleic Hydrazide @ 100 ppm at 90th and 105th days after sowing to enhance the sympodial branches formation.

Harvesting

- Harvest only fully bursted bolls.
- Harvest the crop as 4 - 6 pickings depending on the cultivar.
- Avoid later pickings (after 4 - 5 pickings) for seed purpose.

Ginning

- Gin the crossed kapas in separate gins erected in seed processing units or farm gins under the close supervision of the authorities concerned to ensure purity and avoid damage.
- Remove hard locks and stained kapas.
- After ginning, clean the seeds by hand picking to remove small, shrivelled and broken seeds.

Pre-storage seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed.
- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu (*Albizzia amara*) leaf powder mixed in the ratio of 5:4:1@ 3 g / kg of seed as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 7 - 8 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 6 %.

(ii) JUTE (*Corchorus olitorius* & *Corchorus capsularis*)

CROP MANAGEMENT

Jute can be successfully grown in Coimbatore, Cuddalore, Villupuram, Vellore, Tiruvannamalai, Chengleput and parts of Thanjavur, Tiruvarur, Nagapattinam, Tiruchirapalli, Perambalur, Karur, Pudukkottai and Tirunelveli, Thoothukudi districts where assured supply of irrigation water is available for its cultivation and retting for fibre extraction.

Soil type: Alluvial sandy loam, clay loamy soils are best suited for jute production. Capsularis jute can grow even in standing water especially towards the latter part of its growth, but Olitorius jute will not thrive in standing water. The latter is more drought resistant and is therefore grown on lighter soils.

Season: February
Land Preparation: Fine tilth is required since the seeds are very small
Manures and fertilizer application: Five tonnes of well decomposed farm yard manure is to be applied during last ploughing. Besides 20 kg per ha each of N, P_2O_5 and K O are to be applied. basally. Beds and channels are formed depending on water resources.

Varieties: Capsularis JRC 212, JRC 321, JRC 7447
Olitorius j JRO 524, JRO 878, JRO 7835

Crop duration 120 to 140 Days

Seed rate and sowing: Seeds can be sown either by broadcasting or by line sowing.

Jute type	Seed rate (kg/ha)		Spacing (cm)	No. of Plants/ Sq. Mtr.
	Line Sowing	Broad Casting		
Olitorius	5	7	25 x 5	80
Capsularis	7	10	30 x 5	67

Weed management: Hand weeding twice on 20 - 25 DAS and 35 - 40 DAS. Fluchloralin can be sprayed at 3 days after sowing at the rate of 1.5 kg per hectare and is followed by irrigation. Further one hand weeding can be taken up at 30 - 35 DAS.

Top dressing of fertiliser: Apply 10 kg of N at 20 - 25 days after first weeding and then again on 35 - 40 days after second weeding as top dressing. During periods of drought and fertilizer shortage, spray 8 kg of urea as 2 per cent urea solution (20 g urea in one litre of water) on jute foliage on 40 - 45 as well as 70 - 75 DAS.

Water Management: Jute crop requires 500 mm of water. First irrigation is to be given after sowing and life irrigation on fourth day after sowing. Afterwards irrigation can be given once in 15 days.

Harvest: Jute crop can be harvested from 100 to 110 DAS but can be extended from 120 - 135 DAS depending on local cropping systems. Jute plants are left in the field for 3 - 4 days for leaf shedding. Then thick and thin plants are sorted out and bundled in convenient size.
Yield: The green plant weight yield is 45 to 50 tonnes per hectare whereas the fibre yield is to 2.5 tonnes per hectare.

7. SUGARCANE (*Saccharum officinarum*)

CLIMATE REQUIREMENT

T_Max°C	T_Min°C	Optimum °C	Rainfall mm	Altitude m MSL
42	15	32 - 35	2500 - 3000	1000

Tropical crop. Besides temperature and rainfall, light (day length) plays a very important role in proper growth and development i.e. tillering of cane. A long, warm growing season with a high incidence of solar radiation and adequate moisture is required.Short day length decreases number of tillers. Under long day length conditions, plant produces more dry matter.

CROP IMPROVEMENT

A. PLANTED (MAIN) CROP

I. SEASON AND VARIETIES

Sugarcane is grown chiefly in the main season (December - May) in the entire State. In parts of Tiruchirapalli, Perambalur, Karur, Salem, Namakkal and Coimbatore districts, it is also raised during the special season (June - September). The particulars in respect of each season are given below:

SEASON, PERIOD OF PLANTING

1. **Main season**

   i) **Early : ** Dec - Jan   ii) **Mid :** Feb - March   iii) **Late:** April – May

2. Special season : June - July

All early season varieties are suitable for special season.

II. PARTICULARS OF VARIETIES

Variety	Duration (Month)	Cane Yield (t/ha)	CCS %	CCS (t/ha)
CoG 94077	11	133.2	13.50	17.60
Co 86032	10-12	110.0	13.00	14.30
CoSi (SC) 6	10-11	142.0	13.10	18.60
TNAU SC Si 7	10-11	156.0	13.40	20.90
TNAU SC Si 8	11-12	146.0	12.90	18.00
CoC 25	10-12	145.7	12.77	18.62
CoG 6	10-12	140.6	13.34	18.39

III. Morphological characters

Characters	CoG 94077	Co 86032
Parentage	Co 740 x Co 775	Co 62198 X CoC 671
Leaf size	Medium	Medium
Leaf colour	Dark green	Dark green
Sheath colour	Greenish with Purple tinge	Green with purple
Sheath clasping	Loose	Loose
Spines	Absent	Few, hard, deciduous
Ligular process	Present on one side	Absent
Stem colour	Greenish yellow	Reddish pink (exposed) Greenish yellow (unexposed)
Girth	Medium	Medium
Joint	Slightly staggered	Cylindrical
Bud Groove	Short, shallow	Absent
Size	Medium	Medium

Character	TNAU SC Si 7	TNAU SC Si 8
Parentage	Co 99034 x CoG 93076	CoC 90063 x Co 8213
Leaf Size	Medium	Medium
Leaf colour	Dark green	Green
Sheath colour	Yellowish Green with purple tinge	Green with purple stripes
Sheath clasping	Slightly tight	Loose
Spines	Present (deciduous)	Very few, soft and deciduous
Splits	Absent	Absent
Ligular process	Present	Present (asymmetrical)
Stem colour	Yellowish green (exposed) pinkish yellow (unexposed)	Greenish yellow (exposed) Greenish Yellow (enexposed)
Girth	Medium	Medium
Joint	Straight	Straight
Bud groove	Absent	Present, Shallow
Size	Medium	Big

Characters	CoC 25	CoG 6
Parentage	Co 85002 x HR 83-144	HR 83-144 x CoH 119
Leaf size	Medium	Medium
Leaf colour	Green	Light Green
Sheath colour	Green with pinkish tinge with scarious border	Greenish purle
Sheath clasping	Loose	Loose
Spines	Absent	Decidous spines
Splits	Absent	Absent
Ligular process	Slightly indicated asymmetrical	Present (crescent shaped)

Stem colour	Greenish yellow green (unexposed) Pinkish (exposed)	yellow green (unexposed) Green yellow (exposed)
Girth	Medium	Medium
Joint	Zig zag	Straight
Bud Groove	Deep and extent all over the length of the internode	Present
Size	Medium	Medium

Parameters	Co 0212	Co 06022
Parentage	Co 7201 X ISH 106	GU 92-275 X Co 86249
Maturity group	Mid-late	Early
Year of release	2016	2018
Institute name	ICAR-Sugarcane Breeding Institute, Coimbatore	ICAR-Sugarcane Breeding Institute, Coimbatore
Cane yield (t/ha)	150.56	135.8
CCS %	12.80	13.10
Sugar yield (t/ha)	19.27	17.68
Reaction to red rot	Moderately resistant	Moderately resistance
Special features	Tolerant to drought and salinity A1 quality jaggery Good ratooner Erect and medium thick cane	A1 quality jaggery of golden yellow colour Non lodging, erect, thick cane Tolerant to water deficit stress

Parameters	Co 09004 (Amritha)	Co 11015 (Atulya)
Parentage	CoC 671 X CoT 8201	CoC 671 X Co 86011
Maturity group	Early	Early
Year of release	-	2019
Institute name	ICAR-Sugarcane Breeding Institute, Coimbatore	ICAR-Sugarcane Breeding Institute, Coimbatore
Cane yield (t/ha)	109.85	142.72
CCS %	18.94 (Sucrose %)	20.22 (Sucrose %)
Sugar yield (t/ha)	14.56	20.16
Reaction to red rot	Moderately resistance	Moderately susceptible
Special features	Resistant to smut Less susceptible to borer Tall cane, early fast growth, high tillering, medium thick cane, non-flowering, non lodging, good ratooner	Short duration (8 month) High sugar content Tall (>250 cm) and erect plant Medium thick cane A1 quality jaggery of golden yellow colour

Varieties suitable for Jaggery: CoG 6, Co 0212, Co 06022 and Co 11015

Varieties suitable for different seasons

Early (Dec-Jan) :

CoC 25, CoG 6, TNAU Sugarcane Si 7, Co 09004, Co 06022, CO 91015(Atulya)

Mid-late (Feb-Mar) :

Co 86032, Co 06030, TNAU Sugarcane Si 8, Co 0212

Special season (June-sept)

CoC 25, CoG 6, TNAU Sugarcane Si 7, Co 09004, Co 06022, CO 91015(Atulya)

Source of Seed

For the varieties released from Tamil Nadu Agricultural University for supply of primary seed materials, the Sugarcane Research Stations at Cuddalore, Sirugamani and Melalathur may be contacted. For other varieties promoted by the factories, for seed materials the concerned factories may be contacted.

CROP MANAGEMENT

IV. MAIN FIELD PREPARATION FOR PLANTING SUGARCANE

1. PREPARATION OF THE FIELD

a) **Wetland (Heavy soils):** In wetlands, preparatory cultivation by ploughing the land and bringing the soil to fine tilth could not be done.

i. After harvest of the paddy crop, form irrigation and drainage channels of 40 cm depth and 30 cm width at intervals of 6 m across the field and along the field borders.

ii. Form ridges and furrows with a spacing of 80 cm between rows with spade.

iii. Stir the furrows with hand hoes and allow the soil to weather for 4 to 5 days.

b) **Problem soils with excessive soil moisture:**

In problem soils, with excessive moisture where it is difficult to drain water, form raised beds at 30 cm intervals with Length - 5 m, Width - 80 cm, and Height -15 cm.

c. **Garden lands with medium and light soils:**

In medium and light soil irrigated by flow or lift irrigation adopt the following:

i. The initial ploughing with two disc plough followed by eight disc plough and using cultivator for deep ploughing followed by one time operation of rotovator to pulverize the soil to get a fine tilth, free of weeds and stubbles.

ii. Level the field with laser leveler for effective and proper irrigation management.

iii. Open ridges and furrows with tractor operated victory plough with a depth of 30cm and spacing of 80 cm between the rows for normal planting with furrow irrigation.

iv. Open irrigation channels at 10 m intervals.

2. BASAL APPLICATION OF ORGANIC MANURES:

Apply FYM at 12.5 t/ha or compost 25 t/ha or filter press mud at 37.5 t/ha before the last ploughing under gardenland conditions. In wetlands this may be applied along the furrows and incorporated well.

Preparation of reinforced compost from sugarcane trash and pressmud:

Spread the sugarcane trash to a thickness of 15 cm over an area of 7 m x 3 m. Then apply pressmud over this trash to a thickness of 5 cm. Sprinkle the fertilizer mixture containing mussoorie rock phosphate, gypsum and urea in the ratio of 2:2:1 over these layers at the rate of 5 kg/100 kg of trash. Moist the trash and pressmud layers adequately with water. Repeat this process till the entire heap rises to a height of 1.5 m. Use cowdung slurry instead of water to moist the layer wherever it is available. Cover the heap with a layer of soil and pressmud at 1:1 ratio to a thickness of 15 cm. Leave the heap as such for three months for decomposition. Moist the heap once in 15 days. During rainy season, avoid moistening the heap. After three months, turn and mix the heap thoroughly and form a heap and leave it for one more month. Then turn and mix the heap thoroughly at the end of the fourth month. Moist the heap once in 15 days during 4th and 5th month also. This method increases the manurial value of trash compost by increasing, N, P and Ca content. It also brings down the C:N ratio by 10 times as compared to raw cane trash.

Composition of cane trash, pressmud and cane trash compost

Major nutrients	Cane trash	Pressmud	Cane trash compost
Nitrogen (N)	0.40	Percent 1.90	1.60
Phosphorus (P)	0.13	1.50	1.10
Potassium(K)	0.40	0.50	0.40
Calcium (Ca)	0.56	3.00	1.00
Magnesium (Mg)	0.30	2.00	0.60
Sulphur (S)	0.12	0.50	0.48

Micronutrients	Cane trash	Pressmud	Cane trash compost
		PPM	
Iron (Fe)	360	2240	2710
Manganese (Mn)	110	400	450
Zinc (Zn)	90	360	370
Copper (Cu)	30	130	80
C:N ratio	113:1	16:1	22:1

2. BASAL APPLICATION OF PERTILIZER

(i) If soil test is not done, follow blanket recommendation of NPK @ 300:100:200 kg/ha
Apply super phosphate (625 kg/ha) along the furrows and incorporate with hand hoe.

(ii) Soil test crop response based integrated plant nutrition system (STCR- IPNS) recommendation may be adopted for prescribing fertilizer doses for specified yield targets. (ready reckoners are furnished)

Sugarcane (1)

Soil :	Mixed black calcareous (Perianaickenpalayam series)	FN = 4.17 T - 1.09 SN - 1.11 ON FP_2O_5= 1.01 T - 2.56 SP - 1.01 OP
Target :	125-150 t ha^{-1}	FK_2O = 3.44 T - 0.84 SK - 1.03 OK

Initial soil test values ($kg\ ha^{-1}$)			**Yield target – 125t ha^{-1}**			**Yield target – 150 t ha^{-1}**		
			NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			**NPK ($kg\ ha^{-1}$) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}**		
SN	**SP**	**SK**	**FN**	**FP_2O_5**	**FK_2O**	**FN**	**FP_2O_5**	**FK_2O**
200	15	300	223	50*	113	328	68	199
220	17	350	201	50*	100*	306	63	157
240	19	400	180	50*	100*	284	58	115
260	21	450	158	50*	100*	262	53	100*
280	23	500	150*	50*	100*	240	50*	100*

* Maintenance dose

Sugarcane (2)

Soil : Red coastal alluvium (Gadillum series)

Target : 125 t ha^{-1} and 150 t ha^{-1}

FN =4.06 T-0.74SN-0.87 ON
FP_2O_5=0.71T-1.09 SP-0.72 OP
FK_2O=2.67T-0.57SK-1.33 OK

Initial soil test values (kg ha^{-1})			Yield target – 125 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 150 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	15	200	280	50*	155	381	50*	222
220	17	220	265	50*	143	366	50*	210
240	19	240	250	50*	132	351	50*	199
260	21	260	235	50*	121	337	50*	187
280	23	280	220	50*	109	322	50*	176

*Maintenance dose

Sugarcane (3)

Soil : Red sandy loam (Irugur series)

Target : 100 t ha^{-1} -125 t ha^{-1}

FN =3.42 T-0.56 SN-0.93 ON
FP_2O_5=1.15T-1.94 SP-0.98 OP
FK_2O=3.16T-0.73SK-0.99 OK

Initial soil test values (kg ha^{-1})			Yield target – 100 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}			Yield target – 125 t ha^{-1} NPK (kg ha^{-1}) + FYM @ 12.5 t ha^{-1} + *Azospirillum* @ 2 kg ha^{-1} + PSB @ 2 kg ha^{-1}		
SN	SP	SK	FN	FP_2O_5	FK_2O	FN	FP_2O_5	FK_2O
200	14	200	150	50*	105	236	72	184
220	16	220	139	50*	100*	224	68	169
240	18	240	128	50*	100*	213	64	155
260	20	260	116	50*	100*	202	60	140
280	22	280	105	50*	100*	191	56	126

*Maintenance dose

Note: FN, FP_2O_5 and K_2O are fertilizer N, P_2O_5 and K_2O in kg ha^{-1}, respectively; T is the yield target in t ha^{-1}; SN, SP and SK respectively are available N,P and K in kg ha^{-1} and ON, OP and OK are the quantities of N, P and K supplied through organic manure inkg ha^{-1}.

(i) Apply 37.5 kg Zinc sulphate/ha and 100 kg Ferrous sulphate + 12.5 t FYM/ha to zinc and iron deficient soils.
(ii) Application of sulphur in the form of Gypsum @ 500 kg /ha to sulphur deficient soils to increase the cane yield and juice quality.
(iii) Basal application of 5 kg $CuSO_4$ for copper deficient soil.

V. MANAGEMENT OF MAIN FIELD OPERATIONS

1. PREPARATION OF SETTS FOR PLANTING

a. Take seed material from short crop (6 to 7 months age) free from pests and diseases incidence.
   i) Detrash the cane with hand before setts preparation.
   ii. Use sharp knife or sett cutting machine developed by TNAU to prepare setts without splits.
   iii. Discard setts with damaged buds, sprouted buds, splits etc.
   iv. **Sett treatment with biofertilizers:** Prepare slurry with 2 kg of *Azospirillum/ Gluconacetobacter,* 2kg of phosphobacteria , 2 kg of SSB and dip the setts required for one ha for 30 minutes and plant. (or) sett treatment with powder formulation of *Gluconacetobacter diazotrophicus* and AM fungi each @ 67.5 g/ha along with 75% of recommended dose of N & P.

2. SETT TREATMENT

- Select healthy setts for planting.
- The setts should be soaked in 100 litres of water dissolved with 50g Carbendazim, 200ml malathion and 1 kg urea for 15 minutes.
- Treat setts with Aerated steam at 50°C for one hour to control primary infection of grassy shoot disease.

3. SEED RATE

75000 two-budded setts/ha.

4. PLANTING

Different systems of planting is not found to influence the millable cane population, commercial cane sugar per cent, cane and sugar yield.

a) Irrigate the furrows to form a slurry in wet land condition (Heavy soil)
b) Place the setts along the centre of the furrows, accommodating 12 buds/metre length. Keep the buds in the lateral position and press gently beneath the soil in the furrow.
c) Next day cover the exposed setts with soil to avoid exposure of setts to sunlight.
d) Plant more setts near the channel or double row planting at every 10th row for gap filling, at later stage.
e) In dry/ garden land dry method of planting may be followed. First arrange the setts along the furrows, cover the setts with soil and then irrigate.

Improved technologies on cane planting systems

Mechanisation of planting

- TNAU mechanical planter is useful for cost effective planting with saving of Rs.3750 / ha and it can cover an area of 1.5ha/day
- Reduces the human labour drudgery and seed rate up to 5 tones/ha.
- Paired row system of planting double side planting of sugarcane setts with 150 + 30 cm spacing for Astraf 8000 series (Mechanical harvester) operated areas and 150 + 30 cm spacing for New Holland 4000 series operated areas may be

adopted with single row of cane planting.

- Sugarcane cultivates under subsurface drip system the laterals may be placed 20cm depth in the furrows and setts are placed 5cm above the laterals.
- For sustainable sugarcane initiative system (SSI) transplanting young chip bud seedling raised in portray (25-35 days old) in wide spacing (5x2 feet) in the main field with drip fertigation system.
- Daincha / Sunhemp intercropping in the wider spaced cane cultivated area for improving soil health and reduce the weed infestation. It also reduces early shoot borer incidences and increases cane yield.
- Plant the setts on one side of the ridge for 80 cm spacing in heavy soil to avoid sett rot resulted better germination
- Sow rhizobium treated green manure seeds @ 10kg/ha on the opposite side of ridge with 10cm. Spacing on or before 3 days after planting.
- Incorporate the green manure crop 50-60 days after planting in between interow of wider spaced crop and give partial earthing up with recommended dose of N fertilizer on 90 – 100 day after planting.
- Introduction of power weeder with rotovator for weeding and earthing up with ridger to save the cost on labour and also to reduce human drudgery.

5. FILLING UP GAPS

ii. Fill the gaps, if any, within 30 days after planting with sprouted setts.
iii. Gap filling with two budded setts/ poly bag seedlings within 15 to 20 days after planting to maintain optimum plant stand.
iv. Maintain adequate moisture for 3 weeks for proper establishment of the sprouted setts.

6. TRASH MULCHING

Mulch the ridges uniformly with cane trash to a thickness of 10 cm within a week after planting. It helps to tide over drought, conserves moisture, reduce weed population and minimise shoot borer incidence. Mulch the field with trash after 21 days of planting in heavy soil and wetland conditions. Avoid trash mulching in areas where incidence of termites is noticed.

7. RAISING INTER CROPS

In areas of adequate irrigation, sow one row of soybean or blackgram or greengram along the centre of the ridge on the 3rd day of planting. Intercropping of daincha or sunhemp along ridges and incorporation of the same on the 45th day during partial earthing up helps to increase the soil fertility, and also the cane yield.

8. WEED MANAGEMENT

WEED MANAGEMENT IN PURE CROP OF SUGARCANE

i. Wherever weed menace is higher, one line weeding along the crop row and spade digging of ridges have to be done on 30, 60 and 90 DAP
ii. Spray Atrazine 1 kg or Oxyflurofen 750 ml/ha mixed in 500 liters of water as pre emergence herbicide on the 3rd day of planting, using deflector or fan type nozzle fitted with knapsack sprayer.
iii. Pre emergence application of atrazine @ 1.0 kg ha^{-1} on 3 DAP followed by post emergence directed application of glyphosate @ 10 ml / litre of water on 45 DAP with hood+ one hand weeding on 90 DAP registered the maximum cane yield.

iv. If the parasitic weed striga is a problem, Pre-emergence application of Atrazine 1.0 kg/ha on 3 DAP + hand weeding on 45 DAP with an earthing up on 60 DAP combined with post-emergence sparaying of 2,4-D @ 6 g (0.6%) + Urea @ 20 g (2%) / litre of water on 90 DAP + Trash mulching 5 t/ha on 120 DAP.

v. Pre- plant application of glyphosate at 2.0 kg h^{a-1} along with 2% ammonium sulphate at 21 days before planting of sugarcane followed by post emergence direct spraying of glyphosate at 2.0 kg h^{a-1} along with 2% ammonium sulphate with a special hood on 30 DAP suppressed the nut sedges *(Cyperus rotandus*) and provided weed free environment.

vi. If herbicide is not applied work the junior-hoe along the ridges on 25, 55 and 85 days after planting for removal of weeds and proper stirring. Remove the weeds along the furrows with hand hoe. Otherwise operate power tiller fitted with tynes for intercultivation.

vii. Control of creeper weeds post emergence directed application of fernoxone (2, 4 –D sodium salt) @ 2 gm + 10 gm of urea per liter of water may be sprayed over the creeper weeds.

Weed management in Sugarcane intercropping system

Premergence application of Thiobencarb @ 1.25 kg ai/ha under intercropping system in Sugarcane with Soybean, blackgram or groundnut gives effective weed control. Raising intercrops is not found to affect the cane yield and quality.

9. EARTHING UP

After application of 3rd dose fertilizer (90 days), work victory plough along the ridges for efficient and economical earthing up. At 150 days after planting, earthing up may be done with spade.

10. DETRASHING

Remove the dry cane leaves on 150^{th} and 210^{th} day to avoid borer infestation.

11. PROPPING

Do double line propping with trash twist at the age of 210 days of the crop.

12. TOP DRESSING WITH FERTILIZERS

Apply 275 kg of nitrogen and 112.5 kg of K O/ha in three equal splits at 30, 60 and 90 days

a. Soil application

Coastal and flow irrigated belts (assured water supply areas). In the case of lift irrigation belt, apply 225 kg of nitrogen and 112.5 kg of K O/ha in three equal splits at 30, 60 and 90 days (water scarcity areas). For jaggery areas, apply 175 kg of nitrogen and 112.5 kg of K_2O/ha in three equal splits on 30, 60 and 90 days.

NITROGEN SAVING

a. **Neem Cake Blended Urea**: Apply 67.5 kg of N/ha + 27.5 kg of Neem Cake at 30 days and repeat on 60th and 90th days.

Note: Neem cake blending: Powder the required quantity of neem cake and mix it with urea thoroughly and keep it for 24 hours. Thus, 75 kg of nitrogen/ha can be saved by this method.

b. **Azospirillum**: Mix 12 packets (2400 g)/ha of Azospirillum inoculant or TNAU Biofert –1 with 25 kg of FYM and 25 kg soil and apply near the clumps on 30th day of planting. Repeat the same on 60th day with another 12 packets (2400 gm). Repeat the above on the other side of the crop row on the 90th day (for lift irrigated belt).
c. **Band placement:** Open deep furrows of 15 cm depth with hand hoes and place the fertilisers in the form of band and cover it properly.
d. **Subsurface application:** Application of 255 kg of Nitrogen in the form of urea along with potash at 10cm depth with 15cm intervals by the side of the cane clump will result in the saving of 20 kg N/ha without any yield reduction.

Nutritional Disorders :

Nitrogen deficiency : All leaves of sugarcane exhibit a yellow – green colour and retardation of growth. Cane stalks are smaller in diameter and premature drying of older leaves. Roots attain a greater length but are smaller in diameter.

Phosphorus deficiency : Reduction in length of sugarcane stalks, diameters of which taper rapidly at growing points. The colour of the leaves is greenish blue, narrow and some what reduce length. Reduced tillering, decreased shoot / root ratio with restricted root development.

Potassium deficiency: Depressed growth, yellowing and marginal drying of older leaves and development of slender stalks. An orange, yellow colour appears in the older lower leaves which develop numerous chlorotic spots that later become brown with dead centre. A reddish discoloration which is confined to the epidermal cells of the upper surfaces and midribs of the leaves. The young leaves appear to have developed from a common point giving a "Bunched top" appearance. Poor root growth with less member of root hairs.

Zinc deficiency: Mild zinc deficiency exhibit a tendency to develop anthrocyanin pigments in the leaves. Pronounced bleaching of the green colour along the major veins and also striped effect due to a loss of chlorophyll along the veins. In acute cases of zinc deficiency there is evidences of necrosis and growth ceases at the growing point (meristem).

Iron deficiency: Symptoms of Iron deficiency are generally seen in young leaves where pale stripes with scanty chlorophyll content occur between parallel lines. In advanced stages of deficiency the young leaves turn completely white, even in the veins. Root growth also becomes restricted.

Boron deficiency : Boron deficiency could be seen in the cane by depressed growth, development of distorted and chlorotic leaves and the presence of definite leaf and stalks lesions. In extreme cases of boron deficiency the plant will die.

Importance of Balanced Nutrition

The soil fertility has declined in many sugarcane growing areas of the state due to improper and some times, distorted fertilizer schedules adopted over the years under intensive cultivation of the crop. Hence balanced application of fertilizer based on soil test values and crop requirement is essential.

How to Evaluate fertilizer requirementt

Through STCR fertilizer prescription equations

a. Perianaickenpalayam series (Inceptisols) of Coimbatore and Erode STL Jurisdiction
$FN = 4.17\ T – 1.09\ SN – 1.11\ ON$ $FP_2O_5 = 1.01\ T – 2.56\ SP – 1.01\ OP$ $FK_2O = 3.44\ T – 0.84\ SK – 1.03\ OK$

b. Gadillum series (Red laterite) of Cuddalore STL Jurisdiction
$FN = 4.06\ T – 0.74\ SN – 0.87\ ON$ $FP_2O_5 = 0.71\ T – 1.09\ SP – 0.72\ OP$ $FK_2O = 2.67\ T – 0.57\ SK – 1.30\ OK$

c. Irugur series (Inceptisols) of Coimbatore, Erode, Trichy and Salem STL Jurisdiction
$FN = 3.42\ T – 0.56\ SN – 0.93\ ON$ $FP_2O_5 = 1.15\ T – 1.94\ SP – 0.98\ OP$ $FK_2O = 3.16\ T – 0.73\ SK – 0.99\ OK$

Micro nutrient fertilizers

1. (a) Zinc deficient soils : Basal application of 37.5 kg/ha of zinc sulphate.
   (b) For zinc deficiency symptoms: foliar spray of 0.5% zinc sulphate with 1% urea at 15 days internal till deficiency symptoms disappear.
2. (a) Iron deficient soils: Basal application of 100 kg/ha of ferrous sulphate + 12.5t FYM.
   (b) For Iron deficiency symptoms: Foliar spray of 1% ferrous sulphate + 0.1% citric acid with 1% urea at 15 days interval till deficiency symptoms disappear.
3. Soil application of $CuSO_4$@ 5 kg/ha in copper deficient soils. Alternatively foliar spray of 0.2% $CuSO_4$ twice during early stage of crop growth.

Common Micronutrient mixture : To provide all micronutrients to sugarcane, 50 kg /ha of micronutrient mixture containing 20 kg Ferrous sulphate,10 kg Manganese sulphate, 10 kg Zinc sulphate, 5 kg of Copper sulphate, 5 kg of Borax mixed with 100 kg of well decomposed FYM, can be recommended as soil application prior to planting. (Or)
Application of TNAU MN mixture @ 50 kg/ha as EFYM for higher cane yield.

Recommended dosage of macro and micronutrients Macronutrients

a. Sugarcane – plant crop (meant for sugar mills) 300:100:200 kg N, P_2O_5 and K_2O per ha
b. Sugarcane – Ratoon crop (meant for sugar mills)
   300 + 25% extra N : 100 : 200 kg N, P_2O_5 and K_2O per ha
c. Sugarcane for jaggery manufacture (plant as well as ratoon crop)
   225 : 62.5 : 112.5 kg N, P_2O_5 and K_2O per ha

13. BIOFERTILIZER FOR SUGARCANE

Azospirillum is the common biofertilizer recommended for N nutrition which could colonize the roots of sugarcane and fix atmospheric nitrogen to the tune of about 50 to 75 kg nitrogen per ha per year. Recently, another endophytic nitrogen fixing bacterium, *Gluconacetobacter diazotrophicus* isolated from sugarcane can able to fix more nitrogen than *Azospirillum*. It colonizes throughout the sugarcane and increases the total N content. In soil, it can also colonize the roots and able to solubilize the phosphate, iron and Zn. It can also enhance the crop growth, yield of sugarcane and sugar content of the juice. Since it is more efficient than *Azospirillum*, this new organism was test-verified in various centres and released as new biofertilizer Gluconacetobacter diazotrophicus TNAU Biofert-I. Phosphobacteria as P solubiliser are recommended for sugarcane crop.

Sett treatment with *Gluconacetobacter diazotrophicus*

Before planting the sugarcane setts can be treated with ten packets (2 kg) per ha of *Gluconacetobacter diazotrophicus* prepared as slurry with 250 L of water.

Soil application *Gluconacetobacter diazotrophicus*

Twelve packets (2.4 kg) per ha is recommended for soil application each at 30th, 60th and 90th day after planting under irrigated condition.

Same method of application can be followed for Phosphobacteria.

- If basal application is not followed apply the same with 30th day, 60th day and 90th day after planting and copiously irrigate the field.
- Biofertilizer treatment should be done just before planting.
- Immediately plant/ Irrigate after biofertilizer application
- Do not mix biofertilizer along with chemical fertilizer.
- Reduces 25% of the recommended N to reap the benefits of biofertilizer application

14. WATER MANAGEMENT

Irrigate the crop depending upon the need during different phases of the crop.

Germination phase (0 - 35 days):

Provide shallow wetting with 2 to 3 cm depth of water at shorter intervals especially for sandy soil for enhancing the germination. Sprinkler irrigation is the suitable method to satisfy the requirement, during initial stages.

Later, irrigation can be provided at 0.75, 0.75 and 0.50 IW/CPE ratio during tillering, grandgrowth and maturity phases respectively. The irrigation intervals in each phase are given below:

Stages	**Days of irrigation interval**	
Tillering phase (36 to 100 days)	8	10
Grand growth phase (101 - 270 days)	8	10
Maturity phase (271 - harvest)	10	14

Drip Irrigation:

- Planting setts obtained from 6-7 months old healthy nursery and planted in paired row planting system with the spacing of 30x30x30 / 150 cm. for manual harvest and 30/150 cm for machine harvest
- Eight setts per metre per row have to be planted on either sides of the ridge thus making it as four row planting system.
- 12 mm drip laterals have to be placed in the middle ridge of each furrow with the lateral spacing of 240 cm & 8 'Lph' clog free drippers should be placed with a spacing of 75 cm on the lateral lines. The lateral length should not exceed more than 30-40 m.
- Phosphorus @ 62.5 kg ha^{-1} has to be applied as basal at the time of planting.
- Nitrogen and Potassium @ 275:112.5 kg ha^{-1} have to be injected into the system as urea and muriate of potash by using "Ventury" assembly in 10-12

equal splits starting from 15 to 150- 180 days after planting.
- o Low or medium in nutrient status soil to be given with 50 per cent additional dose of Nitrogen and Potassium.
- o Irrigation is given once in three days based on the evapo-transpiration demand of the crop.
- o The double side planting of sugarcane with lateral spacing of 120+40 cm under subsurface drip fertigation system improves the yield.
- o Application of 125 % recommended NPK (Rec NPK-275 :63:112.5 kg /ha) through fertigation under pit system of planting inprove the yield and yield attributes.

Concept of fertigation

- Fertigation is the judicious application of fertilizers by combining with irrigation water.
- Fertigation can be achieved through fertilizer tank, venturi System, Injector Pump, Non- Electric Proportional Liquid Dispenser (NEPLD) and Automated system.
- Recommended N & K @ of 275 and 112.5 kg. ha^{-1} may be applied in 14 equal splits with 15 days interval from 15 DAP.
- 25 kg N and 8 kg K_2O per ha per split.
- Urea and MOP (white potash) fertilisers can be used as N and K sources respectively
- Fertigation up to 210 DAP can also be recommended.

Advantages of Fertigation

- Ensures a regular flow of water as well as nutrients resulting in increased growth rates for higher yields
- Offers greater versatility in the timing of the nutrient application to meet specific crop demands
- Improves availability of nutrients and their uptake by the roots
- Safer application method which eliminates the danger of burning the plant root system
- Offers simpler and more convenient application than soil application of fertilizer thus saving time, labour, equipment and energy
- Improves fertilizer use efficiency
- Reduction of soil compaction and mechanical damage to the crops
- Potential reduction of environmental contamination
- Convenient use of compound and ready-mix nutrient solutions containing also small concentration of micronutrients.

15. Contingent plan

Gradual widening of furrow:

At the time of planting, form furrow at a width of 30 cm initially. After that, widen the furrow to 45 cm on 45th day during first light earthing up and subsequently deepen the furrow on 90th day to save 35% of water.

Drought Management:

i. Soak the setts in lime solution (80 kg Kiln lime in 400 lit) for one hour.
ii. Plant in deep furrows of 30 cm depth.
iii. Foliar spray of kel and urea each at 2.5 per cent during moisture stress period at 15 days interval.
iv. Foliar spray of Kaolin (60 g in 1 ltr. of water) to alleviate the water stress.
v. Under water scarcity condition, alternate furrow and skip furrow method is beneficial.
vi. Apply 125 kg of MOP additionally at 120 day of planting.
vii. Basal incorporation of coir waste @ 25 tonnes/ha at the time of last ploughing.
viii. Removal of dry trash at 5th month and leave it as mulch, in the field.

16. CROP PROTECTION

A. Pest Management:

SUGARCANE

- Deep plough during summer
- Select scale insect free setts
- Treat the setts with imidacloprid 70% WS **@** 100 ml/ 100 kg to avoid termite
- Adopt early planting (Dec-Jan)
- Plant sugarcane in paired or wider rows for taking effective control measures
- Trash mulch on ridges at 3 DAP
- Intercrop with green gram, black gram and daincha
- Keep bunds free from weeds
- Avoid ratoons in infested fields
- Provide adequate rrigation & avoid excessive use of nitrogenous fertilizers
- Detrash on 150 and 210th DAP
- Drain excess water
- Avoid use of insecticide treated leaves as cattle feed.

Early Shoot borer, *Chilo infuscatellus*	• Release *Sturmiopsis inferens* gravid females @ 125/ha on 30 and 45 DAP **Apply any one of the following insecticides/ha** • Chlorantraniliprole 0.4% G @ 18.75 g • Chlorantraniliprole 18.5 % SC @ 375 ml • Chlorpyriphos 20%EC 1.0 lit • Fipronil 0.3% GR @ 25 kg • Fipronil 5% SC @1.5 lit • Monocrotophos 36 %SL 1.5 lit • NSKE 5 % • Thiamethoxam 75% w/w SG @ 160 g
Internode borer, *Chilo sacchariphagus indicus*	• Release egg parasitoid, *Trichogramma chilonis* at the rate of 2.5 cc / release/ha (Six releases at 15 days interval starting from fourth month).
Top shoot borer, *Scirpophaga excerptalis*	• Collect and destroy egg masses • Release prepupal parasitoid, *Isotima javensis* @ 125 females /ha

	Apply any one of the following insecticides/ha • Carbofuran 3% CG @ 66 kg • Chlorantraniliprole 0.4% G @ 18.75 kg • Chlorantraniliprole 18.5% SC 375 ml
Pyrilla, *Pyrilla perpusilla*	• Release lepidopteran parasitoid, *Epiricrania melanoleuca* @ 8000 -10,000 cocoon /ha (or) 8 - 10 lakh eggs/ha **Spray any one of the following on 150 and 210 DAP /ha after detrashing** • Chlorpyriphos 20 % EC @ 1.50 lit • Monocrotophos 36%SL @ 1.50 lit
Aleurodids, *Aleurolobus barodensis* **Aphid,** *Melanaphis sacchari,* *M. indosacchari* **Scale insect,** *Melanaspis glomerata* **Mealybug,** *Saccharicoccus sacchari*	• Spray monocrotophos 36%SL @ 1.50 lit/ha
Termite, *Odontotermes obesus*	• Flood irrigate the furrows at the time of planting • Drench soil with chlorpyriphos 20% EC @ 6.25 lit/ha Apply any one of the following insecticides/ha • Chlorantraniliprole 18.5% SC @ 500 ml • Clothianidin 50% WDG @ 250 g • Imidacloprid 17.8% SL@ 350 ml • Thiamethoxam 75% w/w SG @160 g
Root grub *Holotrichia consanguinea H. serrata* *Leucopholis lepidophora*	• Set up light trap to collect and destroy adults • Collect and destroy adult beetles present on neem, *Ailanthus* and *Acacia* trees • Imidacloprid 17.8% SL@ 350 ml / ha
Woolly aphid, *Ceratovacuna lanigera*	• Avoid transportation of aphid infested leaves from one location to another • Conserve and augment biocontrol agents like *Dipha aphidivora, Micromus* and coccinellids Spray any one of the following insecticides /ha • Chlorpyriphos 25%EC 1.0 lit • Monocrotophos 36%SL 625 ml
Root borer	Spray any one of the following insecticides /ha • Fipronil 5% SC @1.5 lit • Fipronil 0.3% GR @ 25 kg

B) Disease management

Disease	Recommendations
Red rot: *Colletotrichum fal catum*	• Selection of setts from healthy nursery programme. • Growing of recommended resistant and moderately resistant varieties *viz*., Co 86249, CoC 22, CoC 25, CoG 6 and Co 0212 • Sett treatment with carbendazim before planting (carbendazim 50 WP @ 0.05% along with 1.0% urea for 15 minutes) • The irrigation interval in red rot affected field must be lengthened. Once in 15 days during tillering, growth phases and once in 25 days during maturity phase which restricts the spread • Removal of the affected clumps at an early stage and soil drenching with 0.1 % carbandazim 50 WP • The trash of red rot affected field after harvest may be uniformly spread and burnt • In the red rot affected field crop rotation with rice for one season and other crops for two seasons could be adopted
Sett rot: *Ceratocytis paradoxa*	• Sett treatment with carbendazim before planting (carbendazim 50 WP @ 0.05% along with 1.0% urea for 15 minutes) • Proper drainage and planting of setts in 1-2 cm depth
Smut: *Sporisorium scitamineum* (*Ustilago scitaminea)*	• Sett treatment with carbendazim before planting (carbendazim 50 WP @ 0.05% along with 1.0% urea for 15 minutes) • Treating the setts with Aerated Steam Therapy (AST) at 50 ºC for 1 hour or in hot water at 50 ºC for 30 minutes or at 52 ºC for 18 minutes • Roguing of smut whips with gunny bags/polythene bag and burnt • Discourage ratooning of the diseased crops having more than 10 per cent infection
Grassy shoot disease (GSD): Candidatus Phytoplasma	• Rogue out infected plants in the seed nursery • Treat setts with aerated steam at 50°C for 1 hour to control primary infection • Spray dimethoate @ 0.1 % to control insect vector • Avoid ratooning if GSD incidence is more than 15 % in the plant crop

Leaf spot: *Cercospora longipes*	• pray mancozeb @ 2 kg or carbendazim @ 500 g/ha
Rust: *Puccinia melanocephala*	• pray mancozeb @ 2 kg/ha
Yellow leaf disease: Sugarcane yellow leaf virus (Vector: *Melanopsis sacchari*)	• se of disease free setts for planting • election of tissue culture seedling from meristem tip culture • roper nutritional management

General Recommendations

- Select healthy setts for planting. In the seed crop, select plants which do not show symptoms of red rot, smut, grassy shoot and ratoon stunting. Setts showing red colour at the cut end and hollows should be rejected.
- In fields which had shown high level of red rot disease, follow crop rotation with rice.
- Sett treatment with carbendazim before planting (carbendazim 50 WP @ 0.05% along with 1.0% urea for 15 minutes.
- Treat setts with aerated steam at 50°C for one hour to control primary infection of smut and grassy shoot disease.
  Clumps infected by grassy shoot, smut and ratoon stunting diseases should be uprooted and destroyed.

17. PRE-HARVEST PRACTICES

a. Apply cane ripeners

i. Spray Sodium metasilicate 4 kg/ha in 750 litres of water on the foliage of crop at 6 months after planting.

ii. Repeat the same twice at 8th and 10th months to obtain higher cane yield and sugar percentage.

b. Assessing maturity of crops

i. Assess the maturity by hand refractometer brix survey and 18 to 20 per cent brix indicates optimum maturity for harvest.

ii. Top-bottom ratio of H.R.Brix reading should be 1:1.

18. HARVESTING

i. Early varieties have to be harvested at 10 to 11 months age and mid-season varieties at 11 to 12 months age.

ii. Harvest the cane at peak maturity. Cut the cane to the ground level for both plant and ratoon crops.

B. RATOON CROP

I. MANAGEMENT OF THE FIELD AFTER HARVEST OF THE PLANT CROP

Complete the following operations within 10 days of harvest of plant crop to obtain better establishment and uniform sprouting of shoots.

1. Remove the trash from the field. Do not burn it. Irrigate the field copiously.
2. Follow stubble shaving with sharp spades to a depth of 4 - 6 cm along the ridges at proper moisture.

3. Work with cooper plough along with sides of the ridges to break the compaction.
4. The gappy areas in the ratoon sugarcane crop should be filled within 30 days of stubble shaving. The sprouted cane stubbles taken from the same field is the best material for full establishment. The next best method is gap filling with seedlings raised in polybags.
5. Apply basal dose of organic manure and super phosphate as recommended for plant crop.

II. MANAGEMENT OF THE CROP

1. 25% additional N application on 5-7 days after ratooning.
2. Foliar spray of Ferrous sulphate at 2.5 kg/ha on the 15th day. If chlorotic condition persists, repeat the spray twice further at 15 days interval. Add urea 2.5 kg/ha in the last spray.
3. Hoeing and weeding on 20th day and 40th to 50th day.
4. First top dressing on 25th day, 2nd on 45th to 50th day.
5. Final manuring on 70th to 75th day.
6. Partial earthing up on 50th day. If junior-hoe is worked two or three times upto 90th day, partial earthing up is not necessary.
7. Final earthing up on 90th day.
8. Detrashing on 120th and 180th day.
9. Trash twist propping on 180th day.
10. Harvest after 11 months.

C. SHORT CROP (NURSERY CROP)

SELECTION OF PROPER PLANTING MONTHS FOR RAISING NURSERY CROP IN RELATION TO MAIN FIELD PLANTING

Raise six to seven months old nursery crop prior to main field planting as follows:

Raise nursery crop during	Main field planting
June	December - January (early season)
July	February - March (Mid season)
August	April - May (Late season)
Dec - Apr	June - September (Special season)

II. PRECAUTIONS IN MAINTAINING NURSERY CROP

Adopt similar production techniques for raising short crop with the following modifications.

1. Do not detrash
2. Do not prop
3. Harvest at 6 to 7 months age
4. Remove trash by hand while preparing setts
5. Avoid bud damage
6. Transport the seed material to other places in the forms of full canes with trash intact.
7. Apply 50 kg of urea as top dressing additionally before one month of cutting the seed cane.

CROP PHYSIOLOGY

Foliar spray of TNAU Sugarcane Booster @ 1.0, 1.5 and 2 kg/acre in 200 litres of water at 45,60 and 75 days after planting enhances cane growth and weight, internodal length, cane yield, sugar content and offers drought tolerance.

CROP PROTECTION

A. Pest Management: Economic threshold level for important pests Economic threshold level for important pests

Pests	Management strategies
Shoot borer *Chilo infuscatellus*	< Cultural: Early season planting (Dec-Jan) ; < Trash mulching on ridges on 3DAP < Intercropping with green gram, black gram, daincha effectively checks shoot borer. < Spray Granulosis virus at 1.5 x 10^{12} PIB/ha twice on 35 and 50 days after planting (DAP) or release 125 gravid females of *Sturmiopsis inferens* /ha on 30 and 45 DAP **Apply any one of the following insecticides: Soil application** Lindane 10 G 12.5 kg Carbofuran 3CG 33 kg **Spraying** Chlorantraniliprole 18.5%SC 375 ml/ha Fipronil 5%SC 1500-2000 ml/ha Fipronil 0.3%GR 25-33.3 Kg/ha Quinalphos 25%EC 2000 ml/ha Phosalone 35 EC 1000 ml NSKE 5 % 25 Kg/ha < Daincha intercropped sugarcane recorded the lowest early shoot borer incidence.
	Note: The virus should be applied with teepol (0.05%) during evening hours. The granular application should be immediately followed by irrigation. ‘Granulosis’ virus spraying on sugarcane at 750 Nos. of diseased larvae, crushed and filtered mixed in 500 l of water has been found harmless to parasitoids and predators. A sticker like ‘teepol’ (250 ml for 500 l) can also be added to make the solution stick on to the surface of the crop and it is preferable to use high volume sprayer to be more effective. On cost benefit ratio basis NSKE 5% is recommended.
Internode borer *Chilo sacchariphagus indicus*	< Release egg parasitoid, *Trichogramma chilonis* at the rate of 2.5 cc/release/ha. Six releases fifteen days interval starting from fourth month onwards will be necessary. < During rainy weather and when ants are present, release the parasite through musquito net covered plastic disposable cups. < Detrash the crop on the 150th and 210th day after planting.
Top shoot borer *Scirpophaga excerptalis*	**Spraying any one of the following insecticides:** Carbofuran 3%G 33.3 kg/ha Chlorantraniprole 18.5%SC 375 ml/ha Phorate 10%G 30 kg/ha **Biocontrol:** Release *Isotima javensis* at 100 pairs/ha

Pyrilla *Pyrilla perpusilla*	**Spray any one of the following on the 150th and 210th day (1000 l spray fluid):** Chlorpyrifos 20% EC 1500 ml/ha Dichlorvos 76% EC 376 ml/ha < Detrash on the above days < Avoid excess use of nitrogen.
Aleurodids *Aleurolobus barodensis*	< **Spray any one of the following when the incidence is noticed (1000 l spray fluid):** Fenitrothion 50 EC 2000 ml Monocrotophos 36 WSC 2000 ml < The pest generally occurs in ill drained soil.
White grub *Holotrichia consanguinea*	< Crop rotation, < Deep ploughing during summer, < Avoid ratoons in infested fields, < Provide adequate irrigation, since under inadequate soil moisture conditions, the pest appears in the root zone.
Termite *Odontotermes obesus*	Flood irrigate the furrows to avoid termite attack in the furrows at the time of planting < **Sett treatment:** Dip the setts in imidacloprid 70 WS 0.1% or Chlorpyriphos 20 EC 0.04 % for 5 min. < **Soil application:** Apply lindane 1.3 D 125 kg/ha < **Spray:** Chlorantraniliprole 18.5%SC 500-625 ml/ha Imidacloprid 17.8% SL 350 ml/ha Chlorpyriphos 20%EC 750 ml/ha
Root borer	**Spraying any one the following insecticides:** Fipronil 5% SC 1500-2000 ml/ha Fipronil 0.3% GR 25-33.3 kg/ha Phorate 10% CG 25 kg/ha
Black bug	**Apply any one of the following insecticides** Chlorpyriphos 20% EC 750 ml/ha Quinalphos 25% EC 2000 ml/ha
Mealy bug *Saccharicoccus sacchari*	< Detrash as per schedule < Drain excess water **Apply any one of the following insecticides when the incidence is noticed spray on the stem only:** Methyl parathion 50 EC 1000 ml/ha Malathion 50 EC 1000 ml/ha
Leaf hopper	**Spraying any one the following insecticides:** Quinalphos 25% EC 1200 ml/ha Carbofuran 3% CG 33.3kg/ha

IMPROVED TECHNIQUES IN BIOLOGICAL CONTROL

Improved adult feeding techniques for *Trichogramma*

- *Trichogramma* adult feeding through cotton swabs will trap the adults which get entangled in the sticky cotton lint. To avoid this, a better adult feeding technique is developed.
- Make small dotted holes in a thick mylar film sheet or old film negatives by using a sewing machine, leaving a gap of 1 cm between the dotted holes horizontally. One side of the sheet (7 x 6 cm) will be smooth and the other will be eruptive. Streak 50% honey solution on the smooth side by using a camel hair brush. Then fold the sheet in such a way that the honey-smeared surface is on the inside and the eruptive surface outside and staple it. The gap between the dotted holes will provide free movement for the adults, which imbibe the honey through eruptive surface. In this method, the adults do not get trapped in the honey solution.

Special problem: Woolly aphid (*Ceratovacuna lanigera*)

Attacked plants could be recognized from a distance by the following symptoms:

- White appearance of the lower surface of colonized top leaves; sooty mould growth and the honeydew exudations deposited on the upper surface of lower or adjacent leaves; occasional white woolly deposition on the ground under severe colonization.
- Established colonies, characterized by the presence of members most of which showed white woolly filaments, can be generally observed from the second leaf downward in the grown-up crop. At low numbers, colonization on leaves is restricted to a short perpendicular distance on either side of the midrib for a considerable length of the leaf.
- Among the plants the attack is seen only in patches.
- Since the infestation has become a major cause for concern, major initiatives have been started by the Department of Agriculture and ICAR.

Management strategies:

- Enforcement of compulsory IPM measures against woolly aphid infestation in newly planted and ratoon sugarcane fields by invoking suitable provisions of the State Pest Act of the State.
- Harvesting of the entire matured sugarcane crop on priority for crushing as well burning of the trash.
- Application of granular systemic insecticides after two days of irrigation may reduce the infestation of aphids even up to 30 days.
- Promotion of paired or wider row cultivation of sugarcane for taking effective control measures.
- Conservation and augmentation of identified potential biocontrol agents like *Dipha aphidivora, Micromus and coccinellids* in woolly aphid infested fields.
- Release of *Dipha aphidivora* @ 1000/ha or *Micromus igorotus* @ 2500/ha wherever possible.
- Conservation of lepidopteran predator, *Dipha aphidivora* predator population in limited areas of sugarcane crop for further distribution and use thereof.
- Regular surveillance and monitoring of sugarcane woolly aphid for timely

forewarning and adoption of IPM measures including judicious use of recommended pesticides and bio-pesticides (*Metarhizium anisopliae*, *Beauveria bassiana*, *Verticillium lecanii*).

- Avoiding transportation of aphid infested leaves from one location to another.
- Avoiding use of infested cane for seed purpose.
- Ensuring that the insecticides treated leaves are not used as fodder.
- Insecticide application at low levels or at initial stages of infestation may be restricted to only attackednplants since the attack is seen only in patches

During acute incidence, spray any one of the following insecticides once or twice in affected patches: Acephate 75SP 2gm/lit Chlorpyrifos 25EC 2ml/lit Monocrotophos 36WSC 2ml/lit.

Disease	**Recommendations**
Red rot: *Colletotrichum falcatum*	•Removal of the affected clumps at an early stage and soil drenching with 0.1 % carbandazim 50 WP •The irrigation interval in red rot affected field must be lengthened. Once in 15 days during tillering, growth phases and once in 25 days during maturity phase which restricts the spread •The trash of red rot affected field after harvest may be uniformly spread and burnt •In the red rot affected field crop rotation with rice for one season and other crops for two seasons could be adopted
Smut: *Sporisorium scitamineum* (*Ustilago scitaminea)*	•Roguing of smut whips with gunny bags/polythene bag and burnt •Discourage ratooning of the diseased crops having more than 10 per cent infection
Grassy shoot disease (GSD): Candidatus Phytoplasma	•Spray dimethoate @ 0.1 % to control insect vector •Avoid ratooning if GSD incidence is more than 15 % in the plant crop
Leaf spot: *Cercospora longipes*	•Spray mancozeb @ 2 kg or carbendazim @ 500 g/ha
Rust: *Puccinia melanocephala*	•Spray mancozeb @ 2 kg/ha
Yellow leaf disease: Sugarcane yellow leaf virus (Vector: *Melanopsis sacchari*)	•Proper nutritional management •Selection of tissue culture seedling from meristem tip culture

General Recommendations

- Select healthy setts for planting. In the seed crop, select plants which do not show symptoms of red rot, smut, grassy shoot and ratoon stunting. Setts showing red colour at the cut end and hollows should be rejected.
- In fields which had shown high level of red rot disease, follow crop rotation with rice.
- Sett treatment with carbendazim before planting (carbendazim 50 WP @ 0.05% along with 1.0% urea for 15 minutes.
- Treat setts with aerated steam at 50°C for one hour to control primary infection of smut and grassy shoot disease.

  Clumps infected by grassy shoot, smut and ratoon stunting diseases should be uprooted and destroyed.

NEMATODE MANAGEMENT

Nematode pest	Control measures
Lesion nematode, *Pratylenchus coffeae*	*Apply carbofuran 3 CG at 33 kg/ha at the time of planting or 2 months after or Cartop 1.5 kg ai/ha or apply pressmud at 15 t/ha or poultry manure at 2 t/ha or neem cake 2 t/ha or apply pressmud at 15 t/ha or poultry manure at 1 t/ha before last ploughing in garden lands. * Under wetland conditions, intercropping sunnhemp or marigold or daincha coupled with application of pressmud 25 t/ha or neem cake 2 t/ha.

8. SWEET SORGHUM (Sorghum bicolor)

Climate Requirement

T_Max^{o}C	T_Min^{o}C	Optimum ^{o}C	Rainfall mm	Altitude m MSL
40	7 - 8	27 - 35	300 - 600	up to 2300

Tropical and sub tropical crop. It can tolerate drought conditions as well as water logging condition. Short day plant. Soil temperature should be above 18^{o}C.

CROP MANAGEMENT

1. TREATMENT OF SEED

Step 1: Treat the seeds 24 hours prior to sowing with Captan or Thiram 2 gm/kg of seed or Metalaxyl 4 gm / kg of seed to control downy mildew.

Step 2: Treat the seeds required for one hectare with 3 packets (600gm) of Azospirillum using rice gruel as binder.

Note: Dissolve 0.5 gm of gum in 20 ml of water. Add 4 ml of Chlorpyriphos 20 EC or Monocrotophos 35 WSC or Phosalone 35 EC. To this add 1.0 kg of seed, pellet and shade dry to control shootfly and stemborer.

2. FARM LAND PREPARATION

- Form ridges and furrows at a spacing of 45 cm apart

3. SOWING

- Seed rate of 10 kg/ha
- Adopt a spacing of 45 x 15 cm (population 1,48,000/ha)
- Sow the seeds at a depth of 2 cm and cover with soil

Note: Use increased seed rate upto 12.5 kg per hectare and remove the shoot fly damaged seedlings at the time of thinning or raise nursery and transplant only healthy seedlings.

4. IMPORTANCE OF INM

Application of inorganic nutrients alone in the long run will lead to soil and environmental pollution. Hence integration organic and inorganic fertilizer will sustain the soil heath and improve the cane yield of the sweet sorghum crop.

5. IMPORTANCE OF BALANCED NUTRITION

Application of balanced fertilizer at recommended dose in the right stage of the crop will not only improve the productivity but also improve the soil fertility and reduce the environmental pollution.

6. EVALUATION OF FERTILIZER REQUIREMENT

Soil testing is suggested tool for evaluating the fertilizer requirement. It has to be done before the cropping season well in advance so as to ascertain the native fertility of the soil and to recommend the correct dose of fertilizer which will reduce the fertilizer cost.

7. RECOMMENDED INM

- Apply 12.5 tons of FYM/ha at last ploughing.
- Soil application of Azospirillum @ 10 packets (2.0 kg/ha) after mixing with 25 kg of FYM + 25 kg of soil may be carried out before sowing/planting.
- 12.5 kg /ha of MN mixture mixed with enough sand to make a total quantity of 50 kg and applied over the furrows and on top 1/3 of the ridges.
- Apply NPK fertilizers as per soil test recommendations. If soil test recommendation is not available adopt a blanket recommendation of 120 : 40: 40 kg of NPK/ha

8. STAGES OF APPLICATION OF FERTILIZERS

- Apply azospirillum and MN mixtures as basal
- Apply half dose of N and full dose of P_2O_5 and K_2O basally before sowing.
- Apply the balance N in two splits of 25% each on 15th and 30th day of sowing.

CROP PROTECTION

Downy mildew

- Rogue downy mildew infected plants up to 45 days after sowing
- Spray any one of the fungicides like Metalaxyl 500 g or Mancozeb 1000g/ha after noticing the symptoms of foliar diseases, for both transplanted and direct sown crops.

Leaf diseases: *Cercospora* leafspot, Rust, *Colletotrichum* leaf spot

- Spray Mancozeb @ 1kg/ha. Repeat fungicidal application after 10 days if necessary

Grain mould

- Spray any one of the fungicides like Mancozeb @ 1000g/ha in case of intermittent rainfall during earhead emergence and repeat if necessary another spray 10 days later

Ergot

- Spray any one of the following fungicide at emergence of earhead (5 - 10% flowering stage) followed by a spray at 50% flowering and repeat the spray after a week if necessary

9. TROPICAL SUGARBEET

PRODUCTION TECHNOLOGY

Introduction

Tropical sugarbeet (*Beta vulgaris* spp. *Vulgaris var altissima Doll*) is a biennial sugar producing tuber crop, grown in temperate countries. This crop constitutes 30% of total world production and distributed in 45 countries. Now tropical sugarbeet hybrids are gaining momentum in tropical and sub tropical countries including Tamil Nadu as a promising energy crop and alternative raw materials for the production of ethanol. Apart from sugar production, the value added products like ethanol can also be extracted from sugarbeet. The ethanol can be blended with petrol or diesel to the extent of 10% and used as bio-fuel. The sugarbeet waste material *viz.,* beet top used as green fodder, beet pulp used as cattle feed and filter cake from industry used as organic manure.

Tropical sugarbeet now emerged as commercial field crop because of the favourable characters like (i) tropical sugarbeet hybrids suitable for Tamil Nadu (ii) Shorter duration of 5 to 6 months (iii) needs moderate water requirement of 60-80 cm. (iv) higher sugar content of 12 – 15% (v) improve soil conditions because of tuber crop and (vi) grow well in saline and alkali soil. The harvesting period of sugarbeet coincides with March – June, the human resource of sugar factory in the off season may efficiently utilized for processing of sugarbeet in the sugar mills, which helps in continuous functioning of sugar mills.

Hybrids and duration

The tropical sugarbeet hybrids suitable for cultivation in Tamil Nadu are Cauvery, Indus and Shubhra. The duration of these tropical hybrids will be 5 to 6 months depending on climatic conditions prevailing during crop growth period.

Climate and season

Tropical sugarbeet require good sunshine during its growth period. The crop does not prefer high rainfall as high soil moisture or continuous heavy rain may affect development of tuber and sugar synthesis. Tropical sugarbeet can be sown in September–November coincide with North East monsoon with a rainfall of 300 – 350 mm well distributed across the growing period which favours vegetative growth and base for root enlargement. The optimum temperature for germination is 20 – 25^{0}C, for growth and development 30 - 35^{0}C and for sugar accumulation in 25– 35^{0}C.

Season

Tropical sugarbeet is sown in September to November and harvested during March and May.

Field preparation

Well drained sandy loam and clayey loam soils having medium depth (45” cm) with fairly good organic status are suitable. Tropical sugarbeet require deep ploughing (45 cm) and followed by 2 – 3 ploughing to obtain a good soil tilth condition for favorable seed germination. Ridges and furrows are formed at 50 cm apart.

Manures and Fertilizers

S.No	Manures and Fertilizers	Basal Application	Top dressing
1	**Manures**	12.5 tonnes /ha	-
2	**Biofertilizers** Azospirillum Phosphobacteria	2 kg /acre (10 pockets) 2 kg /acre (10 pockets)	-
3.	**Fertilizers** Nitrogen Phosphorus Potassium	75kg /ha 75kg /ha 75kg /ha	37.5 kg / ha each at 25 & 50 DAS

Seeds and sowing

Optimum population is 1,00,000 - 1,20,000 /ha. Hence use only pellated seeds 1,20,000 Nos /ha which require 6 pockets (3.6kg / ha.-One pocket contains 20000 seeds (600 g)]. The recommended spacing is 50 x 20 cm. The pellated seed is dippled at 2 cm depth in the sides of ridges at 20 cm apart. 45 x 15 cm spacing found to be optimum for higher root yield.

Weeding and Earthing up

The crops should be maintained weed free situation upto 75 days. Pretilachlor 50 EC @ 0.5 Kg ai/ha or Pendimethalin @ 1.0 lit /ha can be dissolved in 300 litres of water and sprayed with hand operated sprayer on 0- 2rd day after sowing, followed by hand weeding on 25th day and 50th day after sowing. The earthing up operations coincides with top dressing of N fertilizer.

Irrigation

Tropical sugarbeet is very sensitive to water stagnation in soil at all stages of crop growth. Irrigation should be based on soil type and climatic condition. Pre-sowing irrigation is essential since at the time of sowing, sufficient soil moisture is must for proper irrigation. First irrigation is crucial for the early establishment of the crop. For loose textured sandy loam soil irrigation once in 5 to 7 days and for heavy textured clay loam soil once in 8 – 10 days is recommended. The irrigation has to be stopped at least 2 to 3 weeks before harvest. At the time of harvest, if the soil is too dry and hard it is necessary to give pre harvest irrigation for easy harvest. Light and frequent irrigation is recommended for maintaining optimum soil moisture. Water requirement is 800 - 850 mm.

Pest and diseases

Pests - Aphids, Tobacco caterpillar and Flea beetles

Diseases- Root and crown rot, *Cercospora* leaf spot and Root knot nematode

Integrated pest and disease management

- Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg of seed
- Summer ploughing and exposing the field to sunlight
- Crop rotation for 3 years with Marigold or gingelly or sunnhemp for root rot and nematode
- Soil application of *Trichoderma viride* or *Pseudomonas fluorescens* @ 2.5 kg/ha mixed with 50 kg of FYM before planting
- Sow castor as trap crop around and within fields to attract adult *Spodoptera*

moth for egg laying

- Set up light traps (1 mercury / 5 ha) for monitoring *Spodoptera litura*
- Setting up pheromone -Pherodin SL @ 12/ha for *Spodoptera litura*
- Removal and destruction of *Spodoptera* egg masses, early stage larvae formed in clusters
- Hand picking and destruction of grown up *Spodoptera* caterpillar

Need based

- Spraying *Spodoptera* nuclear polyhedrosis virus at 1.5 x 1012 POB/ha
- Spray NSKE 5% for aphids flea beetles and for early instar caterpillars
- Use of poison bait pellets prepared with rice bran 12.5 kg, jaggery 1.25 kg, carbaryl 50% WP 1.25 kg in 7.5 lit water for *Spodoptera litura*
- Spray any one of the following insecticides using a high volume sprayer covering the foliage and soil surface
- Chlorpyriphos 20 EC - 2 ml / lit,Dichlorvos 76 WSC - 1 ml/lit, Fenitrothion 50 EC - 1 ml/lit Spray malathion 50 EC (2 ml/lit) for flea beetle and leaf webber Spray Imidacloprid 200 SL (0.2 ml/lit) or methyl demeton 25 EC (2 ml/lit) or dimethoate 30 EC (2 ml/lit) for aphids
- Applying neem cake @ 150 kg/ha for root rot
- Foliar spray of Mancozeb 2.5 g / lit or Chlorothalonil 2 g / litre of water for *Cercospora* leaf spot
- Neem cake @ 1 t/ha or carbofuran @ 33 kg/ha as spot application on 30 days after sowing for nematode management

Harvest and yield

The tropical sugarbeet crop matured in about 5 to 6 months. The yellowing of lower leaf whirls of matured plant, Nitrogen deficiency and root brix reading of 15 to 18% indicate the maturity of beet root for harvest. The average root yield of tropical sugarbeet is 80 – 100 tonnes / ha.

CROP PROTECTION

Integrated disease management

- Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg of seed
- Summer ploughing and exposing the field to sunlight
- Crop rotation for 3 years with Marigold or gingelly or sunnhemp for root rot and nematode
- Soil application of *Trichoderma viride* or *Pseudomonas fluorescens* @ 2.5 kg/ha mixed with 50 kg of FYM before planting

TEMPLATE FOR TECHNOLOGY

Introduction

Tropical sugarbeet (*Beta vulgaris* spp. *Vulgaris var altissima Doll*) is a biennial sugar producing tuber crop, grown in temperate countries. Now tropical sugarbeet hybrids are gaining momentum in tropical and sub tropical belts including Tamil Nadu as a promising alternative energy crop for the production of ethanol and alternate sugar producing crop. The ethanol can be blended with petrol or diesel to the extent of 10% and used as bio-fuel.

The bi-products of sugarbeet *viz.,* beet top can be used as green fodder, green leaf manure and raw material for vermi compost while beet pulp is used as cattle feed and filter cake used as manure.

Right seed

- Use pelleted seed
- Variety -Nil
- Hybrids - Cauvery, Indus and Shubhra
- At present no seed production in India, seeds source-
  Syngenta India Ltd., (Seeds division,) 1170 / 27, Revenue colony Shivaji nagar, Pune-411005 Phone: 020-2553 5996 Fax:020 -2553 7571

Right technology

- Seed Treatment : Already it is treated and marketed as pelleted seed.
- Seeds Rate / ha: One 1.2 lakh pelleted seeds(3.6Kg)
- Land Preparation:
- Thorough land preparation of 45 cm deep ploughing,
- Formation of ridges and furrows with a spacing of 50X20 cm and height of the ridges @15 - 20 cm.
- Sowing: Dibble the seed at 2 cm depth on the top of the ridge with a spacing of 20 cm between plants.
- Weed free environment up to 60th day
- Pre-emergence application of Pretilachlor 50EC 0.5 Kg ai / ha in 500 litre of water or Pendimethalin 30 EC 1.0 Kg ai/ha dissolved in 500 litre of water
- Hand weeding on 25th and 50 th days after sowing

Right nutrition

- Balanced application of organic and inorganic fertilizers
- FYM 12.5 t/ha and basal application of 2 kg of Azospirillum and 2 kg of phosphobacteria
- Based on the soil test value, inorganic fertilizer has to be applied. In the absence of soil test value, blanket recommendation of 150:75:75 NPK kg/ha
- Stages of application of fertilizer: Basal 50% N, full P and full K. Remaining 25% N on 20 to 25 days after sowing and 25% N on 40 to 45 days after sowing.
- Timely and need based, placement of fertilizer and earthing up
- Excess N should be avoided

Right pest and diseases management

- Pests - Aphid, Tobacco caterpillar and Flea beetles
- Diseases- Root and crown rot, *Cercospora* leaf spot and Root knot nematode

Integrated pest and disease management

- Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg of seed
- Summer ploughing and exposing the field to sunlight
- Crop rotation for 3 years with Marigold or gingelly or sunnhemp for root rot and nematode
- Soil application of *Trichoderma viride* or *Pseudomonas fluorescens* @ 2.5 kg/ha

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- Sow castor as trap crop around and within fields to attract adult *Spodoptera* moth for egg laying
- Set up light traps (1 mercury / 5 ha) for monitoring *Spodoptera litura*
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- Fenitrothion 50 EC - 1 ml/lit
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- Spray Imidacloprid 200 SL (0.2 ml/lit) or methyl demeton 25 EC (2 ml/lit) or dimethoate 30 EC (2 ml/lit) for aphids
- Applying neem cake @ 150 kg/ha for root rot
- Foliar spray of Mancozeb 2.5 g / lit for *Cercospora* leaf spot
- Neem cake @ 1 t/ha or carbofuran @ 33 kg/ha as spot application on 30 days after sowing for nematode management

Water management

- Optimum EC of irrigation water upto 1 ds/m
- It can be grown in water containing EC: 1 to 2 ds/m
- Irrigation schedule: Life irrigation on 3rd day
- For vegetative stage(upto 45 DAS) - 4 irrigation, vegetative to tuber initiation (75 DAS) - 4 irrigation, tuber maturation(upto 125 DAS) - 4 irrigation and Maturity - 2 irrigation – upto 15- 20 DAS maintain optimum soil moisture for good germination and population
- Drip fertigation with 100 % recommended dose of Fertilizer 150:75:75 Kg NPK ha^{-1} found to be better for tropical sugarbeet

Post harvest management

- Stop irrigation 15-20 days prior to harvest. This allows sugar accumulation
- Just hand pulling and keeping the tops, store in a shaded conditions
- Roots of sugarbeet reach the factory within 48 hours for processing
- Yield 80 to 100 t/ha, Sugar recovery- 15 -16%

10. FORAGE CROPS

(i) FODDER CHOLAM

MULTICUT FODDER SORGHUM (*Sorghum bicolour*)

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Varieties
Irrigated		
All Irrigated districts	**Jan - Feb and Apr – May**	**CO (FS) 29 and CO31**

II. PARTICULARS OF MULTICUT FODDER SORGHUM VARIETIES

PARTICULARS	CO (FS) 29	CO 31
Parentage	Derivative of the cross TNS 30 x *Sorghum sudanense*	Gamma ray induced mutant of CO (FS) 29
Duration (Days)	Multicut (3 years)	Multicut (3 years)
Average green fodder yield (t/ha)	160-170 (6-7 cuts)	190 (6-7 cuts)
Morphological characters		
Plant height (cm)	260-280	270 -290
Number of tillers	10-15	12-17
Number of leaves/stem	8-10	9-11
Leaf length (cm)	80-90	85-95
Leaf breadth (cm)	3.5-4.6	4.5 - 5.0
Leaf stem ratio	0.2-0.25	0.26
Quality characters		
Protein content (%)	8.64	9.86
Dry matter (%)	23.60	25.90
Crude fibre (%)	21.00	19.80
IVDMD (%)	50.30	52

B. CROP MANAGEMENT

I. GREEN FODDER PRODUCTION

IRRIGATED FODDER CHOLAM

1. Soil

All types of soil with good drainage. Does not come up well in flooded or waterlogged conditions.

2. Preparatory cultivation

Plough with an iron plough once and with a country plough twice. Form ridges and furrows of 6 m long and 60 cm apart and plant on either side of the ridge

3 . Sowing

- Seed treatment: Treat the seeds with 3 packets (600 g)/ha of *Azospirillum* and 3 packets (600g) of *Phosphobacteria* or *Azophos* 6 packets (1200g)
- Seed rate: 5 kg/ha
- Spacing: 30 x 10 -15 cm

4. Nutrient management

- Spread 25 tonnes/ha of FYM or compost on the unploughed field, along with 10 packets of *Azospirillum* inoculants (2000 g) and 10 packets of *Phosphobacteria* (2000g) or 20 packets of *Azophos* (4000g)
- Apply 45 : 40 : 40 kg N,P, K/ha as basal and 45 kg N as top dressing on 30 DAS followed by the application of 45 kg N/ha after every cut. After 4th cut, apply 40 kg P and 40 kg K along with 45 kg N to sustain the fodder yield and quality.
- Application of *Azospirillum* (2000g/ha) and *Phosphobacterium* (2000g/ha) together as a mixture or *Azophos* (4000g/ha) along with 75% required dose of N and P fertilizer will enhance the yield besides saving of 25% of fertilizer dose.

5. Irrigation management

Irrigate immediately after sowing. Life irrigation on the third day and thereafter once in 10 days,

6. Weed management

First hand weeding on the 20th day of sowing and if necessary 2nd hand weeding between 35 - 40 days after sowing. After each harvest a weeding may be given before fertilization**.**

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

Green fodder should be harvested at 50% flowering stage. First harvest at 65-70 days after sowing and there after the ratoon crop may be harvested once in 50 days depending on flowering**.**

9. **Green fodder yield**

As green fodder under irrigated conditions, a pure crop yields about 170 to 190 t/ha (6-7 cuts) of green fodder

(ii) FODDER MAIZE

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Varieties
Irrigated		
All Districts	Throughout the year	African tall

II. PARTICULARS OF FODDER MAIZE VARIETIES

PARTICULARS	African Tall
Parentage	Composite
Duration (Days)	60-70
Green fodder yield (t/ha)	35-40
Morphological characters	
Plant height (cm)	302.00
Number of leaves	13.30
Leaf length (cm)	81.30
Leaf breadth (cm)	8.15
Stem thickness (cm)	1.77
Leaf-stem ratio	0.21
Quality characters	
Crude protein (%)	9.80
Dry matter (%)	17.65

B. CROP MANAGEMENT

1. GREEN FODDER PRODUCTION

1. Soil

All types of soil with good drainage. Does not come up well in flooded or waterlogged conditions.

2. Preparatory cultivation

Plough the field twice with an iron plough and three or four times with country plough. Form ridges and furrows using a ridger, 30 cm apart and form beds of size 10 m^2 or 20 m^2 depending on the availability of water and slope of the land.

3. Nutrient management

- Spread FYM or compost at 12.5 t/ha along with 10 packets of Azospirillum (2000 g) and 10 packets of Phosphobacteria (2000g) inoculum or 20 packets of Azophos (4000g) and incorporate the manure into the soil during ploughing.
- Apply NPK fertilizers as per soil test recommendation as for as possible. If soil testing is not done, follow blanket recommendation of 30: 40: 20 kg N, P_2O_5 and K_2O / ha.

Apply 30 kg N/ha at 30 days after sowing as top dressing.

4. Sowing

- Seed treatment: Treat the seeds with Captan @ 2g/kg + Carbaryl @ 200mg/kg^{-1} of seeds 24 hours before sowing.Then,Treat the seeds with 3 packets (600 g) *Azospirillum* inoculant and 3 packets (600 g) of *Phosphobacteria* or 6 packets of *Azophos* (1200 g) before sowing.
- Spacing : 30 x 15 cm,
- Seed rate : 40 kg/ha

5. Water management

Irrigate immediately after sowing and give life irrigation on the third day and thereafter once in 10 days.

6. Weed management

First weeding at 25th DAS. Next hand weeding may be given as and when necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvest

At 50% flowering (65-70 days after sowing)

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 35 to 40 t/ha of green fodder.

Note:

*Fodder maize can be intercropped with fodder cowpea varieties CO 5 or CO (FC) 8 at 3:1 ratio and harvested together to provide balanced nutritious fodder.

TNAU vermicompost at 5 t/ha + 75% recommended dose of fertilizer for intercropping of maize and cowpea produces green fodder yield of 105 t/ha/yr (3 crops/year) which is sufficient to maintain 7 adults and 3 calf.

2. SEED PRODUCTION

VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production leave a distance of 200 m all around the field from the same and other varieties of the crop.

Seed rate

- 20 kg/ha.

Spacing

- 60 x 20 cm (65 x 15 cm)

Fertilizer requirement

- Apply NPK @ 175 : 90 : 90 kg / ha + 25 kg $ZnSO_4$ / ha as basal application.

Harvest

- Seeds attained physiological maturity on 40^{th} day after anthesis.

Seed treatment

- Treat the seeds with carbendazim @ 2 g / kg of seed along with carbaryl @ 200 mg / kg of seed.
- Treat the seeds with halogen mixture ($CaOCl_2$ + $CaCO_3$ + arappu leaf powder mixed in the ratio of 5:4:1 @ 3 g / kg as eco-friendly treatment.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with seed moisture content of 10 - 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with seed moisture content of 8 - 10 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with seed moisture content of less than 8 %.

(iii) NEELAKOLUKATTAI (BLUE BUFFEL GRASS) - (*Cenchrus glaucus*)

CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Variety
Rain fed (Pasture grass)		
All Districts North-East Monsoon	Oct - Dec	CO 1

II. PARTICULARS OF NEELAKOLUKATTAI VARIETY

PARTICULARS	CO 1
Parentage	Clonal selection from Vellakoil local (FS 391)
Duration (Days)	Perennial
Green fodder yield (t/ha/year)	40 (4 cuts)
Seed yield (kg/ha/year)	55 - 60
Morphological characters	
Plant height (cm)	120-130
Number of tillers	60-65
Number of leaves	550--600
Leaf length (cm)	25-30
Leaf width (cm)	0.8-1.0
Leaf stem ratio	0.93
Quality characters	
Dry matter (%)	28.00
Crude protein (%)	9.06
Crude fibre (%)	34.6
Phosphorus (%)	0.26
Calcium (%)	0.58
Manganese (ppm)	56
IVDMD (%)	49.4

B. CROP MANAGEMENT

1. Soil

Well drained soil with high calcium content is suitable. It can also be grown in saline/alkaline soils.

2. Preparatory cultivation

Plough the field twice or thrice with an iron plough to ensure good tilth and form ridges and furrows at 50 cm spacing.

3. Nutrient management

- Spread FYM or compost at 5 t/ha and incorporate the manure into the soil during ploughing.
- Apply NPK fertilizers as per soil test recommendations. If the soil test is not done, follow the blanket recommendation of 25: 40: 20 kg N, P and K per hectare.

- Basal dressing: Apply full dose of NPK before sowing.
- Top dressing: After every harvest apply 25 kg N/ha during the rainy season.

4. Sowing/planting

- Seed treatment: Fresh seeds have dormancy for 6 - 8 months. To break dormancy, soak the seeds in 1 % potassium nitrate solution for 48 hours prior to sowing.
- Seed rate: 6 - 8 kg/ha or 40,000 rooted slips/ha.
- Spacing: 50 x 30 cm.

5. Weed management

Hand weeding can be done as and when necessary.

6. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

7. Harvest

First cut on 70th or 75th day after sowing and subsequent 4 - 6 cuts depending on growth.

8. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 40 t/ha/year (4 – 6 harvests) of green fodder.

Note:

- Tolerant to drought conditions.
- Kolukattai grass can be intercropped with *Stylosanthes scabra* in the ratio of 3:1.

VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 10 m all around the field from the same and other varieties of the crop.

Pre-sowing seed treatment

- Scarify the seeds with sand at 2:1 ratio for 2 min. for improved seed germination.

Harvest

- Harvest the crop when the panicle dried completely.

Seed grading

- Grade the seeds with BSS 14 x 14 wire mesh sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 10 - 12 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 10 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(iv) **GUINEA GRASS (*Panicum maximum*)**

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/ Season	Month	Varieties
Irrigated		
All Districts	Throughout the year	CO 2 and CO (GG) 3
Rain fed		
All Districts	Jun – Sep / Oct - Nov	CO 2 and CO (GG) 3

II. PARTICULARS OF GUINEA GRASS VARIETIES

PARTICULARS	CO 2	CO (GG) 3
Year of Release	2000	2009
Year of Notification	SO.821(E)/13.09.2000	SO.1919(E)/30.07.2014
Parentage	CO 1 x Centenario	Clonal selection from Mumbasa
Duration (Days)	Perennial	Perennial
Green fodder yield (t/ha/year)	270 (7harvests)	340-360 (7harvests)
Morphological characters		
Plant height (cm)	150-200	210-240
Number of tillers/clump	80-100	40-50
Leaf length (cm)	65-75	97-110
Leaf width (cm)	2.5-2.9	3.2 - 4.5
Leaf-stem ratio	-	0.73
Quality characters		
Dry matter (%)	25.94	20.2
Crude protein (%)	8.92	6.35
Crude fibre (%)	34.6	30.3
Phosphorus (%)	0.29	0.19
Calcium (%)	0.59	-
Magnesium (ppm)	0.38	-
IVDMD (%)	49.5	-

B. CROP MANAGEMENT

1. Soil

All types of soil with good drainage.

2. Preparatory cultivation

Plough 2 to 3 times to obtain a good tilth and form ridges and furrows at 50 cm spacing.

3. Nutrient management

- Spread FYM or compost at 25 t/ha and incorporate the manure into the soil during ploughing.
- Apply NPK fertilizers as per soil test recommendation as for as possible. If soil testing is not done, follow the blanket recommendations of 100:50:40 of NPK in kg/ha. Apply full dose of P, K and 50% N basally before planting.
- Top dressing of 50% N on 30 DAP.
- Repeat the application of 75 kg N/ha after each cut for sustaining higher yield.

4. Planting

i. Irrigate through the furrows and plant one rooted slip per hill.
ii. Spacing 50 x 50 cm and 40,000 rooted slips are required to plant one hectare.

5. Irrigation management

Immediately after planting, give life irrigation on the third day and thereafter once in 10 days.

6. Weed Management

Hoeing and weeding on 30th day after planting. Earthing up should be practiced once after every three harvests.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvest

First harvest on 75 - 80 days after planting. Subsequent cuts at the interval of 45 days.

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 270 to 360 t/ha of green fodder.

Note:

- Guinea grass can be intercropped with *Desmanthus* (Velimasal) at 3:1 ratio and can be harvested together and fed to the animals.
- Rooted slips uprooted from 90 days old crop can be used for further propagation

(v) **DEENANATH GRASS (*Pennisetum pedicellatum*)**

CROP MANAGEMENT

1. **SOIL**
   All types of soil with good drainage. Does not come up well on heavy clay soil or flooded or waterlogged conditions.

2. **PREPARATORY CULTIVATION**
   Plough 2-3 times to obtain good tilth and form beds and channels.

3. **MANURING**
   Basal: FYM 25 t/ha NPK 20 : 25 : 20 kg/ha
   Top dressing: 20 Kg N on 30th day after sowing 50% of this has to be applied for rainfed crop

4. **SEED RATE**
   2.5 kg/ha

5. **SPACING**
   35 x 10 cm or solid sowing in lines 30 cm apart.

6. **AFTER CULTIVATION**
   Hoeing and weeding on 30th day after sowing

7. **IRRIGATION**
   Once in ten days or depending on soil condition

8. **PLANT PROTECTION**
   Generally not recommended

9. **HARVEST**
   55-60th day after sowing.

10. **GREEN FODDER YIELD**
    Irrigated crop : 25-30 t/ha first crop. Ratoon crop : 15-20 t/ha
    Rainfed crop : 15-20 t/ha

SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified quality seed production, leave a distance of 10 m all around the field from the same and other varieties of the crop.

Pre-sowing seed management

- Scarify the seeds in a defluffer followed by soaking in GA_3 (200 ppm) + KNO_3 (0.25 %) solution for 16 hours.
- Pellet the seed with DAP @ 60 g / kg and arappu leaf (*Albizzia amara*) powder @ 500 g / kg of seed to enable easy handling of seed during sowing and also for better establishment.

Harvest

- Harvest the crop when the panicle dried completely.
- Delayed harvesting resulted in shattering loss.

Pre- storage seed treatment

- Treat the seeds with carbendazim @ 4 g / kg of seed.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(vi) CUMBU NAPIER HYBRIDS

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Varieties
Irrigated		
All Districts	Throughout the year	KKM 1, CO 3, CO (CN) 4, CO (CN)5

II. PARTICULARS OF CUMBU NAPIER GRASS VARIETIES

PARTICULARS	KKM 1	CO 3	CO (CN)4	CO (BN) 5
Year of Release	2000		2008	2012
Year of Notification	So.161(E)/04.02.2004			SO.1146(E)/24.04.2014
Parentage	IP 15507 x FD 429	PT 1697 x *Penneisetum purpureum*	CO 8 x FD 461	IP 20594 x Napier grass FD 437
Duration (Days)	Perennial	**Perennial**	**Perennial**	**Perennial**
Green fodder yield (t/ha/yr)	288	350 (7 harvests)	375-400 (7 harvests)	360 (7 harvests)
Morphological characters				
Plant height (cm)	155-160	300 – 360	400-500	400-500 cm
No. of leaves per clump	165-170	400-450	400-450	400-430
No. of tillers per clump	10-15	30 – 40	30 – 40	30-40
Leaf stem ratio	-	0.70	0.71	1.19
Leaf length (cm)	110-115	80 – 95	110-115	100-110 cm
Leaf width (cm)	4.5-5.0	3.0 – 4.2	4.0-5.0	4.0-5.0 cm
Quality characters				
Dry matter yield (t/ha/yr)	47.23	65.12	79.87	79.20
Crude protein yield (t/ha)	4.65	5.40	8.71	11.08
Dry matter (%)	16.4	17.0	21.3	22.0
Crude protein (%)	9.85	10.5	10.71	14.0

B. CROP MANAGEMENT

1. Soil

All types of soil with good drainage.

2. Preparatory cultivation

Plough with an iron plough two to three times to obtain good tilth. Form ridges and furrows of 6 m long and 60 cm apart.

3. Nutrient Management

- Spread FYM or compost at 25 t/ha along with 10 packets of Azospirillum (2000 g) and 10 packets of Phosphobacteria (2000g) inoculum or 20packets of Azophos (4000g) and incorporate the manure into the soil during ploughing.
- Apply NPK fertilizers as per soil test recommendation as for as possible. If soil testing

is not done, follow the blanket recommendations of 150:50:40 of NPK in kg/ha. Apply full dose of P, K and 50% N basally before planting.

- Top dressing of 50% N on 30 DAS.
- Repeat the application of 75 kg N after each cut for sustaining higher yield.
- Application of *Azospirillum* (2000g) and *Phosphobacterium* (2000g) or *Azophos* (4000g) along with 75% of recommended dose of N and P fertilizers enhanced the yield besides saving of 25% of fertilizer dose.

4. Planting

i. Irrigate through the furrows and plant one rooted slip/two budded stem cutting per hill.

ii. Spacing 60 x 50 cm and 33,333 planting material are required to plant one hectare.

5. Irrigation management

Immediately after planting, give life irrigation on the third day and thereafter once in 10 days. Sewage or waste water can also be used for irrigation.

Paried row drip system (60/90 cm x 50 cm) + drip irrigation at 125% PE + nitrogen fertigiation at 100% RDN was found to be suitable for obtaining similar green fodder yield as that of surface irrigation with 12.6% water saving in Bajra Naiper hybrid grass,

6. Weed management

Hand weeding can be done whenever necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

First harvest is to be done on 75 to 80 days after planting and subsequent harvests can be done at intervals of 45 days.

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 360 to 400 t/ha of green fodder.

Note

- Quartering has to be done every year or whenever the clumps become unwidely and large.
- Wherever necessary to alleviate the ill effects of oxalates in this grass, the following steps are suggested.
    i. Feeding 5 kg of leguminous fodder per day per animal along with these grasses or
    ii. Providing calcium, bone meal or mineral mixture to the animal or
    iii. Giving daily half litre of supernatant clear lime water along with the drinking water or sprinkling this water on the fodder
    iv. Cultivation o 14 cents of green fodder (Cumbu Napier hybrid grass: 9 cents and *Desmanthus:* 5 cents) are needed for a milch animal with a milk yield of 10 lit/day/animal.
    v. Cultivation of 2.5 cents of green fodder (Cumbu Napier hybrid grass: 1.5 cents and *Desmanthus:* 1.0 cent) are needed for a goat with average body weight of body weight of 40 kg.

(vii)LUCERNE - KUDIRAI MASAL (*Medicago sativa*)

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Variety
Irrigated		
Coimbatore, Thiruppur, Erode, Krishnagiri and Dharmapuri	Throughout the year	CO 2 and CO3

II. PARTICULARS OF LUCERNE VARIETY

PARTICULARS	CO 2	CO 3
Year of Release	2013	2017
Year of Notification	SO.268(E)/28.01.2015	SO. 1498 (E) / 0104.2019
Parentage	Polycross derivative involving CO 1	Polycross derivative involving CO 1
Duration (Days)	Perennial	Perennial
Green fodder yield (t/ha/year)	120.6 (14 harvests)	115 days (12-14 harvests)
Seed yield (kg/ha)	240-250	200-230
Morphological characters		
Plant height (cm)	70-80	75-80
No. of branches per plant	15-20	15-20
No. of pods per plant	18-20	20-22
No. of seeds per pod	4-6	4-6
Quality characters		
Protein content (%)	23.5	22.4
Dry matter (%)	16.8	17.0
Dry matter yield (t/ha/year)	20.16	21.94
Crude fibre (%)	19.0	19.0
Phosphorous	0.45	0.43
Potassium	3.83	3.75
Calcium	1.89	1.90
Magnesium	0.37	0.32
Iron	420	410
Zinc	288	220

B. CROP MANAGEMENT

I. GREEN FODDER PRODUCTION

1. Soil

Well drained black cotton soils are well suited. It can be raised in alkaline soils also.

2. Preparatory cultivation

Plough three or four times with iron plough to obtain good tilth. Form beds of size 10 m^2 or 20 m^2 depending on the availability of water and slope of land.

3. Nutrient management

- Spread FYM or compost at 25 t/ha and incorporate the manure into the soil during ploughing.

- Apply NPK fertilizers as per soil test recommendations as for as possible. If soil testing is not done, follow the blanket recommendation of 25:120:40 kg NPK/ha.
- Apply the full dose NPK of 25:120:40 basally before sowing.

4. Sowing

- Seed treatment: Treat the seeds with 3 packets (600 g) *of Rhizobium* and 3 packets (600 g) *Phosphobacteria* before sowing.
- Seed rate: 20 kg/ha. Good quality seeds free from *Cuscuta* seeds should be used.
- Spacing: 25 cm x 10 cm

5. Irrigation management

Irrigate immediately after sowing, life irrigation on the third day and thereafter once in a week.

6. Weed management

Hand weeding is given as and when necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

First harvest at 65 – 70 days after sowing. Subsequent harvests are made at intervals of 20 – 25 days.

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 80-130 t/ha/year (14 harvests) of green fodder.

II.SEED PRODUCTION

1. Season

Seed production is practiced once in a year during summer months. The crop should be harvested during first week of March and allowed for seed production in such a way that the peak period of flowering should coincide with summer days.

2. Isolation

Adopt 100 m for certified seed production and 400 m for foundation seed production

3. Foliar spray

- Boron application in the form of Borax (150 ppm) increases the seed quality.
- Foliar spraying of $ZnSO_4$ + Borax at 0.3% improves pod and seed weight and also increases the germination potential and vigour of seeds.

4. Harvesting

Hand picking of pods would be done at physiological maturity stage.

5. Other management practices

As given in crop management techniques for green fodder production.

6. Seed Yield

240 kg/ha

(viii) HEDGE LUCERNE - VELIMASAL (*Desmanthus virgatus*)

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/ Season	Month	Varieties
Irrigated		
All Districts	Throughout the year	CO 1
Rain fed		
All Districts	June - October	CO 1

II. PARTICULARS OF VELIMASAL VARIETY

PARTICULARS	CO 1
Parentage	Introduction
Duration (Days)	Perennial
Green fodder yield (t/ha/year)	90-100 (7 harvests)
Seed yield (kg/ha)	200 - 250
Morphological characters	
Plant height (cm)	110-120
No. of branches per plant	15-20
No. of pods per plant	75-100
No. of seeds per pod	6-10
Quality characters	
Protein content (%)	20 - 22
Dry matter (%)	18 - 20
Dry matter yield (t/ha/year)	16.2-20.0

B. CROP MANAGEMENT

I. GREEN FODDER PRODUCTION

1. Soil

All types of soils with good drainage.

2. Preparatory cultivation

Plough with an iron plough once and three or four times with country plough to obtain good tilth. Form beds of size 10 m^2 or 20 m^2 depending on the availability of water and slope of land.

3. Nutrient management

- Spread FYM or compost at 25 t/ha and incorporate the manure into the soil during ploughing
- Apply NPK fertilizer as per soil test recommendations as far as possible. If the soil testing is not done, follow the blanket recommendations of 25: 40:20 kg NPK/ha.
- Apply full dose of NPK basally before sowing.

4. Sowing

- Seed treatment: To get better germination seeds must be treated in hot water at 80^{o}C for 5 minutes (boiling water removed from the flame and kept for 4 minutes to attain 80^{o}C). After hot water treatment, seeds should be washed with cold water and soaked in cold water over a night. Seeds should be shade dried before sowing.Treat the seeds with 3 packets (600 g) *of Rhizobium* and 3 packets (600 g)

Phosphobacteria before sowing.

- Seed rate: 20 kg/ha
- Spacing; 50 cm x solid row

5. Irrigation management

Irrigate immediately after sowing, life irrigation on the third day and thereafter once in a week.

6. Weed management

Hoeing and weeding are given as and when necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

First cut on 90^{th} day after sowing at 50 cm height and subsequent cuts at intervals of 40 days at the same height.

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 120 t/ha of green fodder.

II.SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Pre-sowing seed treatment

- Scarify the seeds with commercial sulphuric acid @ 200 ml / kg of seed for 15 min. to break the seed coat dormancy.

Spacing

- 60 x 20 cm

Fertilizer

- Apply NPK @ 25:40:20 kg / ha as basal for the first crop.

Foliar application

- Foliar spray of 200 ppm salicylic acid thrice at 10 days interval after 50 per cent flowering to improve seed set.

Harvest

- Harvest the pods in pickings.
- Delayed harvest leads to 100 % shattering loss.

Seed size

- Grade the seed using BSS 14 x 14 sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(ix) FODDER COWPEA (*Vigna unguiculata*)

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/ Season	Month	Varieties
Irrigated		
Erode, Madurai, Dindigul, Theni, Cuddalore, Villupuram and Tiruvannamalai	June-July	CO 9

II. PARTICULARS OF FODDER COWPEA VARIETIES

PARTICULARS	CO 9
Year of Release	2016
Year of Notification	SO.1379(E)/ 27.03.2018
Parentage	Cross derivative from CO 5 x Bundel Lobia 2
Duration (Days)	50-55
Green fodder yield (t/ha)	23
Seed yield (kg/ha)	745
Morphological characters	
Plant height (cm)	130-140
No. of branches	4-5
Leaf length (cm)	12.0
Leaf width (cm)	10.0
Leaf stem ratio	
Quality characters	
Dry matter content (%)	16.86
Crude protein content (%)	21.56

B. CROP MANAGEMENT

I. GREEN FODDER PRODUCTION

1. Soil

All types of soils with good drainage.

2. Preparatory cultivation

Plough twice with an iron plough and three or four times with country plough to obtain good tilth.

Form ridges and furrows of 6 m length and 30 cm apart. If ridges and furrows are not made, form beds of size 20 m^2 depending on the availability of water.

3.. Sowing

- Seed treatment: Treat the seeds with 3 packets (600 g) *of Rhizobium* and 3 packets (600 g) *Phosphobacteria* before sowing.
- Seed rate: 25 kg/ha.
- Spacing: 30 x 15 cm

4. Nutrient management

- Spread FYM or compost at 12.5 t/ha and incorporate the manure into the soil during ploughing
- Apply NPK fertilizer as per soil test recommendations as far as possible. If the soil

testing is not done, follow the blanket recommendations of 25:40:20 kg NPK/ha.

- Apply full dose of NPK basally before sowing.

5. Irrigation management

Irrigate immediately after sowing, life irrigation on third day and thereafter once in ten days.

6. Weed management

Hoeing and weeding are given as and when necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

Harvest 50 - 55 days after sowing (50% flowering stage).

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 18-25 t/ha of green fodder.

II.SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 25 m all around the field from the same and other varieties of the crop.

Sowing Season

- October to January.

Pre-sowing seed treatment

- Scarify the seeds with conc. H_2SO_4 acid @ 200 ml / kg for 4 min.
- After scarification, soak the seeds in KNO_3 @ 0.25 % for 3 h to improve germination.

Seed grading

- Grade the seeds using BSS 16 x 16 wire mesh sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(x) MUYAL MASAL (STYLO) - (*Stylosanthes scabra*)

CROP IMPROVEMENT

1. SEASON

June - July to September - October.

2. VARIETIES

a. *S.hamata* (Annual)
b. *S.scabra* (Perennial)

CROP MANAGEMENT

1. APPLICATION OF FYM

Apply and spread 10 t/ha of FYM or compost

2. FORMING BEDS

Form beds of size 10^2 m or 20^2 m

3. APPLICATION OF FERTILIZER

a. Apply NPK fertilizers as per soil test recommendation as for as possible. If the soil testing is not done, follow the blanket recommendation of 20:60:15 kg NPK/ha.
b. Apply full dose of NPK basally.

4. SOWING

a. Seed are to be treated with 3 pockets rhizobium culture (600 g/ha).
b. For line sowing (30 x 15 cm) the seed rate is 6 kg/ha and for broadcasting 10 kg/ha.
c. Stylo seeds possess hard seed coat. So acid scarification is to be done by dipping the seeds in concentrated sulphuric acid for three minutes and washing thoroughly with tap water and scarified seeds are again to be presoaked in cold water overnight. (or) Seeds can also be scarified in hot water by immersing the seeds for 4 minutes in hot water of 80º C and the seeds are again to be presoaked in cold water overnight.

5. WATER MANAGEMENT

It is a rainfed crop. But during the period of establishment, care should be taken to provide sufficient moisture.

6. WEED MANAGEMENT

Hand weeding may be given as and when necessary.

7. Plant Protection:

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. HARVESTING

First harvest can be taken 75 days after sowing at flowering stage and subsequent harvests depending upon the growth.

9. GREEN FODDER YIELD

It is to be noted that during the first year, the establishment after sowing is very slow and the yield is low. Later on when the crop establishes well due to self seeding, it yields about 30 to 35 t/ha/year from the third year onwards.

VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For certified / quality seed production, leave a distance of 25 m all around the field from the same and other varieties of the crop.

Sowing Season

- October to January.

Pre-sowing seed treatment

- Scarify the seeds with conc. H_2SO_4 acid @ 200 ml / kg for 4 min.
- After scarification, soak the seeds in KNO_3 @ 0.25 % for 3 h to improve germination.

Seed grading

- Grade the seeds using BSS 16 x 16 wire mesh sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 - 10 %.
- Store the seeds in polylined gunny bag for medium term storage (12 - 15 months) with a seed moisture content of 8 - 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content less than 8 %.

(xi) LEUCAENA - SOUNDAL

A. CROP IMPROVEMENT

I. SEASON AND VARIETIES

Zone District/Season	Month	Varieties
Rain fed		
All Districts	June - October	CO 1 (*Leucaena leucocephala*) and Pudia Soundal (*Leucaena diversifolia*)

II. PARTICULARS OF SOUNDAL VARIETIES

PARTICULARS	CO 1	Pudia Soundal (*Leucaena diversifolia*)
Parentage	*Leucaena leucocephala*	*Leucaena diversifolia* K – 186 Introduced from Australia
Duration	Perennial tree	Perennial tree
Green fodder yield (t/ha/yr)	35	40
Morphological characters		
Tree height (cm)	35’	30’ in about six years
Leaf stem ratio	-	1.8
Quality characters		
Dry matter content (%)	24.94	25.02
Crude protein content (%)	26.12	26.00
Crude fat (%)	9.51	9.85
Phosphorous (%)	0.09	0.37
Potassium (%)	6.4	3.2
Calcium (%)	0.9	2.4
Magnesium (%)	0.88	1.32
IVDMD (%)	46.01	46.25
Mimosine content (%)	3.07	3.00
Tannin content (%)	3.04	1.95
Carotene content (mg/100g)	11.39	11.54
Resistance to pests	-	Resistant to Psyllids

B. CROP MANAGEMENT

1. Soil

All types of soils with good drainage.

2. Preparatory cultivation

Plough with an iron plough twice and three or four times with country plough to obtain good tilth. Form ridges and furrows at 6 m x 1 m.

3. Nutrient management

- Spread FYM or compost at 25 t/ha and incorporate the manure into the soil during ploughing
- Apply NPK fertilizer as per soil test recommendations as far as possible. If the soil testing is not done, follow the blanket recommendations of 10:60:30 kg NPK/ha.
- Apply full dose of NPK basally before sowing.

4. Sowing

- Seed treatment: Seeds are hard and require scarification to obtain high and uniform germination. Scarification of seeds can be done by pounding the seeds with sand in mortar. Acid scarification can also be done by dipping the seeds in concentrated sulphuric acid for three minutes and washing thoroughly with tap water. Another easiest method is hot water treatment by soaking the seeds in hot water (80^{o} C) for 4 minutes (boiling water removed from the flame and kept for 4 minutes comes down to 80^{o} C). A still simpler method would be to bring water to boil (100^{o} C) in a vessel, take it out of the flame and immediately pour it over the seeds and keep them for 3 to 4 minutes. Then, the hot water may be poured out and cold water added to steep the seeds over night. Seeds can also be simply soaked in plain water for 72 hrs before sowing. Treat the seeds with 3 packets (600 g) *of Rhizobium* and 3 packets (600 g) *Phosphobacteria* before sowing.

- Seed rate: 10 kg/ha.
- Spacing: 2 m x 1 m

5. Irrigation management

This may be done when the crop is raised under irrigated condition. Once established, this plant can withstand several months of dry weather. However, to ensure rapid seedling growth, the land should be adequately moist up to 5 - 6 months. In summer, irrigation once in 6 weeks is adequate.

6. Weed management : Hoeing and weeding are given as and when necessary.

7. Plant protection

As per CIB&RC, insecticide is not recommended for the management of pests in fodder crops. No insecticide is registered/label claimed against the pests of fodder crops.

8. Harvesting

Plant can be harvested as short as 6 months after planting. However, the initial cutting should not be done until the trunk has attained at least 3 cm diameter or the plant has completed one seed production cycle. Harvests can be repeated once in 40 - 80 days depending upon growth and season. In drought prone areas, allow the trees to grow for two years to ensure deep root penetration before commencing harvest. The trees can be cut at 90 to 100 cm height from ground level. For poles and fuel, allow the tree to grow straight without cutting for 2.5 or 5 years as the case may be.

9. Green fodder yield

As green fodder under irrigated conditions, a pure crop yields about 80 to 110 t/ha of green fodder. Under rainfed conditions 40 t/ha of green fodder is obtained after 2 years of initial growth and pruning to a height of 100 cm.

II.SEED PRODUCTION

1.Sowing

- Spacing: 2 m x 1.5 m
- Seed rate: 10 kg

2. Harvest

Seed attains physiological maturity at 35 days after anthesis, when the pods turn brown and seeds become shiny brown.

3. Other management practices

As in crop management technique for green fodder production.

4. Seed Yield : 500 kg/ha

11. **GREEN MANURE CROPS**

(i) DAINCHA - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For foundation / quality seed production leave a distance of 10 m all around the field from the same and other varieties of the crop.
- For certified quality seed production leave a distance of 5 m all around the field from the same and other varieties of the crop.

Season

- Rabi and summer are best seasons.
- October-November sowing will give better seed yield.

Seed rate

- 20 - 30 kg / ha.

Seed treatment

- Seed pelleting with *Rhizobium* @ 5 packets / ha.
- Treat the seed with carbendazim @ 2 g / kg of seeds.
- For hard seeds treat with sulphuric acid @ 100 ml / kg of seed for 10 - 20 minutes.

Spacing

- 60 x 20 cm (vary based on soil fertility and type of soil as 45 x 20 cm, 75 x 45 cm).

Fertilizer

- 20:40:20 kg / ha (vary based on soil fertility and type of soil as 10:40:30, 10:30:30 and 20:50:70 kg of NPK / ha).

Foliar application

- Spray 2 % DAP at 35 and 45 days after sowing.

Harvest

- Picking as once over harvest.

Seed grading

- Grade the seeds with BSS 7 x 7 wire mesh sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content of 8 %.

(i) SUNNHEMP - VARIETAL SEED PRODUCTION

Land requirement

- Land should be free of volunteer plants. The previous crop should not be the same variety or other varieties of the same crop. It can be the same variety if it is certified as per the procedures of certification agency.

Isolation

- For foundation quality seed production leave a distance of 250 m all around the field from the same and other varieties of the crop.
- For certified quality seed production leave a distance of 100 m all around the field from the same and other varieties of the crop.

Season

- Rabi and summer are best seasons.
- October-November sowing will give better seed yield.

Seed rate

- 20 - 30 kg / ha.

Seed treatment

- Seed pelleting with *Rhizobium* @ 5 packets / ha.
- Treat the seed with carbendazim @ 2 g / kg of seeds.
- For hard seeds treat with sulphuric acid @ 100 ml / kg of seed for 10 - 20 minutes.

Spacing

- 30 x 30 cm (vary based on soil fertility and type of soil as 45 x 20 cm, 75 x 45 cm)

Fertilizer

- 20:40:20 kg / ha (vary based on soil fertility and type of soil as 10:40:30,10:30:30 and 20:50:70 kg of NPK / ha).

Foliar application

- Spray 1% sulphate of potash at 40 and 60 days after sowing.

Harvest

- Picking as once over harvest.

Seed grading

- Grade the seeds with 10 / 64" round perforated sieve.

Storage

- Store the seeds in gunny or cloth bags for short term storage (8 - 9 months) with a seed moisture content of 9 %.
- Store the seeds in 700 gauge polythene bag for long term storage (more than 15 months) with a seed moisture content of 8 %.

12. MUSHROOM CULTIVATION

Agriculture will continue to be the main strength of Indian economy. With the variety of agricultural crops grown today, we have achieved food security by prodding about 240 million tonnes of food grains. However, our struggle to achieve nutritional security is still on. In future, the ever increasing population, depleting agricultural land, changes in environment, water shortage and need for quality food products at competitive rates are going to be the vital issues and secondary agricultural vocations are going to occupy a prominent place to fill the void quality food requirements. The demand for quality food and novel products is increasing with the changes in life style and income. To meet these challenges and to provide food and nutritional security to our people, it is important to diversify the agricultural activities in areas like horticulture. Diversification in any farming systems imparts sustainability. Mushrooms are one such component that not only impart diversification but also help in addressing the problems of quality food, health and environmental sustainability. The present century is going to be a century of functional foods from synthetic chemicals and mushroom cultivation fits very well into this category and is going to be an important vocation.

Mushrooms represent microbial technology that recycles agricultural residues into food and manure. It is solid state fermentation system in which crop residues are converted into valuable food rich in microbial protein. These are important source of quality protein, minerals and various novel compounds of medicinal value, do not compare for land and have very high productivity per unit area and time. These are considered to be the highest protein per unit area and time due to utilization of vertical space and short crop cycle. Due to their cultivation under controlled conditions the water requirements is less than any other crop grown in the field and has all the potentials of being a major crop in coming years.

Mushroom farming today is being practiced in more than 100 countries and its production is increasing at an annual rate of 6-7%. In some developed countries of Europe and America, mushroom farming has attained the status of a high-tech industry with very high levels of mechanization and automation. China leads in mushroom production and China alone is reported to grow more than 20 different types of mushroom at commercial scale and mushroom cultivation has become China's sixth largest industry. The USA is the second largest producer of mushroom sharing 16% of the world output. Presently, three geographical regions- Europe, America and East Asia contribute to about 96% of world mushroom production. With the rise in the income level, the demand for mushrooms at very low costs with the help of seasonal growing, state subsidies and capturing the potential markets in the world with processed mushrooms at costs not remunerative to the growers in other mushroom producing countries.

Commercial production of edible mushrooms represents unique exploitation of the microbial technology for the bio conversion of the agricultural, industrial, forestry and household waste into nutritious and proteinaceous food. Our country can emerge as a major player in mushroom production in wake of availability of plenty of agricultural residues and labour.Integrating mushroom cultivation in wake of availability of plenty of agricultural residues and labour. Integrating mushroom cultivation in the existing farming systems will not only supplement the income of the farmers but also will promote proper recycling of agro-residues thereby improving soil health and promoting organic agriculture. In India, mushroom research started in1960s and the cultivation picked up in 1970s and new varieties were evolved in button and oyster mushroom during 1980s and 1990s. Since the year 2000, our country is progressing keeping in pace with global growth by developing technologies for cultivation of medicinal mushrooms.

India has varied agro-climate, abundance of agricultural residues and plenty of manpower making it suitable for cultivating different mushrooms. Our country produces about 600 million tonnes of agricultural waste per annum and a major part of it is let out to decompose naturally or burnt *in situ*. This can effectively be utilized to produce highly nutritive food such as mushrooms and spent mushroom substrate can be converted into organic manure and vermi-compost. Mushrooms are grown seasonally as well as in state-of-art environment controlled cropping rooms all the year round in the commercial units. Mushroom growing is a highly labour oriented venture and labour availability is no constraint in the country and two factors, that is, availabilities of raw materials and labour make mushroom growing economically profitable in India. Moreover, scope for intense diversification by cultivation of other edible mushrooms like oyster, shiitake, milky and other medicinal mushrooms are additional opportunities for Indian growers.

At present, four mushrooms *viz.*, Button mushroom (*Agaricus bisporus*), Oyster Mushroom (*Pleurotus* spp), Paddy straw mushroom (*Volvariella* spp.) and Milky mushroom (*Calocybe indica*) have been recommended for round the year cultivation in India.

India produces about 600 million tonnes of agricultural by products, which can profitably be utilized for the cultivation of mushrooms. Currently, we are using 0.04% of these residues for producing around 1.29 lakh tons of mushrooms of which 85% is button mushroom. India contributes about 3% of the total world button mushroom production. Even if we use 1% of the residues for mushroom production, we can produce 3.0 million tons of mushrooms, which will be almost equal to current global button mushroom production (current world production 3.4 million tons). To remain competitive it will be important to harness science and modern technologies for solving the problems of production and bio–risk management. Mushroom being an indoor crop, utilizing vertical space offers a solution to shrinking land and better water utility.

Mushrooms have been reported to be capable of transforming agro wastes like paddy straw into protein rich food and have been confirmed to be sources of single cell protein. Mushrooms contain rich source of carbohydrates, proteins, amino acids and dietary fibre. Vitamins such as riboflavin, niacin and pantothenic acid, and the essential minerals selenium, copper and potassium are abundant in mushrooms. The foremost importance is that mushrooms do not have cholesterol, instead contain ergosterol that act as a precursor for vitamin D synthesis in human body. Mushrooms are believed to help fight against cancer, relieves hypertension, imparts protection from heart diseases. Mushroom crop is in fact a boon that can solve several problems like the protein malnutrition, unemployment issues and environmental pollution.

Mushrooms are cultivated indoors and do not require arable land and mushroom is a short duration crop with high yield per unit time. For small farmers and landless workers mushroom cultivation is highly suitable for the economic and social security of this group. This hi-tech horticulture venture relieves the pressure on arable land, because it cultivation is indoors, and is also more suited to the women folk. Mushrooms supplement and complement the nutritional deficiencies and are regarded as the highest producers of protein per unit area and almost 100 times more than the conventional agriculture and animal husbandry.

At present, in Tamil Nadu the annual production of mushroom is around 11,000 tonnes, button mushroom accounts for 7,500 tonnes, Oyster mushroom accounts for 2700 tonnes and milky mushroom contributes for 800 tonnes. During the past two decades, the Mushroom Research and Training Centre of the Department of Plant Pathology, Tamil Nadu Agricultural University, Coimbatore has made tremendous efforts on transfer of mushroom cultivation technology by imparting trainings. By this way it has contributed for the establishment of about 50 spawn producers and 600 oyster mushroom growers accounting for 7- 8 tonnes / day, 50 button mushroom growers producing 18-20 tonnes / day and 35 milky mushroom growers contributing 1-2 tonnes / day in Tamil Nadu. This account for around 8 per cent of total mushroom production of the country.

Mushroom varieties/strains released from TNAU for commercial cultivation

Scientific Name	Variety/strain name	Place of release
Oyster mushroom		
Pleurotus sajorcaju	M2	Dept. of Plant Pathology, TNAU, Coimbatore
P. citrinopileatus	CO1	Dept. of Plant Pathology, TNAU, Coimbatore
P. djamor	MDU 1	Dept. of Plant Pathology,AC&RI, Madurai
P. eous	APK 1	Regional Research Station, Aruppukottai
P. ostreatus	Ooty 1	Horticultural Research Station,Uthagamandalam
P. florida	Pf	Dept. of Plant Pathology,TNAU, Coimbatore
P.platypus	Pp	Dept. of Plant Pathology, TNAU,Coimbatore
P. flabellatus	MDU 2	Dept. of Plant Pathology, AC&RI Madurai
Hypsizygus ulmarius	CO2	Dept. of Plant Pathology,TNAU, Coimbatore
Milky mushroom		
Calocybe indica	APK 2	Regional Research Station, Aruppukottai
Tricholoma giganteum	CO 3	Dept. of Plant Pathology, TANU, Coimbatore
Button mushroom		
Agaricus bisporus	Ooty 1 Ooty 2	Horticultural Research Station, Vijayanagaram Horticultura Research Station, Vijayanangaram

Mushroom Cultivation techniques for Oyster and Milk mushroom

Base culture/ Nucleus culture

Tissue culture technique is used to bring the edible mushroom to pure culture so that the mushroom fungus can further be used to prepare spawn. Which is an essential material for mushroom cultivation.

- This nucleus culture is grown on Potato Dextrose Agar medium in test tubes.

- A small tissue from a well-grown mushroom is aseptically transferred to agar medium in a test tube in a culture room.
- The test tubes are incubated under room temperature for 10 days for full white growth of fungal cultue. This is called base culture/nucleus culture and further used for preparation of Mother Spawn.

Mother spawn

Mother spawn is nothing but the mushroom fungus grown on a grain based medium. Among the several substrate materials tested by TNAU, Coimbatore, sorghum grains are the best substrate for excellent growth of the fungus. Well-filled, disease-free sorghum grains are used as substrate for growing the spawn materials. The various steps involving in preparation of mother spawn are listed below here under.

- The sorghum grains are washed in water thoroughly to remove chaffy and damaged grains.
- The grains are half cooked in an autoclave / vessel for 30 minutes to soften them.
- The half cooked grains are spread evenly over hessian cloth on a platform to remove the excess water.
- Calcium carbonate is mixed thoroughly with the cooked, dried grains@ 20g/Kg.
- The grains are filled inn polypropylene bags up to $3/4^{th}$ height (approximately 300-330 g/bag).
- A one inch diameter PVP ring is inserted on open end of the bag and plugged with non-absorbent cotton wood.
- The bags are arranged inside an autoclave and sterilized under 20-lbs,pressure for 2 hours.
- The bags after cooling are kept inside the culture room under the UV light for 20 min.
- After 20 minutes the UV light is put off and the fungal culture is transferred in to the sterilized cholam bags.
- The inoculated bags are kept in a clean room under temperature for 10 days for further preparation of bed spawn.

Bed Spawn

The method of preparation of bed spawn was same as that of mother spawn. The cooking, filling and sterilization were similar to that of mother spawn. After sterilization, the bags are taken and the fully grown mother spawn is used for inoculation to prepare bed spawn. Thirty bed spawn can be prepared from a single mother spawn. The bags are incubated at room temperature (27±2˚C) for 10 days and used as bed spawn.

Cultivation of Oyster mushroom

The oyster mushrooms can be grown indoors in cropping house where a temperature of 25-30^{o} C and relative humidity of 80-85 per cent can be maintained.

- Paddy straw is used as the raw substrate which has to be soaked in water for 4 hours and boiled or steamed in autoclave for 45 minutes and shade dried until 65-70% moisture.
- Cylindrical beds are prepared using 60x30 cm polythene bags with a thickness of 80 gauge.

- Paddy straw and spawn are filled as alternate layers in polythene bags and 10-12 holes are made in the beds.
- The bags are placed in the cropping house/shed in racks or in hanging rope system. After 15-16 days when the paddy straws in the bags are covered with white mycelia growth, pinheads start emerging where water spray is essential to prevent drying of buds.
- First harvest begins from 3-4 days after in head emergence and likewise at 5-7 days internal three harvests can be done.
- Total cropping cycle is around 40-45 days.
- The average bio efficiency ranges (100-150 per cent) depending on the variety.

Cultivation of Milky mushroom

The milky mushroom requires a temperature of 30-35˚C and relative humidity of 85-90 per cent. For cultivation of this mushroom two shed are needed.

- Thatched shed / cropping house (28±2˚C).for Spawn running
- A sunken blue poly house (For Cropping)
- Three feet deep pit is dug out and sides are lined with hollow blocks and semicircular structure is built with GI pipe of Langley and covered with Blue silpaulin sheet.
- Paddy straw is processed as in oyster mushroom cultivation and cylindrical beds are prepared with 90x30 cm polythene bags and stored at 30^{o} C in thatched sheds (spawn running room).
- After 18-20 days when the paddy straws in the bags are covered with white mycelial growth, the beds are cut in to two halves and casing soil (autoclaved garden soil) is layered on to the cut halves for 2 cm height and sprayed with water.
- The cased beds are placed in poly houses and the required tem premature is maintained.
- The pinheads emerge from the cut halves over the casing soil on 25-26^{th} day.
- First harvest begins on 28^{th} day and likewise three- five harvests can be done. The total cropping cycle is around 45-50 days. The average bio efficiency ranges from 150-160 per cent.

Economics of Spawn Production (100 spawn bags per day)

Sl. No.	Item	Quantity	Rate (Rs.)	Total (Rs.)
A.	**Capital investment**			
1.	Autoclave	1	70,000	70,000
2.	Boiler (GL drum 100 lit. Capacity)	2	2,500	5,000
3.	Culture room with work table (low cost)	1	20,000	20,000
4.	UV lamp with fittings	1	2,500	2,500
5.	Tube light fittings	1	1000	1000
6.	Advance for LPG gas	2	3,000	6,000
7.	Spawn storage room	1	30,000	30,000
8.	Bunsen burner	1	300	300
9.	Hear efficient chulah	1	1000	1000
10.	Glass wares & chemicals			5000
	Total			**1,40,800**

B.	**Fixed cost**			
1.	Interest on capital investment @ 15%			21,120
2.	Depreciation (Item 3&7 @ 5%)			2,500
3.	Depreciation (Item 1 2,4,5,8 & 9, 10-10%)			9,080
	Total			**32,700**
C.	**Recurring cost (100 spawn x300 days)**			
1.	Polypropylene bags	150Kg	140	21,000
2.	Cholam grains	8000Kg	26	2,08,000
3.	Calcium carbonate (commercial grade)	160Kg	25	4000
4.	Non-absorbent cotton (400 g rolls)	600	110/roll	66,000
5.	Electricity & Fuel	--	--	60,000
6.	Labor @ 2 men per day for 300 days	300	360/day	2,16,000
7.	Miscellaneous	--	--	10,000
	Total			**5,85,000**

Total cost of Spawn production / Year (Rs)—

Working expenditure	:	5,85,000
Total fixed cost	:	32,700
Total Cost	:	**6,17,700**
Income (Rs.)		
By sale of 30,000 spawn bags @ Rs. 40 per bag	:	12,00,000
Total cost	:	**6,17,700**
Net income per year	:	**5,82,300**

Economics of Oyster mushroom production (10 Kg/day/300 days)
Low cost Investment

Sl. no.	Item	Quantity	Rate (Rs.)	Total (Rs.)
A.	**Capital Investment**			
1.	Thatched House (15' x 25'	1	30,000	30,000
2.	Chaff cutter (Lever type)	1	2000	2,000
3.	Boiler	1	2,000	2,000
4.	Drum	1	1,000	1,000
5.	Spraying systems	1	1,000	1,000
6.	Biomass stove		1,000	1,000
	Total			**37,000**
B.	**Fixed cost**			
1.	Interest on A @ 15%			5,550
2.	Depreciation (Item 1 @ 30%)			9,000
3.	Depreciation (Item 2,3,4,5,& 6 @ 10%)			700
	Total			**15,250**
C.	**Recurring Cost**			
1.	Paddy straw cost + transport	3.5t	7000	24,500
2.	Spawn @ Rs. 40 / No	2000	40	80,000
3.	Polythene bags for bed & packing	25Kg	135	3,375
5.	Labour @ 1 Per day	300	360/day	1,08,000
6.	Others	--		5,000
	Total			**2,20,875**

Total cost of mushroom production / Year (Rs.)

Working expenditure	:	2.20.875
Total fixed cost	:	15,250
Total Cost	**:**	**2,36,125**

Income (Rs.)

By sale of 10Kg/day @ Rs. 135 for300 days	:	4,05,000
Total cost	:	2,36,125
Net Income per year	:	**1,68,875**

Economics of Milky mushroom production (10 Kg/day/300 days)
Low cost Investment

Sl. No.	Item	Quantity	Rate (Rs.)	Total (Rs.)
A.	**Capital Investment**			
1.	Thatched House (15'x 20') Blue Poly house- 20'x50' area (1000 sq.ft)	1 1	20,000 60,000	80,000
2.	Chasff cutter (Lever typw)	1	2000	2,000
3.	Boiler	1	2,000	2,000
4.	Drum	1	1,000	1,000
5.	Spraying systems	1	1,000	1,000
6.	Biomass stoce		1,000	1,000
	Total			**87,000**
B.	**Fixed cost**			
1.	Interest on A @ 15 %			13,050
2.	Depreciation Item 1 @ 10 %)			8,000
3.	Depreciation (Item 2,3,4,5, & 6 @ 10%)			700
	Total			**21,750**
C.	**Recurring Cost**			
1.	Paddy straw cost + transport	3.5 t	7000	24,500
2.	Spawn @ 40 / day	1600	40	64,000
3.	Polythene bags for bed & packing	25Kg	135	3,375
4.	Labour @ 1 per day	300	360/day	1,08,000
5.	Others	--		5,000
	Total			**2,04,875**

Total cost of mushroom production / Year (Rs.)

Working expenditure	:	**2,04,875**
Total fixed cost	:	**21,750**
Total Cost	**:**	**2,26,625**

Income (Rs.)

By sale of 10Kg/day @ Rs. 145 for 300 days	:	4,35,000
Total cost	:	2,26,625
Net Income per year	:	**2,08,375**

13. COMPOSTING AND INDUSTRIAL WASTE WATER UTILIZATION

I. CROP RESIDUE COMPOSTING

Crop residues are the plant parts that are left in the field after harvest. The harvest refuses include straws, stubble, stover and haulms of different crops. Crop remains are also from thrashing sheds or that are discarded during crop processing. This includes process wastes like groundnut shell, oil cakes, rice husks and cobs of maize, sorghum and cumbu. The greatest potential as a biomass resource appears to be from the field residues of sorghum, maize, soybean, cotton, sugarcane etc. In Tamil Nadu 190 lakh tonnes of crop residues are available for use. These residues will contribute 1.0 lakh ton of nitrogen, 0.5 lakh ton of phosphorus and 2.0 lakh tons of potassium. However crop residues need composting before being used as manure.

Waste collection

Crop residues accumulated in different locations are to be brought to compost yard. The compost yard should be located in anyone corner of the farm with accessibility via good road. Water resource should also be available in sufficient quantity. The crop residues that are brought to compost yard should be heaped in one corner for further processing.

Shredding of waste materials

Particle size is one of the factor that influence the composting. It is advisable to shred all the crop residues that are used for composting. Shredding the waste manually is labour intensive. Shredder machine can be employed to shred all the crop residue biomass. Particle size of 5 cm is recommended for quick composting.

Mixing of green waste and brown waste

Carbon and nitrogen ratio decides the initiation of composting process. If C:N ratio is wide (100:1) composting will not take place. Narrow C:N ratio of 30:1 is ideal for composting. To get a narrow C:N ratio, carbon and nitrogen rich material should be mixed together. Green coloured waste materials like glyricidia leaves, parthenium, freshly harvested weeds, sesbania leaves are rich in nitrogen, whereas brown coloured waste material like straw, coir dust, dried leaves and dried grasses are rich in carbon. In any composting process these carbon and nitrogen rich material is to be mixed together to make the composting quicker rather than putting green waste alone or brown color waste alone for composting. Animal dung is also a good source of nitrogen. While making heap formation, alternative layers of carbon rich material, animal dung and nitrogen rich material are to be heaped to get a quicker result in composting.

Compost heap formation

Minimum 4 feet height should be maintained for composting. The composting area should be elevated one and have sufficient shade. While heap formation, all the crop residues should be mixed together to form a heterogeneous material rather than a single homogenous material. Alternate layers of carbon and nitrogen rich material with intermittent layers of animal dung are essential. After heap formation the material should be thoroughly moistened.

Bioinputs for composting

TNAU Biomineralizer consortium contains groups of microorganisms, which accelerate the composting process. If this inoculum is not added to the composting material, natural microorganisms establish on the waste material on its own and do the composting work. This process takes longtime. But if external source of inoculum is added, the microbial activity starts earlier and composting period will be reduced.

For one tonne of crop wastes 2 kg of TNAU Biomineralizer is recommended. This two kg Biomineralizer should be mixed with 20 liters of water and made slurry. When the compost heap is formed in between layers the slurry should be inoculated, so that it mixes with the waste material thoroughly for uniform coating of microorganism on the waste material. Cow dung slurry is also a good source for microbial inoculum. But it carries unwanted microorganisms also which may compete with composting organism. But when TNAU Biomineralizer is not available, cow dung slurry is a good source material. For one tonne of crop residues 40 kg fresh cow dung is required. This 40 kg fresh cow dung is mixed with 100 litres of water and it should be thoroughly poured over the waste material. Cow dung slurry acts as nitrogen source as well as source of microbial inoculum.

Aerating the compost material

Sufficient quantity of oxygen should be available inside the compost heap. For this external air should be freely get in and comes out of the material. Normally to allow the fresh air to get inside, the compost heap should be turned upside down, once in fifteen days. In this process top layer comes to bottom and bottom layer goes to top. This process also activates the microbial process and compost process is hastened. In some cases air ventilating pipe maybe inserted vertically and horizontally, to allow the air to pass through. The wood chip that is available as waste in wood processing industry may also be used as bulking agent in the composting process. This bulking agent gives more air space to the compost material.

Moisture maintenance

Throughout the composting period 60% moisture should be maintained. On any situation, compost material should not be allowed to dry. If the material becomes dry, all the microorganisms present in the crop residues will die and the compost process gets affected.

Compost maturity

Volume reduction, black colour, earthy odor, reduction in particle size are all the physical factors to be observed for compost maturity. After satisfying with the compost maturity index, the compost heap can be disturbed and spread on the floor for curing. After curing for one day, the composted material is sieved through 4 mm sieve to get uniform composted material. The residues collected after composting has to be again composted to finish the composting process.

Compost enrichment

The harvested compost should be heaped in a shade, preferably on a hard floor. The beneficial microorganisms like *Azotobacter*or, *Azospirillum, Pseudomonas, Phosphobacteria* (0.2%) and rock phosphate (2%) have to be inoculated for one ton of compost. Forty per cent moisture should be maintained for the maximum growth of inoculated microorganism.

This incubation should be allowed for 20 days for the organism to reach the maximum population. Now the compost is called as enriched compost. The advantage of enriched compost overnormal compost is the quality manure with higher nutrient status with high number of beneficial microorganisms and plant growth promoting substances.

Nutritive value of Biocompost

The nutritive value of biocompost varies from lot to lot because of varying input materials. But in general biocompost contains all the macro and micro nutrients required for crops, which is given in the following table. Even though the quantity available is low it covers all the requirements of the crop.

Nutrient content of biocompost prepared from different wastes

Biocompost	Nutrient content (%)		
	Nitrogen	Phosphorous	Potash
Animal refuse			
Cattle dung	0.3 - 0.4	0.1 – 0.2	0.1 – 0.3
Horse dung	0.4 – 0.5	0.3 – 0.4	0.3 – 0.4
Sheep dung	0.5 – 0.7	0.4 – 0.6	0.3 - 0.1
Night soil	1.0 – 1.6	0.8 – 1.2	0.2 – 0.6
Poultry manure	1.8 – 2.2	1.4 – 1.8	0.8 – 0.9
Sewage sludge	2.0 – 3.5	---	---
Cattle urine	0.9 – 1.2	Trace	0.5 – 1.0
Horse urine	1.2 – 1.5	Trace	1.3 – 1.5
Sheep urine	1.5 – 1.7	0.1 – 0.2	0.1 – 0.3
Wood Ash			
Ash coal	0.73	0.45	0.53
Ash wood	0.1 – 0.2	0.8 – 5.9	1.5 – 36.00
Habitation waste & factory waste			
Rural compost	0.5 – 1.0	0.4 – 0.8	0.8 – 1.2
Urban compost	0.7 – 2.0	0.9 – 3.0	0.3 – 1.9
Farmyard manure	0.4 – 1.5	0.3 – 0.9	2.0 – 7.0
Straw and stalk			
Pearl millet	0.65	0.75	2.50
Cotton	0.44	0.10	0.66
Banana pseudo stem	0.61	0.12	1.00
Sorghum	0.40	0.23	2.17
Maize	0.42	1.57	1.65
Paddy straw	0.36	0.08	0.71
Tobacco	1.12	0.84	0.80
Pigeon pea	1.10	0.58	1.28
Sugarcane trash	0.53	0.10	1.10
Wheat	0.53	0.10	1.10
Tobacco dust	1.10	0.31	0.93

Benefits of biocompost

- Quality and enriched manure from the crop and animal residues available in the farm. The manure contains both nutrients and beneficial microorganisms.
- There is improvement in the physical, chemical and biological properties of the soil due to regular addition of biocompost.
- Quality products will be obtained from the crop due to improvement in the soil fertility status.
- Soil organic matter content increased and soil biodiversity also improved due to enhanced soil organic matter content.

Compost application

Organic manures are highly regarded as good source of material to maintain soil health and increasing soil organic carbon content. Organic manures cannot be equated with inorganic fertilizers. But organic manures deliver all the nutrients to the soil but with little quantity. For one hectare of land 5 tons of enriched biocompost is recommended. It can be used as basal application in the field before taking up planting work.

Limitation in biocompost application

- While preparing the biocompost, it should be ensured that the material is composted thoroughly.
- If the materials are not fully composted, the material should be sieved through 4 mm sieve and sieved material will be taken as well as composted one. The residues will be put back for another round of composting.
- It is better avoid woody material like heavy branches from pruned trees and other wooden materials. It will take long time and it interferes with other material for composting.

II. ENRICHED ORGANIC MANURE FROM COIR DUST

Enriched organic manure from coir dust is nutrient rich organic manure obtained by composting coir dust along with poultry manure, rock phosphate and microbial inoculants *Pleurotus sajor-caju*, *Bacillus* sp,*Trichoderma* sp and *Pseudomonas* sp. This is a simple and rapid technique to compost coir dust within 60 days.

Inputs

Coir dust	:	1tonne
Poultry manure	:	200 kg
Rock phosphate	:	10 kg
Pleurotussajor-caju	:	2kg
Microbial inoculants	:	2kg

(*Bacillus* sp + *Trichoderma* sp + *Pseudomonas* sp)

Methodology

A partially shaded area should be selected for composting of coir dust. The floor of the selected area must be hard to prevent leaching of water or nutrients from the compost. Spread one tonne of coir dust over the floor selected for composting. A hard-cemented surface is ideal for composting. Otherwise the floor should be hardened by putting stones and other hardy materials. Poultry manure (200 kg) and rock phosphate (10 kg), *Pleurotus sajor caju* (2 kg), microbial inoculums (2 kg) consists of *Bacillus, Trichoderma,* and *Pseudomonas* are added to the coir dust. All the above materials are mixed together thoroughly with one tonne of coir dust. After thorough mixing it should be sprinkled with water and formed in to a heap. The moisture level should be maintained at 60% level through out the composting period. However water should not be dripped out of the composting material. For uniform composting of coir dust, the compost should be turned once in every 10 days. There will be reduction in volume of coir dust and all the material will be changed to black in color after 60th day with an earthy odor from the composted material. It will have high water holding capacity. The enriched coir dust compost contains the following nutrients.

Nutritive value of Enriched Organic Manure from Coir Dust

S. No.	Parameters	Composition
1.	Carbon	28 %
2.	Nitrogen	1.82 %
3.	Phosphorus	2.34 %
4.	Potassium	0.91 %
5.	Cellulose	4.20 %
6.	Lignin	15.39 %
7.	C/N ratio	15.94 1
8.	Iron	1419 mg kg^{-1}
9.	Manganese	116 mg kg^{-1}
10.	Zinc	169 mg kg^{-1}
11.	Copper	115 mg kg^{-1}

Advantages of Enriched Organic Manure from Coir Dust

1. The enriched organic manure from coir dust is produced with in a period of 60 days, whereas in other methods the compost is produced 90 to 120 days.
2. The enriched organic manure from coir dust is environment friendly organic manure, suitable for all soils and crops. It is processed from natural biomass adopting organic method and utilizing bio-agents for decomposition of coir dust.
3. Application of the enriched organic manure from coir dust improves the physico-chemical properties of the soil by increasing the nutrient availability in the soil and improving the soil structure, aggregation, porosity and water holding capacity. The soil fertility is enhanced.
4. The enriched organic manure from coir dust supplies macronutrients (Nitrogen, Phosphorus and Potassium) as well as micronutrients (Iron, Manganese, Copper and Zinc) to the corps.

5. It is an excellent organic medium and basal manure for application in planting pits for crops and forest trees especially in areas of water scarcity and drought.
6. The enriched organic manure from coir dust is an excellent soil ameliorant and soil conditioner for correcting soil problems.Hence it can be used as a component of biological reclamation system for bringing alkaline, saline and also ill drained soils back to remunerative farming.

III. COMPOSTING OF POULTRY WASTES

Value addition of Poultry Waste through Composting technology

Poultry industry is one of the largest and fastest growing livestock production systems in the world. In India, there are about 3430 million populations of poultry with a waste generation of 3.30 million tonnes per year. The localized nature of poultry production also means that it can represent a large percentage of the agricultural economy in many states or regions. Although economical and successful, the poultry industry is currently facing with a number of highly complex and challenging environmental problems, many of which are related to its size and geographically concentrated nature. From an agricultural perspective, poultry wastes play a major role in the contamination of ground water through nitrate nitrogen. Also, the eutrophication of surface water due to phosphorus, pesticides, heavy metals and pathogens present in the poultry wastes applied to soils are the central environmental issues at the present time.

Among the animal manures, poultry droppings have higher nutrient contents. It has nitrogen (4.55 to 5.46 %), phosphorus (2.46 to 2.82 %), potassium (2.02 to 2.32 %), calcium (4.52 to 8.15 %), magnesium (0.52 to 0.73 %) and appreciable quantities of micronutrients like Cu, Zn, Fe, Mn etc. In addition to this cellulose (2.26 to 3.62%), hermicellulose (1.89 to 2.77 %) and lignin (1.07 to 2.16 %) are also present in poultry waste. These components upon microbial action can be converted to value added compost with high nutrient status. In poultry droppings, nearly 60% of nitrogen which is present as uric acid and urea is lost through ammonia volatilization by hydrolysis. This loss of nitrogen reduces the agronomic value of the product, besides causing atmospheric pollution. Composting with amendment seems promising in conservation of nitrogen in poultry droppings. Nitrogen in poultry waste can be effectively conserved by composting with suitable organic amendment. The technologies developed will be highly useful to the poultry farmers.

Method of preparation of poultry waste compost using coir pith

Inputs required

- Poultry droppings
- Coirpith
- *Pleurotus sajor-caju*

Method

A known quantity of the poultry waste as collected above along with coir pith is inoculated with *Pleurotus sajor-caju*@ 2 packets per tonne *to* speed up the composting process. This mixer should be placed under shade as heap. The moisture content of the heap should be maintained at 50 to 60%. Periodical turning must be given on 21st, 28th and 35th days of composting. Another two packets of *Pleurotus sajor-caju* is to be added during turning given on the 28th day of composting. Good quality compost will be attained after 45th day of composting. The nutrient contents of the composts of poultry litter collected from caged system and deep litter systems are as below;

Nutrient content	
Nitrogen (%)	2.08 – 2.13
Phosphorous (%)	2.40 -2.61
Potassium (%)	2.03-2.94
C:N ratio	13:1-14:1

Points to be remembered

- Elevated shady place is highly suitable.
- Within a period of 10 to 15 days, the temperature of the heap will raise to maximum. If the temperature drops below 50º C, the heaps should be spread and moistened with water to bring the moisture content to 60%.
- Colour of the compost will turn from brown to black.
- The matured compost will be odourless.
- The volume of the compost heap will be reduced to 1/3.
- Temperature of the heap will be same as the ambient air temperature and stable.
- Matured compost will be light and fine textured.
- Moisture content of the heap can be measured using moisture meter or by taking handful of compost from the heap and squeezing it with the fingers. If excess water drips out from the compost, then it is considered to have >60 % moisture. If small quantity of water oozes out as drops, then moisture content is considered to be optimum *i.e.*, 60%.
- Each compost heap should have a minimum of one tonne to retain the heat for post decomposition.

Value

Animal manures especially poultry manure are rich in N and the nutrient value of the manure is reduced by loss of N through ammonia volatilization and denitrification. Good quality poultry manure can be obtained by mixing the poultry waste with selective carbonaceous material such as coirpith and inoculation with suitable microorganism. It can be used as an eco-friendly technique for the conversion of poultry waste into valuable compost.

Benefits

Poultry wastes contain higher concentrations of nitrogen, calcium and phosphorus than wastes of other animal species and the presence of nutrients provides more incentive for the utilization of this resource. The loss of nitrogen from poultry droppings can be effectively conserved by composting with coir pith and serves as a good source of organic nutrients to agricultural fields. To make the organic nutrients present in poultry waste available to plants, the waste has to be composted suitably to minimize the volatilization of ammonia.

Applications

The poultry waste compost will be a very good organic manure@6 ton / ha for all the crops.

Limitations

The uninterrupted availability of the raw materials has to be ensured for continuous production on a commercial scale.

IV. VERMICOMPOST

Earthworms live in the soil and feed on decaying organic material. After digestion, the undigested material moves through the alimentary canal of the earthworm, a thin layer of oil is deposited on the castings. This layer erodes over a period of 2 months. So although the plant nutrients are immediately available, they are slowly released to last longer. The process in the alimentary canal of the earthworm transforms organic waste to natural fertilizer. The chemical changes that organic wastes undergo include deodorizing and neutralizing. This means that the pH of the castings is 7 (neutral) and the castings are odorless. The worm castings also contain bacteria, so the process is continued in the soil, and microbiological activity is promoted.

Vermicomposting is the process of turning organic debris into worm castings. The worm castings are very important to the fertility of the soil. The castings contain high amounts of nitrogen, potassium, phosphorus, calcium, and magnesium. Castings contain: 5 times the available nitrogen, 7 times the available potash, and 1 ½ times more calcium than found in good top soil. Several researchers have demonstrated that earthworm castings have excellent aeration, porosity, structure, drainage, and moisture-holding capacity. The content of the earthworm castings, along with the natural tillage by the worms burrowing action, enhances the permeability of water in the soil. Worm castings can hold close to nine times their weight in water. "Vermiconversion," or using earthworms to convert waste into soil additives, has been done on a relatively small scale for some time. A recommended rate of vermicompost application is 15-20 per cent. Vermicomposting is done on small and large scales.

Materials for preparation of Vermicompost

Any types of biodegradable wastes can be used for vermicomposting

1. Crop residues
2. Weed biomass
3. Vegetable waste
4. Leaf litter
5. Hotel refuse
6. Waste from agro-industries
7. Biodegradable portion of urban and rural wastes

PHASE OF VERMICOMPOSTING

Phase 1	:	Processing involving collection of wastes, shredding, mechanical separation of the metal, glass and ceramics and storage of organic wastes.
Phase 2	:	Pre digestion of organic waste for twenty days by heaping the material along with cattle dung slurry. This process partially digests the material and fit for earthworm consumption. Cattle dung and biogas slurry may be used after drying. Wet dung should not be used for vermicompost production.
Phase 3	:	Preparation of earthworm bed. A concrete base is required to put the waste for vermicompost preparation. Loose soil will allow the worms to go into soil and also while watering, all the dissolvable nutrients go into the soil along with water.

Phase 4	:	Collection of earthworm after vermicompost collection. Sieving the composted material to separate fully composted material. The partially composted material will be again put into vermicompost bed.
Phase 5	:	Storing the vermicompost in proper place to maintain moisture and allow the beneficial microorganisms to grow.

Bedding

Bedding is any material that provides the worms with a relatively stable habitat. This habitat must have the following characteristics:

High absorbency

Worms breathe through their skins and therefore must have a moist environment in which to live. If a worm's skin dries out, it dies. The bedding must be able to absorb and retain water fairly well if the worms are to thrive.

Good bulking potential

If the material is too dense to begin with, or packs too tightly, then the flow of air is reduced or eliminated. Worms require oxygen to live, just as we do. Different materials affect the overall porosity of the bedding through a variety of factors, including the range of particle size and shape, the texture, and the strength and rigidity of its structure. The overall effect is referred to in this document as the material's bulking potential.

Low protein and/or nitrogen content (high Carbon: Nitrogen ratio)

Although the worms do consume their bedding as it breaks down, it is very important that this be a slow process. High protein/nitrogen levels can result in rapid degradation and its associated heating, creating inhospitable, often fatal, conditions. Heating can occur safely in the food layers of the vermiculture or vermicomposting system, but not in the bedding.

VERMICOMPOST PRODUCTION METHODOLOGY

i) Selection of suitable earthworm

For vermicompost production, the surface dwelling earthworm alone should be used. The earthworm, which lives below the soil, is not suitable for vermicompost production. The African earthworm (*Eudrillus engeniae),* Red worms (*Eisenia foetida)* and composting worm (*Peronyx excavatus)* are promising worms used for vermicompost production. All the three worms can be mixed together for vermicompost production. The African worm (*Eudrillus eugeniae)* is preferred over other two types, because it produces higher production of vermicompost in short period of time and more young ones in the composting period.

ii) Selection of site for vermicompost production

Vermicompost can be produced in any place with shade, high humidity and cool. Abandoned cattle shed or poultry shed or unused buildings can be used. If it is to be produced in open area, shady place is selected. A thatched roof may be provided to protect the process from direct sunlight and rain. The waste heaped for vermicompost production should be covered with moist gunny bags.

iii) Containers for vermicompost production

A cement tub may be constructed to a height of 2½ feet and a breadth of 3 feet. The length may be fixed to any level depending upon the size of the room. The bottom of the tub is made to slope like structure to drain the excess water from vermicompost unit. A small sump is necessary to collect the drain water.In another option over the hand floor, hollow blocks / bricks may be arranged in compartment to a height of one feet, breadth of 3 feet and length to a desired level to have quick harvest. In this method, moisture assessment will be very easy. No excess water will be drained. Vermicompost can also be prepared in wooden boxes, plastic buckets or in any containers with a drain hole at the bottom.

iv) Vermiculture bed

Vermiculture bed or worm bed (3 cm) can be prepared by placing after saw dust or husk or coir waste or sugarcane trash in the bottom of tub / container. A layer of fine sand (3 cm) should be spread over the culture bed followed by a layer of garden soil (3 cm). All layers must be moistened with water.

v) Worm Food

Compost worms are big eaters. Under ideal conditions, they are able to consume in excess of their body weight each day, although the general rule-of-thumb is ½ of their body weight per day. They will eat almost anything organic (that is, of plant or animal origin), but they definitely prefer some foods to others. Manures are the most commonly used worm feedstock, with dairy and beef manures generally considered the best natural food for *Eisenia*, with the possible exception of rabbit manure. The former, being more often available in large quantities, is the feed most often used.

Common Worm Feed Stocks

Food	Advantages	Disadvantages
Cattle manure	Good nutrition; natural food, therefore little adaptation required	Weed seeds make pre-composting necessary
Poultry manure	High N content results in good nutrition and a high-value product	High protein levels can be dangerous to worms, so must be used in small quantities; major adaptation required for worms not used to this feedstock. May be pre-composted but not necessary if used cautiously
Sheep/Goat manure	Good nutrition	Require pre-composting (weed seeds); small particle size can lead to packing, necessitating extra bulking material
Hog manure	Good nutrition; produces excellent vermicompost	Usually in liquid form, therefore must be dewatered or used with large quantities of highly absorbent bedding
Rabbit manure	N content second only to poultry manure, there-fore good nutrition; contains very good mix of vitamins & minerals; ideal earth-worm feed	Must be leached prior to use because of high urine content; can overheat if quantities too large; availability usually not good

Fresh food scraps (e.g., peels, other food prep waste, leftovers, commercial food processing wastes)	Excellent nutrition, good moisture content, possibility of revenues from waste tipping fees	Extremely variable (depending on source); high N can result in overheating; meat & high-fat wastes can create anaerobic conditions and odours, attract pests, so should NOT be included without pre-composting
Pre-composted food wastes	Good nutrition; partial decomposition makes digestion by worms easier and faster; can include meat and other greasy wastes; less tendency to overheat.	Nutrition less than with fresh food wastes.
Biosolids (human waste)	Excellent nutrition and excellent product; can be activated or non-activated sludge, septic sludge; possibility of waste management revenues	Heavy metal and/or chemical contamination (if from municipal sources); odour during application to beds (worms control fairly quickly); possibility of pathogen survival if process not complete
Seaweed	Good nutrition; results in excellent product, high in micronutrients and beneficial microbes	Salt must be rinsed off, as it is detrimental to worms; availability varies by region
Legume hays	Higher N content makes these good feed as well as reasonable bedding.	Moisture levels not as high as other feeds, requires more input and monitoring
Corrugated cardboard (including waxed)	Excellent nutrition (due to high-protein glue used to hold layers together); worms like this material; possible revenue source from WM fees	Must be shredded (waxed variety) and/or soaked (non-waxed) prior to feeding
Fish, poultry offal; blood wastes; animal mortalities	High N content provides good nutrition; opportunity to turn problematic wastes into high-quality product	Must be pre-composted until past thermophillic stage

vi) Selection for vermicompost production

Cattle dung (except pig, poultry and goat), farm wastes, crop residues, vegetable market waste, flower market waste, agro industrial waste, fruit market waste and all other bio degradable waste are suitable for vermicompost production. The cattle dung should be dried in open sunlight before used for vermicompost production. All other waste should be predigested with cow dung for twenty days before put into vermibed for composting.

vii) Putting the waste in the container

The predigested waste material should be mud with 30% cattle dung either by weight or volume. The mixed waste is placed into the tub / container upto brim. The moisture level should be maintained at 60%. Over this material, the selected earthworm is placed uniformly. For one-meter length, one-meter breadth and 0.5-meter height, 1 kg of worm (1000 Nos.) is required. There is no necessity that earthworm should be put inside the waste. Earthworm will move inside on its own.

viii) Watering the vermibed

Daily watering is not required for vermibed. But 60% moisture should be maintained throughout the period. If necessity arises, water should be sprinkled over the bed rather than pouring the water. Watering should be stopped before the harvest of vermicompost.

ix) Harvesting vermicompost

In the tub method of composting, the castings formed on the top layer are collected periodically. The collection may be carried out once in a week. With hand the casting will be scooped out and put in a shady place as heap like structure. The harvesting of casting should be limited up to earthworm presence on top layer. This periodical harvesting is necessary for free flow and retain the compost quality. Otherwise the finished compost get compacted when watering is done. In small bed type of vermicomposting method, periodical harvesting is not required. Since the height of the waste material heaped is around 1 foot, the produced vermicompost will be harvested after the process is over.

x) Harvesting earthworm

After the vermicompost production, the earthworm present in the tub / small bed may be harvested by trapping method. In the vermibed, before harvesting the compost, small, fresh cow dung ball is made and inserted inside the bed in five or six places. After 24 hours, the cow dung ball is removed. All the worms will be adhered into the ball. Putting the cow dung ball in a bucket of water will separate this adhered worm. The collected worms will be used for next batch of composting.

xi) Nutritive value of vermicompost

The nutrients content in vermicompost vary depending on the waste materials that is being used for compost preparation. If the waste materials are heterogeneous one, there will be wide range of nutrients available in the compost. If the waste materials are homogenous one, there will be only certain nutrients are available. The common available nutrients in vermicompost is as follows

Organic carbon	:	9.5 – 17.98%
Nitrogen	:	0.5 – 1.50%
Phosphorous	:	0.1 – 0.30%
Potassium	:	0.15 – 0.56%
Sodium	:	0.06 – 0.30%
Calcium and Magnesium	:	22.67 to 47.60 meq/100g
Copper	:	2 – 9.50 mg kg-1
Iron	:	2 – 9.30 mg kg-1
Zinc	:	5.70 – 11.50 mg kg-1
Sulphur	:	128 – 548 mg kg-1

xii) Storing and packing of vermicompost

The harvested vermicompost should be stored in dark, cool place. It should have minimum 40% moisture. Sunlight should not fall over the composted material. It will lead to loss of moisture and nutrient content. It is advocated that the harvested composted material is openly stored rather than packed in over sac. Packing can be done at the time of selling. If it is stored in open place, periodical sprinkling of water may be done to maintain moisture level and also to maintain beneficial microbial population. If the necessity comes to store the material, laminated over sac is used for packing. This will minimize the moisture evaporation loss. Vermicompost can be stored for one year without loss of its quality, if the moisture is maintained at 40% level.

4. Advantages of vermicompost

- Vermicompost is rich in all essential plant nutrients.
- Provides excellent effect on overall plant growth, encourages the growth of new shoots / leaves and improves the quality and shelf life of the produce.
- Vermicompost is free flowing, easy to apply, handle and store and does not have bad odour.
- It improves soil structure, texture, aeration, and waterholding capacity and prevents soil erosion.
- Vermicompost is rich in beneficial micro flora such as a fixers, P- solubilizers, cellulose decomposing micro-flora etc in addition to improve soil environment.
- Vermicompost contains earthworm cocoons and increases the population and activity of earthworm in the soil.
- It neutralizes the soil protection.
- It prevents nutrient losses and increases the use efficiency of chemical fertilizers.
- Vermicompost is free from pathogens, toxic elements, weed seeds etc.
- Vermicompost minimizes the incidence of pest and diseases.
- It enhances the decomposition of organic matter in soil.
- It contains valuable vitamins, enzymes and hormones like auxins, gibberellins etc.

5. Pests and diseases of vermicompost

Compost worms are not subject to diseases caused by micro-organisms, but they are subject to predation by certain animals and insects (red mites are the worst) and to a disease known as “sour crop” caused by environmental conditions.

INDUSTRIAL WASTE UTILIZATION FOR LAND RECLAMATION AND CROP PRODUCTION

Application of Untreated Distillery Effluent (Spentwash) for the Reclamation of Sodic Soils

Amendments generally used to reclaim sodic soils are gypsum, phosphogypsum, iron pyrites and elemental sulphur. All these are inorganic in nature. Some of the organic amendments to reclaim the sodic soils are press-mud, farmyard manure (FYM), coir dust and green manures. The direct discharge of untreated distillery effluent (spentwash) to reclaim and improve the productivity of the sodic soils is now advocated. Untreated distillery effluent (spentwash) is acidic (pH: 3.8 – 4.2) with considerable quantity of potassium, calcium and magnesium and traces of micronutrients. Organic compounds, mainly the humic related melanoidins improve the bio-catalytic potential of the treated soil.

Hence, only one time application of 3.75 to 5.00 lakhs litres of untreated distillery effluent (spentwash) per hectare of sodic soils in summer months is recommended. Natural oxidation can be induced for a period of six weeks with two intermittent dry ploughing at a particular interval. Then, after 45 – 60th day of application, soil is to be irrigated with fresh water and drained. This treatment reduces the pH and exchangeable sodium percentage to normal level and increases the productivity of the sodic soils. After this reclamation practice, rice crop can be raised in the effluent applied field adopting the conventional cultivation technique. Application of this effluent again to the next crop/season or year after year and also to the land nearby drinking water sources is not advocated.

Application of Treated Distillery Effluent to Crops

- Treated distillery effluent contains nitrogen 1200 mg L^{-1}, phosphate 500 mg L^{-1}, potash 12000 mg L^{-1}, calcium 1800 mg L^{-1} and iron 300 mg L^{-1}. Since the effluent has higher dissolved salts, 50 times diluted effluent can be irrigated to sugarcane, banana, ragi, sunflower, grasses, cotton and soybean.

- It can also be used as one time application to fallow land at the rate 20,000 to 40, 000 litres per hectare. It should be allowed for complete drying over a period of 20 to 30 days. The effluent applied field is to be thoroughly ploughed two times for the natural oxidation and mineralization of organic matter. After that, crops can be raised in the effluent applied field adopting the conventional methods. Application of this effluent again to the next crop/season or year after year and also to the land nearby drinking water sources is not advocated.

Irrigation of Pulp and Paper Mill Effluents

Pulp and paper effluents contain lot of dissolved solids and stabilized organic matter. The properly treated effluent with EC less than 1.2 dSm^{-1} as such can safely be used for irrigation with appropriate amendments *viz.,* pressmud @ 5 tonnes ha^{-1} (or) fortified pressmud @ 2.5 tonnes ha^{-1} or daincha as in -situ green manure (6.25 tonnes ha^{-1}).

Though there were perceptible changes in soil pH, EC, available NPK, exchangeable cations, exchangeable sodium per cent and sodium absorption ratio, there is no detrimental effect due to sodium either on soil or plants grown in sandy loam soils with good drainage facilities. This treated effluent can be used for irrigation in these soils for the following crops and varieties along with recommended doses of amendments *viz.,* pressmud @ 5 tonnes ha^{-1}, or fortified pressmud @ 2.5 tonnes ha^{-1} or daincha as in situ green manure (6.25 tonnes ha^{-1}).

Crops		Varieties
Rice	:	IR 20, TRY 1, CO 43.
Maize	:	CO 1
Sunflower	:	CO 2
Groundnut	:	TMV 7
Soybean	:	CO 1
Sugarcane	:	CO 6304, COSi 86071, COC 95071, CO 86032
Tapioca	:	CO (TP) 4, CO 2, CO 3, MVD 1

However, irrigating this treated effluent to oil seed crops like gingelly and castor, pulses like greengram and blackgram is not advocated as they were found to be sensitive for this type of effluent irrigation.

Reclamation of papermill effluent irrigated soil

Application of 7.25 t ha^{-1} of gypsum is recommended to reclaim the TEWLIS area soils of Karur district (Moolimangalam, Pandipalayam, Pazhamapuram, Thadampalayam and Ponniagoundanpudur) where the treated paper mill effluent is being continuously used for irrigation since 1995. Application of pressmud @ 6 t ha^{-1} along with Blue Green Algae (15 kg ha^{-1}) and Gypsum (50% Gypsum requirement) is also effective in reclaiming the saline sodic soil with continuous papermill effluent irrigation and to increase the green fodder yield of *Lucerne.*

Crops and Varieties Suitable for Tannery Waste Affected Soils

Based on the results of field trials conducted at Vellore district, the following crops, trees and their varieties are recommended for the tannery waste affected soils

Crops	<u>Varieties</u>
Cereals	: Rice (TRY 1, CO 43, Paiyur 1, ASD 16)
Millets	: Ragi (CO 12, CO 13)
Oilseeds	: Sunflower (CO 4, Morden) and Mustard
Cash crops	: Sugarcane (COG 94076, COG 88123, COC 771)
Vegetables	: Brinjal, Bhendi, Chillies, Tomato (PKM 1)
Flowering crops	: Jasmine, Neerium, Tuberose
Trees	: Eucalyptus, Casuarinas and Acacia

TNAU constructed wetland technology

TNAU constructed wetland technology is recommended for treating the papermill effluent using species *viz., Typha latifolia, Pharagmitis australis* and *Cyperus pangorei* with plant density of lakhs shoots ha^{-1} (25 shoots m^{-2}). Around 1 ha of wetland area is required to treat 1000 m^3 of wastewater per day with a retention time of 2 – 3 days.

The wetland beds should be lined with an impermeable liner made of PVC or high-density poly ethylene (HDPE). The bottom most layer of wetland should be filled with ½ to 1” pebbles to a depth of 6 cm followed by Pea gravel of 6 cm, coarse sand and fine sand each of 7 cm and the top layer with soil to a depth of 9 cm.

14. SERICULTURE

A. MULBERRY (*Morus* spp.) CULTIVATION

1. IRRIGATED

MULBERRY VARIETIES

Kanva 2 (M 5), MR 2, S 36, S 1635,DD, V1.

SOIL TYPE

Deep red soil or red loamy soil. Avoid saline, alkaline or highly acidic soils.

NURSERY

- Select 800 m^2 area near water source for raising saplings required for planting one hectare of main field.
- Apply 1600 kg of FYM.
- Raise nursery beds of 4 m x 1.5m size. The length can be of convenient size depending upon the slop and irrigation source.
- Semi-hardwood cuttings of 10 to 12 mm diameter, free from pests and diseases are selected from 6 to 8 months old well established garden.
- The cutting should be of 15 to 20 cm length with 3 to 4 active buds and should have 45° slanting sharp clean cut (without splitting the bark) at the bottom end.
- Use power operated mulberry cutter (TNAU stem cutting machine) for quick cutting of propagation material with an output of 1000 cuttings per hour.
- Mix one kg of *Azospirillum* (A_ZP_2 culture) in 40 l of water and keep the bottom ends for 30 minutes in it.
- Apply VAM @ 100 g/m^2 of nursery area and irrigated.
- Plant the cuttings in the nursery at 15 cm x 7 cm spacing at an angle of 45° Ensure exposure of atleast one active bud in each cutting.
- Dust one kg endosulfan 4D or malathion 5D or quinalphos 1.5D to prevent termite attack. Drench the soil with carbendazim 50 WP (2 g/l) or apply *Trichoderma viride* 0.5 g/m^2 to prevent root rot and collar rot.
- After weeding, apply 100 g of urea/m^2 of nursery between 45 and 50 days after planting. Transplant 90 to 120 days old saplings.

MAIN FIELD

- Plough the land with disc plough or mould board plough followed by cultivator and otavator.
- Use chisel plough to break open hard pan by operating the plough in criss-cross direction at 50 cm distance.
- During the last plough, apply 20 tonnes of FYM /ha or 5.63 tonnes of vermicompost / ha.

Planting

- Plant the saplings in ridges and furrows at 90 cm x 90 cm spacing (normal row) or at 75/105 cm x 90 cm spacing under paired row system.
- Planting should coincide with onset of monsoon. Gaps should be filled up to maintain a population of 12,345 plants/ha.

Nutrient management

Manures and Fertilizers FYM : 20 t/ha/yr

Fertilizers : 300 : 120 : 120 kg NPK/ha/yr Apply in split doses after every pruning

Application after pruning	Nitrogen (kg)	Phosphorus (kg)	Potassium (kg)
1st	60	60	60
2nd	60	--	--
3rd	60	60	60
4th	60	--	--
5th	60	--	--
Total	**300**	**120**	**120**

For the variety V1, apply 375 : 140 : 140 kg NPK / ha / yr (in equal splits as above)

Note: Apply the fertilizers based on the Soil Test recommendations to optimize the NPK requirement.

Nitrogen

- Apply *Azospirillum* in five split doses at 4 kg/ha, each time, after every pruning to compensate 25% of inorganic N fertilizer.
- *In situ* growing and incorporation of sunnhemp, combined with bio-fertilizer can save 50% of N.

Phosphorus

- Apply phosphorus solubilizing bacteria at 10 kg/ha/yr in two equal splits.
- Apply phosphorus as Enriched FYM (EFYM) in two equal splits along with first and third application of nitrogen.

Preparation of Enriched Farmyard Manure (EFYM)

Mix 375 kg Single Super Phosphate with 750 kg FYM, moisten and keep it in an anaerobic condition for 45 days.

Micronutrients

Foliar spray of 1% FeSO4 or 0.5% ZnSO4 or both whenever the deficiency symptoms of zinc of iron are noticed.

Inter Crop

After every pruning, grow short duration crops like greens, greengram, blackgram, coriander, cowpea, horsegram and sunnhemp.

Weed Management

Weeding should be done manually or chemically after pruning, based on need. Apply glyphosate at 7.5 ml with 10 g of ammonium sulphate / litre of water.

Water Management

Irrigate the field once in seven to eight days based on the need. Drip irrigation, if followed, can save 40% of water requirement.

Pruning

Once in a year, bottom pruning is done leaving a stem of 10 cm height. Other prunings are done at a height of 30-35 cm from ground level. Totally, five prunings are practiced every year.

Harvesting

First leaf harvest can be made six months after planting. Subsequent leaf harvests can be taken 45 days after pruning. Five harvests can be had in an year.

Varieties	**Leaf yield (t/ha/yr)**
Kanva 2, MR 2, S 36, S 1635	35 - 40
DD	40 – 45
V 1	55 - 60

2. Chawki garden

Maintain a separate Chawki garden for rearing young age worms. Otherwise, a part of the main field (5% area) can be allotted for this purpose.

Varieties : S 36 (More suitable because of high carbohydrate and protein content) FYM : 40 t/ha/yr
Fertilizers : 225:150:150 Kg NPK /ha/yr in eight splits
Irrigation : Once in five days
Yield : 25 t/ha/yr in 12 harvests.

Note: V 1 is also suitable for chawki rearing with high nutrient input.

3. RAINFED

Varieties : S 13, S 34, S 1635, RFS 135, RFS 175,MR2
Spacing : 90 cm x 90 cm in pit system of planting
Manures and Fertilizers :
FYM : 20 t/ha/yr
Fertilizer : 100:50:50 kg NPK/ha/yr Apply in split doses after pruninLeaf yield : 12 -15 t/ha/yr

4. Pest and Disease management

Tukra,Pink mealy bug (*Maconellicoccus hirsutus)*

- Cut and burn the affected shoots
- Spot application of endosulfan 4D or malathion 5D around the bushes to kill the phoretic ants.
- Spray dichlorvos 76 WSC @ 1 ml/litre (safe waiting period – 10 days)
- Release predatory coccinellids, *Cryptolaemus montrouzieri* @ 750 beetles / ha or
- *Scymnus coccivora* @ 1000 beetles / ha, *Chrysoperla carnea* @ 2500 eggs/ha. Spray 3 % neem oil with 0.5 % wetting agent.

Thrips (*Pseudodendrothrips mori*)

Spray dichlorvos 76 WSC @ 2 ml/litre or malathion 50 EC @ 2 ml/litre. Spray 3 % neem oil with 0.5 % wetting agent

Leaf webber (*Diaphania pulverulentalis*)

Irrigate the mulberry field immediately after pruning to expose the leaf webber pupae. Release pupal parasitoid, *Tetrastichus howardi* @ 50,000/ha next day after pruning Egg parasitoid, *Trichogramma chilonis* @ 5cc/ha at 10 days after pruning.

Spray dichlorvos 76 WSC @ 1 ml/litre (500 ml/ha) on 30 days after pruning. Clip and burn the affected shoots.

Black Scale (*Saissetia nigra*)

Scrap with a plate to dislodge the insects. Spray malathion 50 EC @ 2ml/litre.

Diseases

Root rot (*Macrophomina phaseolina, Fusarium* spp.)

- Apply neem cake @ 1 t/ha in five split doses Uproot and burn the diseased plants
- Apply copper oxychloride @ 2 g/ litre in the affected areas
- Application of *Pseudomonas fluorescens* + *Trichoderma viride* + *Trichoderma harzianum* + FYM (1:1:1:20) @100 g/ plant.

Powdery mildew (*Phyllactinea corylea*)

Spray wettable sulphur or carbendazim @ 2g/litre.

B. SILKWORM (*Bombyx mori*) REARING

1. Silkworm races

Multi X Bivoltine (cross breeds):

Irrigated areas	: BL24 x NB4D2, PM x CSR2, PM x NB4D2, APM1 x APS8 (Swarnaandhra), BL43 x NB4D2
Rainfed areas	: PM x C.Nichi, BL23 x NB4D2
Bivoltine hybrids	: CSR2 x CSR 4, CSR 18 x CSR 19, KSO1 x NP 2 Double
hybrids	: DH1- [(CSR 6 X CSR 26) (CSR 2 X CSR 27)], DH2- [(CSR 2 X CSR 27) (CSR 6 X CSR 26)].

2. Rearing house

A well ventilated CSB model rearing house with separate ante room, Chawki room, late age worm rearing room and spinning hall should be used for silkworm rearing.

Avoid rearing in dwelling house and in thatched sheds.

3. General disinfection

- Spray 2% formalin with 0.3% slaked lime or 2.5% chlorine dioxide with 0.5% slaked lime @ 2 litres/m^2 area for disinfecting the rearing house.
- Dip the rearing equipments in 2% bleaching powder solution and sundry before use.
- Dust 5% bleaching powder with slaked lime powder @ 200 g/m^2 around the rearing house and the passages, and sprinkle water @ 1 litre/m^2 floor area.

4. Incubation of eggs

Incubate the eggs at 25°C temperature and 80% humidity. At head pigmentation stage (about 48 hours before hatching), keep in dark condition by wrapping in black paper or by keeping them in a box (black boxing)

5. Optimum rearing conditions

Instar	Period (days)	Temp (°C)	Humidity (%)	Leaf size (cm^2)	Size of the cleaning net (mm)	Quantity of leaves (kg) required for 100 DFLs	
						Cross breeds	Bivoltines
Early							
I	3-4	27-28	85-90	0.5-2.0	2	4-5	6-7
II	2-3	26-27	80-85	2.0-4.0	2	6-8	9-10
Late							
III	4-5	25-26	75-80	4.0-6.0	10	30-35**	35-40**
IV	4-5	24-25	70-75	Entire	20	80-90**	120-150**
V	7-9	22-24	70-75	Entire	20	700-800**	800-950**
Total						**820-938**	**1070-1157***

New CSR breeds / hybrids require 15 to 20% higher quantity of leaves.

** Note: The ratio between stem and leaves in the shoot ranges from 3:2 to 1:1. The shoots can be harvested and used accordingly for shoot rearing.

6. Chawki rearing and cleaning:

In a tray of 120 cm x 90 cm x 10 cm size, 20 DFLs are brushed and reared till the end of second age.

From brushing to the end of second age, the larvae are fed with tender leaves.

The leaves are selected from the largest glossy leaf, 3rd and 4th from the top for I instar larvae. The 5th to 8th leaves are used to rear the second instar larvae.

In the first age, one cleaning is given just a day before the worms settle for moult.

In the second age, two cleanings are given, one after resumption of feeding and the other a day before the second moult.

7. Shoot rearing for late age worms

Provide separate rearing house for shoot rearing in shady areas.

Fabricate the rack stand with wood or steel and the rearing seat with wire mesh/bamboo mat. Shoot rearing rack of 1.2m x 11m size is sufficient to rear 50 DFLs.

Provide 15cm border on all sides of the shelf to prevent the dispersal of the larvae.

Arrange the shelves in three tier system with 50 cm space between the tiers. Clean the bed once in each instar.

For cleaning, place two ropes parallelly on the bed and place the new shoots over the ropes. After all the worms have moved on to the new shoots, take the rope from the bed and remove the remains and refuses.

8. Shoot harvesting and feeding

Harvest the shoots at 1 m height from ground level at 60 to 70 days after pruning.

Store the shoots vertically upwards in dark cooler room.

Provide thin layer of water (3cm) in one corner of storage room and place the cut ends of shoots in the water for moisture retention.

Provide a layer of newspaper in rearing shelf. Spread the shoot in perpendicular to width of the bed.

Place top and bottom ends of the shoots alternatively to ensure equal mixing of different qualities of leaves.

Transfer the third instar larvae to shoots immediately after moulting.

Apply soya flour twice @ 5g / kg of shoots on first day of first feeding during fourth and fifth instars during first feeding.

9. **Pest and diseases management**

Pest

Uzi fly (*Exorista bombycis*)

Provide physical barriers like wire mesh or nylon net in the doors and windows of the rearing rooms.

Spray uzicide (1 % benzoic acid) over the larvae to dislodge the eggs.

Dissolve "uzi tables" in water (2 tablets /l) or Asiphor 15 ml/I of water in white bowls to attract the adults (uzitrap). Keep them near windows or at the entrance of rearing room.

Release hyperparasitiod, *Nesolynx thymus* @ 20,000 adults / 100 dfls.

Spray uzifly ovo repellant @ 5 ml/ 5 litres on 3^{rd}, 4^{th} and 5^{th} instar larvae to ward off uzifly from laying eggs on silkworm larvae.

SILKWORM DISEASES

General precautionary measures to be taken Proper disinfection of rearing room and appliances. Providing good ventillation

Maintenance of proper temperature and humidity in the rearing room or avoiding fluctuation in temperature and humidity conditions.

Feeding worms with good quality leaves.

Avoiding starvation.

Avoiding over crowding

Avoiding any damage to the skin of worms.

Proper disposal of the dead worms in 2 % bleaching powder + 0.3 % slaked lime solution.

Avoiding borrowing of mountages.

9.2.2. General bed disinfection

Keep the rearing bed thin and dry by applying slacked lime at 30 to 50 g/ m^2.

Apply bed disinfections such as Sakthi or Vijetha or Resham Jyothi or Sanjeevini @ 4 kg/100 DFLs to prevent the secondary transmission of diseases.

Disinfection of rearing bed with bed disinfectants at dose of 4 kgs/100 dfls and are to be applied at 3g/sq. feet for chawki worms and 5g/sqfeet for late age worms.

9.2.2. Viral/ Grasserie disease

Treat the mulberry leaves with aqueous leaf extract of *Psoralea corylifolia or Plectranthus amboinicus* at 0.1 % or 1000 ppm and feed to the worms immediately after second and third moult. Gentamycin 100 ppm is to be administered after fourth moult.

Bacterial/Flacherie disease (*Streptococcus,Staphylococcus Bacillus, Serratia*)

Treat the mulberry leaves with aqueous leaf extract of *Aegle marmelos* or *Thuja orientalis* at 0.1% or 1000 ppm after second and third moult and 500 ppm *Streptomycin* sulphate after fourth moult.

Fungal/Muscadine disease(*Beauveria bassiana, Metarhizium anisopliae, Aspergillus flavus*)

10 g of dithane M 45 mixed with 1 kg of slaked lime is dusted over chawki worms at 3 g/sq feet. In case of late age worms, 20 g of dithane M 45 is applied at 5 g/ sq.feet. Disinfect rearing rooms and trays with 4% pentachlorophenol to control Aspergillosis.

Pebrine (*Nosema bombycis*)

This disease is taken care off in the grainages and only disease free eggs are supplied to farmers. If the disease is chance encountered in the rearing, the diseased eggs, larvae, pupae, moths, bed refuses and faecal pellets should be disposed after thorough disinfection. Storage of leaves in a separate room.

10. Moulting care

Apply slaked lime @ 30 to 50 g/m^2 when all the worms settle for moult for uniform moulting. Dust 10 g dithane M 45 mixed with 1 Kg of slaked lime over chawki worms at 3 g/ sq. feet. In case of late age worms, 20 g of dithane M 45 is applied at 5 g/ sq. feet.

11. Mounting

- For early and uniform spinning of cocoons, apply Sampoorna @ 20 ml (dissolved in 4 litres of water) / 100 DFLs over the leaves and feed to silkworms.
- Spray the mulberry leaves with phytojuvenoid, Illamathi and feed to second day old fifth instar. Avoid hiring of mountages.
- Arrange 800 to 900 worms per m^2 on a mountage.
- Mountages should be kept in shade in a well ventilated place in slanting position during spinning. Rotary mountages can also be used (one set of rotary mountage can accommodate 1560 worms).

12. Harvesting

Harvest the cocoons of crossbreeds and bivoltines on 5th and 7th day after spinning respectively.

Cross breeds	:	Rainfed 20-25 kg/100 DFLs
Irrigated 50-60 kg/100 DFLs Bivoltines	:	60-70 kg/100 DFLs

15. AGROFORESTRY

The present trend of growing trees in the farm lands demands for identification of economically potential tree species suitable to different climatic conditions and soil types. The concept of agroforestry implies sustained, combined management of the same piece of land for silvicultural, agricultural and pastural crops leading to an overall increase of production compared to single crop management. This practice is of immense importance to our country for it is intimately linked with the question of increasing wood and food production to meet the needs of burgeoning population and conservation of soil, land ,moisture resources which is vital for the tropical regions. Properly distributed tree growth acts as a foster mother to agriculture. This is particularly true in dry inhospitable climatic conditions. Tree growth in such cases conserves soil moisture, increasing atmospheric humidity, improves soil fertility, protects field crops against the scorching and desiccating effects of winds and generally makes the climate more equable and pleasant, thereby stepping up agricultural production.

The silviculture of important agroforestry tree species viz., *Eucalyptus sp., Casuarina spp, Ailanthus excels*a, *Melia dubia*, *Tectona grandis, Santalum album* , *Pterocarpus santalinus, Neolamarckia cadamba* and *Leucaena leucocephala* are given hereunder .

1. SILVICULTURE OF EUCALYPTUS

Species		:	*Eucalyptus camaldulensis, Eucalyptus tereticornis*
Family		:	Myrtaceae
Common Name		:	Red gum, Mysore gum
Locality Factors	Altitude	:	0 - 1000 m
	Mean Annual Rainfall	:	600 – 1500 mm
	Mean Annual Temperature	:	2°C to 32°C
	Soil type	:	Sandy loams Gravels and Alluvial Soil
Phenology	Flowering	:	Flowering occurs twice a year May to June October to November
	Fruit ripening	:	July December
Silvicultural Characters		:	Strong light demander Coppice - coppices freely and vigorously
Nursery Techniques	Seed Propagation	:	Seeds are raised in the mother bed Germinated seedlings are transplanted Six month old seedlings are ready for planting
	Vegetative Propagation	:	Clonal technology – mini clonal Establishment of clonal mother garden

			management of mother garden with irrigation and fertilizer Induction of micro suits Cutting of 5 to 10 cm Planting in root trainer filled with coir compost Root initiation within 21 days Hardening 45 to 60 days 90 days old plants are ready for planting
Silvicultural Treatment	Spacing	:	3m x 1.35 m
	Pit Size	:	30 cm x 30 cm x 30 cm
	Basal Application	:	250g of Vermi-compost or 2kg of Farmyard manure per pit with 50-100 g DAP
	Planting Time	:	June to October
	Irrigation	:	3 to 6 litres per day
	Fertilizer Application	:	2kg of FYM and 100g – 500g all 19 every six month once
	Weeding	:	Two weedings per annum for 2 years
	Pruning	:	Self Pruning
	Thinning	:	Only dead and diseased
Rate of growth		:	Fast growing short rotation trees. It yield an average of 125 - 150 tonnes / ha in 3 years.
Rotation		:	3 Years for Biomass, 5-6 Years for Pulpwood 6-8 Years for Ply wood
Utilization		:	Eucalyptus is an excellent raw material for pulp and paper production due to higher pulp yield ranged between 44 and 48 percent. Fuel wood and charcoal due to high calorific value of over 4500 kcal / kg. Wood is strong and used in particleboard and hardboard industries. The leaves of the Eucalyptus species are rich in essential oils

2. SILVICULTURE OF CASUARINA

Species		:	*Casuarina equisetifolia, Casuarina junghuhniana*
Family		:	Casuarinaceae
Common Name		:	Beefwood, She-oak
Locality Factors	Altitude	:	0 - 1000 m
	Mean Annual Rainfall	:	900 – 3800 mm
	Mean Annual Temperature	:	10 - 47° C

<table>
<tr><td></td><td>Soil type</td><td>:</td><td>Best in loose, fine coastal sands.
For inland conditions - Well drained sandy soils.
It tolerates Lateritic and red soils and also saline, alkaline and acidic conditions.</td></tr>
<tr><td rowspan="2">Phenology</td><td>Flowering</td><td>:</td><td>Flowering occurs twice a year
February to April
September to October</td></tr>
<tr><td>Fruit ripening</td><td>:</td><td>June
December</td></tr>
<tr><td colspan="2">Silvicultural Characters</td><td>:</td><td>Strong light demander and drought resistant.
Susceptible to fire.
Coppices badly.
Tolerate waterlogged conditions</td></tr>
<tr><td rowspan="2">Nursery Techniques</td><td>Seed Propagation</td><td>:</td><td>Seed treatment – Nil
Seeds are raised in the mother bed
Germinated seedlings are transplanted
Six month old seedlings are ready for planting</td></tr>
<tr><td>Vegetative Propagation</td><td>:</td><td>Clonal technology – mini clonal
Establishment of clonal mother garden
management of mother garden with irrigation and fertilizer
Induction of micro suits
Cutting of 5 to 10 cm
Planting in root trainer filled with coir compost
Root initiation within 21 days
Hardening 45 to 60 days
90 days old plants are ready for planting</td></tr>
<tr><td rowspan="9">Silvicultural Treatment</td><td>Spacing</td><td>:</td><td>1.5 x 1.5 m to 2 x 2 m</td></tr>
<tr><td>Pit Size</td><td>:</td><td>30 cm x 30 cm x 30 cm</td></tr>
<tr><td>Basal Application</td><td>:</td><td>250g of Vermi-compost or 2kg of Farmyard manure per pit with 50-100 g DAP</td></tr>
<tr><td>Planting Time</td><td>:</td><td>South West and North East monsoon</td></tr>
<tr><td>Irrigation</td><td>:</td><td>6 to 8 litters per day</td></tr>
<tr><td>Fertilizer Application</td><td>:</td><td>100 kg urea first year in three dozes
150 kg DAP at four dozes in second year
150 complex in three dozes in third year</td></tr>
<tr><td>Weeding</td><td>:</td><td>Two weedings per annum</td></tr>
<tr><td>Pruning</td><td>:</td><td>Once in every six months</td></tr>
<tr><td>Thinning</td><td>:</td><td>Dead and diseases</td></tr>
<tr><td colspan="2">Rate of growth</td><td>:</td><td>High Yielding short rotation trees.
It yield an average of 100 to 150 tonnes / ha in 3 years.</td></tr>
<tr><td colspan="2">Rotation</td><td>:</td><td>Rotation age of Casuarina is 36 months</td></tr>
<tr><td colspan="2">Utilization</td><td>:</td><td>Pulp wood: Pulp yield is more than 47 %.
Fuel Wood: Calorific value is 4950 Cal/Kg.
Timber: Density is 850 Kg/m^3.</td></tr>
</table>

			Wind breaks Poles

3. SILVICULTURE OF MELIA

Species		:	*Melia dubia*
Family		:	Meliaceae
Common Name		:	Malabar Neem, Melia
Locality Factors	Altitude	:	1500 to 1800 m
	Mean Annual Rainfall	:	800 – 1000 mm
	Mean Annual Temperature	:	32 - 40° C
	Soil type	:	Deep, well drained sandy loam soils
Phenology	Flowering	:	November – January
	Fruit ripening	:	January – February
Silvicultural Characters		:	Light demander Susceptible to damage by fires Saplings suffer from browsing
Nursery Techniques	Seed Propagation	:	Seed has extracted from this stony endocarp and treated with GA at 100 ppm for overnight. Seed are sown sand bed and germination starts in 30 days Germination is only 40% Germinated seedlings are transplanted in poly bags six month old seedlings area ready for planting
	Vegetative Propagation	:	Clonal technology – mini clonal Establishment of clonal mother garden management of mother garden with irrigation and fertilizer Induction of micro suits Cutting of 5 to 10 cm Planting in root trainer filled with coir compost Root initiation within 21 days Hardening 45 to 60 days 90 days old plants are ready for planting
Silvicultural Treatment	Spacing	:	Plywood : 4m x 4m (or) 5m x 5m (or) 6 m x6m Pulp & Plywood : a) 6'x 6' (First two years)
	Pit Size	:	30cm x 30cm x 30cm
	Planting Time	:	June to October
	Irrigation	:	Apply light irrigation once in 7 to 10 days. This could be done through drip irrigation
	Fertilizer Application	:	Mixture of compost and organic fertilizers, bio-fertilizer and planting (25-50 g), respectively, to be applied to the pits
	Weeding	:	Annual
	Pruning	:	Annual

	Thinning	:	Thinning of alternate rows at the beginning of 3rd year Thinning alternate diagonals at the beginning of 5th year Final harvest at the beginning of 7th year
Rate of growth		:	For pulpwood: 100-150 tonnes / ha in three years. For Plywood: 200 tonnes / ha in six years
Rotation		:	For Plywood: 5-7 Years For Pulp and Paper: 24-36 months
Utilization		:	The wood is used for packing cases, cigar boxes, ceiling planks, building purposes, agricultural implements, pencils, match boxes, splints and Catamarans. It is employed for outriggers of boats. It is suitable for musical instruments, tea boxes and ply board. It is a good fuel wood (Calorific value: 3,400 - 4,100 cal.) The fruit of the plant is bitter. It is considered anthelmintic.

4. SILVICULTURE OF TEAK

Species		:	***Tectona grandis***
Family		:	*Verbenaceae*
Common Name		:	Teak
Locality Factors	**Altitude**	:	0 to 1200 m
	Mean Annual Rainfall	:	1000 - 5000 mm
	Mean Annual Temperature	:	2°C to 48°C
	Soil type	:	Deep and well-drained soil Fertile Alluvial-colluvial soil
Phenology	**Flowering**	:	January to April
	Fruit ripening	:	May to July
Silvicultural Characters		:	Strong light demander Sensitive to frost and drought Good coppicer and pollards vigorously
Nursery Techniques	Seed Propagation	:	Seeds are treated with alternate wetting and drying for 14 days Seeds sown in mother bed Germinations stars after 3rd week 9 – 12 months old seedlings are lifted for stump preparation Stump size 2.5 cm shoot portion 22.5 root portion Stump are used directly for planting

	Vegetative Propagation	:	❖ Nil
Silvicultural Treatment	Spacing	:	2m x 2m
	Pit Size	:	30 cm x 30 cm x 30 cm
	Basal Application	:	2 kg of farmyard manure, 100 g DAP
	Planting Time	:	June – July or September - October
	Irrigation	:	
	Fertilizer Application	:	2kg of FYM, 100g of complex fertilizer and 100g – 300g all 19 every six month once
	Weeding	:	Every 3 month once
	Pruning	:	Necessary, Every 6 month once
	Thinning	:	Thinning cycle of 4, 8, 12, 18, 26 and 36 years have been followed for 50 years rotation. In Tamil Nadu the thinning cycle of 5, 10, 18, 25 and 36 years are followed for 50 years rotation. In both the cases, the first two thinning are mechanical and the rest are Silvicultural at C grade thinning.
Rate of growth		:	Teak is a fast grower and attains a height of 15feet to 20 feet in one year under well managed condition.
Rotation		:	20 years under well irrigated and managed condition 40-60 years under forest site condition 15 years under bund plantations.
Utilization		:	Teak is a moderately strong timber with a density of 660 kg/m^3 and is preferred as a most suitable timber both for domestic and industrial utility. Teak is known as a renowned timber due to durability, dimensional stability, working quality and resistant to termites. Teak wood is used in all construction purpose such as beams, columns, doors, windows, flooring, panelling etc. It is one of the best timbers for furniture and cabinet making wagon and railway cadges. For marine construction and ship building teak is preferred due to dimensional stability.

5. SILVICULTURE OF AILANTHUS

Species	:	***Ailanthus excelsa***
Family	:	*Simaroubaceae*
Common Name	:	Tree of heaven

Locality Factors	Altitude	:	0 to 900 m
	Mean Annual Rainfall	:	500 - 1900 mm
	Mean Annual Temperature	:	12.5°C to 47.5°C
	Soil type	:	Porous sandy loams
Phenology	Flowering	:	February to March
	Fruit ripening	:	April to may
Silvicultural Characters		:	Strong light demander Sensitive to drought Moderately frost tender Coppices well and produces root suckers freely. Susceptible to water logging areas
Nursery Techniques	Seed Propagation	:	De winged Seeds soaked in cold water for 5- 7 days by replacing fresh water daily The seeds are dibbled in poly bag and water daily Six month old seedlings are ready for planting.
Silvicultural Treatment	Spacing	:	6m x 6m
	Pit Size	:	30 cm x 30 cm x 30 cm or 45 cm x 45 cm x 45 cm
	Basal Application	:	2 kg of farmyard manure, 100 g DAP
	Planting Time	:	The area is cleared and pits are dug out in the month of February - March and the soil is allowed to weather. The planting in pits is carried out in the month of July.
	Irrigation	:	5 – 8 litters per day
	Fertilizer Application	:	5kg of FYM and 100g – 200g complex fertilizer yearly once
	Weeding	:	Timely and regular weeding for the first two years are very essential
	Thinning	:	The first silvicultural thinning may be carried out in the seventh or eighth year when the tree attains a height of 10-12 m.
Rate of growth		:	Ailanthus is a slow growing tree and attains a height of 10 feet to 15 feet in six year under well managed condition.
Rotation		:	6 – 8 Years
Utilization		:	This species is extensively used for making matchwood boxes and match splints. The wood is extensively used in cottage industries for making wooden toys and cheap quality cricket bats. The tree is used for making packing cases and wooden boxes. The wood is used for packing cases, fishing floats and sword sheaths. The leaves are rated as highly palatable and protein

		rich nutritious fodder for sheep and goats and are said to augment milk production. The stem and branches are used for fuel wood but it gives poor quality fuel as it burns quickly and does not sustain heat for long. The tree is the most adaptable and pollution tolerant. It is suitable for sloppy, degraded and denuded areas and wasteland. It is also yields gum of inferior quality. The bitter and aromatic leaves of the plant show medicinal properties. The leaves are used for the preparation of lotions for scabies. Quassinoids and ailantic acid are isolated from bark. The dried bark is aromatic and burnt as incense.

6. SILVICULTURE OF SUBABUL

Species		:	*Leucaena leucocephala*
Family		:	Fabaceae
Common Name		:	Subabul
Locality Factors	Altitude	:	0 to 1500 m
	Mean Annual Rainfall	:	650 - 3000 mm
	Mean Annual Temperature	:	15°C to 36°C
	Soil type	:	Calcareous soils Saline soils and on alkaline soils up to pH 8
Phenology	Flowering	:	Two flowering season July – November February – May
	Fruit ripening	:	December June
Silvicultural Characters		:	Strong light demander Vigorous coppicer Moderate frost tender Drought resistance
Nursery Techniques	Seed Propagation	:	Seeds are treated with concentrated sulphuric acid and sown directly in the poly bag 3 – 4 months old seedlings are ready for planting
	Vegetative Propagation	:	Seeds sown in mother bed Germinations starts after 3rd week 9 – 12 months old seedlings are lifted for stump preparation

			Stump size 2.5 cm shoot portion 22.5 root portion Stump are used directly for planting
Silvicultural Treatment	Spacing	:	1.5m x 1.5m; 2m x 2m; 3m x 3m
	Pit Size	:	30 cm x 30 cm x 30 cm
	Basal Application	:	2 kg of farmyard manure, 50 g of super phosphate, 50 g DAP
	Planting Time	:	2-4 month old seedlings can be used to planting out in the month of July.
	Irrigation	:	-
	Fertilizer Application	:	2kg of FYM and 100g – 500g all 19 every six month once
	Weeding	:	3 month once for first two years
	Pruning	:	Regular during first one year
Rate of growth		:	High yielding short rotation tree It yield an average of 100 tonnes/ha in 3 to 4 years
Rotation		:	4-6 Years depends on location
Utilization		:	Subabul is a hard heavy wood (about 800 kg/m) and medium density wood. Subabul is one of the highest quality and most palatable fodder trees. Subabul is an excellent firewood species with a specific gravity of 0.45-0.55 and a high calorific value of 4600 cal/kg.

7. SILVICULTURAL OF SANDALWOOD

Species		:	***Santalum album***
Family		:	*Santalaceae*
Common Name		:	East Indian Sandalwood
Locality Factors	Altitude	:	90 – 1500 m
	Mean Annual Rainfall	:	500-2000 mm
	Mean Annual Temperature	:	15-35°C
	Soil type	:	Sandy clayey red lateritic loamy even in black cotton soil Red ferruginous (iron) loam over lying on metamorphic rocks Rocky ground and stony or gravelly soils
Phenology	Flowering	:	Two Flowering Season February – April October - November
	Fruit ripening	:	May December
Silvicultural Characters		:	Shade bearer

			Root suckers freely Coppices fairly well
Nursery Techniques	Seed Propagation	:	The seeds exhibits initial dormancy for 3-4 weeks and after 4 weeks it starts germination which is about 60% The uniform and very good germination can be obtained soaking seeds with 0.05% gibberlic acid over night. The sandal seedlings are transplanted along with host plants *viz.*, Casuarina, Cajanus cajan, Albizia, Alternenthra, Amaranthus etc.
	Vegetative Propagation	:	Root cuttings of sandalwood gives only 20 % success.
Silvicultural Treatment	Spacing	:	3 X 3 m to 5 X 5 m
	Pit Size	:	30 x 30 x 30 cm (or) 40 x 40 x 40 cm (or) 60 x 60 x 60 cm
	Basal Application	:	Soil mixture with neem cake 25-50 g / pits, Chlorpyrifos 2 g powder / pit
	Planting Time	:	Monsoon Season
	Host	:	Sandal has association with over 150 species of host. Albizia, Terminalia, Lagerstroemia, Dalbergia, Casuarina, *Acacia nilotica*, *Pongamia pinnata, Wrightia tinctoria* and *Cassia siamea* are the major host plants.
	Irrigation	:	4-5 litres / Day based on the growth
	Fertilizer Application	:	During soil working periods application of farmyard manure @ 5 kg / plant
	Weeding	:	Yearly once
	Pruning	:	Pruning is essential to get good heartwood formation
Rate of growth		:	Slow growing species The heartwood formation in sandal starts after 10 years. The heartwood forms at the rate of 1 kg/annum after 20 years.
Rotation		:	Physical rotation. The dead and naturally fallen trees are harvested
Utilization		:	**Sapwood:** Sapwood is white and scentless used for manufacture of agarbattis. **Heartwood:** Heartwood of sandal is moderately hard, heavy and strongly scented, wood and oil are used in incense, perfumes, soap making and medicines. **Religious:** Sandal is considered sacred by Hindus. **Essential oil:** Valuable oil, 'the sandal oil', is distilled from the heartwood (4-13%) and is used in perfumery, soap making and medicines. Seed oil: Seeds yield oil that can be used in the manufacture of paint. **Medicinal uses:** The wood is bitter, dry, antipyretic,

		aphrodisiac useful in diseases of the heart, burning sensation, cold, bronchitis, vaginal discharges and small pox.

8. SILVICULTURAL OF REDSANDER

Species		:	***Pterocarpus santalinus***
Family		:	Leguminosaceae
Common Name		:	Red sander, Red sandalwood
Locality Factors	Altitude	:	150 – 900 m
	Mean Annual Rainfall	:	350 -1350 mm
	Mean Annual Temperature	:	12 – 47°C
	Soil type	:	Properly drained red soil Dry rocky soil Quartzite, shale, limestone and lateritic soil
Phenology	Flowering	:	April to June
	Fruit ripening	:	February to March
Silvicultural Characters		:	Strong light demander Resistance to fire Excellent coppice Drought resistance even at juvenile stage Produce root suckers freely
Nursery Techniques	Seed Propagation	:	The seeds are soaked with cow dung slurry for 72 hours or seeds are soaked with water for 72 hours with frequent change of water for every 12 hours. Seed rate per bed: 1 kg Germination Percent: 60-70%
	Vegetative Propagation	:	Stump Planting One year old seedlings are preferred for making stumps. The stumps should contain 25-30 cm of roots and 10 – 15 cm of shoot. Survival percentage is 87 %
Silvicultural Treatment	Spacing	:	3 m X 3 m to 6 m X 6 m
	Pit Size	:	30cm x 30cm x 30cm
	Basal Application	:	250g of Vermi-compost or 2kg of Farmyard manure per pit with 50-100 g DAP
	Planting Time	:	September to December Rainy season
	Irrigation	:	Frequent irrigation is essential for initial three years of planting
	Fertilizer Application	:	Farmyard manure @ 4 kg / plant/ year DAP @ 100 g / Plant
	Weeding	:	3-4 weeding per year based on weed pressure.

	Pruning	:	Pruning is essential to obtain straight pole.
	Thinning	:	Thinning is practiced in every 5 year cycle
Rate of growth		:	Slow growing tree The maximum height and girth were 12 m and 66 cm respectively Wavy grained tree is preferable.
Rotation		:	25 years and above
Utilization		:	Wood is fine red colour and beautifully streaked. Wood weight is 900 – 1265 kg / cum with very strong and extremely hard. Timber: Excellent timber with little shrinkage. Musical instruments: wood is used as manufacturing of special musical instrument "Shamisen" The dye santalin extracted from wood is used as medicinal values.

9. SILVICULTURE OF KADAM

Species		:	***Neolamarckia cadamba***
Family		:	*Rubiaceae*
Common Name		:	Kadamba, Japon, Kalempayam, Vellaikkatambu
Locality Factors	Altitude	:	300-800 m
	Mean Annual Rainfall	:	300 - 1600 m
	Mean Annual Temperature	:	5 - 32 ^{o}C
	Soil type	:	Prefers well drained entisols
Phenology	Flowering	:	May to June
	Fruit ripening	:	September to February
Silvicultural Characters		:	light demander Tree coppices well
Nursery Techniques	Seed Propagation	:	The young seedlings are highly susceptible to weeds and should be weeded regularly. 2-month seedlings can be transplanted in nursery beds or into polythene bags, where they can be retained before planting at the start of the monsoon rains
	Vegetative Propagation	:	Clonal technology – mini clonal Establishment of clonal mother garden management of mother garden with irrigation and fertilizer Induction of micro suits Cutting of 5 to 10 cm Planting in root trainer filled with coir compost with the treatment of IBA 1000 ppm Root initiation within 21 days

			Hardening 45 to 60 days 90 days old plants are ready for planting
Silvicultural Treatment	Spacing	:	3 m x 3 m
	Pit Size	:	30 cm x 30 cm x 30 cm
	Basal Application	:	250g of Vermi-compost or 2kg of Farmyard manure per pit with 50-100 g DAP
	Planting Time	:	June to October
	Irrigation	:	4 to 8 litters per day
	Fertilizer Application	:	2kg of FYM and 100g – 500g all 19 every six month once
	Weeding	:	Two weedings per annum
	Pruning	:	Once in two months
	Thinning	:	Only dead and diseased
Rate of growth		:	Fast growing short rotation trees. It yield an average of 125 - 150 tonnes / ha in 3 years.
Rotation		:	3-4 Years for Pulpwood 6-8 Years for Ply wood
Utilization		:	The wood has a density of 290-560 kg/cu m at 15% moisture content, a fine to medium texture; straight grain; low luster and has no characteristic odour or taste. It is easy to work with hand and machine tools, cuts cleanly, gives a very good surface and is easy to nail The timber is used for plywood, light construction, pulp and paper, boxes and crates, dug-out canoes, and furniture components. Kadam yields a pulp of satisfactory brightness and performance as a hand sheet. A yellow dye can be obtained from the root bark Kadam flowers are an important raw material in the production of 'attar', which are Indian perfumes with sandalwood (*Santalum spp*.) base in which one of the essences is absorbed through hydro-distillation The dried bark is used to relieve fever and as a tonic. An extract of the leaves serves as a mouth gargle

MASS MULTIPLICATION OF BAMBOOS USING ENTIRE CULM

Bamboos are the versatile trees, which flowers only once in its life cycle (40-60 years) and dies what is popularly known as parthenogenesis. Hence, seed availability is very less at the same time the seeds are less viable. This difficulty promoted bamboo propagation through two nodal culm cutting with rooting hormone treatment. This conventional technique accounts only less than 25 percent success rate. The present technology is developed using entire culm without rooting hormone treatment and achieved 90 percent rooting.

Remove one year old culm from the matured mother clump at 5-10 years growth stage. Care should be taken to remove the culm without damaging the culm as well as mother clump. The removed culm should be delimbed carefully by leaving growing buds in the nodes. Then, the culm should be placed it in the raised nursery bed and covered with loose soil and sand mixture for half inch thickness. After providing adequate shade to the culms in the nursery bed with coconut sheaths or rice straw, watering should be done to field capacity. Watering twice a day should be continued and shoot emergence will be observed after one month from all buds present in all nodes of the entire culm. Continue watering up to three months. The root emergence could be observed in 2-3 months. After rooting, the rooted culm should be removed entirely from the soil without any damage. To facilitate uprooting the rooted culms without damage, watering should be done. Each rooted node with shoots should be separated with small hand saw. The separated cutting can be transferred to poly bag.

MINI CLONAL PROPAGATION FOR TREE CROPS

A mini clonal technology has been developed for casuarinas and Melia which is one of the pioneering attempts in the country for these industrial wood species. Under this technology, the superior clonal plants were planted in a mini clonal garden and are provided with regular irrigation and fertilization in order to enhance shoot multiplication. In this method, the mother plants are planted at 10 cm x 10 cm spacing and 60 days old plants are ready for collection of cuttings.

Clonal Garden establishment

The clonal garden can be established at a size of 10 x 1 x 0.6 m or 5 x 1 x 0.6 m or 3 x 1x 0.6m using cement trough or GI trough. The bed should be filled with 20 mm stones upto 25 cm and over which finely sewed river sand can be filled. The trough should have facility for drainage. The raised beds can be covered with 100 micron UV stabilized polyethylene film on the top and covered with insect proof mesh to protect the plants and to ensure its freeness from pest and diseases.

Fertigation

The clonal garden should be maintained with irrigation at an interval of every one hour and supplemented with the following nutrients.

- Urea – 300-400 g/m^2
- SSP – 150-175 g/m^2
- KCL – 175-250 g/m^2
- Micro nutrient mixture – 100 g/m^2

The nutrients can be applied twice or thrice depending on the rate of growth of plants.

Clonal Management

The plants are allowed to grow upto 60 days by applying the required nutrient composition. After 60 days the plants are pruned at required size preferably at half of the plant to induce new shoots. With continuous irrigation and nutrient management the cuttings will start producing shoots from 8-10 days onwards and after 15-20 days the cuttings can be collected and treated with 2% carbendazim solution.

Clonal Treatment

The newly induced shoots were separated from the plants and are treated with or without 1500 ppm IBA (liquid formulation) and planted in 90 cc root trainers filled with decomposed coir pith. The rooting started in 15 days and 25 days old rooted plants are ready for hardening.

Green House Conditions

The root trainers are kept under green house conditions with a temperature regime of 32 – 35°C and a relative humidity of 85-95%. Periodical watering once in every 30 minutes is preferred.

Acclimatization and Hardening

The rooted plants are hardened in a shade house condition with 50% shading for 7-15 days and maintained with adequate irrigations. After hardening chamber, the plants are lifted to open nursery for 30 days. Watering is done 2 times a day and the fertilizer of all 19 (N:P:K) can be applied at the rate of 5g/plant. During this hardening, application of carbendazim (2g/l) or triazophos (2ml/lr) is recommended based on the incidence of diseases and pests.

VALUE ADDITION OF PLANTATION RESIDUES THROUGH BRIQUETTING TECHNLOGY

Briquetting is the process of converting low bulk density biomass into high density and energy concentrated fuel briquettes besides compacting the loose biomass into dense block.

Raw materials for Briquetting

Almost all agro and forest residues can be briquetted. Agro and forest residues include saw dust, rice husk, groundnut shell, cotton stalks, wood chips etc. Forest residues such as plantation residues, mill residues (ply wood and match wood residues) can be used for making briquettes. All these residues can be briquetted individually and in combination with or without using binders. The factors that mainly influence the selection of raw materials are moisture content, ash content, flow characteristics, particle size and availability of raw materials. Moisture content in the range of 10-15% is preferred because high moisture content will pose problems in binding and more energy for drying. The ash content of biomass affects its slagging behaviour, operating conditions and mineral composition of ash. Biomass feedstock having upto 4% of ash content is preferred for briquetting. Granular homogeneous materials which can flow easily in conveyers are suitable for briquetting.

Briquetting Process

The series of steps involved in the briquetting process are as follows:

i. Collection of raw material

In general, any material that will burn but is not in a convenient shape, size or form to be readily usable as fuel is a good candidate for briquetting.

ii. Preparation of raw materials

Preparation of raw materials includes size reduction, drying, mixing of raw materials in correct proportion, mixing of raw materials with binders etc.

iii. Size reduction

Raw material is first reduced in size by chopping, crushing, breaking, rolling, hammering, milling, grinding, cutting etc. until it reaches a suitably small and uniform sized material (1 to 10). For some material which are available in the size range of 1 to 10 mm need not be sized reduced. Since the size reduction process consumes a good deal of energy, this should be as short as possible. Biomass with irregular size which was difficult to handle. This size of biomass reduced with the help of Shredder machine. Biomass passed through the shredding machine for size reduction and powder of uniform size is made. Shredder size 22” and sharp rotating blades rotated at a speed of 3300 RPM is used. Generally sieve size is 1/16 inch.

Drying

Raw materials are available with high moisture content than what is required for briquetting. Drying can be dome in open air (sun), with a heater or with hot air. At the time of harvesting, the biomass contains more than 40 percentage of moisture content. For briquetting we need the moisture content of biomass to be in the range of 10-12 percent. Moisture reduction is done using a solar dryer.

Raw material mixing

It is done to make briquettes from more than one raw material. Mixing has to be done in proper way so that the product should have good compaction and high calorific value.

Mixing of raw material with binders

Mixing of raw materials with binders in correct proportion is important for the production of briquettes with good compaction. This is best accomplished by a trial and error method of making several briquettes with different mixtures of binding material and testing each for its mechanical strength and burning characteristics. The cost of binding material can be a critical factor for economic success of the project. Natural or synthetic resins, tar, animal manure , molasses, lingo-sulphates, sewage mud, fish water, algae, starch, slime, clay, mud, and cement are some of the binders used in briquetting process.

Compaction

Compaction process takes place inside the briquetting machine and the process depends on the briquetting technology adopted.

Cooling and storage of briquettes

Briquettes extruding out of the machine are hot with temperatures exceeding 200°c and hence they have to be cooled and stored.

Parameters considered for design of the briquette machine:

Bio waste used: saw dust / ground nut / coffee husk / agricultural / forest waste

Max Moisture Level in Bio waste: 8 - 10 %

Output Size	40 mm	50 mm	60 mm	70 mm	90 mm
Capacity kg/hr	200/250	250/350	650/750	900/1100	1500/1750
Length	50 - 300	50 - 300	50 - 300	50 - 300	50 - 300
Power Required	25 HP	30 HP	50 HP	70 HP	90 HP

MULTIFUNCTIONAL AGROFORESTRY

Agroforestry is one of the oldest land use practices which combines the components of agriculture and forestry production within the same unit area of the land. This agroforestry combines the production of diverse but essential resources for local subsistence. The traditional agroforestry has not seen as an intensive or highly optimised production concept. However, its importance and strength is located within its diverse usability in long term production system and comparatively sustainable impacts in environment and ecosystem functions. The traditional agroforestry system has been utilized as a land use system to increase the livelihood security and reduce the vulnerability to climate and environmental change. The agroforestry systems have provided food, fuel, and fodder besides protecting the natural resources. But today, the rapid phases of population growth coupled with developmental projects have necessitated large volume of wood and wood products from tree outside forests particularly from agroforestry plantations. The industrialization and globalization has made production process more intensified and highly specialized. However, the agroforestry practices are unable to keep up with the higher economic and mono-culture oriented production.

In India, the agroforestry supports nearly 72% of fuel wood demand, over 70% of plywood, 60-80% of pulpwood and around 11% of fodder needs besides satisfying the domestic needs of the society. In case of timber demand the agroforestry is able to meet 2/3rd of the requirement. The role of agroforestry in soil water conservation, Biodiversity conservation and mitigation and adaptation to climate change are very well established. Hence, agroforestry has played a significant role in extending the multifunctional benefits to the society and to the environment. Against this back drop Forest College and Research Institute has conceived and developed a Multifunctional agroforestry model for adoption to double the farmer’s income through enhancing productivity per unit area.

MULTIFUNCTIONAL AGROFORESRY SYSTEM – A PILOT MODEL

The Forest College and Research Institute of Tamil Nadu Agricultural University has designed and established a multifunctional agroforestry model on a pilot basis in 0.75 acre of land. This multifunctional model has been conceptualized with an idea of ensuring monthly income to the farmers practicing the model. This model is conceived based on the fact that the government officials are getting salary on 30th of every month and it is based on this idea the model is conceived and implemented that every month the farmer may also get income from one or other components of multifunctional agroforestry systems. Accordingly the model has been established and it consists of the following components.

Table 1. Components of Multi-functional Agroforestry Model

Tree Components	
High value trees	*Santalum album, Pterocarpus santalinus, Aquilaria agallocha and Dalbergia latifolia*
Timber species	*Tectona grandis, Terminalia tomentosa, Pterocarpus marsupium*
Plywood trees	*Melia dubia, Neolamarkia cadamba, Swietenia macrophylla, Melia volkensii*

TBO's	*Jatropha curcus, Pongamia pinnata, Madhuca longifolia, Callophyllum inophyllum.*
Medicinal Trees	*Terminalia arjuna, Annona reticulata, Strychnos nuxvomica, Aegle marmelos*

Horticultural components	
Fruit crops	Guava, Custard apple, Amla, Jamun, lemon
Leaf crops	Moringa, Curry leaf
Flower crops	Jasmine, Marie gold
Medicinal plants	*Senna auriculata, Hemidesmus indicus*
Agricultural components	
Agricultural Crops	Greens, Pulses, Vegetable, Oilseeds
Pastoral components	
Grasses	CO-3 and CO-4

Intercrops	**No of plants**	**per plant yield (kg)**	**total yield (kg/ year**	**Cost Rs./kg**	**total cost Rs.**
Jasminum grandiflorum	100	2	200	200	40000
Curry leaf	350		80	20	1600
Jasminum officinale	72	0.8	57.6	200	11520
Nerium oleander	28	0.35	9.8	50	490
Fodder	300	8	2800	2	5600
Moringa oleifera	16	8	128	50	6400
Total Income					**65610**

Beneficiary	:	Family Farming (4 Member Family)
Cost of establishment	:	1 lakh / acre including micro irrigation facility
Revenue	:	Atleast Rs. 550 – 1000 / day

All these components have been incorporated in a circular model and the same .After three to six months of development, the annual crop components start generating income and after one year, the entire model has become functionally active to generate income on monthly basis to the growers. The functionality and its economic impact are monitored from the inception of the model and will bring good database for promoting multifunctional agroforestry system in a long term approach.

16.INTEGRATED FARMING SYSTEMS RESEARCH IN TAMIL NADU

The marginal and small farmers constitute 78.2 per cent of the farming community in India. The unique Indian situation of small fragmented holdings and lack of capital investments is not suitable for single commodity farming being practiced in developed countries. So, the integrated farming system appears to be a viable solution to the Indian agriculture for increasing productivity and income of the small and marginal farmers with constrained resources. Efforts for a holistic integration of different farming enterprises with cropping were carried out for Western, Cauvery delta, Southern and North western zones out of seven agro climatic zones of Tamil Nadu with the objectives of increasing income and recycling of farm wastes and by-products to sustain the soil productivity since 1985. The approaches were to find out viable components for wetland, irrigated upland and rainfed situations existing in different ecological zones.

INTEGRATED FARMING SYSTEMS

1. WESTERN ZONE

WETLAND

Integrated farming systems experiments were conducted at wetlands from 1987-onwards at Tamil Nadu Agricultural University, Coimbatore involving different components *viz*., poultry, pigeon, goat, fishery and mushroom.

Crop + poultry/ pigeon + fish + mushroom

In this system, the component of integrated farming system involved were crop + fish + mushroom, crop + poultry + fish + mushroom and crop + pigeon + fish + mushroom and was taken from 1993–1995. The efficiency of the component linkages was evaluated predominantly on the basis of productivity, its income and employment generation with the possibility of utilizing recycled organic wastes as nutrient to enrich the soil fertility.

To enhance and sustain the productivity, economic returns, employment generation for the family labour round the year and soil fertility with environmental protection, integration of rice-gingelly- maize and rice- soybean -sunflower cropping each in 0.45 ha with recycled poultry manure as fish pond silt to rice and 75 per cent of the recommended NPK to each crop in the system + poultry (50 layers) + fish (1000 polyculture fingerlings in 0.10 ha of ponded water) comprising catla (20 per cent), silver carp (20 per cent), rohu (20 per cent), mrigal (15 per cent), common carp (15 per cent) and grass carp (10 per cent) fed with poultry dropping + oyster mushroom (5kg/day) for the lowland farmers having one hectare farm.

Cropping + poultry / pigeon / goat + fishery

During 1998-2001, the study involved cropping, poultry, pigeon, goat and fishery enterprises in all possible combinations, with a view to recycle the residue and by-products of one component over the other. In one hectare farm, an area of 0.75 ha was assigned for crop activity, 0.10 ha for growing fodder grass to feed the goat unit (20+1), 0.03 ha allotted to goat shed and the remaining 0.12 ha allotted to 3 fish ponds. Three integrated farming systems *viz.,* crop + fish + poultry (20 Bapkok layer birds), crop + fish +

pigeon (40 pairs) and crop + fish + goat (Tellicherry breed of 20 female and 1 male maintained in 0.03 ha deep litter system) were tried for three years. Polyculture fingerlings of 400 numbers (catla, rohu, mirgal/ common carp and grass carp) in the ratio of 40:20:30:10, respectively, reared in 3 ponds of size 0.04 ha (depth of 1.5 m) each.

Integration of crop with fish, poultry, pigeon and goat resulted in higher productivity than cropping alone under lowland. Crop + fish + goat integration recorded higher rice grain equivalent yield of 39610 kg/ha. The highest net return of Rs.131118 and per day return of Rs.511 ha^{-1} were obtained by integrating goat + fish + cropping applied with recycled fishpond silt enriched with goat droppings. Higher net return of Rs.3.36 for every rupee invested was obtained by integration of pigeon + fish + cropping applied with recycled fishpond silt enriched with pigeon droppings. The poultry, pigeon and goat droppings were utilized as feed initially and at the end of a year after the fish harvest, about 4500 kg of settled silt from each pond were collected. The pond silt was utilized as organic sources to supply sufficient quantity of nutrients to the crops.

IRRIGATED UPLAND

Crop + Dairy + Biogas + Mushroom + Fish

Integration of 0.32 ha each of sorghum + red gram - sunflower + coriander, maize + fodder cowpea - cotton + coriander and perennial fodder CO3 grass + legume fodder (Lucerne) with dairy (6 cows + 4 calves), biogas (2 m^3 capacity) and mushroom (2 kg day^{-1}) + spawn production (10 bottles day^{-1}) recorded higher productivity than the cropping alone with sorghum - cotton (0.50 ha) and maize - cotton (0.50 ha) cropping systems. Cropping + dairy + biogas + fish + mushroom integration recorded the highest gross, net and per day returns. It also registered the highest benefit cost ratio of 2.41 during 2000-2001.

On farm study was conducted during 2004-2007. The crop activity in integrated farming system consisted of field crop, vegetable crop and fodder crops. The livestock kept were two cross bred milch cows + one calf, ten female tellicherry does + one buck and twenty guinea fowls. Improved farming system gave the maximum maize grain equivalent yield of 22,754 kg/ac/year which was 47.9 % higher than the traditional farming system. The improved system was able to generate employment of 235 man day's $acre^{-1}$ which was higher than traditional farming system (105 man day's $acre^{-1}$). Through recycling of crop residues and livestock manure about 3.72 tonnes of bio-compost and 1.59 tonnes of vermicompost were obtained. This could able to supply 26.0, 22.3, and 26.0 kg N P K to field and fodder crops through biocompost application and 39.4,10.5 and 18.0 kg NPK to vegetable crops as vermicompost application in an acre land area. The returns per rupee of investment from the ratio of gross value of output to total cost (GVCR) was 3.62 and ratio of net value of the products to total cost (NVCR) was 2.80.

RAINFED LAND

Crop + Tree + Goat

Integrated farming system model involving crop + tree + goat was taken from 1999-2001. Experimental results on integrated farming system revealed that (i) integration of sorghum + cowpea (grain), sorghum + cowpea (fodder) and *C. glaucus* each in 0.33 ha intercropped in *E. officinalis* with Tellichery goat component (5+1) in 0.01 ha resulted in higher productivity, economic returns and provided better employment opportunity and improved soil fertility than raising sole sorghum alone (ii) coir pith mulching and pitcher irrigation increased the tree seedling growth than the control, (iii) tied ridges conserved more moisture and improved the productivity of the crops, (iv) application of 50 per cent N through fertilizer and 50 per cent N through goat manure increased the productivity, enhanced the soil fertility and provided better opportunity for recycling of manure to the crops. Results of on-farm field experiments conducted during 2009 - 2011, revealed that, integration of *Cenchrus setigerus* + *Stylosanthes hamata* and fodder sorghum + *Pillipesara* with sheep (5+1) and buffalo (2 No.'s) could be the best silvipastoral farming system with the application of recommended dose of 25: 45: 19 kg ha^{-1} NPK for *Cenchrus* based system and 30: 20: 10 kg ha^{-1} NPK for fodder sorghum based system along with FYM (10 t ha^{-1}) for dry land of Western Zone of Tamil Nadu.

2. CAUVERY DELTA ZONE

Cropping + fish + poultry/duck/goat/dairy

An experiment was conducted during 1992-1994 in rice based farming system as a demonstration trial at Aduthurai. The components were cropping, fish culture and poultry. An area of 0.40 ha was selected for the farming system study, considering the small and marginal farmers of the state. Conventional cropping as practised by farmers was taken up in an area of 0.96 hectare. In the fish pond with 400 m^2, fingerlings belonging to the species *viz.,* Catla (*Catla catla*) (200) Rohu (*Labeo rohita*) (100), Mirgal (*Cirrbinus mrigala*) (100) were stocked.

The economics worked out for the system as a whole was Rs.28.983, in which cropping system contributed Rs.23,709, Poultry and Fisheries contributed additional income of Rs.5,274. Poultry droppings added to the fish pond as feed was 3 tonnes $year^{-1}$ (100g/birds). Mean number of egg production was 262 $year^{-1}$ $bird^{-1}$. In the case of fish pond (0.04) yield recorded was 234 kg. Of the income obtained from the integrated farming system, 78% was from cropping system and poultry cum fisheries generated additional income and employment.

Cropping + **duck + fish culture**

Integrated farming system with duck-cum-fish culture as a component was tried during the year 1989. Two farm holdings each with the size of one ha were selected for conducting the study. In one holding, conventional cropping as practiced by farmers was followed. In another one hectare, cropping was practiced in an area of 0.973 ha and an area of 0.027 ha was allotted for duck-cum-fish culture. Economics of IFS was compared with existing cropping systems. Net income of Rs.13790/- was obtained from existing cropping system (*Kuruvai-thaladi* rice - pulse) and a net income of Rs.22676/- was obtained from the modified cropping system (*Kuruvai - thaladi* rice - cotton and maize) with an area of 0.973 ha allotted for cropping.

The additional profit from modified cropping alone was Rs.8886/-. From duck-cum-fish culture as a component in mixed farming system, a net profit of Rs.1441/- was obtained from an area of 0.027 ha. Totally an additional income of Rs.10327/- was obtained from the mixed farming system over existing cropping system.

3. SOUTHERN ZONE

WETLANDS OF TAMBIRAPARANI COMMAND AREA

Evaluation of integrated farming system for wetland farms was conducted during 1990-92. The components of the traditional and integrated farming systems are as follows:

Traditional system (1 ha)	:	Rice – Rice – Fallow Integrated farming system (1
ha)	:	Crop + dairy + fishery
0.4 ha	:	Rice-rice + soybean (bund) – blackgram
0.2 ha	:	Ragi + sunflower (border) – fodder maize + cowpea – cotton + Greengram
0.2 ha	:	Bajra napier hybrid fodder grass
0.1 ha	:	Desmanthus
0.04 ha	:	Fish pond
0.06 ha	:	Cattle shed for 3 jersey cows and 2 work bullocks

The integrated farming system provided a net income of Rs.25,215/- which was 100.7 per cent more than the income from the traditional rice farming (Rs.12,662/-

On-farm studies were also undertaken to assess the economic benefits of integrated farming systems actually practised by wetland farmers of Tirunelveli-Kattabomman district during 1990. The study covered four farms in four villages with a farm size of 1-2 ha, raising rice, banana and pulses. The allied activities includes dairy, goat rearing, poultry and fishery. The average monthly income of the farm family practicing the integrated farming system varied from Rs.1,850 to 2,560.

WETLANDS OF PERIYAR VAIGAI COMMAND AREA

In the Periyar – Vaigai command area, nearly one lakh hectares are raised with a single crop of rice during August – January. The lands are usually left fallow after rice harvest in January. To assess the potential of IFS in such single crop wetlands, experiments were conducted at Agricultural College and Research Institute, Madurai during 1989-91. The results revealed that by crop intensification, diversified cropping and by inclusion of fishery and poultry, the farm income per acre could be increased by Rs.5435 to Rs.6235 per year.

DRY LANDS OF SOUTHERN ZONE

To identify suitable integrated farming for the dry lands, experiments were conducted at Regional Research Station, Aruppukottai and Agricultural Research Station, Kovilpatti. In the rainfed black soils, the common crops are sorghum, pulses, cotton and sunflower. Introduction of tree legumes like subabul, *Acacia senegal* and *Prosopis cineraria* and perennial fodder grass *Cenchrus ciliaris* and inclusion of goat rearing were evaluated at Regional Research Station, Aruppukottai. Five female and one male goat of Tellicherry breed were raised in deep litter system. The results revealed that the IFS yielded an additional income of Rs.2163 to Rs.2556 per year from a farm area of 1.6 ha. In another study at Aruppukottai proved the IFS system of crop + horticulture + goat proven to be successful in the black soils and increased the profit by Rs.2363 to Rs.4706 per ha over cropping alone.

At Agricultural Research Station, Kovilpatti, studies were taken in farmers' holdings in the dryland red soils. IFS with crop+goat yielded an annual income of Rs.8410 per ha compared to Rs.4654 per ha under traditional cropping alone.

4. NORTH WESTERN ZONE

The studies were made under garden land condition. The results revealed that in both Paiyur and Yercaud Centres dairy linked farming system was more remunerative, with more employment generation. The next successful farming system under rainfed condition at Paiyur was sericulture.

POTENTIAL ALTERNATIVES

Western zone

Wetlands Crop + poultry/ pigeon + fish + mushroom Crop + poultry/ pigeon + goat + fishery

Upland with supplemental irrigation

Crop + dairy + biogas + silviculture

Crop + dairy + biogas + mushroom + fish

Rainfed lands Crop + goat, Crop + goat + tree

Cauvery Delta zone

Crop + poultry + fish Crop + duck + fish Crop + milch animals Crop + goat + dairy

Southern zone

Wetlands of Tambirabarani Command: Crop + dairy + fishery Wetlands of Periyar Vaigai Command: Crop + fish + poultry Dry lands: Crop + orchard + goat

North Western zone

- Crop + dairy + poultry
- Crop + dairy + poultry + sericulture

Adoption of improved farming system models can result in the advantages listed below.

- Higher food production to equate the demand of the exploding population of our nation
- Increased farm income through proper residue recycling and allied components
- Sustainable soil fertility and productivity through organic waste recycling
- Integration of allied activities will result in the availability of nutritious food enriched with protein, carbohydrate, fat, minerals and vitamins
- Integrated farming will help in environmental protection through effective recycling of waste from animal activities like piggery, poultry and pigeon rearing
- Reduced production cost of components through input recycling from the byproducts of allied enterprises
- Regular stable income through the products like egg, milk, mushroom, vegetables, honey and silkworm cocoons from the linked activities in integrated farming
- Inclusion of biogas & agro forestry in integrated farming system will solve the prognosticated energy crisis
- Cultivation of fodder crops as intercropping and as border cropping will result in the availability of adequate nutritious fodder for animal components like milch cow, goat / sheep, pig and rabbit
- Firewood and construction wood requirements could be met from the agroforestry system without affecting the natural forest
- Avoidance of soil loss through erosion by agro-forestry and proper cultivation of each part of land by integrated farming
- Generation of regular employment for the farm family members of small and marginal farmers

17. WEEDS

MANAGEMENT OF PROBLEM, PERENNIAL AND PARASITIC WEEDS

I. *Cynodon dactylon* (*Arugu*) and *Cyperus rotundus* (*Koarai*)

Management of perennial weeds like *Cynodon dactylon* and *Cyperus rotundus* by the application of Glyphosate 15 ml + Ammonium sulphate 20g + activator 1 ml / lit of water

Approach	:	Post emergence, total, translocative herbicide
Stage of weed	:	Active growing, pre flowering stage
Sprayer	:	Hand operated *Knapsack / Backpack*
Nozzle	:	WFN 24 & ULV 50 with 30 Psi
Spray volume	:	200-250 litre / ha

Application technology

Non-Crop Situation/Crop Fallow Situation - Blanket application
Cropped Situation - Pre-sowing / planting- Stale seed bed (Blanket application). Established Crops - Directed application using hoods.

Note: Rain free period / waiting period: 48 hours

II. Solanum elaegnifolium (Kattu kandan kathiri)

Post-emergence application of Glyphosate 20 ml + Ammonium sulphate 20g/ha+ activator 1ml/litre of water or Glyphosate 10 ml in combination with 2,4-D sodium salt 6 g + activator 1ml / litre

Note: The application should be during the active growth / vegetative phase of weed

III. Parthenium hysterophorus (Parthenium natchu chedi)

- Manual removal and destruction of Parthenium plants before flowering using hand glouse / machineries **(or)**
- Pre-emergence application of atrazine 4 g / litre in 500 litres of water / hectare **(or)**
- Uniform spraying of sodium chloride 200g + 2 ml soap oil / litre of water **(or)**
- Spraying of 2,4-D sodium salt 8 g or glyphosate 10 ml + 20g ammonium sulphate + 2 ml soap solution / litre of water before flowering **(or)**
- Post-emergence application of metribuzin 3 g / litre of water under non crop situation.
- Raising competitive plants like *Cassia serecea* and *Abutilon indicum* on fallow lands to replace Parthenium **(or)**
- Biological control by Mexican beetle which is very active during only in monsoon seasons.

Note: Parthenium can be decomposed well before flowering and used as organic manure.

IV. Ipomoea carnea (Neyveli kattamanakku)

- Foliar application of 2,4-D sodium salt 8 g + urea 20g + soap oil 2 ml / litre of water and then removal and burning of dried weeds **(or)**
- Manual / mechanical removal of grownup plants in channels during summer.

Note: Composted *Ipomoea carnea* can be used as organic manure preferably in rice fields.

V. Eichhornia crassipes (Agaya thamarai)

- Manual / Mechanical removal and drying
- Application of 2,4-D sodium salt 8g + urea 20g or Paraquat 6 ml / litre of water
- Application of Glyphosate 15ml+Ammonium sulphate 20g/ litre of water

Note: Vermi-composting and composting of dried water hyacinth and can be used as organic manure in irrigated upland ecosystems.

VI. Portulaca quadrifida (Shiru pasari)

- Post-emergence tank mix directed application of Glyphosate 10 ml / ha + 2, 4-D sodium salt 5g / lit to control *Portulaca quadrifida* in cropped fields.

Note: Not to use above herbicides in broadleaved crops particularly cotton and bhendi or other vegetables and pulses as well as oilseed crops.

VII. Striga asiatica (Sudu malli)

- Pre-emergence application of atrazine 1.0 kg/ha on 3rd DAP + hand weeding on 45 DAP with an earthing up on 60 DAP combined with post-emergence spraying of 2,4-D 6 g (0.6%) + urea 20 g (2%) / litre of water on 90 DAP + trash mulching 5 t/ha on 120 DAP

VIII. Orabanche (Pukaielai kalan)

- Pre-emergence application of pendimethalin 1.0 kg/ha or oxyfluorfen 0.30 kg/ha on 3 DATP in tobacco, tomato and brinjal and 3 DAP in potato.
- Plant hole application of neem cake 25 g / plant or drenching of copper sulphate 5% provides partial control of *Orabanche* in tobacco.
- Directed application of paraquat 6 ml/litre of water or glyphosate 8 ml/ litre of water or imazethapyr 3 ml/ litre of water on the Orbanche shoots

18. SOIL RELATED CONSTRAINTS AND THEIR MANAGEMENT

A constraint free soil environment is very important for achieving higher food production. The major soil constraints affecting the crop production in Tamil Nadu are

a) Chemical constraints : salinity, sodicity, acidity and nutrient toxicities
b) Physical constraints : high or low permeability, sub soil hard pan, surface crusting, fluffy paddy soils, sandy soils etc.

1. Saline soils

Saline soils are characterised by higher amount of water soluble salt, due to which the crop growth is affected. For these soils with electrical conductivity of more than 4 dS m^{-1}, provision of lateral and main drainage channels of 60 cm deep and 45 cm wide and leaching of salts could reclaim the soils. Application of farm yard manure at 5 t ha^{-1} at 10 - 15 days before transplanting in the case of paddy crop and before sowing in the case of garden land crops can alleviate the problems of salinity.

2. Sodic soils

Sodic soils are characterised by the predominance of sodium in the complex with the exchangeable sodium percentage exceeding 15 and the pH more than 8.5 .To reclaim the sodic soils, plough the soil at optimum soil moisture regime, apply gypsum at 50% gypsum requirement uniformly, impound water, provide drainage for leaching out the soluble salts and apply green manure at 5 t ha^{-1} 10 to 15 days before transplanting in the case of paddy crop.

3. Acid soils

Acid soils are characteristically low in pH (< 6.0). Predominance of H^+ and Al^{3+} cause acidity resulting in deficiency of P, K, Ca, Mg , Mo and B. This soils are prevalent in a) hilly tracts of Ooty, Kodaikkanal and Yercaud b) Laterite soils of Pudukkottai, Kanyakumari etc Application of lime **(**as per the lime requirement test**)** uniformly by broadcast and incorporation is recommended. The alternate amendments like dolomite, basic slag, flue dust, wood ash, pulp mill lime may also be used on lime equivalent basis**.**

4. Iron and Aluminum toxicity

These are characterized by the presence of higher concentration of Fe^{2+} and Al^{3+} more specifically in flooded soils. Prevalent in Kanyakumari and Pudukkottai Districts. Application of lime as per the lime requirement along with the recommended dose of NPK and organic manure will suppress the toxicity.

For ` Ela ` soils of Kanyakumari district (Alfisols, pH : 4-5) ,. application of lime as per lime requirement with recommended NPK + foliar spray of 0.5 % $ZnSO_4$ + 0.2% $CuSO_4$ +1% DAP + 1% MOP thrice during AT to PI will help to overcome the problem in rice. Based on the screening tests, the rice cultivars of the region have been rated for their tolerance to Fe toxicity

Highly susceptible : ADT 36
Mod. susceptible : ADT 42, IR 50, CORH 1
Less susceptible : TPS 1, ASD 16 & 18, IR 64, JJ 92, TKM 9, CO 37 & CO 41

5. Fluffy paddy soils

The traditional method of preparing the soil for transplanting rice consists of puddling which results in substantial break down of aggregates with uniform structures less mass. Under continuous flooding and submergence of soil in a rice-rice-rice cropping system, the soil particles are always in a state of flux and the mechanical strength is lost leading to the fluffy ness. This is further aggrevated by *insitu* application of rice stubbles and weeds during puddling. They are characterized by low bulk density of the top soil resulting in the sinking of farm animals and labourers as well as poor anchorage to paddy seedlings. For such soils, passing of 400 kg stone roller or oil drum with sand inside eight times at proper moisture level (moisture level at friable condition of soil which is approximately 13 per cent) once in two to three years.

6. Sandy soils

Sandy soils are containing predominant amounts of sand resulting in higher percolation rates and nutrient losses. Campacting the soil with 400 kg stone roller or oil drum with sand / stones inside eight times at proper moisture level (moisture level at friable condition of soil which is approximately 13 per cent) once in two to three years could reduce the percolation losses. Addition of tank silt for coastal sandy soils is recommended for enhancing their productivity.

7. Sub Soil hard pan

Hard pan occurs in red soil areas due to the movement of clay and iron hydroxides and calcium carbonate and settling at shallow depth, which increases the soil bulk density to more than 1.8 g/cc, their by preventing the root proliferation. These soils can be reclaimed by chiselling the soils with chisel plough at 0.5 m interval first in one direction and then in the direction perpendicular to the previous one, once in two to three years. Applications of FYM or composted coir pith at 12.5 t ha-1could bring additional yields of about 30 per cent over control. Deep ploughing of the field during summer season can be followed to open up the sub soil. Cultivation of deep rooted crops like tapioca, cotton and semipereneal crop like mulberry encourage natural breaking of the hardpan.

8. Surface crusted soils

Surface crusting occurs due to the presence of colloidal oxides of iron and aluminium in Alfisols which binds the soil particles under wet regimes. On drying it forms a hard mass on the surface and prevents the emerging seedlings and arrest the free exchange of gases between the soil and atmosphere. The surface crust can be easily broken by harrowing or cultivator ploughing and its formation can be prevented by improving the aggregate stability by the application of lime or gypsum at 2 t ha^{-1} and FYM at 12.5 t ha^{-1}. Sprinkle water at periodic intervals. Bold grain crops like cowpea may be grown

9. Heavy textured clay soils

The clay soils are containing major amounts of clay resulting in the poor permeability and nutrient fixation. Such soils can be reclaimed by the addition of river sand at 100 t ha^{-1} or managed by deep ploughing the field with mould board plough or disc plough during summer or forming contour and compartmental bunds and also adoption of ridges and furrows to enhance the infiltration and percolation.

10. Low permeable black soils

These soils are having infiltration rate less than 6 cm per day due to high clay content. The amount of water entering in to the soil is reduced, resulting in high run off encouraging the erosion of surface soil with nutrients. Heavy clay and high capillary porosity results in impeded drainage and reduced soil conditions. Application of 100 cart loads of red loam soil or river sand and deep ploughing the field with mould board plough or disc plough during summer to enhance the infiltration and percolation. Application of FYM , composted coir pith or pressmud at 25 t ha^{-1} per year will improve the physical properties and internal drainage of the soil.

11. High permeable red soils

These soils are having sand exceeding 70 per cent and are not able to retain water and nutrients. These soils are devoid of finer particles and organic matter, thus aggregates are weekly formed; presence of high non capillary pores results in poor soil structure. Compacting the soil with 15 passes after 24 hours of irrigation, application of tank silt or black soil @ 25 t ha^{-1} per year along with FYM, composted coir pith to improve the water holding capacity of the soil. Providing asphalt, polythene sheet etc below the soil surface will reduce infiltration

19. CHISEL TECHNOLOGY

The occurrence of hard pans at shallow depth is the most prevalent soil physical constraint in soils. The agricultural crops are denied of the full benefits of the soil fertility and nutrient use due to this constraint. The sub-soil hard pans are characterized by high bulk density (1.8 g cc.') which in turn lowers infiltration, water storage capacity, available water and movement of air and nutrients, with concomitant adverse effect on the yield of crops. This problem is predominantly present in six districts of Tamil Nadu viz., Coimbatore, Erode, Dharmapuri, Tiruchirappalli, Madurai and Salem particularly under rainfed farming affecting a total of 3.8 lakh hectares of land.

TECHNOLOGY

Plough the field with chisel plough at 50 cm interval in both the directions viz., horizontally and vertically. Chiselling helps to break the hard pan in the sub soil. Besides, it ploughs upto 45cm depth. Chisel plough is a heavy iron plough which goes up to 45 cm depth, thereby shattering the hard pans. It is usually drawn by the tractor. Fabrication of chisel plough has been done by the Department of Farm Machinery, Tamil Nadu Agricultural University, Coimbatore.

- Spread 12.5 t of FYM / pressmud / composted coir pith per hectare evenly on the surface.
- Give two ploughings using a country plough for incorporating the added manures. The broken hard pan and incorporation of manures make the soil to conserve more moisture.

Vegetative barriers for soil moisture conservation

For better in-situ moisture conservation in drylands of Vertisols, raise vegetative barriers of vettiver or lemon grass across the slope and along the contours at 0.5 m vertical interval.

20. SURGE IRRIGATION

Even as advanced pressure irrigation method, such as drip and sprinkler systems are in vogue the traditional gravity surface irrigation methods still remain inevitable due to their simplicity in layouts and low installation and operational expenses. However the short strip furrow and check basin layouts (the primary surface irrigation methods in Tamilnadu) warrant division of the irrigated fields into a number of square or rectangular (2 m x 2 m to 6 m x 6 m) plots encompassed by criss- cross ridges and feeder channels for facilitating irrigation flow from head to tail end of the field. This eventually results in prolonged irrigation application time and reduced irrigation efficiencies of 55 - 65% only due to excessive seepage, deep percolation and runoff losses (35-45%). Besides, the criss- cross layout with cross ridges and feeder channels leads to land loss of 15 -25%. In view of minimizing the land and water loss and to accomplish high level of irrigation and water use efficiencies a relatively new surface irrigation method called "surge irrigation" was introduced in TNAU with extensive experimental trials on it's hydraulic performance evaluation and crop compatibility during 1992-95.

Features of Surge irrigation

The term " Surge irrigation" refers to the delivering irrigation flows into individual long furrows (more than 25 m upto 200 m) in an intermittent fashion of predetermined ON-OFF time cycles (5 minute to 10 minutes) with the design duration of irrigation. During the ON time water front advances into the furrow over a certain length and during the subsequent OFF time the water applied partially saturates the soil and infiltration rate gets reduced on the advanced length. When water is delivered in the succeeding ON time, the water front advance gets accelerated due to the reduced intake rate and eventually it reaches the tail end of long furrow with in 30 - 50% of the design duration of irrigation. This process of ON-OFF water supply and cutoff results in highly minimized deep percolation and runoff losses (hardly exceeding 20%). Hence, high uniformity of soil moisture distribution with in the effective root zone is achieved over the entire furrow length resulting in enhanced irrigation efficiencies of more than 85% to 95%. In addition due to the series of long furrows emanating from a single head channel, the criss - cross ridges and feeder channel of division are eliminated thereby limiting the land loss within 5% only.

Contributions of TNAU in surge irrigation research

- Manual semi automated and automated surge irrigation layouts were designed and the irrigation parameters such as the individual furrow discharges (30 lit/min to 120 lit/min), surge cycle ON-OFF times (5 min to 30 min), surge cycle ratio (0.25 to 0.66), furrow gradients (0.1% to 0.6%), furrow size (30-120cm) and furrow length (50-200m) could be optimized through mathematical models.
- A significant contribution from TNAU is the development of an original emprical model for the prediction of waterfront advance times and resulting in irrigation water distribution efficiencies.

Soil suitability	:	Sandy clay loam and loamy soils only
		Crops tested maize sunflower and sorghum
Water saving	:	25-40%
Land saving	:	15-25%
Labour saving	:	40%

Limitations

Surge irrigation systems do not show marked differences in land and water saving in extremely clay or sandy soils. Besides, surge irrigation technology is still in the infant stage in India and requires popularization through extension methods.

21. MICRO IRRIGATION

Micro irrigation is a modern method of irrigation; by this method water is irrigated through drippers, sprinklers, foggers and by other emitters on surface or subsurface of the land. Major components of a micro irrigation system is as follows.

Water source, pumping devices (motor and pump), ball valves, fertigation equipments, filters, control valves, PVC joining accessories (Main and sub main) and emitters. In this system water is applied drop by drop nearer the root zone area of the crop. The drippers are fixed based on the spacing of crop. Many different types of emitters are available in the market. They are classified as Inline drippers, on line drippers, Micro tubes, Pressure compensated drippers.

Drip irrigation is most suitable for wider spacing crops. Micro sprinkler irrigation system is mostly followed in sandy or loamy soils. This system is most suitable to horticultural crops and small grasses. In this method water is sprinkled in a lower height at various directions.

Portable micro sprinklers are also available. They distribute slightly more water than drippers and micro sprinklers. They spray water in not more than one meter. It is used for preparing nursery and lawns in soils with low water holding capacity.

Advantages of drip irrigation system

- Water saving and higher yield
- High quality and increased fruit size
- Suitable for all types of soil
- Easy method of fertigation and chemigation
- Saving in labour and field preparation cost

Disadvantage of drip irrigation system

- High initial investment
- Clogging of emitters
- Possible damage of system components due to animals, etc.,

Investment cost mostly differs based on spacing of the crops

- Generally, the reasons for clogging are solid particles (sand, rust), soft dirt (organic matter, algae, micro organism, salt), sediments (salt in the fertilizers).
- Filtration is the main key factor to the success or failure of the system. The aim of filtration is to stop dirt particles which damage any components of the system.

- To remove salt encrustation, 30 per cent commercial hydrochloric acid can been used at the rate of one liter per one m^3 of system discharge.
- To remove algae and fungal clogging 5 to 500 ppm sodium hydrochloride (10 per cent chlorine) can be used.

Maintenance of drip system

- Back washing and sand filters has to be cleaned
- Frequent cleaning of emitters and drippers
- Flushing at every irrigation
- Cleaning of sub main and main pipes
- Cleaning of PVC pipes and laterals and acid or chlorine may be used to remove clogging.

Water used and yield of crops in micro and conventional irrigation methods

Crop	Methods of irrigation	Water require-ment (cm)	% water saving	Yield kg ha^{-1}	% increase in yield	Water use efficiency (kg ha mm^{-1})
Banana	Drip	97.00	45.00	87500	52.00	90.20
	Surface	176.00	-	57500	-	32.67
Sugarcane	Drip	94.00	56.00	170000	33.00	180.85
	Furrow	215.00	-	128000	-	59.53
Grapes	Drip	27.80	48.00	32500	23.00	116.90
	Surface	53.20	-	26400	-	49.62
Aerobic rice	Conventional aerobic rice	74.30	38.10	4747	-	6.39
	Surface drip	61.90	48.40	5940	14.20	9.60
	Sub-surface drip	61.90	48.40	6227	19.80	9.74
	Conventional transplanted	120.00	-	5200	-	4.33
Cotton	Drip	28.00	66.27	3250	25.00	116.10
	Furrow	83.00	-	2600	-	31.33
Beetroot	Drip	17.70	79.34	887	55.34	50.11
	Surface	85.70	-	571	-	6.66
Radish	Drip	10.80	75.72	1186	13.49	109.80
	Surface	46.40	-	1045	-	22.52
Papaya	Drip	73.88	67.89	23490	69.47	0.32
	Surface	225.80	-	13860	-	0.06
Mulberry	Drip	20.00	60.00	71400	3.03	3570
	Surface	50.00	-	69300	-	1386
Tomato	Drip	18.40	39.00	48000	50.00	260.86
	Surface	30.00	-	32000	-	106.66

(WTC Annual Reports 1985-2003)

Affordable micro irrigation systems

Affordable micro irrigation system is mostly suitable to kitchen garden, nursery and ornamental crops.

1. **Bucket kit system**

Bucket kit system is designed for kitchen garden suitable for women, marginal and small farmers. It consists of a bucket (15 lit.) 10 metre long lateral (12mm) fitted with drippers (4 LPH), which can irrigate about 100 plants in approximately 15 m^2 area. The bucket is placed at a height of 1m (3 feet) and water is filled for 4 to 5 times daily.

2. **Drum kit system**

This system is ideally suitable to kitchen garden and small commercial vegetable growers. The drum is having 200 liter capacity which would supply water approximately 500 plants by filling the drum twice daily. It consists of lateral (16mm and 12mm). One number of 16mm lateral and five 12mm laterals are used. This system could cover an area of 120 m^2 (3 cents).

3. **Micro sprinkler system**

Micro sprinkler kit is suitable for farmers with access to pressurised water. It is very useful for groundnut, vegetables, nurseries home gardens, and lawns etc. It can be connected with a tap from an overhead tank or a domestic water pump. It consists of 15 micro sprinklers with pipes irrigating an area of 250 m^2 (6 cents). Fertigation can also be done through this method.

Sl. No.	Item	Selling Cost/Unit	Area covered by the kits
1.	Bucket Kit (Drip system)	Rs.225	20 m^2 (0.5 cent)
2.	Drum Kit (Drip system)	Rs.600 (Excluding Drum Cost)	120 m^2 (3.0 cents)
3.	Micro sprinkler kit	Rs.900	240 m^2 (6.0 cents)

Fertigation

Fertigation is a method of fertilizer application in which fertilizer is incorporated within the irrigation water by the drip system. In this system fertilizer solution is distributed evenly in irrigation. The availability of nutrients is very high therefore the efficiency is more. In this method liquid fertilizer as well as water soluble fertilizers are used. By this method, fertilizer use efficiency is increased from 80 to 90 per cent.

Fertilizer efficiencies of various application methods

Nutrient	Fertilizer use efficiency (%)	
	Soil application	Fertigation
Nitrogen	30-50	95
Phosphorous	20	45
Potassium	50	80

Advantages of fertigation

- Nutrients and water are supplied near the active root zone through fertigation which results in greater absorption by the crops.
- As water and fertilizer are supplied evenly to all the crops through fertigation there is possibility for getting 25-50 per cent higher yield.
- Fertilizer use efficiency through fertigation ranges between 80-90 per cent, which helps to save a minimum of 25 per cent of nutrients.
- By this way, along with less amount of water and saving of fertilizer, time, labour

and energy use is also reduced substantially.

Water saving, yield and profit under drip and drip fertigation systems

Crops	Water Saving (%)	Yield (t/ha)			Profit (Rs/ha)		
		Conventional	Drip	Drip+ Fertgn	Conventional	Drip	Drip + Fertgn
Banana	35	26	30	37	81000	98000	120000
Sugarcane	29	120	160	207	30000	47000	68000
Tomato	32	45	56	65	56000	77000	95000
Aerobic rice	48	4.75	5.58	6.23	47470	55760	62270

Fertilizer used in fertigation

- Urea, potash and highly water soluble fertilizers are available for applying through fertigation.
- Application of super phosphorus through fertigation must be avoided as it makes precipitation of phosphate salts. Thus phosphoric acid is more suitable for fertigation as it is available in liquid form.
- Special fertilisers like mono ammonium phosphate (Nitrogen and Phosphorus), poly feed (Nitrogen, Phosphorus and Potassium), Multi K (Nitrogen and Potassium), Potassium sulphate (Potassium and Sulphur) are highly suitable for fertigation as they are highly soluble in water. Fe, Mn, Zn, Cu, B, Mo are also supplied along with special fertilisers.

Fertilizers commonly used in fertigation

Name	N – P_2O_5 – K_2O content	Solubility (g/l) at 20^0 C
Ammonium nitrate	34-0-0	1830
Ammonium sulphate	21-0-0	760
Urea	46-0-0	1100
Monoammonium phosphate	12-61-0	282
Diammonium phosphate	18-46-0	575
Potassium chloride	0-0-60	347
Potassium nitrate	13-0-44	316
Potassium sulphate	0-0-50	110
Monopotassium phosphate	0-52-34	230
Phosphoric acid	0-52-0	457

Special water soluble fertilizers

Name	N %	P_2O_5 %	K_2O %
Polyfeed	19	19	19
Polyfeed	20	20	20
Polyfeed	11	42	11
Polyfeed	16	8	24
Polyfeed	19	19	19
Polyfeed	15	15	30
MAP	12	61	0

Multi-K	13	0	46
MKP	0	52	34
SOP	0	0	50

N fertigation

Urea is well suited for injection in micro irrigation system. It is highly soluble and dissolves in non-ionic form, so that it does not react with other substances in the water. Also urea does not cause precipitation problems. Urea, ammonium nitrate, ammonium sulphate, calcium ammonium sulphate, calcium ammonium nitrate are used as nitrogenous fertilizers in drip fertigation.

P fertigation

Application of phosphorus to irrigation water may cause precipitation of phosphate salts. Phosphoric acid and mono ammonium phosphate appears to be more suitable for fertigation.

K fertigation

Application of K fertilizer does not cause any precipitation of salts. Potassium nitrate, Potassium chloride, Potassium sulphate and mono potassium phosphate are used in drip fertigation.

Micro nutrients

Fe, Mn, Zn, Cu, B, Mo could be used as micro nutrients in drip fertigation.

Fertigation equipments

Three main groups of equipments used in drip system are :

- Ventury
- Fertilizer tank
- Fertilizer pump

Ventury

Constriction in the main water flow pipe causes a pressure difference (Vaccum) which is sufficient to suck fertilizer solution from an open container into the water flow. It is very easy to handle and it is affordable even by small farmers. This equipment is most suitable for smaller area.

Fertilizer tank

A tank containing fertilizer solution is connected to the irrigation pipe at the supply point. Part of the irrigation water is diverted through the tank diluting the nutrient solution and returning to the main supply pipe. The concentration of fertilizer in the tank thus becomes gradually reduced.

Fertilizer pump

The fertilizer pump is a standard component of the control head. The fertilizer solution is held in non-pressurised tank and it can be injected into the irrigation water at any desired ratio. Therefore the fertilizer availability to each plants is maintained properly.

Cost of fertigation equipments

Sl.No.	Fertigation devices	Cost (Rs.)
1.	Ventury type	1200
2.	Fertilizer Tank	3000
3.	Injectors	12000

Economics of drip irrigation system

The initial investment in drip irrigation system is mainly depends upon the spacing of crops. The initial cost will be almost 75 - 85 thousand rupees per hectare for wider spacing crops such as coconut, mango, grapes and for orchard crops. The initial cost is approximately 1 - 1.25 lakh rupees per hectare for close spacing crops such as sugarcane, banana, papaya, mulberry, turmeric, tapioca, vegetables and flower crops.

Drip fertigation technology for aerobic rice Surface drip fertigation

Under aerobic rice conditions, provision of surface drip fertigation (with 0.8 m lateral spacing provided with drippers at 0.3 m distance) scheduled at 125 % Pan Evaporation value for clay soil / 150 % open Pan Evaporation value for sandy soil + STCR based NPK fertigation + biofertigation of *Azophosmet* @ 500 mL ha^{-1} during panicle initiation and flag leaf stages is recommended.

Sub Surface drip biogation

Under aerobic rice conditions, provision of sub-surface drip fertigation (10 cm depth with 0.8 m lateral spacing provided with drippers at 0.3 m distance) scheduled at 125 % Pan Evaporation value for clay soil / 150 % open Pan Evaporation value for sandy soil + STCR based NPK fertigation + biofertigation of *Azophosmet* and seaweed extract each @ 500 mL ha^{-1} during panicle initiation and flag leaf stages is recommended.

22. AGROMETEOROLOGY

CROP PLANNING AND MANAGEMENT

DRYLAND

1. Length of Growing Period

Length of growing period is defined as a period in which the available soil moisture is enough to meet the evapotranspiration requirement of dry land crops and hence the dry land productivity is assured. Based on scientific study (Jeevananda Reddy, 1983), length of growing period for different rain gauge stations of each district of different agroclimatic zones of Tamil Nadu have been computed. The length of growing period is given as 'G' with starting and ending of length of growing period in terms of Meteorological standard weeks. If the G is less than 5 weeks period it means that always crop failures will occur. The G period must be a minimum of 14 weeks (98 days) which permit the dry land crop to attain its potential productivity. If the growing period is 14 weeks, a single dry land crop can be cultivated. If G period is between 14 to 20 weeks, suitable inter cropping system can be recommended. If the G period is more than 20 weeks long duration crop / double crop can be organized.

The following information indicates length of growing period for different district of Tamil Nadu. Based on the G period, suitable dry land crop may be selected.

1. North Eastern Zone

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Thiruvaalur	Athipettu	34-52	19
	Ponneri	33-52	20
	Poonamallee	32-52	21
	Saidapet	32-52	21
	Tirutani	31-50	20
	Tiruvallur	31-51	21
Kanchipuram	Chengalpattu	30-52	23
	Cheyur	33-52,1	21
	Covelong	31-52	22
	Kanchipuram	29-51	23
	Madurantakam	30-52	23
	Sriperumudur	31-51	21
	Uttiramerur	30-51	22
	Vayalur	34-52	19
Vellore	Ambur	33-46	14
	Arakkonam	29-51	13
	Gudiyattam	33-47	15
	Sholingnur	31-49	19
	Tiruppattur	31-45	15
	Vaniyambadi	32-45	14

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
	Vellore	30-50	21
	Walajapet	30-50	21
Tiruvannamalai	Arani	30-50	21
	Chengam	31-49	19
	Cheyyar	30-50	21
	Polur	30-50	21
	Tiruvannamalai	31-50	20
	Vandavasi	29-51	23
Viluppuram	Gingee	30-51	22
	Tindivanam	31-52	22
	Tirukkovilur	30-50	21
	Ulundurpettai	32-51	20
	Vanur	32-52,1	22
	Viluppuram	31-51	21
Cuddalore	Cuddalore	32-52,1,2	23
	Kurinjippadi	32-52,1	22
	Marakkanam	32-52	21
	Panruti	31-52	22
	Porto Novo	33-52, 1,2	22
	Srimushnam	33-52	20
	Tittagudi	31-51	21
	Vriddhachalam	31-51	21
Perambalur	Chettikulam	35-48	14
	Jayamkonda cholapuram	35-52	18
	Uppiliyapuram	38-48	11
Chennai	Nungambakkam	32-52	21

2. North Western Zone

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Dharmapuri	Denkanikota	32-46	15
	Dharmapuri	32-46	15
	Harur	33-47	15
	Hosur	33-45	13
	Krishnagiri	33-45	13
	Palacode	32-46	15
	Pennagaram	33-45	13
	Rayakottai	33-46	14
	Thalli	31-44	14
	Uttangarai	31-46	16
Salem	Attur	33-48	16
	Omalur	29-45	17

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
	Salem	27-45	19
	Sankari Durg	33-45	13
	Tammampatti	34-49	16
	Valapadi	33-46	14
Namakkal	Namakkal	33-46	14
	Paramathi	35-45	11
	Rasipuram	30-45	16
	Sendamangalam	32-45	14
Perambalur	Ariyalur	35-50	16
	Perambalur	35-50	16

3. Western Zone

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Coimbatore	Annur	38-47	10
	Avanashi	38-47	10
	Coimbatore	41-47	7
	Mettupalayam	39-50	12
	Palladam	41-47	7
	Periyanaiakampalayam	38-49	12
	Pollachi	24-31, 41-47	8, 7
	Sulur	41-46	6
	Tiruppur	38-47	10
	Udumalaipettai	41-48	8
Erode	Bhavani	34-47	14
	Dharapuram	40-47	8
	Erode	34-47	14
	Gopichettipalayam	35-47	13
	Kangayam	38-47	10
	Kodumudi	38-44	7
	Perundurai	35-47	13
	Sathyamangalam	35-47	13

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Karur	Aravakkurichi	39-46	8
	Karur	39-45	7
Dindigul	Nilakottai	36-47	12
	Palani	40-49	10
Theni	Periakulam	38-49	12
	Uttamapalayam	40-48	9
Madurai	Usilampatti	36-49	14

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Tiruchchirapalli	Manaparai	36-48	13

4. Cauvery Delta Zone

District	Station	G period Met. Standard week)	No. of G period (No. of weeks)
Thanjavur	Atirampattinam	34-52	19
	Kattuumvadi	39-51	13
	Kumbakonam	34-52	19
	Papanasam	35-52	18
	Pattukottai	35-52	18
	Thanjavur	35-51	17
	Tirukkatupalli	35-49	15
	Vallam	33-50	18
Thiruvarur	Kudavasal	35-52,1	19
	Mannargudi	34-52,1	20
	Muttupet	35-52,1,2	20
	Nannilam	35-52,1	19
	Neidavasal	35-52,1,2	20
	Nidamangalam	35-52,1	19
	Thiruvaiyaru	35-50	16
	Thiruvarur	35-52,1,2	20
	Tirutturaippundi	35-52,1,2	20
	Valangiman	35-52	18
Nagapattinam	Mayuram	35-52,1	19
	Nagapattinam	37-52,1,2	18
	Sirkazhi	34-52,1,2	21
	Tarangambadi	35-52,1,2	20
	Tiruppundi	36-52,1,2,3	20
	Vedaranniyam	35-52,1,2,3	21
Tiruchchirapalli	Kulattur	36-48	13
	Kulittalai	38-47	10
	Lalgudi	38-49	12
	Manapparai	36-48	13
	Musiri	38-47	10
	Tattayyangarpettai	36-47	12
	Tiruchchirapalli	36-48	13
Perambalur	Turaiyur	36-47	12
Cuddalore	Chidambaram	33-52,1,2	22
	Kattumannarkovil	33-52,1	21
Pudukkottai	Arantangi	34-50	17

5. Southern Zone

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Ramanathapuram	Kamudi	41-47	7
	Mudukulattur	41-49	9
	Pamban	42-52,1	12
	Paramakudi	40-48	9
	Ramanathapuram	41-52	12
	Theethanathanam	41-51	11
	Tiruvadanai	41-50	10
	Vattaram	41-51	11
Virudunagar	Arupukottai	39-48	10
	Sattur	41-48	8
	Sivakasi	41-48	8
	Srivilliputtur	41-49	9
	Virudunagar	38-48	11
	Watrap	39-50	12
Tuticorin	Arasadi	43-49	7
	Kayattur	41-49	9
	Kovilpatti	41-49	9
	Kulasekarapatnam	42-52	11
	Morekulam	42-51	10
	Ottappidaram	41-48	8
	Sattankulam	42-50	9
	Srivaikuntam	42-50	9
	Tiruchchendur	42-52	11
	Tuticorin	43-50	8
Tirunelveli	Ambasamudram	42-52,1	12
	Ayikudi	42-51	10
	Kadaiyam	42-52,1	12
	Kadaiyanallur	42-51	10
	Kirnurnam	43-51	9
	Nanguneri	42-51	10
	Palayamkottai	42-50	9
	Radhapuram	42-49	8
	Sankarankovil	41-49	9
	Shencottah	41-51	11
	Sivagiri	41-52	12
	Tenkasi	41-52	12
	Tirunelveli	42-51	10
Sivaganga	Manamadurai	39-48	10
	Sivaganga	35-48	14
	Tirupattur	33-48	16
Madurai	Cholavandan	36-48	13

	Madurai	34-48	15
	Melur	33-49	17
	Nattam	33-49	17
	Peraiyur	36-49	14
	Tirumangalam	34-49	16
Pudukkottai	Adanakottai	37-49	13
	Alangudi	36-50	15
	Annavasal	36-47	12
	Ilupur	36-48	13
	Karambakkudi	38-50	13
	Kilanilai	38-49	12
	Marungapuri	35-49	15
	Ponnamaravati	34-48	15
	Pudukkottai	35-49	15
	Tirumayam	36-48	13
	Udyalipatti	37-47	11
	Viralimalai	38-48	11
Dindigul	Chattrapatti	38-50	13
	Dindigul	36-49	14
	Vedasunthur	38-48	11
Theni	Bodinayakanur	39-48	10

6. High Rainfall Zone

District	Station	G period (Met. Standard week)	No. of G period (No. of weeks)
Kanyakumari	Aramboli	41-49	9
	Eranial	19-29, 39-48	11, 10
	Kalial	14-50	37
	Kolachel	21-30, 40-48	10, 9
	Kottaram	42-48	7
	Kulasegraram	12-50	39
	Kuzhitturai	15-48	34
	Meycode adanadi	15-49	35
	Mulakumood	15-31, 40-49	17, 10
	Nagerkoil	20-26, 40-48	7, 9
	P.P. Channel	16-29, 40-50	14, 11
	Pechiaprai	13-50	38
	Puthendam	13-50	38
	Rajakkammangalam	19-27, 40-48	9, 9
	Seetapal	19-26, 40-48	8, 9
	Shorlakode	18-25, 36-47	8, 12
	Thadikarekonam	14-49	36
	Thamaraikulam	42-47	6
The Nilgiris	Devala	17-50	34
	Glen Morgan	16-50	35

	Gudalur	17-49	33
	Kallatty	17-51	35
	Ketty	18-52,1	36
	Kodanadu	18-52,1,2	37
	Kotagiri	14-52,1-3	40
	Kundha (Kailkund)	21-52,1	33
	Naduvattam	16-49	34
	Ootacamund	16-51	36
Coimbatore	Anaimalai	32-52,1	22
Dindigul	Kodaikanal	32-52,1	22
Salem	Yercaud	34-50	17

2. Climate of Tamil Nadu

South West Monsoon

Arid	Semi arid	Humid
(MI = (-) 66.7 to (-) 100)	(MI = (-) 66.7 to (-) 33.3)	(MI = 20 to 80)
Coimbatore, Erode, Tiruchirapalli, Karur, Perambalur, Madurai, Theni, Dindugal, Ramanathapuram, Sivagangai,Viruudunagar, Tirunelveli, Tuticorin	Kanchipuram, Thrivallur Cuddalore, Villupuram Dharmapuri , Krishnagiri Salem ,Namakal Pudukottai Thanjavur ,Nagai & Thiruvarur Kanyakumari Velore, Thiruvannamalai	**The Nilgiris**

North East Monsoon

Dry sub humid MI= (-) 33 to 0	Moist sub humid MI= 0 to 20	Per humid MI=100 and above.
Coimbatore, Erode, Dharmapuri, Krishnagiri. Salem, Namakal	Triuchirapalli, Karur, Perambalur, Pudukottai, Madurai, Theni, Dindugal, Sivagangai, Vridunagar, Ramanathapuram, Kanchipuram, Thiruvallur, Vellore, Tiruvannamalai, Cuddalore,Villupuram, Thanjavur, Nagai, Thiruvarur, Kanyakumari, Thirunelveli, Tuticorin	The Nilgiris

3. Rain fall pattern in Tamil Nadu

The rainfall pattern of Tamil Nadu based on the criteria of rainfall quantity and Seasons of precipitation is given below: (NCA, 1976)

A = > 30 cm rainfall per month
B = 30-20 cm rainfall per month
C = 20-10 cm rainfall per month
D = 10-5 cm rainfall per month
E = < 5 cm rainfall per month

Three distinct seasons have been considered
Pre monsoon season :
February to May Monsoon season :
June to September Post monsoon season :
October to January

Considering the distribution of rainfall within a season, a standard pattern is developed.

This is explained through the following example

$$A_2\ B_2\ (C_1\ B_1\ A_1\ E_1)\ C_2\ D_1\ E_1$$

Where in,

i) Alphabets in bracket denote rainfall in monsoon season months.
ii) Left to bracket denotes rainfall in pre-monsoon months.
iii) Right to bracket denotes rainfall in post monsoon months.
iv) Numerical suffix gives the number of months.

Rainfall pattern in Tamil Nadu

Rainfall	Taluks in which the pattern is seen
$E_4\ (E_4)\ C_2\ E_2$	Aruppukottai, Paramakudi, Muthukulathur, Thiruvadanai, Sathur, Srivilliputhur, Kovilpatti, Vilathikulam, Sankarankovil, Thoothukudi, Srivaikuntam, Udumalpet, Coimbatore, Dharapuram, Palladam, Gobichettipalayam, Bhavani, Erode, Avinashi, Uthamapalayam, Palani, Kodaikanal, Vedasandur, Dindugul, Nilakottai, Usilampatti, Thirumangalam, Periakulam, Karur
$E_4\ (E_4)\ B_1\ C_2\ E_1$	Tiruchendur, Nanguneri, Tirunelveli, Ambasamudram, Ramanathapuram $E_4\ (E_4)\ A_1\ B_2\ E_1$ Nagapattinam, Thiruthuraipoondi, Lalgudi, Musiri

E_4 (C_1 E_3) C_2 E_2	Thuraiyur, Kulithalai, Tiruchirappalli, Manapparai, Pollachi, Agastheswaram
E_4 (C_1 E_3) A_1B_1 C_1 E_1	Mayavaram, Nannilam
E_4 (C_2 E_2) C_1 E_3	Vaniyambadi, Thirupathur, Uthankarai, Thirukoilur, Kallakurichi, Perambalur, Kulathur, Alangudi, Thirumayam, Harur, Athur, Thanjavur, Aranthangi, Arakkonam, Walajapet, Cheyyar, Arani, Polur, Chengam, Thiruvannamalai,,Gudiyatham, Vellore, Thiruthani, Madurai North, Madurai South, Melur, Thirupathur, Sivaganga, Chengam, Wandavasi.
E_4 (C_2 E_2) B_1C_2 E_1	Virudhachalam, Ariyalur, Udayarpalayam, Kumbakonam, Papanasam, Mannargudi, Pattukottai, Orathanadu, Tenkasi, Shencottai, Thiruvallur, Sriperumpudur, Kanchipuram, Chengalpattu, Maduranthagam, Tindivanam, Villupuram.
E_4 (C_2 E_2) A_1 B_1C_1 E_1	Ponneri, Saidapet, Chidambaram, Sirkazhi. C_1 E_3(C_1 E_3) C_1 E_3 Hosur, Denkanikottai,
C_1 E_3(C_2 E_2) C_1 E_3	Omalur, Krishnagiri, Dharmapuri, Mettur, Salem, Rasipuram, Sangagiri, Thiruchengodu, Namakkal

4. Pre monsoon sowing

Based on the probability of receiving sowing rains, pre monsoon dry seeding weeks have been identified for the different districts of Tamil Nadu, which is feasible in Vertisols.

Name of the Districts	Sowing STD week	Dates
1. Coimbatore & Erode	37 to 38	Sep 10 to 23
2. Dharmapuri	38 to 39	Sep 17 to 30
3. Vellore	36 to 37	Sep 3 to 16
4. Ramanathapuram	40 to 41	Oct 1 to 14
5. Thoothukudi	39 to 40	Sep 24 to Oct 7
6. Thrinelveli	39 to 40	Sep 24 to Oct 7
7. Virudhunagar	38 to 39	Sep 17 to 30

5. Water balance study

Water balance study was conducted for Tamil Nadu based on the Water Requirement Satisfaction Index (WRSI). It is suggested that sorghum can be sown during 36^{th} Std week against 16^{th} Std week. The data from the Table indicate that if it is sown during 16^{th} Std week, the crop may suffer with soil moisture stress. This result is valid for sorghum crop for Manapparai Taluk sowing by 36^{th} standard week is recommended.

Manapparai - Sorghum crop

Manapparai (16th week sowing)			Manapparai (36th week sowing)		
STD week	Date	WRSI	STD week	Date	WRSI
16	April 16 - 22	100.00	36	Sep 3 - 9	100
17	April 23 - 29	100.00	37	Sep 10 - 16	100
18	April 30-May 6	97.78	38	Sep 17 - 23	100
19	May 7 - 13	95.81	39	Sep 24 - 30	100
20	May 14 - 20	92.11	40	Oct 1 - 7	100
21	May 21 - 27	87.59	41	Oct 8 - 14	100
22	May 28 -Jun 3	81.80	42	Oct 15 - 21	100
23	June 4 - 10	74.58	43	Oct 22 - 28	100
24	June 11 - 17	66.13	44	Oct 29 - Nov 4	100
25	June 18 - 24	54.06	45	Nov 5 - 11	100
26	June 25-July 1	48.29	46	Nov 12 - 18	100
27	July 2 - 8	43.93	47	Nov 19 - 25	100
28	July 9 - 15	43.93	48	Nov 26 - Dec 2	100

Simila study was undertaken for Namakkal Taluk for Groundnut sowing:

The result indicates that, rainfed groundnut sowing can be taken in the order of 28th Std week, 26th Std week, 23rd Std week.

Further studies were made from water balance for rainfed crops of Virudhunagar district and the information are presented in the Table *

District	Location	Crop	Soil	Sowing week (MSW)	Final harvest (MSW)	Moisture stress period(MSW)
Virudhunagar	Aruppukottai	Cotton	Black	36	4	1 to 4
Virudhunagar	Rajapalayam	Cotton	Black	36	4	3 to 4
Virudhunagar	Sattur	Cotton	Black	36	4	1 to 4
Virudhunagar	Srivilliputtur	Cotton	Black	36	4	2 to 4
Virudhunagar	Tiruchuli	Cotton	Black	36	4	1 to 4
Virudhunagar	Virudhunagar	Cotton	Black	36	4	1 to 4
Virudhunagar	Aruppukottai	Cotton	Black	37	5	1 to 5
Virudhunagar	Rajapalayam	Cotton	Black	37	5	2 to 5
Virudhunagar	Sattur	Cotton	Black	37	5	1 to 5
Virudhunagar	Srivilliputtur	Cotton	Black	37	5	2 to 5
Virudhunagar	Tiruchuli	Cotton	Black	37	5	51to 5
Virudhunagar	Virudhunagar	Cotton	Black	37	5	2 to 5
Virudhunagar	Aruppukottai	Cotton	Black	38	6	1 to6
Virudhunagar	Rajapalayam	Cotton	Black	38	6	2 to 6
Virudhunagar	Sattur	Cotton	Black	38	6	1 to 6
Virudhunagar	Srivilliputtur	Cotton	Black	38	6	2 to 6
Virudhunagar	Tiruchuli	Cotton	Black	38	6	51to 6
Virudhunagar	Virudhunagar	Cotton	Black	38	6	1 to 6
Virudhunagar	Aruppukottai	Cotton	Black	39	7	1 to 7
Virudhunagar	Rajapalayam	Cotton	Black	39	7	2 to 7

Virudhunagar	Sattur	Cotton	Black	39	7	1 to 7
Virudhunagar	Srivilliputtur	Cotton	Black	39	7	2 to 7
Virudhunagar	Tiruchuli	Cotton	Black	39	7	1 to 7
Virudhunagar	Virudhunagar	Cotton	Black	39	7	1 to 7
Virudhunagar	Aruppukottai	Cotton	Red	36	4	49 to 4
Virudhunagar	Rajapalayam	Cotton	Red	36	4	51 to 4
Virudhunagar	Sattur	Cotton	Red	36	4	52 to 4
Virudhunagar	Srivilliputtur	Cotton	Red	36	4	50 to 4
Virudhunagar	Tiruchuli	Cotton	Red	36	4	49 to 4
Virudhunagar	Virudhunagar	Cotton	Red	36	4	51 to 4
Virudhunagar	Aruppukottai	Cotton	Red	37	5	50 to 5
Virudhunagar	Rajapalayam	Cotton	Red	37	5	51 to 5
Virudhunagar	Sattur	Cotton	Red	37	5	50 to 5
Virudhunagar	Srivilliputtur	Cotton	Red	37	5	50 to 5
Virudhunagar	Tiruchuli	Cotton	Red	37	5	49 to 5
Virudhunagar	Virudhunagar	Cotton	Red	37	5	51 to 5
Virudhunagar	Aruppukottai	Cotton	Red	38	6	50 to 6
Virudhunagar	Rajapalayam	Cotton	Red	38	6	51 to 6
Virudhunagar	Sattur	Cotton	Red	38	6	50 to 6
Virudhunagar	Srivilliputtur	Cotton	Red	38	6	51 to 6
Virudhunagar	Tiruchuli	Cotton	Red	38	6	49 to 6
Virudhunagar	Virudhunagar	Cotton	Red	38	6	50 to 6
Virudhunagar	Aruppukottai	Cotton	Red	39	7	51 to 7
Virudhunagar	Rajapalayam	Cotton	Red	39	7	51 to 7
Virudhunagar	Sattur	Cotton	Red	39	7	51 to 7
Virudhunagar	Srivilliputtur	Cotton	Red	39	7	51 to 7
Virudhunagar	Tiruchuli	Cotton	Red	39	7	50 to 7
Virudhunagar	Virudhunagar	Cotton	Red	39	7	50 to 7
Virudhunagar	Aruppukottai	Pulses	Black	36	48	-
Virudhunagar	Rajapalayam	Pulses	Black	36	48	-
Virudhunagar	Sattur	Pulses	Black	36	48	-
Virudhunagar	Srivilliputtur	Pulses	Black	36	48	-
Virudhunagar	Tiruchuli	Pulses	Black	36	48	-
Virudhunagar	Virudhunagar	Pulses	Black	36	48	-
Virudhunagar	Aruppukottai	Pulses	Black	37	49	-
Virudhunagar	Rajapalayam	Pulses	Black	37	49	-
Virudhunagar	Sattur	Pulses	Black	37	49	-
Virudhunagar	Srivilliputtur	Pulses	Black	37	49	-
Virudhunagar	Tiruchuli	Pulses	Black	37	49	-
Virudhunagar	Virudhunagar	Pulses	Black	37	49	-
Virudhunagar	Aruppukottai	Pulses	Black	38	50	-
Virudhunagar	Rajapalayam	Pulses	Black	38	50	-
Virudhunagar	Sattur	Pulses	Black	38	50	-
Virudhunagar	Srivilliputtur	Pulses	Black	38	50	-
Virudhunagar	Tiruchuli	Pulses	Black	38	50	-
Virudhunagar	Virudhunagar	Pulses	Black	38	50	-
Virudhunagar	Aruppukottai	Pulses	Black	39	51	-
Virudhunagar	Rajapalayam	Pulses	Black	39	51	-
Virudhunagar	Sattur	Pulses	Black	39	51	-

Virudhunagar	Srivilliputtur	Pulses	Black	39	51	-
Virudhunagar	Tiruchuli	Pulses	Black	39	51	-
Virudhunagar	Virudhunagar	Pulses	Black	39	51	-
Virudhunagar	Aruppukottai	Pulses	Red	36	48	-
Virudhunagar	Rajapalayam	Pulses	Red	36	48	-
Virudhunagar	Sattur	Pulses	Red	36	48	-
Virudhunagar	Srivilliputtur	Pulses	Red	36	48	-
Virudhunagar	Tiruchuli	Pulses	Red	36	48	-
Virudhunagar	Virudhunagar	Pulses	Red	36	48	-
Virudhunagar	Aruppukottai	Pulses	Red	37	49	-
Virudhunagar	Rajapalayam	Pulses	Red	37	49	-
Virudhunagar	Sattur	Pulses	Red	37	49	-
Virudhunagar	Srivilliputtur	Pulses	Red	37	49	-
Virudhunagar	Tiruchuli	Pulses	Red	37	49	-
Virudhunagar	Virudhunagar	Pulses	Red	37	49	-
Virudhunagar	Aruppukottai	Pulses	Red	39	51	-
Virudhunagar	Rajapalayam	Pulses	Red	39	51	-
Virudhunagar	Sattur	Pulses	Red	39	51	-
Virudhunagar	Srivilliputtur	Pulses	Red	39	51	-
Virudhunagar	Tiruchuli	Pulses	Red	39	51	-
Virudhunagar	Virudhunagar	Pulses	Red	39	51	-
Virudhunagar	Aruppukottai	Redgram	Black	36	01	-
Virudhunagar	Rajapalayam	Redgram	Black	36	01	-
Virudhunagar	Sattur	Redgram	Black	36	01	-
Virudhunagar	Srivilliputtur	Redgram	Black	36	01	-
Virudhunagar	Tiruchuli	Redgram	Black	36	01	-
Virudhunagar	Virudhunagar	Redgram	Black	36	01	-
Virudhunagar	Aruppukottai	Redgram	Black	37	02	-
Virudhunagar	Rajapalayam	Redgram	Black	37	02	-
Virudhunagar	Sattur	Redgram	Black	37	02	-
Virudhunagar	Srivilliputtur	Redgram	Black	37	02	-
Virudhunagar	Tiruchuli	Redgram	Black	37	02	-
Virudhunagar	Virudhunagar	Redgram	Black	37	02	-
Virudhunagar	Aruppukottai	Redgram	Black	38	03	-
Virudhunagar	Rajapalayam	Redgram	Black	38	03	-
Virudhunagar	Sattur	Redgram	Black	38	03	03
Virudhunagar	Srivilliputtur	Redgram	Black	38	03	-
Virudhunagar	Tiruchuli	Redgram	Black	38	03	-
Virudhunagar	Virudhunagar	Redgram	Black	38	03	-
Virudhunagar	Aruppukottai	Redgram	Red	36	01	-
Virudhunagar	Rajapalayam	Redgram	Red	36	01	-
Virudhunagar	Sattur	Redgram	Red	36	01	-
Virudhunagar	Srivilliputtur	Redgram	Red	36	01	-
Virudhunagar	Tiruchuli	Redgram	Red	36	01	-
Virudhunagar	Virudhunagar	Redgram	Red	36	01	-
Virudhunagar	Aruppukottai	Redgram	Red	37	02	-
Virudhunagar	Rajapalayam	Redgram	Red	37	02	-
Virudhunagar	Sattur	Redgram	Red	37	02	02
Virudhunagar	Srivilliputtur	Redgram	Red	37	02	-

Virudhunagar	Tiruchuli	Redgram	Red	37	02	02
Virudhunagar	Virudhunagar	Redgram	Red	37	02	01 to 02
Virudhunagar	Aruppukottai	Redgram	Red	38	03	02 to 03
Virudhunagar	Rajapalayam	Redgram	Red	38	03	03
Virudhunagar	Sattur	Redgram	Red	38	03	01 to 03
Virudhunagar	Srivilliputtur	Redgram	Red	38	03	03
Virudhunagar	Tiruchuli	Redgram	Red	38	03	01 to 03
Virudhunagar	Virudhunagar	Redgram	Red	38	03	01 to 03
Virudhunagar	Aruppukottai	Sorghum	Black	36	52	48 to 52
Virudhunagar	Rajapalayam	Sorghum	Black	36	52	-
Virudhunagar	Sattur	Sorghum	Black	36	52	49 to 52
Virudhunagar	Srivilliputtur	Sorghum	Black	36	52	-
Virudhunagar	Tiruchuli	Sorghum	Black	36	52	48 to 52
Virudhunagar	Virudhunagar	Sorghum	Black	36	52	49 to 52
Virudhunagar	Aruppukottai	Sorghum	Black	37	01	49 to 01
Virudhunagar	Rajapalayam	Sorghum	Black	37	01	-
Virudhunagar	Sattur	Sorghum	Black	37	01	50 to 01
Virudhunagar	Srivilliputtur	Sorghum	Black	37	01	-
Virudhunagar	Tiruchuli	Sorghum	Black	37	01	49 to 01
Virudhunagar	Virudhunagar	Sorghum	Black	37	01	50 to 01
Virudhunagar	Aruppukottai	Sorghum	Black	38	02	50 to 02
Virudhunagar	Rajapalayam	Sorghum	Black	38	02	-
Virudhunagar	Sattur	Sorghum	Black	38	02	51 to 02
Virudhunagar	Srivilliputtur	Sorghum	Black	38	02	02
Virudhunagar	Tiruchuli	Sorghum	Black	38	02	49 to 02
Virudhunagar	Virudhunagar	Sorghum	Black	38	02	51 to 02
Virudhunagar	Aruppukottai	Sorghum	Black	39	03	52 to 03
Virudhunagar	Rajapalayam	Sorghum	Black	39	03	01 to 03
Virudhunagar	Sattur	Sorghum	Black	39	03	51 to 03
Virudhunagar	Srivilliputtur	Sorghum	Black	39	03	02 to 03
Virudhunagar	Tiruchuli	Sorghum	Black	39	03	50 to 03
Virudhunagar	Virudhunagar	Sorghum	Black	39	03	51 to 03
Virudhunagar	Aruppukottai	Sorghum	Red	36	52	48 to 52
Virudhunagar	Rajapalayam	Sorghum	Red	36	52	-
Virudhunagar	Sattur	Sorghum	Red	36	52	49 to 52
Virudhunagar	Srivilliputtur	Sorghum	Red	36	52	-
Virudhunagar	Tiruchuli	Sorghum	Red	36	52	48 to 52
Virudhunagar	Virudhunagar	Sorghum	Red	36	52	49 to 52
Virudhunagar	Aruppukottai	Sorghum	Red	37	01	49 to 01
Virudhunagar	Rajapalayam	Sorghum	Red	37	01	51 to 01
Virudhunagar	Sattur	Sorghum	Red	37	01	50 to 01
Virudhunagar	Srivilliputtur	Sorghum	Red	37	01	52 to 01
Virudhunagar	Tiruchuli	Sorghum	Red	37	01	49 to 01
Virudhunagar	Virudhunagar	Sorghum	Red	37	01	49 to 01
Virudhunagar	Aruppukottai	Sorghum	Red	38	02	49 to 02
Virudhunagar	Rajapalayam	Sorghum	Red	38	02	51 to 02
Virudhunagar	Sattur	Sorghum	Red	38	02	50 to 02
Virudhunagar	Srivilliputtur	Sorghum	Red	38	02	51 to 02
Virudhunagar	Tiruchuli	Sorghum	Red	38	02	49 to 02

Virudhunagar	Virudhunagar	Sorghum	Red	38	02	49 to02
Virudhunagar	Aruppukottai	Sorghum	Red	39	03	50 to 03
Virudhunagar	Rajapalayam	Sorghum	Red	39	03	51 to 03
Virudhunagar	Sattur	Sorghum	Red	39	03	50 to 03
Virudhunagar	Srivilliputtur	Sorghum	Red	39	03	51 to 03
Virudhunagar	Tiruchuli	Sorghum	Red	39	03	49 to 03
Virudhunagar	Virudhunagar	Sorghum	Red	39	03	49 to 03
Virudhunagar	Aruppukottai	Sunflower	Black	43	03	-
Virudhunagar	Rajapalayam	Sunflower	Black	43	03	-
Virudhunagar	Sattur	Sunflower	Black	43	03	02 to 03
Virudhunagar	Srivilliputtur	Sunflower	Black	43	03	-
Virudhunagar	Tiruchuli	Sunflower	Black	43	03	02 to 03
Virudhunagar	Virudhunagar	Sunflower	Black	43	03	02 to 03
Virudhunagar	Aruppukottai	Sunflower	Black	44	04	02 to 04
Virudhunagar	Rajapalayam	Sunflower	Black	44	04	04
Virudhunagar	Sattur	Sunflower	Black	44	04	02 to 04
Virudhunagar	Srivilliputtur	Sunflower	Black	44	04	04
Virudhunagar	Tiruchuli	Sunflower	Black	44	04	01 to 04
Virudhunagar	Virudhunagar	Sunflower	Black	44	04	02 to 04
Virudhunagar	Aruppukottai	Sunflower	Red	43	03	52 to 03
Virudhunagar	Rajapalayam	Sunflower	Red	43	03	01 to 03
Virudhunagar	Sattur	Sunflower	Red	43	03	52 to 03
Virudhunagar	Srivilliputtur	Sunflower	Red	43	03	01 to 03
Virudhunagar	Tiruchuli	Sunflower	Red	43	03	51 to 03
Virudhunagar	Virudhunagar	Sunflower	Red	43	03	52 to 03
Virudhunagar	Aruppukottai	Sunflower	Red	44	04	01 to 04
Virudhunagar	Rajapalayam	Sunflower	Red	44	04	01 to 04
Virudhunagar	Sattur	Sunflower	Red	44	04	52 to 04
Virudhunagar	Srivilliputtur	Sunflower	Red	44	04	01 to 04
Virudhunagar	Tiruchuli	Sunflower	Red	44	04	51 to 04
Virudhunagar	Virudhunagar	Sunflower	Red	44	04	52 to 04
Virudhunagar	Aruppukottai	Groundnut	Red	26	41	33 to 41
Virudhunagar	Rajapalayam	Groundnut	Red	26	41	31 to 41
Virudhunagar	Sattur	Groundnut	Red	26	41	30 to 41
Virudhunagar	Srivilliputtur	Groundnut	Red	26	41	29 to 41
Virudhunagar	Tiruchuli	Groundnut	Red	26	41	32 to 41
Virudhunagar	Virudhunagar	Groundnut	Red	26	41	30 to 41
Virudhunagar	Aruppukottai	Groundnut	Red	27	42	33 to 42
Virudhunagar	Rajapalayam	Groundnut	Red	27	42	31 to 42
Virudhunagar	Sattur	Groundnut	Red	27	42	31 to 42
Virudhunagar	Srivilliputtur	Groundnut	Red	27	42	31 to 42
Virudhunagar	Tiruchuli	Groundnut	Red	27	42	32 to 42
Virudhunagar	Virudhunagar	Groundnut	Red	27	42	32 to 42

Virudhunagar	Aruppukottai	Groundnut	Red	28	43	33 to 43
Virudhunagar	Rajapalayam	Groundnut	Red	28	43	31 to 43
Virudhunagar	Sattur	Groundnut	Red	28	43	32 to 43
Virudhunagar	Srivilliputtur	Groundnut	Red	28	43	32 to 43
Virudhunagar	Tiruchuli	Groundnut	Red	28	43	33 to 43
Virudhunagar	Virudhunagar	Groundnut	Red	28	43	33 to 43

* Note: During moisture stress period suitable agro- techniques may be adopted. If moisture stress period is long concerned sowing week may not be viable.

6. Weather Based Management Technologies

i) Nutrient management for *thaladi* season rice

Application of 200:75:75kg NPK/ha for November 15th transplanted crop (Co45 or Co43) under split application of N at 40, 20, 20 and 20% respectively during basal, active tillering, panicle initiation and flowering while 75 percent P and K as basal and 12.5 percent P and K as foliar spray twice at panicle initiation and flowering stages.

ii) Acceptable *insitu* moisture conservation practice for rainfed groundnut – sunflower and maize

During South West monsoon season groundnut sowing along the contour and ridging to be done three weeks after sowing. During NEM, especially for sunflower, the same technology of contour sowing followed by ridging three weeks latter can be adopted. In respect of maize, sowing and tieing alternate furrows with mulching of locally available material can be practiced.

iii) Sustainable dryland management for hybrid maize (UMH 28)

Sowing of dry land hybrid maize at 38th meteorological standard week (17th – 23 Sept.) with modified crop production recommendation based on medium range weather forecast is suggested.

iv) Time of sowing and nutrient level for sorghum under different rainfall situations in dryland (black soil) of western agro climate zone of Tamil Nadu

Sowing of sorghum variety CSV15 before the receipt of monsoon rainfall (Premonsoon sowing) is recommended with 60:30:0 kg NPK / ha during above average rainfall year and 40:20:0 Kg NPK / ha during below average rainfall year. The result is applicable when seasonal climate forecast information is available in advance.

v) Technical feasibility of introducing new irrigated cropping system of Greengram – Maize – Sunflower against the outdated cropping system of Cotton – Sorghum – Finger millet of western agro climatic zone of Tamil Nadu

Sowing of crops at normal sowing of concerned crops viz.; 33 Meteorological Standard Week (MSW) for (Aug 13-19) greengram, 48th MSW (Nov 26-Dec2) for maize and 15th MSW (April 9-15) for sunflower with 100 percent inorganic source of recommended nutrients for green gram (12.5:50:0 kg NPK / ha) and sunflower (40:20:20 kg NPK/ ha) and 25% organic N alone and 75 percent inorganic source of nutrient recommended to maize (135:62.5: 50 kg NPK/ ha) for the new tailored cropping system of Greengram –Maize – Sunflower.

vi) Potential season and sowing window for CoH3 Hybrid Maize under irrigated condition

Sowing of irrigated Maize hybrid CoH3 in the second fortnight of August during *Kharif* season with integrated application of both organics and inorganic at 50:50 either as blanket (135:625:50 kg NPK/ha) or as soil test based recommendation.

vii) Potential transplanting window for hybrid rice

Planting hybrid rice CORH2 either on 26th September or at 3rd October as compared to planting in normal date of planting of 19th September which is recommended for planting rice variety especially for the variety ADT39.

viii) Polyethylene film mulch for irrigated groundnut

Spreading of seven micron thickness black polyethylene film as mulch to irrigated groundnut along with pre-plant incorporation of fluchloralin @ 1.0 kg ai/ha under flat bed system.

ix) Forewarning disease incidence in groundnut

Forewarning model was developed against late leaf spot and rust diseases in groundnut. The model was validated and the deviation is around 10 percent. The model was developed for both for Aliyarnagar (mountain climate) and Vridhachalam (Marine climate) domain.

7. Basic information

i. Crop – weather studies

Rice grain yield of *Kuruvai* and *Thaladi* seasons over 30 years (1961 – 1990) were correlated with concerned weather data. Reproductive stage was very critical to prevalence weather parameters both for *Kuruvai* and *Thaladi* seasons. In addition maturity stage of *Kuruvai* and Vegetative stage of *thaladi* season were also critical to weather.

During *Thaladi* season, correlation study indicated the positive relationship for maximum temperature at vegetative and reproductive stages.

ii. Management response to seasonal climate forecast in cropping system

Two locations viz. Avinashi and Thiruchengodu were considered for the study. Model to simulate the yield of crops (Groundnut, Cotton) was done.

The chance of achieving (65%) at least 1000 kg/ha of peanut occur, when the Southern Oscillation Index (SOI) phase is positive for April / May. Conversely there is only 32% chance of achieving such a yield in years when the SOI is falling. Similar analysis was conducted for cotton and economic performance of both systems was compared on gross margin basis. Results indicate that in positive SOI years, peanuts out performed in cotton in 70 percent of years, but income difference can still range from Rs.(-)15,000 to (+) 15,000 / ha. However under falling SOI conditions peanut only had minor advantage in 40% of years (up to Rs.3,800/ ha).

iii. Seasonal rainfall Vs El-Nino

Analyses of long term average of Southwest monsoon rainfall during El-Nino years revealed that during El-Nino years, the amount of rainfall found decreased in all the locations of Tamil Nadu as compared to normal rainfall of this season, except Northeastern parts of Tamil Nadu. Analyses of long term Northeast monsoon rainfall indicate that during El-Nino years there was increase in amount of rainfall than normal in all the locations of Tamil Nadu.

iv. Tamil years Vs annual rainfall forecast

The annual rainfall of a particular Tamil year in a cycle of 60 years was not the same for the corresponding Tamil years on the forth coming cycle and one can expect an opposite event.

v. Stars Vs Seasonal rainfall forecast

The star Revathi had greater influence on rainfall during hot weather period (March- May) while during Southwest monsoon (June – Sept) and Northeast monsoon seasons (Oct – Dec), stars Maham and Uthiram respectively did influence seasonal rainfall.

In the monthly analysis at 30% probability, the star Uthiram had influenced in getting rainfall of > 20mm during July and November months. While during other months the stars viz. Maham, Pooradam, Kettai, Swathi and Moolam showed their influence to get < 20mm of rainfall.

vi. Pest and weather relationship study in cotton

When maximum and minimum temperature got increased, the infestation from American bollworm also got decreased. In contrast, positive relationship existed for pink bollworm for the above weather parameters. In the case of aphid, maximum temperature, diurnal variation, Relative Temperature Disparity, bright sunshine hours, and wind speed, had negative relationship, while positive correlation was observed for minimum temperature.

vii. Study on the weather relationship of *eriophyid* mite in coconut

The maximum temperature had negative correlation with nuts affected in all the varieties (Tall (east coast), Dwarf (yellow), Tall X Dwarf, Orange, and Dwarf X Tall) at three months after spathe emergence; where as positive correlation was obtained for maximum temperature one to two months before spathe emergence in respect of Tall (east coast) and Dwarf x Tall varieties. In general *eriophyid* mite affected nuts were either positively and negatively influenced by minimum temperature and relative humidity respectively (0722 IST and 1422 IST). From the stepwise regression analysis made, one to two months earlier or one to two months after spathe emergence, wind speed had higher influence on the nuts affected with mite irrespective of varieties except Tall x Dwarf .

viii. Probing the association of lunar phases "*Thithies*" with rainfall at Coimbatore

Based on the interaction between earth and moon in relation to sun, each month is governed by both new moon and full moon. In between these two, there are fourteen *thithies* covering the 14 days interval. A study was undertaken to find out the association between rainfall and the different *thithies*. Results revealed that the first eight *thithies* succeeding new moon, and eight *thithies* preceding the new moon did relate to annual rainfall events. Higher rainfall occurred normally during the eight *thithies* preceding the new moon as compared to *thithies* succeeding the new moon. Almost similar results could be noticed for both Southwest and Northeast monsoon seasons. Analysis also indicated that towards full moon phase, the *thithi* Shasthi (sixth phase) is associated with high rainfall while such effect was noticed at Ekadasi (eleventh phase) *thithi* towards new moon. High intensity events occurred frequently during new moon phase as compared to full moon phase.

8. Medium range weather forecast

In Tamil Nadu, about 55.4 per cent of the arable land depends entirely on rainfall for its crop productions. Since rainfall varies in space and time, there is risk in farming for dry land crop production. Proper understanding of the climate and issuing weather forecast based on the dynamic nature of atmosphere would help in multiple ways. Four different weather forecasts are presently made. They are now casting, short range, medium range and long range.

Among the forecasts, the weather forecast given under medium range seems to serve the purpose of the farmers, since it provides enough time to the farmers to change the agricultural operations based on anticipated weather change under dry land environment.

In this context, a project on the establishment of National Centre for Medium Range Weather Forecast (NCMRWF) and Development of Agro-meteorological service was approved by the Government of India and implemented by the Department of Science and Technology (DST) in mission mode. Currently local weather forecast based on Direct Model output of General Circulation Model (GCM) is prepared by NCMRWF and given to Agromet Advisory Service units located at different State Agricultural Universities (SAU) including seven in Tamil Nadu, four under TNAU (Coimbatore, Pechiparai, Kovilpatti and Aduthurai) and two under Tamil Nadu Veterinary and Animal Sciences University (Chennai and Namakkal) and one at Kannivadi (MSSRF). In turn the SAU prepares weather based agro advisory bulletin and communicate to the farmers for making decisions on agricultural activities based on anticipated weather change. The forecast covers, cloud cover, rainfall, wind speed, wind direction, maximum temperature, and minimum temperature. This forecast is given for four days from Tuesday to Friday and again from Friday to Monday and thus it covers a whole week.

Presently TNAU installs Automatic Weather Station at block level and once completed, block level weather forecast with agro advisories will be given.

9. Seasonal climate forecast

Seasonal climate forecast is being given to all districts of Tamil Nadu through TNAU Research Stations both for South-west and North-east monsoon seasons with a lead time of 15 days. This forecast contains the seasonal rainfall both in temporal and spatial dimensions. This forecast is based on probability analysis made through Australian Rainman Software. The inputs are location specific past rainfall data more than twenty one years and real time southern oscillation index and sea surface temperature. This type of forecast is being given from 1999 onwards and presently institutionalized by the TNAU. Based on the verification of the forecast, the accuracy goes up to 70 per cent. Since the forecast is given with a lead time the information is highly useful for farm planning and hence it becomes response farming in nature.

10. Climate change and crop production

a) Model result on Temperature and Rainfall

The results of the projected climate change over Cauvery basin of Tamil Nadu for A1B scenario using PRECIS and RegCM3 regional climate models showed an increasing trend for maximum temperature, minimum temperature and rainfall. Decadal means of maximum and minimum temperatures were generated to understand the variation more clearly and the results revealed that the increase in maximum temperature in PRECIS was 3.7^{o}C and in RegCM3, it was 3.1^{o}C. The increase in minimum temperature in PRECIS model was 4.2^{o}C and in RegCM3, it was 3.7^{o}C during the same period. The increase in minimum temperatures is higher than maximum temperatures in both models.

b) Model result on rice productivity

The study on the yield of ADT 43 rice over Cauvery Delta Zone as simulated by Decision support System for Agricultural technology Transfer (DSSAT) under CO2 fertilization, the result had shown that a reduction of 135 Kg ha^{-1} decade^{-1} for PRECIS (Providing Regional Climates for Impact
Studies) model, while there was an increase in yield of24 Kg ha^{-1} decade^{-1} for RegCM3 (Regional Climate Model System 3) model, thus indicating the possibility of change in rice yield under climate change scenario.

c) Impact of Climate change on crops

Analysis on the maize crop yield indicated reduction in yield by 3.0, 9.3 and 18.3 per cent respectively during 2020, 2050 and 2080 from the current yield levels in the major maize growing districts of Tamil Nadu with increase in minimum temperature. Sorghum crop yield is expected to decline by 4.5, 11.2 and 18.7 per cent during 2020, 2050 and 2080 from the current yield levels if no management intervention is made in the major sorghum growing districts of Tamil Nadu. This is due to nighttime temperature increase.

d) Adaptation strategies developed under ClimaRice project for sustaining rice productivity in Cauvery Delta Zone (CDZ) against climate change

- Introduction of System of Rice Intensification (SRI) under non-rainy season
- Introduction of temperature tolerant rice varieties
- Seed treatment with bio-fertilizer (Azospirillum), application of blue green alage (BGA) and growing azolla as dual crop in rice. This reduces methane emission from the rice field

Table of Meteorological Standard Week

Std. Week. No.	Month	Dates	Std. Week. No.	Month	Dates
1	January	1-7	27	July	2-8
2		8-14	28		9-15
3		15-21	29		16-22
4		22-28	30		23-29
5		29-4	31		30-5
6	February	5-11	32	August	6-12
7		12-18	33		13-19
8		19-25	34		20-26
9		26-4*	35		27-2
10	March	5-11	36	September	3-9
11		12-18	37		10-16
12		19-25	38		17-23
13		26-1	39		24-30
14	April	2-8	40	October	1-7
15		9-15	41		8-14
16		16-22	42		15-21
17		23-29	43		22-28
18		30-6	44		29-4
19	May	7-13	45	November	5-11
20		14-20	46		12-18
21		21-27	47		19-25
22		28-3	48		26-2
23	June	4-10	49	December	3-9
24		11-17	50		10-16
25		18-24	51		17-23
26		25-1	52		24-31**

23. FARM IMPLEMENTS AND MACHINERY

i. LAND PREPARATION IMPLEMENTS

TRACTOR OPERATED CHISEL PLOUGH

Purpose	Suitable for deep tillage upto a depth of 40 cm for breaking hard soil pan.
Type of the implement	Tractor operated
Field capacity	1.4 ha / day at a spacing of 1.5m between rows
Cost of the implement	Rs.12000/- (Approximately)
Salient Features	Operated by any 35-45 hptractor. The implement is simple in construction and has only three components namely frame, standard and share. The share has a lift angle of 20 degree, width of 25mm and length of 150mm. The implement is protected by shear pin which prevents damage from overloading.

TRACTOR OPERATED SUBSOIL COIRPITH APPLICATOR

Purpose	The sub soil coir pith mulch is applied at 15-30 cm deep
Type of the implement	Tractor operated
Field capacity	0.60 ha/day
Cost of the implement	Rs.40,000/- (Approximately)
Salient Features	Ensures higher moisture retention, crop growth and yield

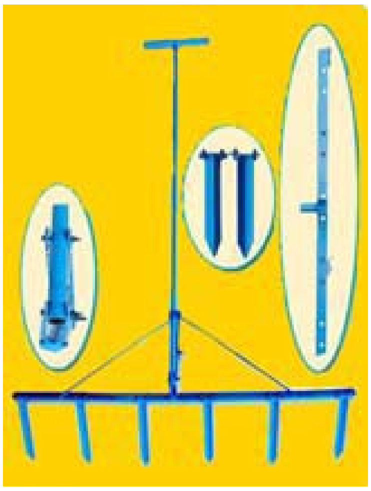
TRACTOR OPERATED ROTARY SPADING MACHINE

Purpose	Primary tillage tools suitable for all soil conditions including wet clay soil and recommended for intercultural operation in coconut orchards
Type of the implement	Tractor operated
Field capacity	1.5 ha per day
Cost of the implement	Rs. 1,20,000 /- (Approximately)
Salient Features	High energy utilization efficiency

TWIN ROW PRECISION ORGANIC MANURE CUM FERTILIZER APPLICATOR

Purpose	Suitable for accurate and controlled application of organic manure/mulch directly below the root zone.
Type of the implement	Tractor operated
Field capacity	1.0 ha per day
Cost of the implement	Rs. 1,25,000 /- (Approximately)
Salient Features	• Helps in improving the soil nutrient use efficiency, crop yield and soil quality. Adjustable spacing between furrows enables the use at different row spacing • Simultaneous precise placements of organic manure and inorganic fertilizer.

MANUALLY OPERATED MARKER



Purpose	Marker for line sowing
Type of the implement	Manually operated
Cost of the implement	Rs. 1800 /- (Approximately)
Salient Features	• Easy to operate • Reduced seed rate • Suitable for small and marginal farmers • Reduction in cost of cultivation • Maintenance of optimum plant population • Easy for inter cultural operation (Weeding, Spraying)

TRACTOR OPERATED PIT DIGGER FOR SUGAR CANE PLANTING

Purpose	Pit Digger for Sugarcane Planting
Type of the implement	Tractor operated
Field capacity	250 to 300 holes per hour
Cost of the implement	Rs. 1,20,00 /- (Approximately)
Salient Features	• At a time two pits of dia 90 cmcan be made • Spacing between the auger is adjustable

SYSTEM FOR CONTROLLED LEVEL OF PUDDLING

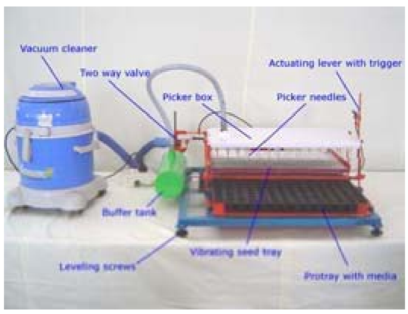
Purpose	To control the level of puddling in wetland
Type of the implement	Tractor mounted
Field capacity	1.5 ha / day
Cost of the implement	Rs.50,000/-(Additional Hydraulic cylinder to the existing laser leveller)
Salient Features	• Constant depth of puddle can be maintained accurately, reducing the tillage energy expended by deep puddling. • Reduce water requirement for puddling. • Establishment of a strong sub soil layer that will reduce deep percolation. • Ensures good level surface layer and subsurface layer enabling easy transplanting

ii. SOWING IMPLEMENTS

SEED CUM FERTILIZER DRILL FOR PADDY

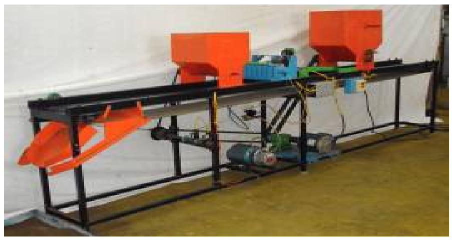
Purpose	For direct sowing of paddy and simultaneous application of fertilizer
Type of the implement	Tractor mounted
Field capacity	3 ha/day
Cost of the implement	Rs. 75,000 /- (Approximately)
Salient Features	• For dry sowing of paddy seeds in 9 rows at uniform depth and spacing

NEEDLE TYPE TRAY SEEDER FOR VEGETABLE NURSERY



Purpose	To sow in pro tray cavities
Field capacity	Can sow seeds in 60 trays/ hour
Cost of the unit	Rs. 35,000 /- (Approximately)
Salient Features	• To mechanize placement of seeds in the pro-tray cells • Singulated raw/ pelleted seeds placed in all the cells in a single stroke • Cost of operation is Rs. 280 for sowing 750 trays /day • Saving in time is 300% and saving in labour is 60

AUTOMATIC PROTRAY SOWING MACHINE FORVEGETABLE NURSERY PRODUCTION



Purpose	Filling medium and sowing seeds in protray
Field capacity	Can sow seeds in 200 trays/ hour
Cost of the implement	Rs. 1,00,000 /- (Approximately)
Salient Features	• The automated protray sowing machine provides for automating all the steps involved in the sowing of vegetable seeds in protrays. • The machine is able to provide above 100 per cent saving in cost.

TRACTOR DRAWN TURMERIC RHIZOME PLANTER

Purpose	To plant turmeric rhizomes on ridges
Type of the implement	Tractor mounted
Field capacity	1.2 ha/day
Cost of the implement	Rs. 60,000 /- (Approximately)
Salient Features	• Three rows can be planted at a time in the required spacing. • Row spacing is adjustable • Saving in quantity of 500 kg/ ha seed rhizomes

SUGARANE SETT CUTTER

Purpose	Cutting sugarcane sett with single bud
Field capacity	1500 setts/hour
Cost of the implement	Rs. 5,000 /- (Approximately)
Salient Features	• Damage is less when compared to manual cutting •

MANUALLY OPERATED CARROT SEEDER

Purpose	Suitable for sowing pelletized carrot seeds
Type of the implement	Manually operated
Field capacity	0.064 ha/day
Cost of the implement	Rs. 30,000 /- (Approximately)
Salient Features	• Pelletized carrot seeds can be sown in six rows. • Specially designed conical foam pad metering mechanism to avoid seed damage. • A single person can easy push the seeder easily on the seed beds.

iii. INTERCULTURAL EQUIPMENTS

TWO ROW FINGER TYPE PADDY ROTARY WEEDER

Purpose	For weeding in rice crops
Type of the implement	Manually operated
Field capacity	0.35 ha/day.
Cost of the implement	Rs. 1400 /- (Approximately)
Salient Features	• Two row paddy weeder • Row to row spacing is adjustable

BATTERY OPERATED PORTABLE WETLAND WEEDER

Purpose	For weeding in rice crop
Type of the implement	Manually operated
Field capacity	0.3 ha/day
Cost of the implement	Rs. 12,000 /- (Approximately)
Salient Features	• Single row paddy weeder • No drudgery involved for the operator

MULTI ROW POWER WEEDER FOR SRI

Purpose	For weeding in rice crop
Type of the implement	Self-propelled
Field capacity	Can weed 0.75 to 1.0 ha per day
Cost of the implement	Rs. 40,000 /- (Approximately)
Salient Features	• Weeding done by two rotary weeding units powered by 1.7 HP engine • Can be operated and lifted by one person easily to change rows. • Complete cutting of weeds at a depth of 3 to 4 cm with less than 1% plant damage.

SUGARCANE DETRASHER

Purpose	For detrashing sugarcane
Type of the implement	Manually operated
Cost of the implement	Rs. 1200 /- (Approximately)
Salient Features	• Labour requirement is less • Easy for handling • Reduced cost of de-trashing • Used for all varieties of cane • Also removes the sprouted buds

iv. HARVESTING AND THRASHING MACHINES

MINI COMBINE HARVESTER FOR PADDY

Purpose	For harvesting paddy
Type of the implement	Self-propelled
Field capacity	1 ha/day
Cost of the implement	Rs. 3,00,000 /- (Approximately)
Salient Features	• It performs different operations like paddy Harvesting, threshing, winnowing simultaneously • Suitable for small and medium farms

v. HORTICULTURAL IMPLEMENTS

TRACTOR OPERATED FRUIT-SHAKE HARVESTER

Purpose	For harvesting tamarind, and citrus fruits
Type of the implement	Tractor operated
Cost of the implement	Rs. 6000 /- (Approximately)
Salient Features	• Harvesting efficiency is 85 per cent • Savings in time is 95 per cent

TRACTOR OPERATED SINGLE / TWO ROW CASSAVA HARVESTER

Purpose	To dig cassava tubers in single / two rows
Type of the implement	Tractor operated
Field capacity	0.65 ha/day for single row and 0.96 ha/day for double row
Cost of the implement	Rs. 45,000/- for single row and Rs. 55,000/- for double row (Approximately)
Salient Features	• It is suitable for both single row and two rows operations. • The blade angle of 20 deg is provided for easy penetration in to the soil. • The row spacing can be altered by moving the shanks in the main frame. • The depth wheels are provided to the depth of operation.

COCONUTTREE CLIMBER

Purpose	To climb coconut trees for harvesting cleaning and other operations
Type of the implement	Manual operation
Field capacity	5-6 trees /hour
Cost of the implement	Rs. 5000 /- (Approximately)
Salient Features	• Even unskilled workers can use it to climb the tree with more stability and comfort. • Seating arrangement provides added comfort and safety. • It eliminates the severe bruises caused in traditional method of climbing due to use of climbing ropes.

IMPROVED COCONUT TREE CLIMBER

Purpose	To climb coconut trees for harvesting cleaning and other operations
Type of the implement	Manual operation
Field capacity	6 trees /hour
Cost of the implement	Rs. 5500 /- (Approximately)
Salient Features	• Lesser weight of the lower unit (3.0 kg) then existing model (6.0 kg) Lower unit is lifted simultaneously by leg and hand force for continuous operation • Comfortably designed upper frame

PALMYRAH TREE CLIMBING DEVICE

Purpose	To climb palmyrah trees for harvesting cleaning and other operations
Type of the implement	Manual climbing
Cost of the implement	Rs. 9000 /- (Approximately)
Salient Features	• Even unskilled workers can use it to climb the tree with increased stability and comfort. • The grippers are so positioned that while ascending/descending up/down, the upper frame accommodating the operator is always horizontal to the ground, irrespective of the girth variations in the tree. • Eliminates the high work stress, severe neck and back pain disorders caused in traditional method of climbing.

ARECANUT HARVETSER

Purpose	To climb arecanut trees for harvesting nuts
Type of the implement	Manual climbing
Cost of the implement	Rs. 9000 /- (Approximately)
Salient Features	• Even unskilled workers can use it to climb the tree with increased stability and comfort. • Lightweight aluminium pole with improved configuration of cutting edge of the knife for easy harvesting • Seating arrangement (adjustable and pivotable) with back rest for safe and secure operations Rotatable unit to facilitate harvesting of bunches form surrounding trees.

AERIAL ACCESS HOIST FOR COCONUT HARVESTING

Purpose	For harvesting coconut
Type of the implement	Tractor mounted
Cost of the implement	Rs. 8.50 lakh /- (Approximately)
Salient Features	• First machine of its kind in tractor mounted form • A full length chassis from front to rear of the tractor provides support • The entire weight of the hoist and moments transmitted through the chassis to the stabilizers without transferring to the tractor chassis. • Four trees can be accessed from a single position. • The time required for locating unit and operating stabilizers - 1 min. • The time required for positioning against a tree of 10 m height is 2 min. • The positioning of the operator platform can be done by the operator himself

TRACTOR OPERATED MULTI-PURPOSE HOIST

Purpose	Amenable for fruit plucking, coconut harvesting, training, pruning, lopping and spraying tree crops.
Type of the implement	Tractor mounted
Cost of the implement	Rs. 55,000 /- (Approximately)
Salient Features	• The equipment is attached to the back of a 45 hp agricultural tractor. • Two labourers can stand on the platform and do operations • Platform can reach a maximum height of 8.1 m

WORKER FRIENDLY ARECANUT STRIPPER

Purpose	Suitable for stripping both green and ripened arecanut.
Type of the implement	Electric motor/engine operated
Field capacity	Can strip 650-950 Kg of arecanut per hour
Cost of the implement	Rs. 25,000 /- (Approximately)
Salient Features	• Damage caused to the stripped arecanut is eliminated. • Results in 66 and 77 per cent saving in cost and time when compared to conventional arecanut stripping • Stripping efficiency of 99.5 per cent is achieved

TRACTOR OPERATED CLUSTER ONION HARVESTER CUM COLLECTOR

Purpose	To dig and collect cluster onion
Type of the implement	Tractor mounted
Field capacity	1.2 ha/day
Cost of the implement	Rs. 75,000 /- (Approximately)
Salient Features	• The cluster onion harvester has a special profile of blade to ensure shallow cut of soil and riddle conveyer for separating the soil from the onion bulbs. • Cross conveyer and also elevating conveyer are provided for easy and continuous movement of onion. • A bag is provided for collection of onion.

vi. MISCELLANEOUS MACHINERY

POWER TILLER OPERATED SLASHER CUM INSITU SHREDDER

Purpose	Suitable for shredding vegetable residues of brinjal, chillies, bhendi, etc. left after harvest and parthenium, etc.
Type of the implement	Power tiller operated
Field capacity	0.8 hectare can be shredded per day
Cost of the implement	Rs. 25,000 /- (Approximately)
Salient Features	• Suitable for any make of power tiller of 10-15 HP • Saving in time - 73 % • Saving in cost - 75%

SUBSOILER ATTACHMENT FOR STUMP REMOVAL

Purpose	Removes stumps in dryland
Type of the implement	Tractor operated
Cost of the implement	Rs. 14,000 (including subsoiler)
Salient Features	• Savings in cost by Rs. 1.50 per stump • Savings in time is 10-12 minutes per stump

TRAILER MOUNTED STEERING FOR POWER TILLER – TRAILER SYSTEM

Purpose	Steering of power tiller
Type of the implement	Power tiller operated
Cost of the steering system	Rs. 2,000 Approximately
Salient Features	• Avoids the operator getting down and turn the power tiller trailer system. • All the controls are well within the reach of the operator. • Shorter turning radius, enabling the operator to take turns even in very narrow space • Operator feels comfortable while taking a turn. • Reduced discomfort to the operator through elimination of lateral and vertical swing of the handle.

24. AGRICULTURAL PROCESSING EQUIPMENTS

TAMARIND DESEEDER

Salient Features:

- Suitable for deseeding dried dehulled tamarind fruits
- Various sizes of the dehulled tamarind fruits can be used for deseeding
- The roller gap can be adjusted as per the size of the tamarind fruits
- The deseeded fruits are separated into pulp strip, seeds and broken pieces
- The cost of the unit: Rs.1,00,000/-
- Capacity of the machine : 30-40 kg/h
- Cost of operation: Rs. 2.5/kg

Improved TNAU Dhal Mill

The salient features of the improved TNAU dhal mill:

- Suitable for splitting, cleaning and grading of pulses into dhal at the rate of 25-30 kg / hour
- Capable of dry milling of cereals into powder by changing into cast iron rolls
- Easy to operate and run by one H.P. single phase motor
- The unit has pitting unit for enhancing the preconditioning process
- Reduced conditioning time of 4-6 hours compared to 12 hours in conventional method
- Milling and grading efficiencies are more than 80%
- The cost of the unit is Rs.50000/-
- The cost of operation is Rs.2 per kg

25. RENEWABLE ENERGY ENGINEERING

(i) SINGLE POT CHULHA

The single pot chulha has a double wall with a gap of 2.5 cm. It has a grate at the bottom of the combustion chamber. Legs have been provided in the four corners of the chulha (5 cm height) as the ash can be collected below the grate. The outer wall has two rectangular secondary air openings at the lower portion on both sides for air entry. The inner wall has 1 cm diameter holes which maintain a triangular pitch of approximately 3 cm. The secondary air enters through the rectangular opening in the outer wall gets heated in the annular chamber and while moving up it passes through the holes in the combustion chamber. The preheated air helps in proper burning of the fuel. The efficiency of single pot improved chulha is 24%. The cost of the unit is Rs.350/-.

(ii) DOUBLE POT CHULHA

The TNAU double pot portable chulha (chimneyless) is made with two walls Around the first pot, an annular chamber having a width of 2.5 cm is left and the outer wall is constructed. The outer wall is also extended to cover the second pot in which case the annular chamber width is 3.5 cm, because of the smaller diameter of the second pot hole. Two secondary air inlets are made, one on the outer wall with rectangular shape (17 cm x 1 cm) near the combustion chamber and the other at the bottom of the second pot hole with round shape having a diameter of 5 cm. At the bottom of the first pot hole in the base, a hole of diameter 14 cm is made and a grate (C.I.) is placed over it. For the entry of secondary air to the first pot hole, 1 cm dia holes are made with a triangular pitch of 3 cm on the inner side of first pot hole and also on the tunnel projecting into the second pot hole. The efficiency of double pot improved chulha is 26%. The cost of the unit is Rs.600/-.

(iii)BIOMASS GAS STOVE

The biomass gas stove has been developed for small scale thermal application in agriculture and allied industries. This stove widens the market for agro wastes, makes possible a higher efficiency and in some cases reduces the time and investment, all by comparison with combustion. The biomass gas stove is a natural convection type updraft gasifier consisting of a cylindrical body made of clay, sand and paddy husk with its top open and bottom closed. The diameters and height of the stove are 290 mm and 630 mm respectively. This can be reduced depending on the applications. An iron grate to hold the biomass is fixed at 50 mm from the base of the reactor. The bottom is provided with an air opening cum ash removal door. At the top, provision is made to place vessel for cooking, boiling etc. The biomass viz., wood chips, agricultural residues like coconut shell, groundnut shell, arecanut husk, tree barks and leaves can be used in this biomass gas stove. The feedstock materials used should preferably be in the form of small chips, splinters and small logs. The fuel consumption is 5 kg/h and its thermal efficiency is 23%. The cost of the unit is Rs.1000/-.

(iv) DOWNDRAFT GASIFIER FOR WATER PUMPING

A downdraft gasifier along with a gas cleaning system retrofitted with a 5 Hp diesel engine coupled with a centrifugal pump for pumping water for irrigation applications. In order to supply producer gas for running the 5 Hp engine on dual fuel mode, a 10 kg/h capacity downdraft gasifier is required. A maximum of 50 percent diesel can be saved by substituting 50 percent of diesel with the producer gas obtained from biomass. And the cost of operating the gasifier coupled water pumping system is around Rs.50/h.

(v) SOLAR TUNNEL DRIER

The solar tunnel dryer is commercially used dryer for drying agro food products. Solar tunnel dryer is working based on Green House Effect principle. It results that 15-20°C increase in temperature above ambient temperature inside the solar tunnel dryer to dry the product from initial moisture to safe storage moisture. It mainly consists of semi-cylindrical tunnel structure, solar collector and cement concrete floor with absorber surface (Table 1). The semi cylindrical pipe structure covered with U.V. stabilized semi-transparent polyethylene sheet of 200-micron thickness. Drying floor with absorber surface is a cement concrete flooring laid at a gradient of 5° along the length with special black coating. The dryer is walk-in type and to facilitate loading and unloading of the product to be dried. The trays made of 8 SWG thick galvanized iron mesh with dimension of 1m x 1m can be used to increase the capacity. The dimensions of drying chamber (LxWxH) is 18m x 3.75m x 2.0m. The length can be varied depending upon capacity of the products to be dried.

(vi) BIOMASS HOT AIR GENERATION SYSTEM INTEGRATED WITH SOLAR TUNNEL DRYER

Renewable energy integrated drying system using solar thermal and biomass hot air with controlled environment can be used for continuous drying of agro-products.

- Solar mode is used for drying during sunshine hours and biomass mode is used during off-sunshine hours, cloudy weather.
- Suitable to dry coconut, turmeric, chillies, medicinal plants, vadam (food) products etc., with hygienic environment and enhanced quality compared to conventional open sun drying.
- Efficient (19 %) biomass combustor with heat exchanger is suitable for various biomass fuel such as coconut husk, coconut shell and wood logs.
- The integrated drying system attains the drying temperature ranged from 45 to 65^{o}C and has controls to maintain desired RH and temperature.
- Loading capacity is 500 kg to 2 tonnes per batch
- Drying time for coconut in integrated dryer is 48-52 h and 4-5 days in Solar tunnel dryer
- Cost of installation of integrated dryer is Rs.6.0 lakhs for 500 kg and Rs.8.0 lakhs for 2 tonnes/batch
- Reduces 35% drying time over solar tunnel dryer and 70 % over conventional open sun drying method.

(vii) NIGHT SOIL BIOGAS PLANT

Night Soil cum Kitchen Waste based Biogas Plant is the integration of modern biogas technology with disposal of night soil and food wastes generated from hostel kitchens and dining halls for harnessing bio-based energy.

- The hydraulic retention time of the plant is 45-55 days and human waste from 40 persons will be required for the production of 1 m^3.
- Biogas contains 65-70% methane and 30-35% carbon dioxide with traces of H_2S.
- Available at varying capacities from 1 m^3 to any higher capacity, based on the feedstock availability.
- Cost of installation is Rs. 30,000 for 1 m^3 and about Rs. 6.5 lakhs for 20 m^3 capacity.
- Biogas is utilised for cooking, lighting through mantle lamps, running engines, electricity generation, etc.
- Biogas burners, lamps and engines are available with hourly capacities from 8 to 100, 4-5 and from 50 cubic feet, respectively.
- Major advantages are sanitation and hygiene, environmental improvement, energy generation, provision of community facilities, etc.

(viii) BIODIESEL PILOT PLANT

The bio-diesel pilot plant consists of a transesterification reactor with heater, a stirrer, chemical mixing tank, glycerol settling tank (2 Nos.) and washing tank. The capacity of pilot bio-diesel plant is 250 litres/day. The cost of the pilot plant is Rs.5 lakhs. oil is mixed with alcohol and catalyst mixture in transesterification reactor. The reactor is kept at reaction temperature for specific duration with vigorous agitation. After reaction, the bio-diesel and glycerol mixture is sent to the glycerol settling tank. The crude bio-diesel is collected and washed to get pure bio-diesel. Depending upon the need, the size of the unit can be scaled up to get higher production.

26. HOME SCIENCE

FOOD PROCESSING TECHNOLOGIES

Utilization of rice bran in traditional breakfast foods

- Rice bran is a byproduct of rice, obtained on polishing. The bran is a rich source of protein, essential fats, B and E vitamins, minerals, fiber and antioxidants.
- Rice bran extracts in water can be used as a fiber free nutrient that has a lot of health giving properties.
- The rice oil is high in mono- and polyunsaturated fatty acids.
- Rice bran extracts can be used in a variety of preparations like bakery products and beverages.
- The Home Science College has standardised the preparation of flavored milk with the incorporation of rice bran extract.
- This beverage has antioxidant properties besides the soluble vitamins and proteins of bran.
- The heat stabilized bran can be incorporated in putt mix, ready to cook idiappam mix and spagetti, thus increasing the nutritive value of these products.

Sorghum Flakes

- Nutritionally superior to rice flakes.
- Ideal breakfast and snack food.
- Simple and low cost processing technique.
- Quick to cook and easily digestible.
- Good source of minerals and fibre.
- Sorghum flakes (100 g) contains 8.6 g protein, 3.7 g fat, 1.5 g fibre, 69 mg calcium and 16 mg iron.

Samai Biscuits

- Little millet (samai) is a good source of minerals, B vitamins and fibre.
- Fat, iron and niacin content are higher in little millet, than in other cereals.
- Calcium, phosphorus and iron content of samai biscuits are 25.8, 150 and 4.21 mg/100g respectively.
- High in fibre (1.40 %), and is important as health food.
- Has a shelf life of upto nine months.

Health Mix for Geriatrics

- The health mix designed for the aged was formulated from cereals, millets, pulses, and vegetables.
- In combination with milk powder and jaggery improved nutritional status of selected elderly subjects.
- Significant increase in haemoglobin level.
- The health mix (100 g) furnished 12 g protein, 2 g fat and 482µg β carotene.

Millet based health food mix

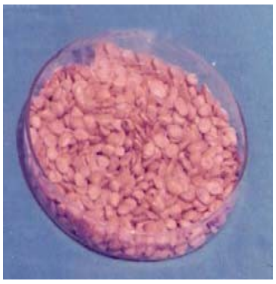
- Can be used for preparing nutritious balls and beverages.
- Easily digestible and good for children and aged persons.
- Health mix (100g) contains 9.75% moisture, 16.61% protein, 3.69% crude fibre,
  6.19 % reducing sugar 9.67 % total sugar, 20.68mg calcium, 244.00 mg phosphorous, 5.35 mg, potassium, 7.57 mg iron and 5.32 mg zinc.
- Has good storage stability.

Khakra

- Khakra is Indian flat bread.
- Traditional food of Gujarat, simulating chapathi in terms of product formulation and is further toasted to a moisture content of 5-6%.
- Incorporated with defatted soya flour (25%), the product is a rich source of protein.
- Addition of defatted soya flour increases protein (70%), iron (50%) and calcium (36%)

The product has a shelf life upto 90 days at room temperature.

Puffed Soya



- Puffing of soya reduces antinutrients and improves bioavailability of nutrients.
- Rich source of protein.
- Easily digestible and ideal protein source for vulnerable groups.
- Provides 32%protein.
- Low moisture content (6%).
- Has storage stability upto two months.

Okara Mix

- ❖ Okara is a by-product obtained during the processing of soy milk.
- ❖ Cheap source of protein and can be used to enrich traditional food products.
- ❖ Fresh okara contains 80% moisture, 13.7% protein, 1.5% fat, 4 % carbohydrate and 2% fibre.
- ❖ The processed okara in the ready to use form finds application in the preparation of traditional foods, bakery and confectionery products.
- ❖ Okara mix has a shelf life of 4 months.

Soya milk Fruit Juice

- ❖ Blending fruit juice and soya milk improves nutritive value and sensory quality.
- ❖ Rose and mango flavoured soya milk is highly acceptable.
- ❖ Contains 4 g protein, 2 g fat, 78 mg calcium, 21 mg phosphorous and 1.5 mg iron per 100 g of the fruit blended soya milk.
- ❖ Has a shelf life of 15 days under refrigerated condition.

  Best used in nutrition intervention programmes for the vulnerable groups.

Extruded products from texturised soya protein

- ❖ Extruded products (noodles) incorporated with texturised soya protein are richin protein.
- ❖ Addition of tomato juice further improved the quality of the noodles.
- ❖ Has good sensory appeal
- ❖ Texturised vegetable protein incorporated noodles is suitable for children.

Texturised soya noodles contains 20.3 g protein and 72 mg calcium

Millet based value added products

- ❖ Millets are miles ahead of rice and wheat in terms of their nutritional content. Millets are good source of minerals and dietary fibre.
- ❖ The nutrients present in the millets have the capacity for reducing the risk of life style diseases.
- ❖ Due to urbanization, climatic changes, erratic rainfall etc, the farmers are forced to seek alternative crops for rice.
- ❖ Forecasting the future need of our country millet based products like multigrain adai mix, multipurpose snack mix, health mix, samosa mix, karasev mix, pongal mix, priyani mix, flakes etc., were developed.
- ❖ The processed millet based products has six months shelf life and it suits the convenience seeking farmers.

Sugarcane Syrup

- Concentrated sugar cane juice.
- Alternative natural sweetener in the place of refined sugar.
- Convenient and ready to use, has wide application in the preparation of traditional, bakery and confectionary products.
- Sugarcane syrup (100 g) contains 43 mg calcium, 42 mg phosphorus and 2 mg iron.
- Has a shelf life of twelve months at room temperature.

At Home Science College and Research Institute, a Food Processing Training Center was established by installing the processing equipments namely Murukku machine, Sieving machine, Mixture machine, Steaming machine, Tray wrapping machine, Handy induction sealing machine, Continuous sealing machine, Nitrogen flush vaccum packaging machine, Pulveriser, Extruder, Milk Extraction unit, Cabiner drier and other processing accessories to train the entreprenuers, farmers, SHGs on processing of millets based value added products. The equipments are available to the budding entrepreneurs on rental basis to utilize the facilities.